



UPSC Prelims

Science & Tech

Class 1

Orientation, Biotechnology Basics



First Class : Science and Technology for UPSC Prelims 2023



- Orientation : Basic details about the course
- What you can expect from me
- What I expect from you
- Doubts / Queries and Everything else
- Academic : Basics of Biotechnology
- Some PYQs

Orientation : Teaching methodology and notes



- Teaching methodology : Slides on the digital board
- Static First, for conceptual clarity.
- Current Affairs integration
- PYQs : 2013 to 2022
- Notes : Slides will be shared, without annotation
- One Pager Notes at the end of the class
- Both of these can be downloaded from the portal.

Orientation : Class Schedule



Date	Day	Subject
15th March	Wednesday	Orientation and Biotechnology Basics
16th March	Thursday	Basic Biology: Cells and Genetics
17th March	Friday	Principles of Biotech and their applications
18th March	Saturday	Biotechnology Current Affairs
20th March	Monday	Health Issues
21st March	Tuesday	Nanotechnology
22nd March	Wednesday	Space Technology

23rd March	Thursday	Space and Défense 1
24th March	Friday	Défense 2
25th March	Saturday	Emerging Technology and Computers 1
27th March	Monday	Emerging Tech and Computers 2
28th March	Tuesday	Energy : Basics
29th March	Wednesday	IPR and Role of Scientists
30th March	Thursday	Reserve Class
31st March	Friday	Reserve Class

Orientation : Class timing



- Classes will be held from Monday to Saturday.
- No Class on Sunday : Your time to revise
- Class timing : **12 NOON**

Orientation : Doubts and Queries



- Answering doubts is extremely important.
- Completing the course on time is also very important.
- In the live class, I will ask you to participate. Put all your comments / queries / suggestions in the discussions board below. Whatever I can address during the class, I will.

Orientation : Doubts and Queries



- Remaining questions will be dealt with on the discussion board, and you will get the answers to your questions.

What you can expect from me



- Firstly, Thank you for placing your trust in me. I will give my best here.
- Coverage of all the static themes in science and technology.
- Barring any force majeure, the course will be finished on time and as per schedule.

Topics To Be Covered



Biotechnology



Health Issues



Nanotechnology



Space Technology



Defence Sector



Computers and
Information Technology



Energy



Intellectual
Property Rights



Role of Indian
Scientists

What you can expect from me



- Coverage of All Current Affairs : You DO NOT NEED TO READ PT 365 : I will cover the PT-365 of past 2 years.
- Coverage of ALL PYQs from 2013 to 2022
- Relevant Content : No mushrooming of Content. Only what is relevant. Our target here is to get a good percentage of the questions right.
- Our goal is not to become a scientist, but to get 100+ marks in prelims.

Utilizing this course effectively



- 15th March to 31st March : Science and Technology
- 3rd April to 20th April : Environment

March

2023

Sun	Mon	Tue	Wed	Thu	Fri	Sat
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

April

2023

Sun	Mon	Tue	Wed	Thu	Fri	Sat
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

Utilizing this course effectively



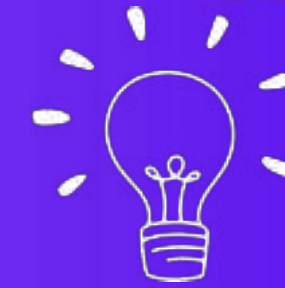
- Attend the courses live
- Spend 2-3 Hours here, after that DO NOT take a break.
- Spend at least 15 minutes revising what we have learnt from the slides and one pager notes.
- Next morning : Revise the previous session before coming here.
- Out of your 8 hours of daily study time, I'm asking for 4 hours.

What I expect from you



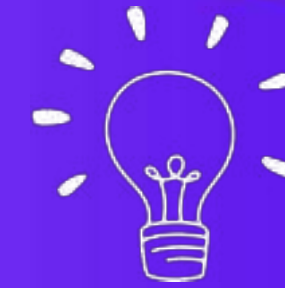
- Work hard and listen effectively.
- Use Active Recall to Revise.
- This is what your ideal schedule looks like :
- Revise yesterday's class in the morning
- 12pm onwards : Attend the class here
- After the class : Revise from the one pager for 15-20 minutes.
- Post lunch : Study a different subject / give test
- Sunday : Revise whatever has been taught the entire week.

What I expect from you



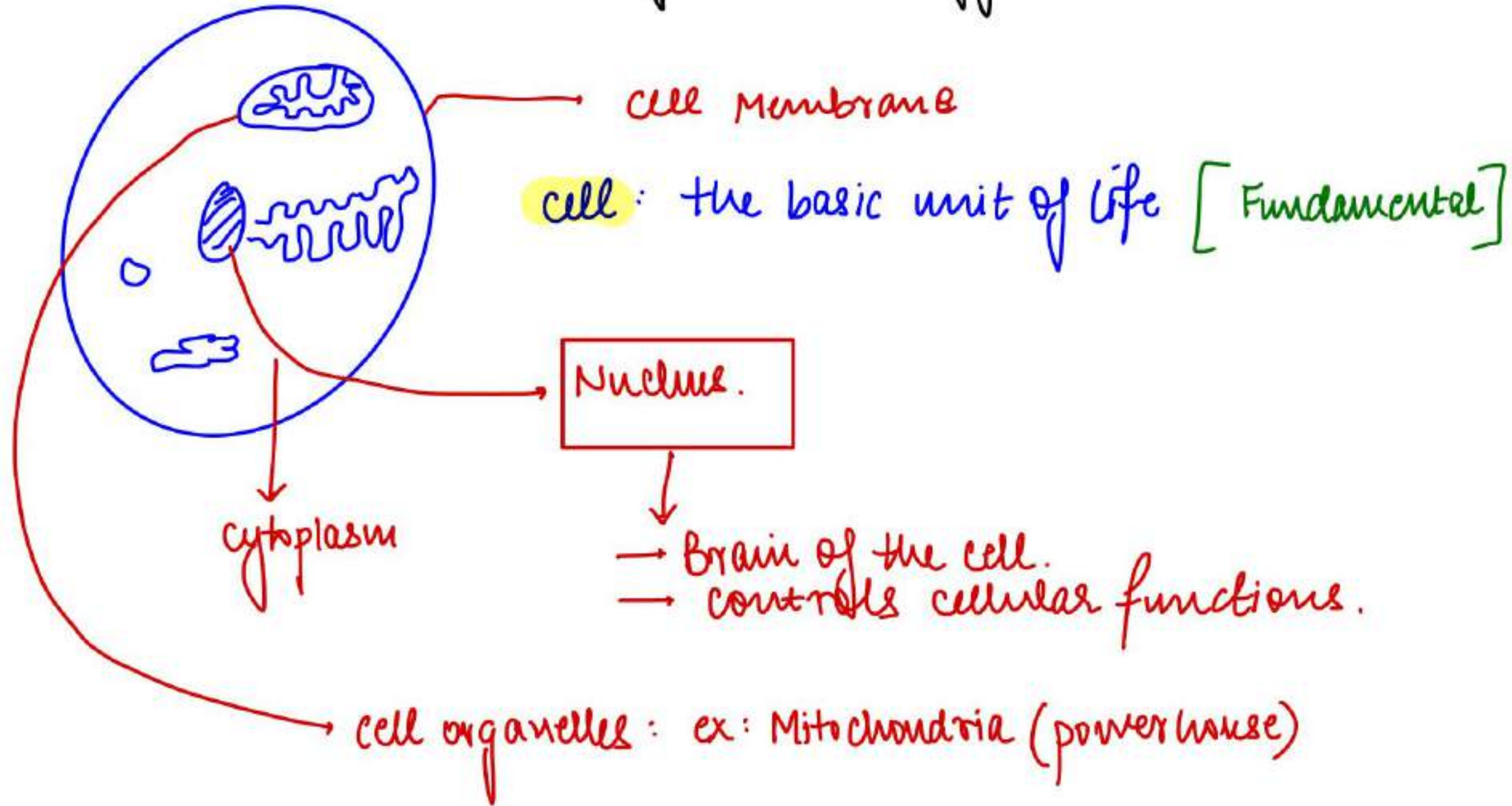
- Live attendance of class Regularly
- Proper timely revision, daily and on Sunday
- Result : From today till 20th April, you will be done with Science and tech and Environment. Your 2 biggest fears will be sorted.
- If you follow the plan, you will have a set of one pagers which you will have revised from.
- 50-60 pages with relevant content that you can effectively revise in one day.

LET'S GET STARTED!

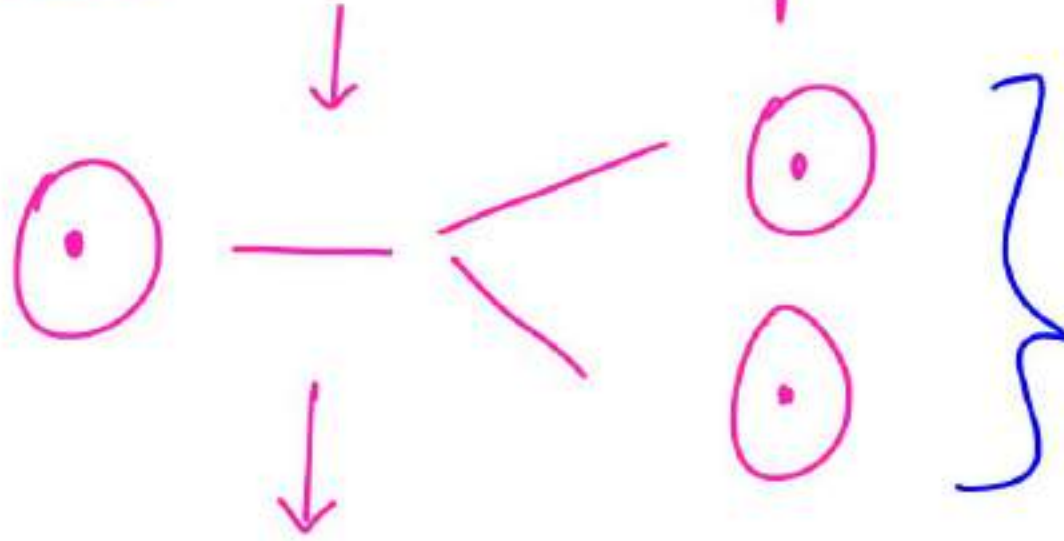


- First topic to cover : Biotechnology Basics.
- Some of this was covered in the demo class, but I will start assuming that you have not seen any of it.
- So if you have not read any science/ environment before today, don't worry. Leave it to me.
- Just stay attentive.
- First we will study biotechnology.

Basics of Biotechnology.



cell has to reproduce



information for transmission
has to be stored somewhere.

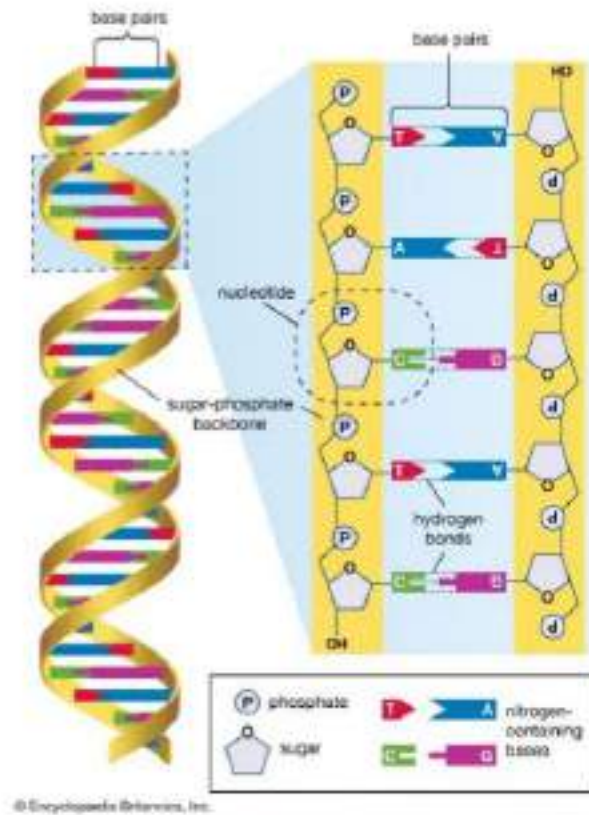
growth can occur only when cells
reproduce.

important information to
run the cell has to go from
one cell to the next.

This information needs to be stored
somewhere.

'information' → DNA
(~~deoxy~~ribo nucleic acid)
↓ ↓
Ribose acidic
Nucleus

DNA looks like:
double helix
structure.



Let's see the condensation

↑
Need to condense

↑
solution

↑
every cell has complete DNA
Forensic

↑
problem: very long (all is small)

What do we know so far?

cell has to reproduce.

For reproduction, information has to pass to daughter cells.

DNA is the information inside a cell.

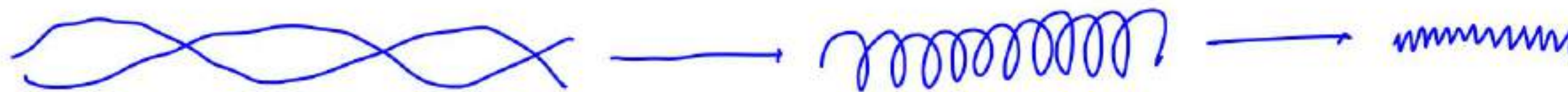
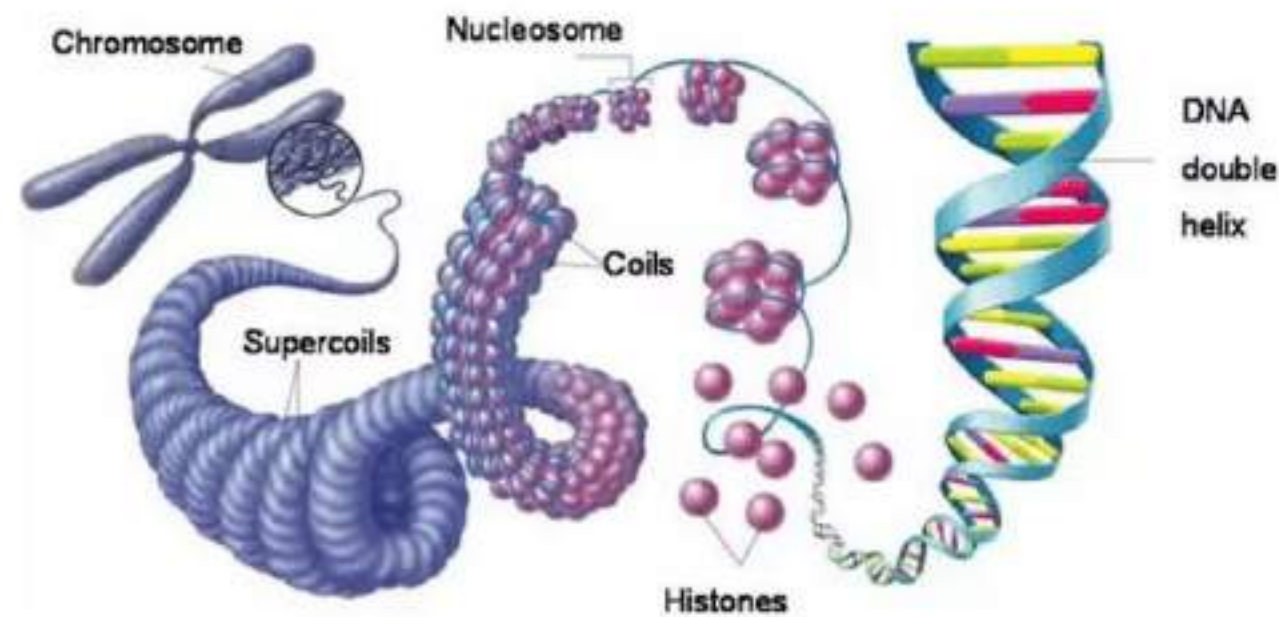
every cell has DNA

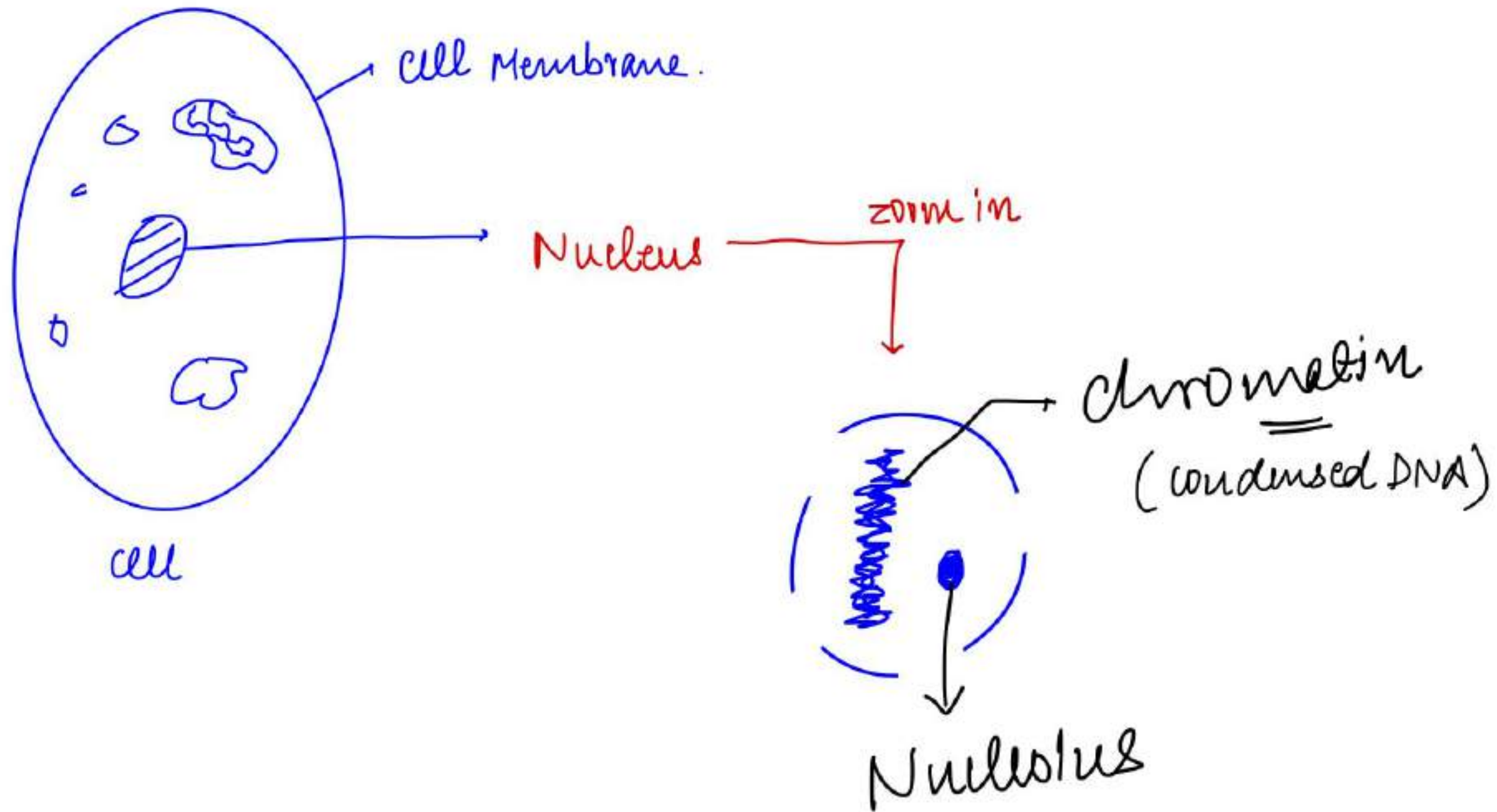
DNA is present in the NUCLEUS.

DNA has to be condensed as it is very long.

DNA is a double helix.

DNA $\xrightarrow[\text{Histones}]{\text{condensation}}$ Chromatin $\xrightarrow[\text{condense}]{\text{further}}$ Chromosomes.





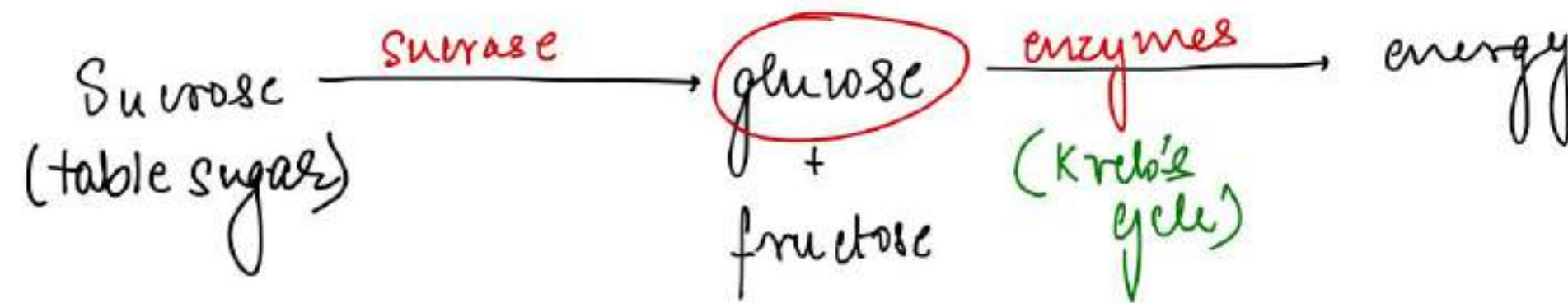
We know: DNA is the information.

⇓
But WHY?

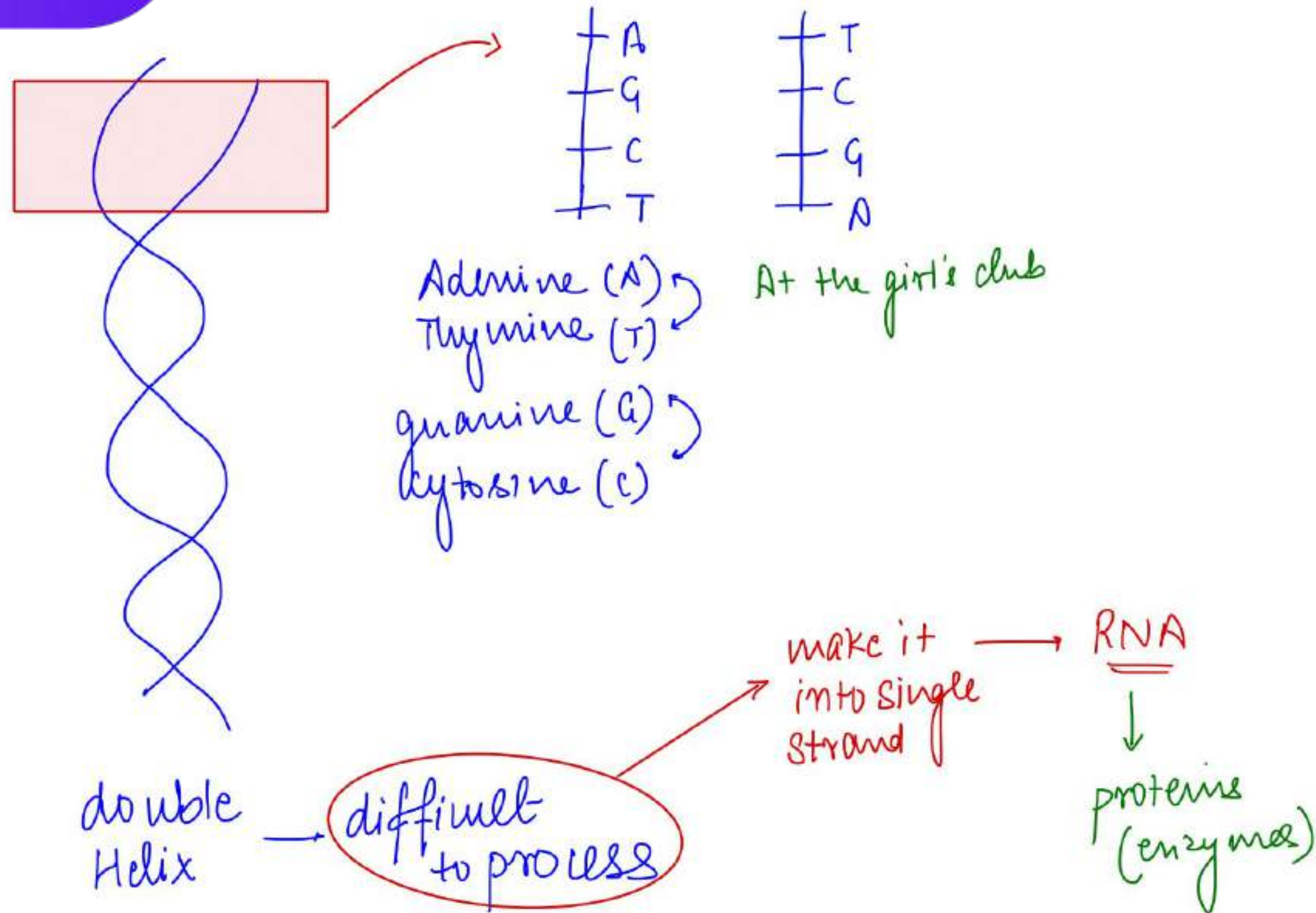
DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ proteins (enzymes)
↓
Ribonucleic acid.

2 questions {
→ why RNA?
→ why protein?

Why proteins (enzymes)?



For a cell to reproduce $\rightarrow$ New cell has to form $\rightarrow$ New cell must have all essential enzymes
enzymes come from DNA.

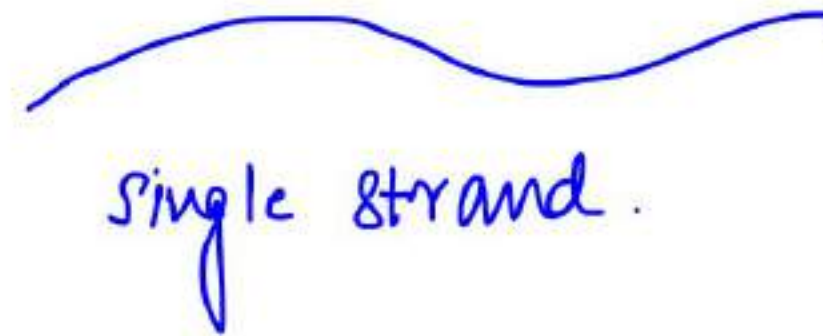


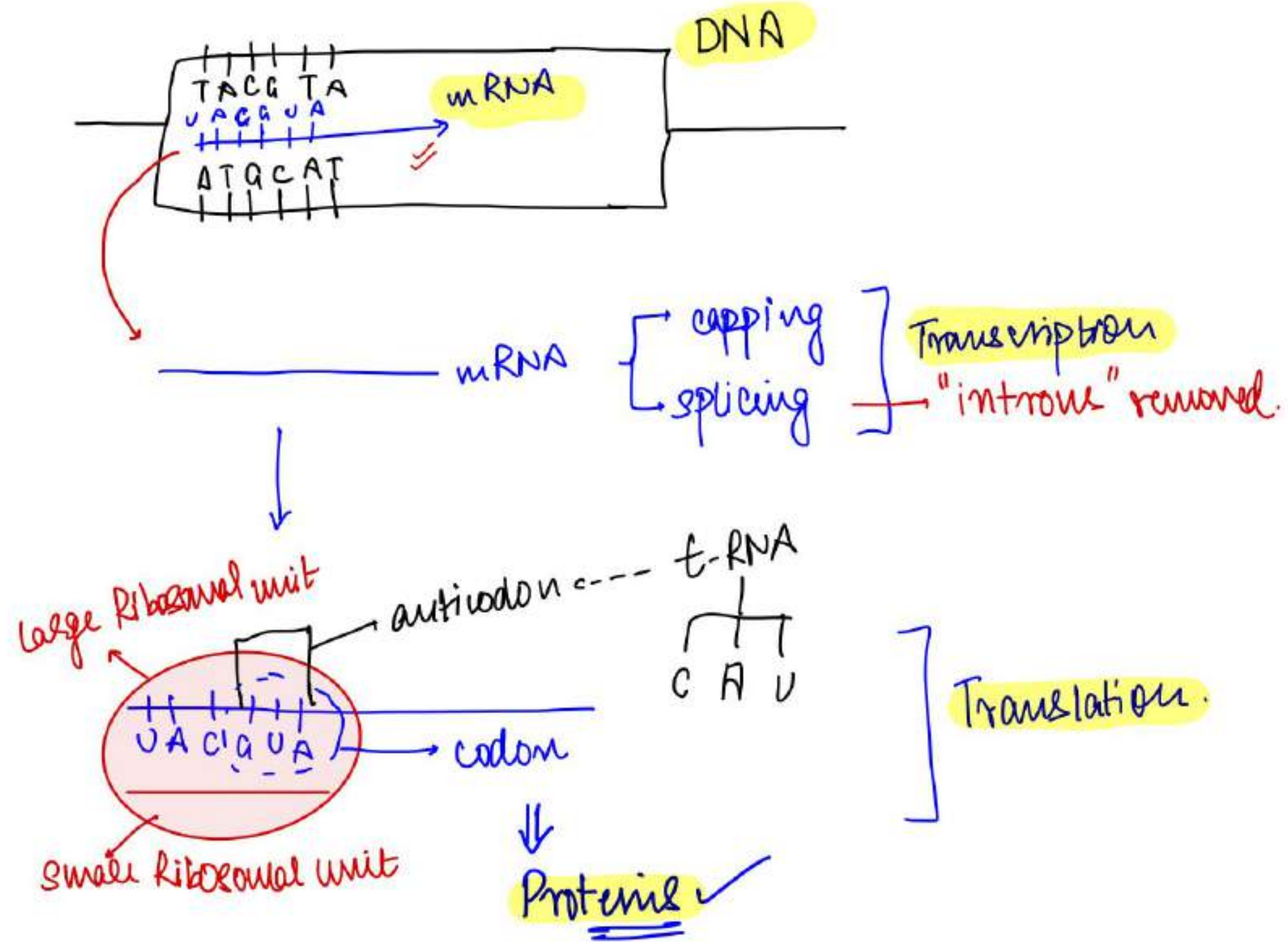
DNA

double
HelixDNA $\xrightarrow{\text{transcription}}$ RNA

v/s

RNA

RNA $\xrightarrow{\text{translation}}$ proteinsU $\rightarrow$ Uracil.



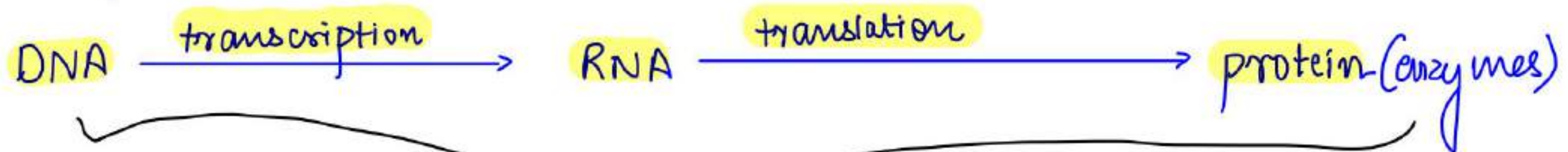
What do we know?

cell has to reproduce.

For reproduction, information has to pass to daughter cells.

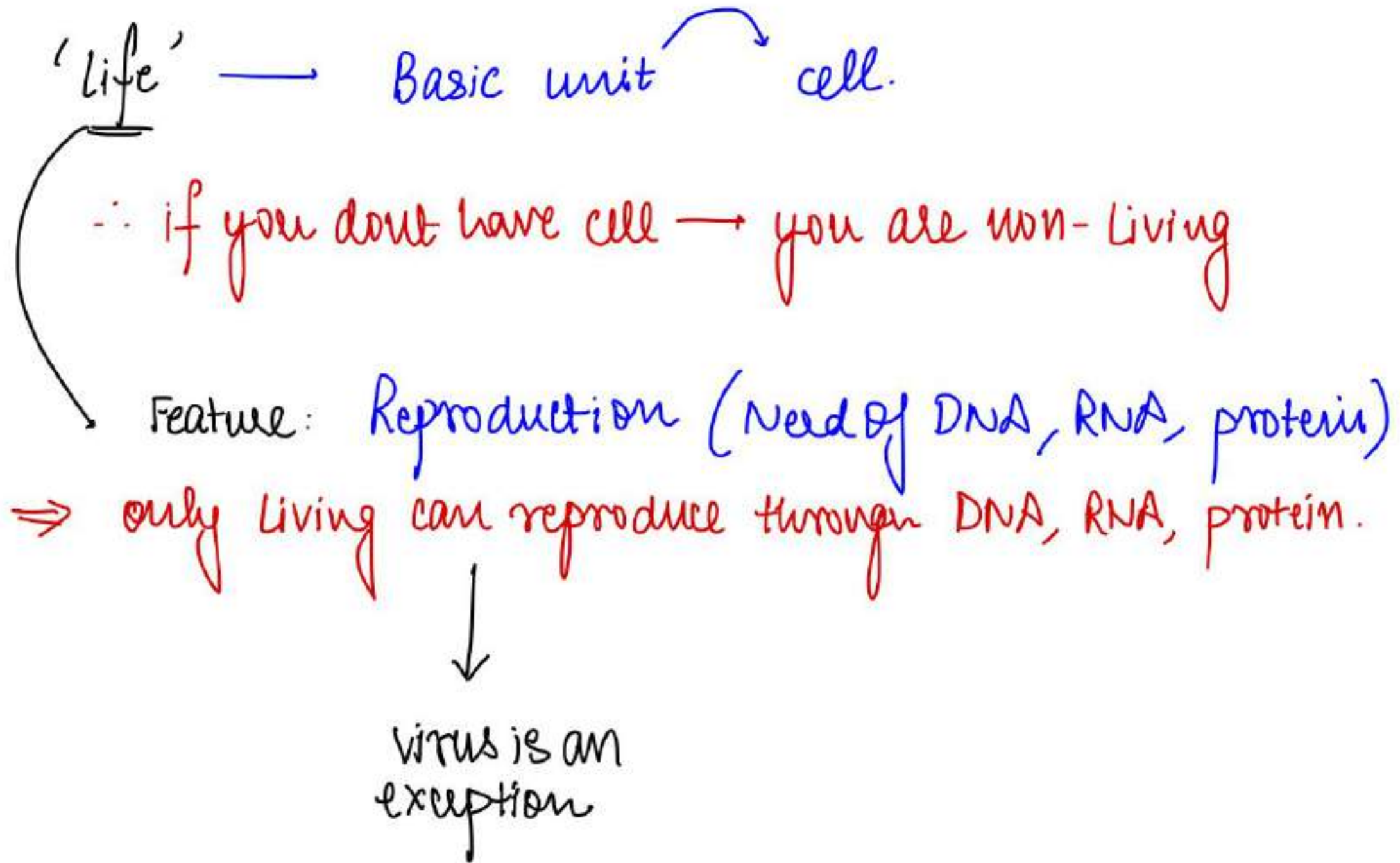
DNA is the information inside a cell.

every cell has DNA



central law of genetics : living cells.

Recap



Viruses.

- cell → Basic unit of life.
- virus → Borderline between living and non living.

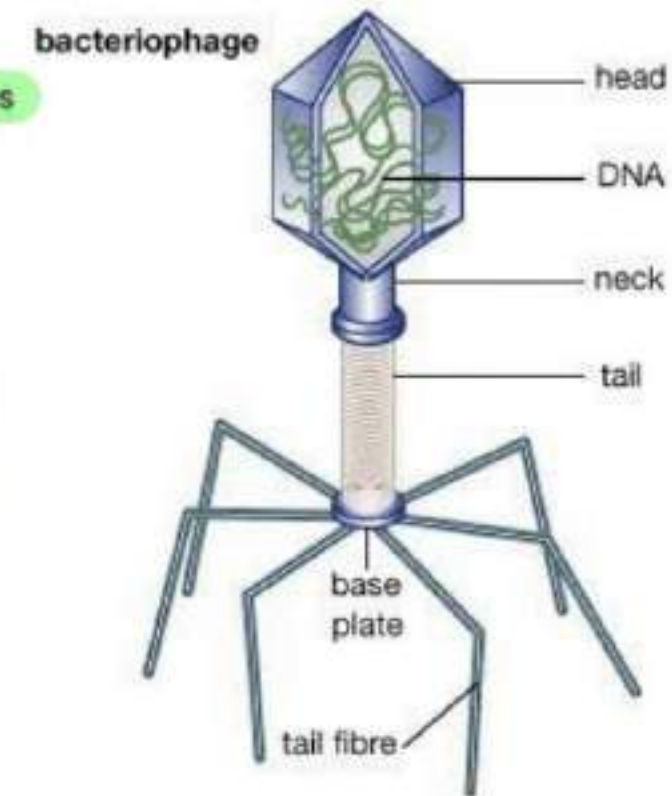
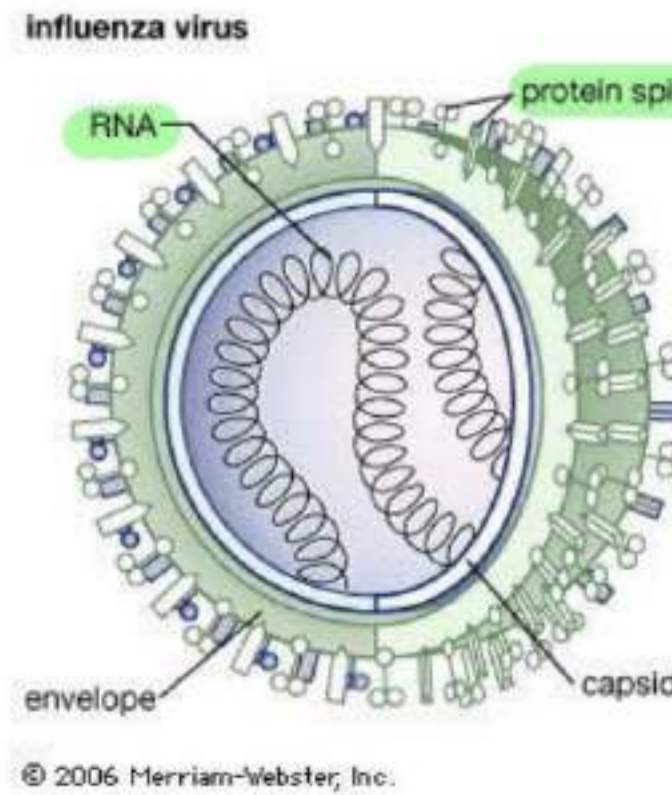
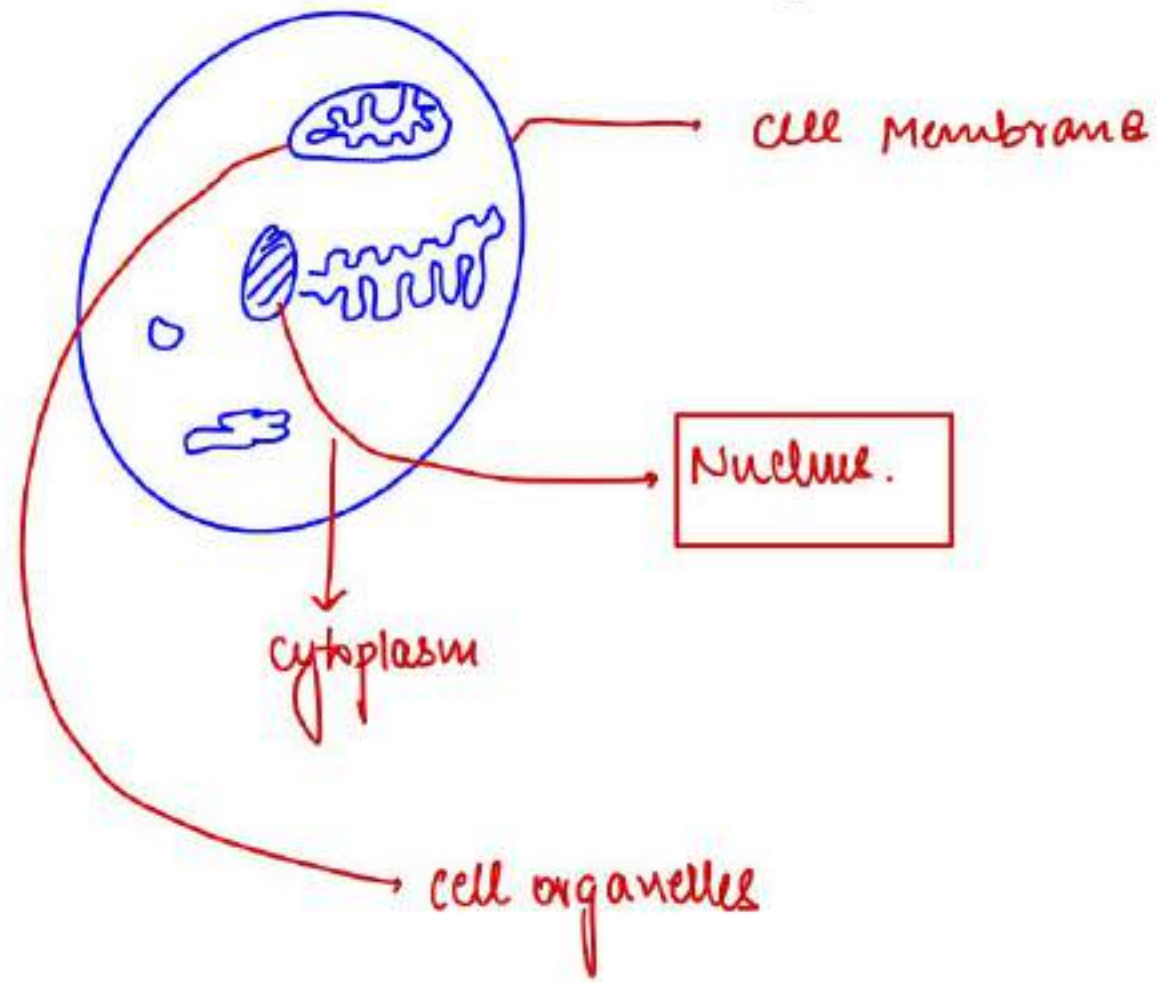
↓
Why?

- viruses have DNA or RNA and some proteins
 - They can reproduce in host cell.
- They don't have their own cellular structure.

Let's compare.

cell

virus.



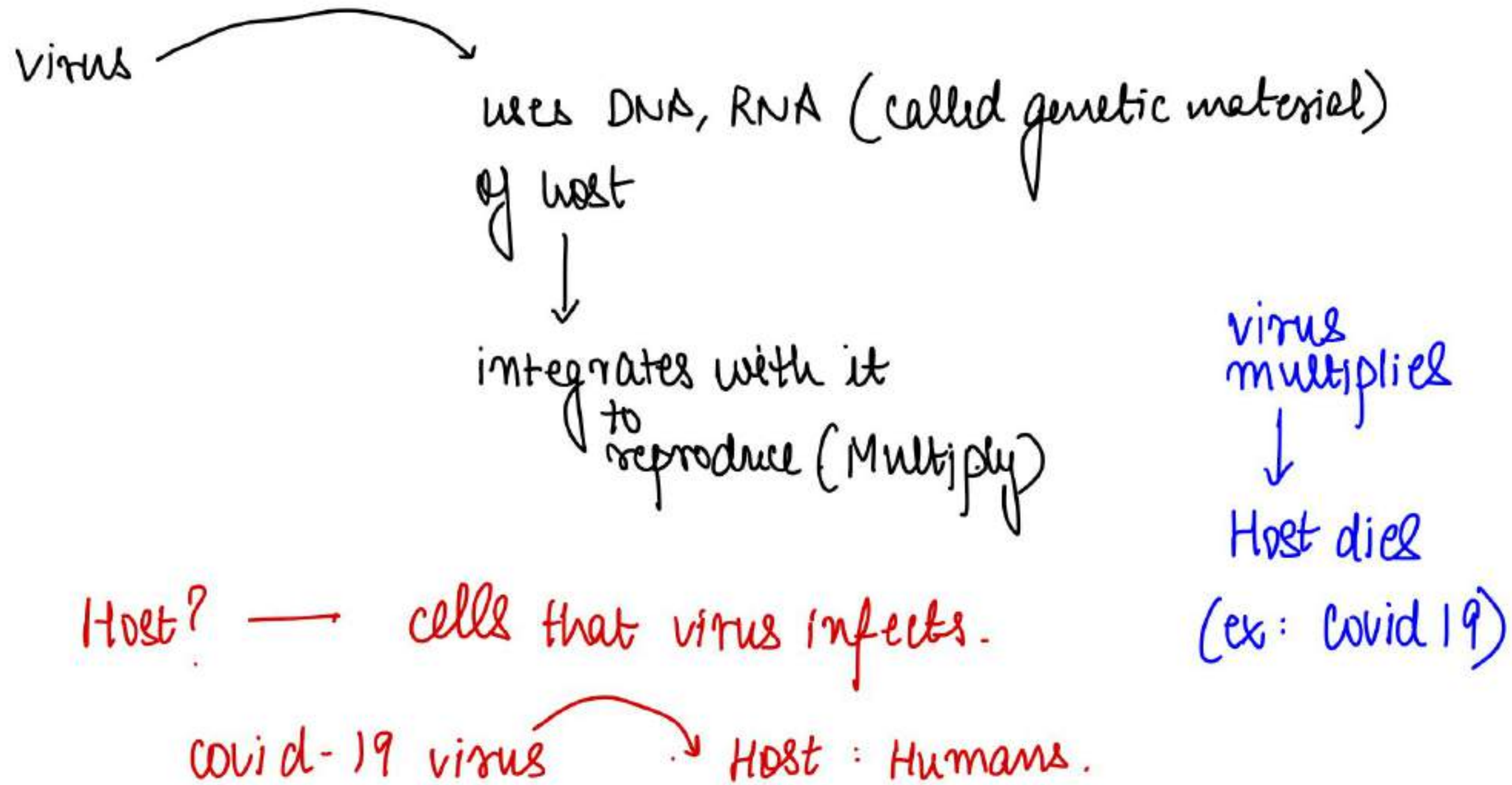
Features of a virus.

- ① Has DNA, RNA, protein
 - ② Can Reproduce, inside a host cell.
 - ③ No cellular structure.
- } *Borderline.*

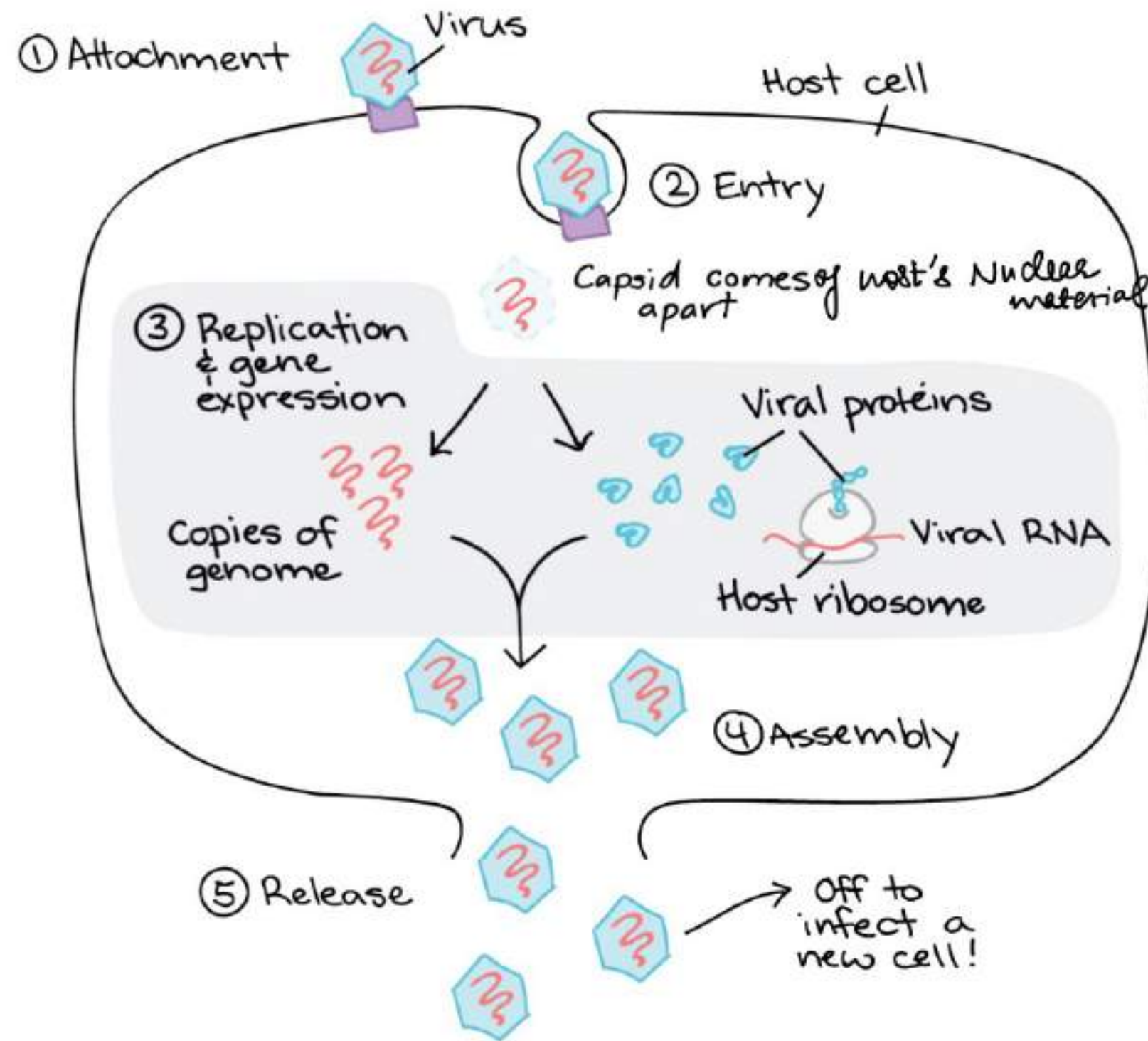
How do they reproduce inside
host cell?

↳ think

if you cannot.....
you ask your friend.



GENERAL DIAGRAM of a VIRUS LIFECYCLE



original infected

DNA ← --- viral DNA

↓

RNA ← --- virus RNA

↓

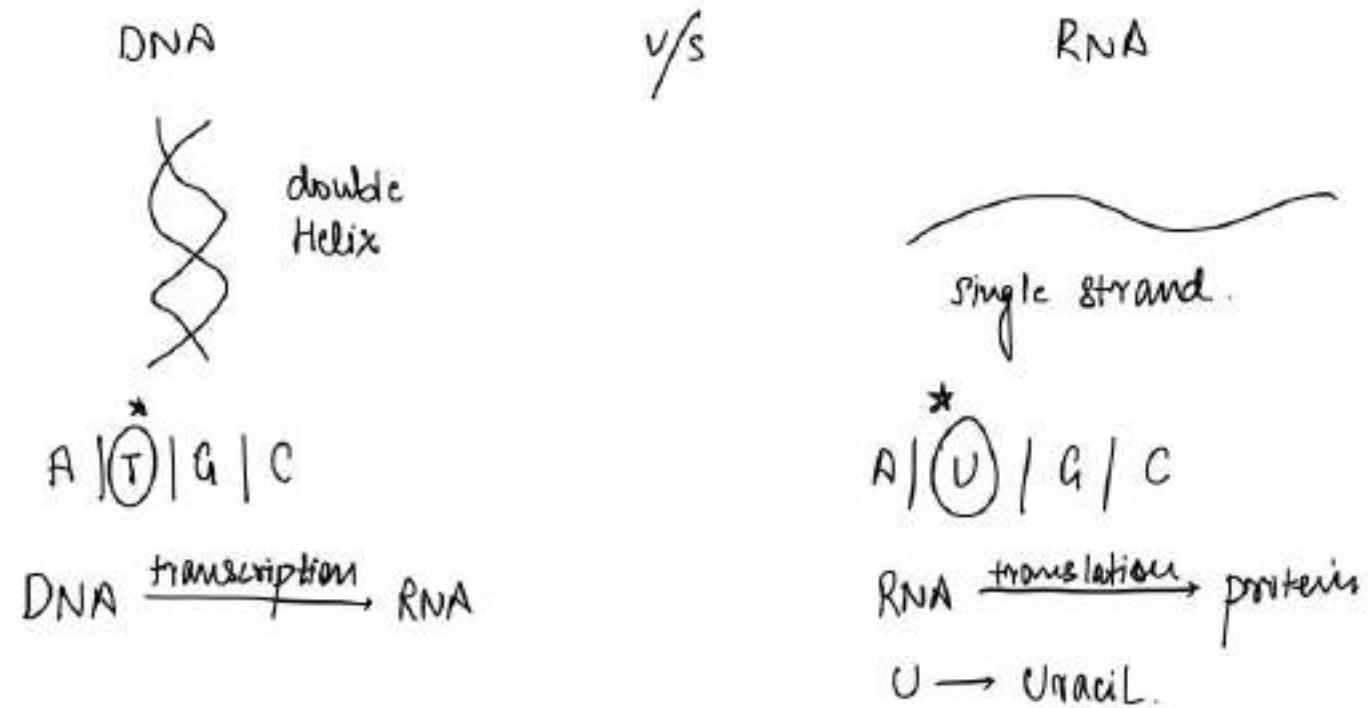
protein ← --- virus proteins

Original Process

DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ Protein

Virus: Starts with RNA $\rightarrow$ But needs to integrate with DNA so that it can use Host cell mechanisms

Remember:

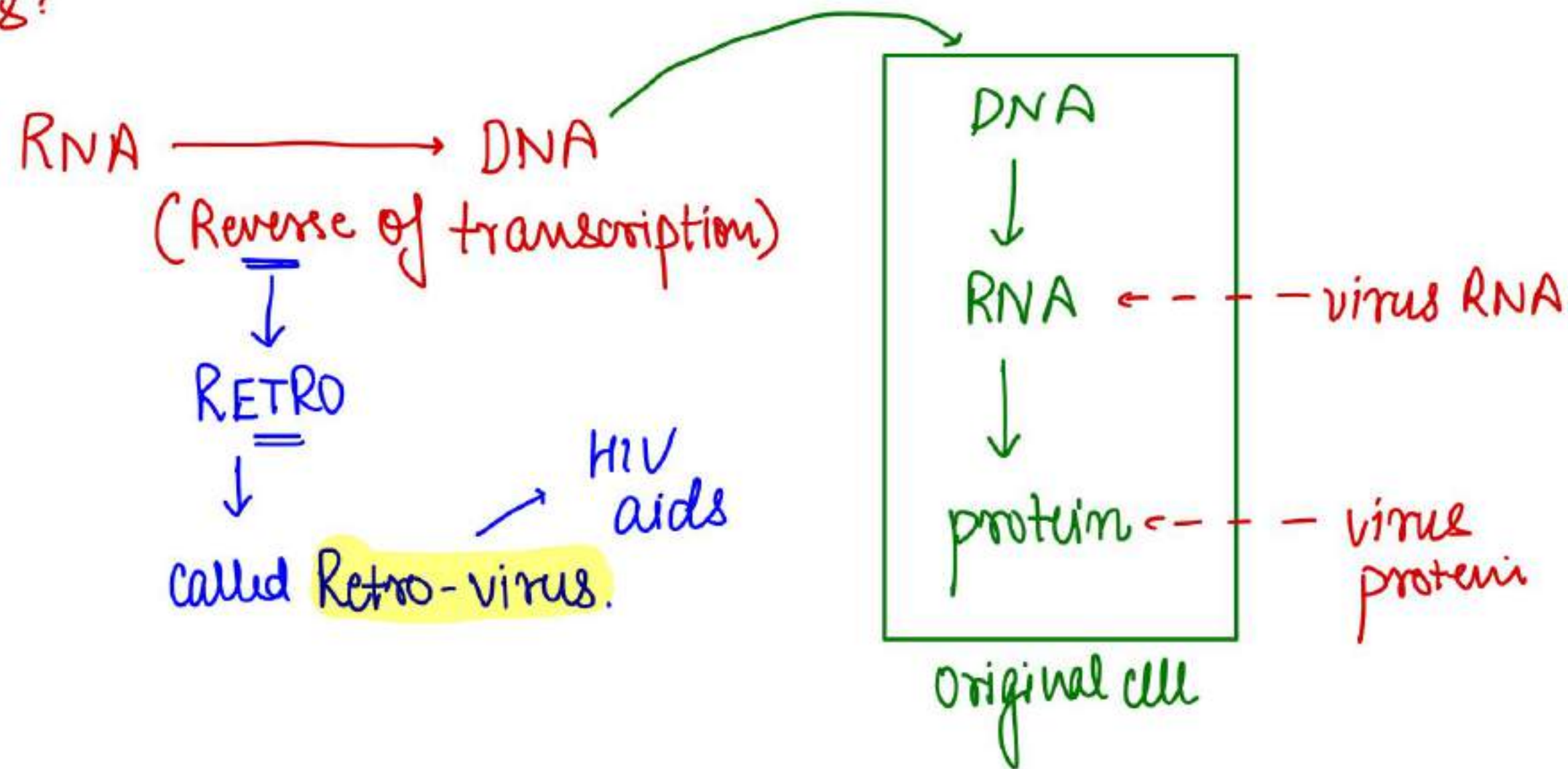


What does virus need to do?

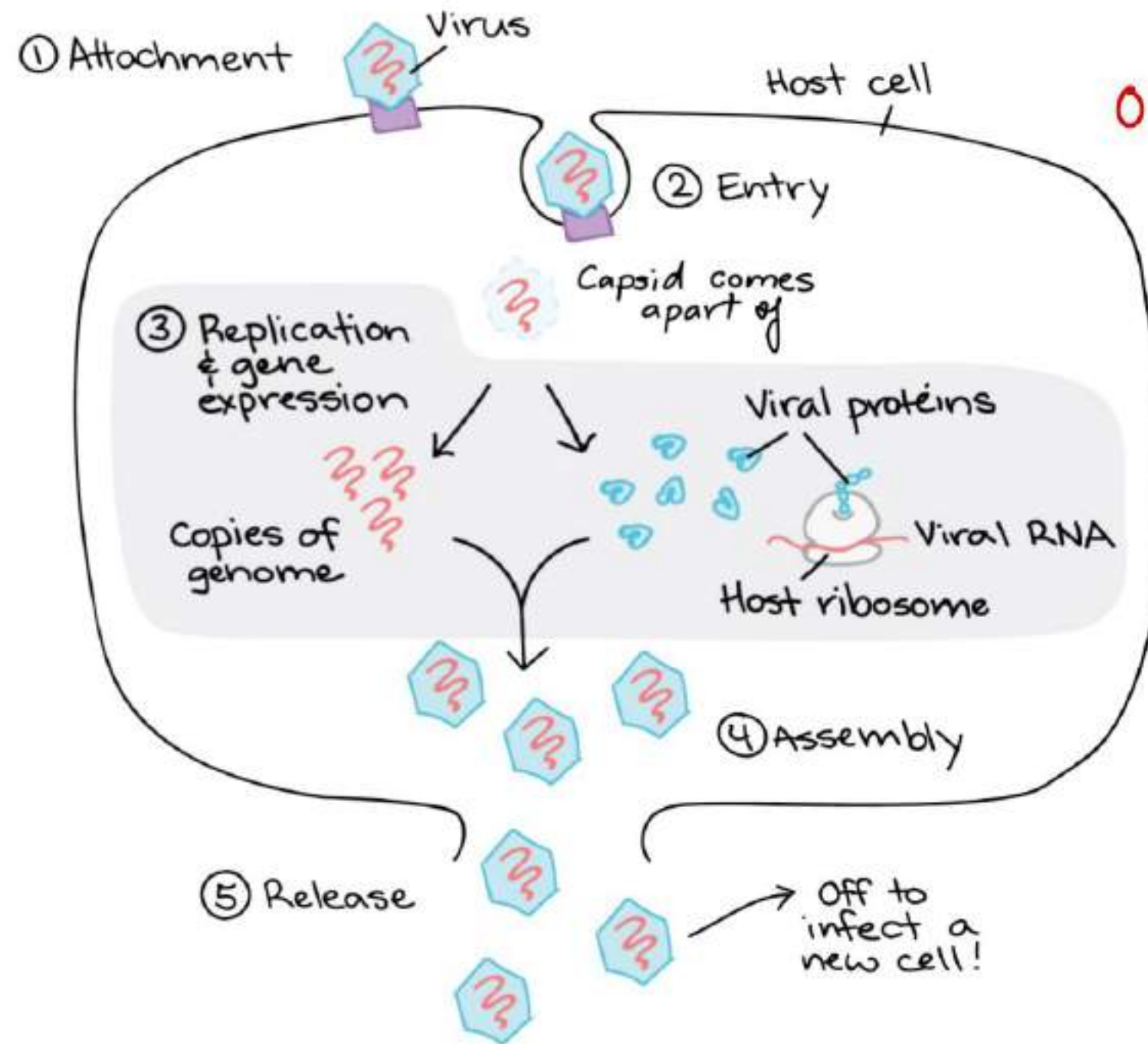
$\downarrow$
think (dies w/o petrol)

- virus needs to convert its own RNA $\rightarrow$ DNA
- then virus can easily integrate $\bar{c}$ host cell.

Virus:



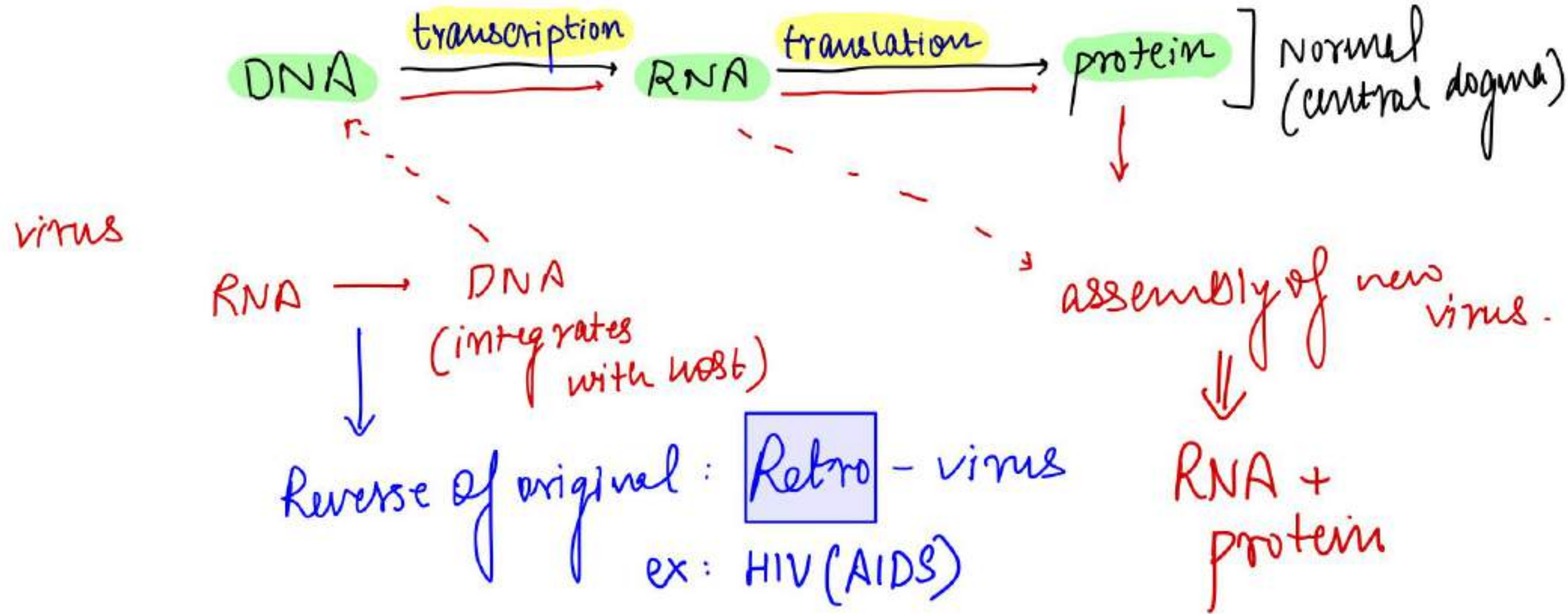
GENERAL DIAGRAM of a VIRUS LIFECYCLE



original virus:
RNA +
protein

through
HOST
cell

Many RNA +
Many protein
↓
Many virus.



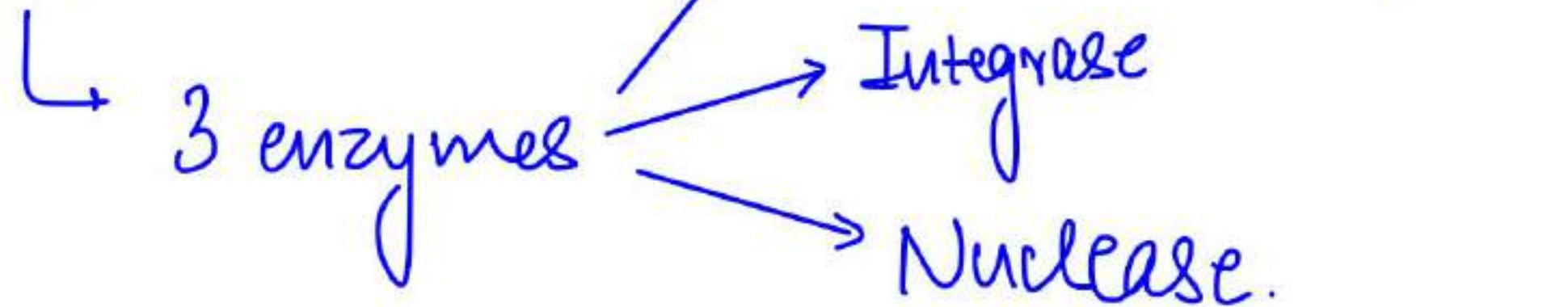
★ Remember: virus can do this only inside HOST, NOT independently.

Now, I will show you a video of HIV infecting a Cell

HIV

- infects CD4+ T Lymphocyte.
- CCR5 / CXCR4 Receptor

· Enters the cell



- we can block it at any stage to prevent HIV infection.

Let's Solve Some Previous Year Questions!

Previous Year Questions: 2016

Q. In the context of the developments in bioinformatics, the term 'transcriptome', sometimes seen in the news, refer to -

- a) a range of enzymes used in genome editing
- b) the full range of mRNA molecules expressed by an organism
- c) the description of the mechanism of gene expression
- d) A mechanism of genetic mutation taking place in cells

Answer: **B**

- The transcriptome is the set of all RNA molecules in one cell or a population of cells.
- It is sometimes used to refer to all RNAs, or just mRNA, depending on the particular experiment.
- It differs from the exome in that it includes only those RNA molecules found in a specified cell population, and usually includes the amount or concentration of each
- RNA molecule in addition to the molecular identities

Previous Year Questions: 2016

Q. Which of the following statements is/are correct?

Viruses can infect -

1. Bacteria
2. Fungi
3. Plants

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Answer: D

- A virus is a small infectious agent that replicates only inside the living cells of other organisms. Viruses can infect all types of life forms, from animals and plants to microorganisms, including bacteria and fungi.
- According to the type of the host they infect, viruses are classified mainly into the following four types:
 - a. Plant viruses including algal viruses-RNA DNA
 - b. Animal viruses including human viruses- DNA/RNA
 - c. Fungal viruses(Mycoviruses)-ds RNA
 - d. Bacterial viruses (Bacteriophages) including cyanophages-DNA

Previous Year Questions: 2021

Q.Consider the following:

1. Bacteria
2. Fungi
3. Virus

Which of the above can be cultured in an artificial/synthetic medium?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Answer: A

- Bacteria and Fungi can be cultured in an artificial/synthetic medium. Whereas viruses require a living host cell for replication. Infected host cells (eukaryotic or prokaryotic) can be cultured and grown, and then the growth medium can be harvested as a source of the virus. Hence, 1 and 2 are correct and 3 is not correct.
- Therefore, option (a) is the correct answer

Previous Year Questions: 2013

Q. Which of the following statements is/are correct?

1. Viruses lack enzymes necessary for the generation of energy.
2. Viruses can be cultured in any synthetic medium.
3. Viruses are transmitted from one organism to another by biological vectors only.

Select the correct answer using the codes given below

- a) 1 only
- b) 1 and 2 only
- c) 1 and 3 only
- d) 1, 2 and 3

Answer: A

- Since Viruses lack organelles and are totally dependent on the host cells metabolic machinery for replication, (hence) they cannot be grown in synthetic media.
- In the laboratory, animal viruses are grown in animals or animal cell culture and plant viruses are grown in plants or plant cell culture.
- There are two categories of Vectors: biological and mechanical.
- As was stated earlier, if the virus increases its number while in association with a vector, then that vector is termed as being biological.
- Conversely, the vector is termed to be mechanical, if the virus doesn't increase its number while in association with that vector.
- Arthropods such as wasps, which repeatedly can sting multiple animals, could serve as mechanical vectors by transporting viruses on the surface of their stingers.

Previous Year Questions: 2021

Q. Consider the following statements:

1. Adenoviruses have single-stranded DNA genomes whereas retroviruses have double-stranded DNA genomes.
2. Common cold is sometimes caused by an adenovirus whereas AIDS is caused by a retrovirus.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Answer: B

- Adenovirus is a type of virus that has no envelope whereas retroviruses are characterized as enveloped viruses. Adenoviruses have double-stranded linear DNA and are associated with two major core proteins. A retrovirus is a virus that uses RNA as its genetic material. When a retrovirus infects a cell, it makes a DNA copy of its genome that is inserted into the DNA of the host cell. Hence, statement 1 is not correct.
- Adenoviruses are common viruses that cause a range of illnesses. They can cause cold-like symptoms, fever, sore throat, bronchitis, pneumonia, diarrhoea, and pink eye (conjunctivitis). Whereas, retroviruses can cause several human diseases such as some forms of cancer and AIDS. Hence, statement 2 is correct.
- Therefore, option (b) is the correct answer.

Previous Year Questions: 2022

Q. Consider the following statements: DNA Barcoding can be a tool to:

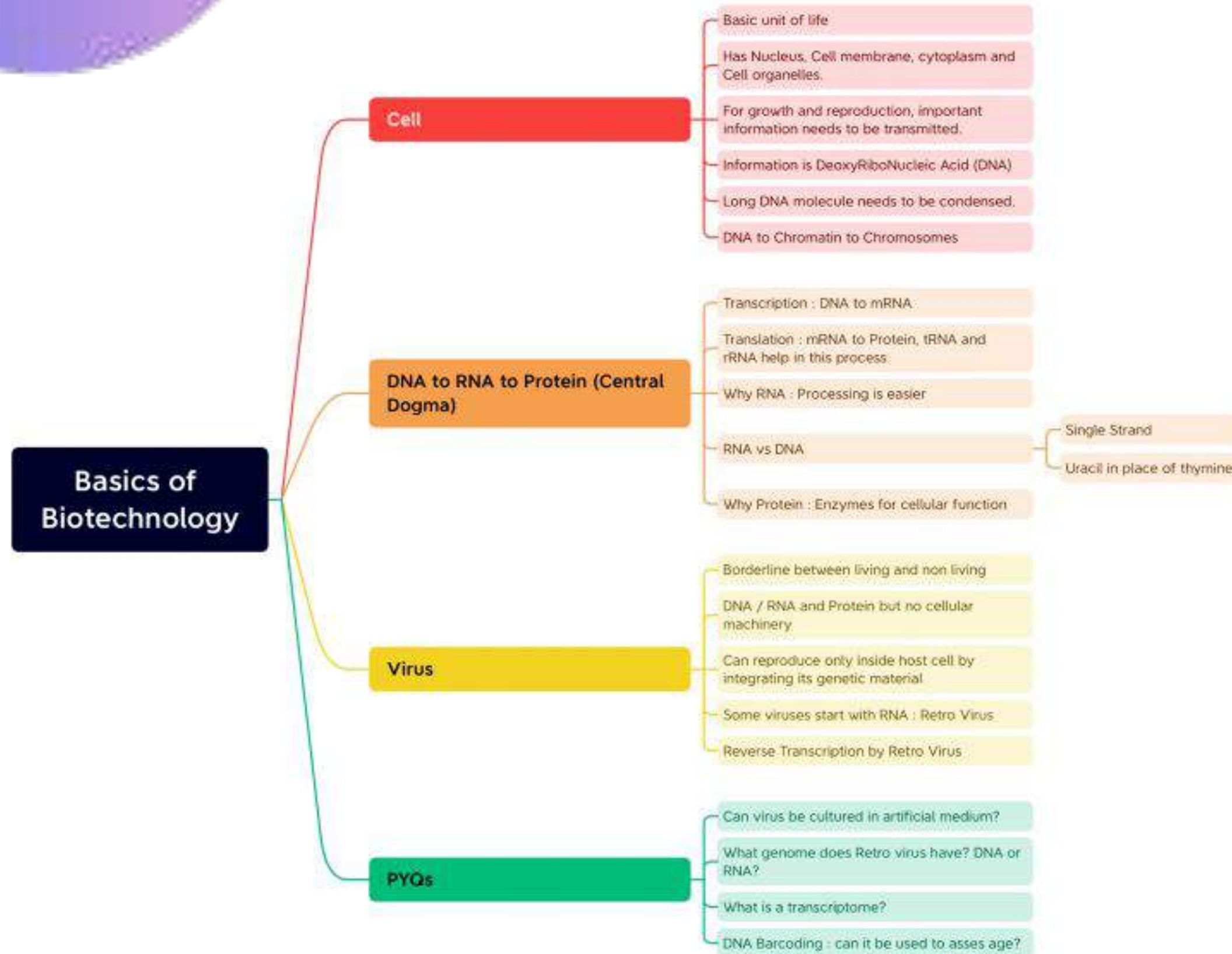
1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. Identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3

Answer: D

- DNA barcoding is a system for species identification focused on the use of a short, standardized genetic region acting as a “barcode”
- DNA barcoding has got no association with determining the age of an organism.
- DNA barcoding is a system for fast and accurate species identification that makes ecological system more accessible by using short DNA sequences instead of whole genome and is used for eukaryotes.
- In recent times, DNA barcoding has emerged as the most potent tool for detection of adulteration of food with unwanted plant and animal material.



Contact Us



95690 93856



www.edsarrthi.com



ask@sarrthi.com



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Clarity for UPSC by Dr. Shivin





UPSC Prelims

Science & Tech

Class 2

Basic Biology: Cells and Genetics



Cell : Fundamental Unit of Life



- Unicellular organisms are capable of
 - (i) independent existence and
 - (ii) performing the essential functions of life.
- Anything less than a complete structure of a cell does not ensure independent living. Hence, cell is the fundamental structural and functional unit of all living organisms.
- Anton Von Leeuwenhoek first saw and described a live cell.
- Robert Brown later discovered the nucleus.

Cell Theory



- Schleiden and Schwann, Modified by Virchow
 - (i) all living organisms are composed of cells and products of cells.
 - (ii) all cells arise from pre-existing cells.

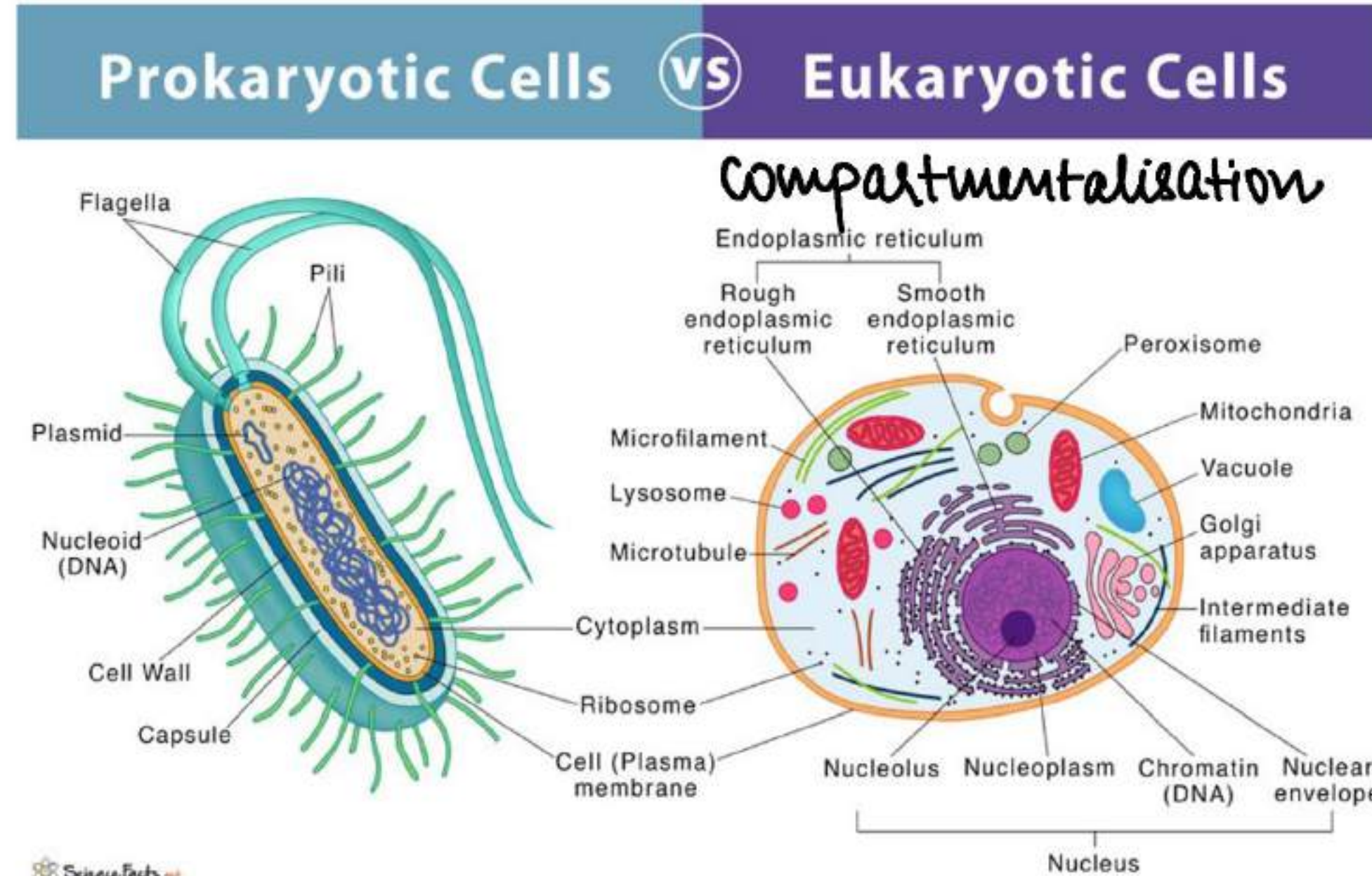
Basic Structure of a Cell



We have already covered this:

- Inside each cell is a dense membrane bound structure called nucleus. This nucleus contains the chromosomes which in turn contain the genetic material, DNA. Cells that have membrane bound nuclei are called eukaryotic whereas cells that lack a membrane bound nucleus are prokaryotic. In both prokaryotic and eukaryotic cells, a semi-fluid matrix called cytoplasm occupies the volume of the cell. The cytoplasm is the main arena of cellular activities in both the plant and animal cells.
- Prokaryotic cells : in monera (will explain later)
- Ribosomes are non-membrane bound organelles found in all cells
- Learn from word breakdown : Pro and Eu + karyon

Eukaryotes vs Prokaryotes



Prokaryotic Cells



- Bacteria, blue-green algae, mycoplasma and PPLO
 - prokaryotes have a cell wall surrounding the cell membrane except in mycoplasma.
- The fluid matrix filling the cell is the cytoplasm. There is no welldefined nucleus. The genetic material is basically naked, not enveloped by a nuclear membrane. In addition to the genomic DNA (the single chromosome/circular DNA), many bacteria have small circular DNA outside the genomic DNA. These smaller DNA are called plasmids.
- The plasmid DNA confers certain unique phenotypic characters to such bacteria. One such character is resistance to antibiotics.
 - Nuclear membrane only in Eukaryotic cells.

Eukaryotic vs. Prokaryotic Cells

Characteristics	Eukaryotic Cells	Prokaryotic Cells
Definition	Any cell that contains a clearly defined nucleus and membrane bound organelles	Any unicellular organism that does not contain a membrane bound nucleus or organelles
Examples	Animal, plant, fungi, and protist cells	Bacteria and Archaea
Nucleus	Present (membrane bound)	Absent (nucleoid region)
Cell Size	Large (10-100 micrometers)	Small (less than a micrometer to 5 micrometers)
DNA Replication	Highly regulated with selective origins and sequences	Replicates entire genome at once
Organism Type	Usually multicellular	Unicellular
Chromosomes	More than one	One long single loop of DNA and plasmids
Ribosomes	Large	Small
Growth Rate/Generation Time	Slower	Faster
Organelles	Present	Absent
Ability to Store Hereditary Information	All eukaryotes have this ability	All prokaryotes have this ability
Cell Wall	Simple: Present in plants and fungi	Complex: Present in all prokaryotes
Plasma Membrane	Present	Present
Cytoplasm	Present	Present

Eukaryotic Cells



- extensive compartmentalisation of cytoplasm through the presence of membrane bound organelles
- Eukaryotic cells possess an organised nucleus with a nuclear envelope
- Their genetic material is organised into chromosomes.

5 Kingdoms on basis of features:

Monera

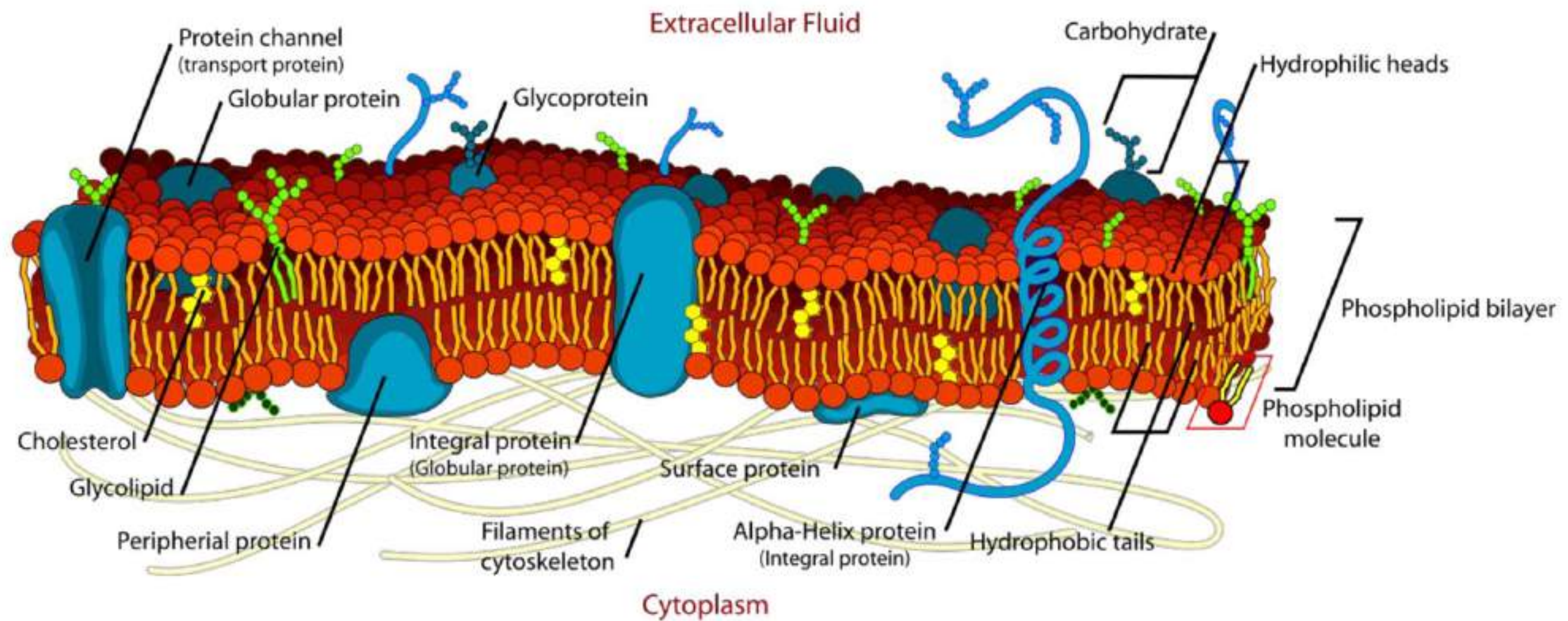
Protista

Fungi

Plantae

Animalia

Cell Membrane : Fluid Mosaic Model



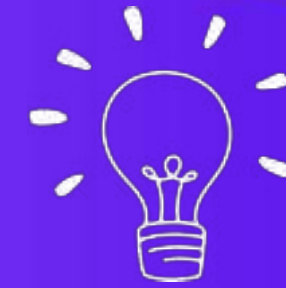
Cell Membrane Continued



mainly composed of lipids and proteins.

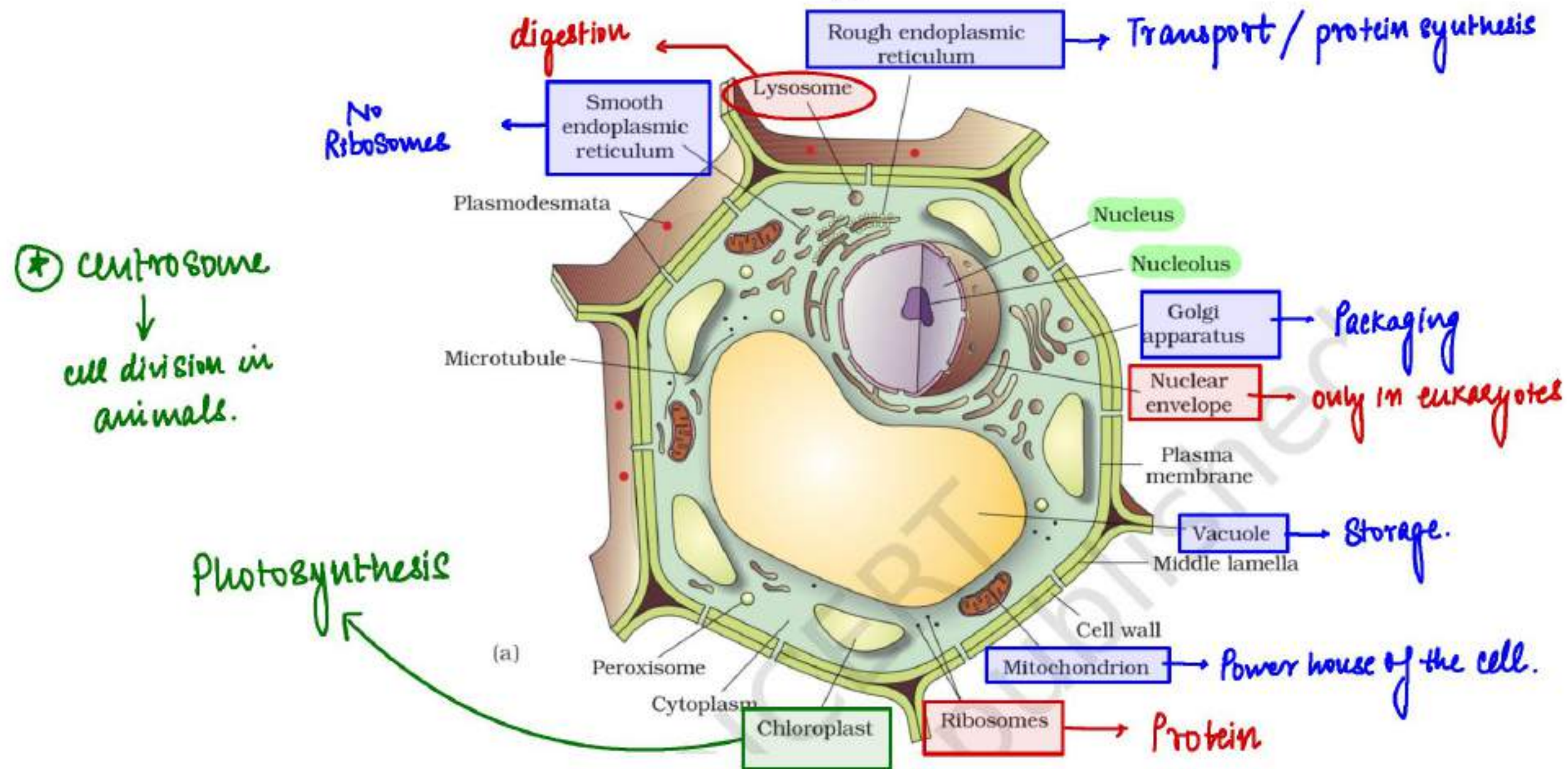
- the major lipids are phospholipids that are arranged in a bilayer.
- Depending on the ease of extraction, membrane proteins can be classified as integral and peripheral. Peripheral proteins lie on the surface of membrane while the integral proteins are partially or totally buried in the membrane.
- quasi-fluid nature of lipid enables lateral movement of proteins : cell growth, formation of intercellular junctions, secretion, endocytosis, cell division etc.
- The membrane is selectively permeable to some molecules present on either side of it.
- Passive transport
- Active transport

Cell Wall



- non-living rigid structure called the cell wall forms an outer covering for the plasma membrane of fungi and plants.
- shape to the cell
- protects the cell from mechanical damage and infection
- cell-to-cell interaction and provides barrier to undesirable macromolecules.
- Algae have cell wall, made of cellulose, galactans, mannans
- Plants : it consists of cellulose, hemicellulose, pectins and proteins.

Cell Organelles



MITOCHONDRIA

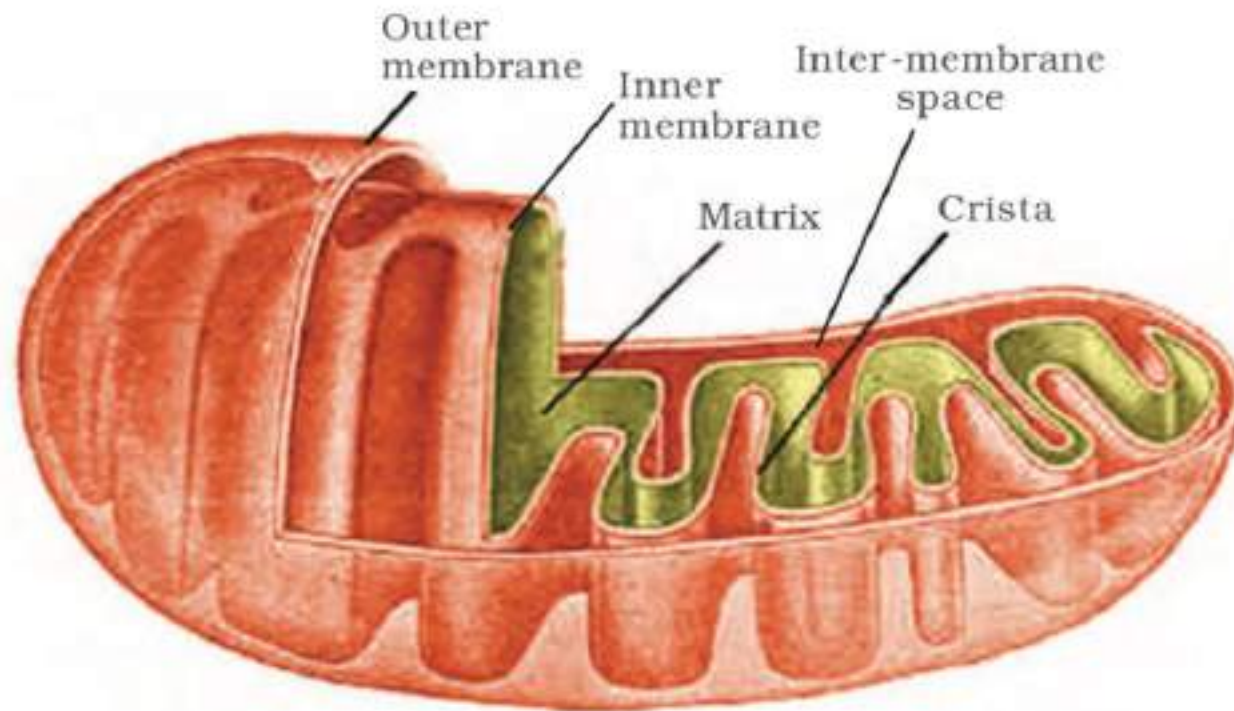
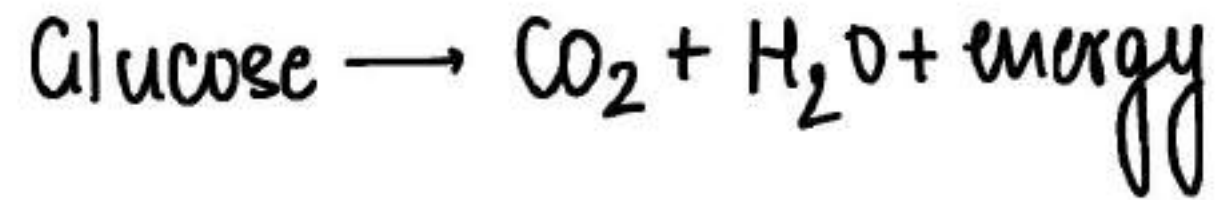
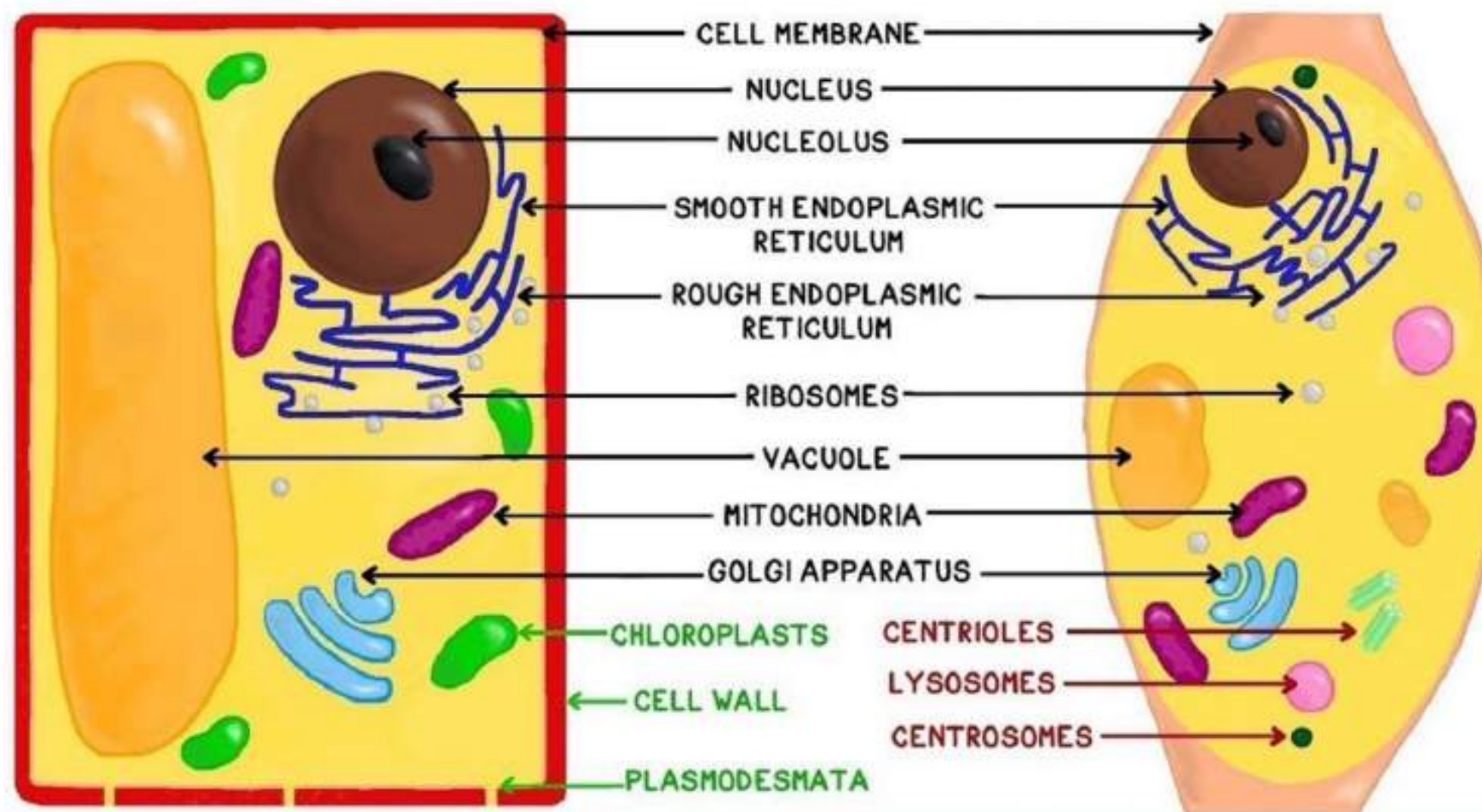


Figure 8.7 Structure of mitochondrion (Longitudinal section)



- Power house of cell
 - Produces energy in form of ATP
 - Possesses a single circular DNA molecule.
- [Important]

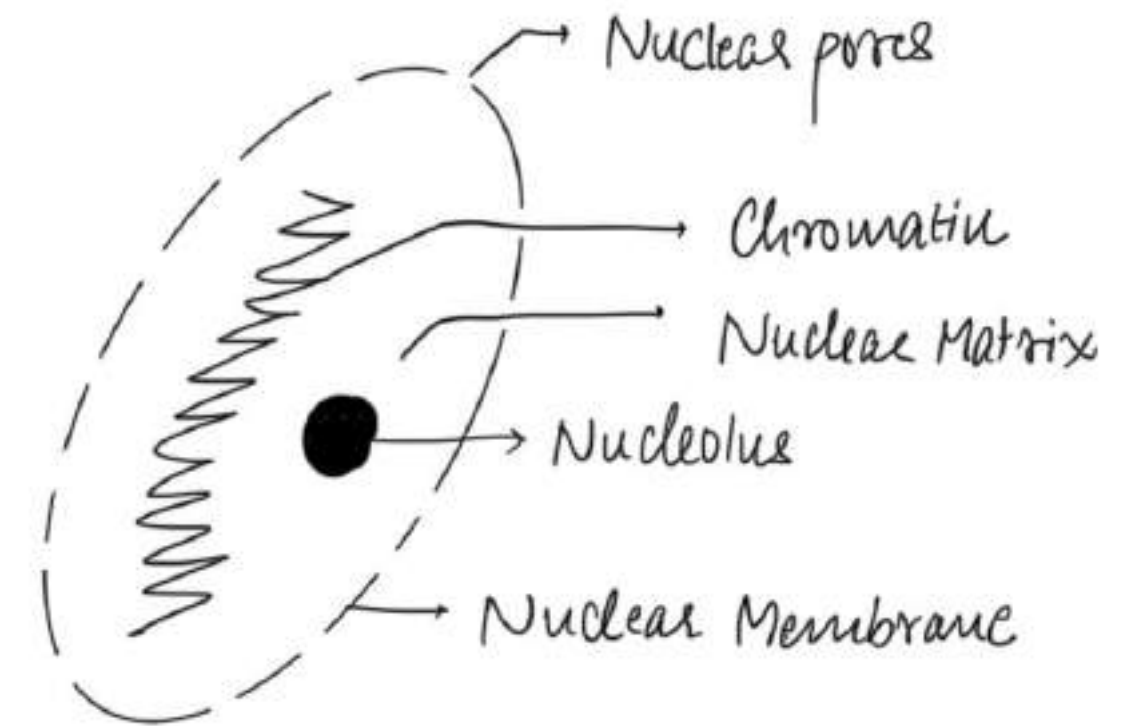
PLANT VS. ANIMAL CELLS

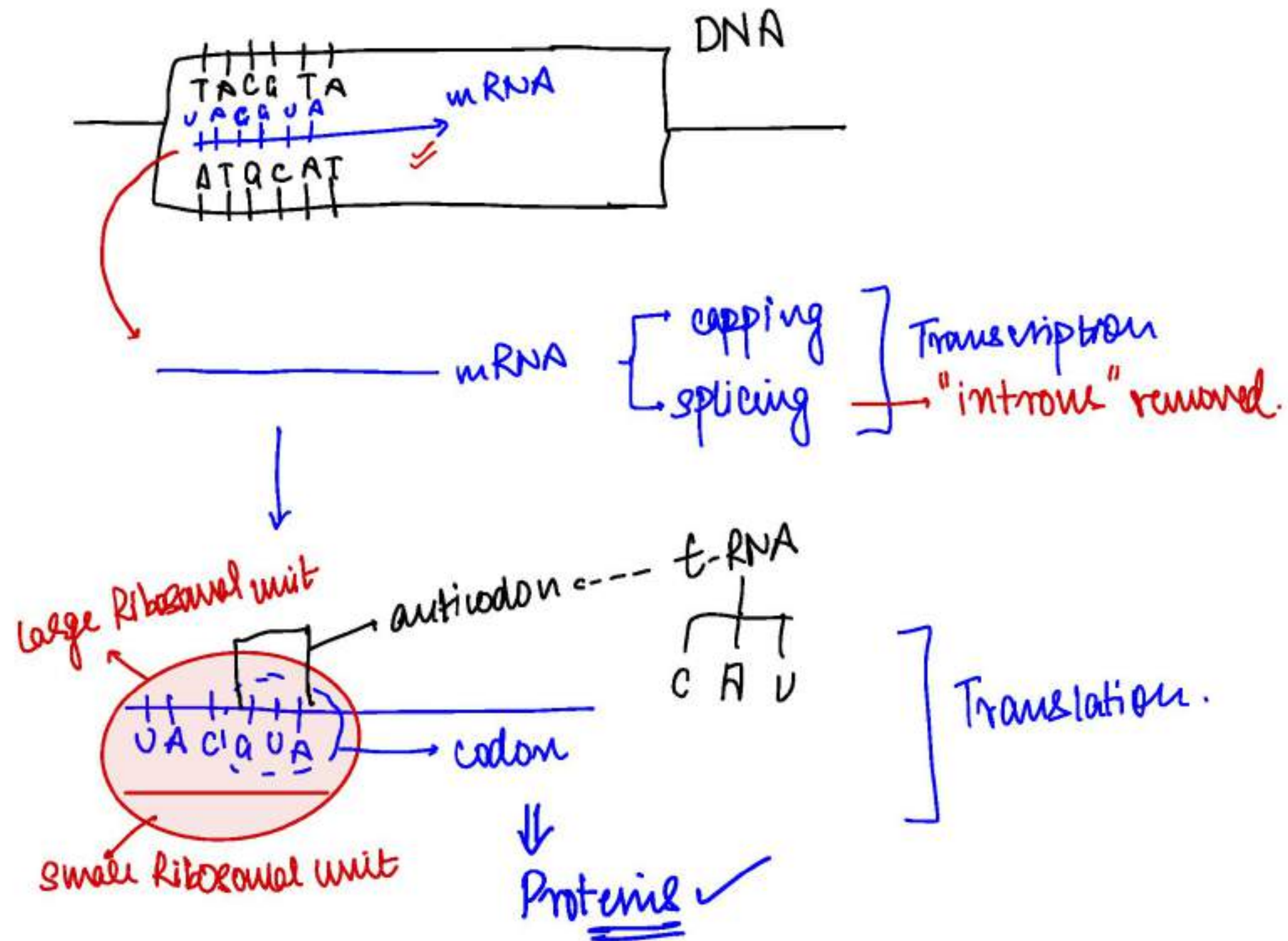


Nucleus



- nuclear pores are the passages through which movement of RNA and protein molecules takes place in both directions between the nucleus and the cytoplasm.
- nuclear matrix or the nucleoplasm contains nucleolus and chromatin.
- The content of nucleolus is continuous with the rest of the nucleoplasm as it is not a membrane bound structure.
- It is a site for active ribosomal RNA synthesis.
- during different stages of cell division, cells show structured chromosomes in place of the nucleus.





T rna, Mrna, codons and proteins : explain here.

Human Reproduction

- Humans → Made up of cells
- Cells → DNA → Chromatin
Chromosomes.
- DNA → characteristic for a species.
↓
therefore chromosomes are
also characteristic for
a species

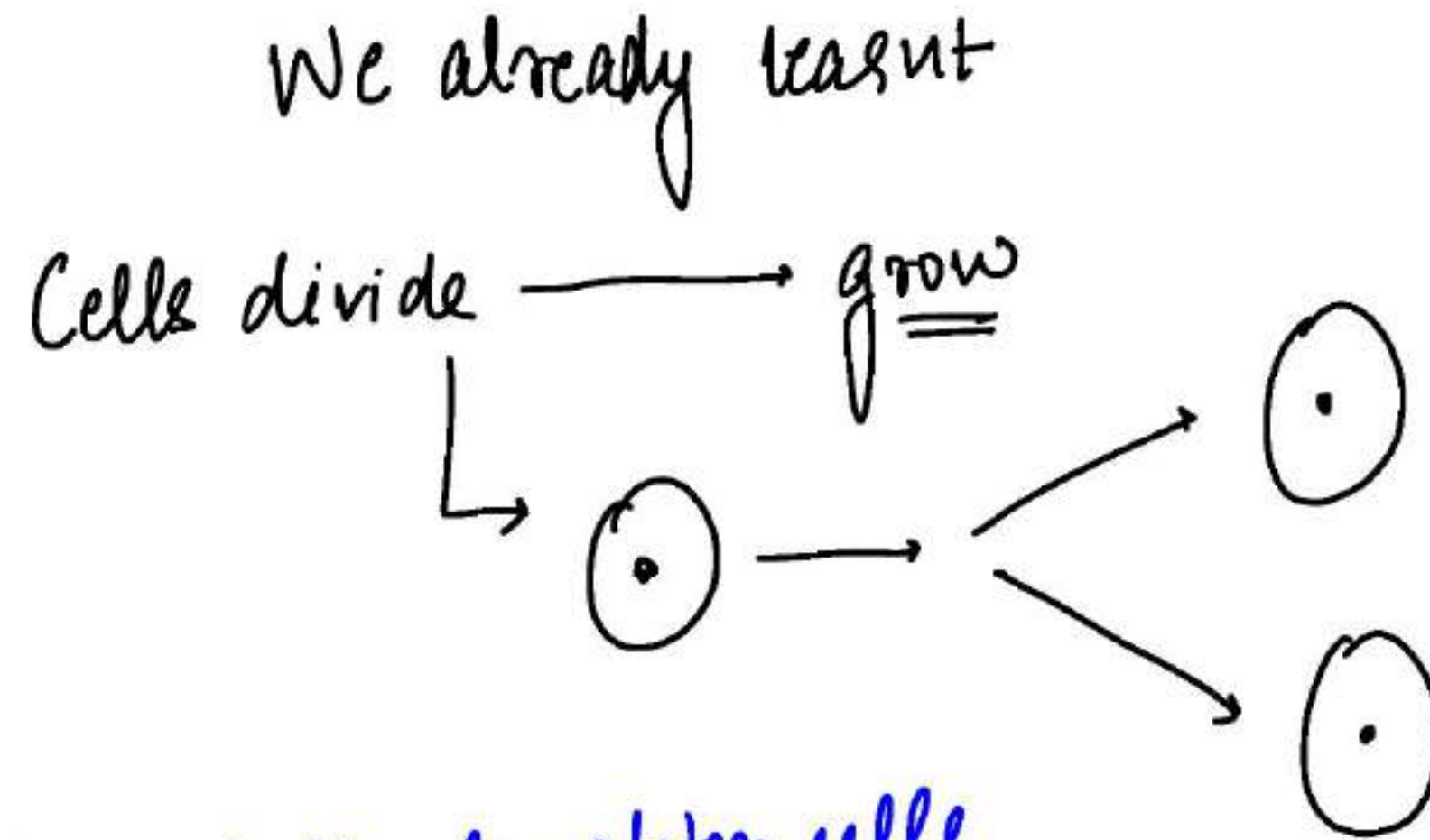
Species	<i>Parascaris equorum</i>	<i>Oryza sativa</i>	<i>Homo sapiens</i>	<i>Pan troglodytes</i>	<i>Canis familiaris</i>
Chromosome #	4	24	46	48	78
Common Name	 Roundworm	 Rice	 Human	 Chimpanzee	 Dog

Chromosomes

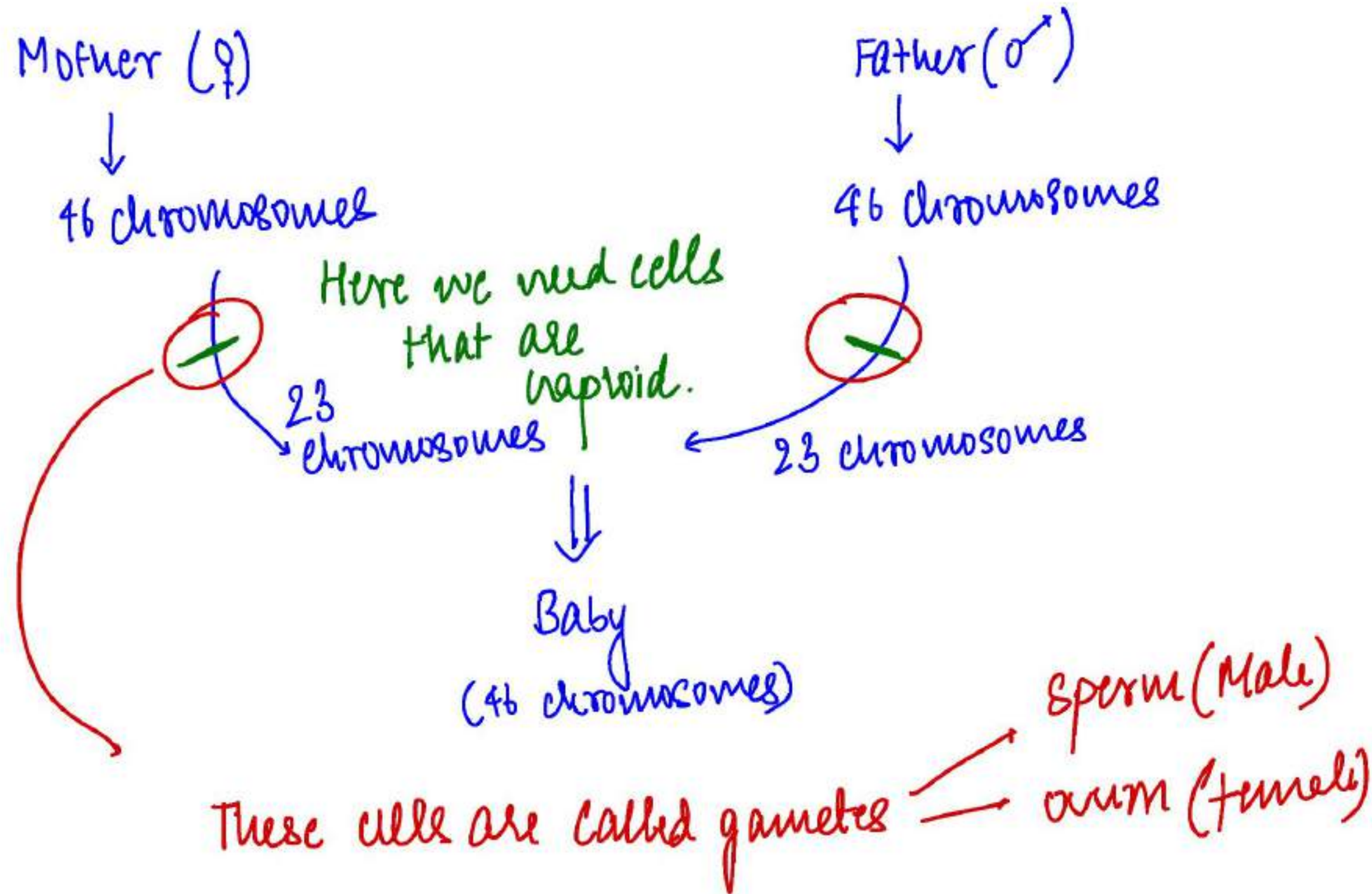
- exist as a pair

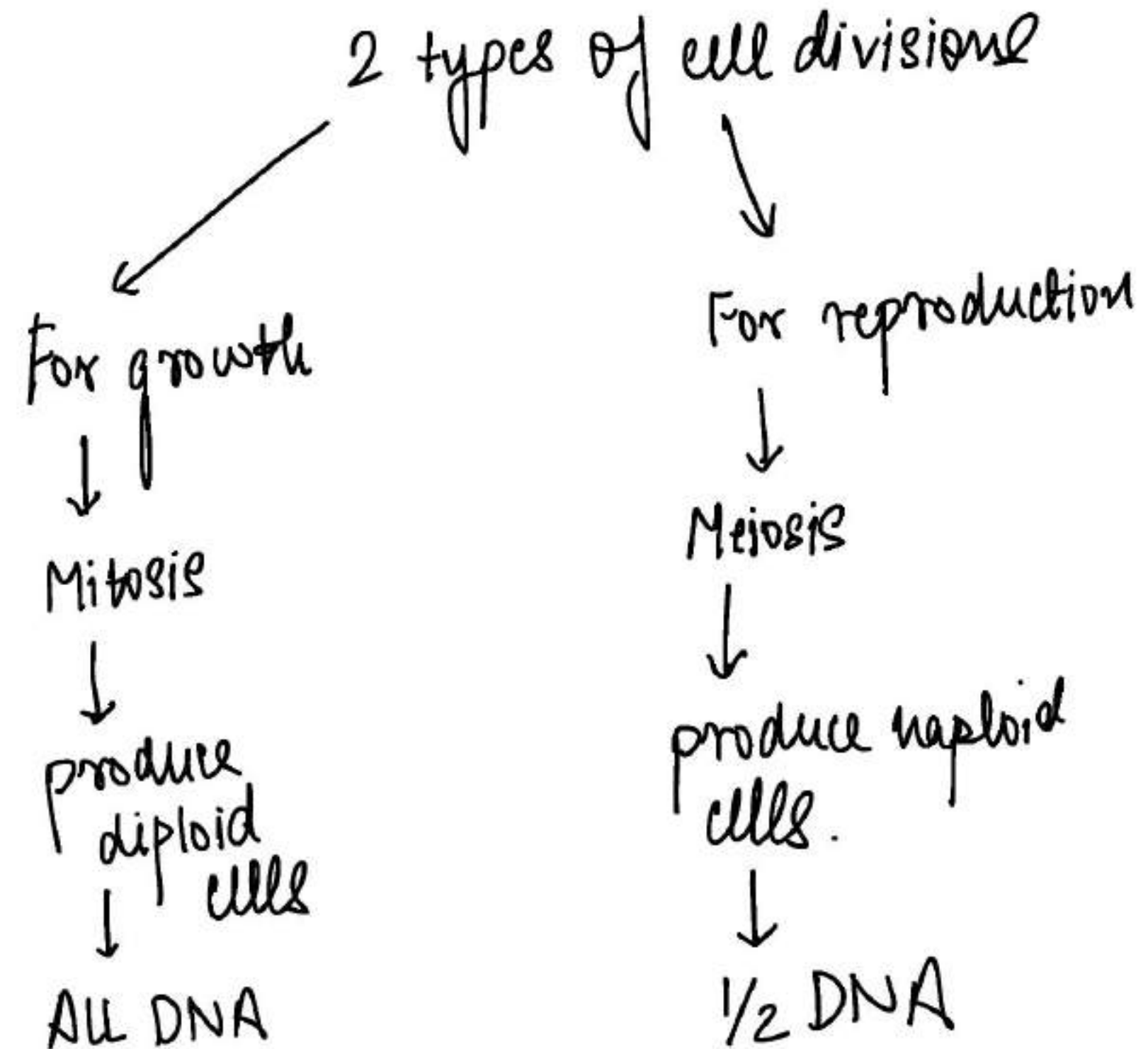
- 46 chromosomes $\rightarrow$ 23 pairs [DNA]

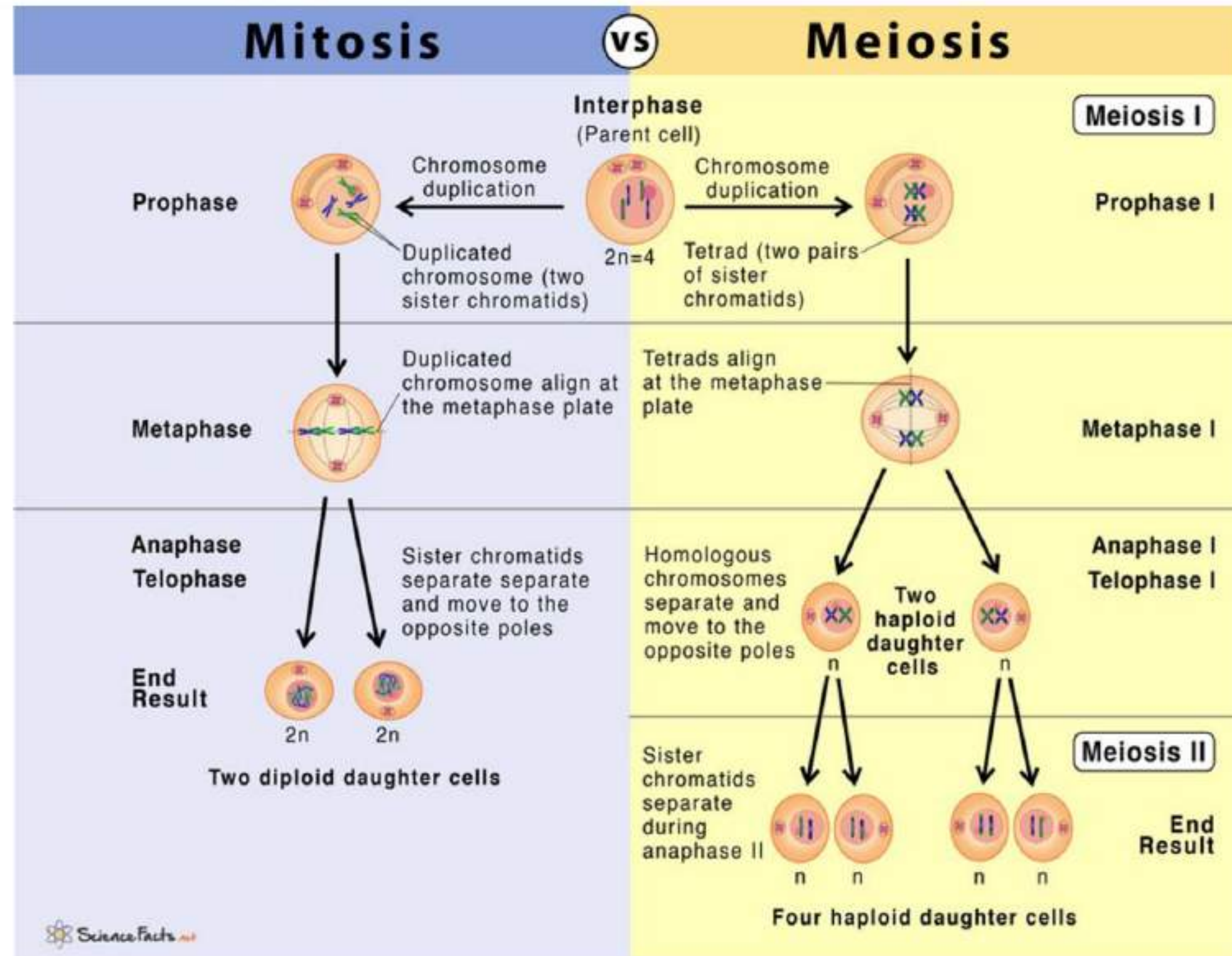
- This is k/a diploid
 - $\swarrow$
Represented as $2(n)$
where $(n) = 23$
- $\rightarrow$ if a cell has only one set
 - $\downarrow$
called haploid (n)
 - $\downarrow$
Why?



Here, both daughter cells must have same no. of chromosomes as parent.



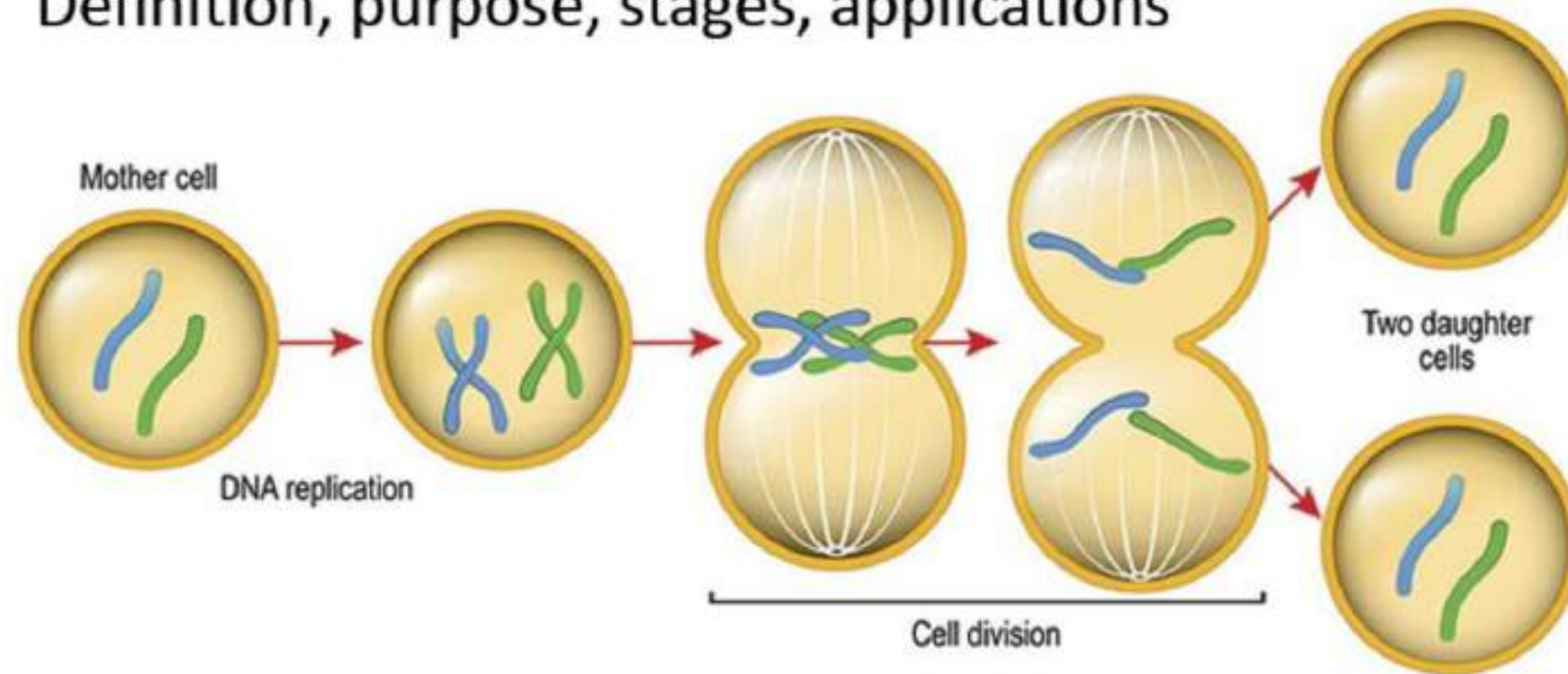




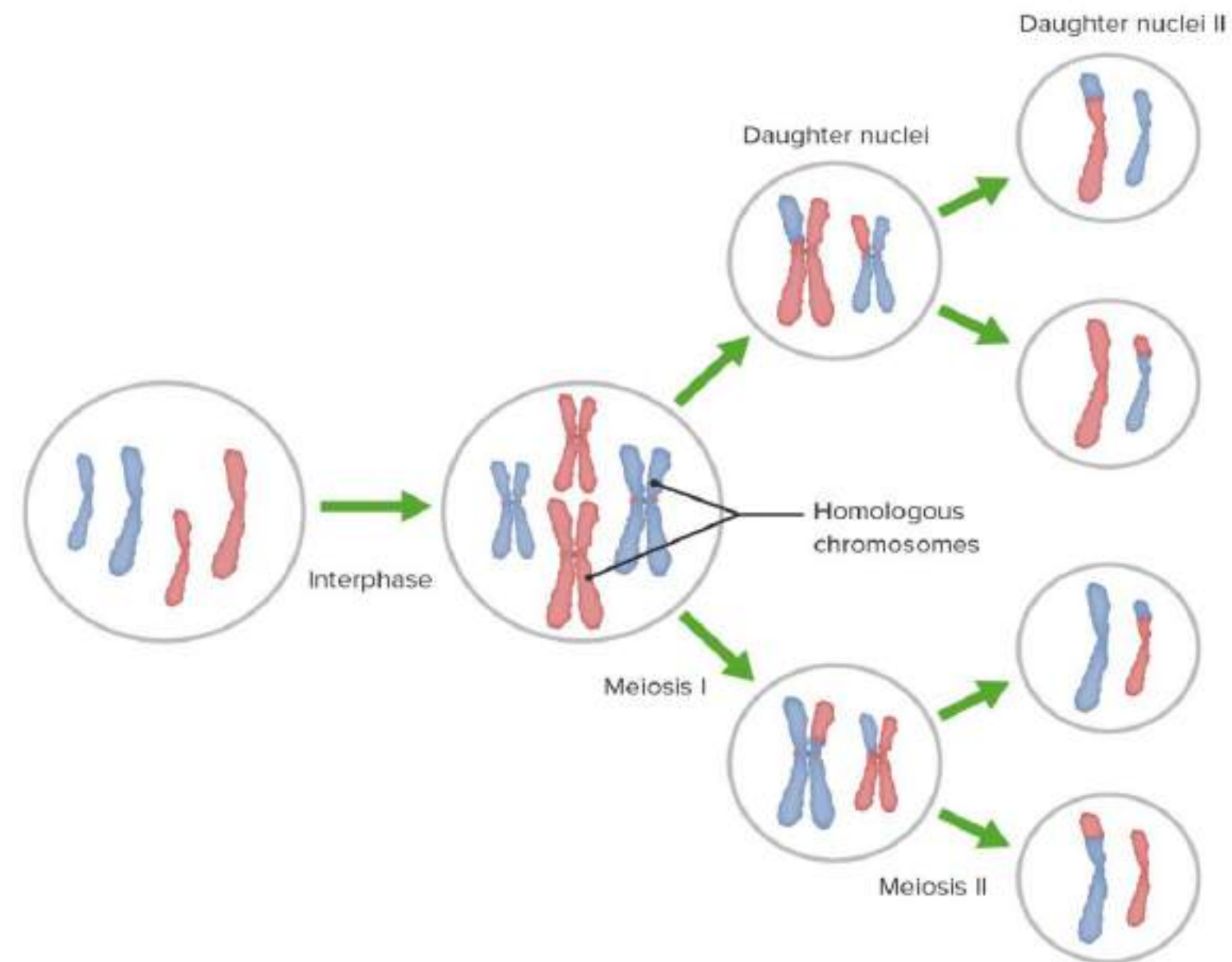
Mitosis

Mitosis

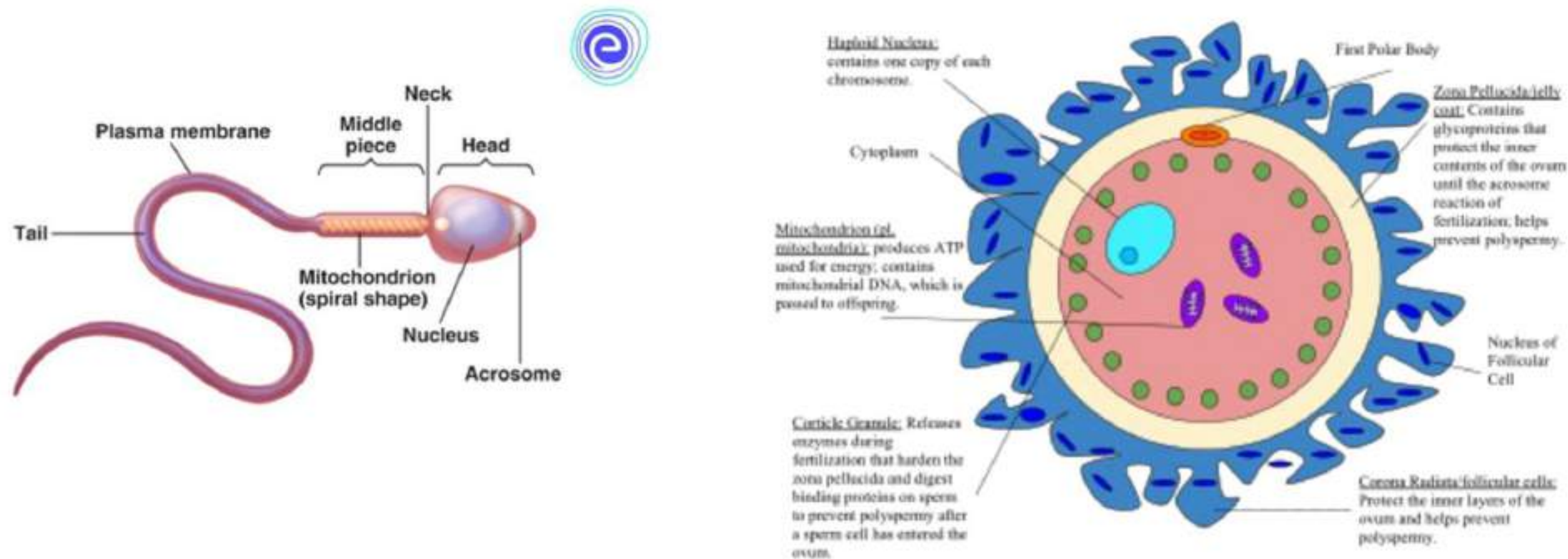
Definition, purpose, stages, applications



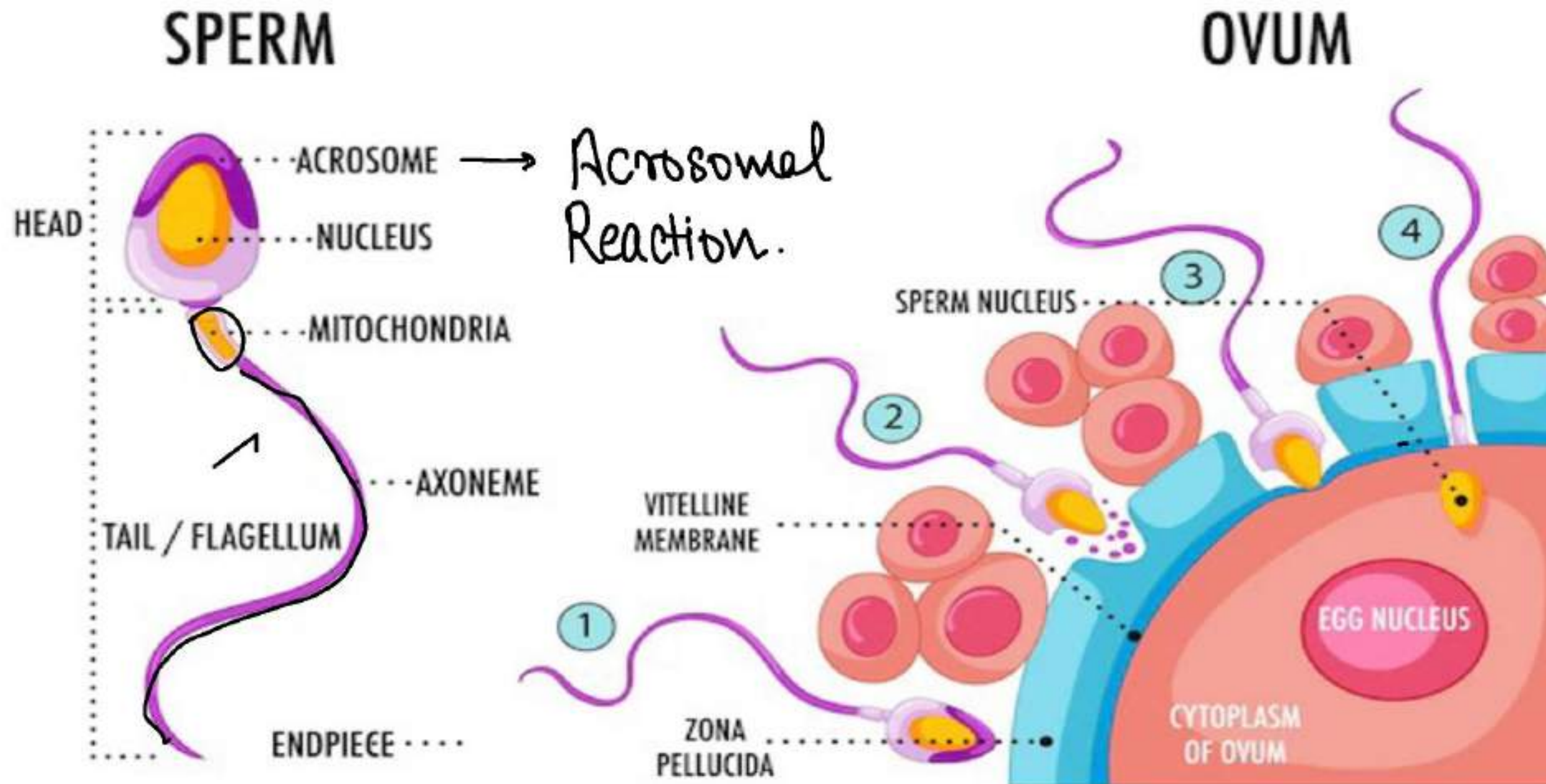
Meiosis



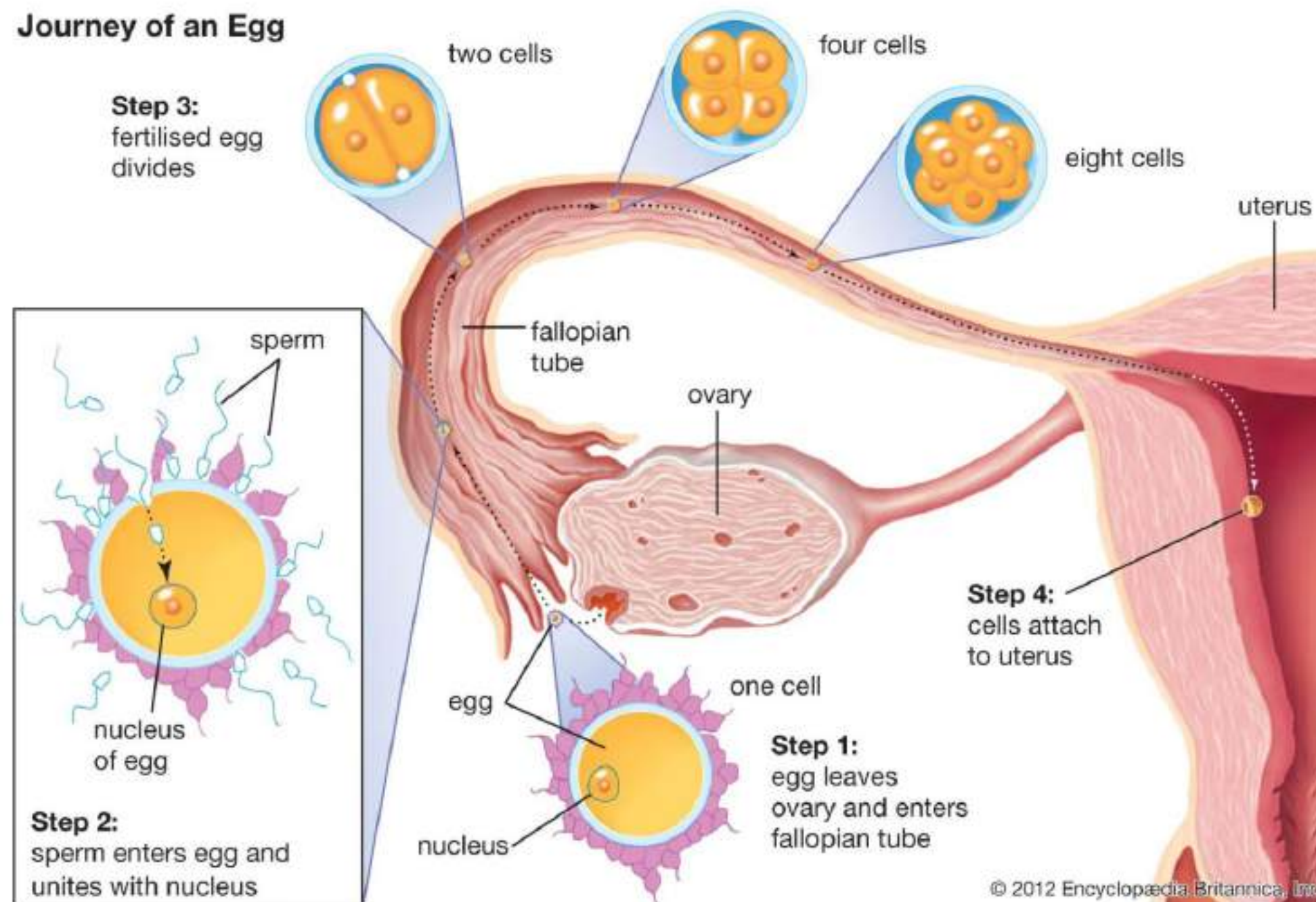
Gametes : Sperm and Ovum



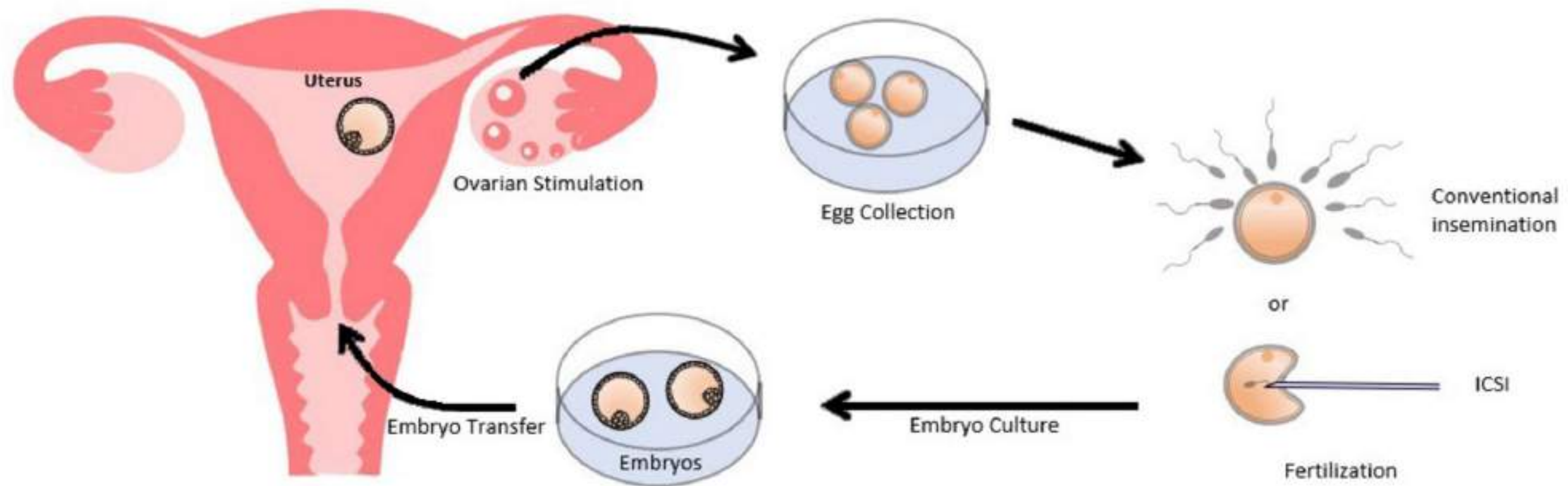
FERTILIZATION



Normal Fertilisation

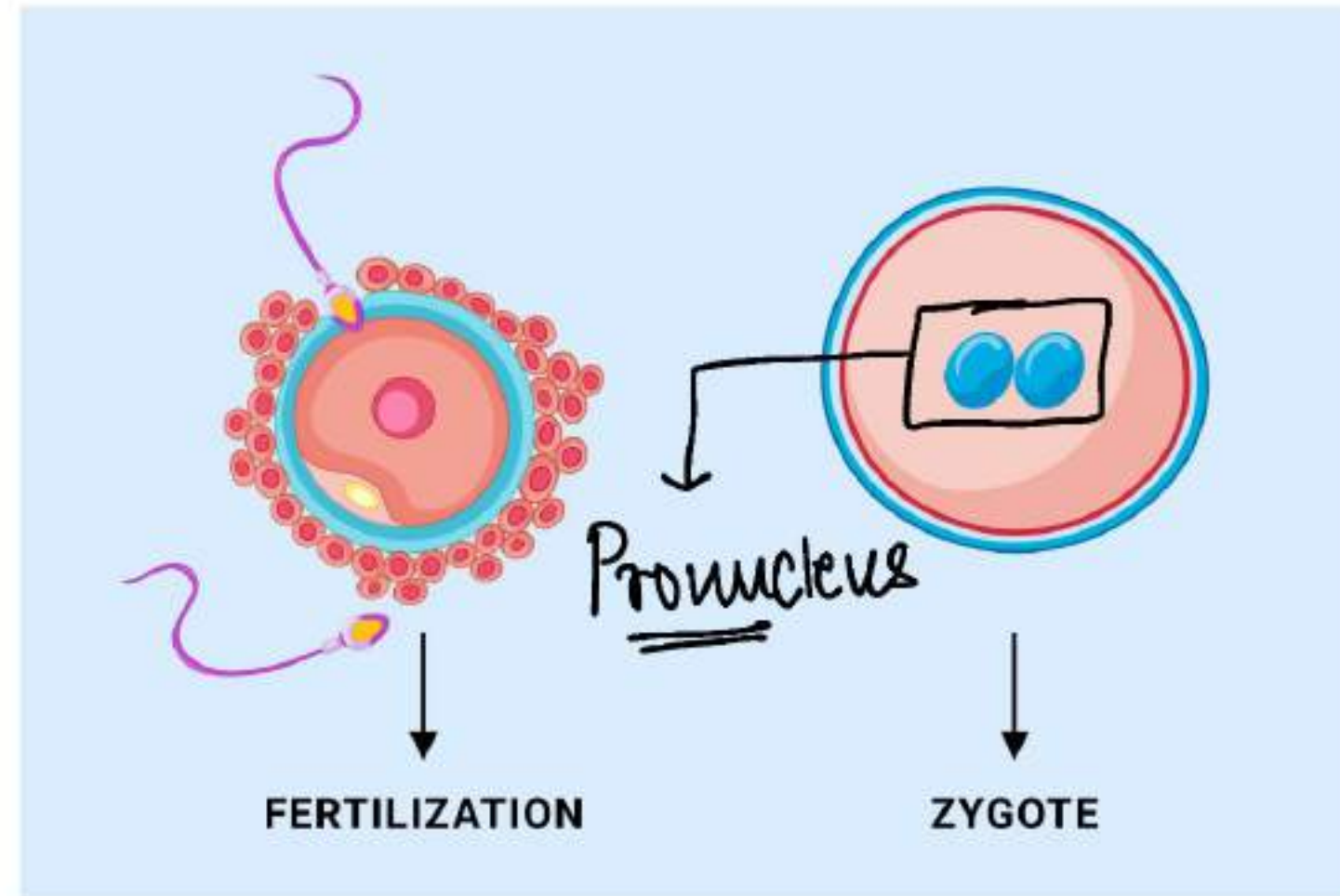


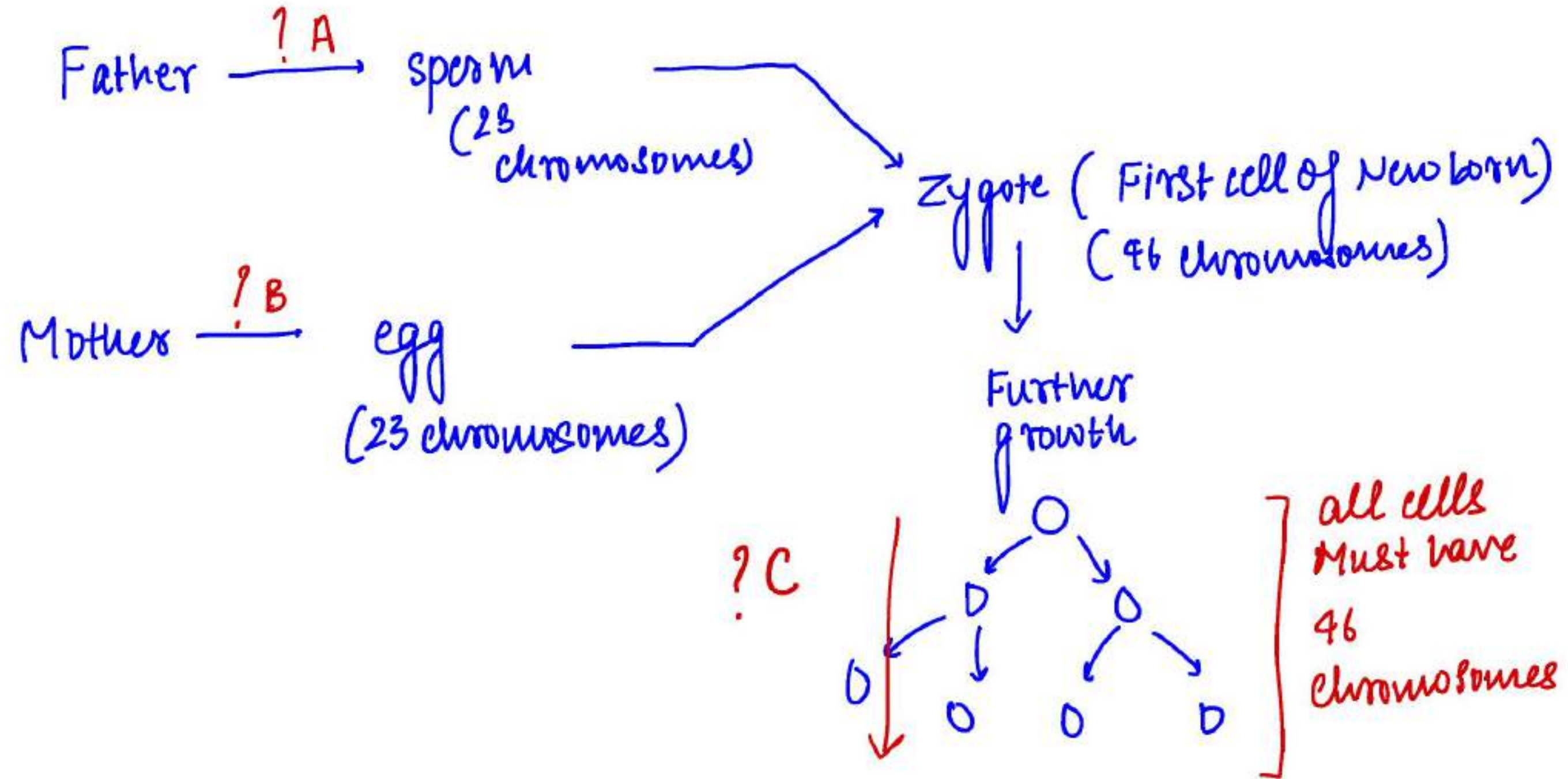
In vitro fertilisation



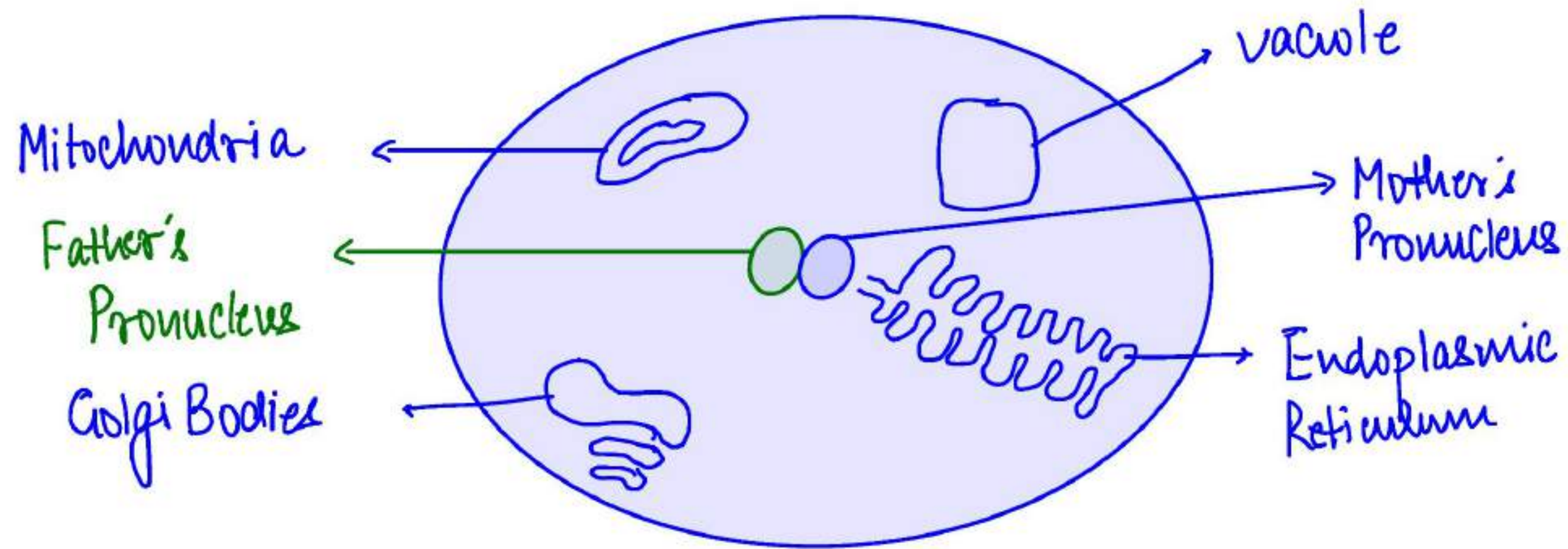
Zygote features

- It has nuclei containing chromosomes from both parents.
- It has organelles only from the mother (ovum)
- Is there any other place other than the nucleus where DNA is present? (try and remember)

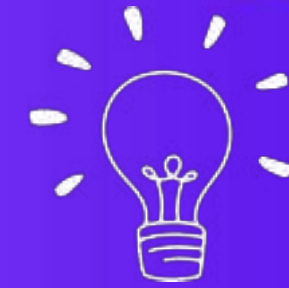




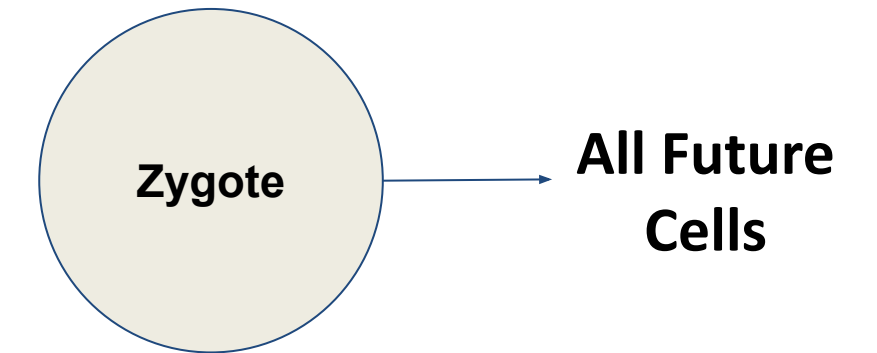
Components of zygote



What do we do if a mother's mitochondria is defective?

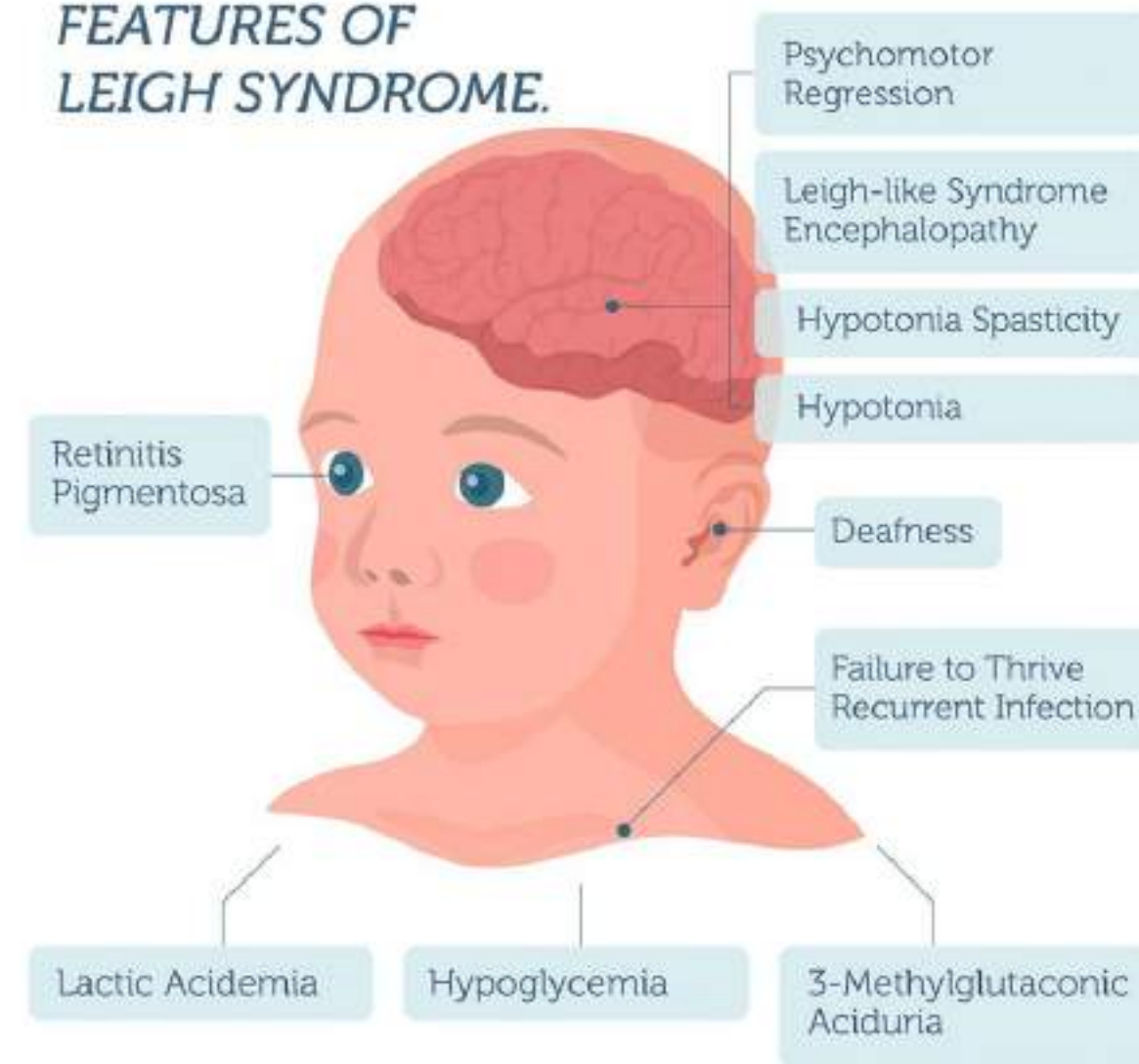


- Replace the mitochondria with healthy mitochondria
- Mitochondrial replacement therapy (MRT), aka mitochondrial donation, is the replacement of mitochondria in one or more cells to prevent or ameliorate disease. MRT originated as a special form of in vitro fertilization in which some or all of the future baby's mitochondrial DNA (mtDNA) comes from a third party. This technique is used in cases when mothers carry genes for mitochondrial diseases.

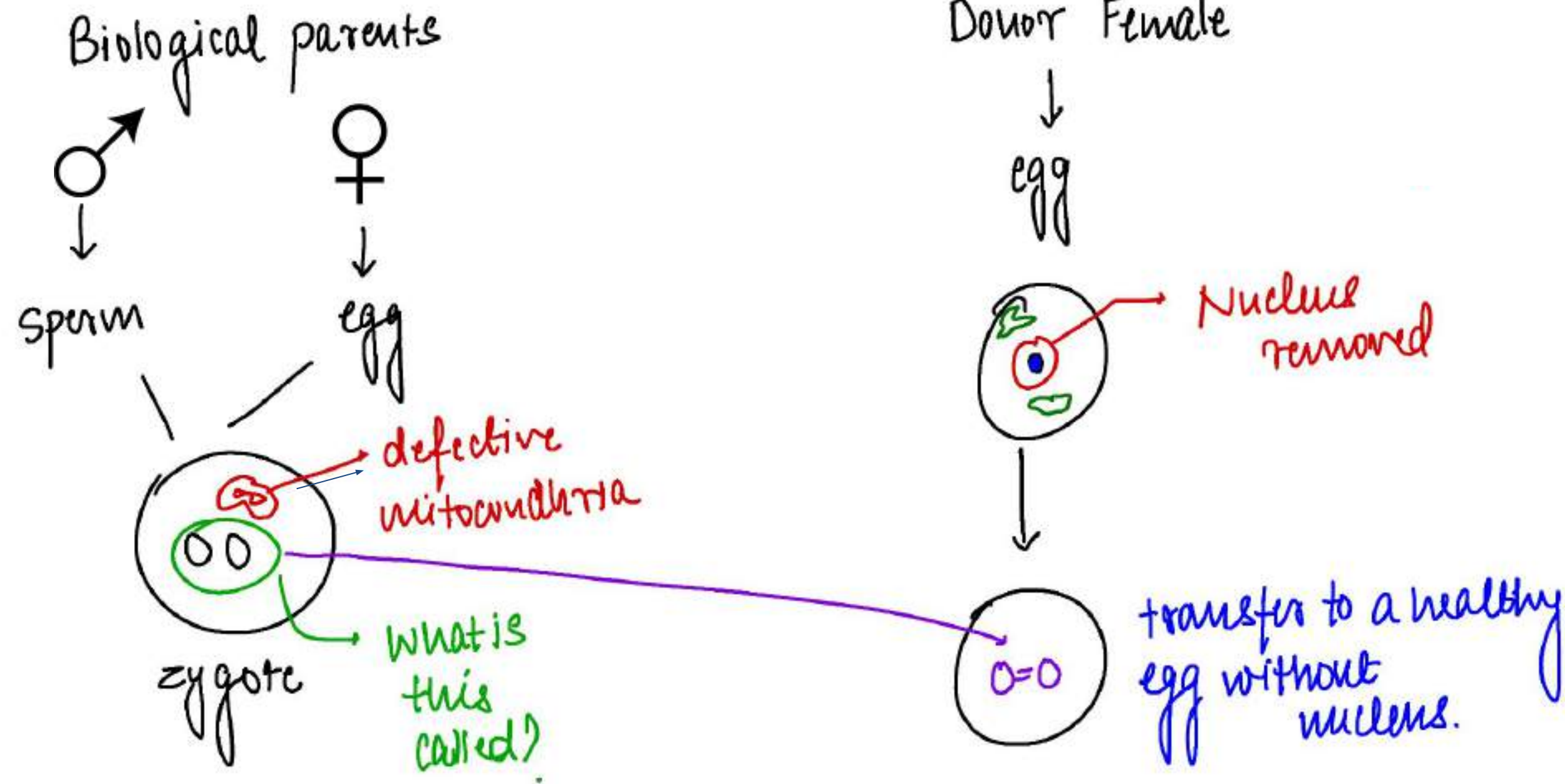


- We need to replace mother's mitochondria with a healthy mitochondria
- Defective mitochondria in mother can cause a disease
↓
eg: LEIGH'S disease

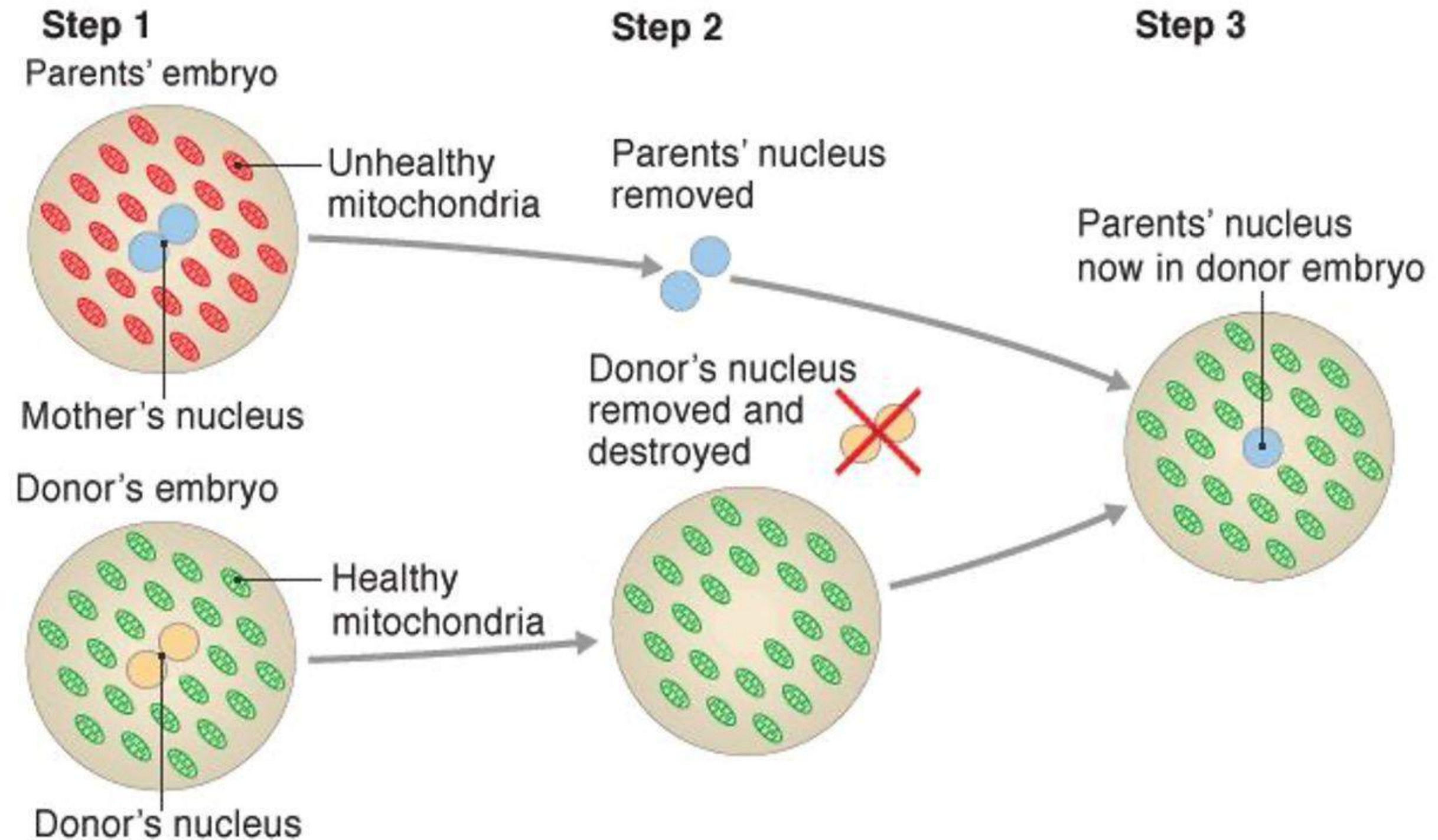
FEATURES OF LEIGH SYNDROME.



Method - 1



How pronuclear transfer works



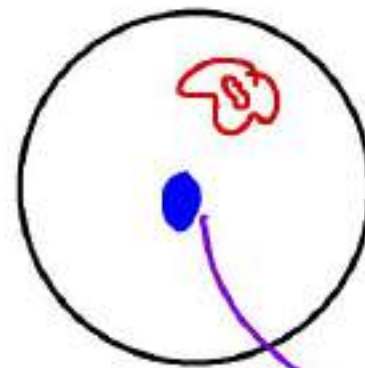
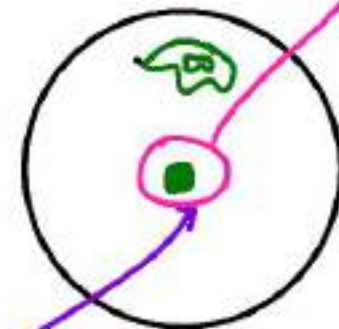
Pronuclear Transfer

After Fertilisation

Method - 2

Biological

♀
↓
egg
↓
defective
mitochondria

Donor
female
↓Healthy
egg
↓

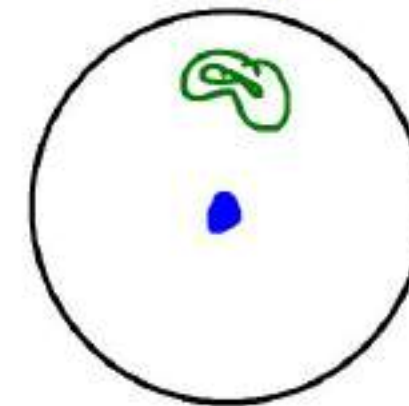
Removed

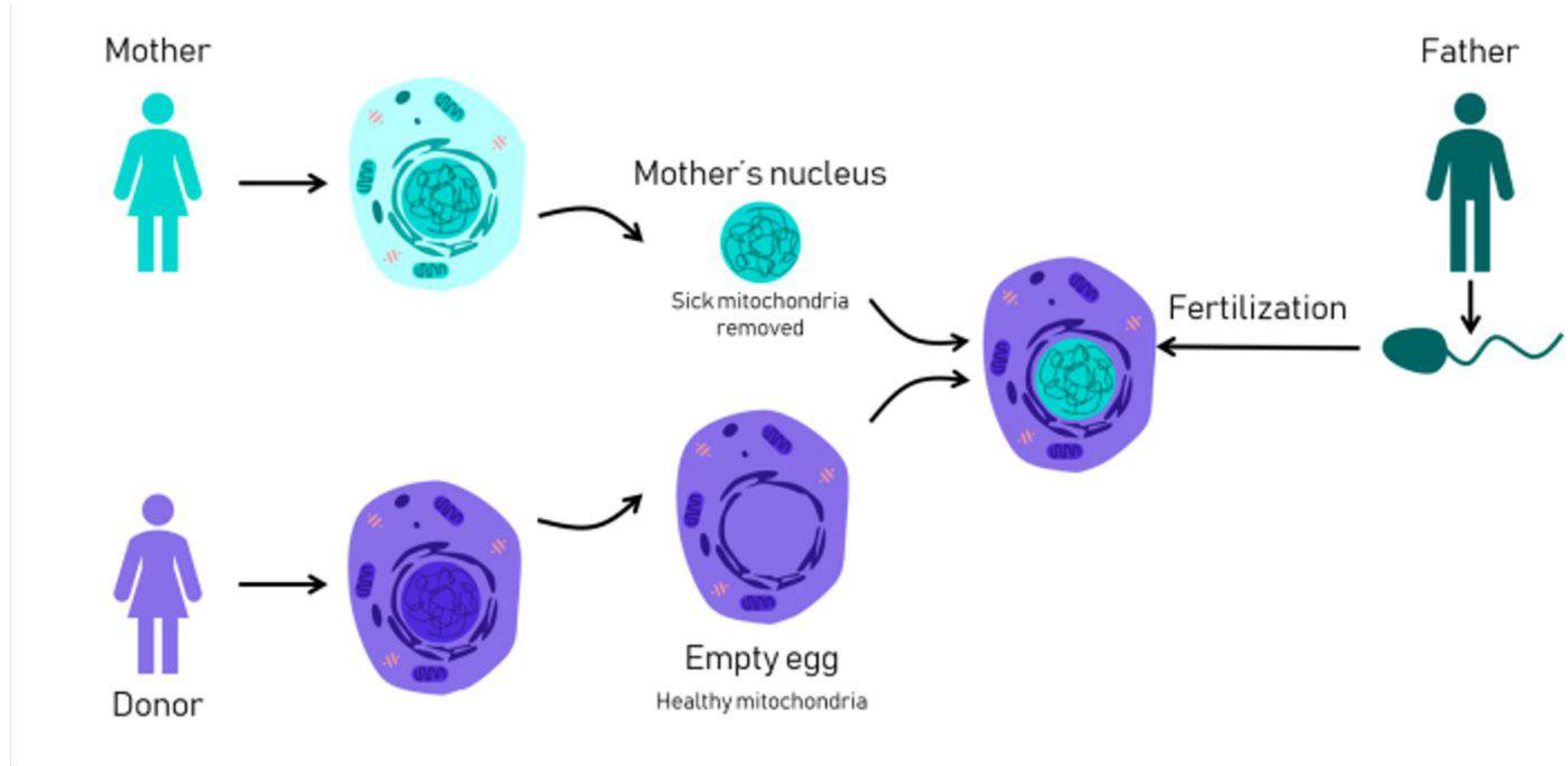
transfer

Biological
♂
↓

Sperm

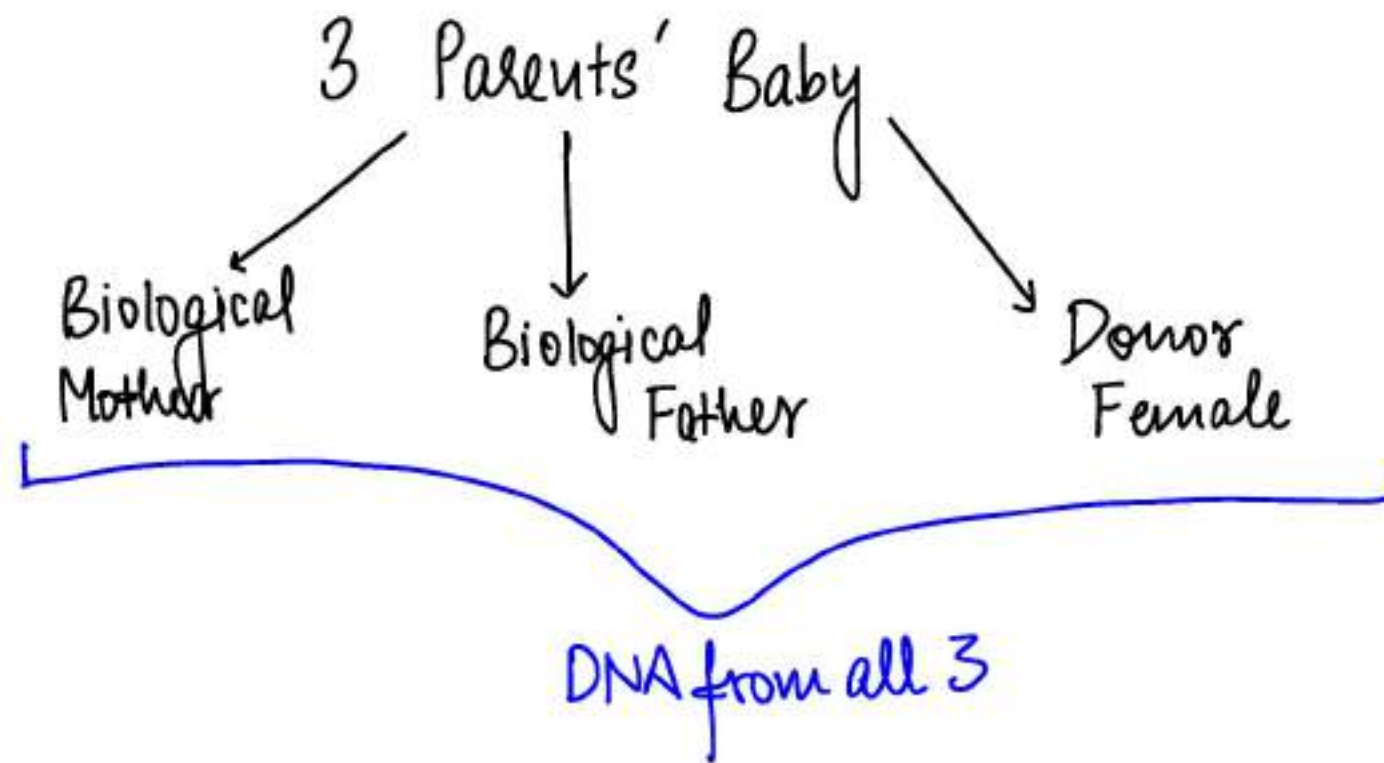
Fertilisation



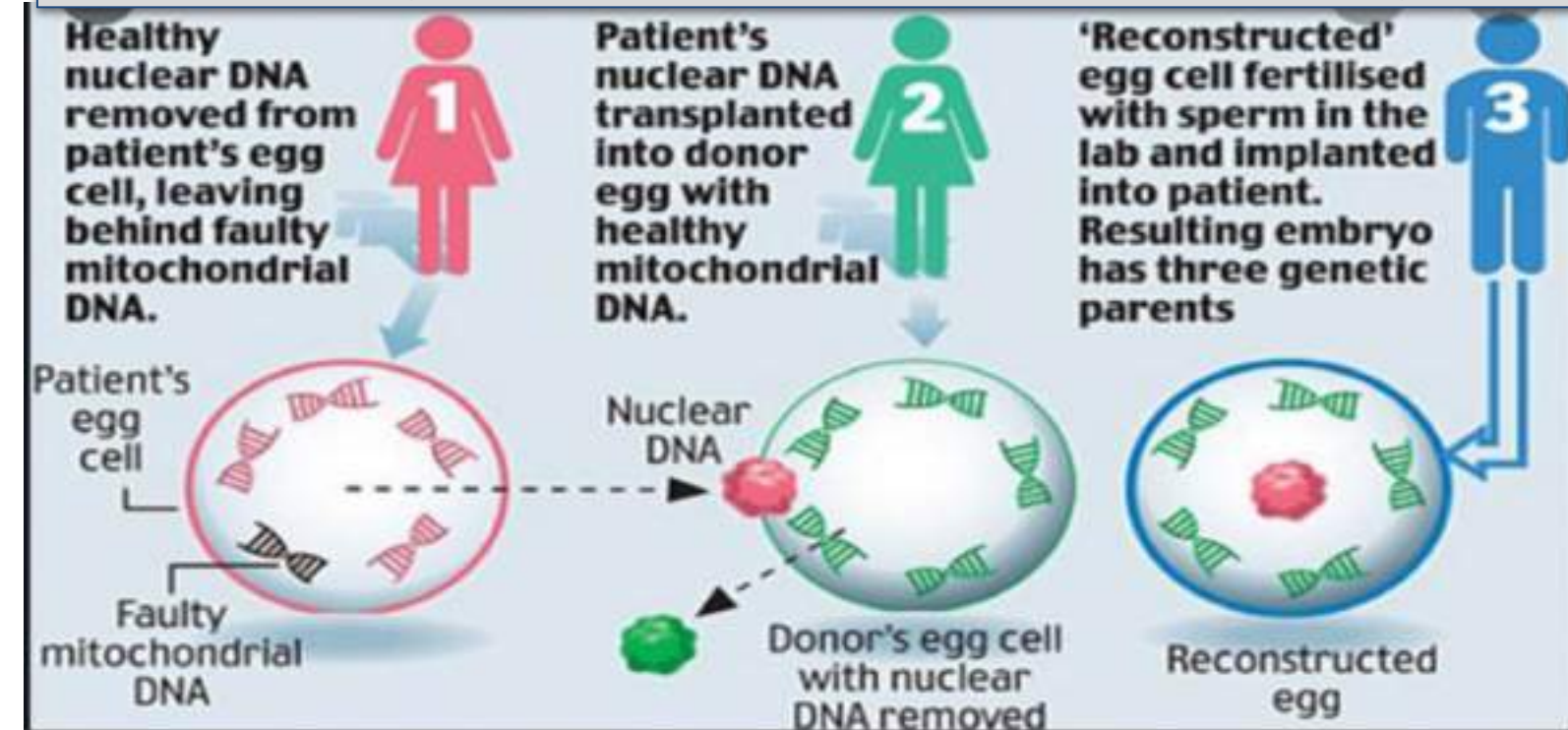


Reconstructed Egg Technology!

Before Fertilisation



Making a Three Parent Embryo



Approved
in
UK,
Mexico.

Test of 6 Questions!

Cut Off: 6/6



Question 1

Q. In the context of hereditary diseases, consider the following statements:

1. Passing on mitochondrial diseases from parent to child can be prevented by mitochondrial replacement therapy either before or after in vitro fertilization of the egg.
2. A child inherits mitochondrial diseases entirely from mother and not from father.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Question 2

Q. In the context of recent advances in human reproductive technology, “Pronuclear Transfer” is used for:

- a) fertilization of egg in vitro by the donor sperm
- b) genetic modification of sperm producing cells
- c) development of stem cells into functional embryos
- d) prevention of mitochondrial diseases in offspring

Question 3

Q.7) Which of the following statements are correct regarding the general difference between plant and animal cells?

1. Plant cells have cellulose cell walls whilst animal cells do not.
2. Plant cells do not have plasma membrane unlike animal cells which do.
3. Mature plant cell has one large vacuole whilst animal cell has many small vacuoles.

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Question 4

Q.Which of the following cell organelles play the most significant role in protein synthesis?

- A.Lysosome and Centrosome
- B.Endoplasmic reticulum and Ribosome
- C.Golgi apparatus and Mitochondria
- D.Lysosome and Mitochondria

Question 5

Q. Which organelle in the cell, other than the nucleus, contains DNA?

- (a) Centriole
- (b) Golgi apparatus
- (c) Lysosome
- (d) Mitochondrion

Question 6

Q. Consider the following statements: DNA Barcoding can be a tool to:

1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. Identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3

Congratulations on giving the test!

Cut-Off Figure —→ Only for Pressure

Let's Discuss The Answers.



Question 1 (Year 2021)

Q. In the context of hereditary diseases, consider the following statements:

1. Passing on mitochondrial diseases from parent to child can be prevented by mitochondrial replacement therapy either before or after in vitro fertilization of the egg.
2. A child inherits mitochondrial diseases entirely from mother and not from father.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2**
- d) Neither 1 nor 2

Answer: C

- The technological development allows nearly complete replacement of the cytoplasm of egg/embryo, eliminating the transmission of undesired defective mitochondria for patients with inherited mitochondrial diseases, which is called mitochondrial replacement therapy (MRT). These processes use assisted reproductive technology, including in vitro fertilization. The father's sperm can fertilize the patient egg before the maternal genetic material is removed or after the maternal genetic material has been transferred to the donor egg. Hence, statement 1 is correct.
- Mitochondrial diseases are chronic (long-term), genetic, often inherited disorders that occur when mitochondria fail to produce enough energy for the body to function properly. As the mitochondrial genome is inherited exclusively from the mother, the child inherits mitochondrial diseases entirely from mother and not from father. Hence, statement 2 is correct.
- Therefore, option (c) is the correct answer.

Question 2 (Year 2020)

Q. In the context of recent advances in human reproductive technology, “Pronuclear Transfer” is used for:

- a) fertilization of egg in vitro by the donor sperm
- b) genetic modification of sperm producing cells
- c) development of stem cells into functional embryos
- d) prevention of mitochondrial diseases in offspring**

Answer: D

- Pronuclear transfer involves the transfer of pronuclei from one zygote to another. This technique first requires fertilisation of healthy donated egg/s (provided by the mitochondrial donor) with the intending male parent sperm. Simultaneously, the intending mother's affected oocytes are fertilised with the intending father's sperm. Both sets of fertilised oocytes are allowed to develop to the early zygote stage where the pronuclei are visible.
- Pronuclear transfer has the potential to 'treat' human mtDNA disease at a genetic level. D is correct.
- Mutations in mtDNA are a cause of mitochondrial disease, a heterogeneous group of diseases that can lead to premature death, sometimes in infancy or childhood. Most mitochondrial diseases lack specific treatments, and women who carry the causative mutations are at high risk of transmitting the diseases to their offspring. Risk of transmission is greatest for women with high heteroplasmy—women whose total mtDNA content in affected cells or tissues is made up of between 60 and 90 percent mutated mtDNA.

Question 3 (Year 2020)

Q. Which of the following statements are correct regarding the general difference between plant and animal cells?

1. Plant cells have cellulose cell walls whilst animal cells do not.
2. Plant cells do not have plasma membrane unlike animal cells which do.
3. Mature plant cell has one large vacuole whilst animal cell has many small vacuoles.

Select the correct answer using the code given below:

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only**
- d) 1, 2 and 3

Answer: C

- Plant and animal cells are different as the former possess cell walls, chloroplasts, plastids and a large central vacuole which are absent in animal cells. Hence, statement 1 and 3 are correct.
- On the other hand, animal cells have centrioles which are absent in almost all plant cells. Both plant and animal cells have plasma membranes. Hence, statement 2 is not correct.
- Therefore, option (c) is the correct answer.

Question 4 (Year 2001)

Q.Which of the following cell organelles play the most significant role in protein synthesis?

A.Lysosome and Centrosome

B.Endoplasmic reticulum and Ribosome

C.Golgi apparatus and Mitochondria

D.Lysosome and Mitochondria

Answer: **B**

As explained in the class.

Question 5 (Year 2001)

Q. Which organelle in the cell, other than the nucleus, contains DNA?

- (a) Centriole
- (b) Golgi apparatus
- (c) Lysosome
- (d) Mitochondrion**

Answer: D

As explained in the class.

Question 6 (Year 2022)

Q. Consider the following statements: DNA Barcoding can be a tool to:

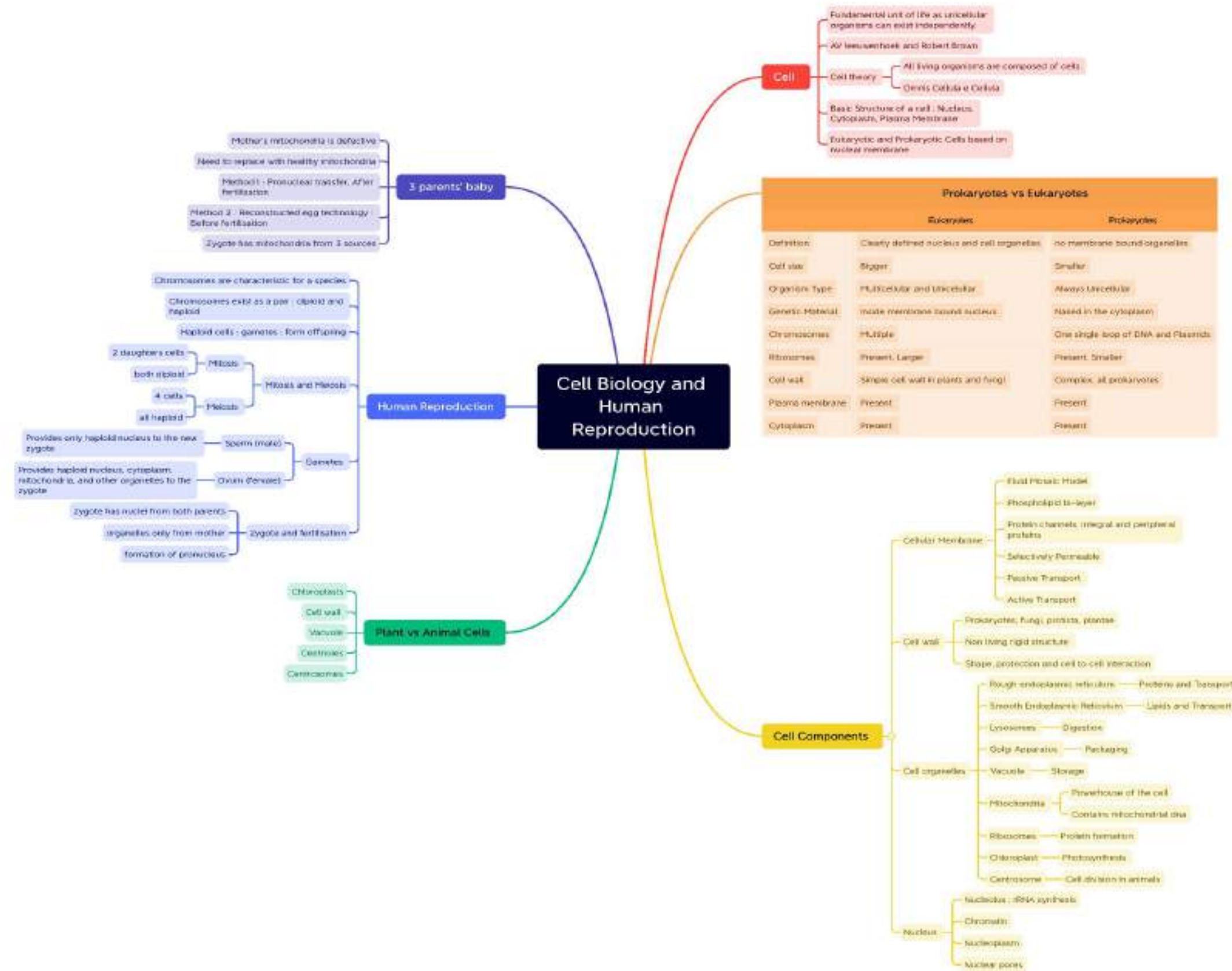
1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. Identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

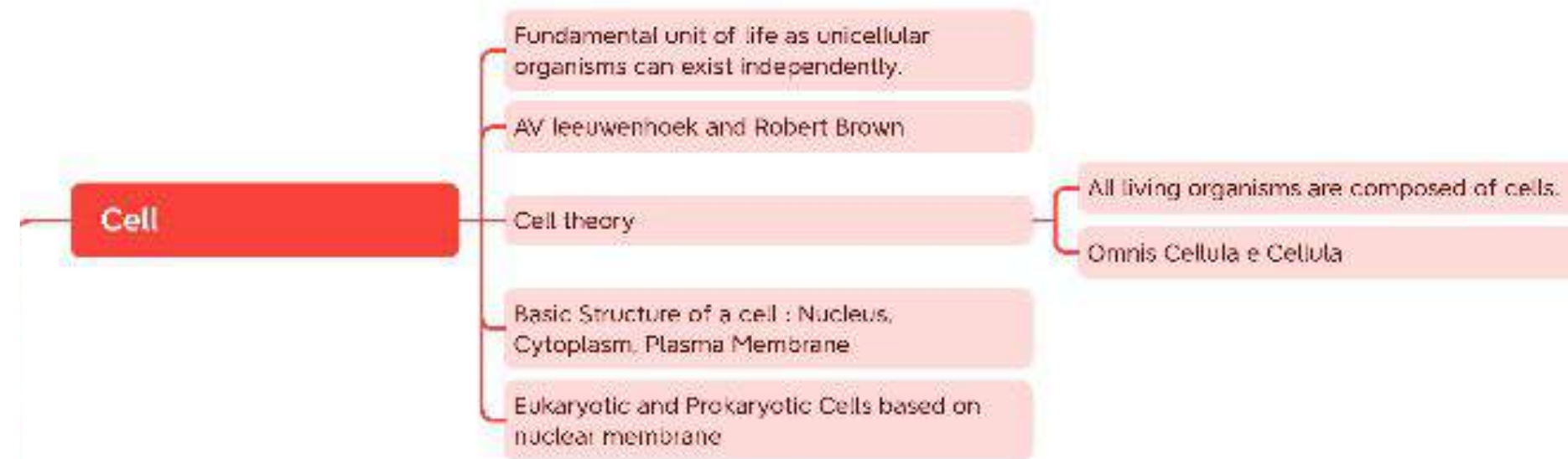
- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3**

Answer: D

- DNA barcoding is a system for species identification focused on the use of a short, standardized genetic region acting as a “barcode”
- DNA barcoding has got no association with determining the age of an organism.
- DNA barcoding is a system for fast and accurate species identification that makes ecological system more accessible by using short DNA sequences instead of whole genome and is used for eukaryotes.
- In recent times, DNA barcoding has emerged as the most potent tool for detection of adulteration of food with unwanted plant and animal material.



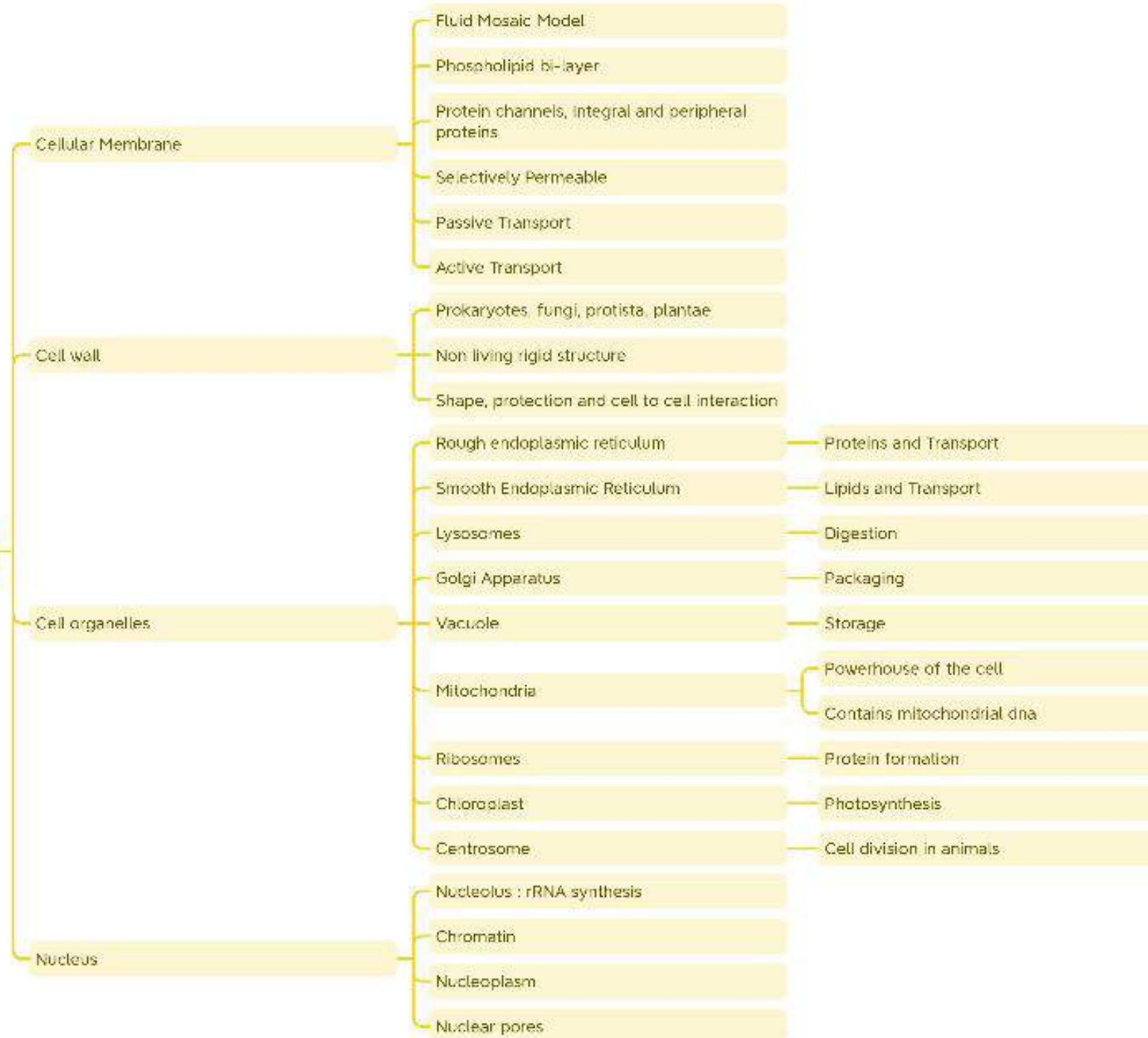
Cell Biology and Human Reproduction



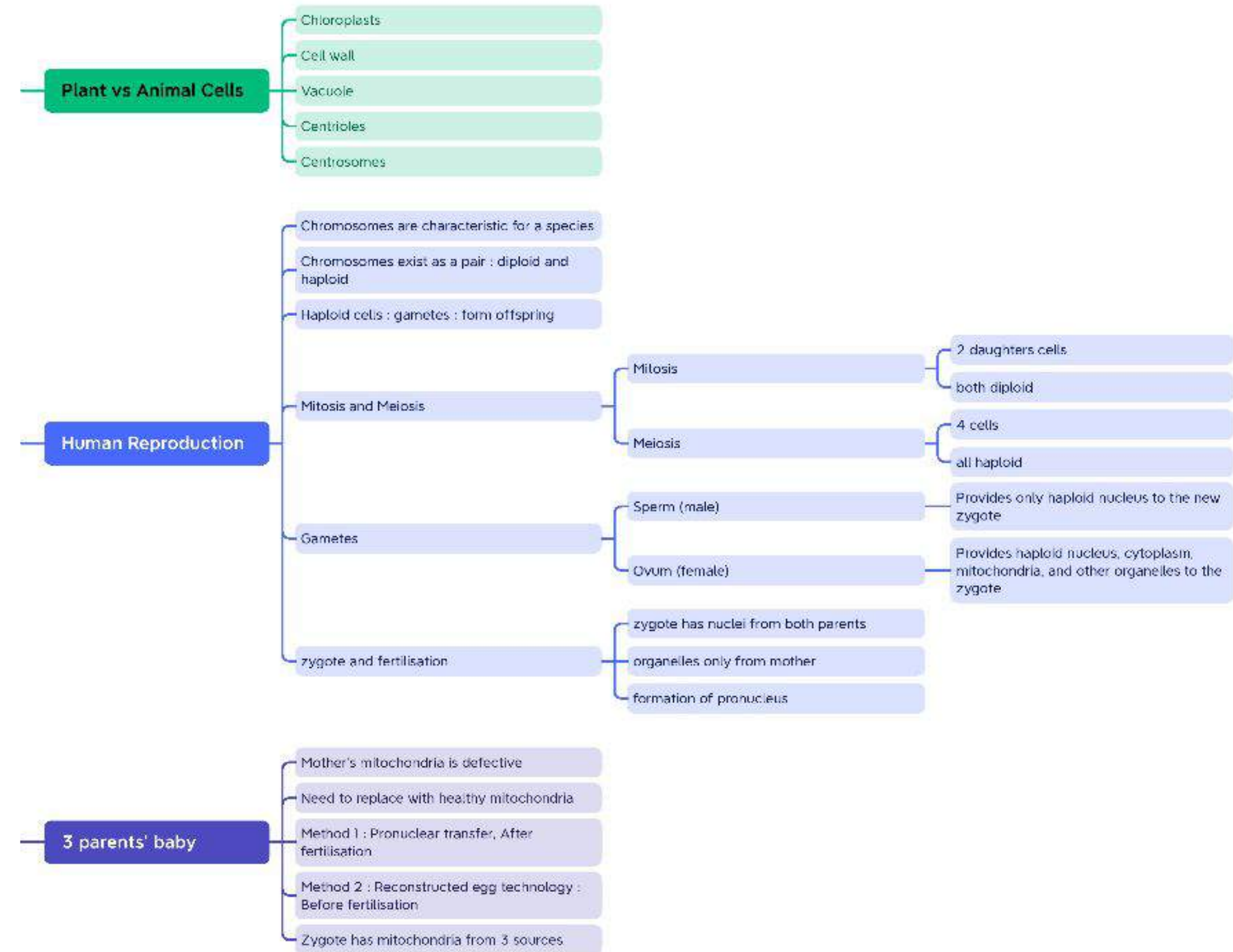
Prokaryotes vs Eukaryotes		
	Eukaryotes	Prokaryotes
Definition	Clearly defined nucleus and cell organelles	no membrane bound organelles
Cell size	Bigger	Smaller
Organism Type	Multicellular and Unicellular	Always Unicellular
Genetic Material	Inside membrane bound nucleus	Naked in the cytoplasm
Chromosomes	Multiple	One single loop of DNA and Plasmids
Ribosomes	Present, Larger	Present, Smaller
Cell wall	Simple cell wall in plants and fungi	Complex, all prokaryotes
Plasma membrane	Present	Present
Cytoplasm	Present	Present

Cell Biology and Human Reproduction

Cell Components



Cell Biology and Human Reproduction



Contact Us



95690 93856



www.edsarrthi.com



ask@sarrthi.com

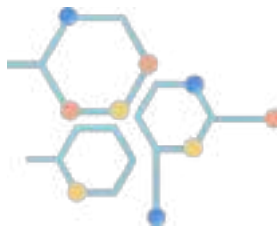


Edsarrthi IAS by Varun Jain!



Clarity for UPSC by Dr. Shivin





UPSC Prelims

Science & Tech

Class 3

Biotechnology

Common Doubts



- Enzymes for protein synthesis vs enzymes for energy creation
- DNA 'change' after viral integration
- Why RNA and why single stranded
- Proteins eaten vs Proteins manufactured
- Other viruses and How they work

Basics of Biotechnology



- DNA to RNA : Transcription
- RNA to protein : Translation
- Creation of mRNA
- Gene vs Genome
- DNA fingerprinting

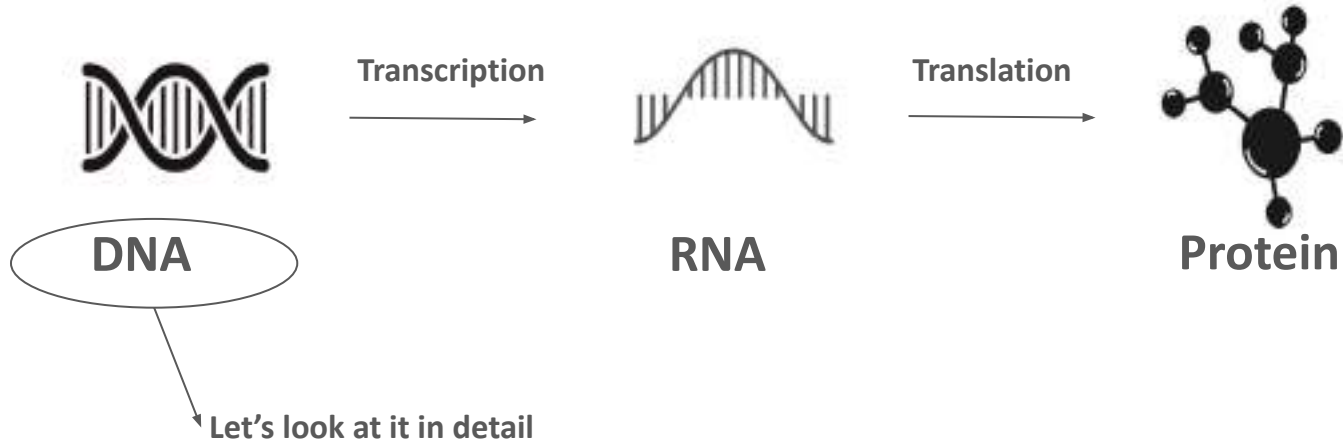


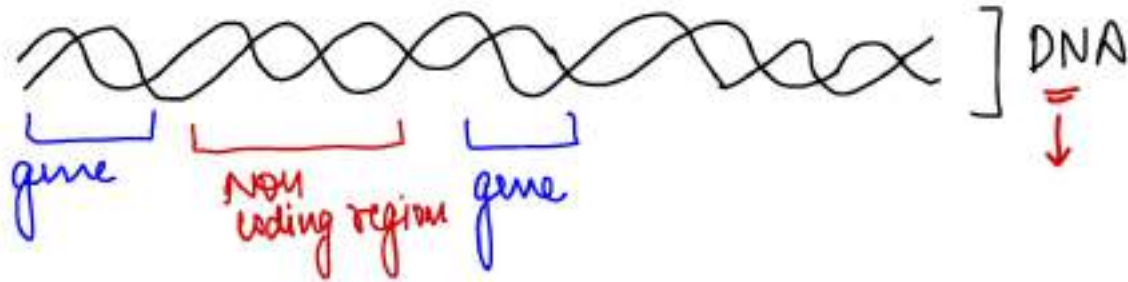
- Nucleic acid is the genetic material of all organisms without any exception.
- Configuration of nuclear material : number or strands / type : DNA or RNA varies.

Basics of Biotechnology



- DNA to RNA : Transcription
- RNA to protein : Translation
- Creation of mRNA
- Gene vs Genome
- DNA fingerprinting





Not all parts of DNA → code for a protein
coding part → genes.
→ code for RNA — code for protein



Question

Between two humans, what will be more similar ?

Genes (Coding Region)

Non Coding Region

Non coding Region

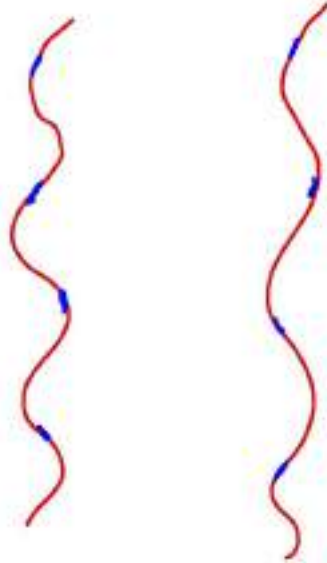
Many differences in between humans

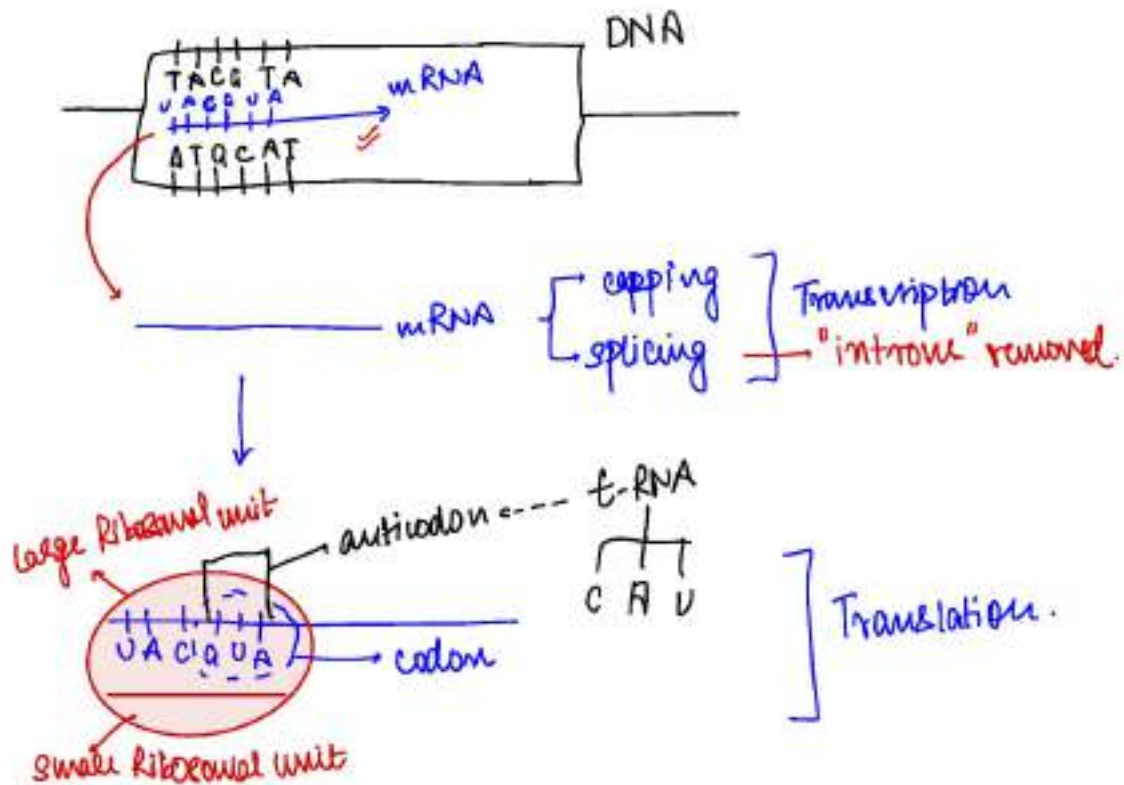


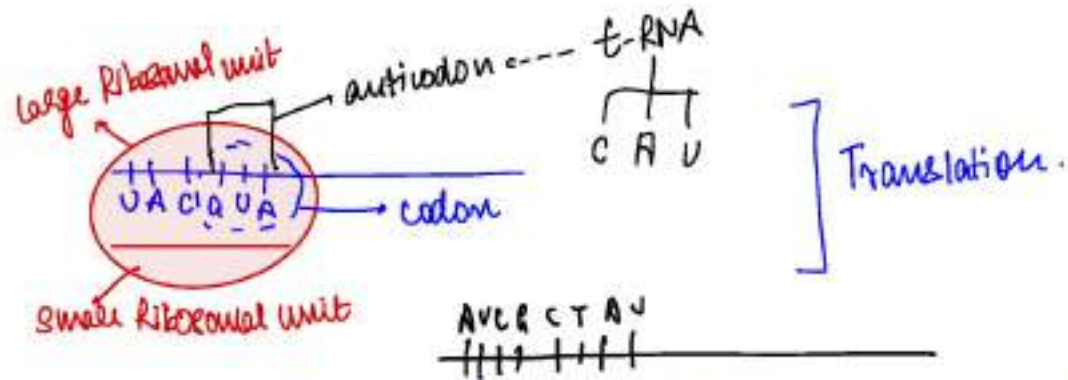
application



DNA fingerprinting



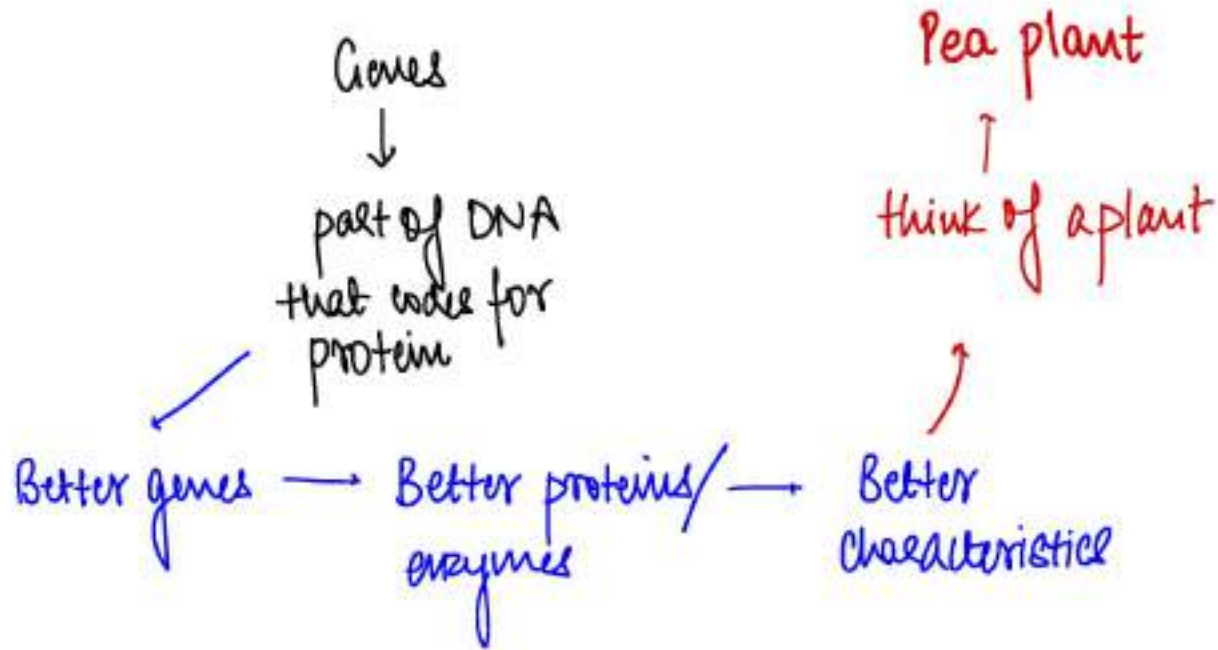


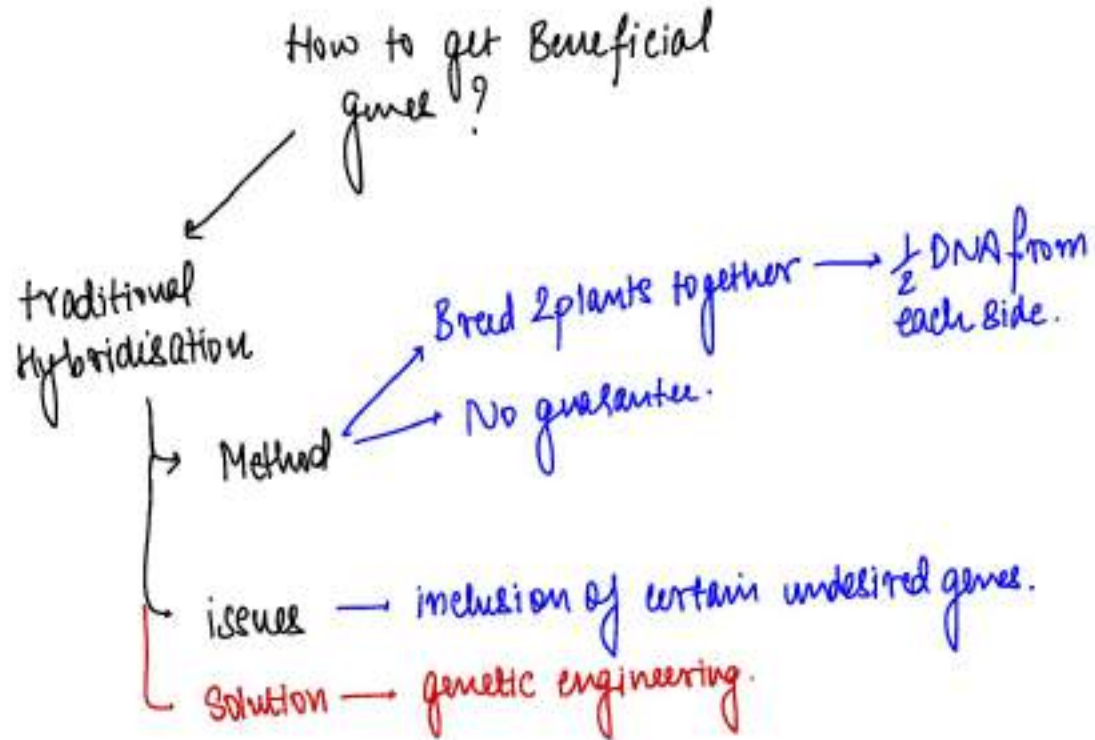


every 3 nucleotides (A, U, C, T) $\rightarrow$ code for an amino acid

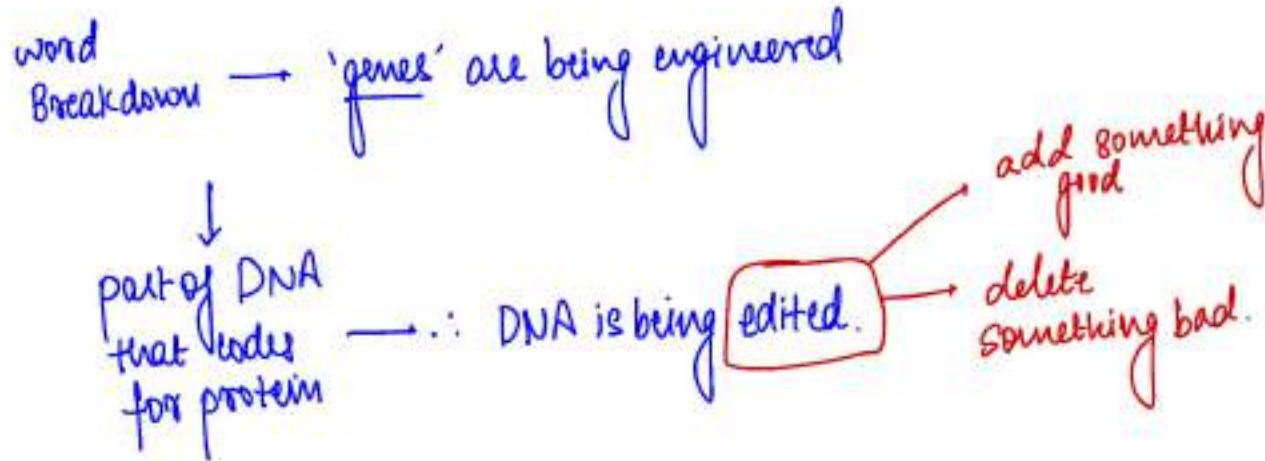
amino acid $\rightarrow$ Make up proteins

These amino acids are brought by 't' $\rightarrow$ transfer RNA.

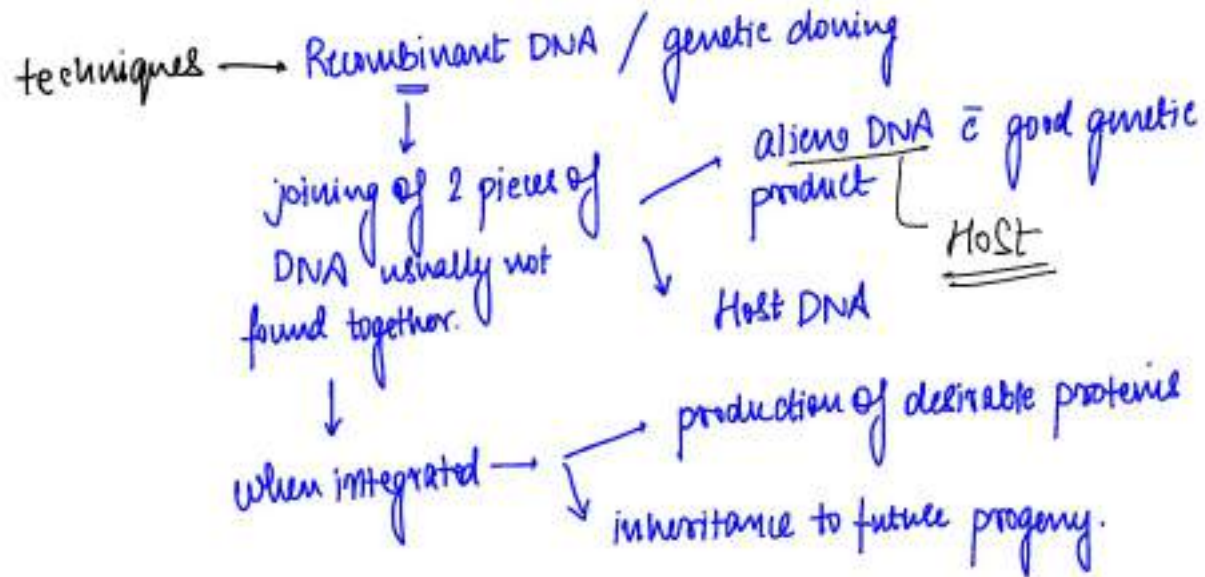




Genetic engineering



Genetic Engineering



Biotechnology : Introduction



- Biotechnology deals with techniques of using live organisms or enzymes from organisms to produce products and processes useful to humans.
- Used in a restricted sense today, to refer to such of those processes which use genetically modified organisms to achieve the same on a larger scale.
- European Federation of Biotechnology (EFB) definition is as follows : The integration of natural science and organisms, cells, parts thereof, and molecular analogues for products and services.

2 principles of Biotechnology



Genetic engineering : Techniques to alter the chemistry of genetic material (DNA and RNA), to introduce these into host organisms and thus change the phenotype of the host organism.

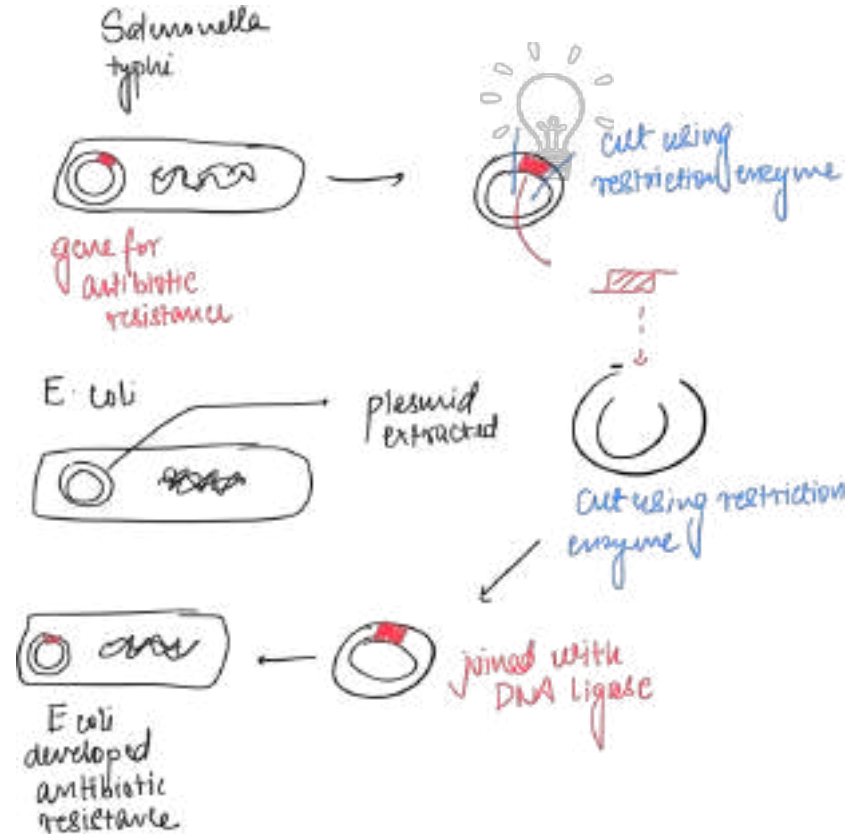
Bioprocess engineering : Maintenance of sterile (microbial contamination-free) ambience in chemical engineering processes to enable growth of only the desired microbe/eukaryotic cell in large quantities for the manufacture of biotechnological products like antibiotics, vaccines, enzymes, etc.

Recombinant DNA

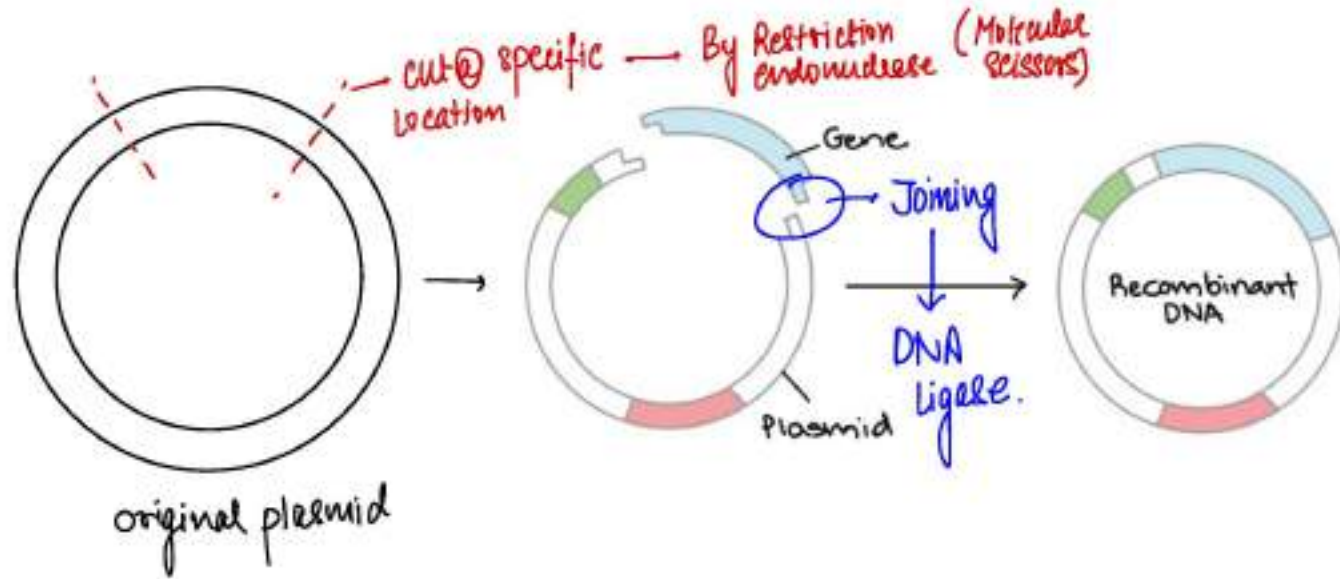


- first instance of the construction of an artificial recombinant DNA molecule : linking a gene encoding antibiotic resistance with a native **plasmid** (autonomously replicating circular extra-chromosomal DNA) of *Salmonella typhimurium* by Stanley Cohen and Herbert Boyer in 1972.

Cohen and Boyer experiment



Plasmid { autonomously replicating circular extra chromosomal DNA
acts as a vector. → Helps in Integration.

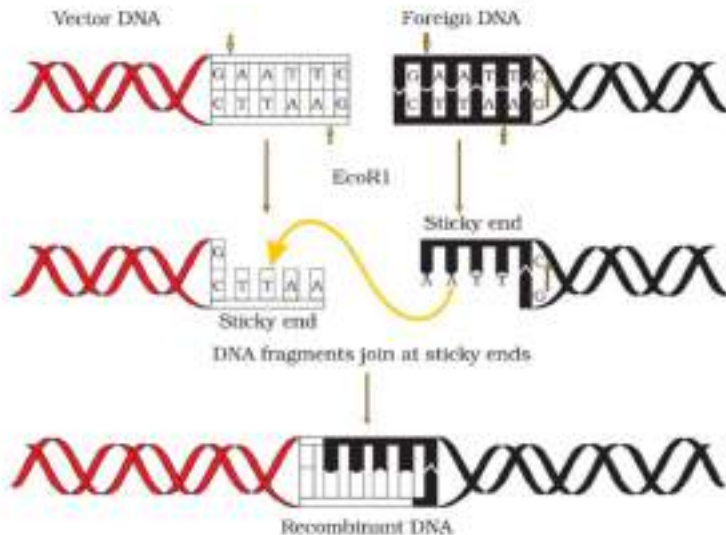


Action of Restriction enzyme

Scissor — cut

The enzyme cuts both DNA strands at the same site

EcoRI cuts the DNA between bases G and A only when the sequence GAATTC is present in the DNA



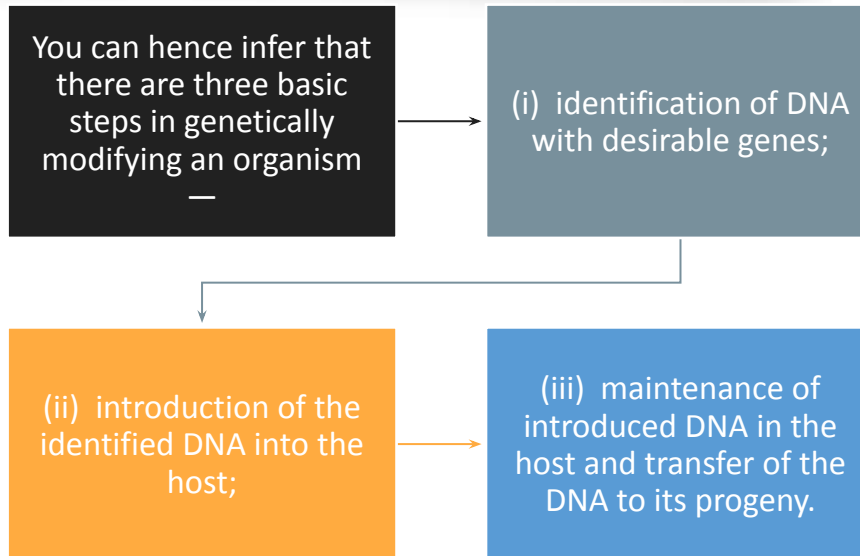
Recombinant DNA technology



- A plasmid can be used as vector to deliver an alien piece of DNA into the host organism.
- The linking of antibiotic resistance gene with the plasmid vector became possible with the enzyme DNA ligase
- This makes a new combination of circular autonomously replicating DNA created *in vitro* and is known as recombinant DNA.
- it could replicate using the new host's DNA polymerase enzyme and make multiple copies.



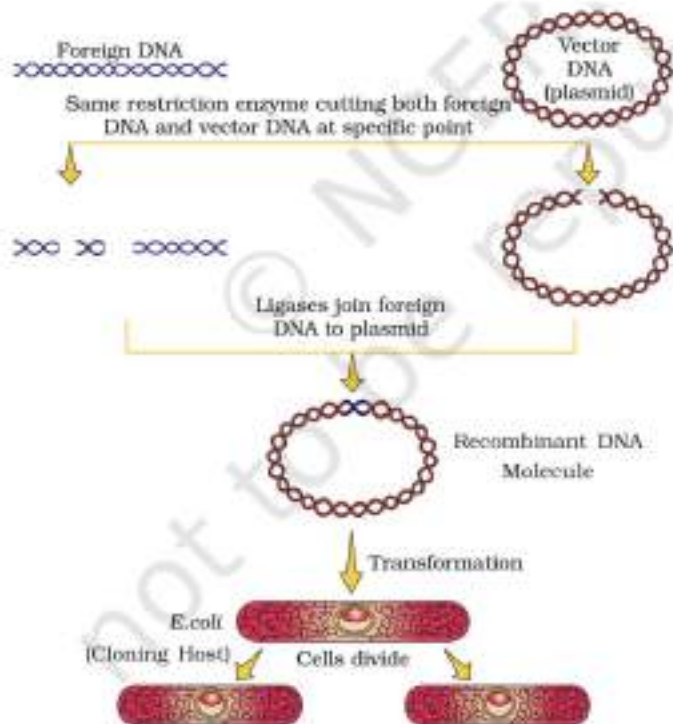
Genetic Modification



Tools of RDT : Restriction Enzymes



- The first restriction endonuclease was Hind II
- It was found that Hind II always cut DNA molecules at a particular point by recognising a specific sequence of six base pairs : this specific sequence was called Recognition Sequence
- Restriction enzymes belong to a larger class of enzymes called nucleases. These are of two kinds; exonucleases and endonucleases.
- Exonucleases remove nucleotides from the ends of the DNA whereas, endonucleases make cuts at specific positions within the DNA.
- Each restriction endonuclease recognises a specific palindromic nucleotide sequences in the DNA.
- When cut by the same restriction enzyme, the resultant DNA fragments have the same kind of 'sticky-ends' and, these can be joined together (end-to-end) using DNA ligases
- Separation and isolation of DNA fragments : Use of Agarose gel electrophoresis



Need of Vectors



- The likely fate of a piece of DNA, which is somehow transferred into an alien organism would not be able to multiply itself in the progeny cells of the organism.
- when it gets integrated into the genome of the recipient, it may multiply and be inherited along with the host DNA.
- This is because the alien piece of DNA has become part of a chromosome, which has the ability to replicate.



Tool of RDT : Cloning Vectors



- Plasmids and Bacteriophages
- Bacteriophages because of their high number per cell, have very high copy numbers of their genome within the bacterial cells.
- If we are able to link an alien piece of DNA with bacteriophage or plasmid DNA, we can multiply its numbers equal to the copy number of the plasmid or bacteriophage.

Steps of RDT

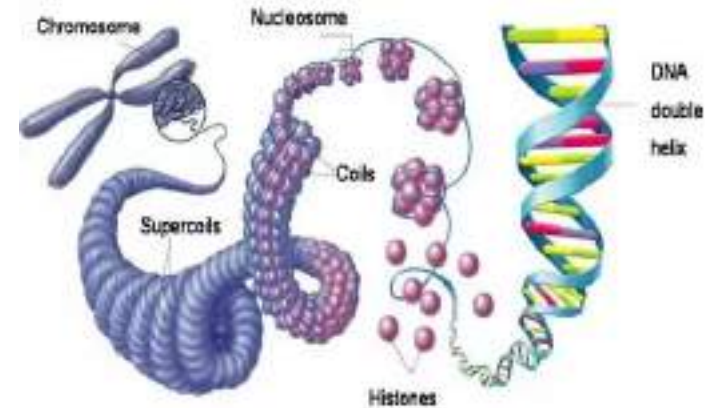


1. Isolation of DNA : pure form
2. Cutting at specific places : RE + gel electrophoresis
3. Polymerase Chain Reaction : amplification of gene of interest
4. Ligation
5. Insertion into host : selectable marker
6. Recombinant Protein production
7. Downstream Processing

Isolation of DNA fragments



1. We need pure form of DNA : break open the cell and release DNA using certain enzymes.
2. This can be achieved by treating
3. the bacterial cells/plant or animal tissue with enzymes such as lysozyme (bacteria), cellulase (plant cells), chitinase (fungus).
4. You do not need to remember these enzymes.
5. Genes are located on DNA that is wound into chromosomes using histones.



Cutting at specific places : RE + gel electrophoresis



1. Cut the source DNA and vector DNA with the same enzyme
2. Agarose gel electrophoresis : separates the DNA fragments by size.
3. This checks progression of restriction enzyme work

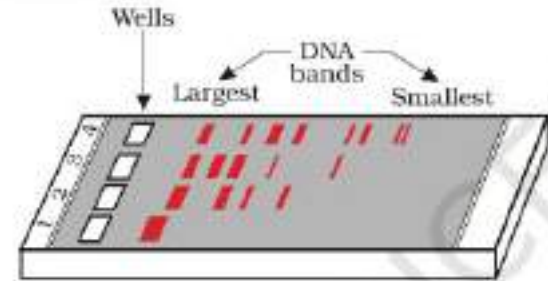
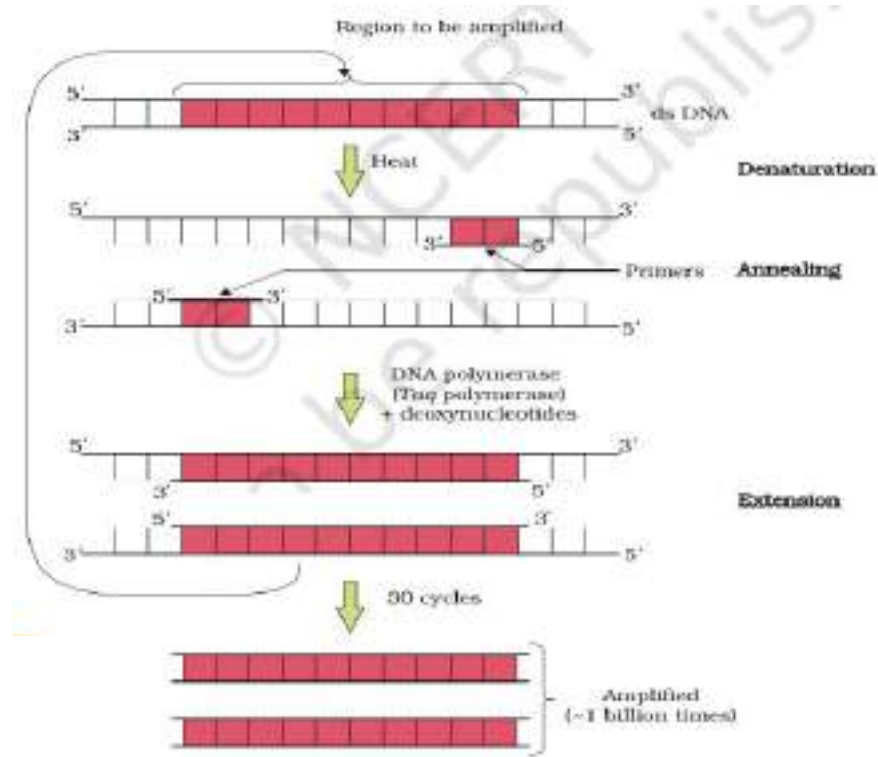
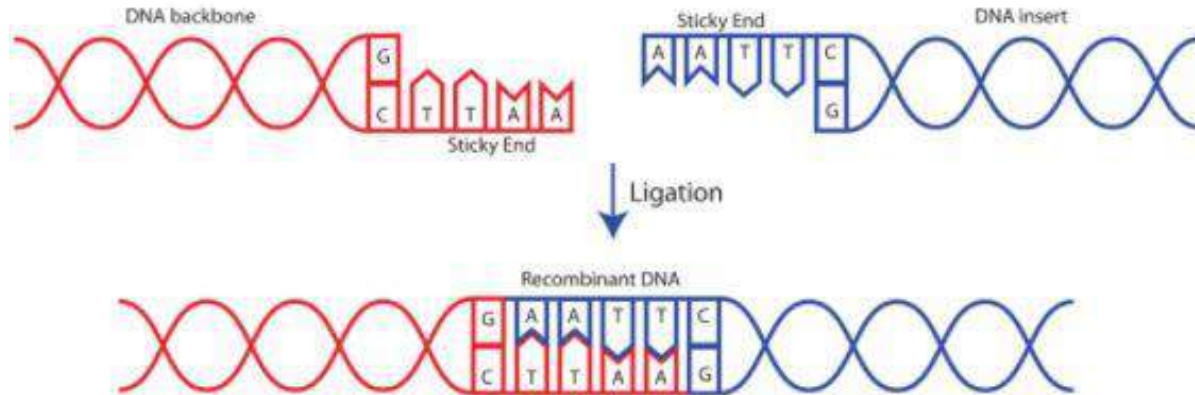


Figure 11.3 A typical agarose gel electrophoresis showing migration of undigested (lane 1) and digested set of DNA fragments (lane 2 to 4)

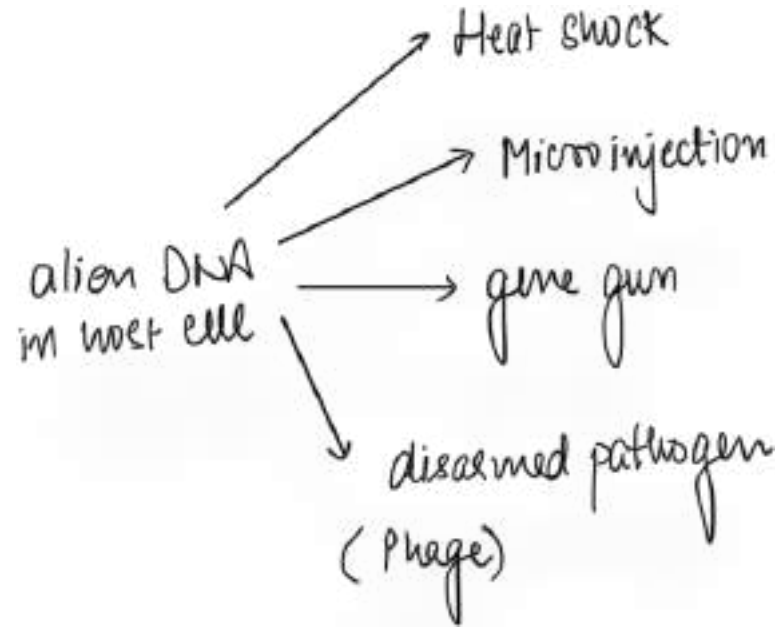
Polymerase chain reaction



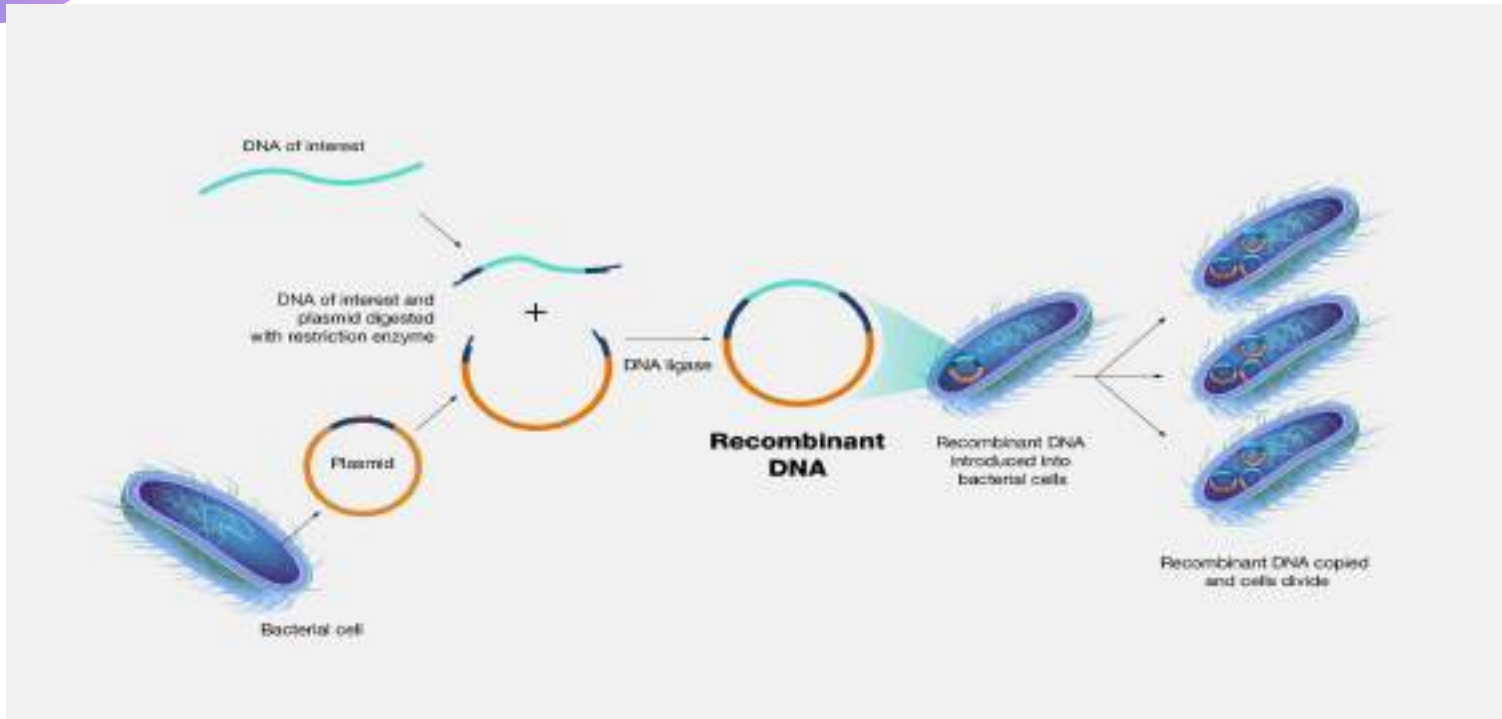
Ligation



Insertion into Host and
production of recombinant
protein



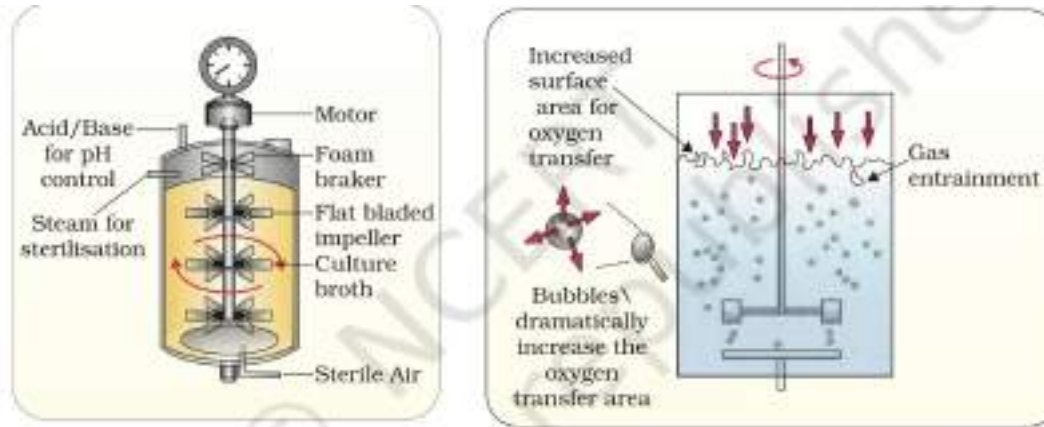
Production of Recombinant Protein



Obtaining foreign gene product



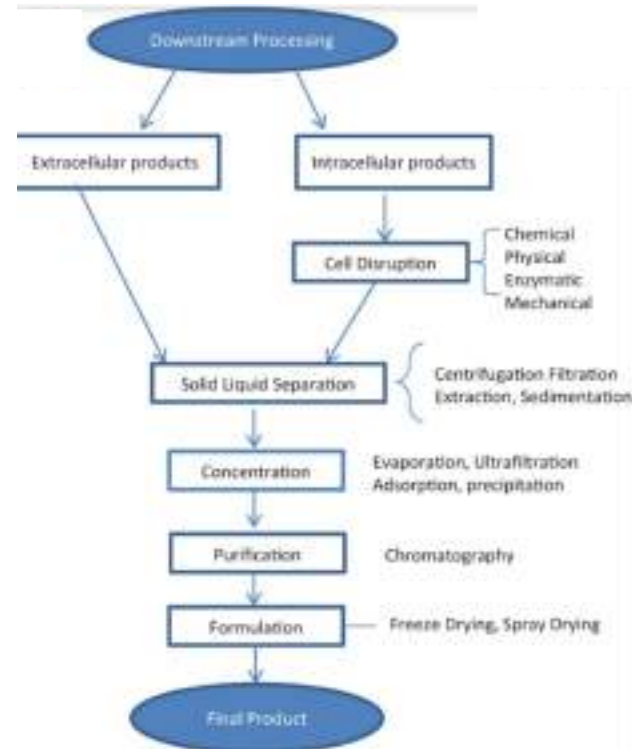
- Use of bioreactors to grow bacteria at a large scale

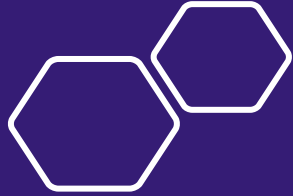


Downstream processing



- Use of bioreactors





Time for Doubts and Questions

5 minutes break

Take this time to revise what
you have learnt so far.

Applications of Biotechnology : Agriculture



- Question : what are the issues of Agrochemical based agriculture?



Solution : GM crops



Applications of genetic modification :

- (i) made crops more tolerant to abiotic stresses (cold, drought, salt, heat).
- (ii) reduced reliance on chemical pesticides (pest-resistant crops).
- (iii) helped to reduce post harvest losses.
- (iv) increased efficiency of mineral usage by plants (this prevents early exhaustion of fertility of soil).
- (v) enhanced nutritional value of food, e.g., golden rice, i.e., Vitamin 'A' enriched rice.
- (vi) Tailor made plants to deliver certain products : pharmaceuticals and fuels.

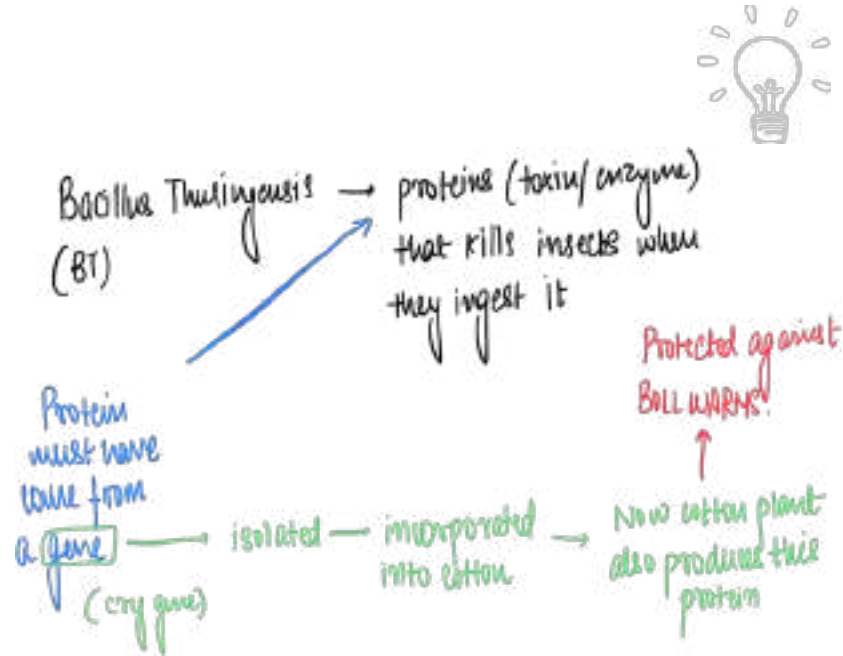
Ex : GM Algae for biofuels

Pest resistance in Detail



- Some strains of bacillus thuringensis produce proteins that kill certain insects (ex : bollworm). This protein is called BT toxin.
- Remains as protoxin in bacteria, but becomes ingested once ingested by a pest.
- Bt toxin gene has been cloned from the bacteria and been expressed in plants to provide resistance to insects without the need for insecticides.
- Specific bt toxin genes isolated from specific strains of bacteria and incorporated into plants. Choice of gene depends on crop and targeted pest.
- in effect this has created a bio-pesticide. Examples are Bt cotton, Bt corn, rice, tomato, potato and soyabean etc.
- Cry genes : CryIAC, cryIIAB : cotton bollworms
- CryIAB : corn borer

BT cotton and Bollworms



Method 2 : RNA interference method



RNAi takes place in all eukaryotic organisms as a method of cellular defense.

The main idea here is to silence the mRNA of the pathogen so that it cannot express its proteins.

This method involves silencing of a specific mRNA due to a complementary dsRNA molecule that binds to and prevents translation of the mRNA (silencing).

A nematode *Meloidogyne incognita* infects the roots of tobacco plants and causes a great reduction in yield.

Using *Agrobacterium* vectors, nematode-specific genes were introduced into the host plant. This allowed host (tobacco) to produce mRNA complementary to the mRNA of pathogen.

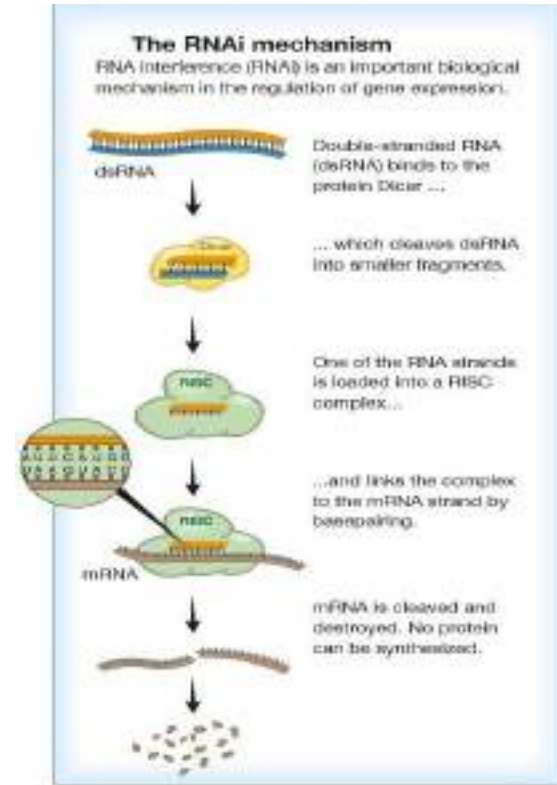
hence two RNA's being complementary to each other formed a double stranded (dsRNA) that initiated RNAi and thus, silenced the specific mRNA, of the nematode. Because of the formation of dsRNA due to binding of the 2 strands, the process of translation was stopped.

The consequence was that the parasite could not survive in a transgenic host expressing specific interfering RNA.

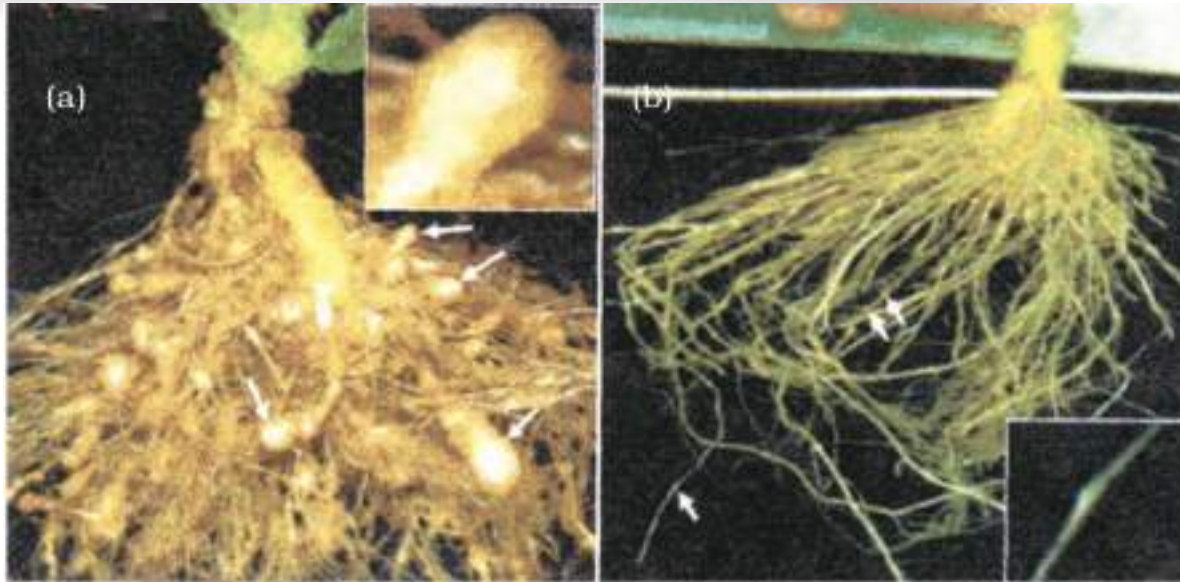
Rna Interference Mechanism



- Dicer protein
- RISC protein
- Formation of dsRNA due to introduction of nematode specific genes
- Stopping translation



RNA interference in tobacco : infection gone!



GM crops : use of mycorrhizza



- A mycorrhiza is a symbiotic relationship between a fungus and a root system.
- A network of fungus filaments surrounds the developing root or reaches the root cells directly.
- The hyphae possess a relatively broad surface area, which allows them to collect water and mineral ions from a significantly more amount of soil than the root.
- Mycorrhizal plants have tolerance to adverse soil, PH, high temp., drought and toxic heavy metals.
- It also helps plants resist certain diseases.
- The roots receive water and minerals from the fungus, and the mycorrhizae receive sugars and N-containing components from the roots.

GM Mustard : DMH 11



- It is a Genetically Modified Mustard
- DMH = Dhara Mustard Hybrid
- It is herbicide tolerant mustard
- Developed by the Centre for Genetic Manipulation of Crop Plants of Delhi University
- It was created by using “barnase/barstar” technology for genetic modification by adding genes from soil bacterium that makes mustard self-pollinating plant. It also allows for cross pollination and hybridisation.
- DMH -11 contains three genes viz. Bar gene, Barnase and Barstar sourced from soil bacterium.
- The bar gene had made plant resistant to herbicide named Basta.

GM Crops Challenges



1. Biosafety : elimination of wild species and pollution of gene pool
2. Safety with respect to human consumption
3. Legal disputes with respect to trait fees : Monsanto developed bt cotton technology called ?? (reveal later). Now seed production companies have to pay royalty to Monsanto.
4. Terminator gene technology : when such plants are planted, they will produce crops only once. Their resultant crop's seeds will be sterile. Farmer forced to buy new seeds every time
5. Genetic Divide : small and large farmers

GM crops : Regulation in India



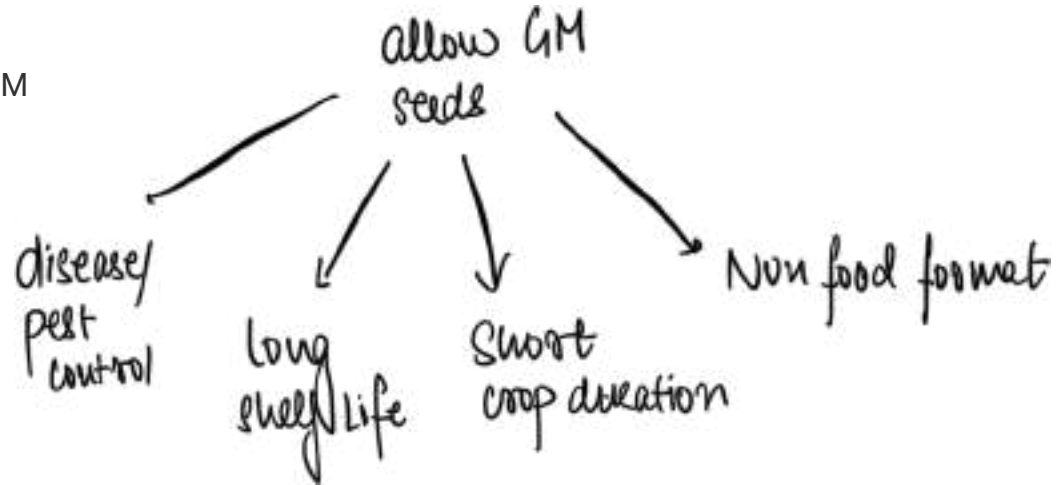
Under Environment Protection Act, 1986 : GEAC (genetic engineering appraisal committee) conducts studies and recommends approving GM crop in India

Final permission : Ministry of Environment, Forests and Climate Change.

Economic Survey 2017 suggestions:



1. Allow GM seeds : low cost, without terminator genes
2. Encourage Domestic production of GM crops to end MNC monopoly



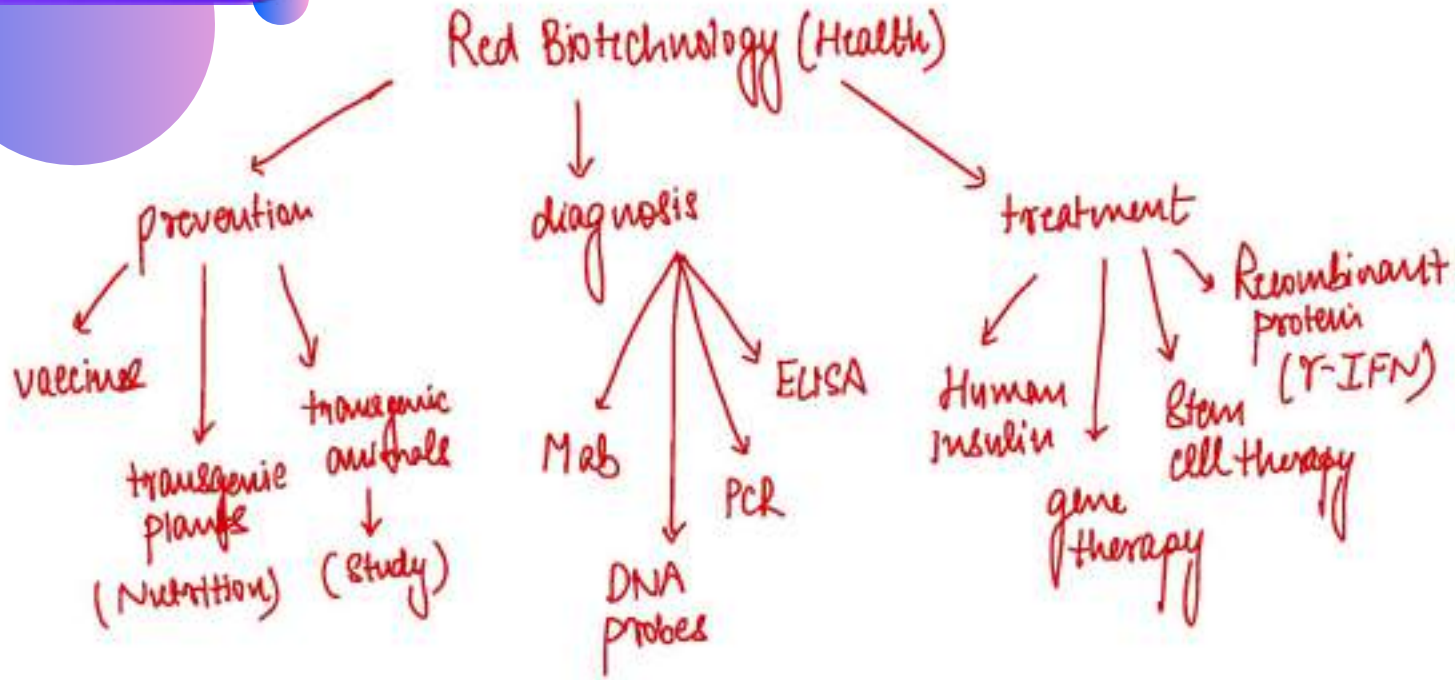
Applications of RDT/Biotechnology [India]

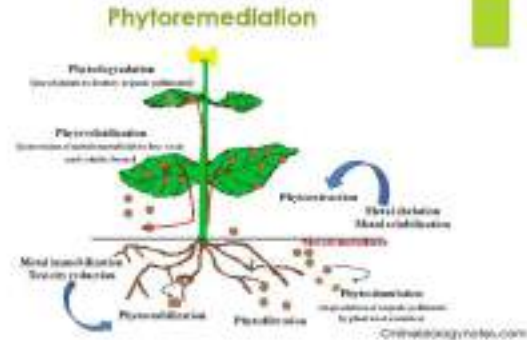
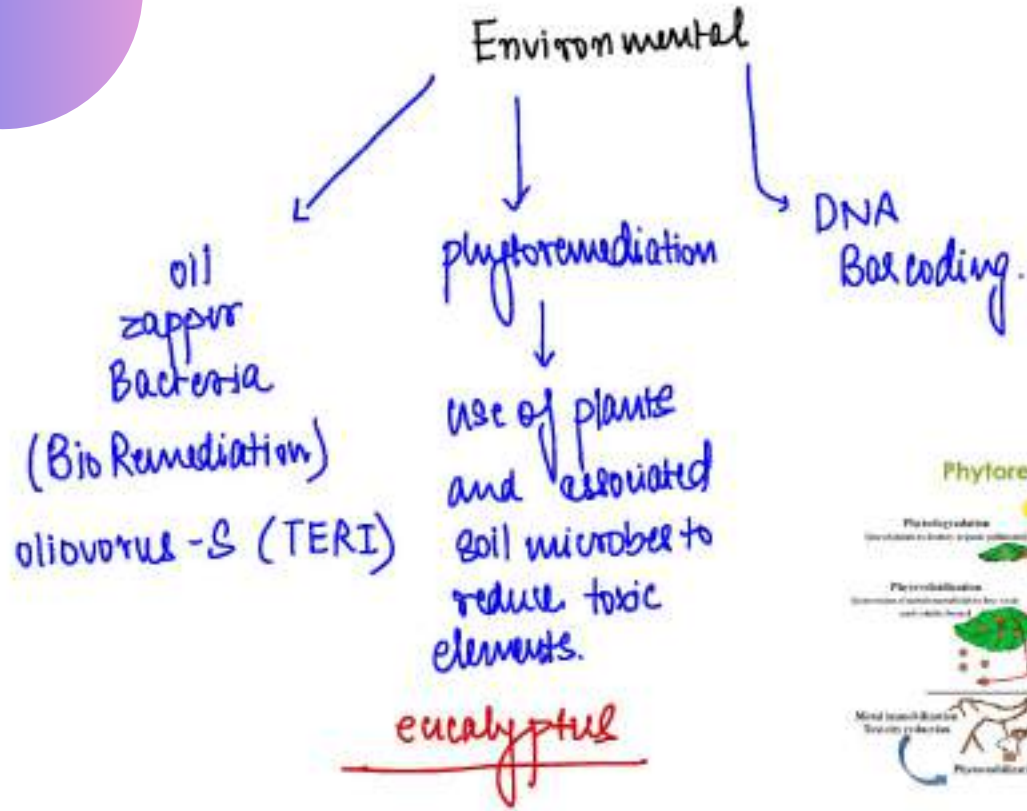
green Biotech (agriculture)

- (i) Yield
- (ii) Biopesticides - BT Brinjal
- (iii) HYV → GM Mustard
- (iv) Biofortification → golden Rice (vit-A)
- (v) Shelf life → Flavr Savr tomato (export)
- (vi) disease (Samba Meenri Rice)
- (vii) RNAi → plant vaccination
- (viii) cold tolerant strawberry
- (ix) Better Breed → cows, etc.

Blue Biotechnology (Marine)

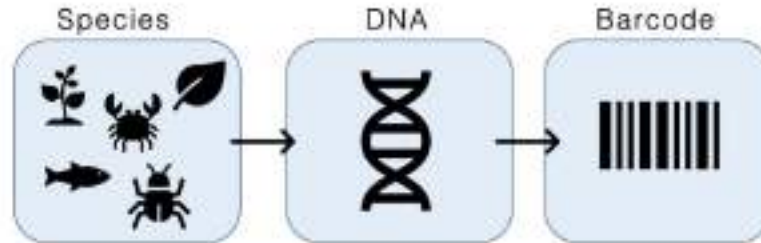
- aquaculture (sea-food cultivation)
- Transgenic fish → freeze resistant fish
- Conservation of species
- seaweed cultivation → ex: agar-agar.





DNA Barcoding

DNA barcoding is a method of species identification using a short section of DNA from a specific gene or genes. The premise of DNA barcoding is that by comparison with a reference library of such DNA sections (also called "sequences"), an individual sequence can be used to uniquely identify an organism to species, just as a supermarket scanner scans barcodes.

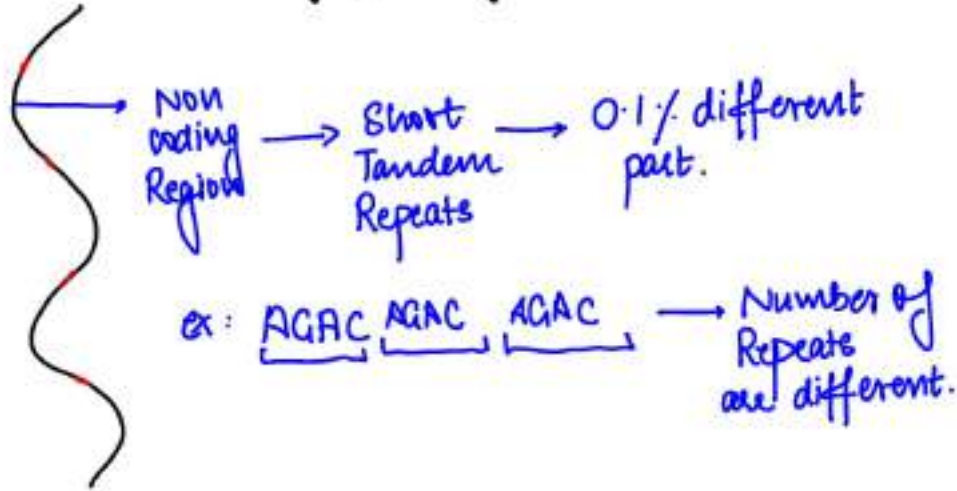


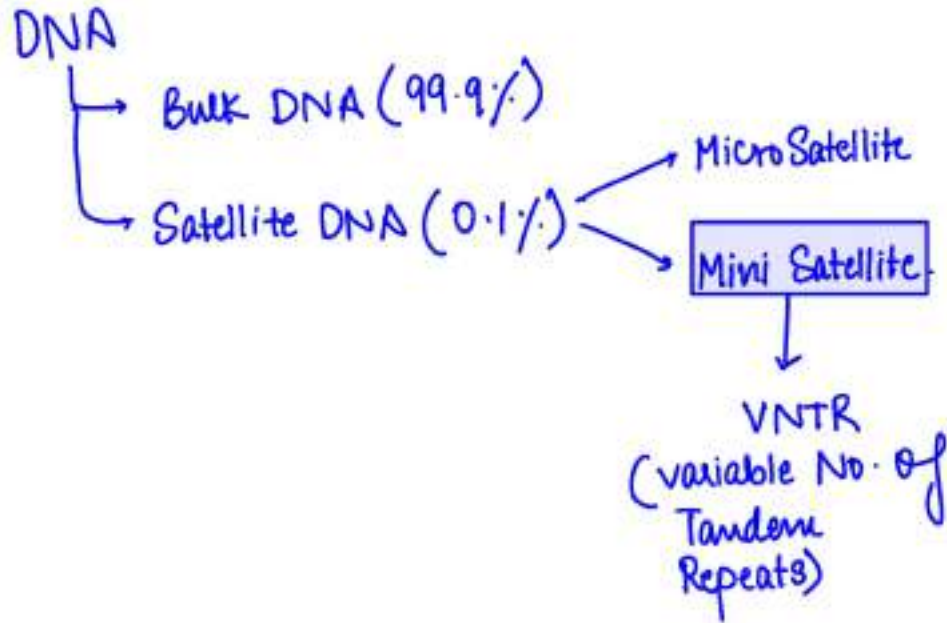
Forensic Biotechnology

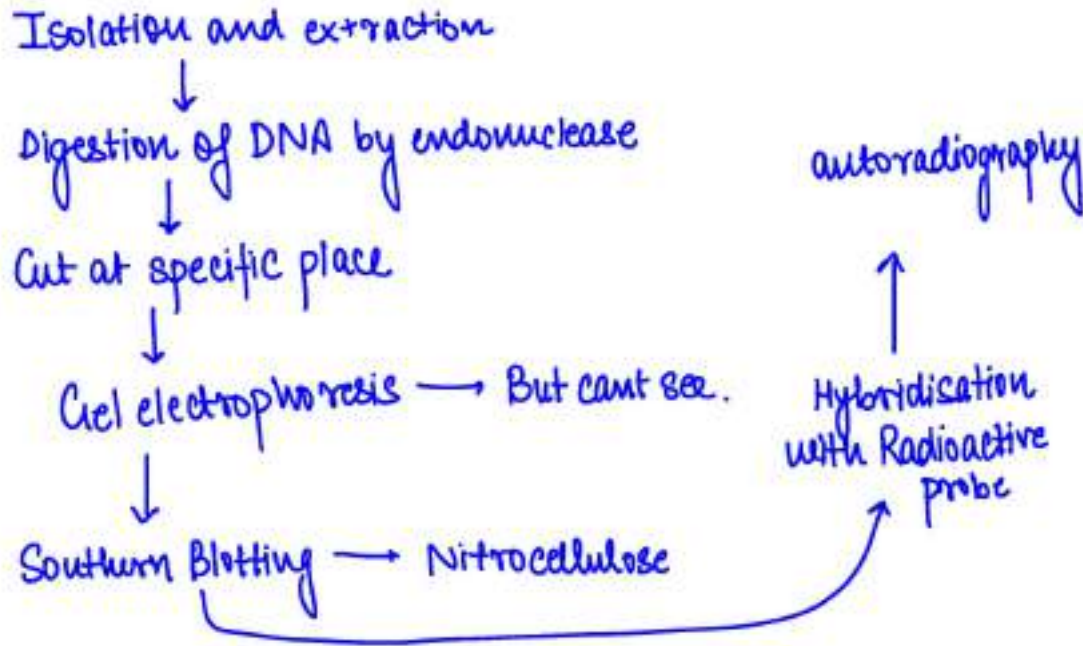


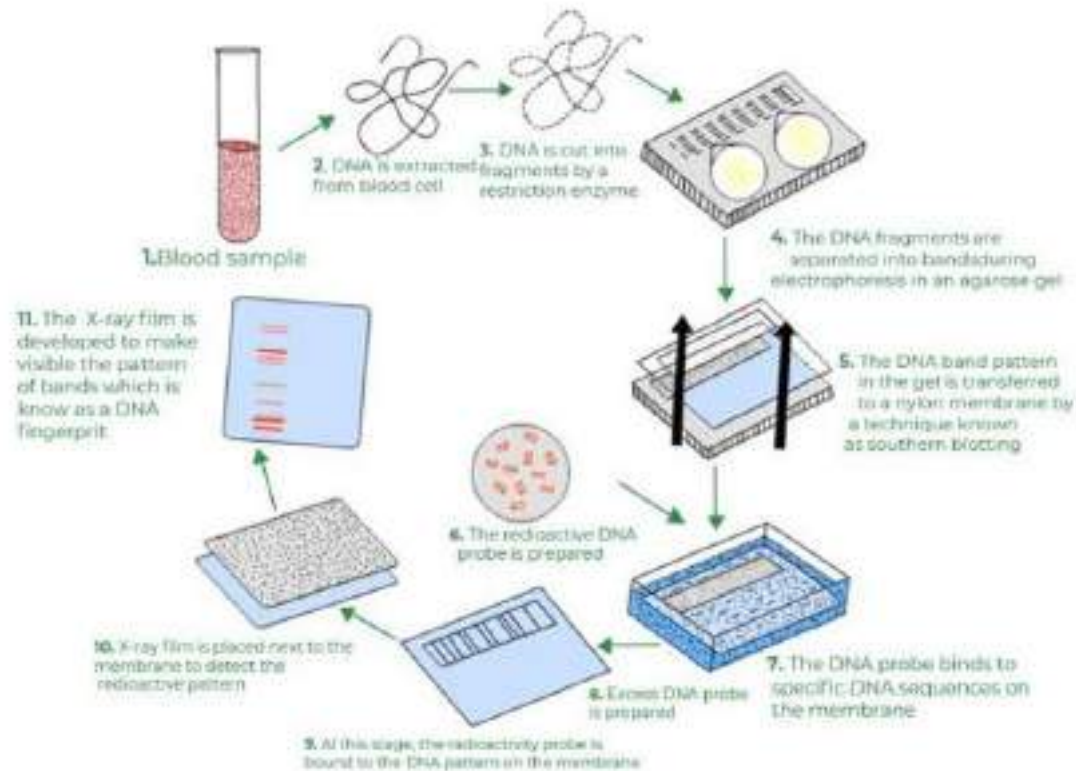
- DNA fingerprinting : identification of criminal by use of biotechnology.
- Here we use something called variable number of tandem repeats (VNTR)
- Use to develop DNA bands in Gel electrophoresis and identify a person.
- Can be used in :
 - Criminal Investigation
 - Paternity testing
 - Immigration Cases
 - Child swapping cases
 - Disaster victim identification

DNA fingerprinting









Test of 6 Questions!

Treat this as a learning exercise!





Question 1

Q. With reference to the Genetically Modified mustard (GM mustard) developed in India, consider the following statements :

1. GM mustard has the genes of a soil bacterium that gives the plant the property of pest-resistance to a wide variety of pests.
2. GM mustard has the genes that allow the plant cross-pollination and hybridization.
3. GM mustard has been developed jointly by the IARI and Punjab Agricultural University.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 only
- c) 2 and 3 only
- d) 1, 2 and 3



Question 2

Q. 'RNA interference (RNAi)' technology has gained popularity in the last few years. Why?

1. It is used in developing gene silencing therapies.
2. It can be used in developing therapies for the treatment of cancer.
3. It can be used to develop hormone replacement therapies.
4. It can be used to produce crop plants that are resistant to viral pathogens.

Select the correct answer using the code given below:

- a) 1, 2 and 4
- b) 2 and 3
- c) 1 and 3
- d) 1 and 4 only



Question 3

Q. Recombinant DNA technology (Genetic Engineering) allows genes to be transferred

1. Across different species of plants.
2. From animals to plants.
3. From microorganisms to higher organisms.

Select the correct Solution using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3



Question 4

Q. Mycorrhizal Biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to -

1. Resist drought and increase absorptive area
2. Tolerate extremes of pH
3. Resist disease infestation

Select the correct answer using the codes given below:

- a) 1 only
- b) 1 and 2 only
- c) 1, 2 and 3 only
- d) 2 and 3 only



Question 5

Q. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future?

1. Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
2. This technique helps in reducing the time required to develop new varieties of crop plants.
3. It can be used to decipher the host-pathogen relationships in crops.



Question 6

Q. Bollgard I and Bollgard II technologies are mentioned in the context of:

- A. clonal propagation of crop plants
- B developing genetically modified crop plants
- C. production of plant growth substances
- D. production of biofertilizers

Congratulations on giving the test!

Let's Discuss The Answers.





Question 1 (Year 2018)

Q. With reference to the Genetically Modified mustard (GM mustard) developed in India, consider the following statements :

1. GM mustard has the genes of a soil bacterium that gives the plant the property of pest-resistance to a wide variety of pests.
2. GM mustard has the genes that allow the plant cross-pollination and hybridization.
3. GM mustard has been developed jointly by the IARI and Punjab Agricultural University.

Which of the statements given above is/are correct?

- a) 1 and 3 only
- b) 2 only**
- c) 2 and 3 only
- d) 1, 2 and 3

Answer: B

- Dhara Mustard Hybrid-11, otherwise known as DMH - 11, is a genetically modified hybrid variety of the mustard species *Brassica juncea*.
- It was developed by Professor Deepak Pental from the University of Delhi, with the aim of reducing India's demand for edible oil imports. DMH - 11 was created through transgenic technology, primarily involving the Bar, Barnase and Barstar gene system.
- The Barnase gene confers male sterility, while the Barstar gene restores DMH - 11's ability to produce fertile seeds.
- The insertion of the third gene Bar enables DMH - 11 to produce phosphinothricin-Nacetyl-transferase, the enzyme responsible for Glufosinate resistance.
- This hybrid mustard variety has come under intense public scrutiny, mainly due to concerns regarding DMH - 11's potential to adversely affect the environment as well as consumer health.
- DMH - 11 was found not to pose any food allergy risks, and has demonstrated increased yields over existing mustard varieties. Conflicting details and results regarding the field trials and safety evaluations conducted on DMH - 11 have delayed its approval for commercial cropping



Question 2 (Year 2019)

Q. 'RNA interference (RNAi)' technology has gained popularity in the last few years. Why?

1. It is used in developing gene silencing therapies.
2. It can be used in developing therapies for the treatment of cancer.
3. It can be used to develop hormone replacement therapies.
4. It can be used to produce crop plants that are resistant to viral pathogens.

Select the correct answer using the code given below:

- a) 1, 2 and 4**
b) 2 and 3
c) 1 and 3
d) 1 and 4 only

Answer: A

- RNA interference (RNAi) or Post-Transcriptional Gene Silencing (PTGS) is a conserved biological response to double-stranded RNA that mediates resistance to both endogenous parasitic and exogenous pathogenic nucleic acids, and regulates the expression of protein-coding genes
- Although most research is currently looking into the applications of RNAi in cancer treatment, the list of possible applications is extensive.
- RNAi could potentially be used to treat viruses, bacterial diseases, parasites, maladaptive genetic mutations, control drug consumption, provide pain relief, and even modulate sleep
- RNA interference has been used for applications in biotechnology and is nearing commercialization in other fields.
- RNAi has resulted in the invention of novel crops such as nicotine-free tobacco, decaffeinated coffee, nutrient fortified vegetation, and hypoallergenic crops.
- The genetically-engineered Arctic apples received FDA approval in 2015.
- The apples were produced by RNAi suppression of the PPO (polyphenol oxidase) gene, making apple varieties that will not undergo browning after being sliced.
- PPO-silenced apples are unable to convert chlorogenic acid into the standard quinone product



Question 3 (Year 2013)

Q. Recombinant DNA technology (Genetic Engineering) allows genes to be transferred

1. Across different species of plants.
2. From animals to plants.
3. From microorganisms to higher organisms.

Select the correct Solution using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3**



Answer: D

- Recombinant DNA technology is a major DNA-based tool that opens a new age for modern biotechnology.
- With this technology, a gene or multiple genes can be identified, cut, and inserted into the genome of another organism.
- Golden rice is developed by transferring carrot genes to rice. the new variety of rice contains Vitamin "A".
- Hence Statement 1 correct (across different species of plants)
- In 1953, scientists discovered the structure of DNA, and in 1972, researchers developed a method for cutting and splicing DNA.
- That method became known as recombinant DNA, or rDNA. Since then, researchers have learned how to move genetic material in the form of DNA from one plant or animal to another.
- They can also take DNA from an animal and place it in a plant, and vice versa



Question 4 (Year 2013)

Q. Mycorrhizal Biotechnology has been used in rehabilitating degraded sites because mycorrhiza enables the plants to -

1. Resist drought and increase absorptive area
2. Tolerate extremes of pH
3. Resist disease infestation

Select the correct answer using the codes given below:

- a) 1 only
- b) 1 and 2 only
- c) 1, 2 and 3 only
- d) 2 and 3 only



Answer: C

- Mycorrhizal plants have tolerance to adverse soil, PH, high temp., drought and toxic heavy metals.
- The most important role of Mycorrhizal fungi is to absorb nutrients and water from the soil and transfer them to their hosts.
- Mycorrhizae help plants withstand root diseases.
- Oxalic acid production by mycorrhizal fungi contributes to disease suppression.



Question 5 (Year 2017)

Q. With reference to agriculture in India, how can the technique of 'genome sequencing', often seen in the news, be used in the immediate future?

1. Genome sequencing can be used to identify genetic markers for disease resistance and drought tolerance in various crop plants.
2. This technique helps in reducing the time required to develop new varieties of crop plants.
3. It can be used to decipher the host-pathogen relationships in crops.



Answer: D

Genome sequencing is figuring out the order of DNA nucleotides, or bases, in a genome—the order of As, Cs, Gs, and Ts that make up an organism's DNA.

Genome sequencing of wild varieties of plants can be used to identify disease resistance and drought tolerance genes in various plants and develop new varieties of crop plants in lesser time. Genome sequencing of crop plants can be helpful in deciphering and understanding the host-pathogen relationship in crops.

Therefore, all statements are correct.



Question 6 (Year 2021)

Q. Bollgard I and Bollgard II technologies are mentioned in the context of:

- A. clonal propagation of crop plants
- B developing genetically modified crop plants**
- C. production of plant growth substances
- D. production of biofertilizers

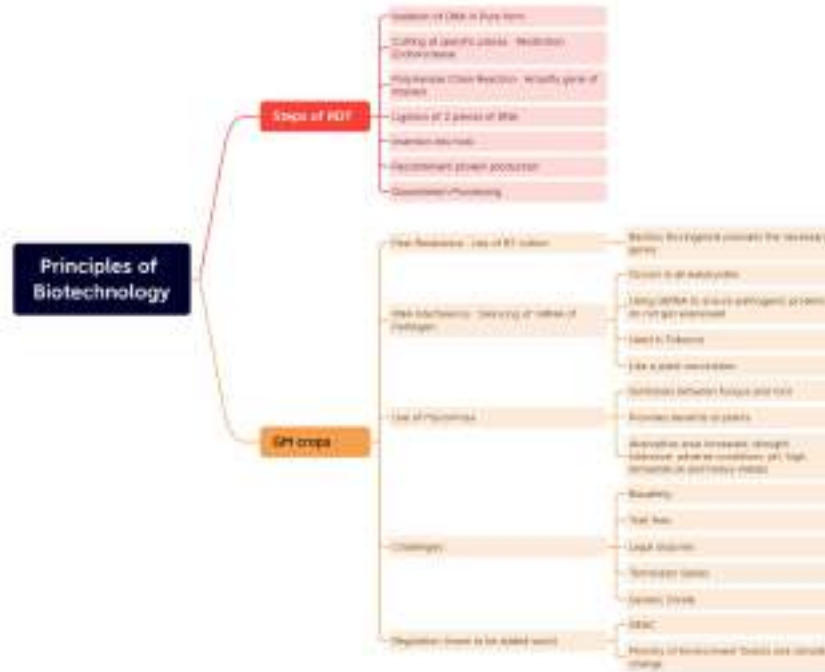


Answer: B

Bollgard I and Bollgard II technologies:

- Bollgard I Bt cotton (single-gene technology) is India's first biotech crop technology approved for commercialization in India in 2002, followed by Bollgard II - double-gene technology in mid-2006, by the Genetic Engineering Approval Committee (GEAC), the Indian regulatory body for biotech crops. Hence, option B is the correct answer.
- Bollgard I cotton provides in-built protection for cotton against destructive American Bollworm *Heliothis Armigera* infestations and contains an insecticidal protein from a naturally occurring soil microorganism, *Bacillus thuringiensis* (Bt).
- Bollgard II technology contains a superior double-gene technology - Cry1Ac and Cry 2Ab which provides protection against bollworms and *Spodoptera* caterpillar, leading to better boll retention, maximum yield, lower pesticides costs, and protection against insect resistance.
- Both, Bollgard II and Bollgard I insect-protected cotton is widely planted around the world as an environmentally friendly way of controlling bollworms.





Prepared with:

Contact Us



95690 93856



www.edsarrthi.com



ask@sarrthi.com

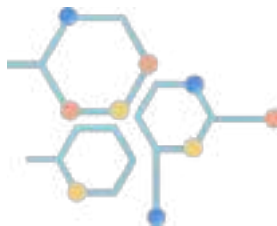


Edsarrthi IAS by Varun Jain!



Clarity for UPSC by Dr. Shivin





UPSC Prelims

Science & Tech

Class 4

Applications of Biotechnology

Class 4 : topics to be covered



- GM crops and Transgenic Animals
- RDT in Medicine : Human Insulin, Gene Therapy
- Crispr CAS – 9
- Animal Cloning
- Stem Cell Therapy
- Vaccination
- Various Applications of Biotechnology

GM crops in India



- GEAC then MoEFCC
- DMH-11 : GM mustard : GEAC vacillating
- BT brinjal : immunity / diversity / side effects (33)
- Issue of Pest Resistance
- BT brinjal was banned
- New issues with Bt cotton : white fly attacks
- MS Swaminathan : called BT crops a failure, govt should resort to Genetic Engineering as a last resort only.

GM plants Regulation : Global



- Protocols to the Convention on Biological Diversity :
- Nagoya protocol
- Cartagena Protocol
- Nagoya Kuala Lumpur protocol
- We will study these when we study environment

GM Crops in India



- Regulatory Framework : Guided by the Environment Protection Act, 1986 and the Rules for the Manufacture/Use/Import/Export and Storage of Hazardous Microorganisms, Genetically Engineered Organisms or Cells, 1989 under the act
- Rules are broad in scope and cover most elements relating to GM crops including sale, storage, export, import production etc.
- Many committees are there for various functions relating to GM crops.

Competent Authorities



1. The Recombinant DNA Advisory Committee (RDAC)



Advisory

2. Institutional Biosafety Committee (IBSC)

3. Review Committee on Genetic Manipulation (RCGM)

4. Genetic Engineering Appraisal Committee (GEAC)



Approval

5. State Biotechnology Coordination Committee (SBCC)

6. District Level Committee (DLC)



Monitoring

Various Committees



- Recombinant DNA Advisory Committee : under DBT, review developments in biotechnology at national and international levels.
- Institutional BioSafety Committee : at every institute, ensures that research in GM crops happens safely. Approves low risk experiments.
- Review Committee on Genetic Modification : Regulatory body inside DBT, responsible for bringing out manuals/ guidelines specifying procedures for research/ use /applications in GMOs. Reviews all projects involving high risk field experiments.
- Genetic Engineering Appraisal Committee : Apex regulatory committee under MOEFCC and gives approval for commercial release of GM crops.

Various Committees



- State Biotechnology Coordination Committee: Monitors at the state level that activities relating to GM crops happen as per regulations. State level nodal agency to assess damage due to GM crops.
- District Level Committee : Monitors at the district level activities related to GM crops. Headed by DC.

Various Committees



- State Biotechnology Coordination Committee: Monitors at the state level that activities relating to GM crops happen as per regulations. State level nodal agency to assess damage due to GM crops.
- District Level Committee : Monitors at the district level activities related to GM crops. Headed by DC.

GEAC : Most important body



- Import, export, manufacture, store, process, use, sell GMOs only with the approval of GEAC.
- Under MoEFCC
- It can also revoke approvals in case of :
 - New information on harmful effects of GMOs
 - GMOs causing damage to environment as was not seen at time of approval
 - Non Compliance of any conditions stipulated by GEAC.

Genetic Engineering Appraisal Committee (GEAC)

Functions under: Ministry of Environment, Forest and Climate Change (MoEF&CC).

Objectives: Responsible for approval of

- **Large scale use of hazardous microorganisms and recombinants** in research and industrial production from environmental angle.
- **Release of genetically engineered organisms and products into environment** including experimental field trials.
- Taking **punitive action under the Environment (Protection) Act 1986.**

Composition: Presently, it has **24 members** and it meets every month. **Chaired by Special Secretary/Additional Secretary of MoEF&CC.**



Exemption from the rules

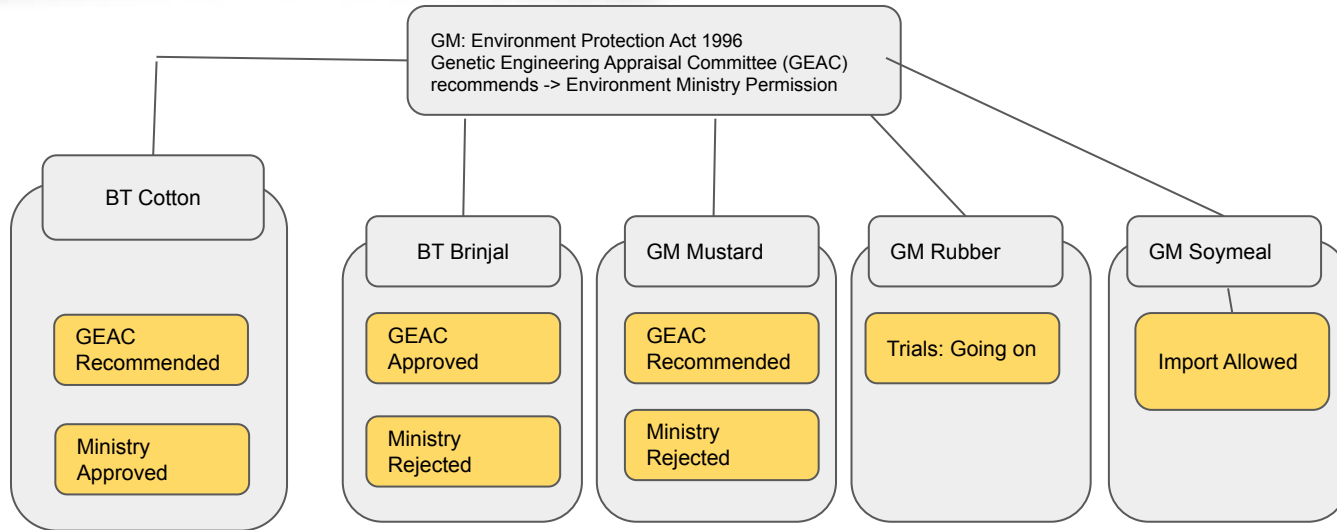


- MoEFCC can grant exemption to certain GMO handlers from the provisions of the rules
- Exemption can be granted for :
 - Indigenous development of recombinant pharmaceuticals that have less risk (risk group 1 and 2)
 - Import and marketing of products derived from LMOs where the end product is not an LMO.
 - Event based approval mechanism for Bt Cotton Hybrids : BT cotton hybrids expressing approved events developed through conventional hybridization are exempted from conduct of detailed biosafety trials.
- Recently, GEAC has given approvals for biosafety trials of Event 142 : a new variety of Bt Brinjal.

GM crop Regulation in India

Mandate of Ministries/Departments	
Ministry of Environment, Forest and Climate Change	• Primarily responsible for conservation and protection of environment, ensuring environmental and human health safety before release of GMOs / LMOs • Nodal agency for implementing Rules, 1989 and the Cartagena Protocol on Biosafety
Department of Biotechnology (Ministry of Science & Technology)	• Nodal department for promoting biotechnology programs • Provides scientific support in implementation of biosafety regulations • Provide services in areas of research, infrastructure, generation of human resource
Ministry of Agriculture	• Policies aimed at agriculture growth • Indian Council of Agricultural Research (ICAR) responsible for monitoring agronomic benefits of GM technology • Monitoring post-release performance of GM crops
Ministry of Health and Family Welfare	• Policies aimed at protecting and monitoring human health • Food Safety and Standards Authority of India responsible for regulating genetically engineered foods
Ministry of Commerce and Industries	• Enhance trade with other countries through export/import policies • Nodal agency for implementing DGFT notification on GMOs
Central Board of Excise and Customs, Department of Revenue, Ministry of Finance	• Enforcement of regulation pertaining to transboundary movement of GMOs/LMOs at point of entry

GM Crops in India



GM crops in India



- Import of GM soymeal allowed : GEAC was bypassed as MoEFCC gave exemption on the basis of the final product being non living. Soyameal is protein rich solid leftover raw material after extracting oil from soybean seed. Major ingredient of poultry feed.
- GM rubber : trials going on. World's first GM rubber plant was planted in assam. Developed specifically for the northeast. Additional copies on gene MnSOD : Manganese Containing superoxide dismutase. This helps plant tide over cold conditions of winter. Developed by kerala based rubber research institute of india.
- Natural rubber : grows in amazon rainforest where there is no cold season and climate is hot and humid.

GM crops in India



- Agharkar research institute (under DST) has mapped 2 genes in wheat called Rht14 and Rht18 : these lead to better seedling vigour and longer coleoptiles (protecting sheath)
- This can allow deeper planting of wheat seeds and can reduce crop residue burning.
- GM food imports : GEAC clears from environmental angle and FSSAI clears from food consumption human health angle. So GM imports are regulated under both EPA, 1986 and Food safety and Standards act, 2006.

2023 current affairs



- DBT has issued guidelines for safety assessment of Genome edited Plants, 2022 easing norms for research into GM crops.
- They have exempted Genome edited plants falling under category of SDN-1 and SDN-2 from 1989 rules. This exemption was earlier granted by MoEFCC.
- These guidelines apply to plants that are genome edited using SDN.
- SDN = Site directed nuclease. It is a type of Gene editing technology. We will study it in detail soon. Word breakdown : Similar to Restriction Endonuclease.

GM plants



- Flavr Savr Tomato : first commercial GM food crop
- Golden Rice : Provitamin A
- BT Crops : Cry 1 AC gene : Protection against Bollworms



Genome editing, Transgenesis and Genetic Modification



- Genome editing : Making changes in the genome but no introduction of foreign genetic material. Only 'cuts' are made, using enzymes such as Site Directed Nuclease and Site Specific Nuclease.
- Genetic Modification : Modification of Organism using Genetic Material from a foreign source. External (alien) DNA is added.
- Transgenesis and Genetic Modification are mostly similar. There is a small difference.
- Although both have altered genomes, a transgenic organism is a GMO containing a DNA sequence or a gene from a different species. Thus, all transgenic organisms are GMOs, but not all GMOs are transgenic.

Recombinant DNA Tech



- Introduction of foreign DNA into host
- Recombinant Human Insulin : Eli Lilly
- Recombinant HGH
- Recombinant Vaccines : we will study in detail
- Recombinant Factor VIII : Haemophilia
- Golden Rice : not currently in use
- Crops

Transgenic Animals



Animals that have had their DNA manipulated to possess and express an extra (foreign) gene are known as transgenic animals.

Why?

<i>normal physiology and development</i>	<i>Study of disease</i>	<i>Biological products : example : Rosie and lactalbumin milk</i>	<i>Vaccine safety</i>	<i>Chemical safety testing</i>
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Transgenic Animals : Applications



- Human alpha 1 anti tripsin : in sheep
- Transgenic pigs : tests for organ transplantation

RDT applications in Medicine



- Management of Adult onset diabetes : earlier we had to take insulin from other animals by killing them.
- This lead to unwanted immune responses (more in the class on immunity)
- So we needed human insulin. But we couldn't kill other humans right? :P
- Solution : Genetic engineering of bacteria to induce them to produce human insulin.
- Produced by Company called Eli-Lily in 1983.

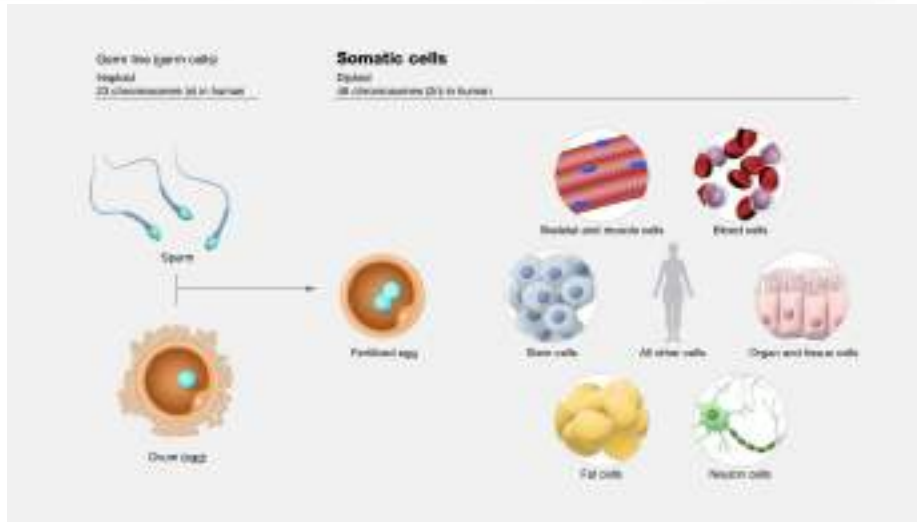
Genetically Engineered Insulin



Human Insulin Production



BASICS : Somatic and Germline cells



Gene Therapy



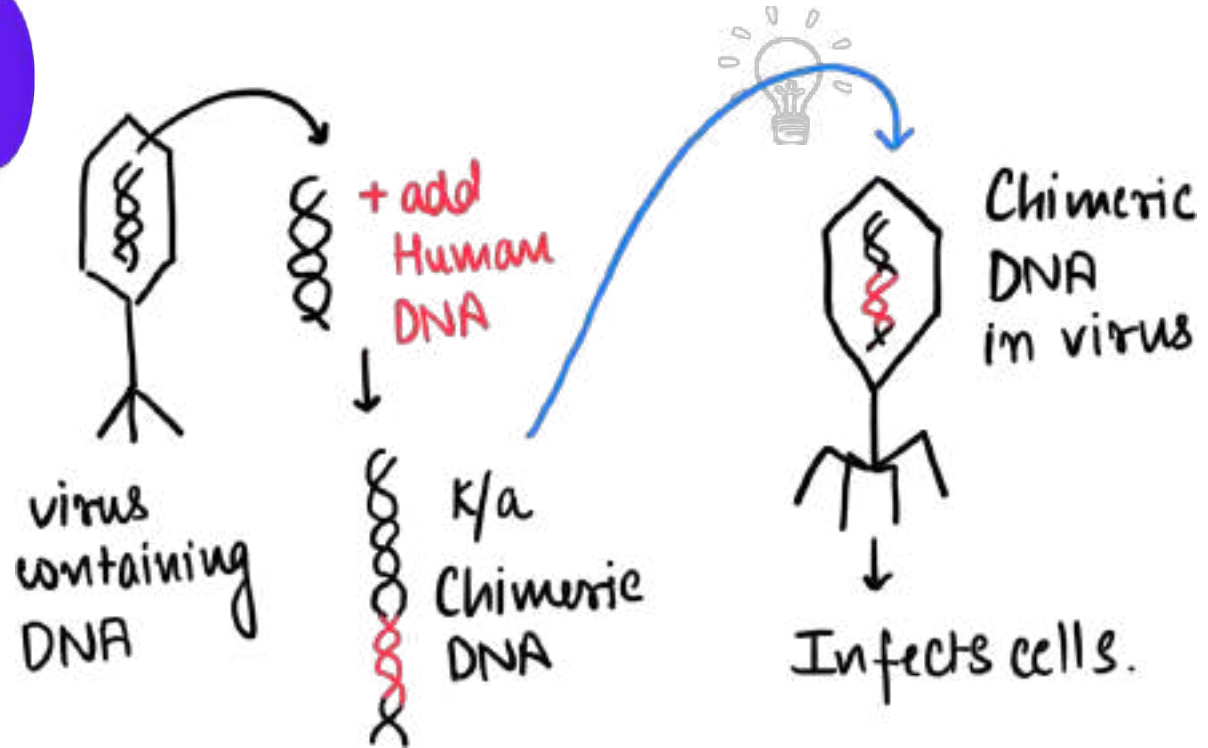
- A child is born with genetic defects. One important gene that he has is non functional. What do we do?
- Delivery of normal genes through baby with gene therapy.
- Gene therapy is a collection of methods that allows correction of a gene defect that has been diagnosed in a child/embryo.
- Here genes are inserted into a person's cells and tissues to treat a disease.
- Correction of a genetic defect involves delivery of a normal gene into the individual or embryo to take over the function of and compensate for the non-functional gene.

Gene Therapy

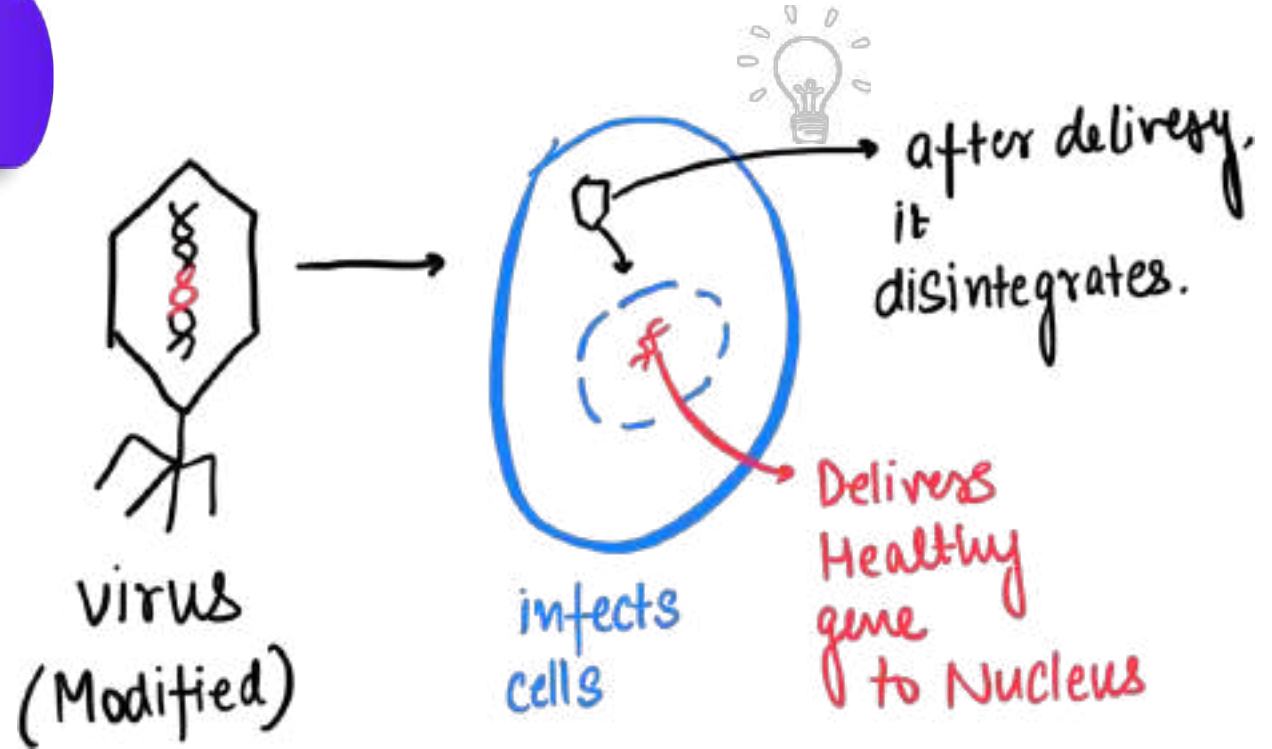


- Biotechnological method to replace defective gene with correct functioning gene.
- 2 types : Somatic cell gene therapy and Germline Gene therapy
- Somatic Gene therapy : Genetic Disorders in individuals can be cured.
- Ex : Haemophilia, Sickle Cell Anaemia (important this year : WHY?)
- Germ Cell Gene therapy : Offspring (child) can be prevented from getting disease.
- Can be used to fight cancer as well : CAR-T cell therapy.

How is it done ?



How is it done (2)



LET'S SEE A VIDEO ON GENE THERAPY



- [click here](#)
- [click here for video 2](#)

Gene Therapy



Can be used to :

- 1. Replace bad gene with healthy gene
- 2. Add a new gene to perform function of missing / non functional gene
- 3. Turn off a problem gene.
- Can be done inside the body (using virus) called in vivo
- We are using modified human virus, for example adenovirus.
- Can be done outside the body : in vitro

In vitro gene therapy : Example : ADA - SCID

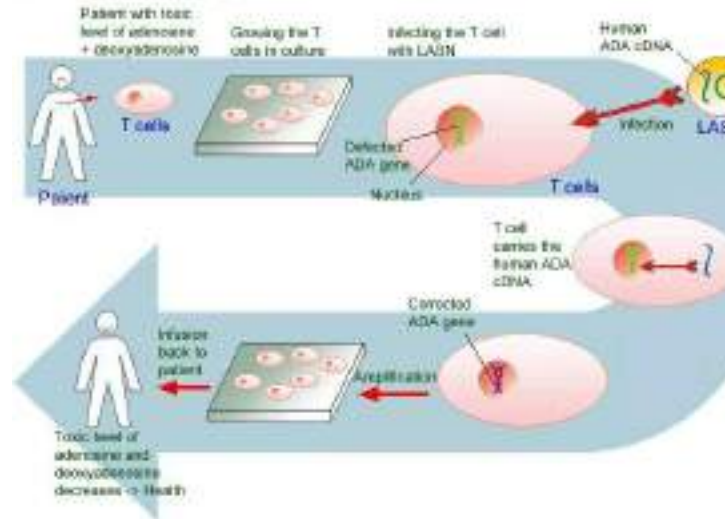


- Adenosine Deaminase Deficiency leading to severe combined immunodeficiency
- T cells of the body need ADA.
- When ADA is absent, patient develops SCID.
- We need to deliver ADA to patient's T cells.
- We use functional ADA gene inserted with viral vector. But virus has to be genetically edited to be made safe.
- T cells are first removed from the patient, edited with viral vector containing new gene, then sent back into the patient.

Gene therapy for ADA- SCID



Gene Therapy for ADA-SCID

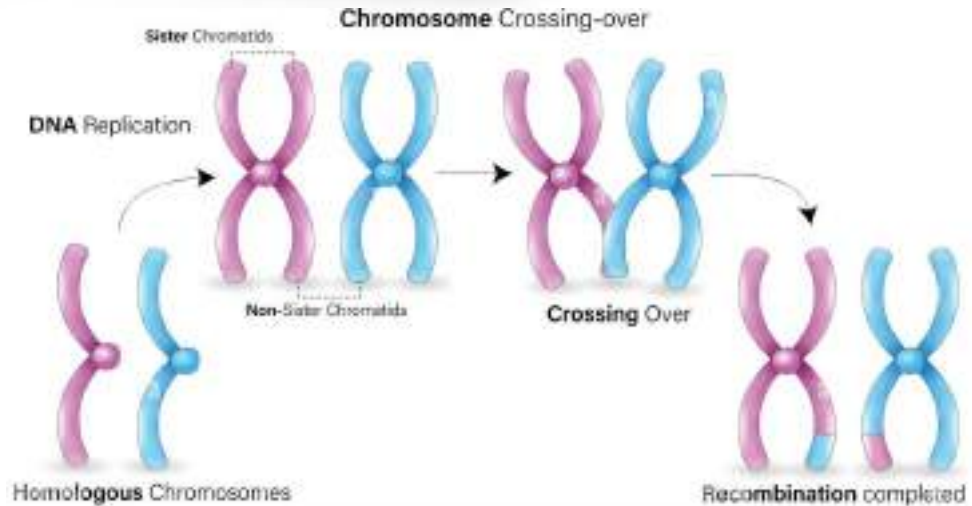
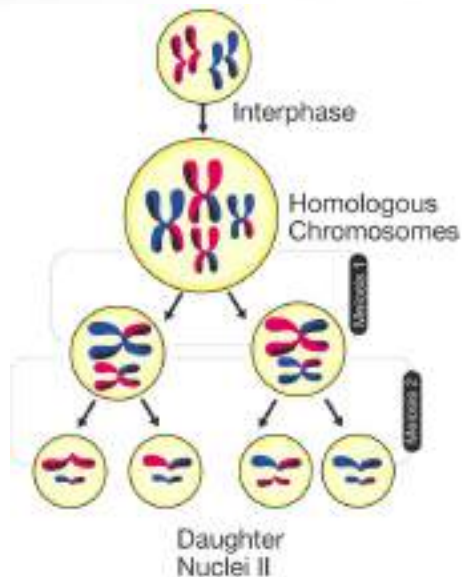


Issues of Gene Therapy



- Short Lived Nature : multiple rounds needed : example : ADA SCID
- Possible solution : Do at early stage.
- Immune response possible, other side effects possible.
- Cannot be used to correct chromosomal disorders, only disorders with single bad gene. (explained later)
- Virus alteration and manipulation : Containment

Chromosomal Disorders : defects during mieosis

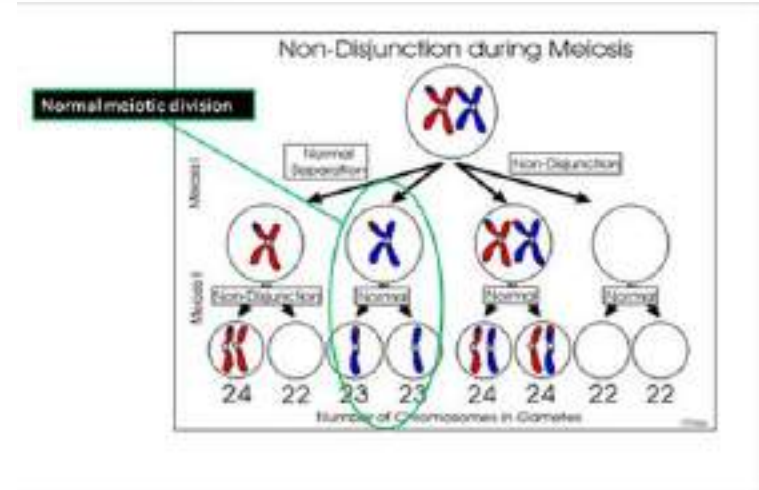


Chromosomal Recombination

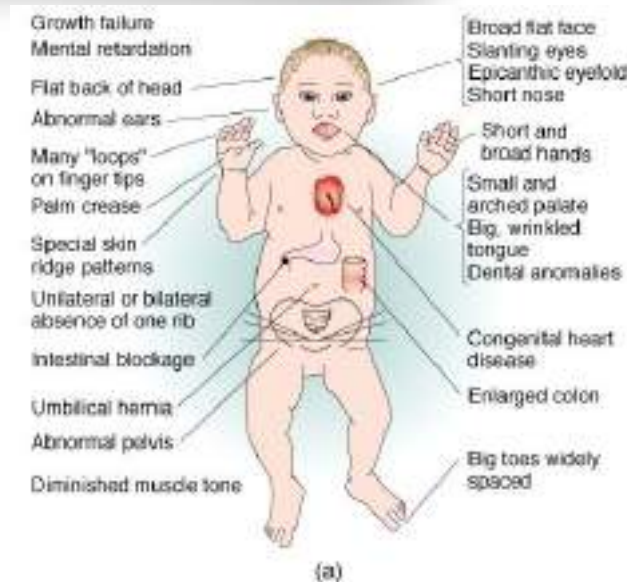
Chromosomal disorders : Down's Syndrome



- Patient has 3 copies of Chromosome number 21
- Leads to severe genetic defects.
- This cannot be cured by gene therapy.



Down's Syndrome : treat them with sympathy

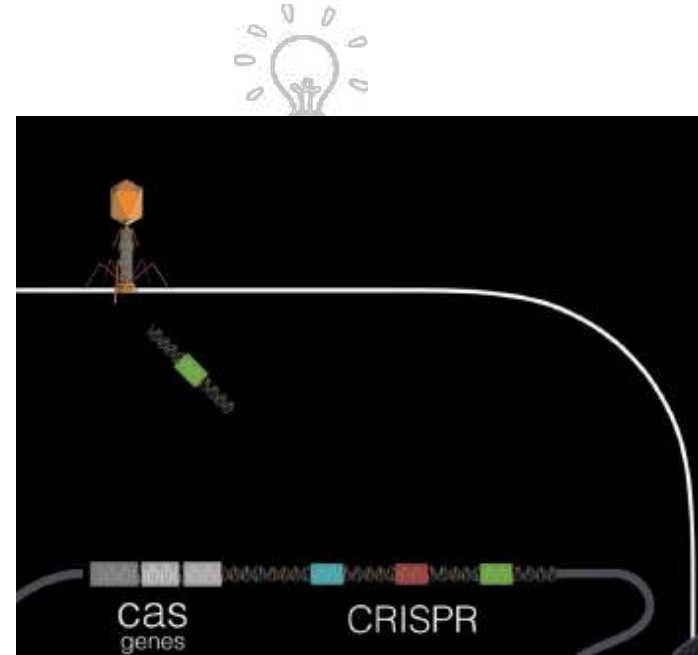


Crispr Cas – 9 and Gene Editing

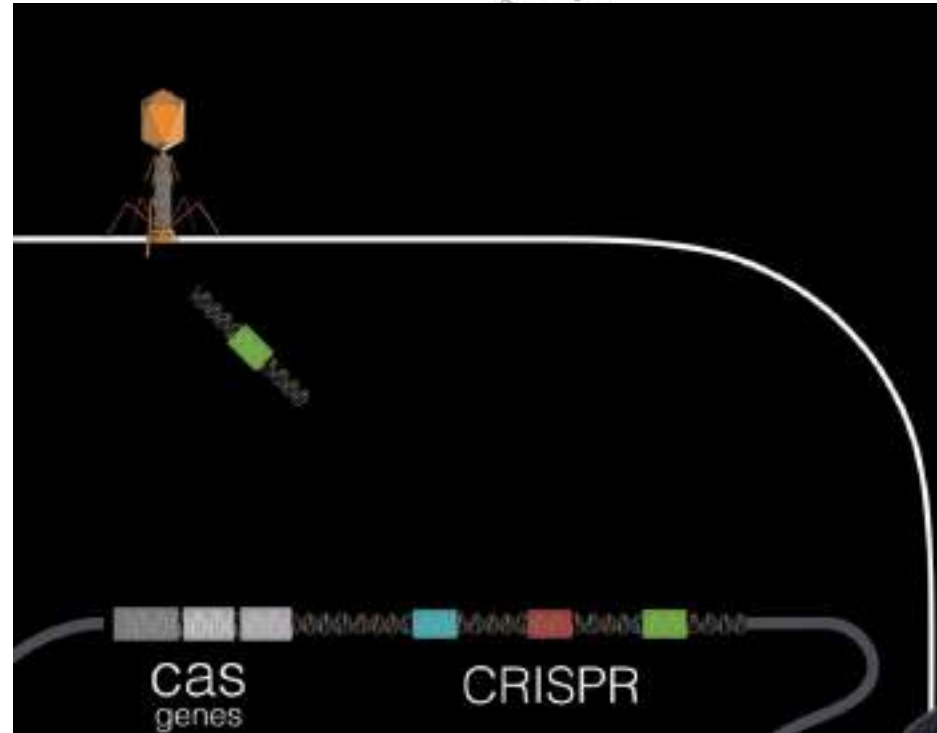


- What are genes?
- Gene Editing : changing genes in humans
- CRISPR : Clustered Regularly Interspaced short Palindromic Repeats
- CAS : crispr associated protein -9
- Originally found in bacteria
- Molecular level scissors and glue (cut and paste mechanism)

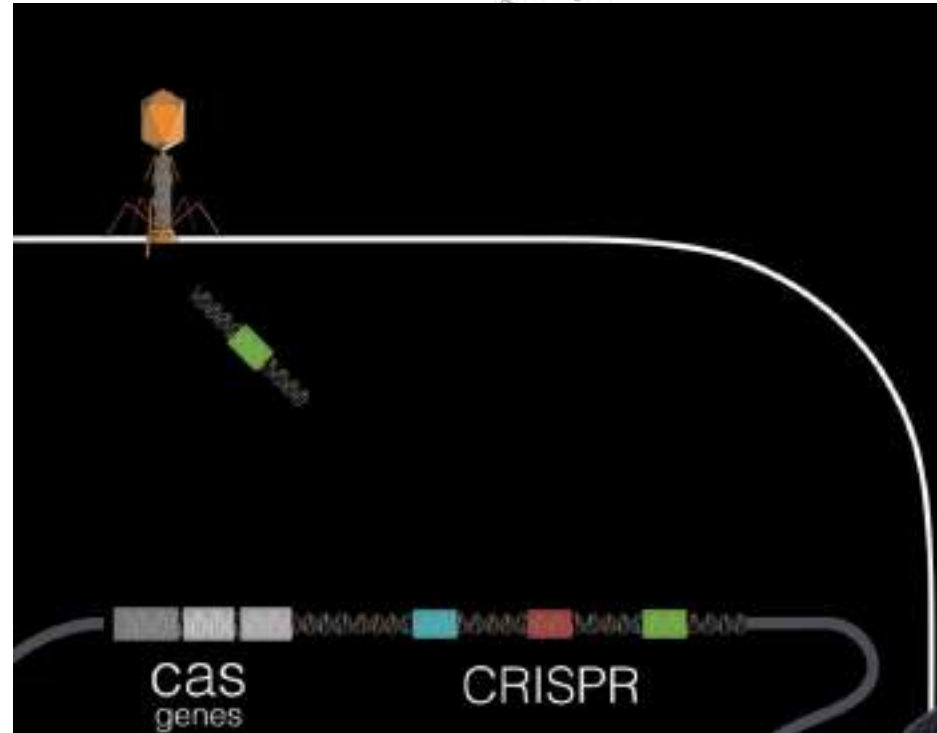
- Palindromes interspaced by spacer DNA
- Cas gene : Cas proteins : Helicases (unwind DNA) Nucleases (Cut DNA)
- It is a defence system that bacteria uses to defend against viruses that infect bacteria. These viruses are called bacteriophage.



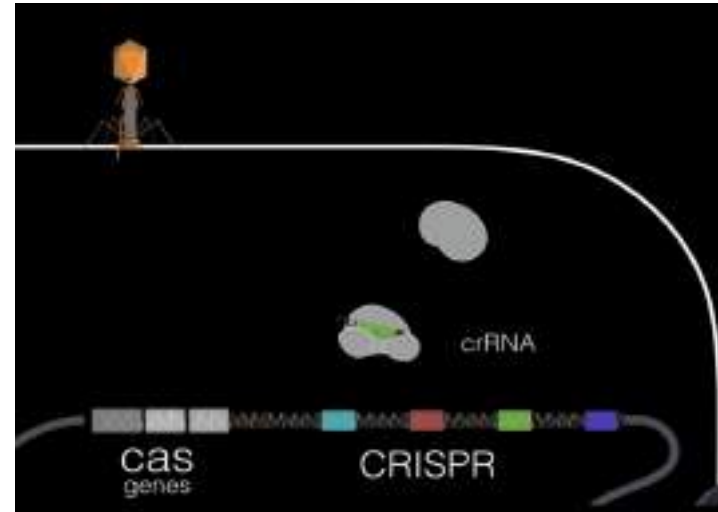
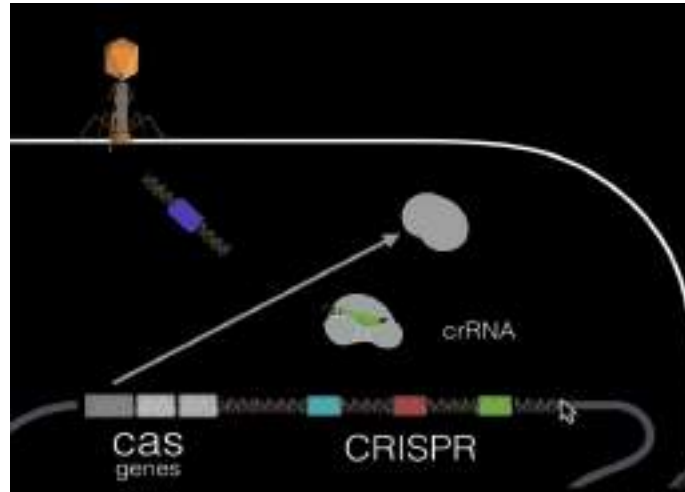
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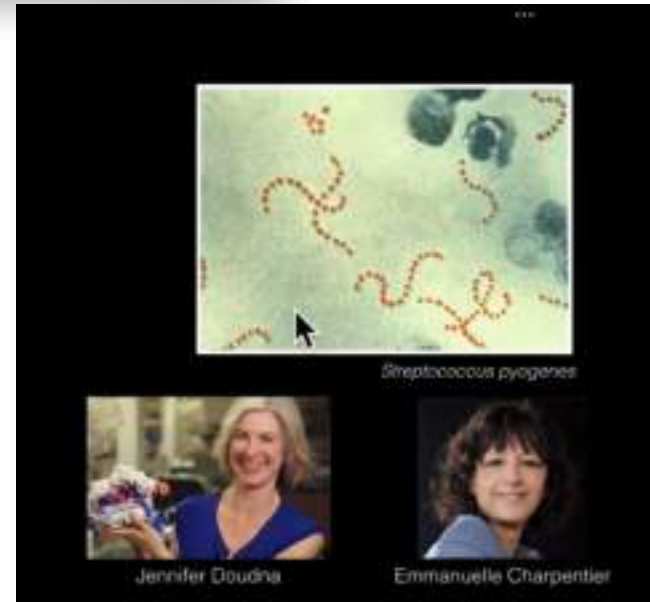
Crispr Cas-9 : Working explained



Crispr



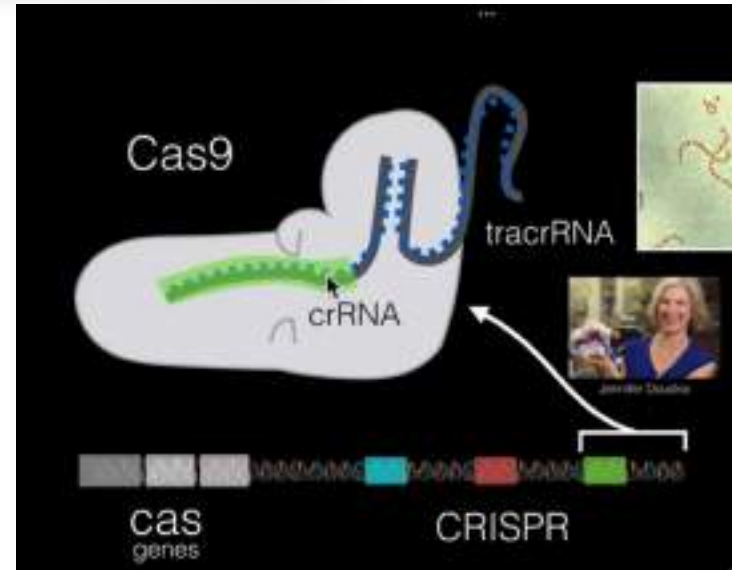
- Spacer – repeat – Spacer– Repeat
- Similar to our immune system.
- Scientists thought of a new use for this in genetic editing.



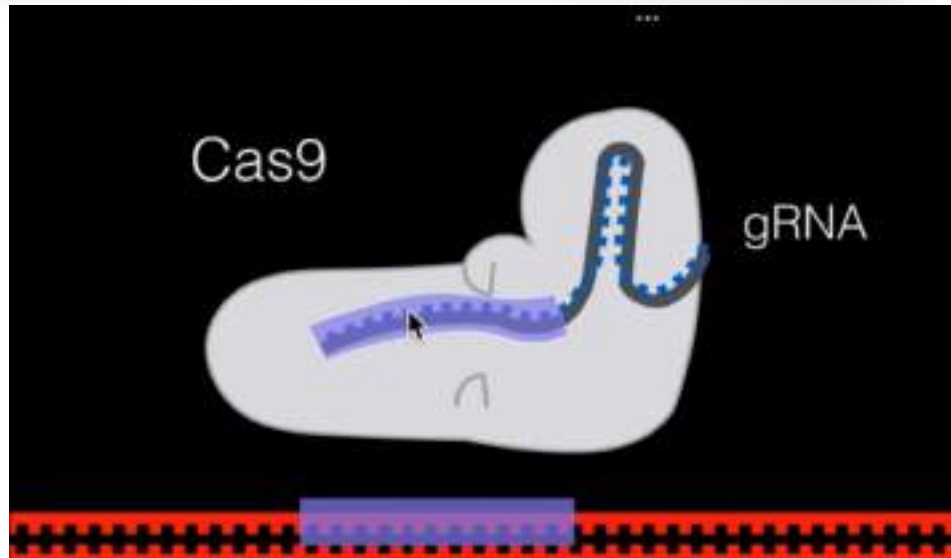
Crispr in genetic editing



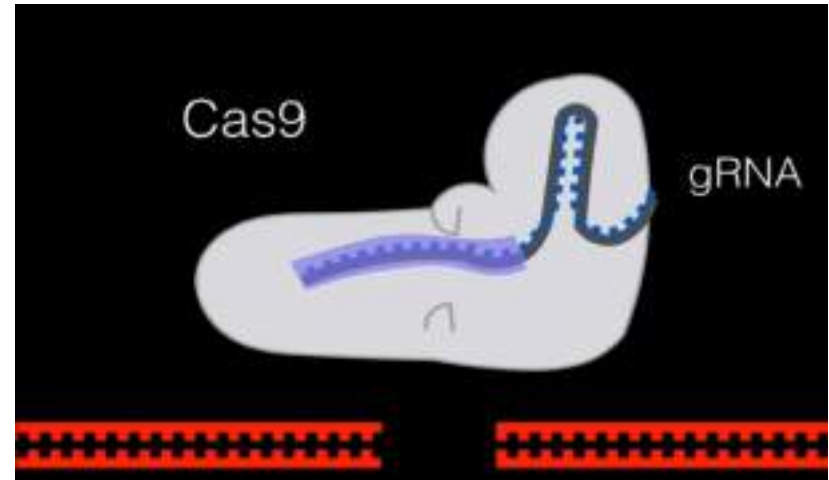
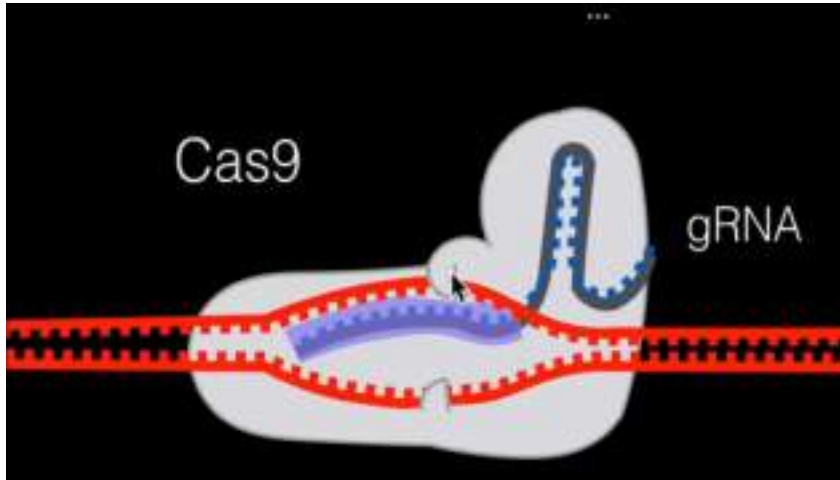
- Nuclease : enzyme responsible for cutting DNA



We modified CRISPR to work in our cells.



We modified CRISPR to work in our cells.



CRISPR Cas 9



- Consists of single guide RNA (sgRNA) : complementary to the part of DNA to be cut
- Cas 9 : crispr associated protein 9 : cuts specific part of dna
- It is like using Microsoft word for a document
- Ctrl + F = Find (SgRNA)
- Replace = Delete and change (Cas-9)
- Crispr cas 9 is molecular scissors

Possible Applications



- Gene editing and gene therapy
- Creation of Genetically Modified Organisms
- Designer babies : Chinese Scientist Jailed for this
- Germline editing of reproductive cells

Issues



- Increased risk of mutations and cancer
- Ethical concerns and designer embryos
- Impact on human variation
- Abuse of gene editing
- Safety and off target effects
- Informed Consent
- Justice and equity

Gene Doping



Changing DNA to enhance performance by athletes

Ex vivo (in vitro)

In vivo

WADA issues

Animal Cloning

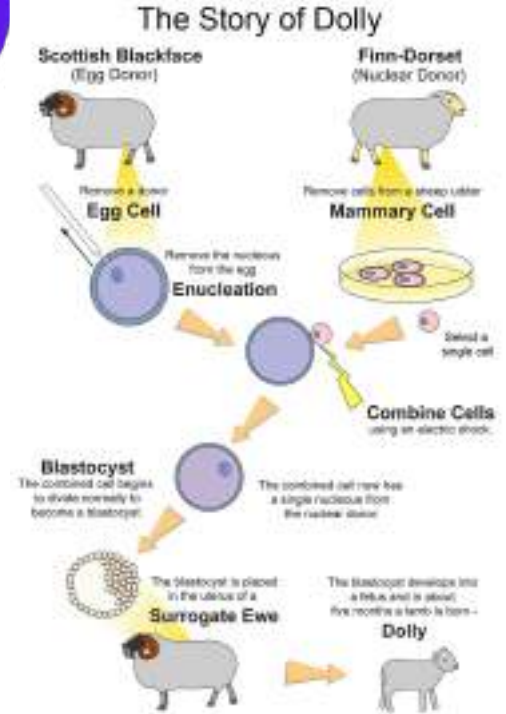


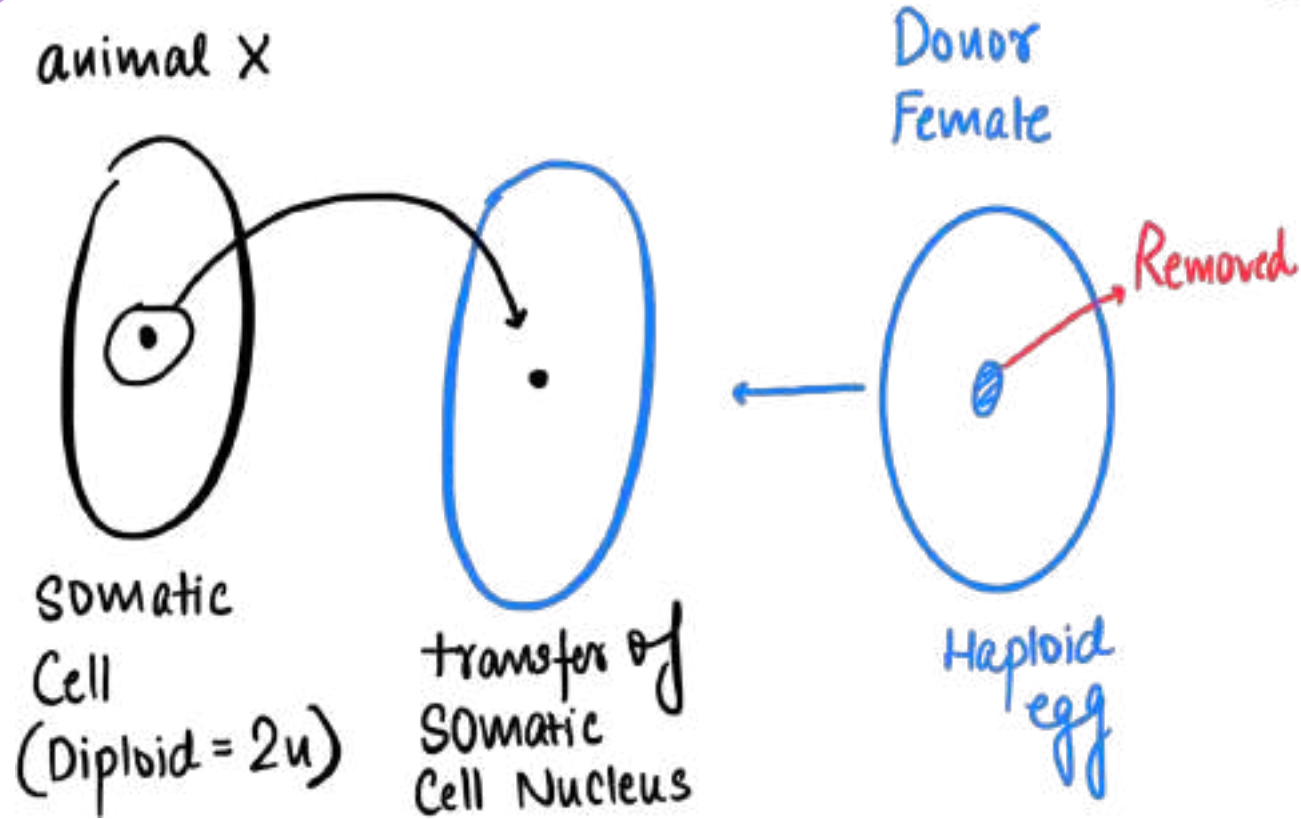
- Identical twins : from 2 parents
- Cloning is the development of cloned animal from a single parent. Also known as Reproductive Cloning.
- Method : Somatic Cell Nuclear Transfer Technique

Animal Cloning and Somatic Cell Transfer



- Cloning : exact same genes of progeny
- Dolly the sheep : Ian Wilmut
- Reproductive cloning
- We will study **Therapeutic Cloning after stem cells**





Immunity



- We get exposed to many disease causing pathogens daily, but this does not cause disease everytime
- Body has a system to defend against foreign organisms
- Remember : Was there such a system in bacteria?

Try doing this question



Q. Consider the following statements: DNA Barcoding can be a tool to:

1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. Identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3

Types of Immunity : Innate and Acquired



- Innate Immunity
- non-specific type of defence, that is present at the time of birth
- Innate immunity consist of four types of barriers.
- Physical barriers : skin / mucus
- Physiological barriers : acid in stomach
- Cellular barriers : WBC (white blood cells) : B cells and T cells
- Cytokine barriers : Interferon

Acquired Immunity

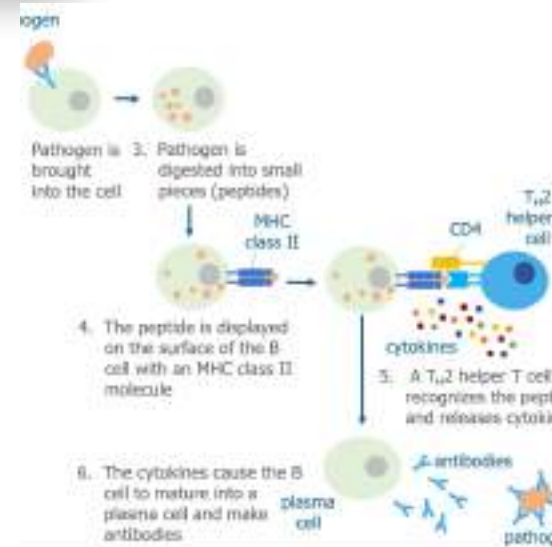


- Why were you able to solve the DNA barcoding question?
- You had done it before : effect of memory : same principle works in immune system
- Acquired immunity : pathogen specific, works on the principle of memory
- our body when it encounters a pathogen for the first time produces a response called primary response which is of low intensity. Subsequent encounter with the same pathogen elicits a highly intensified secondary or anamnestic response. This is ascribed to the fact that our body appears to have memory of the first encounter.
- primary and secondary immune responses are carried out with the help of two special types of lymphocytes present in our blood, i.e., B-lymphocytes and T-lymphocytes

B cells



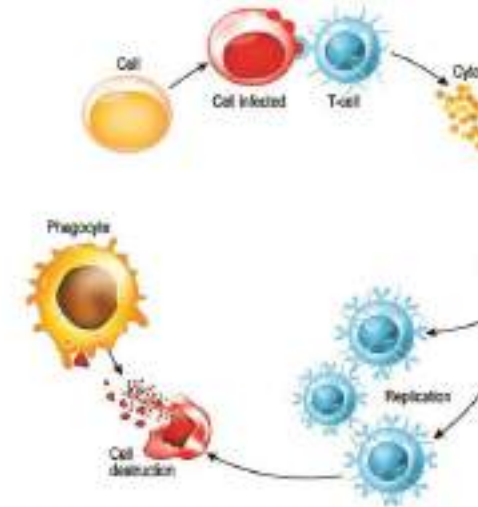
- The B-lymphocytes
- produce an army of proteins in
- response to pathogens into our
- blood to fight with them. These
- proteins are called antibodies.



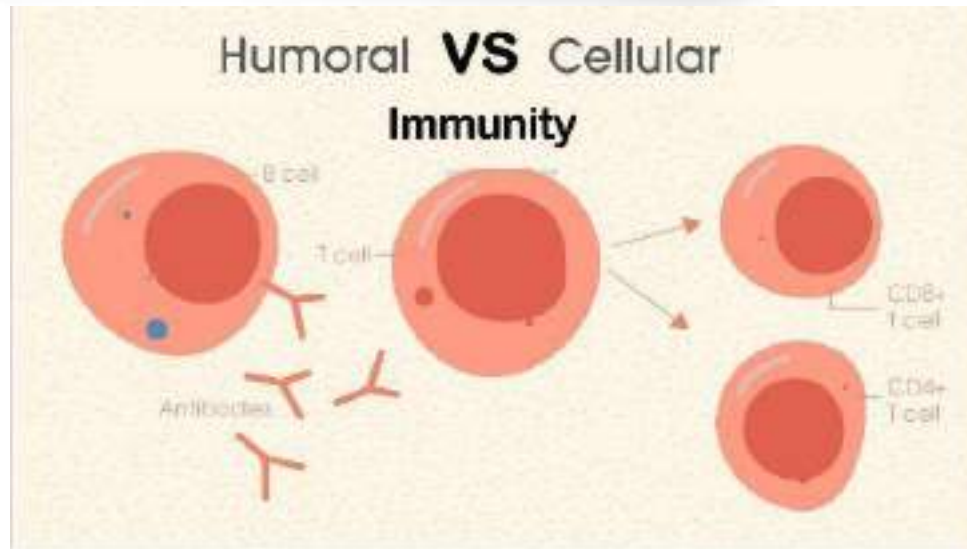
T cells



- T-cells themselves do not secrete antibodies but help B
- cells produce them. Each antibody molecule has four peptide chains, two small called light chains and two longer called heavy chains. Hence, an antibody is represented as H₂L₂.
- Because these antibodies are found in the blood, the response is also called as humoral immune response.
- The second type is called cell-mediated immune response or cell-mediated immunity (CMI). The T-lymphocytes mediate CMI.



Humoral and Cell Mediated Immunity

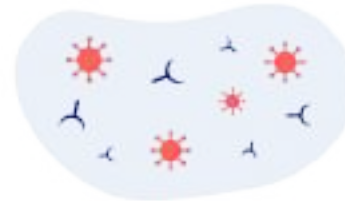


Active vs Passive Immunity



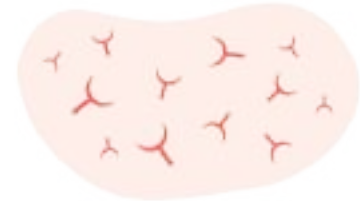
- When a host is exposed to antigens, which may be in the form of living or dead microbes or other proteins, antibodies are produced in the host body. This type of immunity is called active immunity.
- Active immunity is slow and takes time to give its full effective response.
- When ready-made antibodies are directly given to protect the body against foreign agents, it is called passive immunity.
- Ex : Mother's milk and colostrum (IgA)

ACTIVE IMMUNITY



Administering living (weakened) or dead pathogens creates a protective immune response

PASSIVE IMMUNITY



The transfer of antibodies from an immunized donor to a non-immune recipient

Auto Immunity

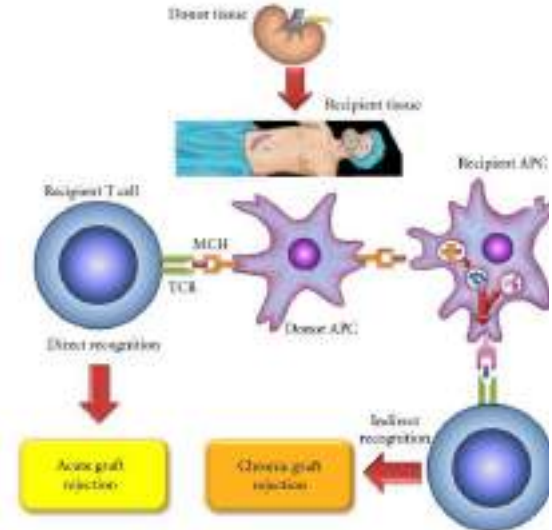


- Memory-based acquired immunity evolved in higher vertebrates based on the ability to differentiate foreign organisms (e.g., pathogens) from self- cells
- sometimes, due to genetic and other unknown reasons, the body attacks self-cells.
- This results in damage to the body and is called auto-immune disease. Ex :Rheumatoid arthritis

Immune system and Grafts



- Graft rejection happens due to our own immune system.
- What could be the possible solution?
- We have read some technologies that help prevent this, try and recall.



Vaccination



- Why am I making you solve prelims questions in a controlled environment with guidance?
- If you can solve easy questions in this setting, you will be able to solve tough ones also in a difficult setting later as you would have developed a certain memory and technique
- Same exact principle applies in the case of Vaccination : based on the property of 'memory' of the immune system.

Active Vaccination



- In vaccination, a preparation of antigenic proteins of pathogen or inactivated/weakened pathogen (vaccine) are introduced into the body.
- The antibodies produced in the body against these antigens would neutralise the pathogenic agents during actual infection.
- The vaccines also generate memory – B and T -cells that recognise the pathogen quickly on subsequent exposure and overwhelm the invaders with a massive production of antibodies.

Passive Vaccination



- If a person is infected with some deadly microbes to which quick immune response is required as in tetanus
- we need to directly inject the preformed antibodies, or antitoxin (a preparation containing antibodies to the toxin).
- Even in cases of snakebites, the injection which is given to the patients, contain preformed antibodies against the snake venom. This type of immunisation is called passive immunisation.

Active vs Passive Vaccination



	Active vaccination	Passive vaccination
Immunization with	Antigen	Pre-formed antibodies
Required time until protection occurs	Weeks-months	Minutes-hours
Duration of protection	Ideally for a lifetime	Few weeks
Role of host immune system	Develops own protective immunity	Only effector mechanisms are activated by administered antibodies
Cells participating	Ideally all cells of the immune system	No cells are affected or only effector cells are activated
Induction of immunological memory	Yes	No

AEFI



- There is no such thing as a "perfect" vaccine which protects everyone who receives it AND is entirely safe for everyone.
- Effective vaccines (i.e. vaccines inducing protective immunity) may produce some undesirable side effects which are mostly mild and clear up quickly.
- EFIs can be related to the vaccine itself (product or quality defect-related reactions), to the vaccination process (error or stress related reactions) or can occur independently from vaccination (coincidental) and are classified as:
 - Vaccine product-related reaction
 - Vaccine quality defect-related reaction
 - Immunization error-related reaction
 - Immunization anxiety-related reaction
 - Coincidental event

Types of Vaccines



- Based on method of Administration
- Im
- Iv
- Sc
- Nasal
- Oral

Types of vaccines : based on pathogen



- Live Attenuated
- Inactivated
- Subunit
- Toxoid
- Recombinant
- mRNA
- DNA

Inactivated Vaccine



- Inactivated vaccines use the killed version of the germ that causes a disease.
- Inactivated vaccines usually don't provide immunity (protection) that's as strong as live vaccines. So you may need several doses over time (booster shots) in order to get ongoing immunity against diseases.
- Inactivated vaccines are used to protect against:
 - Hepatitis A
 - Flu (shot only)
 - Polio (shot only)
 - Rabies

Live Attenuated Vaccine



- Live vaccines use a weakened (or attenuated) form of the germ that causes a disease.
- Because these vaccines are so similar to the natural infection that they help prevent, they create a strong and long-lasting immune response. Just 1 or 2 doses of most live vaccines can give you a lifetime of protection against a germ and the disease it causes.
- But live vaccines also have some limitations. For example:
- Because they contain a small amount of the weakened live virus, they may cause disease in some people, especially those with a compromised immune system
- They need to be kept cool, so they don't travel well.
- Live vaccines are used to protect against:
- [Measles, mumps, rubella](#) (MMR combined vaccine)
- [Rotavirus](#)
- [Smallpox](#)
- [Chickenpox](#)
- [Yellow fever](#)

Subunit Vaccine



- A subunit vaccine is a vaccine that contains purified parts of the pathogen that are antigenic, or necessary to elicit a protective immune response.
- A "subunit" vaccine doesn't contain the whole pathogen, unlike live attenuated or inactivated vaccine, but contains only the antigenic parts such as proteins, polysaccharides or peptides.
- Because the vaccine doesn't contain "live" components of the pathogen, there is no risk of introducing the disease, and is safer and more stable than vaccine containing whole pathogens. Other advantages include being well-established technology and being suitable for immunocompromised individuals
- Disadvantages include being relatively complex to manufacture compared to some vaccines, possibly requiring adjuvants and booster shots, and requiring time to examine which antigenic combinations may work best.

Recombinant Viral Vector Vaccine



- Uses a safe virus to deliver specific sub parts of a pathogen.
- Ex : Spike protein of coronavirus
- Immune response is generated without causing a disease
- Genetic instructions of making such a protein are inserted into a safe virus (through RDnaTech)
- Several different viruses have been used as vectors, including influenza, vesicular stomatitis virus (VSV), measles virus, and adenovirus, which causes the common cold. Adenovirus is one of the viral vectors used in some COVID-19 vaccines being studied in clinical trials.

Conjugate vaccine



- A conjugate vaccine is a type of subunit vaccine which combines a weak antigen with a strong antigen as a carrier so that the immune system has a stronger response to the weak antigen.
- Most vaccines contain a single antigen that the body will recognize. However, the antigen of some pathogens does not elicit a strong response from the immune system, so a vaccination against this weak antigen would not protect the person later in life.
- In this case, a conjugate vaccine is used in order to invoke an immune system response against the weak antigen. In a conjugate vaccine, the weak antigen is covalently attached to a strong antigen, thereby eliciting a stronger immunological response to the weak antigen.
- Most commonly, the weak antigen is a polysaccharide that is attached to strong protein antigen. However, peptide/protein and protein/protein conjugates have also been developed.

Polysaccharide Vaccine



- Polysaccharides : poly carbohydrates
- Vaccines that consist of Poly carbohydrate molecule that generates immune response
- Generally vaccines are made of protein, that's why polysaccharide vaccines are special.

mRNA vaccine



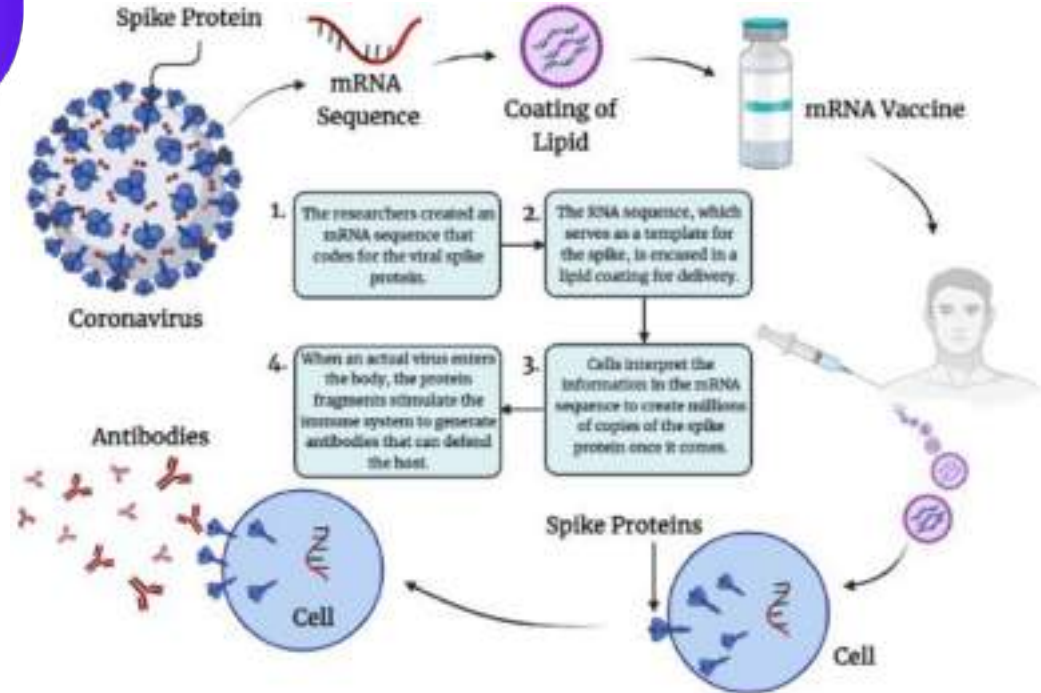
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- What is the vaccine principle?
- Now you try and tell me what a mRNA vaccine will do.

mRNA Vaccines : type of nucleic acid vaccine



- An mRNA vaccine is a type of vaccine that uses a copy of a molecule called messenger RNA (mRNA) to produce an immune response. The vaccine delivers molecules of antigen-encoding mRNA into immune cells, which use the designed mRNA as a blueprint to build foreign protein that would normally be produced by a pathogen (such as a virus) or by a cancer cell. These protein molecules stimulate an adaptive immune response that teaches the body to identify and destroy the corresponding pathogen or cancer cells.

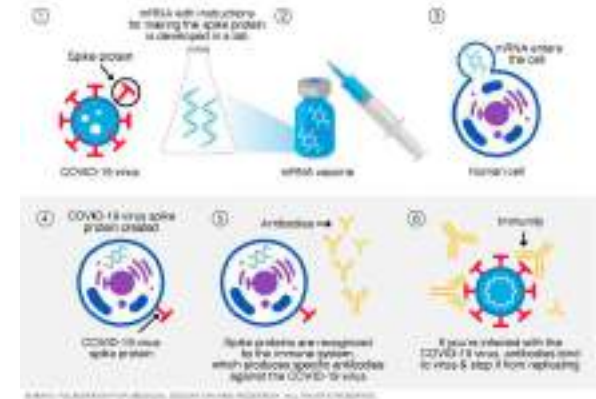
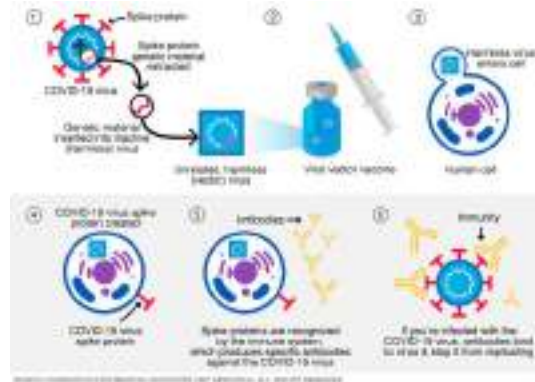
mRNA vaccine

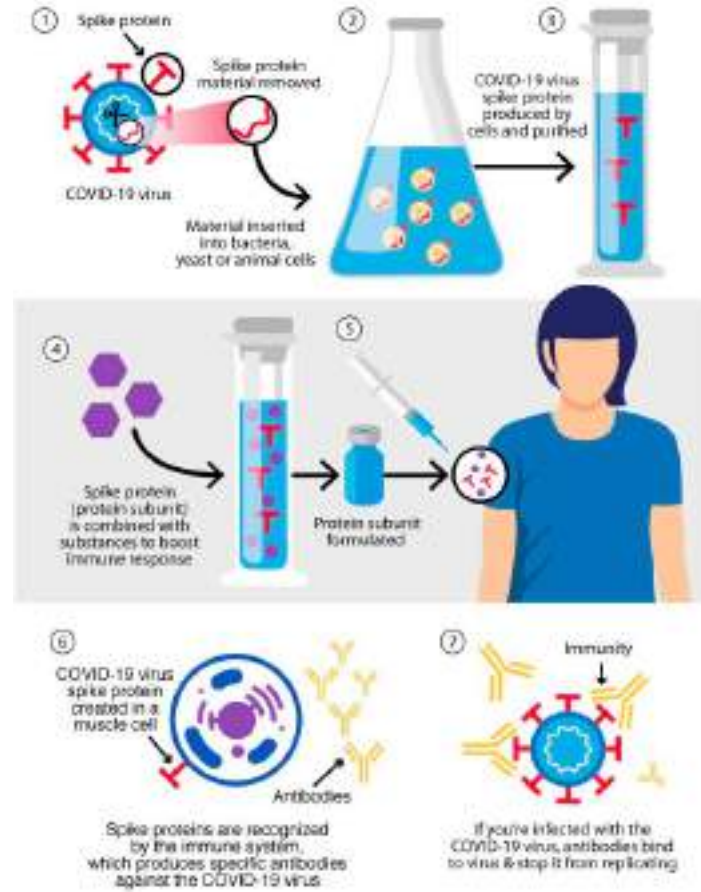


Covid 19 Vaccines



- mRNA
- Viral Vector
- Subunit





Covid 19 Vaccines in India

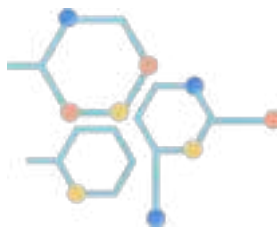


- Covishield : Oxford and Astra Zeneca : viral vector vaccine
- In this type of vaccine, genetic material from the COVID-19 virus is placed in a modified version of a different virus (viral vector). When the viral vector gets into your cells, it delivers genetic material from the COVID-19 virus that gives your cells instructions to make copies of the S protein. Once your cells display the Spike proteins on their surfaces, your immune system responds by creating antibodies and defensive white blood cells. If you later become infected with the COVID-19 virus, the antibodies will fight the virus.

Covid 19 Vaccines India



- Covaxin : bharat biotech
- Inactivated virus technology
- Sputnik V
- Similar to covishield



UPSC Prelims

Science & Tech

Class No. 5

Reserve class (biotechnology)



Recap



- Innate and Acquired Immunity
- B cells (humoral immunity) and T cells (cell mediated immunity)
- Active, Passive and Auto-Immunity
- Principle of Memory for Vaccination
- Anti- gen vs Patho – gen
- Active and Passive Vaccination

AEFI : Adverse Effects Following Immunization



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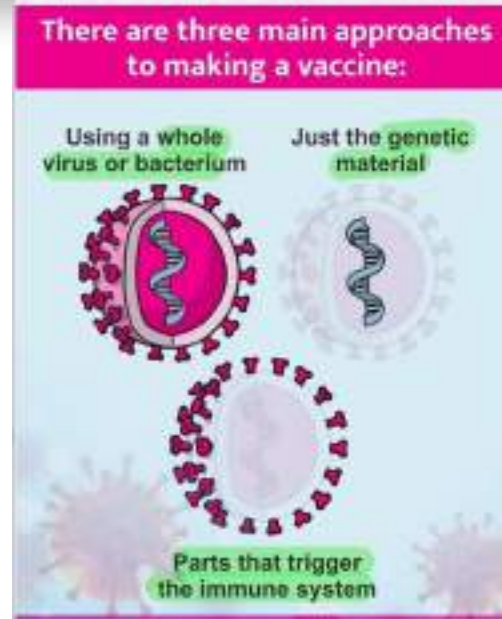


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- Viral Vector vaccines produce immunity from both B and T cells (how? Not important)

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Polysaccharide Vaccine



- Polysaccharides : poly carbohydrates
- Vaccines that consist of Poly carbohydrate molecule that generates immune response
- Generally vaccines are made of protein, that's why polysaccharide vaccines are special.

Nucleic Acid Vaccines



- Using genetic material from a pathogen to stimulate an immune response.
- Could be Using DNA (dna vaccine) or RNA (rna vaccine).

mRNA vaccine

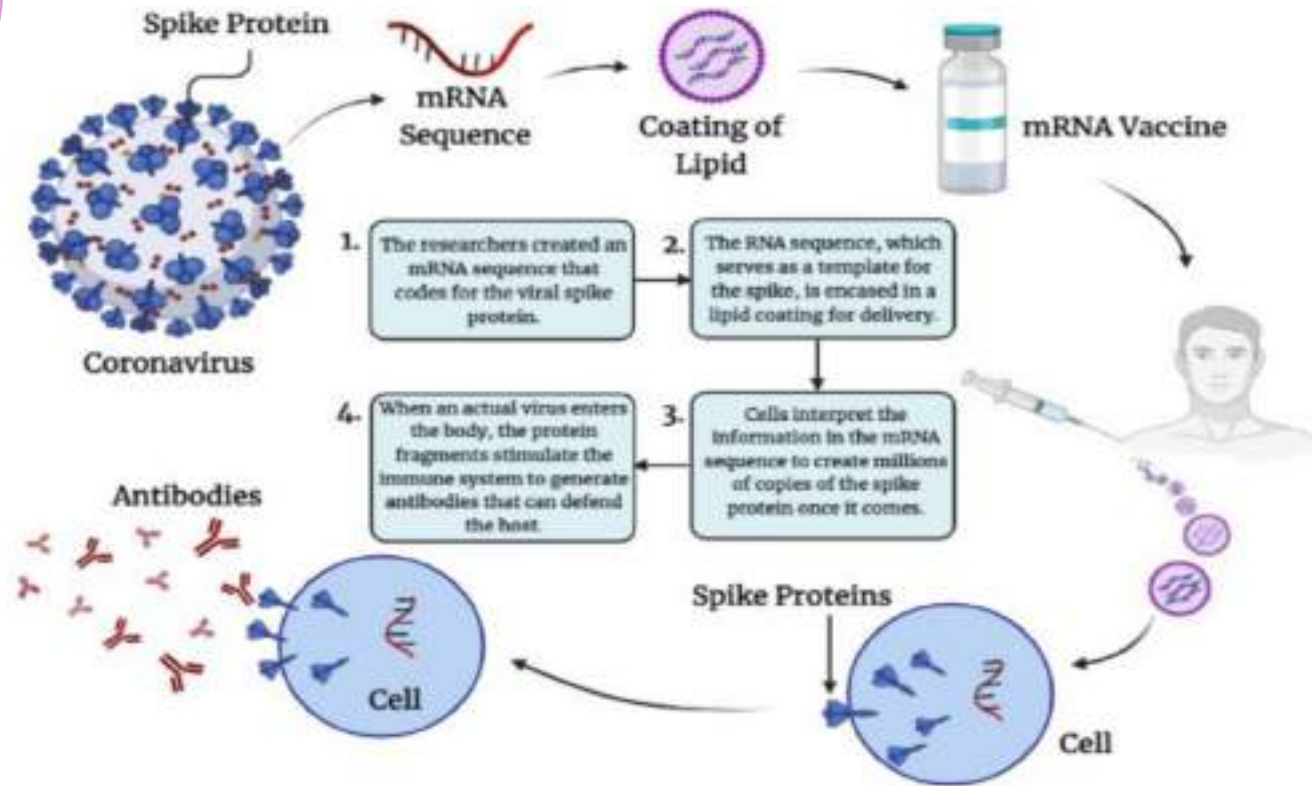


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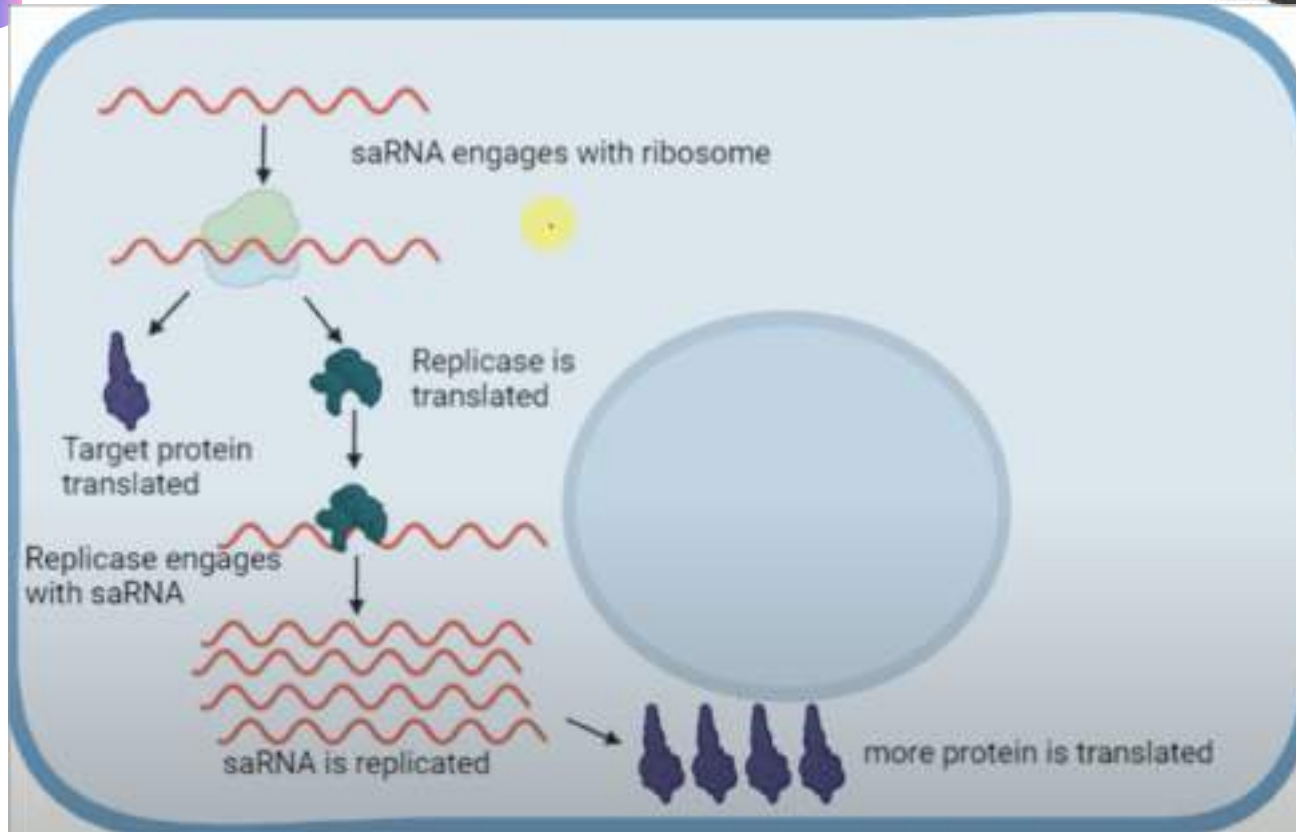
New Type : Self Amplifying mRNA vaccine



- A Self-amplifying messenger RNA (mRNA) vaccine, ARCT-154, developed by Arcturus Therapeutics Holdings (US), showed promising results against Covid- 19 .
- Conventional mRNA : codes for spike protein of Covid-19 virus.
- Self amplifying mRNA vaccine codes for 4 additional proteins that enable amplification of the mRNA strand.
- Stronger Immunity, Storage is easier, minimises dose of mRNA, cost reduced.

Self Amplifying mRNA





DNA Vaccines



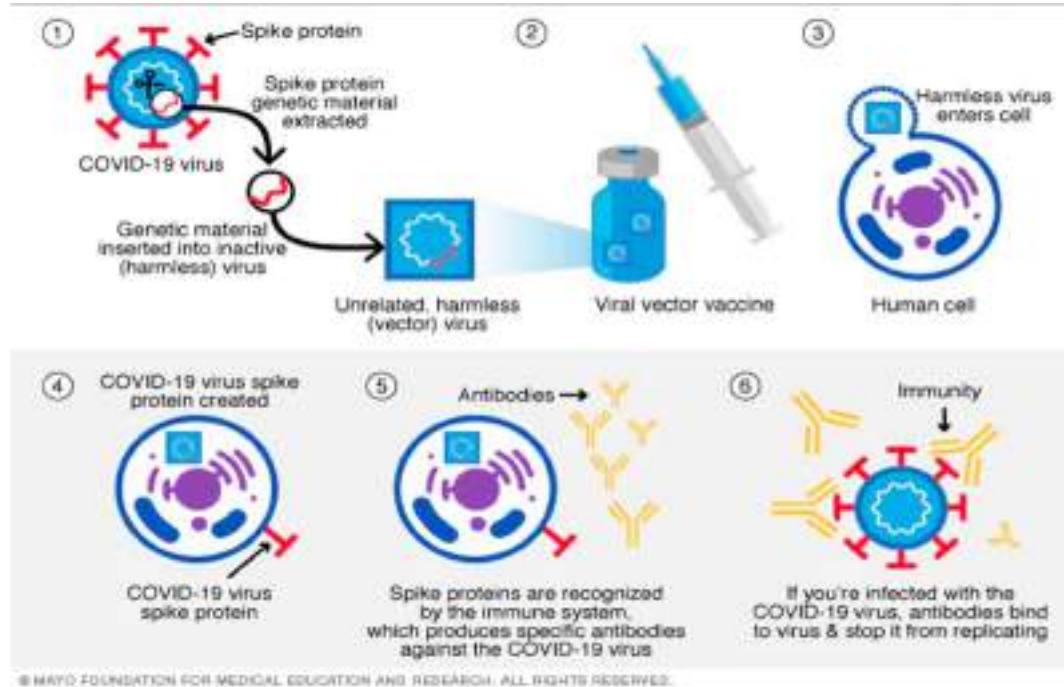
- Piece of DNA encoding the antigen is inserted into the bacterial plasmid.
- DNA plasmids are inserted intra muscularly and driven into cells with help of electroporation.
- DNA based vaccines are more stable : don't require ultra cold storage so they are more cost effective.
- Video : How to DNA vaccines work (Elsevier)

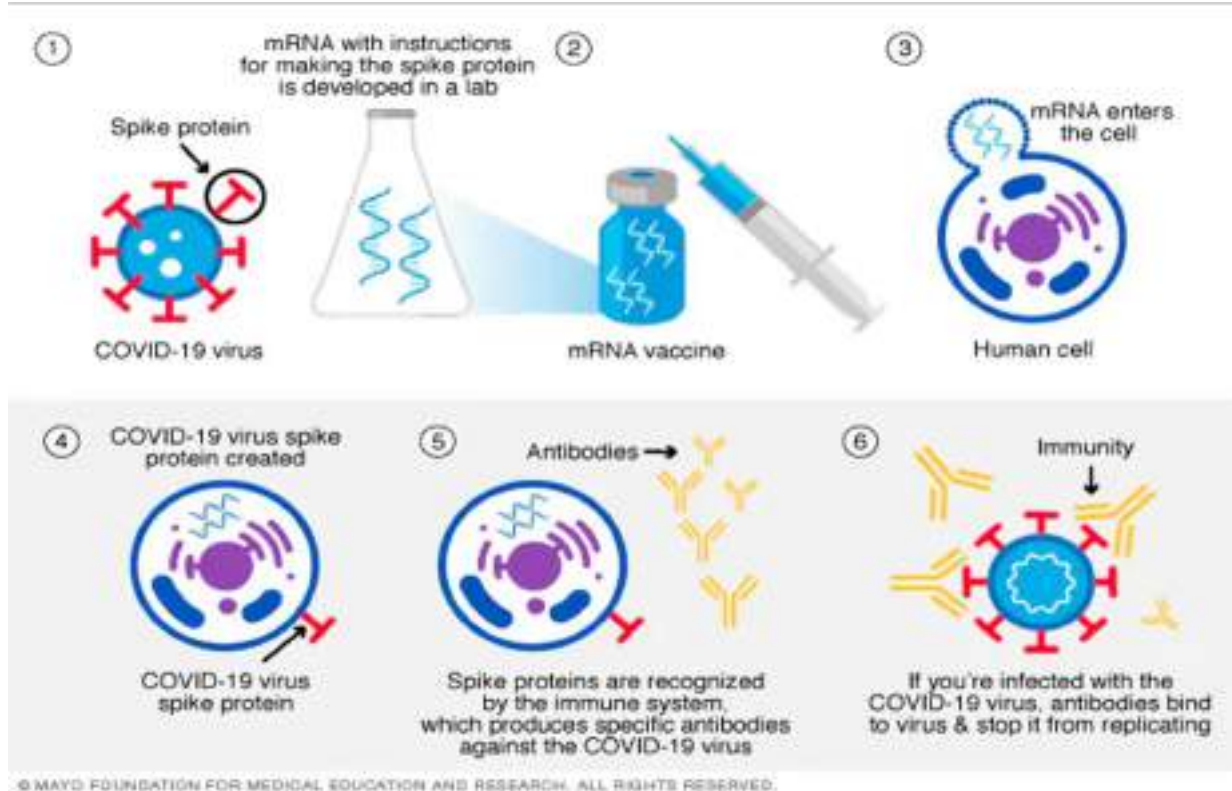


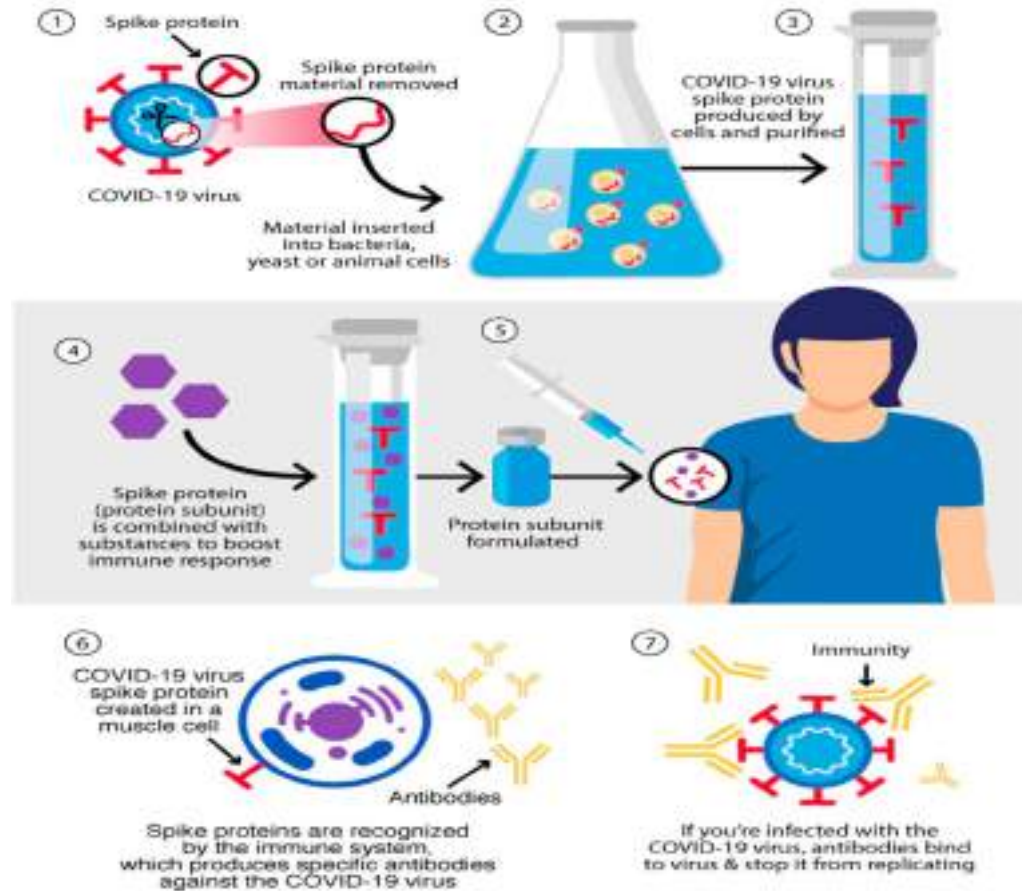
Covid 19 Vaccines



- mRNA
- Viral Vector
- Subunit
- DNA (new) : Zycov-d
- Nasal (new)







Covid 19 Vaccines in India



- Covishield : Oxford and Astra Zeneca : viral vector vaccine
- In this type of vaccine, genetic material from the COVID-19 virus is placed in a modified version of a different virus (viral vector). When the viral vector gets into your cells, it delivers genetic material from the COVID-19 virus that gives your cells instructions to make copies of the S protein. Once your cells display the Spike proteins on their surfaces, your immune system responds by creating antibodies and defensive white blood cells. If you later become infected with the COVID-19 virus, the antibodies will fight the virus.



Covid 19 Vaccines India



- Covaxin : bharat biotech
- Inactivated virus technology
- Sputnik V
- Similar to covishield
- Zycov – d
- Inocovacc

ZyCoV -D



- Developed by Zydus Cadilla, got emergency use authorization
- World's first DNA vaccine against covid 19, it has been granted emergency use authorization.
- Developed in partnership with DBT under Mission Covid Suraksha
- Can be administered without a needle : special intra dermal system called tropis that uses precise stream of jet to penetrate the skin.
- For adults and those above 12 years old
- 3 doses needed.

iNCOVACC



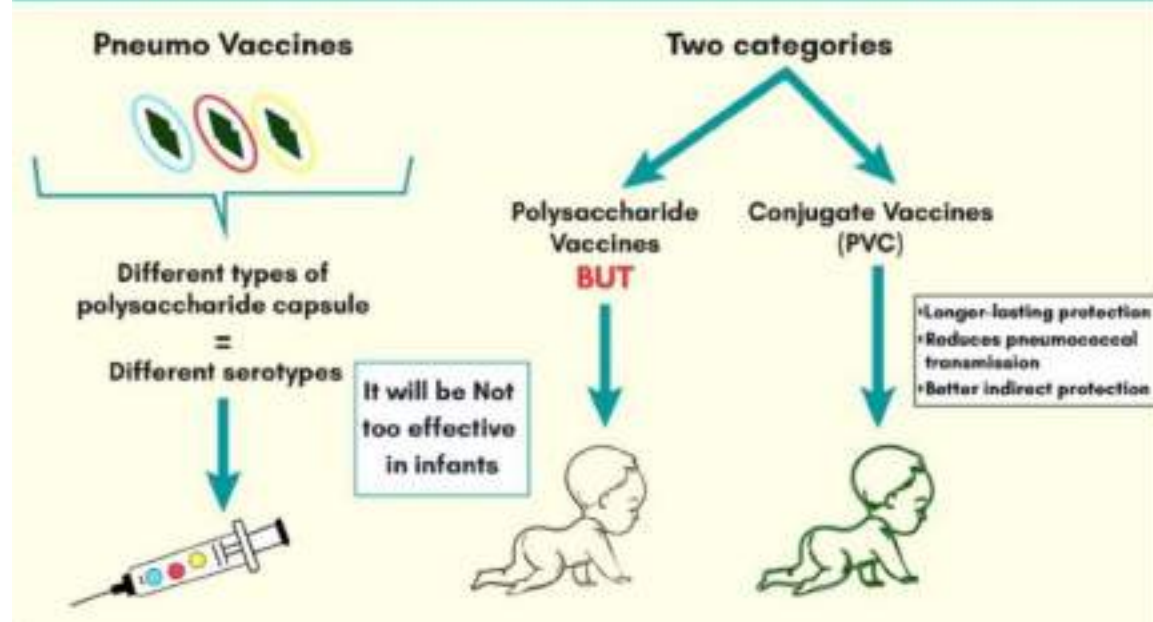
- Bharat biotech has developed it as the world's first intranasal vaccine for Covid-19.
- Received emergency use authorization.
- Adenovirus vector based vaccine.

Pneumococcal Vaccine



- Pneumococcal Conjugate vaccine has been included under Universal Immunisation Programme.
- UIP : 12 Vaccine preventable diseases : Diphtheria, Pertussis, Tetanus, Polio, Measles, Rubella, severe form of Childhood Tuberculosis, Rotavirus diarrhoea, Hepatitis B and Meningitis & Pneumonia caused by Haemophilus Influenzae type B (10 national) and Pneumococcal Pneumonia (Bihar, himachal) and Japanese Encephalitis (endemic districts)
- PCV : provides protection against Streptococcus Pneumonia, which causes pneumonia, meningitis, sinusitis, bronchitis and middle ear infection.

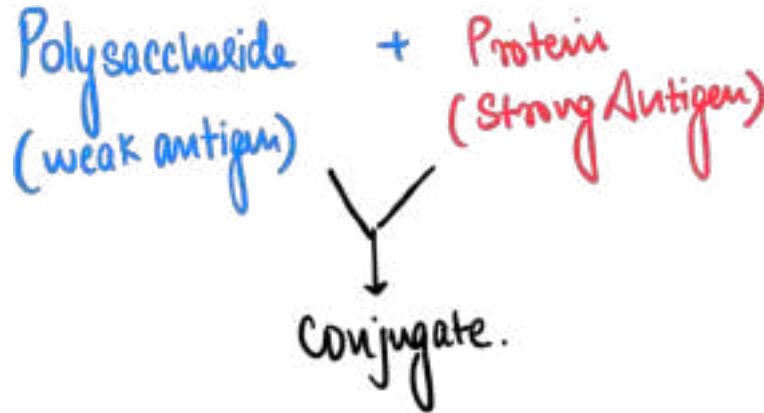
Types of pneumococcal vaccines



PCV



- If only polysaccharide is present (bacterial capsule) : the immunogenicity is less.
- Carrier protein is attached to make the immunogenicity higher : stronger immune response.



BCG vaccines (Bacillus Calmette Guerin)



- It has been 100 years since the BCG vaccine was first used in humans in 1921 against tuberculosis (TB). Only licensed vaccine available.
- Helps prevent severe childhood tuberculosis.
- Mycobacterium Tuberculosis is the causative organism, transmission is through respiratory pathway.
- Efficacy is higher farther from the equator because tropical regions have many environmental strains of Mycobacterium that lead to interference with vaccine action.
- Also used for Other mycobacterial diseases like leprosy, Buruli ulcer and immunotherapy against urinary bladder cancer and malignant melanoma.
- TB in India : India has 27% of global disease burden.
- ICMR : trials of recombinant BCG (vpm1002) and Mycobacterium Indicus Pranii

Toxoid Vaccine



- A toxoid is an inactivated toxin (usually an exotoxin) whose toxicity has been suppressed either by chemical (formalin) or heat treatment, while other properties, typically immunogenicity, are maintained. Toxins are secreted by bacteria, whereas toxoids are altered form of toxins; toxoids are not secreted by bacteria. Thus, when used during vaccination, an immune response is mounted and immunological memory is formed against the molecular markers of the toxoid without resulting in toxin-induced illness. Such a preparation is also known as an anatoxin.
- Example : Diphtheria Toxoid Vaccine

Tetanus Vaccine



- Active Tetanus Vaccination : Tetanus Toxoid : Given during childhood
- Also given to those who have an injury with uncertain immunization history
- Passive Vaccination : high risk : given to those who don't have immunity and need immediate help

Test of 4 Questions!

This is not a learning exercise. You must get all of these correct.

Cut off : 4/4





Question 1 (Previous Year Question 2020)

Q. What is the importance of using Pneumococcal Conjugate Vaccines in India?

1. These vaccines are effective against pneumonia as well as meningitis and sepsis.
2. Dependence on antibiotics that are not effective against drug-resistant bacteria can be reduced.
3. These vaccines have no side effects and cause no allergic reactions.

Which of the statements given above is/are correct?

- A. 1 only
- B. 1 and 2 only
- C. 3 only
- D. All of the above



Question 2 (Previous Year Question 2021)

Q. With reference to recent developments regarding 'Recombinant Vector Vaccines', consider the following statements:

1. Genetic engineering is applied in the development of these vaccines.
2. Bacteria and viruses are used as vectors.

Which of the statements given above is/are correct?

- A. 1 only
- B. 2 only
- C. Both
- D. none



Question 3 (Previous Year Question 2022)

In the context of vaccines manufactured to prevent COVID-19 pandemic, consider the following statements :

1. The Serum Institute of India produced COVID-19 vaccine named Covishield using mRNA platform.
2. Sputnik V vaccine is manufactured using vector based platform.
3. COVAXIN is an inactivated pathogen based vaccine.

Which of the statements given above are correct?

- A. 1 and 2 only
- B. 1 and 3 only
- C. 2 and 3 only
- D. All 1, 2 and 3



Question 4 (Previous Year Question 2022)

Q.10 – Which one of the following statements best describes the role of B cells and T cells in the human body ?

- (a) They protect the body from environmental allergens.
- (b) They alleviate the body's pain and inflammation.
- (c) They act as immunosuppressants in the body.
- (d) They protect the body from the diseases caused by pathogens.

Congratulations on giving the test!

Let's Discuss The Answers.





Question 1 (Previous Year Question 2020)

Q. What is the importance of using Pneumococcal Conjugate Vaccines in India?

1. These vaccines are effective against pneumonia as well as meningitis and sepsis.
2. Dependence on antibiotics that are not effective against drug-resistant bacteria can be reduced.
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Which of the statements given above is/are correct?

- A. 1 only
- B. 1 and 2 only**
- C. 3 only
- D. All of the above



Answer: B

- Pneumococcal Conjugate Vaccines (PCVs) prevent pneumococcal diseases. Pneumococcal disease refers to any illness caused by pneumococcal bacteria. *Streptococcus pneumoniae* (pneumococcus) is a leading cause of bacterial pneumonia, meningitis and sepsis in children.
- PCVs could potentially prevent a substantial proportion of episodes of bacteremic disease, pneumonia, meningitis, sepsis and otitis media, especially in young children. Hence, statement 1 is correct.
- The growing resistance of pneumococcal bacteria to commonly used antibiotics underlines the urgent need for vaccines to be used to control pneumococcal disease. PCV prevents antibiotic resistant pneumococcal infections. Hence, statement 2 is correct.
- Redness, swelling, pain, or tenderness where the shot is given, and fever, loss of appetite, fussiness (irritability), feeling tired, headache and chills can happen. Hence, statement 3 is not correct.
- Therefore, option (b) is the correct answer.



Question 2 (Previous Year Question 2021)

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1. Genetic engineering is applied in the development of these vaccines.
2. Bacteria and viruses are used as vectors.

Which of the statements given above is/are correct?

- A. 1 only
- B. 2 only
- C. **Both**
- D. none



Answer: C

- Recombinant vector vaccines are made through genetic engineering. The gene that creates the protein for a bacteria or virus is isolated and placed inside another cell's genes. Hence, statement 1 is correct.
- Numerous viral vectors are available for vaccine development, such as vaccinia, modified vaccinia virus Ankara, adenovirus, adeno-associated virus, retrovirus/lentivirus, alphavirus, herpes virus, etc. Hence, statement 2 is correct.
- Therefore, option (c) is the correct answer.



Question 3 (Previous Year Question 2022)

In the context of vaccines manufactured to prevent COVID-19 pandemic, consider the following statements :

1. The Serum Institute of India produced COVID-19 vaccine named Covishield using mRNA platform.
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3. COVAXIN is an inactivated pathogen based vaccine.

Which of the statements given above are correct?

- A. 1 and 2 only
- B. 1 and 3 only
- C. 2 and 3 only**
- D. All 1, 2 and 3



Answer: C

- Covisheild is a recombinant, replication-deficient chimpanzee adenovirus vector encoding the SARS-CoV-2 Spike (S) glycoprotein. Following administration, the genetic material of part of corona virus is expressed which stimulates an immune response.
- Sputnik V vaccine is based on a well-studied human adenovirus vector platform.
- COVAXIN is developed using Whole-Virion Inactivated Vero Cell derived platform technology. Inactivated vaccines contain dead virus, incapable of infecting people but still able to instruct the immune system to mount a defensive reaction against an infection.



Question 4 (Previous Year Question 2022)

Q.10 – Which one of the following statements best describes the role of B cells and T cells in the human body ?

- (a) They protect the body from environmental allergens.
- (b) They alleviate the body's pain and inflammation.
- (c) They act as immunosuppressants in the body.
- (d) They protect the body from the diseases caused by pathogens.**



Answer: D

- T cells protect us from infection. In our daily lives, we're constantly exposed to pathogens, such as bacteria, viruses and fungi. Without T lymphocytes, also called T cells, every exposure could be life-threatening.
- B cells create antibodies. B lymphocytes, also called B cells, create a type of protein called an antibody. These antibodies bind to pathogens or to foreign substances, such as toxins, to neutralize them. For example, an antibody can bind to a virus, which prevents it from entering a normal cell and causing infection.

Concept of Mutations



- A mutation is a change in the DNA sequence of an organism. Mutations can result from errors in DNA replication during cell division, exposure to mutagens or a viral infection
- Change in DNA = New genes = Can be beneficial or harmful
- Happens in normal process of cellular repair
- Hiroshima / Nagasaki : Mutagens
- All living cells can mutate

Gene Therapy



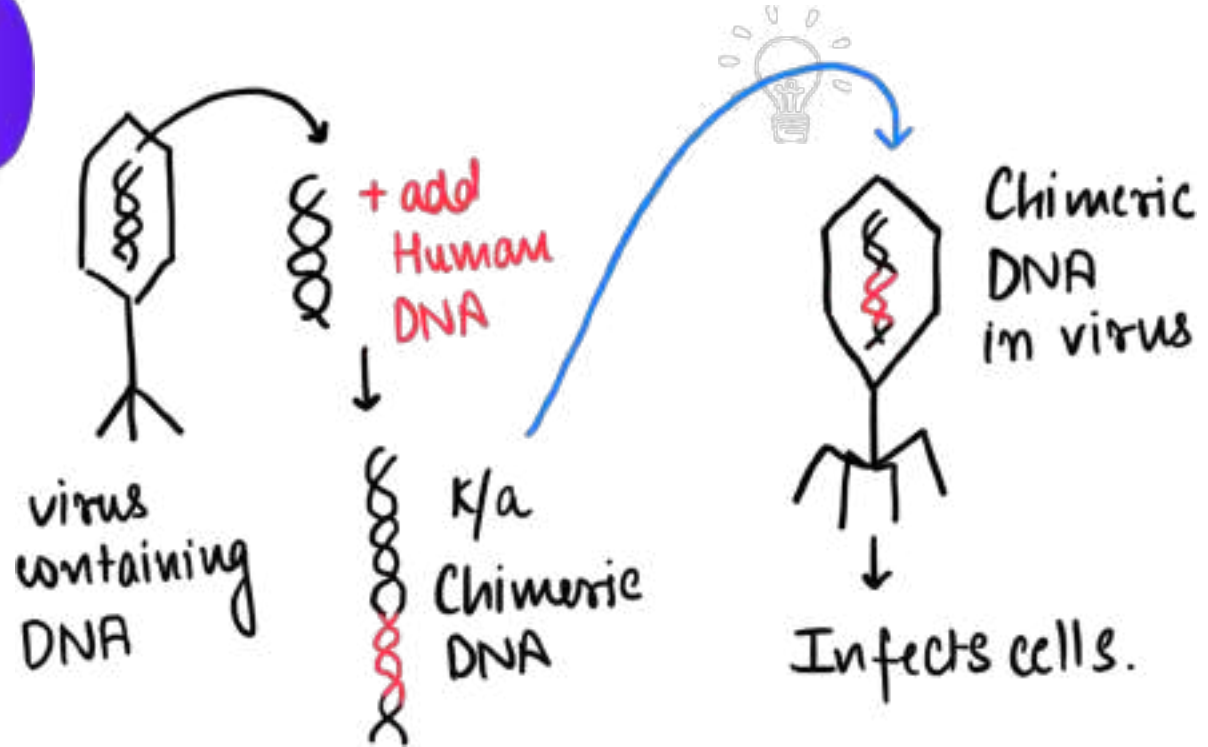
- A child is born with genetic defects. One important gene that he has is non functional. What do we do?
- Delivery of normal genes through baby with gene therapy.
- Gene therapy is a collection of methods that allows correction of a gene defect that has been diagnosed in a child/embryo.
- Here genes are inserted into a person's cells and tissues to treat a disease.
- Correction of a genetic defect involves delivery of a normal gene into the individual or embryo to take over the function of and compensate for the non-functional gene.

Gene Therapy

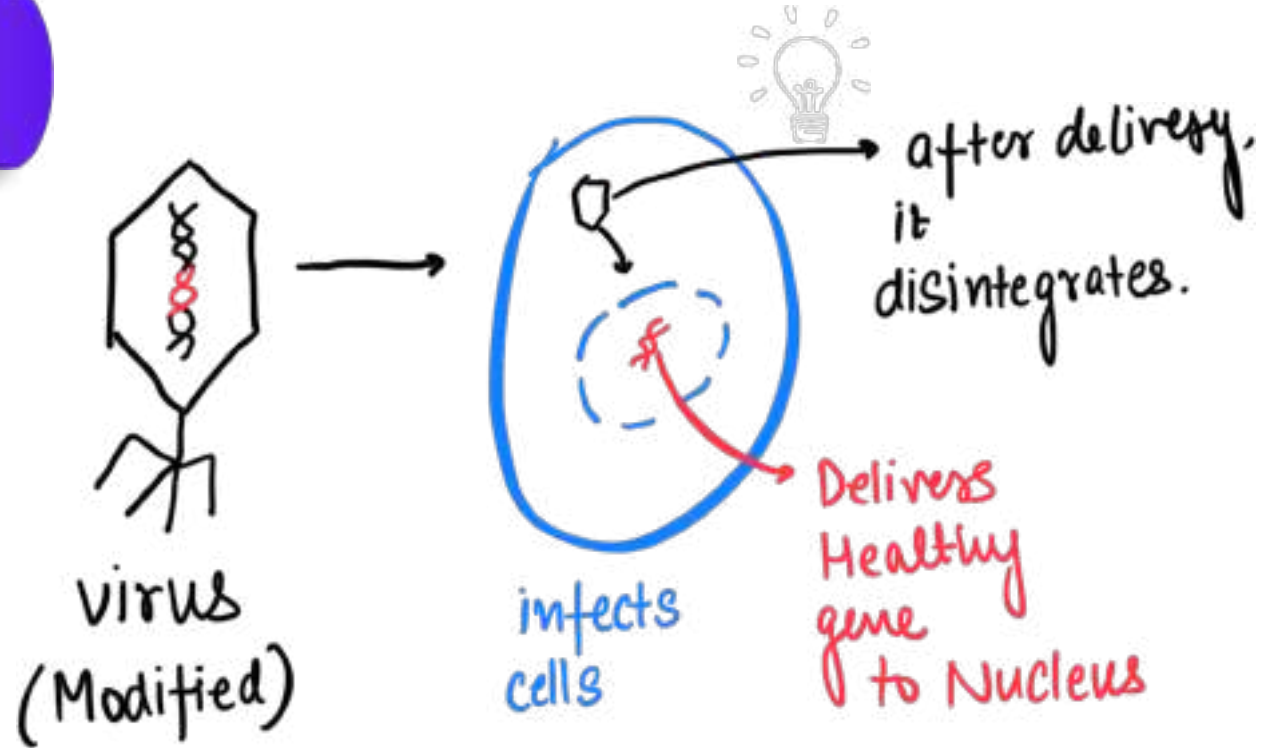


- Biotechnological method to replace defective gene with correct functioning gene.
- 2 types : Somatic cell gene therapy and Germline Gene therapy
- Somatic Gene therapy : Genetic Disorders in individuals can be cured.
- Ex : Haemophilia, Sickle Cell Anaemia (important this year : WHY?)
- Germ Cell Gene therapy : Offspring (child) can be prevented from getting disease.
- Can be used to fight cancer as well : CAR-T cell therapy.

How is it done ?



How is it done (2)



LET'S SEE A VIDEO ON GENE THERAPY



- [click here \(nationwide children SMA type 1\)](#)
- [click here for video 2](#)

Gene Therapy



Can be used to :

- 1. Replace bad gene with healthy gene
- 2. Add a new gene to perform function of missing / non functional gene
- 3. Turn off a problem gene.
- Can be done inside the body (using virus) called in vivo
- We are using modified human virus, for example adenovirus.
- Can be done outside the body : in vitro

In vitro gene therapy : Example : ADA - SCID

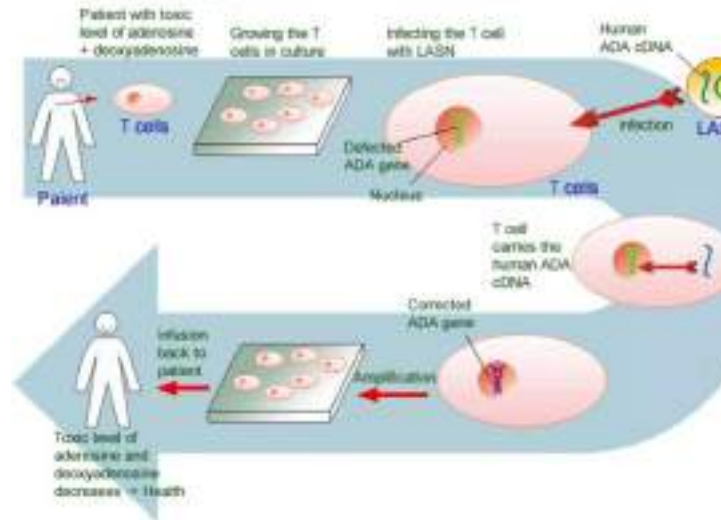


- Adenosine Deaminase Deficiency leading to severe combined immunodeficiency
- T cells of the body need ADA.
- When ADA is absent, patient develops SCID.
- We need to deliver ADA to patient's T cells.
- We use functional ADA gene inserted with viral vector. But virus has to be genetically edited to be made safe.
- T cells are first removed from the patient, edited with viral vector containing new gene, then sent back into the patient.

Gene therapy for ADA- SCID



Gene Therapy for ADA-SCID

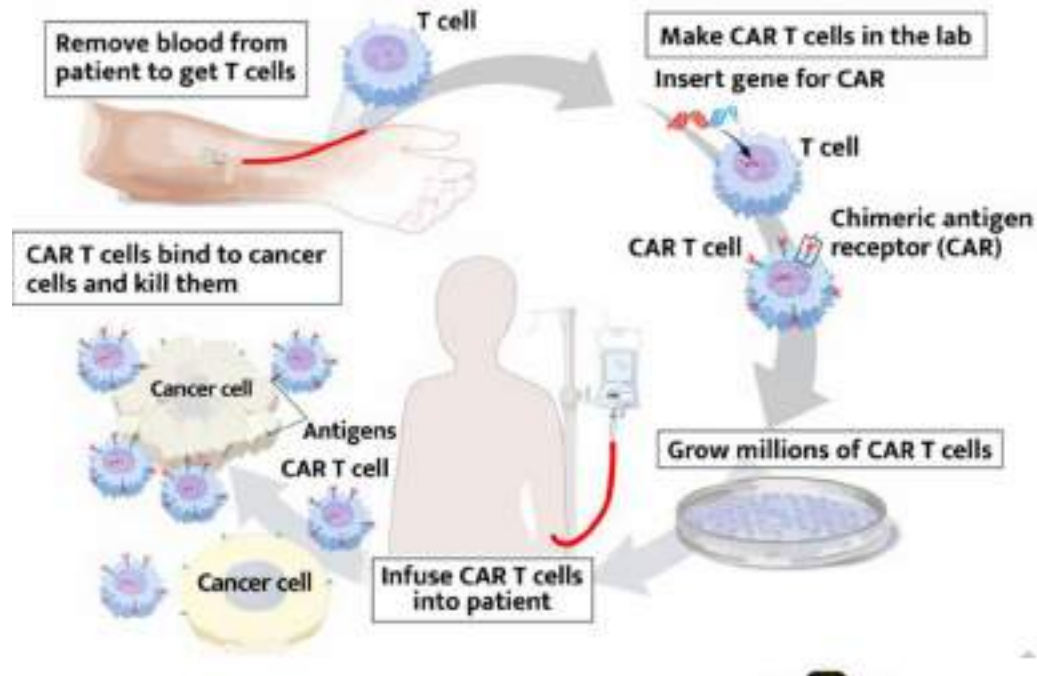


CAR-T Cell Therapy



- Immune T cells to fight cancer by editing them in the Lab.
- T cells are taken from the patient's blood and are edited by adding a gene for man made receptor : CAR (chimeric antigen receptor)
- CAR is a receptor that binds to certain proteins on cancer cells.
- CAR added T cells attack cancer cells.

CAR T-CELL THERAPY



CAR-T Cell Therapy : Benefits



- Can cure certain types of cancers.
- Unlike chemotherapy (current method of treating cancer) CAR-T cell therapy is used only once.
- Shorter Treatment
- Faster Recovery

Issues of Gene Therapy



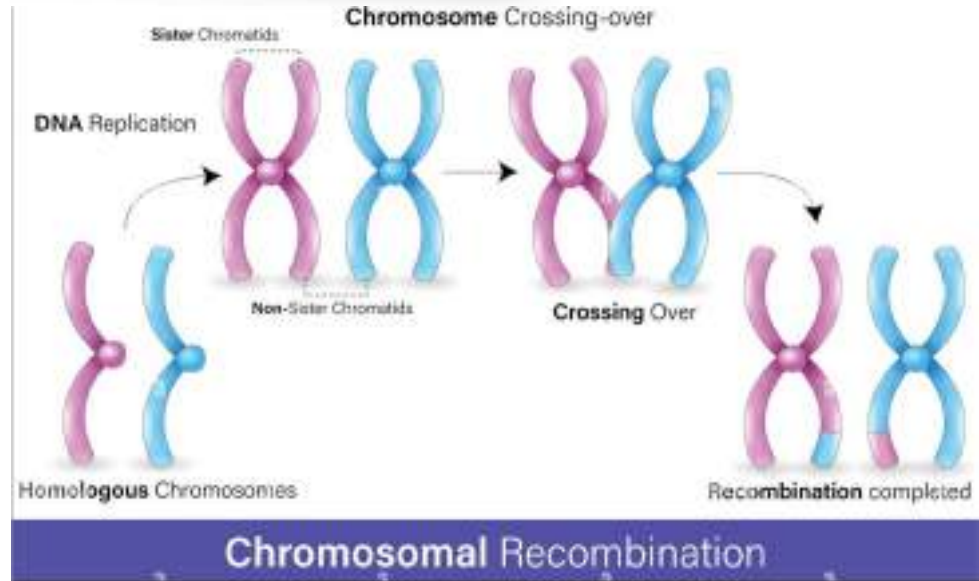
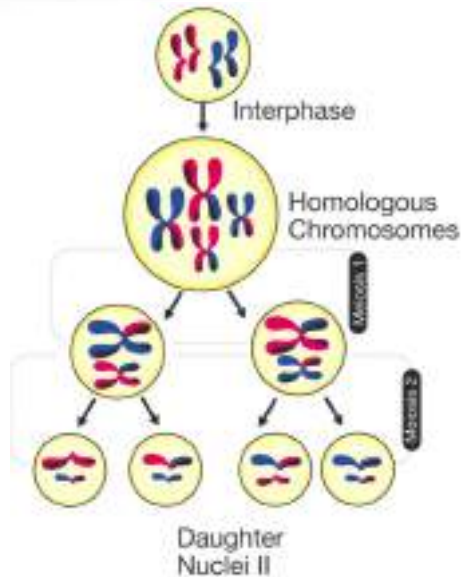
- Short Lived Nature : multiple rounds needed : example : ADA SCID
- Possible solution : Do at early stage.
- Immune response possible, other side effects possible.
- Cannot be used to correct chromosomal disorders, only disorders with single bad gene. (explained later)
- Virus alteration and manipulation : Containment

Issues of Gene Therapy



- Short Lived Nature : multiple rounds needed : example : ADA SCID
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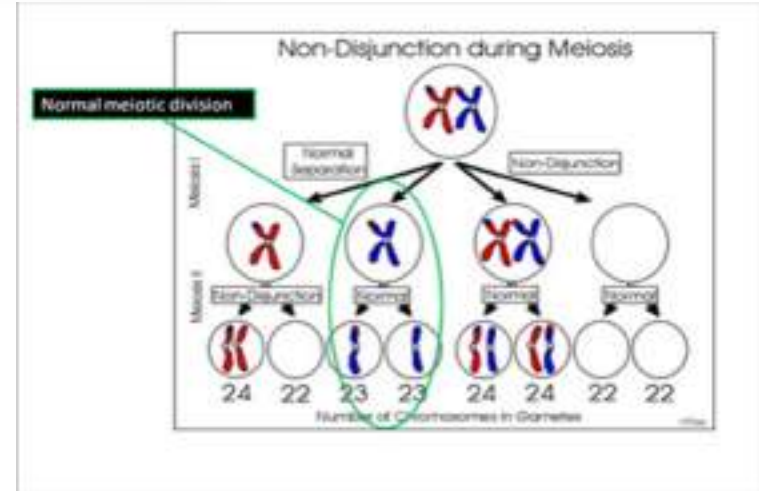
Chromosomal Disorders : defects during mieosis



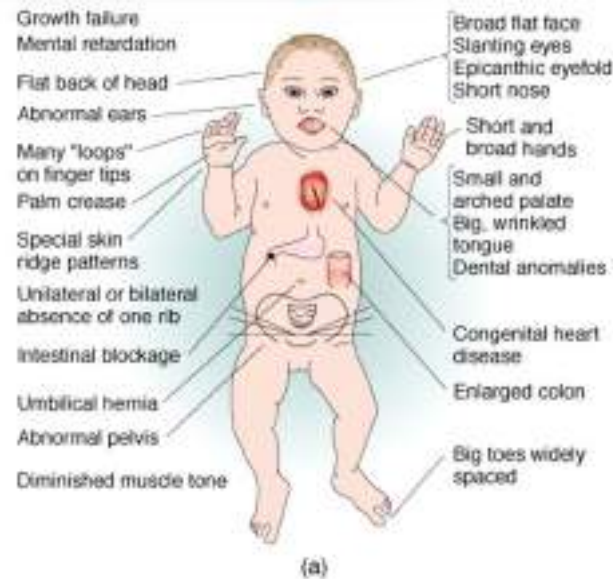
Chromosomal disorders : Down's Syndrome

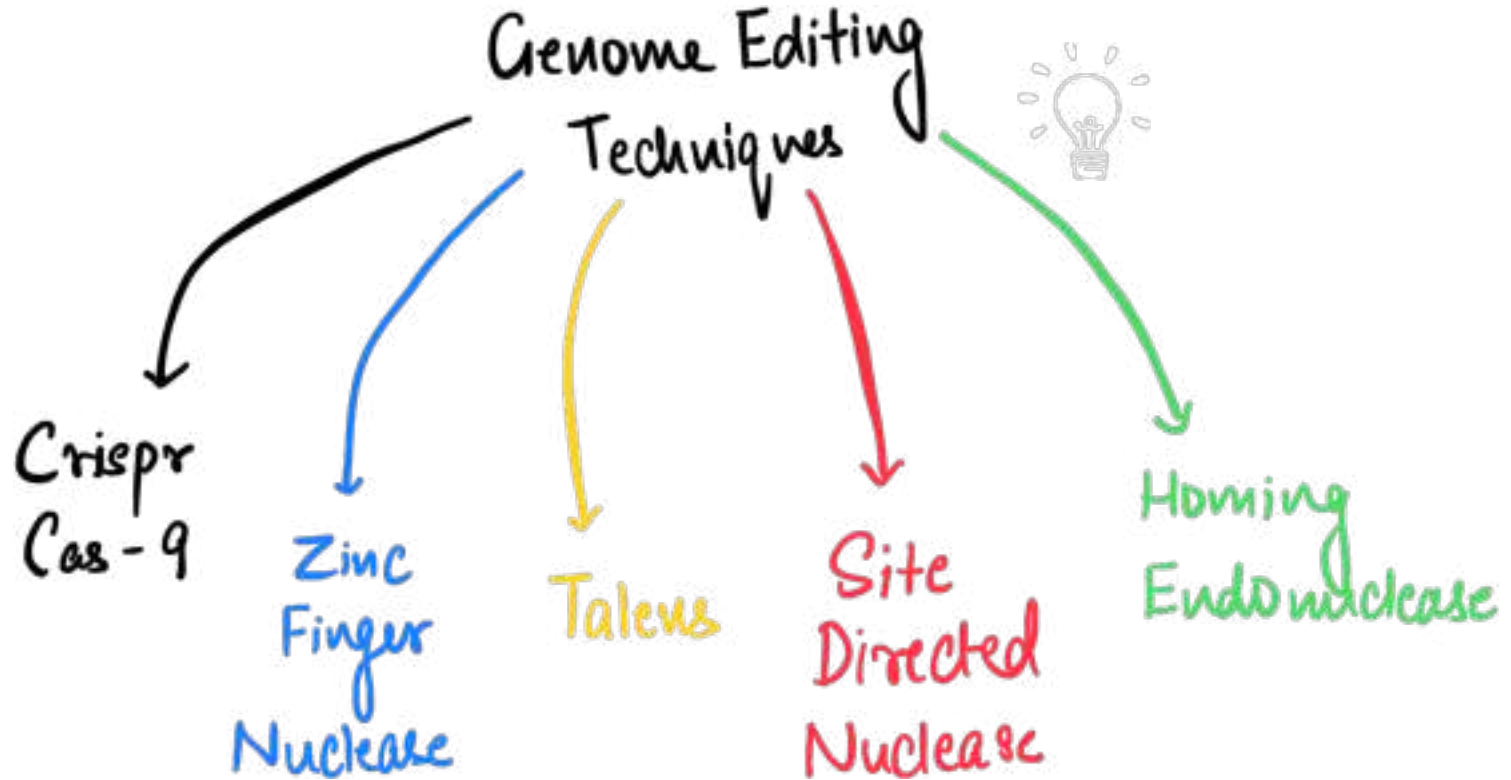


- Patient has 3 copies of Chromosome number 21
- Leads to severe genetic defects.
- This cannot be cured by gene therapy.



Down's Syndrome : treat them with sympathy





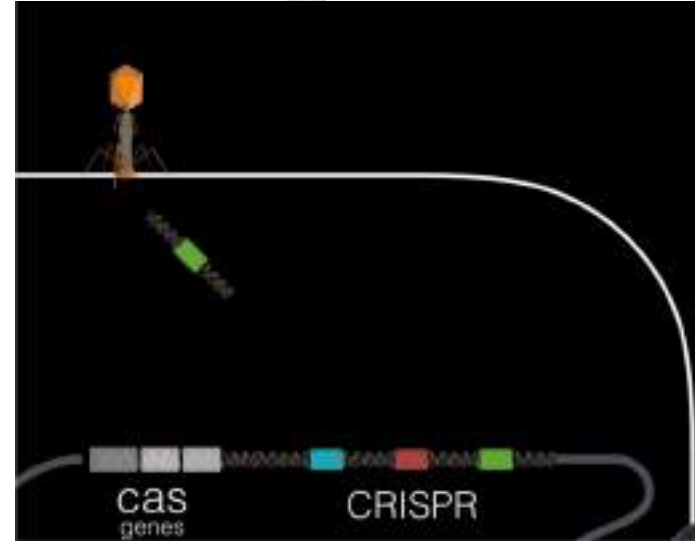
Crispr Cas – 9 and Gene Editing



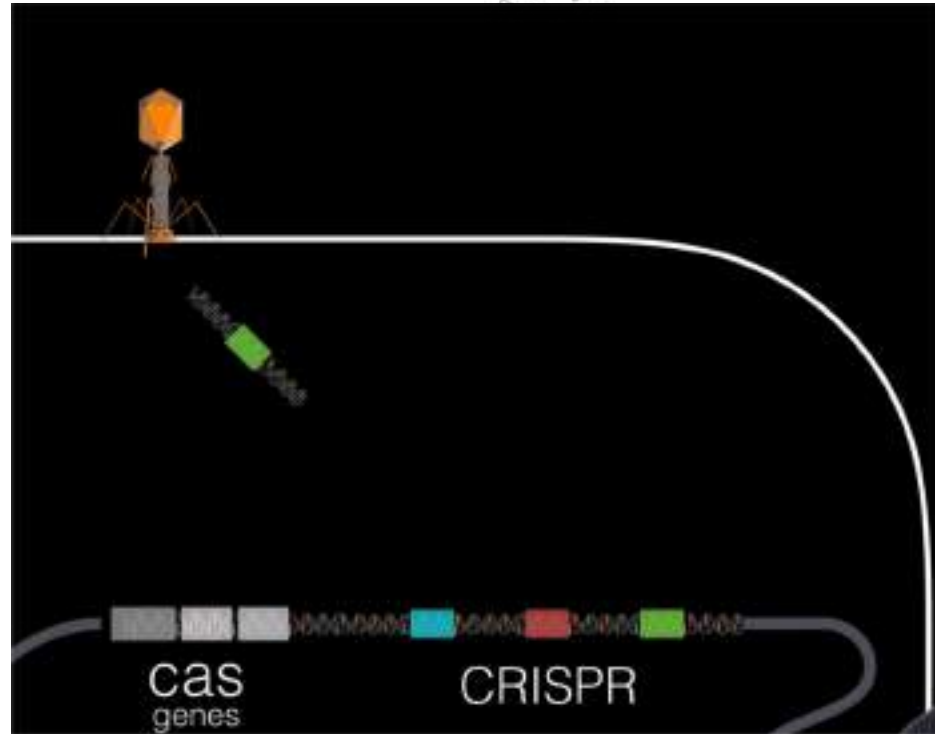
- What are genes?
- Gene Editing : changing genes in humans
- CRISPR : Clustered Regularly Interspaced short Palindromic Repeats
- CAS : crispr associated protein -9
- Originally found in bacteria
- Molecular level scissors and glue (cut and paste mechanism)



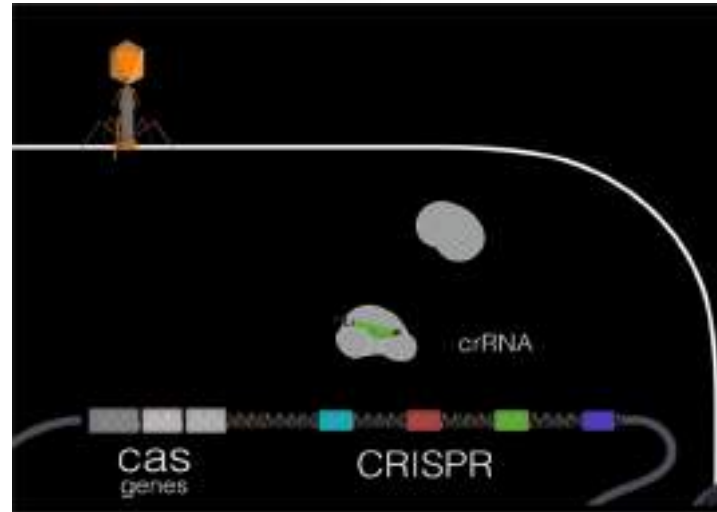
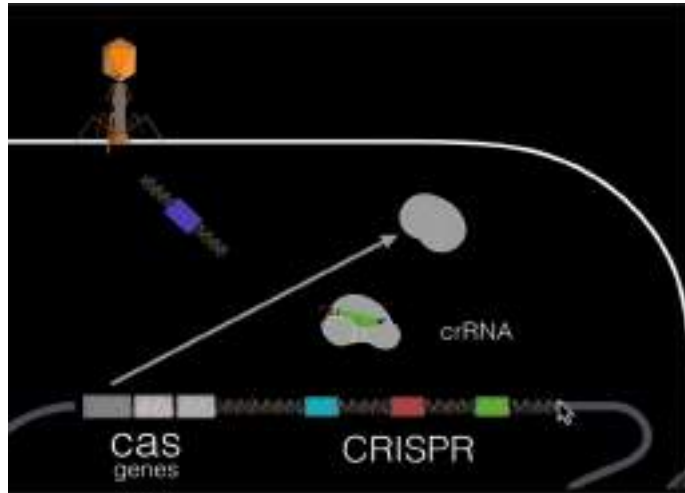
- Palindromes interspaced by spacer DNA
- Cas gene : Cas proteins :
Helicases (unwind DNA)
Nucleases (Cut DNA)
- It is a defence system that bacteria uses to defend against viruses that infect bacteria. These viruses are called bacteriophage.



- It is a defence system that bacteria uses to defend against viruses that infect bacteria. These viruses are called bacteriophage.



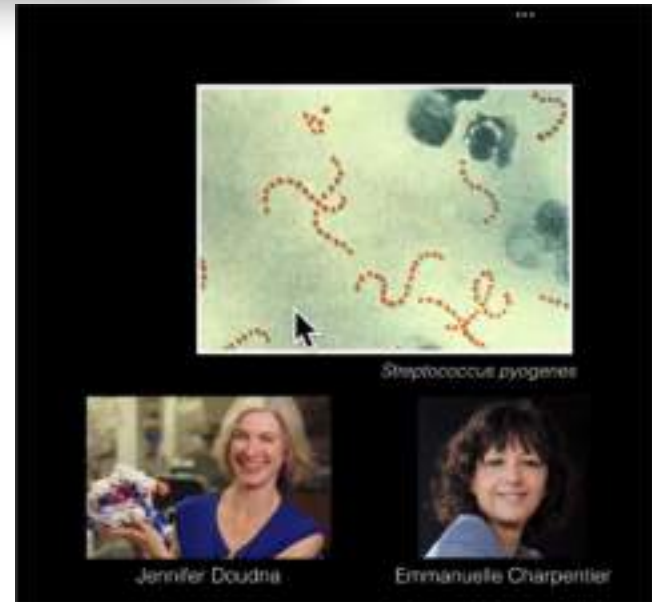
Crispr Cas-9 : Working explained



Crispr



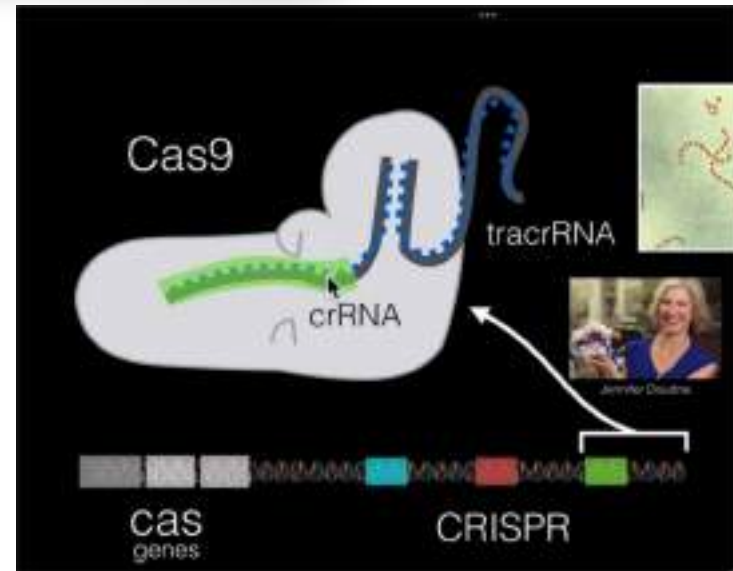
- Spacer – repeat – Spacer– Repeat
- Similar to our immune system.
- Scientists thought of a new use for this in genetic editing.



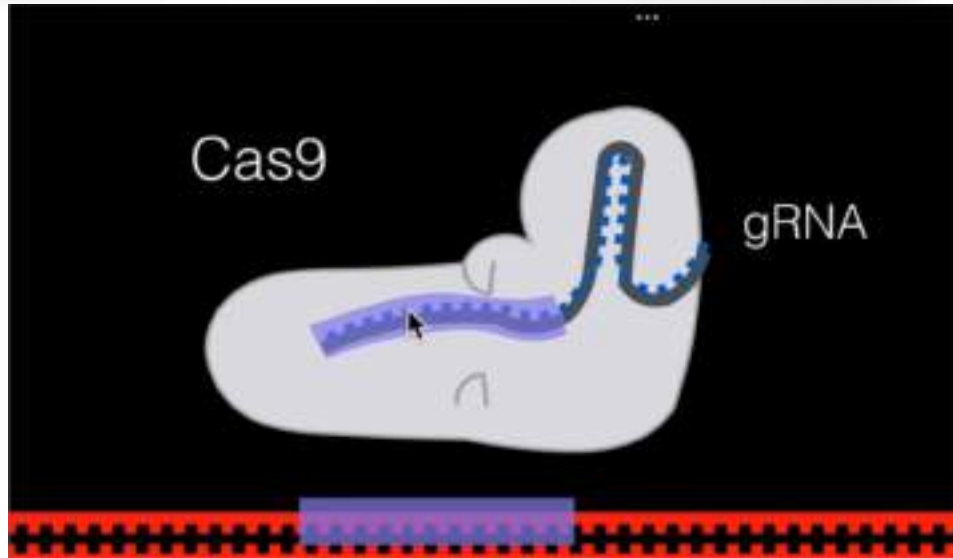
Crispr in genetic editing



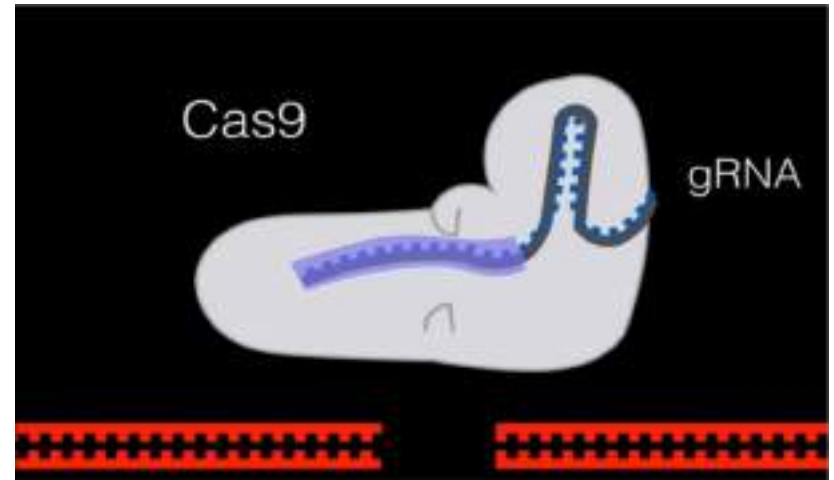
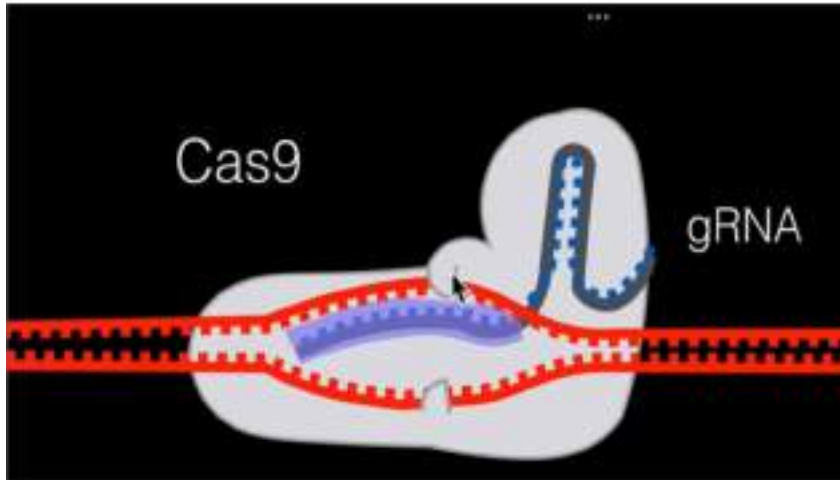
- Nuclease : enzyme responsible for cutting DNA



We modified CRISPR to work in our cells.



We modified CRISPR to work in our cells.



CRISPR Cas 9



- Consists of single guide RNA (sgRNA) : complementary to the part of DNA to be cut
- Cas 9 : crispr associated protein 9 : cuts specific part of dna
- It is like using Microsoft word for a document
- Ctrl + F = Find (SgRNA)
- Replace = Delete and change (Cas-9)
- Crispr cas 9 is molecular scissors

Possible Applications



- Gene editing and gene therapy
- Creation of Genetically Modified Organisms
- Designer babies : Chinese Scientist Jailed for this
- Germline editing of reproductive cells

Issues



- Increased risk of mutations and cancer
- Ethical concerns and designer embryos
- Impact on human variation
- Abuse of gene editing
- Safety and off target effects
- Informed Consent
- Justice and equity

Precision Guided Sterile Insects



- CRISPR-based system was developed to safely restrain mosquito vectors via sterilization.
- precision-guided sterile insect technique (pgSIT) was used to alter genes in *Aedes aegypti*, the species responsible for spreading malaria, dengue, chikungunya and Zika.
- pgSIT uses CRISPR to sterilise male mosquitoes and render female mosquitoes (which spread disease) flightless.
- "Gene drive" systems could suppress disease vectors by passing desired genetic alterations indefinitely from one generation to the next.
- Unlike 'gene drive' the pgSIT system is self-limiting and is not predicted to persist or spread in the environment.

Zinc finger nuclease

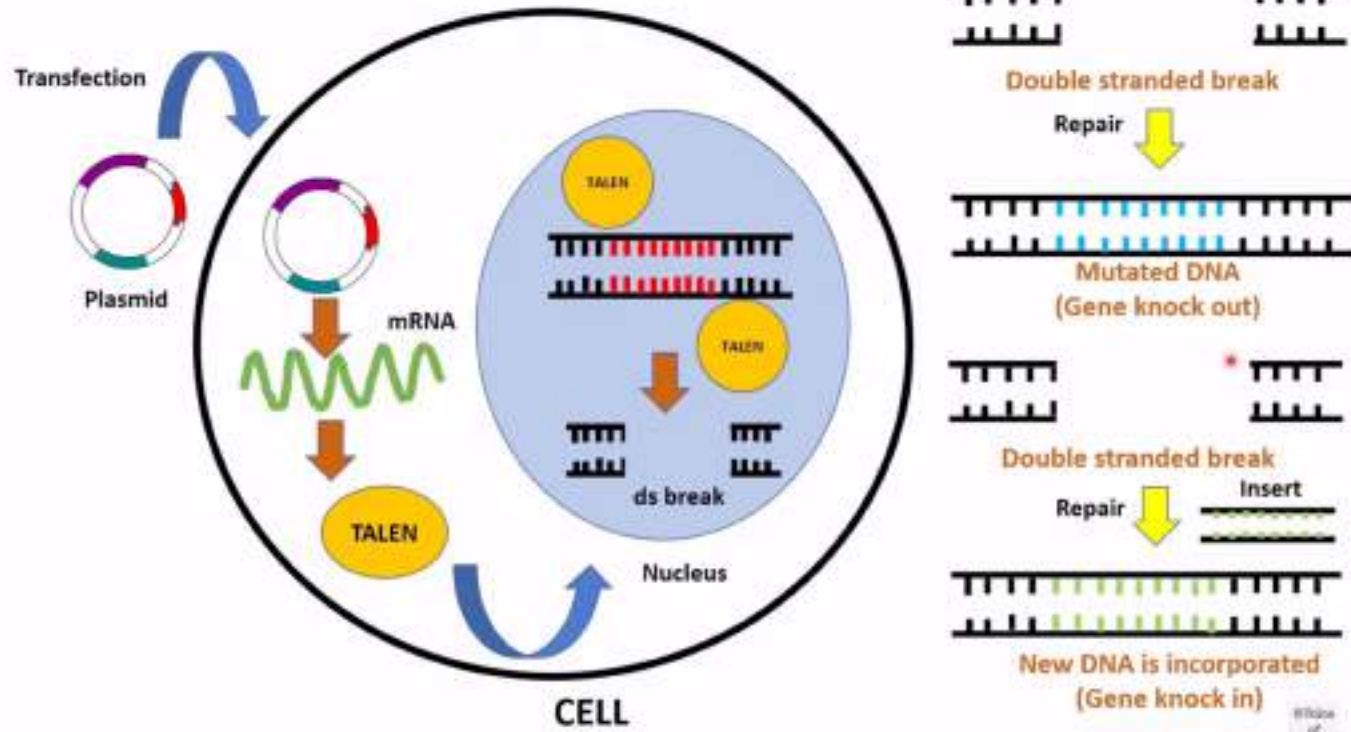


- DNA Scissors. This is all you actually need to remember.
- DNA binding part is zinc finger protein. It will recognise portion of DNA to be cut.
- Nuclease part of ZFN (required to cut) is FOK1 Nuclease. (easy to remember? 🧠)
- Success rate is low, 1-20%
- Older Technology than Crispr cas – 9.

TALENs



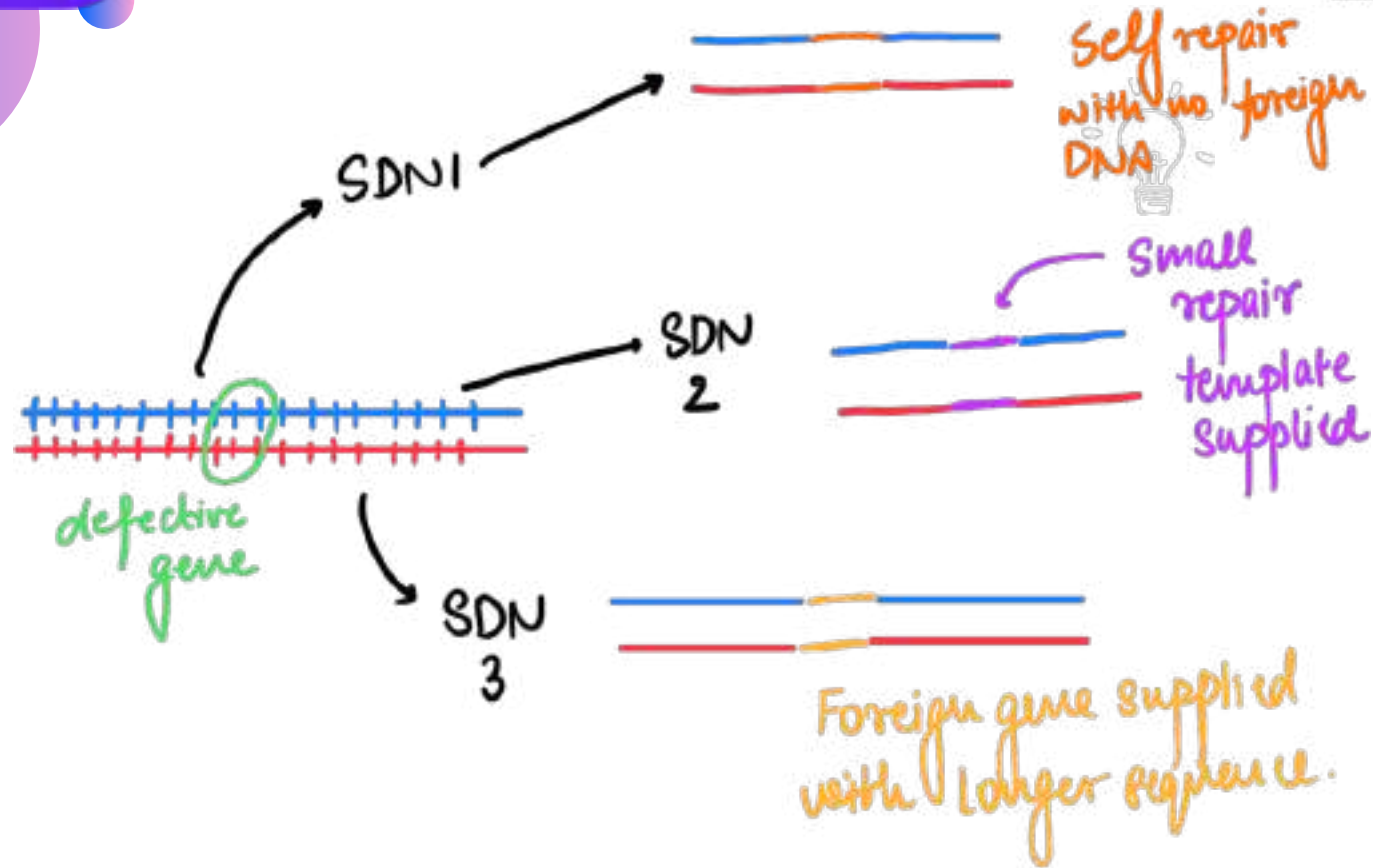
- DNA binding domain of TALENs is made of Transcription activator like effector (TALE) domain
- Nuclease Part is same, Fok1 nuclease.
- Also DNA scissors.
- Taken from xanthomonas bacteria.



Site Directed Nuclease



- Site Directed Nuclease are genome editing techniques
- All 3 have different functions. (next slide)
- SDN refers to cleaving DNA strands to affect subsequent genome editing, taking advantage of natural repair mechanism to introduce small changes.
- Main SDN technologies : ZFN and TALENs



Site Directed Nuclease



- SDN 1 and SDN 2 : No alien genetic material is being introduced.
- Result is indistinguishable from conventionally bred crop varieties.
- SDN 3 : Genes of Foreign origin are introduced.
- SDN 1 and SDN 2 : have been exempted from 1989 rules on GM crops.
- SDN 3 : Will come under the rules if Foreign DNA **exceeding 20 base pairs** in inserted.

Homing Endonuclease / Meganuclease



- Another type of molecular scissors
- Cuts the DNA and then you can choose to add or not add any gene.



Gene Doping



Changing DNA to enhance performance by athletes

Ex vivo (in vitro)

In vivo

WADA issues

Animal Cloning

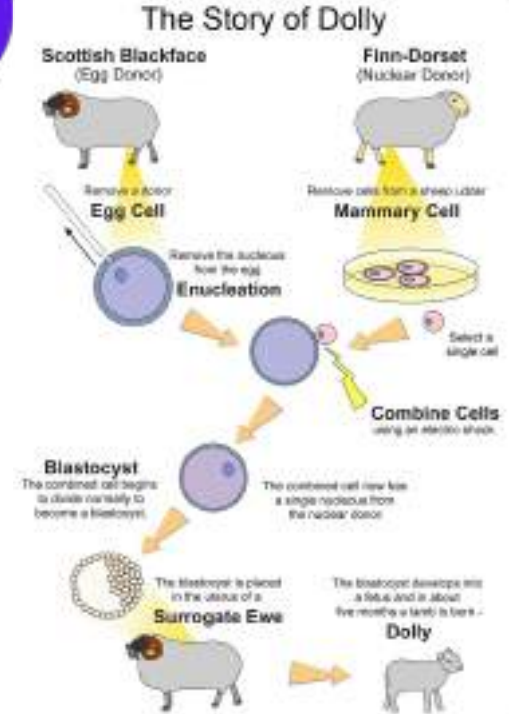


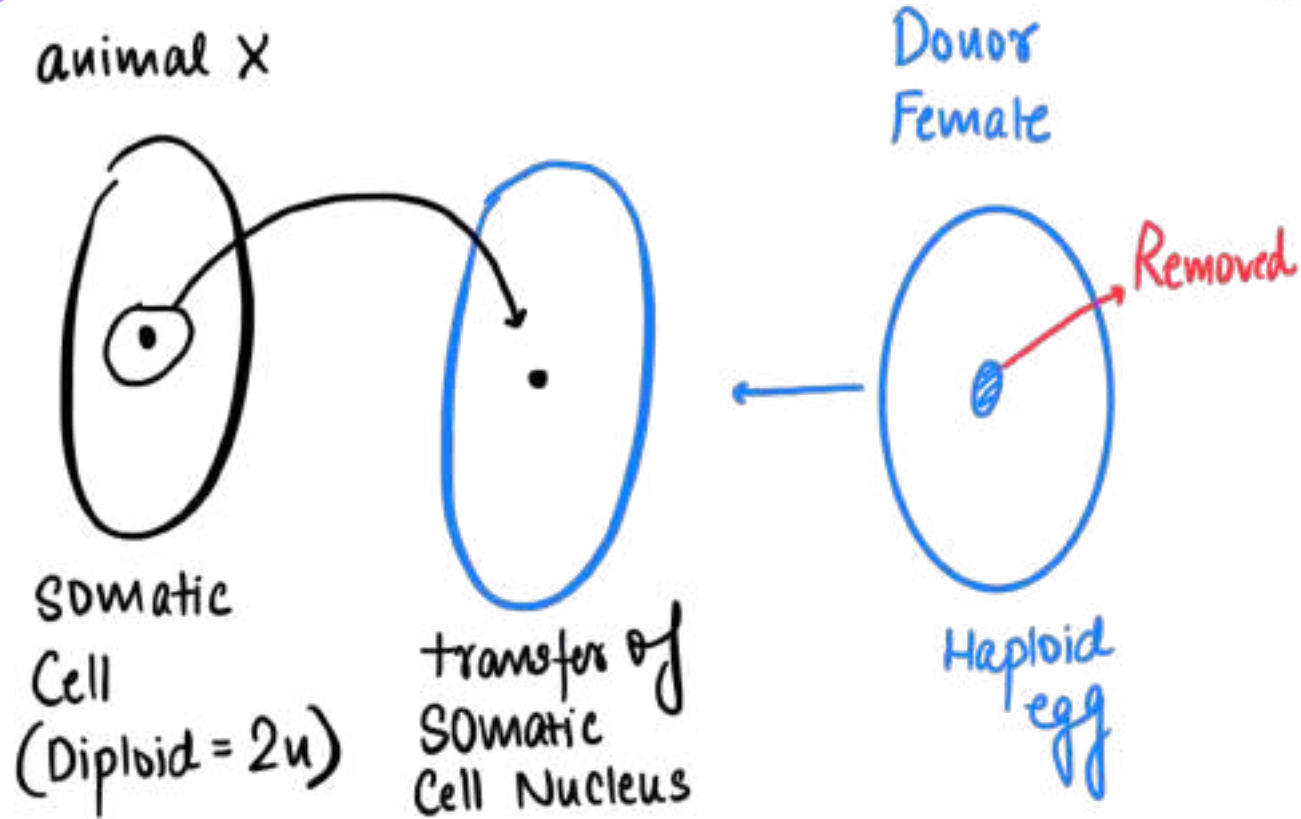
- Identical twins : from 2 parents
- Cloning is the development of cloned animal from a single parent. Also known as Reproductive Cloning.
- Method : Somatic Cell Nuclear Transfer Technique

Animal Cloning and Somatic Cell Transfer



- Cloning : exact same genes of progeny
- Dolly the sheep : Ian Wilmut
- Reproductive cloning
- We will study Therapeutic Cloning after stem cells

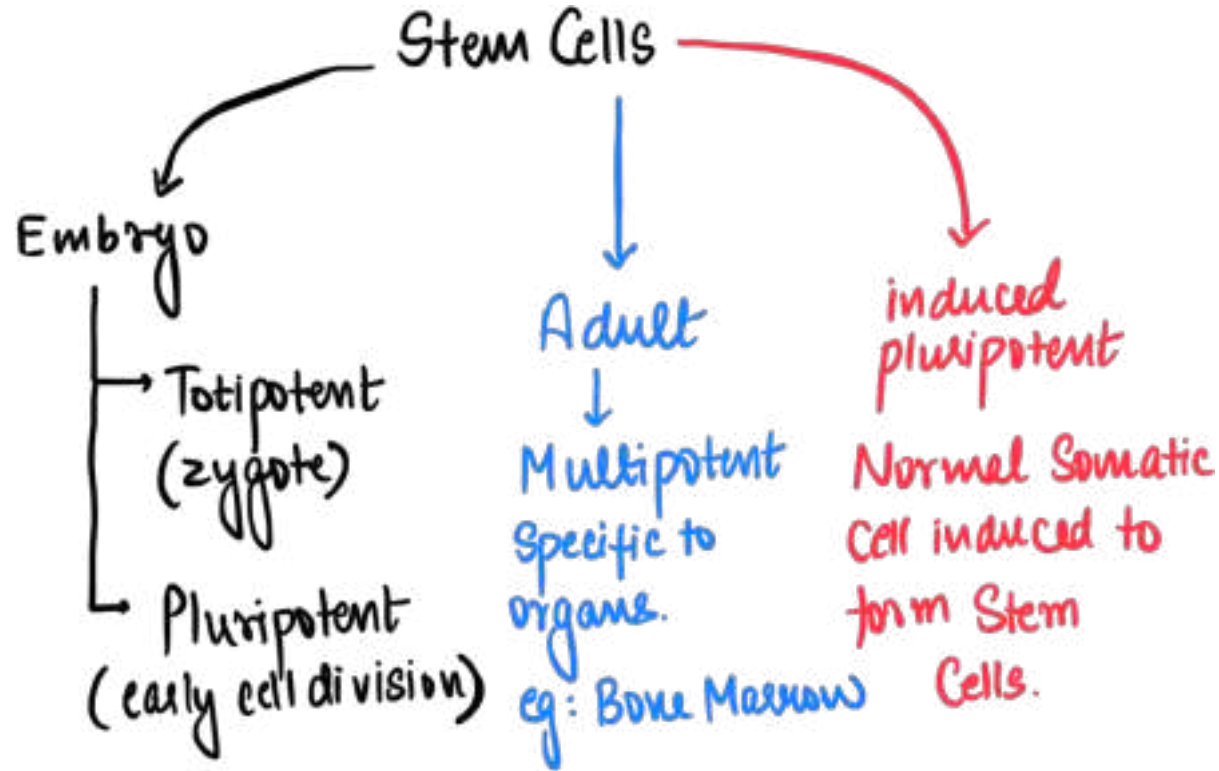




Cloning



- Recently China has cloned a wild arctic wolf first time in the world
- Clone is NOT completely identical : does not always look exactly the same as environment also plays a role in determining physical features of an individual.
- 3 types of artificial cloning :
- Gene/DNA cloning : DNA fragment from one organism to a self replicating genetic element, such as bacterial plasmid.
- Reproductive Cloning : Transferring nuclear material isolated from somatic cell into enucleated egg.
- Therapeutic cloning.



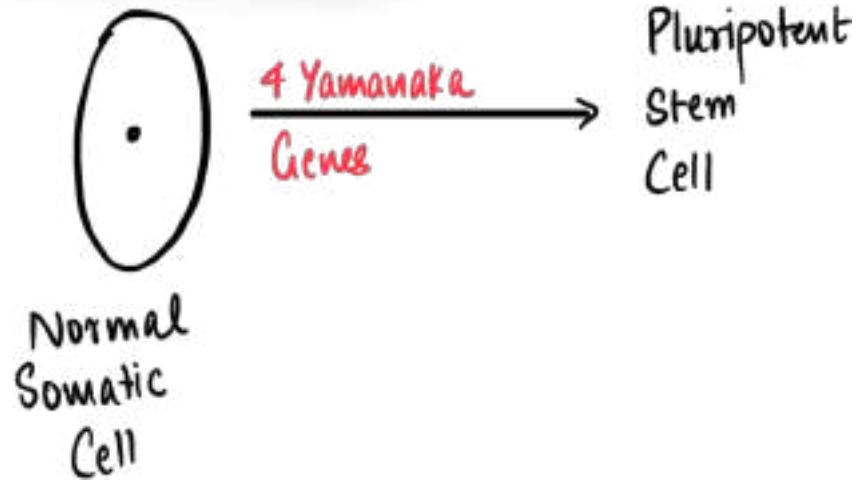
Stem Cells

STAGE OF EMBRYOGENESIS*	EVENTS	IMAGES (NOT TO SCALE)	TOTIPOTENT or PLURIPOTENT?
Fertilization Carnegie Stage 1	Ovum and sperm fuse to form a diploid zygote.		Totipotent
Cleavage Carnegie Stage 2	The zygote undergoes cleavage via mitosis to increase the number of cells within the zona pellucida.	Cells have a distinctly round shape. 	Totipotent
Compaction** Carnegie Stage 2	Blastomeres form junctions with one another, so the cells lose their round shape. The resulting formation is a mulberry-shaped mass of cells within the zona pellucida called a morula, which appears when there are 16 cells.	Cell boundaries are less clear. 	Totipotent
Differentiation Carnegie Stage 2	Cells are divided into distinct lineages: the inner cell mass (epiblast and hypoblast) and the trophoblast.	 epiblast hypoblast trophoblast	Epiblast = Pluripotent Hypoblast and trophoblast are neither totipotent nor pluripotent.

Induced Pluripotent Stem Cells



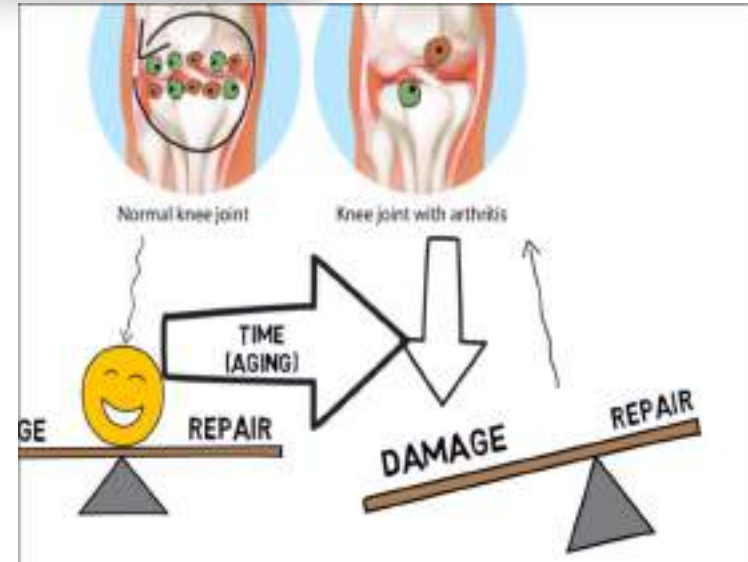
- Conversion of Somatic Cells to form Pluripotent Stem Cells.
- 4 specific genes (Yamanaka factors) when introduced into normal somatic cell, turn it into stem cell.
- Shinya Yamanaka : Nobel prize in 2012



Stem Cell Therapy



- Osteoarthritis of Knee : degenerative disease
- Where damage overtakes normal repair.
- Here, we can use patient's own stem cells to provide
- Regenerative support and relieve the injuries and the pain.
- Therefore, Stem Cells are a form of regenerative medicine



Stem Cell Therapy



- Can be used to replace dead / damaged tissues by normal functioning cells produced by stem cells.
- Can be used for diseases like :
 - Thalassemia
 - Leukemia
 - Osteoarthritis and Osteoporosis
 - Blindness
- Advantages : Minimal Immune rejection because cells are derived from the body itself.

Future prospects



- Body organs can be grown in labs if we combine Stem Cells and 3D printers.
- ICMR : guidelines for stem cells in India.
- Stem Cell Banking : storage of umbilical cord stem cells at -170°C with liquid nitrogen to be used for future therapy.

Challenges



- Synthetic biology : seen as interference with nature
- Biosafety concerns
- Abuse of Technology
- Genetic Divide
- Ethical Concerns

Therapeutic Cloning



- Producing a healthy replica of a human organ
- Better for transplanting organs : no immunosuppressant
- It is like reproductive cloning till production of embryo. Produced embryo is allowed to grow in lab.
- Embryo is allowed to grow and then **stem cells are extracted**
- Treat diseases by replacing damaged cell
- Applications : Ageing, cancer, implant surgeries.
- Ethical issues : Divine command theory, life begins at conception (Christian ethics); Violative of Natural laws

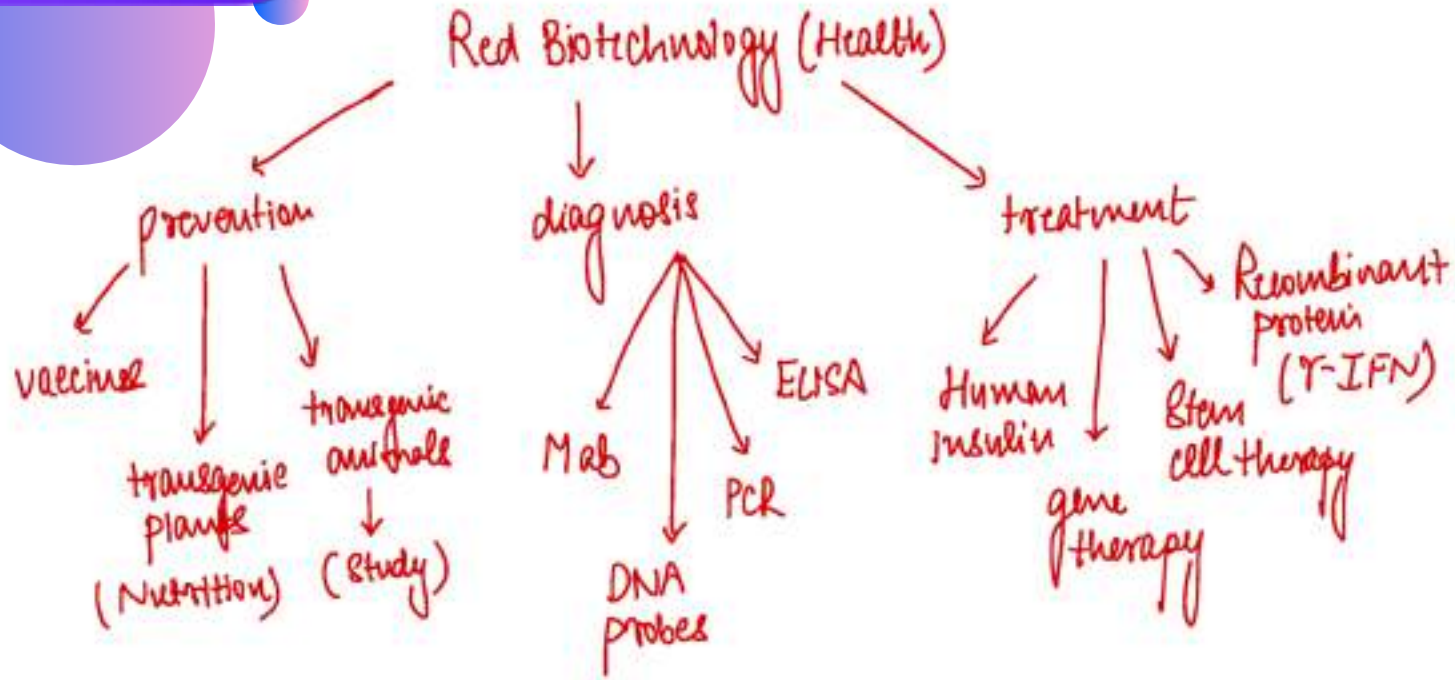
Applications of RDT/Biotechnology [India]

green Biotech (agriculture)

- (i) Yield
- (ii) Biopesticides: BT Brinjal
- (iii) HYV → GM Mustard
- (iv) Biofortification → golden Rice (vit-A)
- (v) Shelf life → Flavr Savr tomato (export)
- (vi) disease (Samba Masuri Rice)
- (vii) RNAi → plant vaccination
- (viii) cold tolerant Strawberry
- (ix) Better Brinde → cows, etc.

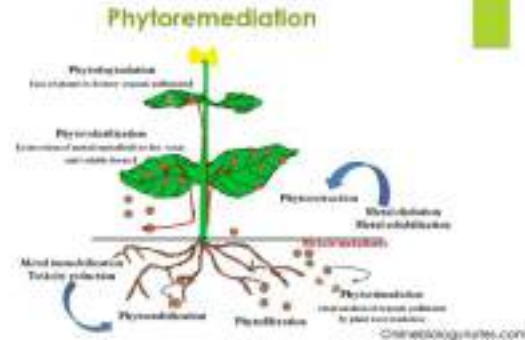
Blue Biotechnology (Marine)

- aquaculture (sea-food cultivation)
- Transgenic fish → freeze resistant fish
- Conservation of species
- seaweed cultivation → ex: agar-agar.



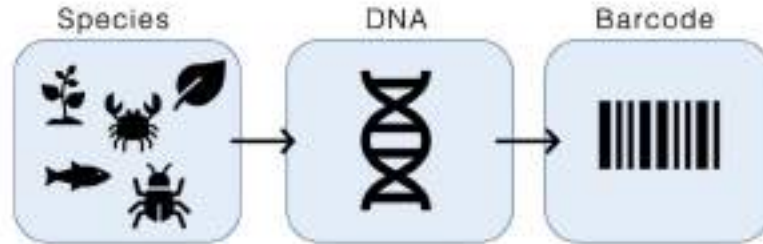
use of plants and associated soil microbes to reduce toxic elements.

eucalyptus



DNA Barcoding

DNA barcoding is a method of species identification using a short section of DNA from a specific gene or genes. The premise of DNA barcoding is that by comparison with a reference library of such DNA sections (also called "sequences"), an individual sequence can be used to uniquely identify an organism to species, just as a supermarket scanner scans barcodes.

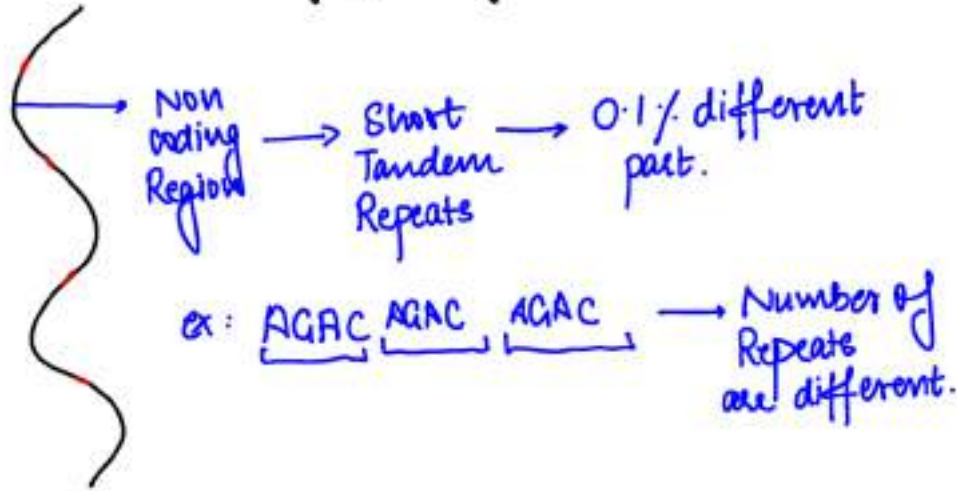


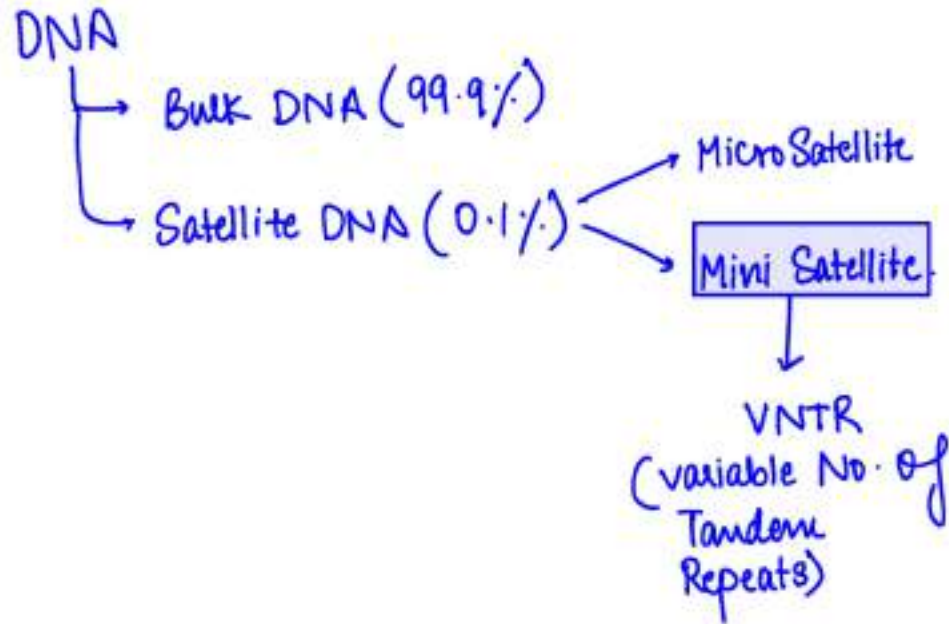
Forensic Biotechnology

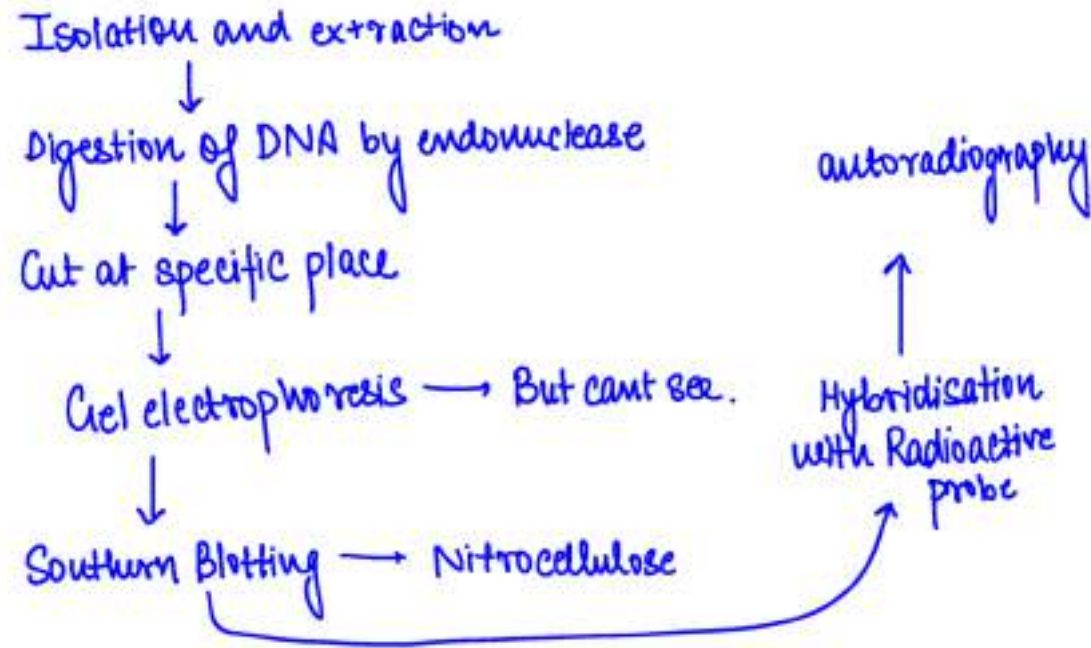


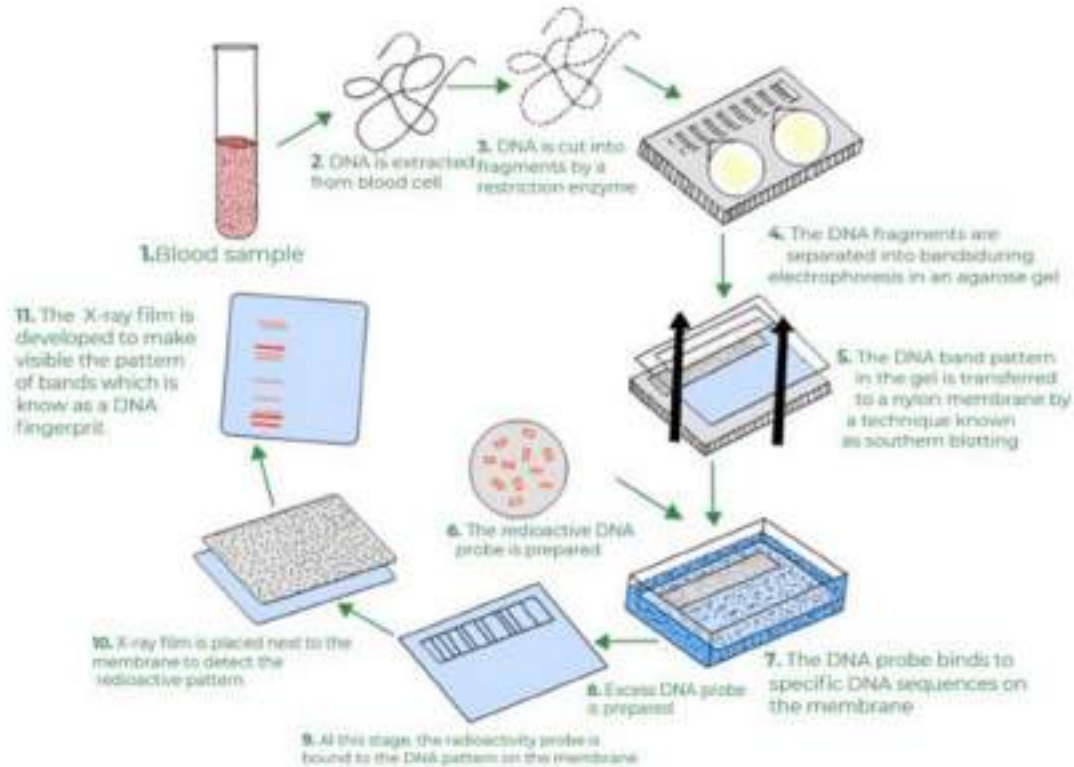
- DNA fingerprinting : identification of criminal by use of biotechnology.
- Here we use something called variable number of tandem repeats (VNTR)
- Use to develop DNA bands in Gel electrophoresis and identify a person.
- Can be used in :
- Criminal Investigation
- Paternity testing
- Immigration Cases
- Child swapping cases
- Disaster victim identification

DNA fingerprinting











UPSC Prelims

Science & Tech

Class No. 6

Health Issues

Genome Sequencing



- Recently, an international team described the first ever sequencing of complete human genome.
- Previous effort : couldn't read 8% of the genome.
- Previous effort was called Human Genome Project

Human Genome Project



- Started in 1990
- 3 billion base pairs of human DNA
- Ethical legal and social issues were addressed
- 97% Junk DNA, only 3% codes for protein
- HGP read : India did not participate
- Built a standard human resource consortium called Genome reference consortium build 38 (GRCch38)
- HGP write: started in 2016 : build an artificial human genome

Human Genome Project : uses



- Disease diagnosis : Identification of genes, rare diseases, neurodegenerative conditions.
- Viral Diseases and their impact (we now have a reference)
- Study of Genetic Variations
- Racism : Debunked
- Gene Therapy
- Evolutionary Biology : Our relationship with primates

Telomere to Telomere Project



- HGP did not do sequencing for 8% of human genome. This was heterochromatin. It was thought to be junk DNA and difficult to sequence.
- Now T2T project has done sequencing for heterochromatin region also.
- Significance :
- Genetic Variations in heterochromatin
- Understanding how genome as a whole works.
- Genetic Diseases
- Pharmacogenetics.

CHROMOSOME PARTS

HETEROCHROMATIN:

- More condensed
- Silenced genes (methylated)
- Gene poor (high AT content)
- Stains darker

EUCHROATIN

- Less condensed
- Gene expressing
- Gene rich (high GC content)
- Stains lighter



Telomere

Indian Initiatives



- CSIR : Indigen Project : Whole genome sequencing of 1000 Indian individuals representing diverse ethnic groups.
- DBT : Genome India project : 10000 genetic samples from citizens across india to build a reference genome.

Viruses



- Borderline between living and non living
- The viruses are non-cellular organisms that are characterised by having an inert crystalline structure outside the living cell.
- A virus is a protein and genetic material that is infectious.
- In general, viruses that infect plants have single stranded RNA and viruses that infect animals have either single or double stranded RNA or double stranded DNA.
- Bacterial viruses or bacteriophages (viruses that infect the bacteria) are usually double stranded DNA viruses

Viruses



- The protein coat called capsid made of small subunits called capsomeres, protects the nucleic acid.
- These capsomeres are arranged in helical or polyhedral geometric forms.
- Viruses can cause many diseases in plants and animals.
- **Viroids** : In 1971, T.O. Diener discovered a new infectious agent that was smaller than viruses and caused potato spindle tuber disease. It was found to be a free RNA; it lacked the protein coat that is found in viruses, hence the name viroid. The RNA of the viroid was of low molecular weight.

Viral Vector Vaccines vs DNA Vaccines



- Viral Vector Vaccine : Delivers genetic Material (DNA) that can be transcribed by the recipient's host cells as mRNA to later be translated into antigen to elicit an immune response.
- DNA Vaccine : nucleic acid vaccines that are genetically engineered into a plasmid. The DNA sequence codes for the antigen against which immunity is sought.

DNA Vaccine vs Recombinant Vaccine



DNA Vaccine	Recombinant Vaccine
Description	
DNA vaccines are prepared by inserting a DNA sequence (that codes for the antigen) into a plasmid and transfected to the host cell.	Recombinant vaccines are protein or DNA recombinants that are inserted into the host cell to trigger immune response.
DNA/Protein	
DNA-based.	Both DNA and protein-based.
Incorporation	
The DNA sequence gets incorporated into the host genome.	A gene of interest is incorporated into a vector which in turn is included in the host genome.
Approval	
ZyCoV-D is a DNA vaccine approved for humans.	Hepatitis B and HPV vaccines have been approved and are readily used.

Nutrients and their roles



- Major nutrients : Carbohydrates , Proteins, Fats
- Doubt regarding proteins
- Major deficiencies and issues
- Kwashiorkor and Marasmus

Kwashiorkor and Marasmus



10 Differences between Kwashiorkor and Marasmus

Kwashiorkor

Protuberant belly

Itchy rash

Xerosis

Poor wound healing

Comparison Table

Kwashiorkor	Marasmus
It develops in children whose diets are deficient of protein.	It is due to deficiency of proteins and calories.
It occurs in children between 6 months and 3 years of age.	It is common in infants under 1 year of age.
Subcutaneous fat is preserved.	Subcutaneous fat is not preserved.
Oedema is present.	Oedema is absent.
Enlarged fatty liver.	No fatty liver.
Ribs are not very prominent.	Ribs become very prominent.
Lethargic.	Alert and irritable.
Muscle wasting mild or absent.	Severe muscle wasting.
Poor appetite.	Voracious feeder.
The person suffering from Kwashiorkor needs adequate amounts of proteins.	The person suffering from Marasmus needs adequate amount of protein, fats and carbohydrates.

MD

Marasmus

Prominent bones

Decrease in subcutaneous fat

Loose skin

Kwashiorkor vs Marasmus

Micronutrients and their deficiencies



- Certain other nutrients are also important for the body
- Called micro-nutrients because they are needed in much more quantities than macro-nutrients
- Include vitamins and minerals
- We need to remember certain names of micronutrients and diseases caused by them : no option but to mug this up

Micronutrient Diseases



Vitamin	Chemical Name	Deficiency Diseases
Fat soluble Vitamins		
A	Retinol, Retinal, Retinoic acid	Night-blindness and keratomalacia
D	Ergocalciferol (D ₂), Cholecalciferol (D ₃)	Rickets and Osteomalacia
E	Tocopherol	Mild hemolytic anemia in newborn infants
K	Phylloquinone (K ₁), Menaquinones (K ₂)	Bleeding diathesis
Water soluble vitamins		
B ₁	Thiamine	Beriberi
B ₂	Riboflavin	Ariboflavinosis
B ₃	Niacin, Niacinamide	Pellagra
B ₅	Pantothenic acid	Paresthesia
B ₆	Pyridoxine, Pyridoxamine, Pyridoxal	Anemia peripheral neuropathy
B ₇	Biotin	Dermatitis
B ₉	Folic acid, Folinic acid	Neural tube defects
B ₁₂	Cyanacobalamine	Megaloblastic anemia
C	Ascorbic acid	Scurvy

Diseases Caused by various Pathogens



- Question : Differentiate between Patho-gen and Anti-gen
Hint : word association will help
- Diseases :
- Viral
- Bacterial
- Protozoan
- Fungi
- Role of Vectors : Question : what is the meaning of vector?

Types of Transmissions



- Feco – oral
- Droplet
- Blood / serum / sexual
- Zoonotic
- Wound breaks



Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Influenza	RNA virus	Respiratory passages: epithelial lining of trachea and bronchi.	Droplet Infection	AVAILABLE : constantly developed due to mutations
Common cold	rhino-virus (RNA Virus)	Respiratory Passage	Think? –do-	AVAILABLE : Flu shots
Small pox	RNA virus	Skin (visible) and Respiratory pathway	-do-	First ever vaccine : Edward Jenner, Live attenuated, now not needed.
Chickenpox	Virus	Blistering Skin rash	Air-borne droplets	Living attenuated virus
Mumps	Virus (RNA virus)	Respiratory Passage	Droplet	LAV
Measles	RNA virus	Lungs, then spread to skin	-do-	-do-
Rubella (german measles)	Virus	Respiratory passage and lymph nodes	Droplet	MMR vaccine for all 3

Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Polio	Rna virus	Nervous system : leads to paralysis	Feces and droplets : Role of contamination of water and hygiene	Oral Polio Vaccine (OPV) IPV : injectable
Yellow Fever	RNA virus	Mild : fever, later fatal heart liver and kidney conditions	Vector based spread : aedes mosquito	LAV
AIDS	RetroVirus (RNA)	Human Immunodeficiency : destroys immune system	Sexual Intercourse, Blood, Vertical Transmission	NONE
Ebola	Ebola Virus (RNA)	Chills, Dehydration, fever. Ultimately : Fatal	Zoonotic Disease and later through contact, even with dead	NONE
Zika	Virus (RNA)	Mild illness, dengue like symptoms	Vector Bite : aedes, sometimes sexual	NONE
Rotavirus	RNA	Diarrhea : GI tract	Think?	Yes, recently developed
Hepatitis : Important, next slide	Word Breakdown			

DIFFERENCE BETWEEN IPV AND OPV

IPV	OPV
Killed formolised virus	Live attenuated virus
Given IM/SC	Given orally
Induces circulating antibody; no local immunity	Both humoral and intestinal immunity
Prevents paralysis; does not prevent reinfection by wild polio viruses	Prevents paralysis and intestinal reinfection
Not useful in epidemics	Effective in controlling epidemics
Content is 10,000 times more than OPV; Costlier	Cheaper
Does not require stringent conditions during storage and transportation	Requires to be stored and transported at sub-zero temperature, unless stabilised

Type of Hepatitis

	A	B	C	D	E
Source of virus	feces	blood/ blood-derived body fluids	blood/ blood-derived body fluids	blood/ blood-derived body fluids	feces
Route of transmission	fecal-oral	percutaneous permucosal	percutaneous permucosal	percutaneous permucosal	fecal-oral
Chronic infection	no	yes	yes	yes	no
Prevention	pre/post- exposure immunization	pre/post- exposure immunization	blood donor screening; risk behavior modification	pre/post- exposure immunization; risk behavior modification	ensure safe drinking water

Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Japanese Encephalitis(JE)	ssRNA	Brain swelling : severe headache. Fatal in children	Culex	Available, Part of National Schedule
KFD	ssRNA	High grade fever and chills	Tick, but no human to human	yes

Viral Diseases in News



- **Zombie Virus** : Zombie virus is virus emerged due to thawing of permafrost as global temperature is rising. The oldest virus is 48,500 years old, called Pandoravirus yedoma. Pandoravirus was discovered below the bottom of a lake in Yukechi Alas in Yakutia, Russia.
- **Marburg Virus Diseases** : Ghana first case. It's a highly infectious viral haemorrhagic fever with a fatality ratio of up to 88 percent. Fruit bats of Pteropodidae Family are key carriers of the disease. Human to h
- **African Swine Flu** : viral disease affecting pigs, not humans.
- **Tomato Flu**
- **Lumpy Skin Disease** : Lumpiprovac-ind developed by ICAR, Live Attenuated Vaccine, Vector Borne Diseases in Cattle, Not Zoonotic

Viral Diseases in News



- White Spot Syndrome Virus : DNA virus, high level of mortality in shrimp
- Porcine virus : Death due to Xenotransplantation (pigs)
- Canine Distemper Virus : Vaccine developed for Asiatic lion.
- Acute Encephalitis Syndrome : Fever and change of Mental State

Covid 19 Mutation



- Covid Virus has been mutating and new variants are forming.
- We know what mutations are : lead to changes in gene of virus.
- New Variant : Functional properties different from original virus.
- When new variant gets established in population, it is referred to new strain. A strain behaves differently from parent virus.
- All strains are variants but not all variants are strains.

Covid 19 Mutation



- Variant classifications :
- Variant of Concern : Increased transmissibility, increased fatality, significant decrease in vaccine or therapy effectiveness. Alpha, Beta, Gamma, Delta, Omicron
- Variant of Interest : Disease severity, immune escape, transmissibility, diagnostic escape, and increased volume of community transmission. Eta, Kappa, Lambda
- Variant Under Monitoring : Genetic changes are there but epidemiological impact is unclear.

Covid 19 Mutation



- 2 important things : Receptor that covid binds to : ACE 2 : Angiotensin Converting Enzyme 2 : Mutation here leads to change in covid infection ability
- Place where antibody binds to covid : Mutation leads to antibody ability
- Site 486 : Controls both things at once
- XBB, XBB.1 : Change in nucleotide : Less bind, More escape.
- XBB 1.5 : 2 mutations simultaneously, Decrease in bind is much less, but antibody escape is much higher. Very dangerous: in china
- Related term : Hybrid Immunity (i'll explain)

Influenza Virus



- Cause respiratory infection.
- 4 types of seasonal viruses : A,B,C,D. A and B cause seasonal epidemics.
- Influenza A : classified on basis of proteins on surface of virus. Hemagglutinin and Neuraminidase. H1N1 and H3N2 are currently circulating. These are swine influenza.
- H9N2, H5N1 are avian influenza.

Bacterial Diseases



Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Diphtheria	Corynebacterium diphtheria	Respiratory Tract	Use logic	Toxoid vaccine
Tuberculosis (TB)	Mycobacterium tuberculosis	Mainly lungs	Droplet infection, Drinking milk from infected cattle.	BCG living attenuated bacteria. Antibiotics e.g. streptomycin.
Pertusis (whooping cough)	Bordetella Pertusis	Respiratory Tract	Droplet	Killed Vaccine
Gonorrhea	N. Gonorrhea	Reproductive Tract, blindness in infants	STD, vertical	No vaccine, but treatable through AMR
Syphilis	T. Pallidum	Reproductive Tract, later other organs including brain	STD	-do-
Tetanus	Clostridium Tetani	Blood : Toxin : later lock jaw	Wound Breaks	Toxoid : done already
Cholera	Vibrio Cholera	GI tract : diarrhoea,	Oro-faecal : water / objects / vector	Antibiotics

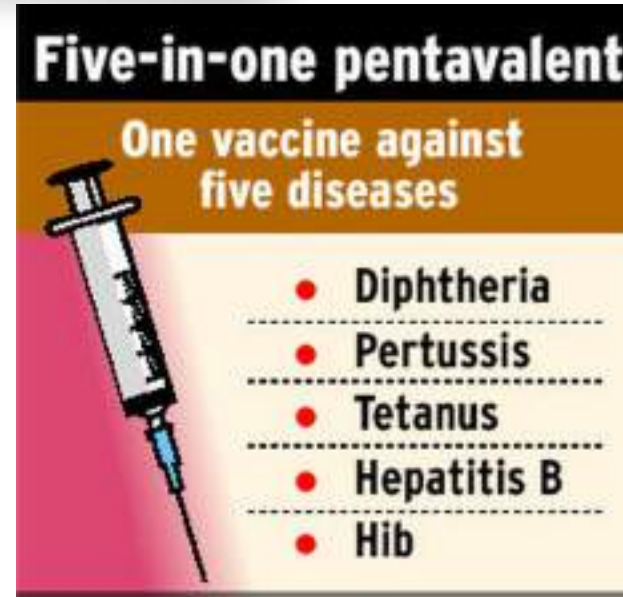


Disease Name	Causative Organism	Symptoms / Organs Affected	Transmission	Vaccination
Typhoid	Salmonella Typhi	GI tract then other organs	Oro Faecal : water / objects / vector	Ty21a and Vi capsular polysaccharide vaccine
Bacterial Dysentery	Shigella	GI tract	-do-	None, but treatable
Haemophilus Influenza type b (HiB) disease	H. Influenzae	Lungs : pneumonia	Respiratory droplets	YES, in national schedule

Pentavalent Vaccine



- Again name breakdown : Penta means 5



PROTOZOAN DISEASES



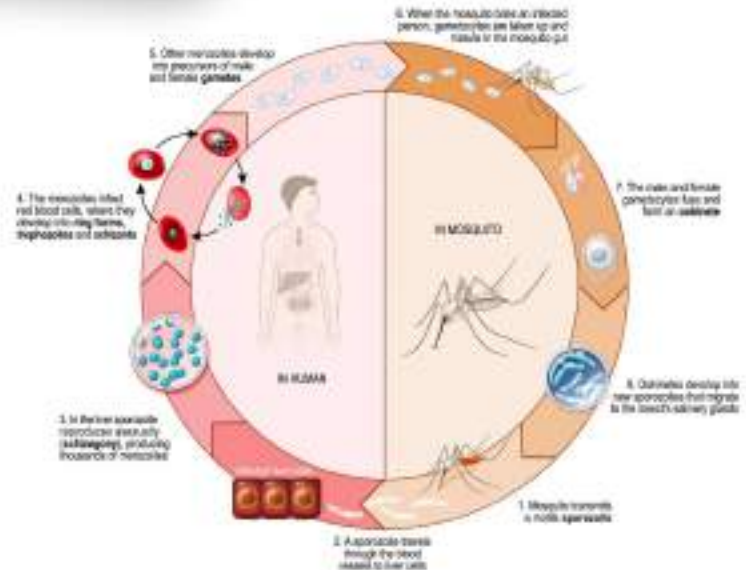
- Filariasis : caused by roundworms
- Vector : Culex, Aedes and Anopheles mosquitoes
- Swelling of lymph nodes : also called elephantiasis
- Lymph?



Malaria



- Caused by plasmodium
- Vector : anopheles
- 4 types : vivax / malaria / falciparum / ovale
- Most severe : falciparum, vivax
- Both found in India



Malaria



- World Malaria Report : Released by WHO
- Malaria Vaccine
- WHO has recommended broad use of the RTS,S/AS01 (RTS,S) malaria vaccine among children living in regions with moderate to high *P. falciparum* malaria transmission.
- Known by its brand name **Mosquirix**, this recombinant protein vaccine has been shown to significantly reduce malaria, and deadly severe malaria among young children, and also reduced the overall hospital admissions.
- India related findings
 - o India accounted for 83% of cases in the WHO South-East Asia Region.

Initiatives for Malaria



- **Malaria Elimination Research Alliance (MERA)- India:** Its purpose is to identify, articulate, prioritise and respond to the research needs of the country to eliminate malaria from India by 2030.
- **E-2025 initiative:** Under this, WHO has identified group of 25 countries with the potential to eradicate malaria by 2025.
- **High Burden to High Impact (HBHI) initiative:** In 2019,WHO has initiated the HBHI initiative in 11 high malaria burden countries, including **India** (West Bengal and Jharkhand, Chhattisgarh and Madhya Pradesh).
- **National Strategic Plan for Malaria Elimination (2017-22)**
- **Ministerial Declaration on Accelerating and Sustaining Malaria Elimination in South-East Asia Region (SEAR)**

Kala Azar



- Aka Leishmaniasis
- Vector : Sandfly
- Endemic to Indian Subcontinent
- Hepato-Splenomegaly is the most common feature

Fungi



- **Diseases caused by Fungi-**
- Ringworm,
- Athlete's foot
- Dhobie itch
- Madura foot

Common Vectors



- **Aedes** – Chikungunya, Dengue fever, Lymphatic filariasis, Rift Valley fever, Yellow fever, Zika
- **Anopheles** – Malaria, Lymphatic filariasis.
- **Culex** – Japanese encephalitis, Lymphatic filariasis, West Nile fever

Remember

- Sandfly?
- Tick?

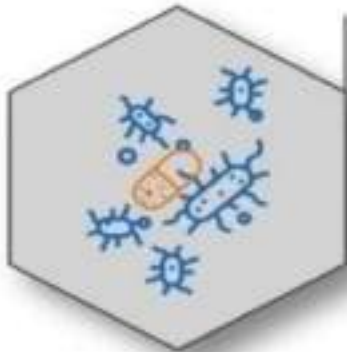
Antimicrobial Resistance



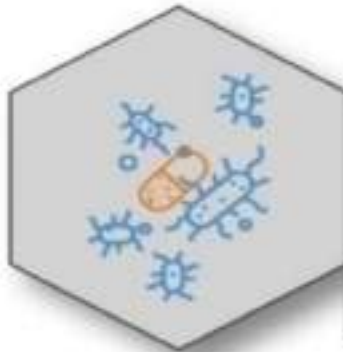
- According to a study by Lancet, Antimicrobial resistance (AMR) is now the leading cause of death across the world.
- **AMR** is the ability of a microorganism (like bacteria, viruses, and some parasites) to stop an antimicrobial (such as antibiotics, antivirals and antimalarials) from working against it.
- While resistance levels continue to grow, **antibiotic development has not.**
 - While 63 new antibiotics were approved between 1980 and 2000, **only 15 new drugs have been approved between 2000 and 2018.**

Out of the seven deadliest drug-resistant bacteria, **vaccines are only available against bacteria causing pneumonia and tuberculosis.**

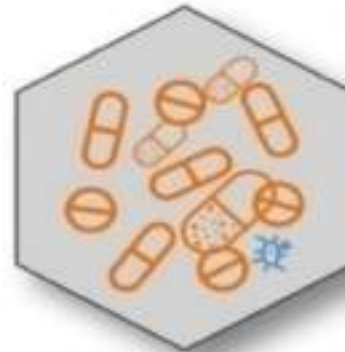
HOW ANTIBIOTIC RESISTANCE HAPPENS



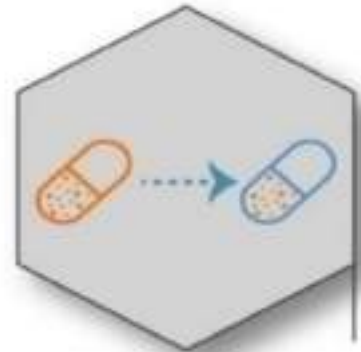
Our body is **home to countless microbes**. Some may be resistant to antibiotics



Antibiotics kill the bacteria
Causing the infections as well as the good bacteria



The **antibiotic-resistant bacteria** are now able to **grow and take over**

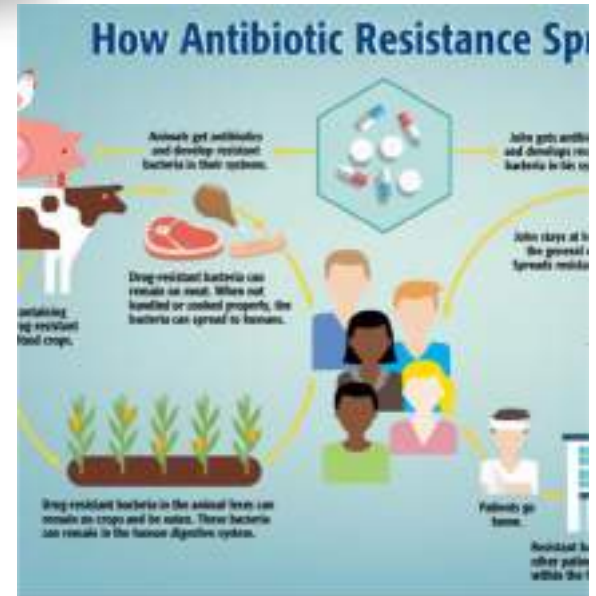


Some bacteria may **give** their antibiotic resistance to other **bacteria**

Causes of AMR



- **Poor hygiene** and infection prevention and control.
- **Overuse of antimicrobials**
- **Over prescription** by doctors and **Non- prescription purchase** in many countries.
- If a person **does not complete a course of antimicrobial drugs**, some microbes may survive and develop resistance to the drug.
- Practice of **adding antibiotics to agricultural feed** promotes drug resistance.
- **Environment** plays a significant role wherein **waste from farms, factories, community (such as mass bathing) and healthcare settings** can contribute to AMR through environmental routes.



Indian Initiatives against AMR



- **National programme on AMR containment**
- **National Action Plan on Antimicrobial Resistance (NAP-AMR)**
- **Red Line campaign**
- **Integrated One Health Surveillance Network for Antimicrobial Resistance**
- **FSSAI has set certain guidelines limiting the antibiotics in food products**
- **India's medical societies adopted the Chennai Declaration**
- **separate Schedule H-1**
- **Use bacteriophages, or 'phages', to kill bacterial cells.**

One health approach



- **FAO, the World Organisation for Animal Health (OIE), WHO and UNEP (quadrupartite)**
- These 4 organizations are **working together to mainstream One Health, they have recently launched the One Health Joint Plan of Action (2022-2026)**
- One Health is an **integrated, unifying approach** that aims to sustainably balance and **optimise the health of people, animals and ecosystems.**
- **Benefits of One Health:**
- **Promote coherence** in action across sectors and areas of expertise in health sector.
- Address the full spectrum of disease control - **from disease prevention to detection, preparedness, response, and management.**

One Health

Human health and animal health are interdependent.
At the same time, both depend on the environment.



One Health Approach : India's initiatives



- One of the priorities of our G20 Presidency
- One Health Support Unit (OHSU) by Department of Animal Husbandry and Dairying, to develop a national One Health Framework.
- Resource allocation and policy ecosystem of zoonotic diseases.
- National Mission on Biodiversity and Human Well-being: Launched by Prime Minister's Science, Technology and Innovation Advisory Council (PM-STIAC).
- One Health' project: By Department of Biotechnology to carry out surveillance of zoonotic as well as transboundary pathogens in India.

One Health Approach : Other global initiatives



- Pilanesberg Resolution : governments and donors
- One world One Health : wildlife conservation society with 12 Manhattan principles.
- National Framework for one health : by FAO

Traditional Medicine



- WHO Director-General laid the foundation stone for world's first and only Global Centre for Traditional Medicine (GCTM) at Jamnagar in Gujarat.
- Traditional medicine (TM) is the sum of the knowledge, skill, and practices based on the theories, beliefs, and experiences indigenous to different cultures, for treatment of physical and mental illness.
- Complementary medicine (CM) or alternative medicine refers to healthcare practices that are not part of that country's own tradition or conventional medicine and are not fully integrated into the dominant health-care system.

Traditional Medicine : Indian Initiatives



- Indian Initiatives :
- National AYUSH Mission (NAM)
- Champion Services Sector Scheme for Medical value Travel
- Ayush Export Promotion Council (AEPC)
- The AYUR SWASTHYA YOJANA
- AYUSH Information Cell

Regulations for Ayurveda Aahara



- FSSAI and MoAYUSH
- Marketing of quality food products to help in expanding the international market for for Make in India
- strict Food Safety and Standards (Ayurveda Aahara) Regulations, 2022 rules.
- special logo has been created for “Ayurveda Aahara”
- ‘Ayurveda Aahara’ will not include Ayurvedic drugs or proprietary Ayurvedic medicines and medicinal products, cosmetics, narcotic or psychotropic substances and herbs.

Viral Diseases in News



- **Monkeypox** : PHEIC, renamed as mpox, zoonotic disease, belongs to orthopoxvirus which also includes variola virus (smallpox), cowpox.
- Animal to human transmission : direct contact with blood, body fluids, lesions.
- Human to Human : Sexual, blood borne.
- HPV : Cervavac', the country's first quadrivalent human papillomavirus vaccine (qHPV) approved for market authorization. **Manufactured by**: Serum Institute of India.
- HPV is a sexually transmitted virus that may cause cervical cancer. It can be eliminated.

Neglected Tropical Diseases



- Communicable diseases in tropical and subtropical countries.
- Populations living in poverty, without adequate sanitation, close contact with infectious vectors and domestic animals.
- India : world's largest burden.
- India has eliminated several NTDs including Guinea worm, Trachoma, Yaws.
- Eradication : only smallpox.

Elimination and Eradication and Control



- Disease control : No longer a public health problem.
- Elimination : Temporary phenomenon, interruption of transmission. Newer cases have stopped but continuous efforts are needed.
- Eradication : permanent, complete eradication of the agent.

Test of 13 Questions!

Test.

Cut off : 10/13





Question 1 (Previous Year Question 2016)

Q. 'Mission Indradhanush' launched by the Government of India pertains to:

- a) Immunization of children and pregnant women
- b) Construction of smart cities across the country
- c) India's own search for Earth-like planets in outer space
- d) New Educational policy



Question 2 (Previous Year Question 2017)

Q. Consider the following statements:

1. In tropical regions, Zika virus disease is transmitted by the same mosquito that transmits dengue.
2. Sexual transmission of Zika virus disease is possible.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2



Question 3 (Previous Year Question 2019)

Q. Which of the following are the reasons for the occurrence of multi-drug resistance in microbial pathogens in India?

1. Genetic predisposition of some people
2. Taking incorrect doses of antibiotics to cure diseases
3. Using antibiotics in livestock farming
4. Multiple chronic diseases in some people

Select the correct answer using the code given below:

- a) 1 and 2
- b) 2 and 3 only
- c) 1, 3 and 4
- d) 2, 3 and 4



Question 4 (Previous Year Question 2019)

Q.What is Cas9 protein that is often mentioned in the news?

- a) A molecular scissors used in targeted gene editing
- b) A biosensor used in the accurate detection of pathogens in patients
- c) A gene that makes plants pest-resistant
- d) A herbicidal substance synthesized in genetically modified crops



Question 5 (Previous Year Question 2019)

Q. Which one of the following statements is not correct?

- a) Hepatitis B virus is transmitted much like HIV.
- b) Hepatitis B, unlike Hepatitis C, does not have a vaccine.
- c) Globally, the number of people infected with Hepatitis B and C viruses are several times more than those infected with HIV.
- d) Some of those infected with Hepatitis B and C viruses do not show the symptoms for many years.



Question 6 (Previous Year Question 2020)

Q. Consider the following statements:

1. Genetic changes can be introduced in the cells that produce eggs or sperms of a prospective parent.
2. A person's genome can be edited before birth at the early embryonic stage.
3. Human /induced pluripotent stem cells can be injected into the embryo of a pig.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 2 only
- d) 1, 2 and 3



Question 7 (Previous Year Question 2021)

Q.The term 'ACE2' is talked about in the context of

- a) genes introduced in the genetically modified plants
- b) development of India's own satellite navigation system
- c) radio collars for wildlife tracking
- d) spread of viral diseases



Question 8 (Previous Year Question 2022)

Q. Consider the following statements:

1. Biofilms can form on medical implants within human tissues.
2. Biofilms can form on food and food processing surfaces.
3. Biofilms can exhibit antibiotic resistance.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3



Question 9 (Previous Year Question 2022)

Q. Consider the following statements in respect of probiotics :

1. Probiotics are made of both bacteria and yeast.
2. The organisms in probiotics are found in foods we ingest but they do not naturally occur in our gut.
3. Probiotics help in the digestion of milk sugars.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) 1 and 3
- (d) 2 and 3



Question 10 (Previous Year Question 2013)

Q. Which of the following diseases can be transmitted from one person to another through tattooing?

1. Chikungunya
2. Hepatitis B
3. HIV-AIDS

Select the correct answer using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only
- d) 1, 2 and 3



Question 11 (Previous Year Question 2014)

Q. Consider the following diseases

1. Diphtheria
2. Chickenpox
3. Smallpox

Which of the above diseases has/have been eradicated in India?

- a) 1 and 2 only
- b) 3 only
- c) 1, 2 and 3
- d) None



Question 12 (Previous Year Question 2014)

Q. Consider the following pairs :

Vitamin Deficiency disease

1. Vitamin C : Scurvy
2. Vitamin D : Rickets
3. Vitamin E : Night blindness

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 3 only
- c) 1, 2 and 3
- d) None



Question 13 (Previous Year Question 2015)

Q. H1N1 virus is sometimes mentioned in the news with reference to which one of the following diseases?

- a) AIDS
- b) Bird flu
- c) Dengue
- d) Swine flu

Congratulations on giving the test!

Let's Discuss The Answers.





Question 1 (Previous Year Question 2016)

Q. 'Mission Indradhanush' launched by the Government of India pertains to:

- a) **Immunization of children and pregnant women**
- b) Construction of smart cities across the country
- c) India's own search for Earth-like planets in outer space
- d) New Educational policy



Answer: **A**

- Mission Indradhanush is a health mission of the government of India. It was launched by Union Health Minister J. P. Nadda on 25 December 2014.
- It aims to immunize all children under the age of 2 years, as well as all pregnant women, against seven vaccine-preventable diseases.
- The diseases being targeted are diphtheria, whooping cough, tetanus, poliomyelitis, tuberculosis, measles and Hepatitis B.
- In addition to these, vaccines for Japanese encephalitis and Haemophilus influenzae type B are also being provided in selected states.



Question 2 (Previous Year Question 2017)

Q. Consider the following statements:

1. In tropical regions, Zika virus disease is transmitted by the same mosquito that transmits dengue.
2. Sexual transmission of Zika virus disease is possible.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 only
- c) Both 1 and 2**
- d) Neither 1 nor 2



Answer: C

- Zika virus (ZIKV) is a member of the virus family Flaviviridae.
- It is spread by daytime-active Aedes mosquitoes, such as *A. aegypti* and *A. albopictus*.
- Its name comes from the Zika Forest of Uganda, where the virus was first isolated in 1947.
- Zika virus is related to the dengue, yellow fever, Japanese encephalitis, and West Nile viruses.
- Zika can be transmitted from men and women to their sexual partners



Question 3 (Previous Year Question 2019)

Q. Which of the following are the reasons for the occurrence of multi-drug resistance in microbial pathogens in India?

1. Genetic predisposition of some people
2. Taking incorrect doses of antibiotics to cure diseases
3. Using antibiotics in livestock farming
4. Multiple chronic diseases in some people

Select the correct answer using the code given below:

- a) 1 and 2
- b) 2 and 3 only**
- c) 1, 3 and 4
- d) 2, 3 and 4



Answer: C



Question 4 (Previous Year Question 2019)

Q.What is Cas9 protein that is often mentioned in the news?

- a) **A molecular scissors used in targeted gene editing**
- b) A biosensor used in the accurate detection of pathogens in patients
- c) A gene that makes plants pest-resistant
- d) A herbicidal substance synthesized in genetically modified crops



Answer: **A**

- Cas9 (CRISPR associated protein 9) is a protein which plays a vital role in the immunological defense of certain bacteria against DNA viruses, and which is heavily utilized in genetic engineering applications.
- Cas9 is a bacterial RNA-guided endonuclease that uses base pairing to recognize and cleave target DNAs with complementarity to the guide RNA.
- The programmable sequence specificity of Cas9 has been harnessed for genome editing and gene expression control in many organisms.
- Its main function is to cut DNA and therefore it can alter a cell's genome



Question 5 (Previous Year Question 2019)

Q. Which one of the following statements is not correct?

- a) Hepatitis B virus is transmitted much like HIV.**
- b) Hepatitis B, unlike Hepatitis C, does not have a vaccine.
- c) Globally, the number of people infected with Hepatitis B and C viruses are several times more than those infected with HIV.
- d) Some of those infected with Hepatitis B and C viruses do not show the symptoms for many years.



Answer: **A**

- The hepatitis B vaccine is a safe and effective vaccine that is recommended for all infants at birth and for children up to 18 years.
- The hepatitis B vaccine is also recommended for adults living with diabetes and those at high risk for infection due to their jobs, lifestyle, living situations, or country of birth.
- Since everyone is at some risk, all adults should seriously consider getting the hepatitis B vaccine for a lifetime protection against a preventable chronic liver disease.
- The hepatitis B vaccine is also known as the first “anti-cancer” vaccine because it prevents hepatitis B, the leading cause of liver cancer worldwide. Thus option b is the answer.



Question 6 (Previous Year Question 2020)

Q. Consider the following statements:

1. Genetic changes can be introduced in the cells that produce eggs or sperms of a prospective parent.
2. A person's genome can be edited before birth at the early embryonic stage.
3. Human /induced pluripotent stem cells can be injected into the embryo of a pig.

Which of the statements given above is/are correct?

- a) 1 only
- b) 2 and 3 only
- c) 2 only
- d) 1, 2 and 3**



Answer: D

- Germline gene therapy is the replacement of genes in egg or sperm cells with which an offspring inherits a new trait. It allows for the correction of disease- causing gene variants that are certain to be passed down from generation to generation. Hence,
- statement 1 is correct.
- CRISPR (Clustered Regularly Interspaced Short Palindromic Repeats) technology is employed to modify human embryos before they are transferred into women's uterus.
- Recently, researchers had successfully created the world's first genetically- edited baby. By using CRISPR technology, the genomes of embryos were edited to disable a gene, CCR5, which allows HIV to infect cells. Hence, statement 2 is correct.
- Due to certain anatomical and physiological features shared with humans, the pig is considered an important animal model of human diseases with unique advantages in surgery and xenotransplantation studies. Hence, statement 3 is correct.
- Therefore, option (d) is the correct answer.



Question 7 (Previous Year Question 2021)

Q.The term 'ACE2' is talked about in the context of

- a) genes introduced in the genetically modified plants
- b) development of India's own satellite navigation system
- c) radio collars for wildlife tracking
- d) spread of viral diseases**



Answer: **D**

- 'ACE2' is a proteinaceous enzyme on the surface of many cell types that stands for Angiotensin Convertase Enzyme-2. It is an enzyme that generates small proteins by cutting up the larger protein angiotensinogen that then goes on to regulate functions in the cell.
- Using the spike-like protein on its surface, the SARS-CoV-2 virus binds to ACE2 like a key being inserted into a lock – prior to entry and infection of cells. Hence, ACE2 acts as a cellular doorway or a receptor for the virus that causes COVID-19.
- Therefore, option (d) is the correct answer.



Question 8 (Previous Year Question 2022)

Q. Consider the following statements:

1. Biofilms can form on medical implants within human tissues.
2. Biofilms can form on food and food processing surfaces.
3. Biofilms can exhibit antibiotic resistance.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3**



Answer: **D**

- Biofilms are a collective of one or more types of microorganisms that can grow on many different surfaces.
- Microorganisms that form biofilms include bacteria, fungi and protists.
- One common example of a biofilm dental plaque, a slimy build-up of bacteria that forms on the surfaces of teeth.
- They can grow on plant tissues and animal tissues, on food and food processing surfaces and on implanted medical devices such as catheters and pacemakers.
- For microorganisms, living as a part of a biofilm comes with certain advantages. They are seen to exhibit antibiotic resistance.



Question 9 (Previous Year Question 2022)

Q. Consider the following statements in respect of probiotics :

1. Probiotics are made of both bacteria and yeast.
2. The organisms in probiotics are found in foods we ingest but they do not naturally occur in our gut.
3. Probiotics help in the digestion of milk sugars.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 only
- (c) 1 and 3**
- (d) 2 and 3



Answer: C

- Probiotics are a combination of live beneficial bacteria and/or yeasts that naturally live in your body.
- Probiotics, live bacteria in our gut, can also be bought as supplements. They're renowned for their ability to boost nutrient absorption, fight inflammation, and enhance digestive health
- Lactobacillus genus of bacteria produces lactase, the enzyme that breaks down lactose, or milk sugar



Question 10 (Previous Year Question 2013)

Q. Which of the following diseases can be transmitted from one person to another through tattooing?

1. Chikungunya
2. Hepatitis B
3. HIV-AIDS

Select the correct answer using the codes given below.

- a) 1 only
- b) 1 and 2 only
- c) 2 and 3 only**
- d) 1, 2 and 3



Answer: C

- Hepatitis B is an infection of the liver. It can cause scarring of the organ, liver failure, and cancer.
- It can be fatal if it isn't treated.
- It's spread when people come in contact with the blood, open sores, or body fluids of someone who has the hepatitis B virus.
- HIV stands for human immunodeficiency virus.
- It harms your immune system by destroying the white blood cells that fight infection.
- This puts you at risk for serious infections and certain cancers. AIDS stands for Acquired Immunodeficiency Syndrome.
- It is the final stage of infection with HIV. Not everyone with HIV develops AIDS.
- HIV most often spreads through unprotected sex with an infected person.
- It may also spread by sharing drug needles or through contact with the blood of an infected person. Women can give it to their babies during pregnancy or childbirth.



Question 11 (Previous Year Question 2014)

Q. Consider the following diseases

1. Diphtheria
2. Chickenpox
3. Smallpox

Which of the above diseases has/have been eradicated in India?

- a) 1 and 2 only
- b) 3 only**
- c) 1, 2 and 3
- d) None



Answer: **B**

- Government introduced Pentavalent vaccine which also fights Diphtheria. It means Diphtheria is also prevalent.
- Chickenpox is still prevalent
- Smallpox has been eradicated from India



Question 12 (Previous Year Question 2014)

Q. Consider the following pairs :

Vitamin Deficiency disease

1. Vitamin C Scurvy
2. Vitamin D Rickets
3. Vitamin E Night blindness

Which of the pairs given above is/are correctly matched?

- a) 1 and 2 only
- b) 3 only
- c) 1, 2 and 3
- d) None



Answer: **A**

- | | |
|--------------|---|
| 1. Vitamin C | Scurvy is right. |
| 2. Vitamin D | Rickets is right |
| 3. Vitamin E | Night blindness is wrong. Vitamin A causes Night blindness. |



Question 13 (Previous Year Question 2015)

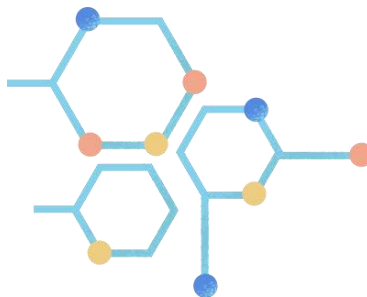
Q. H1N1 virus is sometimes mentioned in the news with reference to which one of the following diseases?

- a) AIDS
- b) Bird flu
- c) Dengue
- d) Swine flu**



Answer: **D**

- Influenza (H1N1) virus is a subtype of influenza A virus that was the most common cause of human influenza (flu) in 2009 and is associated with the 1918 outbreak known as the Spanish Flu.
- It is an orthomyxovirus that contains the glycoproteins haemagglutinin and neuraminidase. For this reason, they are described as H1N1, H1N2 etc. depending on the type of H or N antigens they express with metabolic synergy.
- Haemagglutinin causes red blood cells to clump together and binds the virus to the infected cell.
- Neuraminidase is a type of glycoside hydrolase enzyme which helps to move the virus particles through the infected cell and assist in budding from the host cells
- In June 2009, the World Health Organization (WHO) declared the new strain of swine-origin H1N1 as a pandemic.



UPSC Prelims

Science & Tech

Class No. 7

Health and Nanotechnology

Tuberculosis



- Recently, World Health Organization (WHO) released new guidance to support National Strategic Planning (NSP) for the tuberculosis (TB).
- NSP for TB : guides how to comprehensively address TB epidemic through goals and strategies.
- Specific, Measurable, Attainable, Realistic, Time bound criteria.
- Update to 2015 document.

India's efforts



- National Strategic Plan 2017 – 2025 : Eliminate TB by 2025.
- NTEP : Earlier this program called RNTCP.
- Nikshay Portal : National TB information system, National data repository
- Nikshay Poshan Yojana : Provides DBT to all TB patients towards nutritional support, by MoHFW, financed by GoI and World Bank.
- TB mukt bharat : PM's initiative by 2025
- Patho detect kit : Indigenous TB kit Indian lab

Other Efforts



- Moscow Declaration, 2017.
- WHO end TB strategy : incidence reduction by 80%, TB deaths by 90%, eliminate catastrophic costs.
- Find all, Treat all, End TB : Partnership between WHO, Stop TB partnership and Global Fund to end TB.
- BPaL : 6 month oral regimen for TB using 3 drugs, Bedaquilline, Pretomanid, Linezolid
- Reduced time of treatment : 18 months to 6 months. Made by TB alliance, a Not for profit organisation.



Tuberculosis

TB is caused by bacteria (*Mycobacterium tuberculosis*) and it most often affects the lungs

Types of TB



Pulmonary TB
(affects lungs)



Extra Pulmonary TB
(affects both lungs and other organs)



For prevention of TB

BCG (Bacille Calmette-Guérin) vaccine is used among children at the time of birth



Drug resistant TB



Multi- Drug Resistance TB (MDR- TB)

Does **not** respond to at least **isoniazid and rifampicin** (2 of the most powerful anti-TB drugs)



Extensively Drug Resistant TB (XDR- TB)

Does **not** respond to at least four of the core anti-TB drugs.



Totally drug-resistant tuberculosis (TDR-TB)

TB which is resistant to all the anti- TB drugs.

Fats



- We have to understand :
- Transfats
- Saturated vs Unsaturated
- Good Lipoproteins vs Bad Lipoproteins.

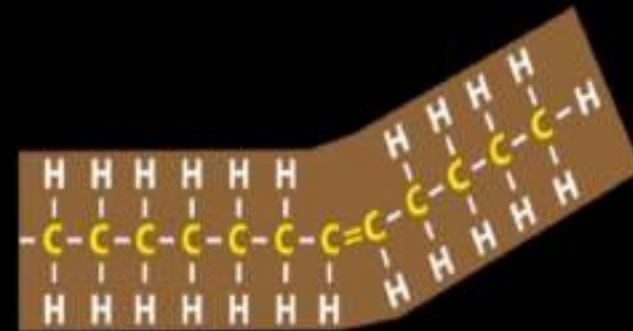
Fats



Saturated Fats	Unsaturated	Transfats
Solid @ room temperature	Liquid @ room temperature	Solid @ room temperature.
Even Natural are bad for you	Plant oils : good for you	artificially formed by hydrogenation of unsaturated fats. Some natural sources.
No double bonds	Have double bonds	Have very few double bonds, very bad for you
Animal Origins	MUFA, PUFA Omega 3 fatty acids, Omega 6 fatty acids.	Partially hydrogenated oils. This is done to increase the shelf life.



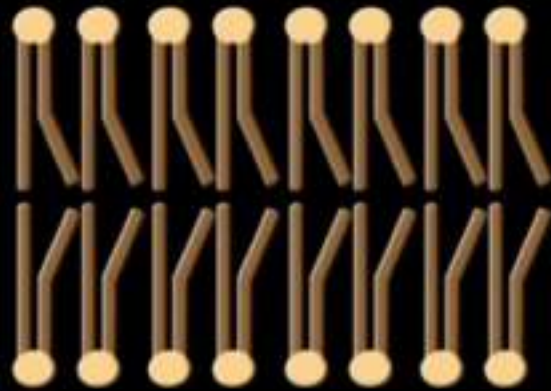
Saturated



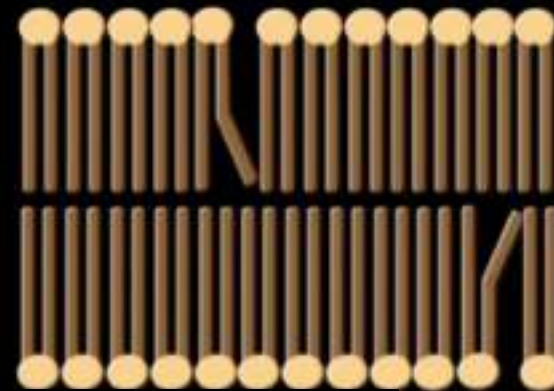
Unsaturated



Optimal unsaturated/saturated ratio

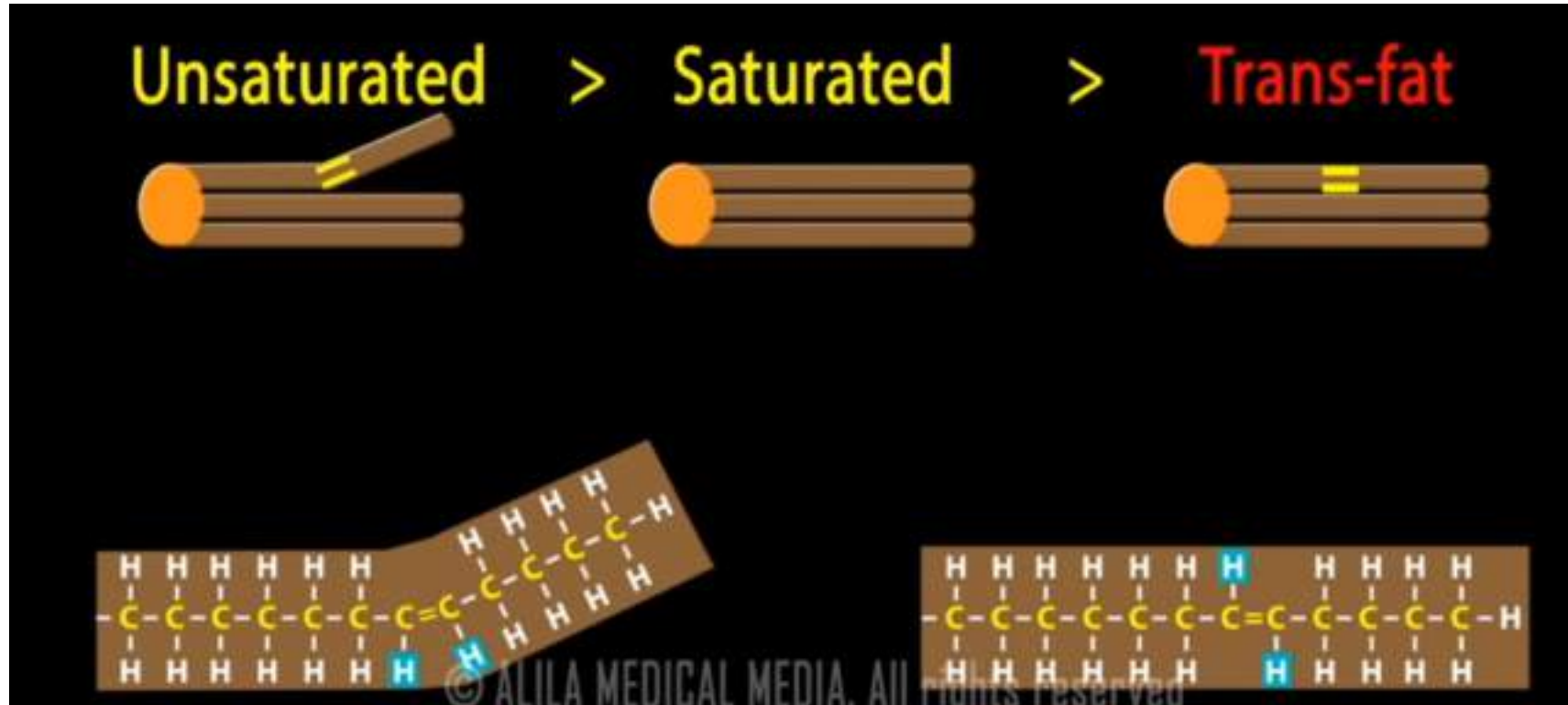


Too much saturated fat



Unsaturated > Saturated







Trans-Fats



- Transfat, or trans-fattyacids, are unsaturated fatty acids that come from either natural or industrial sources:
- Naturally occurring trans-fat come from ruminants (cows and sheep).
- Industrially produced trans-fat are found in hardened vegetable fats such as margarine and are often present in snack foods, baked goods and fried foods.
- Industrially produced trans-fats are formed in a process that adds hydrogen to vegetable oil converting the liquid into a solid, resulting in “partially hydrogenated” oil (PHO).
- Manufacturers often use them as they have a longer shelf life and are cheaper than other fats.
- Health Impacts: Intake of TFA is associated with increased risk of heart attacks and death from coronary heart disease.
- Approximately 540,000 deaths each year globally and 60,000 in India can be attributed to intake of industrially produced TFA.

What they do to the body

Trans fats (also known as partially hydrogenated oils) are created by adding hydrogen to liquid vegetable oil, giving them longer shelf life and making them more suitable for frying and other uses. They also raise bad (LDL) cholesterol levels while lowering good (HDL cholesterol).

Good cholesterol
High-density lipoproteins (HDL) pick up excess cholesterol and transport it back to the body's liver for processing.

Bad cholesterol
Low density lipoproteins (LDL) transport cholesterol throughout the body. As cholesterol builds up in the walls of the body's arteries, the arteries become narrow and hardened, reducing blood flow and leading to an increased chance of heart attack and stroke.

Cholesterol build-up

Artery

Cholesterol

Heart

Liver

Where it comes from

Trans fats are created by hydrogenation, a chemical process that hardens vegetable oil. The more solid the fat, the more trans fat it contains.

Low trans fat				
				
Vegetable oil	Semi-liquid margarine	Soft margarine	Stick margarine	Shortening

Initiatives around transfats



- WHO : trans fat elimination report called Countdown to 2023.
- Replace action framework : to eliminate industrially produced TFA from food.
- WHO recommendation : Transfat intake should be less than 1% of energy uptake.
- Part of Triple billion target of WHO :
 - 1 billion more people benefitting from universal health coverage
 - 1 billion more people better protected from health emergencies
 - 1 billion more people enjoying better health and well-being.

Indian Initiatives



- Target to make country TFA Free
- Phased target : reduce TFA in all food products containing all oils and fats less than 2% from 1 January 2022.
- Trans fat free logo : Food Contains less than 0.2g per 100g or 100ml and if edible oils or fats contain less than 1g of TFA per 100ml or 100g.
- Multilingual Mass media campaign called heart attack rewind campaign.

Foodborne Diseases



- South-East Asia Region contributes a quarter of the global burden of FBD morbidity and nearly 42% of FBD mortality
- FBD costing India almost 15 billion USD annually

Classification of Foodborne Diseases



- Food-borne infections – caused by consuming foods or liquids contaminated with bacteria, viruses, or parasites.
- These pathogens cause infection by invading and multiplying in the lining of the intestines and/or other tissues
- Food-borne intoxications - caused by consuming foods or beverages contaminated with a toxin.
Sources of toxins: Certain Bacteria, Poisonous chemicals, Natural toxins found in animals, plants, and fungi

Steps taken by FSSAI for Food Safety



- Eat right movement
- Blissful Hygienic Offering to God (BHOG)
- Hygiene Rating Scheme
- Food Safety Compliance System (FoSCoS)

Antioxidants



- Regular intake of fruits and vegetables neutralizes the free radical produced in the body during metabolism

State Food Safety Index



- Released by FSSAI annually: no need to study components

**Food Safety and Standards
Authority of India (FSSAI)**


New Delhi, India

**About:** Established, under administration **Ministry of Health & Family Welfare**, for laying down science based standards for articles of food to ensure safe food for human consumption.

**Concerned Law:** Food Safety and Standards Act, 2006.

**Mandate:**

- ▶ Laying down mechanisms and guidelines for accreditation of certification bodies.
- ▶ Provide scientific advice and technical support to Central Government and State Governments.
- ▶ Collect data regarding food consumption, incidence and prevalence of biological risk, identification of emerging risks and introduction of rapid alert system.
- ▶ Contribute to the development of international technical standards for food, sanitary and phyto-sanitary standards.



Food Fortification



- Consumer Affairs Ministry had issued uniform specifications for fortified rice kernels (FRK) for Grade A and common rice for procurement of fortified rice stocks, wherein 1% of FRK should be blended with normal rice
- Fortified rice: distributed under public distribution system and midday meals
- FSSAI norms: fortified rice contain iron, folic acid and Vitamin B-12. Rice may also be fortified with micronutrients such as Zinc, Vit A, Vit B1, Vit B2, Vit B3 and Vit B6

About Fortification



- Regulation: Under Food Safety Standards (Fortification of Foods) Regulation, 2018.
- Deliberately increasing the content of an essential micronutrient, i.e vitamins and minerals.
- Successful for iodized salt: Iodine-deficient countries has decreased from 54 to 32 since 2003.
- In India, wheat flour and rice are fortified with Iron, Vit B12 and folic acid, Milk and Edible Oil with Vit A and D and Double Fortified Salt with Iodine and Iron.

Standard Operating Procedure



- Issued by Department of food and public distribution to maintain desired quality standards for implementation of **rice fortification**.
- **Rice Specification:** Milled rice in which Fortified Rice Kernels (FRK) will be blended should comply with rice specification as per regulation.
- **Packaging of fortified food:** Fortified food package shall contain: **Fortified with (name of the fortificant)** and **+F logo** on the label.

About Standard Operating Procedure - Cont'd

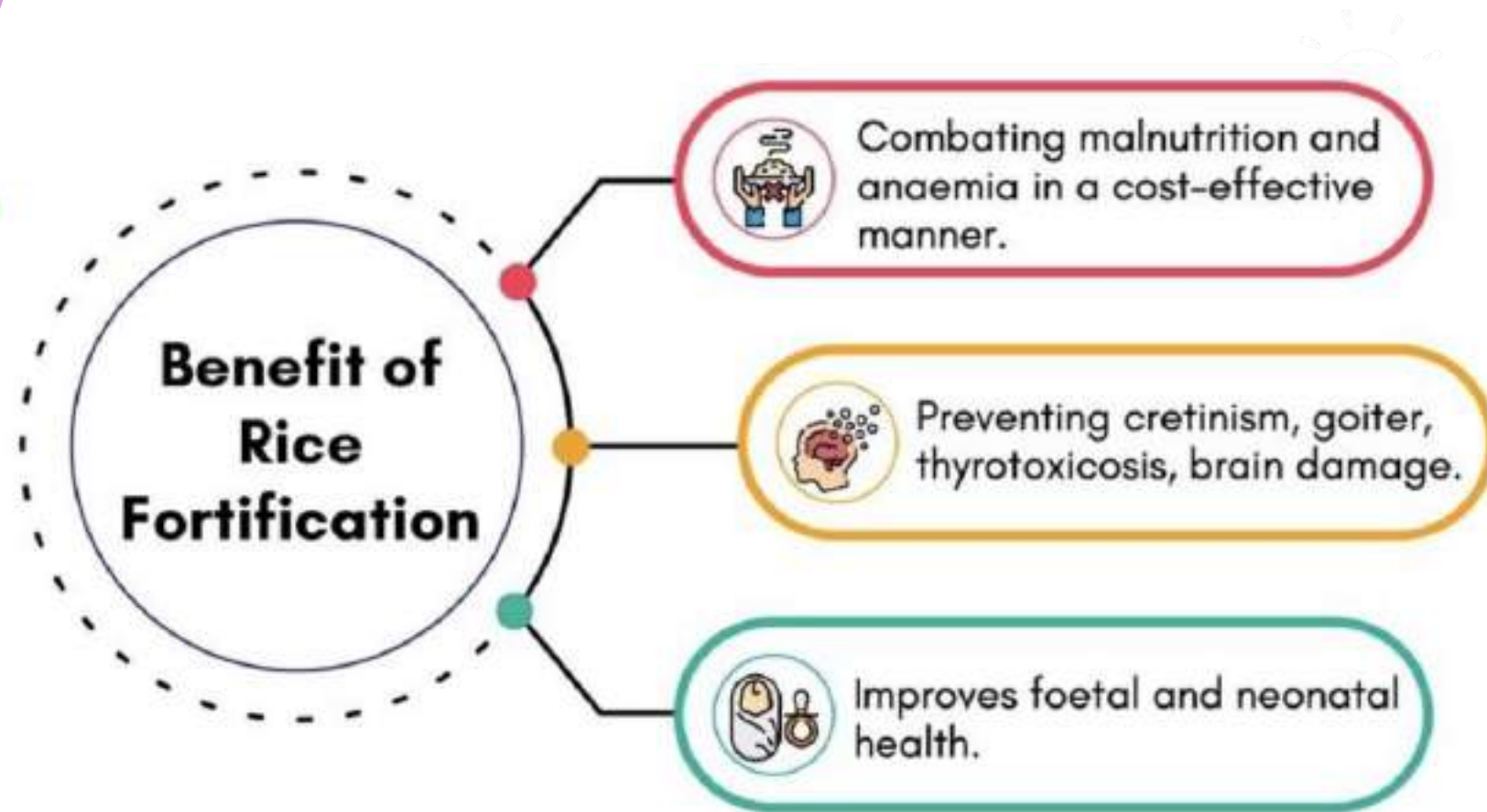


- May also carry a tagline Sampurna Poshan Swasth Jeevan
- Food fortified with Iron shall carry a warning statement for People with Thalassemia
- Rice when fortified shall contain mandatory micronutrients (Iron, Folic acid and Vit B12), or optional (Zinc, Vit A, Thiamine, Riboflavin, Niacin and Pyridoxine as per levels given by Food safety and Standards (Fortification of Foods) Regulations 2018

Technologies to add micronutrients



- Dusting: Adds micronutrients onto the surface of the rice grains using electrostatic force. Provides limited nutrient protection when rice is washed, soaked or cooked in excess water
- Coating and extrusion: Two step process. coating or extrusion technology is used to produce fortified kernels and the fortified kernels are blended with non-fortified rice at a ratio of 0.5% to 2%.
- In India, rice is fortified using extrusion technology in which milled rice is pulverized and mixed with a premix. FRK is added to traditional rice in ratio ranging from 1:50 to 1: 200



Forms of food fortification



- **Mass fortification:** micronutrients to foods commonly consumed by the general public
- **Targeted fortification:** specific subgroups of the population
- **Market-driven fortification:** business-oriented initiative.

Benefits of food fortification



- Addressing Hidden hunger
- Safe method
- Does not require any changes in food habits of people
- Does not alter the characteristics of the food—the taste, the feel, the look.

Steps towards fortification in India



- Food fortification in India began in the 1950s with vegetable oil fortification and salt iodization
- 10th, 11th ,12th Five Year Plans, POSHAN Abhiyan (National Nutrition Mission) and Anaemia-Mukt Bharat Mission
- In 2016, FSSAI imposed regulations on fortification for staples
- Other Steps taken by FSSAI include:
 - A dedicated unit called the Food Fortification Resource Centre
 - capacity building trainings and developed Information Education & Communication (IEC) material specifically for POSHAN Maah
 - Eat Right India movement

Oral Rehydration Solution



- Dr Dilip Mahalanabis who pioneered the proper practical, emergency use of oral rehydration solution, for diarrhoeal diseases passed away recently.
- Dr Dilip Mahalanabis was working in refugee camps during the 1971 Bangladesh Liberation war
- ORS: mixture of electrolytes (salts) and carbohydrates (in the form of sugar) dissolved in water. Used to replace salts and water that the body loses during dehydration caused by gastroenteritis, diarrhoea or vomiting. Contains potassium and sodium.
- These components maximize fluid absorption in the gastrointestinal tract.
- Gastrointestinal tract relies on sodium-glucose cotransporters (SGLTs)
- Cotransporters help move substances across membranes.

Indian Biological Data Center



- Indian Biological Data Center (IBDC) was inaugurated at Faridabad, Haryana
- First national repository for life science data
- Storage: Four-petabyte supercomputer '**Brahm**'
- Supported by: Department of Biotechnology (DBT).
- **Data sharing:** FAIR (Findable, Accessible, Interoperable and Reusable)
- **FAIR Data Principles:** support the reusability of digital assets.

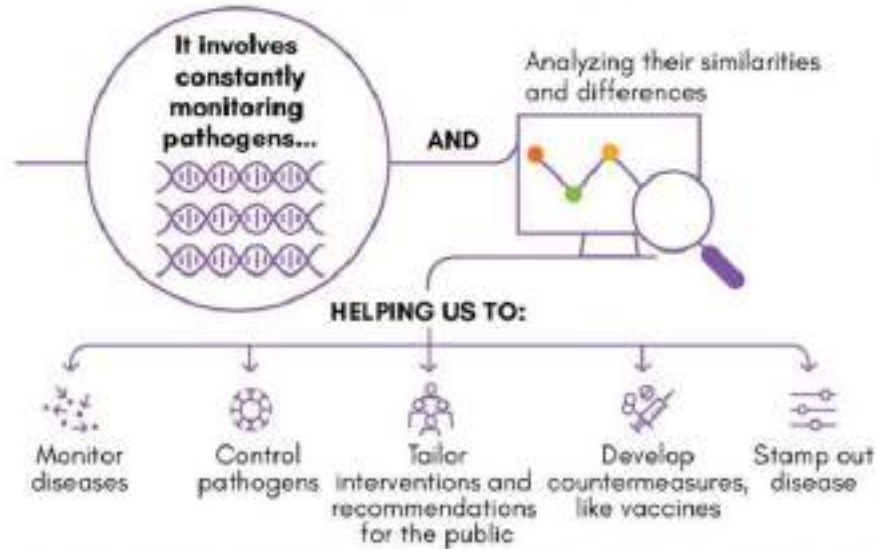
Indian Biological Data Center - Cont'd



- Also contains genomes sequenced by the Indian Sars-CoV-2 Genomic Consortium (INSACOG)
- Two sections have been developed:
 - Indian Nucleotide Data Archive (INDA) shall archive data generated from publicly funded research projects in India and provide internationally accepted data accession numbers
 - Indian Nucleotide Data Archive – Controlled Access (INDA-CA)

GENOMIC SURVEILLANCE

WHAT IS GENOMIC SURVEILLANCE?



APPLICATIONS OF GENOMICS:

-  **Human genomics for medical purposes** (Helping to treat genetic disorders)
-  **Microbial genomics for medical purposes** (Identification of infectious agents, vaccine design etc.)
-  **Agriculture and aquaculture genomics** (Identification of new traits and disease susceptibility etc.)
-  **Biological and medical research** (drug development, discovery of genes influencing disease etc.)
-  **Others** (forensic science, assessment of ancestry etc.)

GENOMIC SURVEILLANCE IN INDIA



Regional Genome Sequencing Laboratories (RGSL) serve as the regional hub for genome sequencing.



Indian SARS-CoV-2 Genomics Consortium (INSACOG) was established to expand Whole Genome Sequencing of SARS-CoV-2 across the nation*.



Indian Tuberculosis Genomic Surveillance Consortium (InTGS) is proposed on the lines of INSACOG.

*INSACOG, jointly initiated by Ministry of Health & Family Welfare, and Department of Biotechnology (DBT) with Council for Scientific & Industrial Research (CSIR) and Indian Council of Medical Research (ICMR), is a consortium of over 50 laboratories.

Dengue

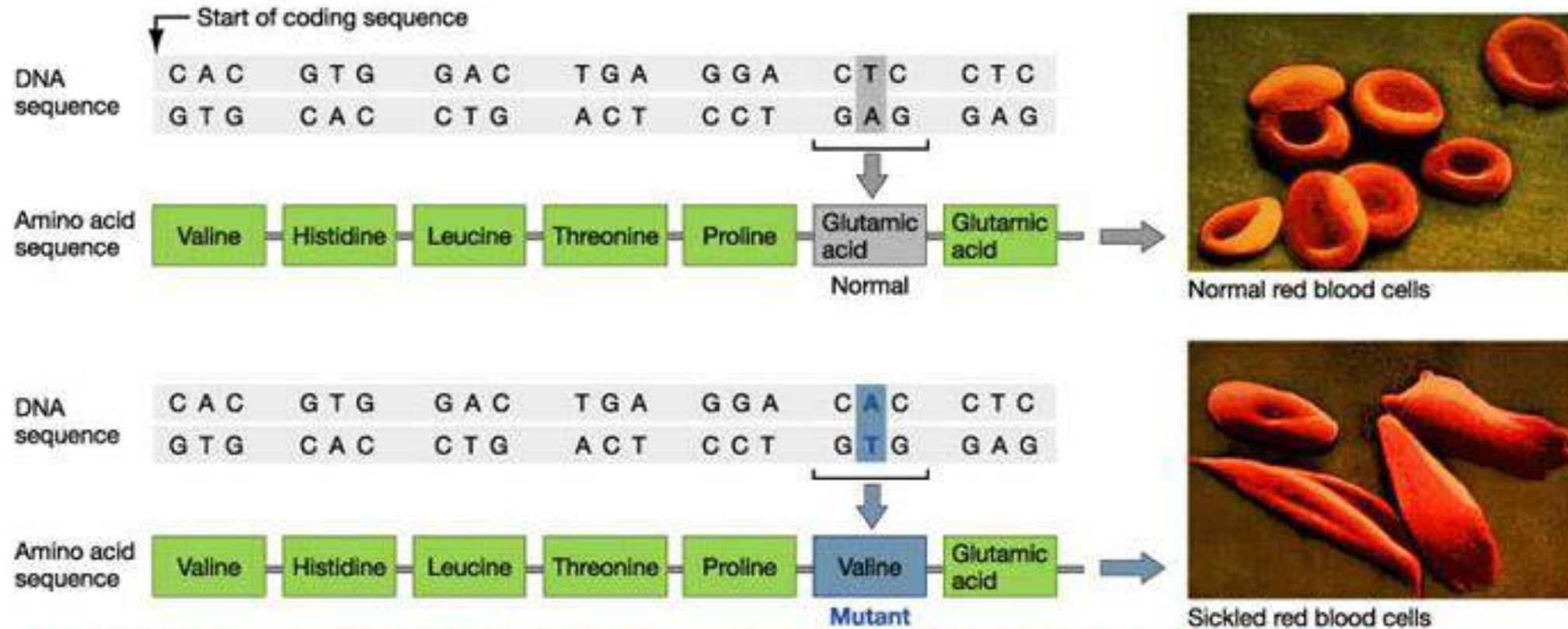


- Mosquito borne tropical disease caused by dengue virus (flavivirus)
- Spread through bite of infected aedes mosquito.
- First dengue vaccine : dengvaxia : approved by USFDA in 2019

Sickle Cell Anaemia



- Important this year : National Mission to Eliminate sickle cell anaemia launched in budget.
- Occurs due to genetic mutation that causes hemoglobin in RBCs to clump together.
- RBC shape changes to Sickle shape : flexibility lost.



The change in amino acid sequence causes hemoglobin molecules to crystallize when oxygen levels in the blood are low. As a result, red blood cells sickle and get stuck in small blood vessels.

Other Important News



- Trachoma : first country in southern Africa to eliminate trachoma is Malawi. India eliminated trachoma in 2017. Trachoma is an infection of the eye caused by Chlamydia Trachomatis. Can cause blindness.
- Diabetes : recent case of infection from diabetes medication. Diabetes is of 2 types : type 1 and type 2.
- Type 1 : insulin dependent, juvenile or childhood onset.
- Type 2 : Non insulin dependent : adult onset. (most common)
- Shigella : bacteria causing shigellosis. Leads to diarrhoea in children. No vaccine available.

Nanotechnology



- What we need to cover :
- Conceptual understanding of nanotech
- Current developments in nanotechnology



Nanotechnology : Basic Concepts



- Term Nanotech comes from the word nanometer
- Nanometer means 10^{-9}
- $10^{-1} = \frac{1}{10}$
- When we bring things to this size, it leads to certain special effects in them.

Special Effects in Nanomaterials



- Electrical Conductivity
- Color Effects
- Chemical Reactivity
- Melting Point
- Wettability

Special Effects : why and how

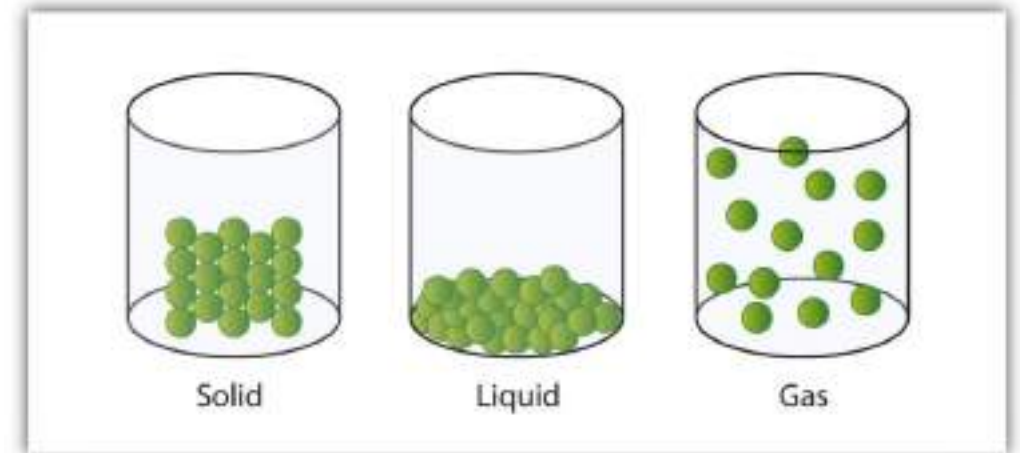


- Disclaimer : you don't need to know these concepts directly
- I am teaching this so that you can develop an understanding
- What do we already know?
- Atoms are constituent elements of materials
- Atoms have electrons, protons and neutrons.
- Solids liquids and gases have different arrangement of atoms.

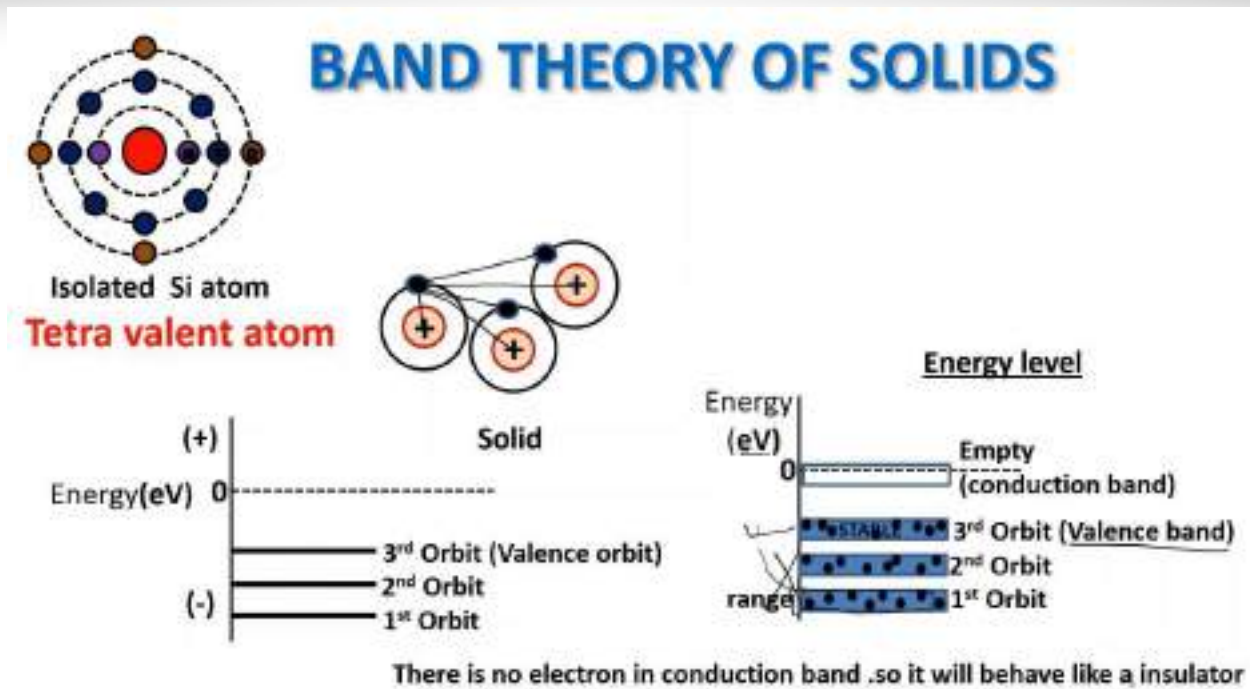
Concepts of Nanotech



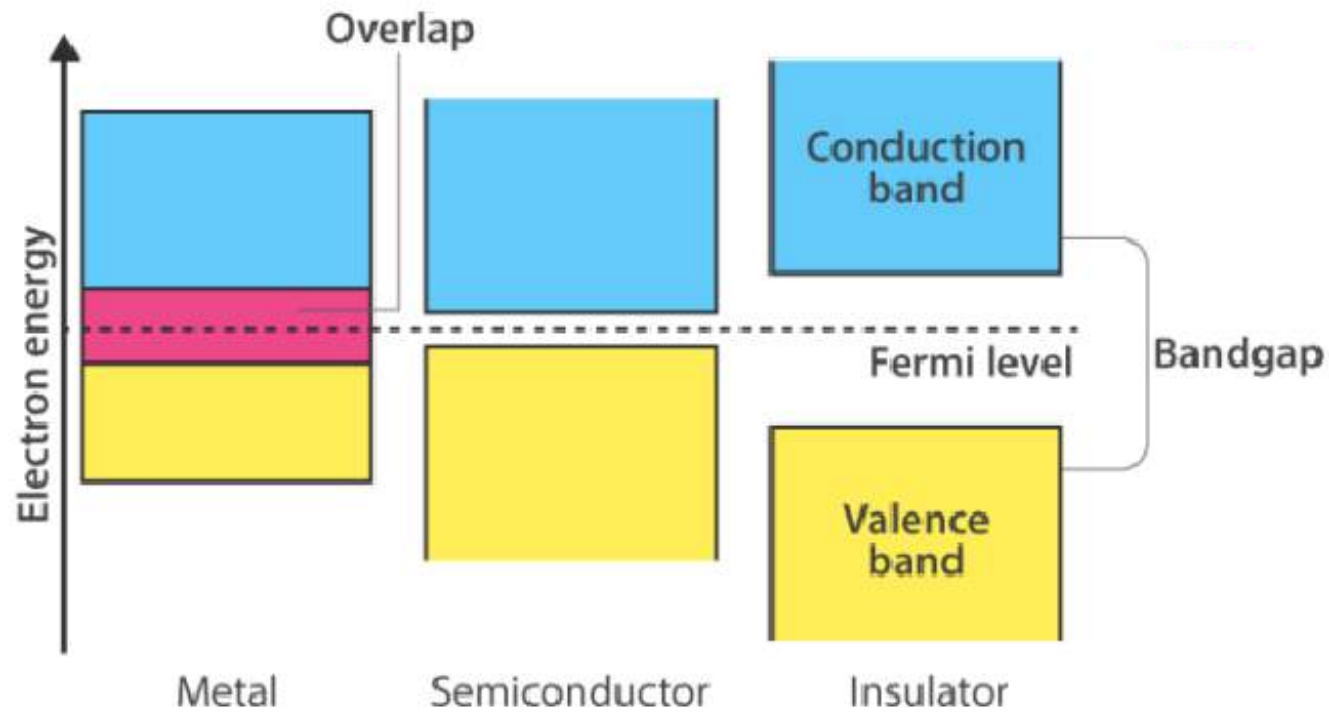
- We are concerned mainly with solids
- When atoms are packed together, their electrons come together into something known as bands
- This band arrangement is size dependent and it gets altered as size changes
- This is known as band theory of solids



Band theory of Solids



Bands

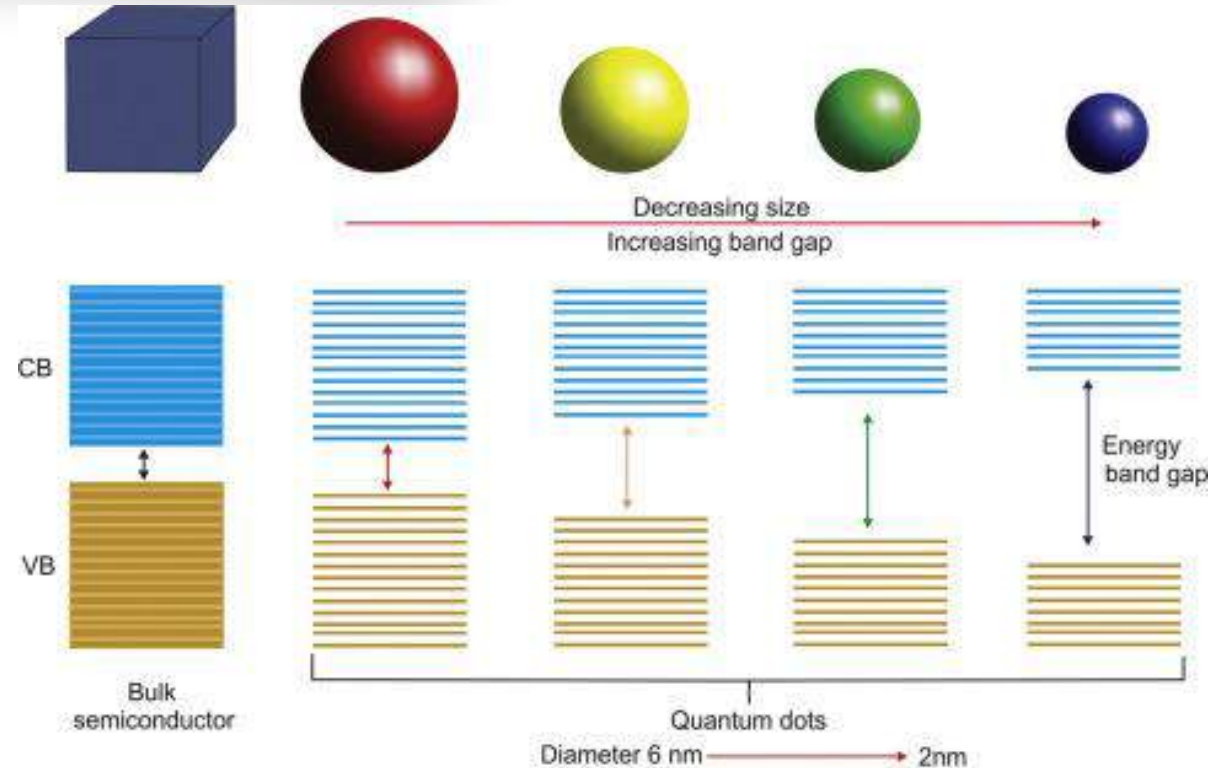


© Byjus.com

Bands change at nano level



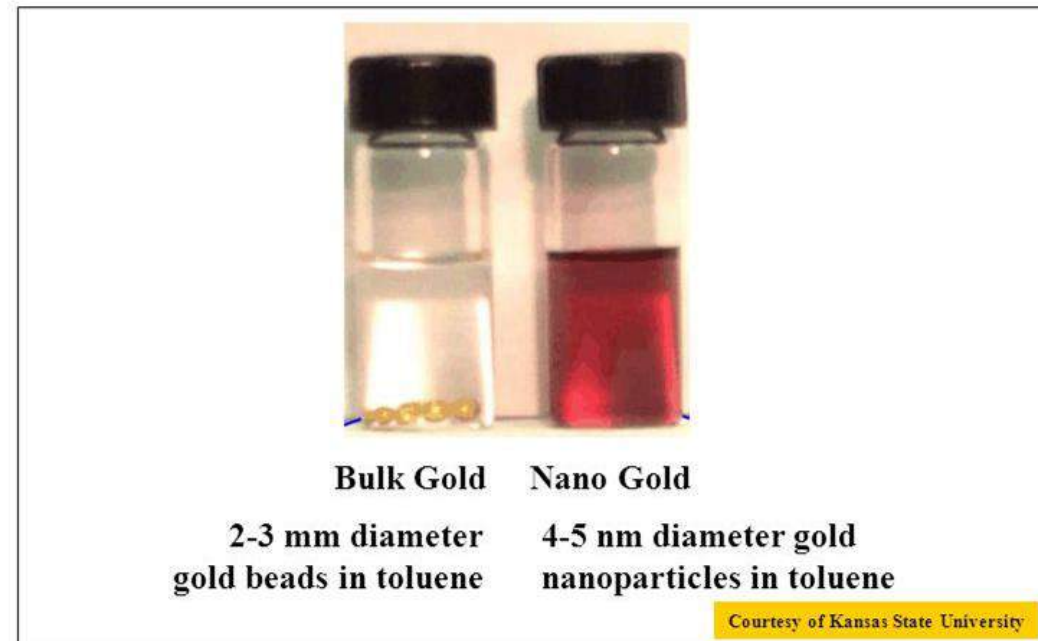
- Band arrangement changes at the level of nanomaterials
- As the bandgap increases, certain properties change
- Change in electrical conductivity : decrease in semi conductors
- Discrete energy levels lead to emission of pure wavelength of light : change in color



GOLD Particles



- Color is size dependent due to band gap alterations.



Extendable Structures: Melting Point, Color, Conductivity
© McREL 2009

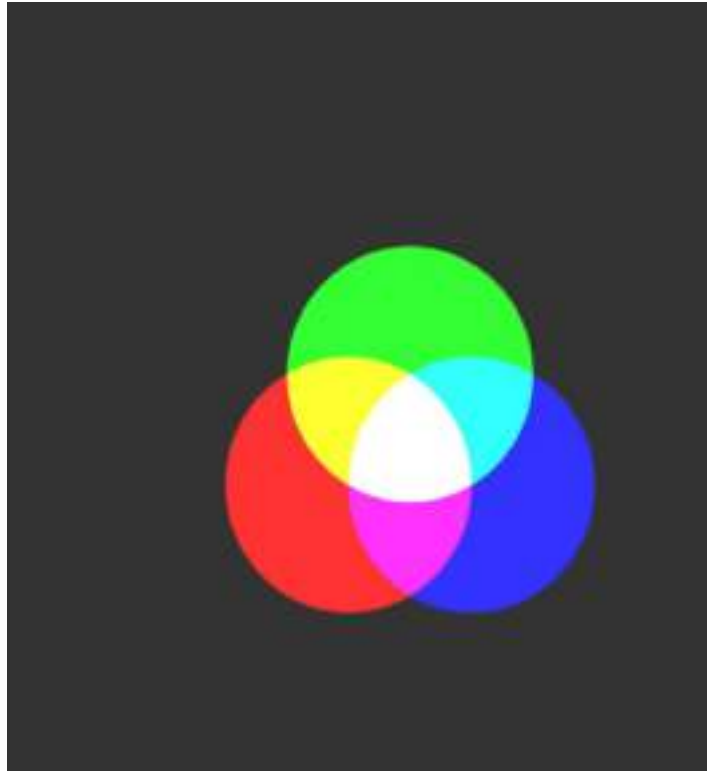
NanoLeap

13

Change in optical properties



- From bands (combined) the energy states become discrete
- Hence the wavelengths emitted are pure
- Quantum Dots : This idea is used in the formation of Quantum LED (QLED)

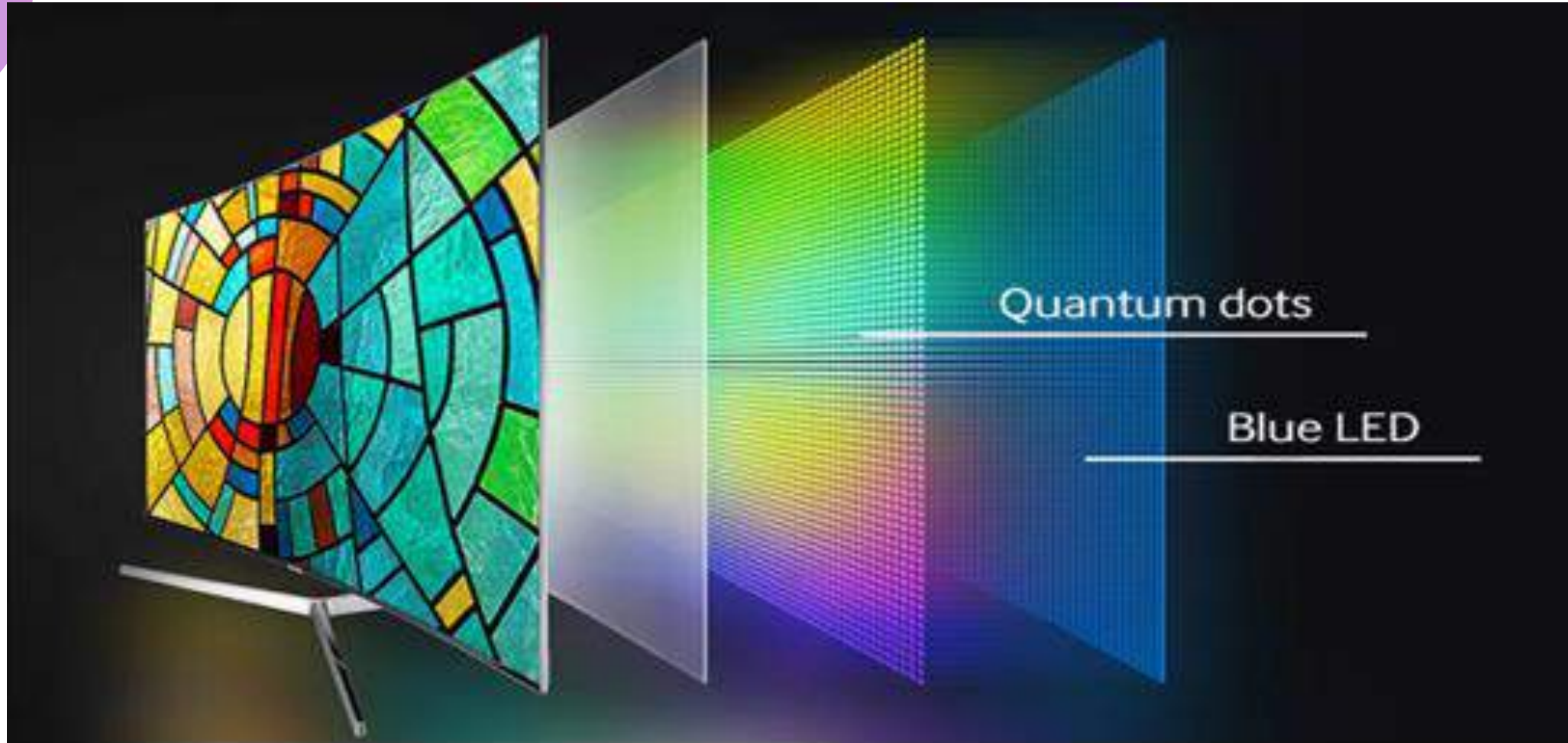


QLED

- Stands for Quantum dot Light Emitting Diode
- Uses quantum dot technology to emit light. Add an additional layer in front of a LED light panel
- Delivers 100% colour volume with unmatched saturation levels
- Not as power efficient as LED

LED

- Stands for Light Emitting Diode
- Uses LED to backlit the display
- Not as good as QLED in colour volume or picture quality
- Delivers optimal brightness level without consuming more power



Chemical Reactivity : change in nano materials



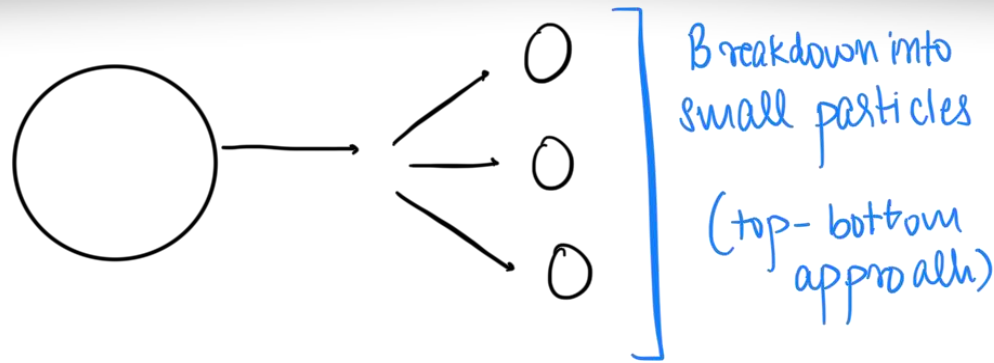
- Live demonstration : Disprin (solid vs crushed) in water
- More surface area : more reaction likelihood
- More reactivity
- Cube with side 4cm vs cube with side 1cm

Change in properties : why?

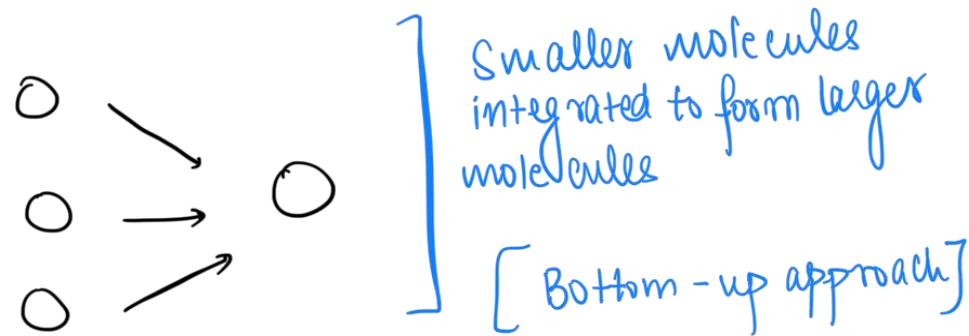


- These are known as quantum effects
- Quantum effects are size dependent
- Another concept : Quantum Confinement
- However, if you can just remember that there is a change in the properties of materials, it is sufficient.

2 approaches to make nanomaterials

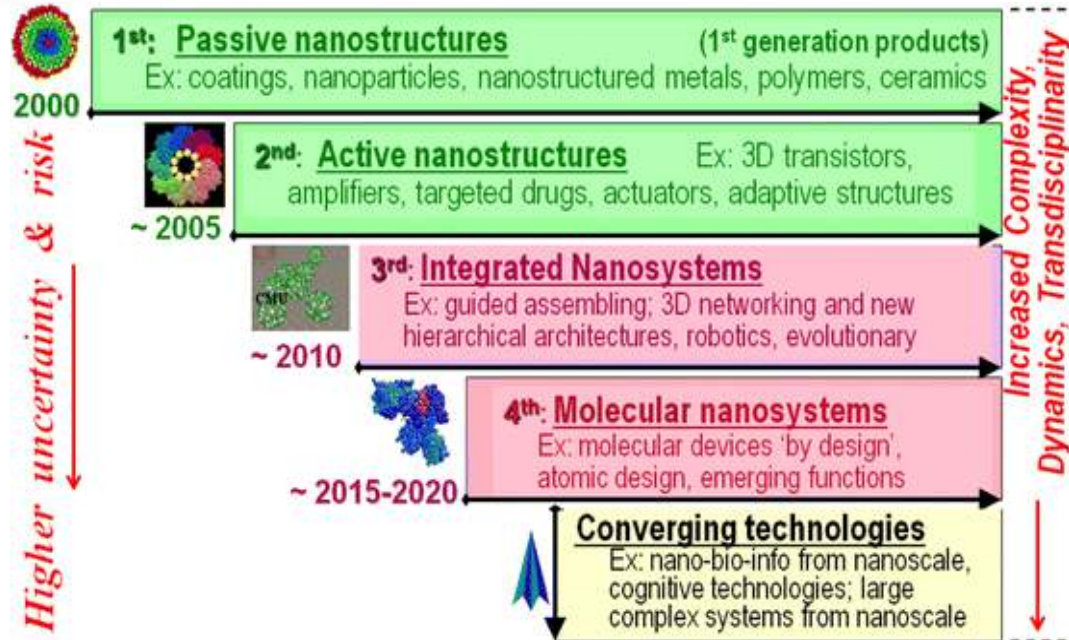


Example : nano catalyst,
nano silicon based solar
panels



Example :
Nanoelectronics and
integrated circuits

Generations of Nanomaterials



1. Passive Structures
1. Active : interact with environment and drug delivery
1. Nanosystems : Complete systems of nanomaterials (nobel prize of 2016)
1. Molecular nanosystems : bottom up approach and concept of additive manufacturing (3d printer)

2016 Chemistry Nobel Prize

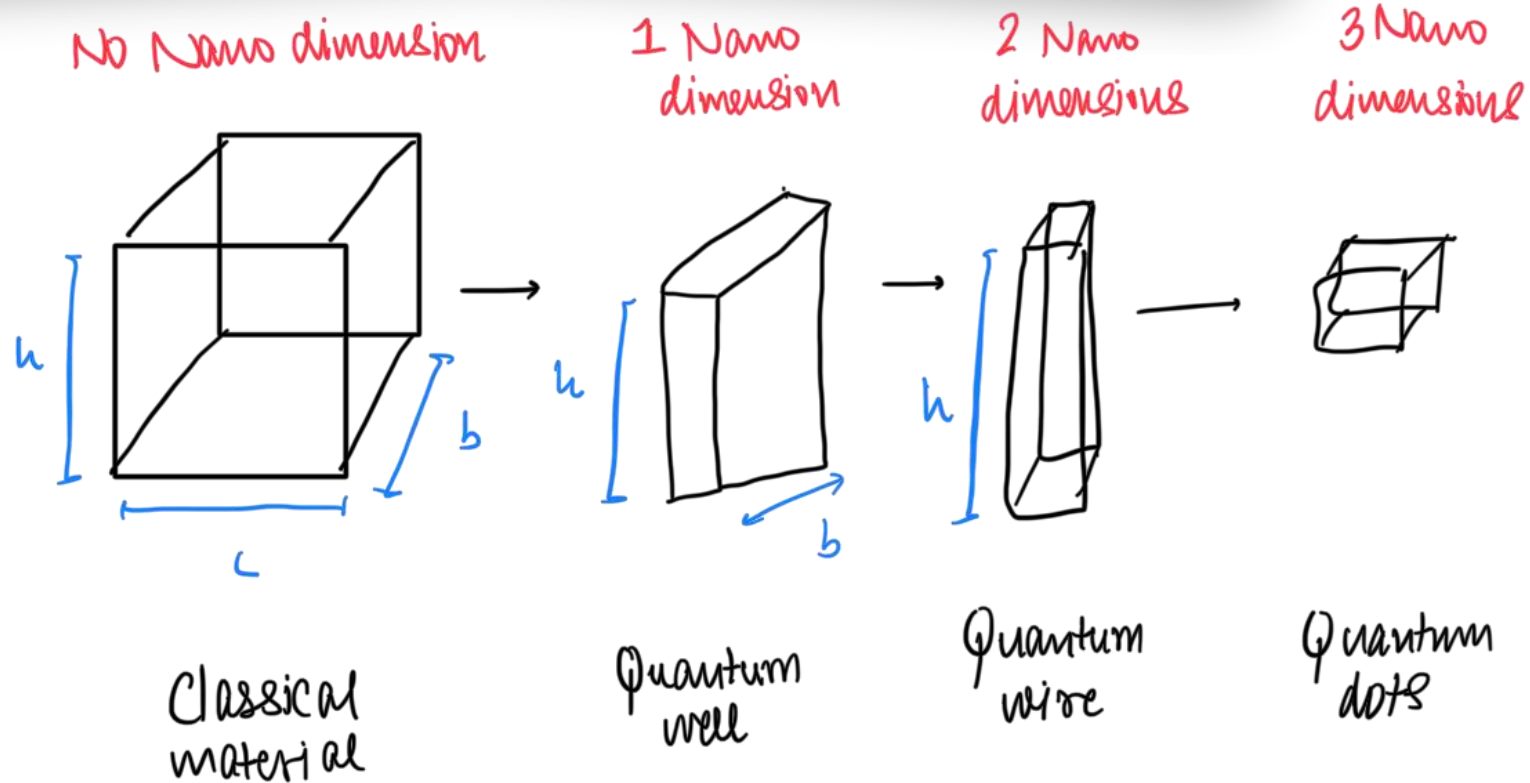


Nobel Prize in Chemistry / Winners (2016)

Fraser Stoddart, Jean-Pierre Sauvage, Ben Feringa

The **Nobel Prize in Chemistry 2016** is awarded to Jean-Pierre Sauvage, Sir J. Fraser Stoddart and Bernard L. Feringa for their design and production of molecular machines. They have developed molecules with controllable movements, which can perform a task when energy is added. Oct 5, 2016

Nanomaterials : Types



Applications of Nanotechnology



- This is where questions are mostly asked.
- Now that you have understood what nanotechnology is, understanding applications is easy.

NanoFertiliser



- Nano Urea Liquid : National Fertilisers Limited (NFL) and Rashtriya Chemicals and Fertilisers Ltd (RCF) signed MoU with IFFCO for 'transfer of technology' of Nano Urea Liquid (NUL) fertiliser
- NUL contains nano-scale nitrogen particles which have more surface area and number of particles making it more impactful.
- India has become the first country globally to start commercial production of NUL.

NanoFertiliser



Significance:

- Reduces the requirement of conventional Urea by 50% or more.
- Improves Soil, Air & Water quality.
- Cheaper than conventional urea. ***

Triboelectric NanoGenerators



- Scientists from the Centre for Nano and Soft Matter Sciences (CNSMS) have fabricated a cost-effective, bio-compatible nanogenerator that can generate electricity from mechanical energy in the form of vibrations present everywhere.
- CNSMS is an autonomous institute under the Department of Science & Technology.
- The energy harvesting TENG works on the principle of creation of electrostatic charges (deficiency or excess of electrons which occurs on ungrounded or insulating surfaces).
- Applications include optoelectronics, self-powered devices, and other biomedical applications.
- major application : pacemakers

Nanomicelles



- Recently, Researchers have found that Nanomicelles can be used for Cancer treatment.
- Nanomicelles are globe-like structures with a hydrophilic outer shell and a hydrophobic interior. This dual property makes them a perfect carrier for delivering drug molecules.
- Advantage: Low toxicity, ability to minimize drug degradation, ability to permeate tissues easily
- for drug delivery, and lower adverse drug side effects.
- Hydrophilic and hydrophobic configuration matches with phospholipid bilayer thereby allowing targeted drug delivery

Gold NanoParticles

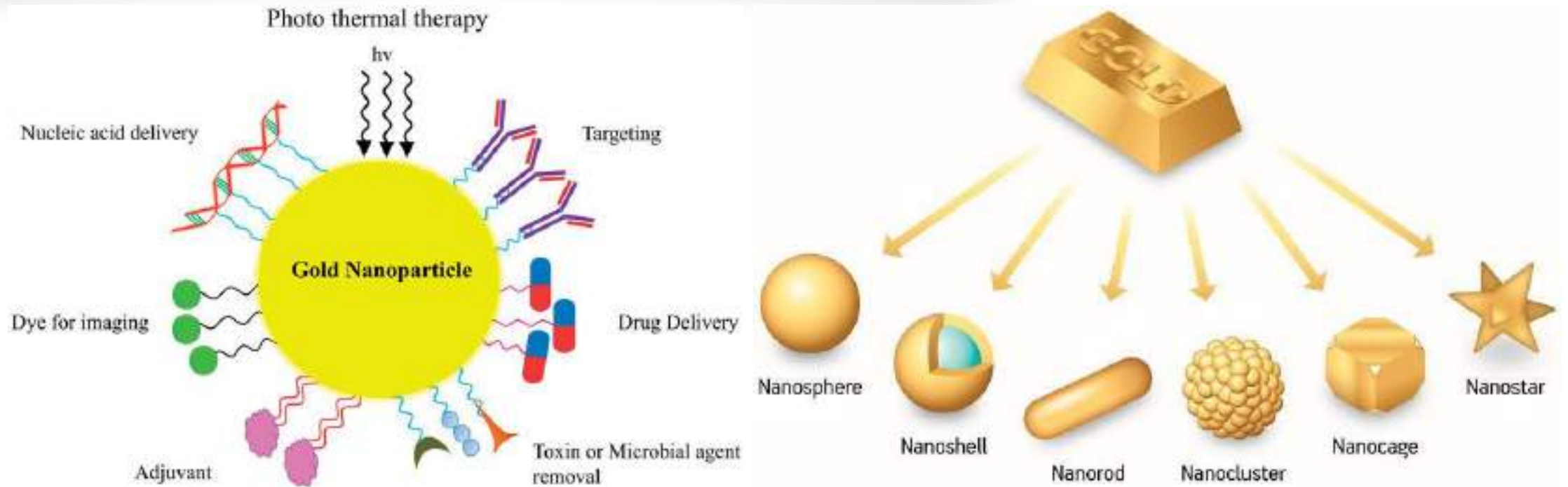


- National Centre for Polar and Ocean Research and the Goa University has successfully synthesized GNPs using psychrotolerant Antarctic bacteria.
- GNP's are biocompatible, have high surface area, more stability, and are non-toxic. GNPs are melted at much lower temperatures (300 °C) than bulk gold (1064 °C).
- Lets understand each of these words

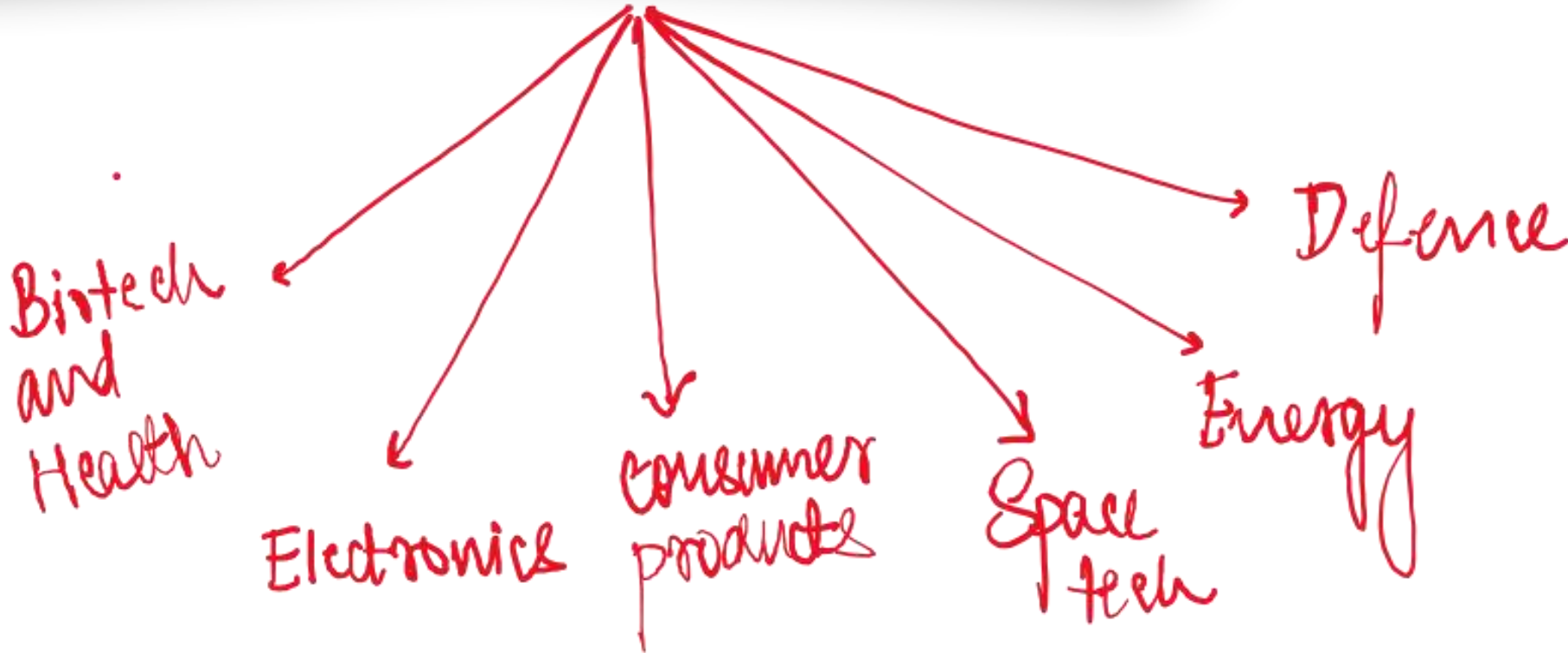
Advantages of GNP :

- Greater solar radiation absorbing ability : photovoltaic cells
- Unique optical properties : therapeutic imaging for better resolution
- Useful in the electronics industry : NOMFET (Nanoparticle Organic Memory Field-Effect Transistor) using GNPs. Mimics human synapse

Gold NanoParticles



Applications of Nanotech : enabling tech



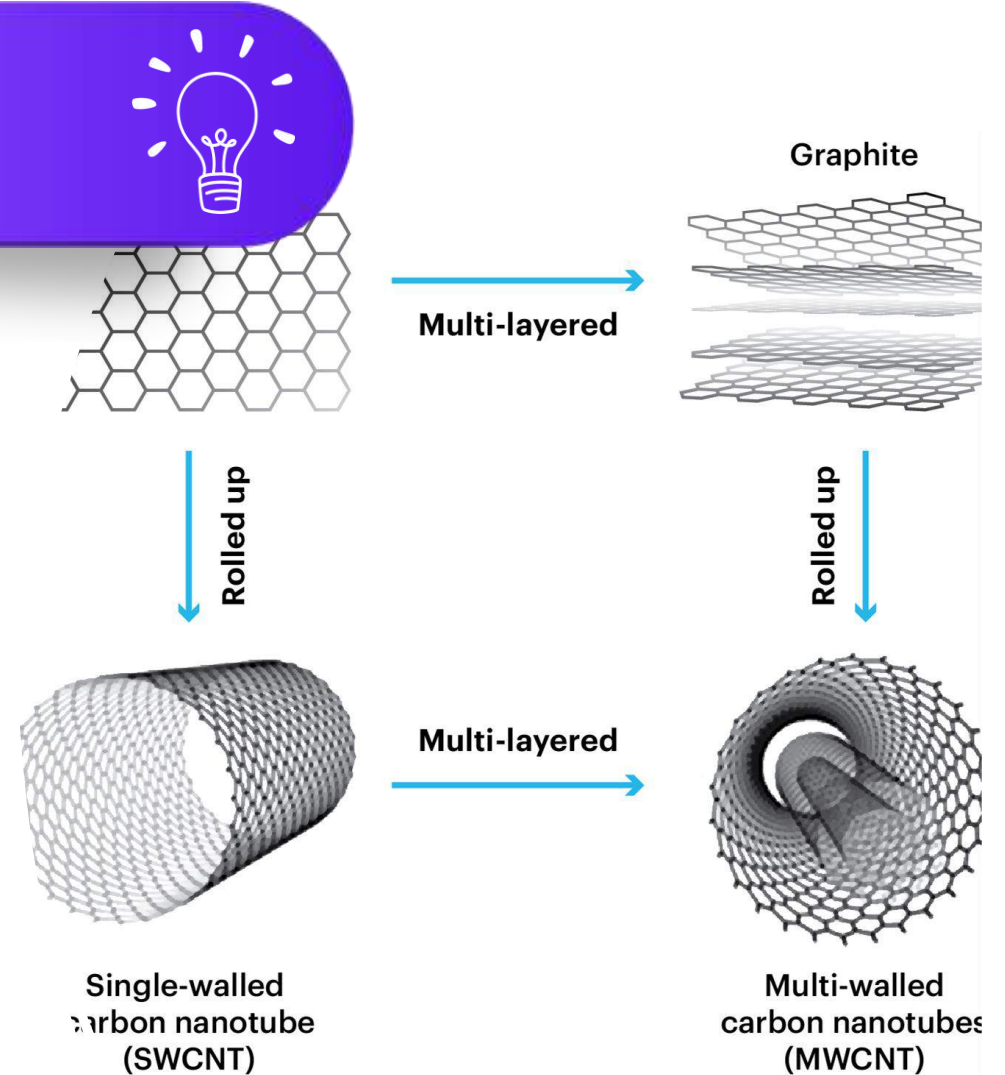
Carbon Nanomaterials



- C-60 Buckminster fullerene
- Carbon Nanotubes
- Graphene : 2d Graphite : Flexible touchscreens

Carbon Nanotubes

- Cylindrical Hollow figures with sheets of graphite
- Novel properties, lightweight
- Tensile strength : load bearing
- Electrical conductivity
- Used in electronics, automobiles, sports, space, aerospace industry
- However, not biocompatible : Immune Reactions



Applications : Industrial Products



- Sunscreens : Titanium dioxide and zinc oxide np help block uv radiations
- Non flammable furniture : coating
- Fabrics and Body armour : lightweight, wettability : easy clean textiles
- Films on glasses, computers and camera displays
- Quantum dots and electrical effects : QLED tv

Environmental Remediation



- Nanosensors : neutralize the effect of pullutants
- Wastewater treatments
- Nano membranes: desalinization

Energy Sector



- Biofuel production : nanobioengineering of enzymes
- Solar panels : TENG
- Hydrogen economy and storage systems
- Nanotube scrubbers and electrostatic precipitators for better efficiency
- GNP : solar panels
- CNT : PV cell
- Graphne : production of green hydrogen

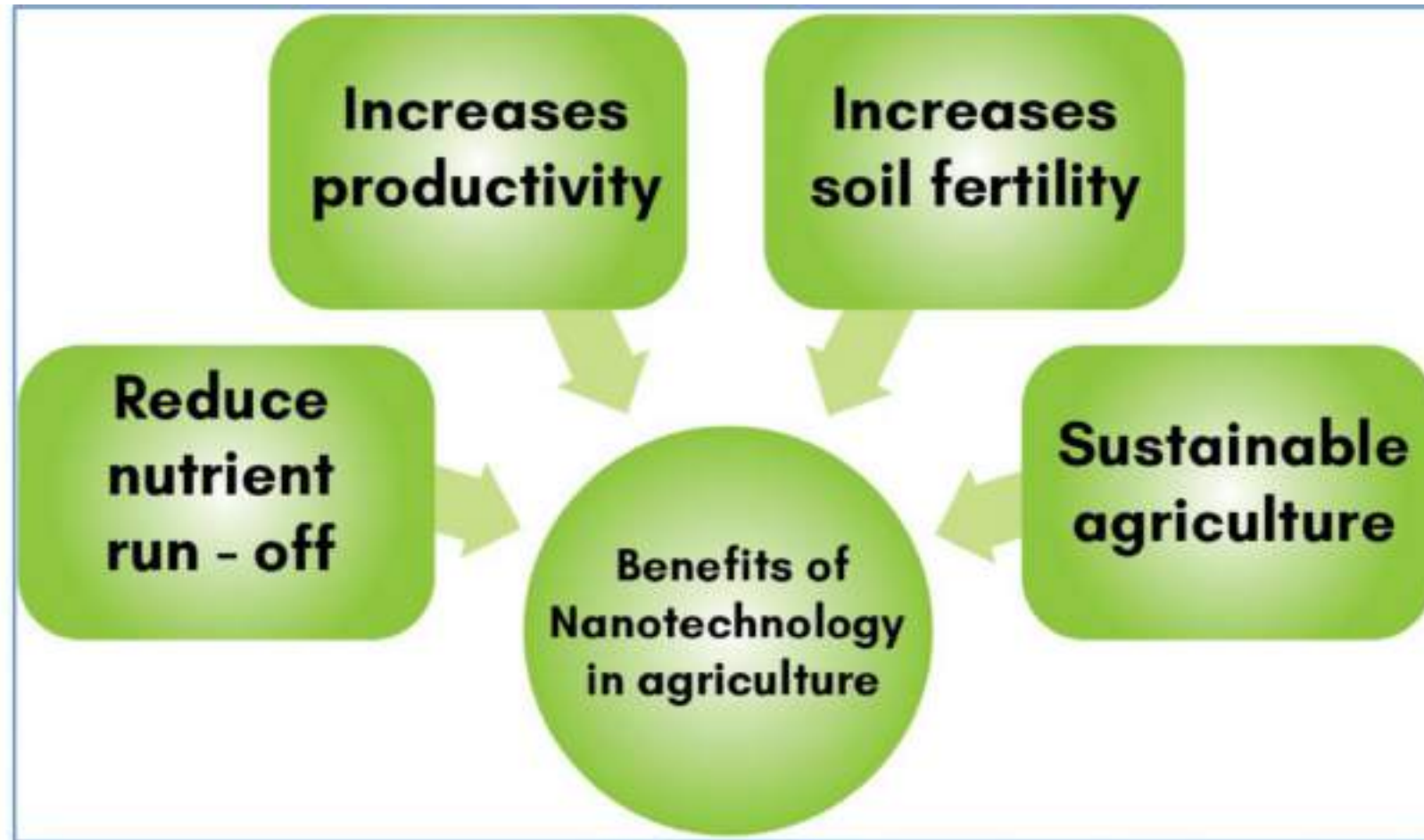
Agriculture Sector



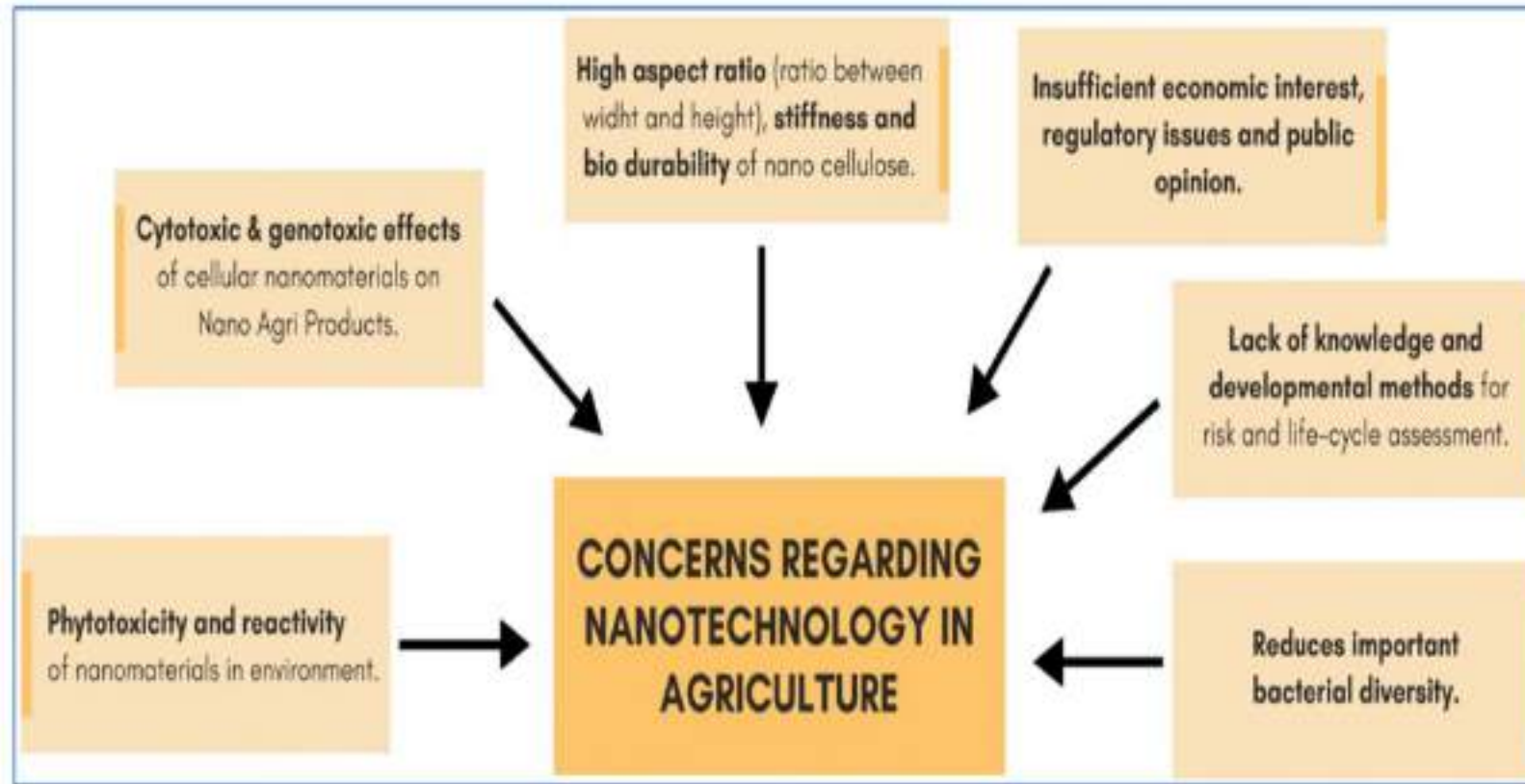
- Nanoherbicides and Nanopesticides : targeted delivery and enhancement of productivity
- Nanophosphorous fertilizer : Millet and cluster beans
- Silver Nanoparticles : against phytopathogens
- Carbon Nanoparticles : seed germination
- Nickel / ferrite : Antifungals
- Zinc oxide/ chitosan : Viral Diseases
- Precision farming : nanosensors

Nanotechnology in agriculture :

advantages :



Nanotechnology in agriculture : disadvantages :



Health Sector



- Antigen Antibody tests : ex : ELISA test (lets understand in detail)
- Prosthetic Devices : nanocomposite materials
- Targeted drug delivery : nanomicelles
- Nano sensors : HIV and Cancer detection
- GNP : medical imaging
- Nanohydrogel based eye drops

Defence



- Tank polymer and Camouflage
- Lightweight materials
- Precision guided missiles

Electronics



- Quantum dots
- Graphene : used in flexible touch screens
- Optoelectronics
- CNT : replace silicon based chips

Risks associated with nanotechnology



- Health Risks : generation of Free Radicals
- Biomagnification in the foodchain of nano particles
- Environmental Contamination
- High levels of complexity
- Lack of risk assessment methodologies
- Fears of self-replicating materials

India and Nanotechnology



- Nano Mission by Dr. CNR Rao in 2007
- The Government of India launched the Nano Mission in 2007 as an “umbrella capacity-building programme”.
- It is being implemented by the Department of Science and Technology (DST) under the Ministry of Science and Technology.
- The objectives of the mission are:
 - Basic research promotion
 - Infrastructure development
 - Nano applications and technology development

India and Nanotechnology



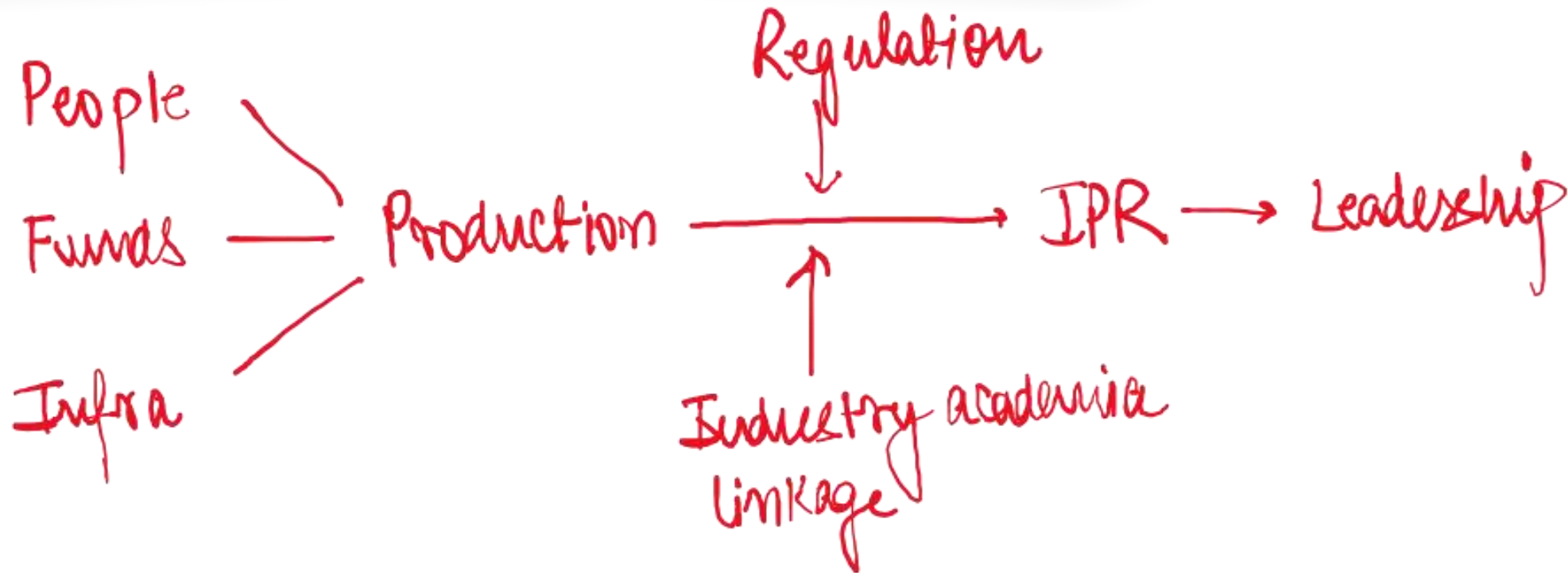
- Human Resource development
- International collaborations
- As a result of the efforts led by the Nano Mission, today, India is amongst the top five nations in the world in terms of scientific publications in nanoscience and technology (moving from 4th to the 3rd position).
- The Nano Mission has established national dialogues to promote R&D in the development of standards for nanotechnology and for laying down a National Regulatory Framework Road-Map for Nanotechnology (NRFR-Nanotech).

India and Nanotechnology



- ICONSAT : International Conference on NanoScience and NanoTechnology : biennial conferences under the aegis of nano mission.
- Center of Excellence in Nano Electronics : IIT Bombay and IISc bangalore

Way Forward: Life Cycle Approach



Test of 5 Questions!

Cut off : 5/5



Question 1 (2011)

Q : A company marketing food products advertises that its items do not contain trans-fats. What does this campaign signify to the customers?

1. food products are not made out of hydrogenated oils.
2. The food products are not made out of animal fats/ oils.
3. The oils used are not likely to damage the cardiovascular health of consumers.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Question 2 (2020)

- With reference to carbon nanotubes, consider the following statements
 1. They can be used as carriers of drugs and antigens in the human body.
 2. They can be made into artificial blood capillaries for an injured part of the human body.
 3. They can be used in biochemical sensors.
 4. Carbon nanotubes are biodegradable.

Q Which of the statements given above are correct?

- A) 1 and 2 only
- B) 2, 3 and 4 only
- C) 1, 3 and 4 only
- D) 1, 2, 3 and 4

Question 3 (2015)

- With reference to the use of nanotechnology in the health sector, which of the following statements is/are correct?

1. Targeted drug delivery is made possible by nanotechnology.
2. Nanotechnology can largely contribute to gene therapy.

Q : Select the correct answer using the code given below.

- A) 1 only
- B) 2 only
- C) Both 1 and 2
- D) Neither 1 nor 2

Question 4 (2012)

- Graphene is frequently in news recently. What is its importance?
 1. It is a two-dimensional material and has good electrical conductivity.
 2. It is one of the thinnest but strongest materials tested so far.
 3. It is entirely made of silicon and has high optical transparency.
 4. It can be used as 'conducting electrodes' required for touch screens, LCDs and organic LEDs.

Which of the statements given above are correct?

- a) 1 and 2 only
- b) 3 and 4 only
- c) 1, 2 and 4 only
- d) All of the above

Question 5 (2022)

Consider the following statements:

1. Other than those made by humans, nanoparticles do not exist in nature.
2. Nanoparticles of some metallic oxides are used in the manufacture of some cosmetics.
3. Nanoparticles of some commercial products which enter the environment are unsafe for humans.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3

Congratulations on giving the test!

Let's Discuss The Answers.



Question 1 (2011)

Q : A company marketing food products advertises that its items do not contain trans-fats. What does this campaign signify to the customers?

1. food products are not made out of hydrogenated oils.
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Which of the statements given above is/are correct?

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(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Question 2 (2020)

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- C) 1, 3 and 4 only**
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- c) **1, 2 and 4 only**
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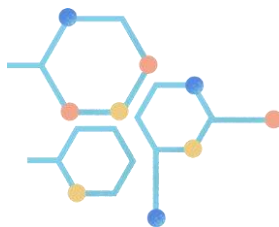
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Which of the statements given above is/are correct?

- (a) 1 only
- (b) 3 only
- (c) 1 and 2
- (d) 2 and 3**



UPSC Prelims

Science & Tech

Class No. 8

Space Technology

Space Technology



- Weightage : 1 or maximum 2 questions in prelims
- So we will study as per its importance
- Today : let's do basics of space tech

Gravitational Force



- Gravity is bending of space time
- However, we will understand it like a force for now.

$$F_{Gravity} = G \frac{M_1 \times M_2}{R^2}$$

- G = Gravitational Constant = $6.6743 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$
- As you can see, this is a very small number
- So for gravity to operate, you need a very large mass
- Otherwise, Gravitational force will be extremely weak.

Orbits

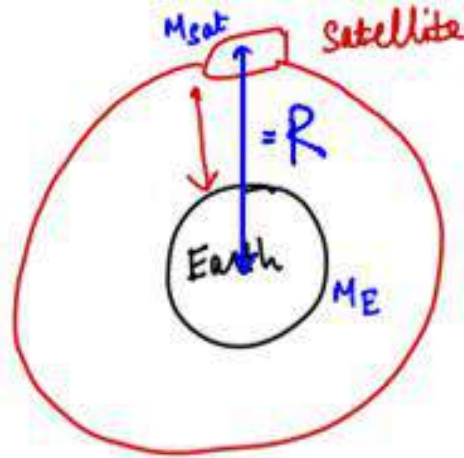


- Definition
- Classification on the basis of :
 - shape
 - revolution
 - Distance from earth

Orbits



Imaginary path taken by a celestial body or spacecraft centripetal force due to gravitational force is the driving force behind the orbit.



Classification of Orbits



1. On basis of shape

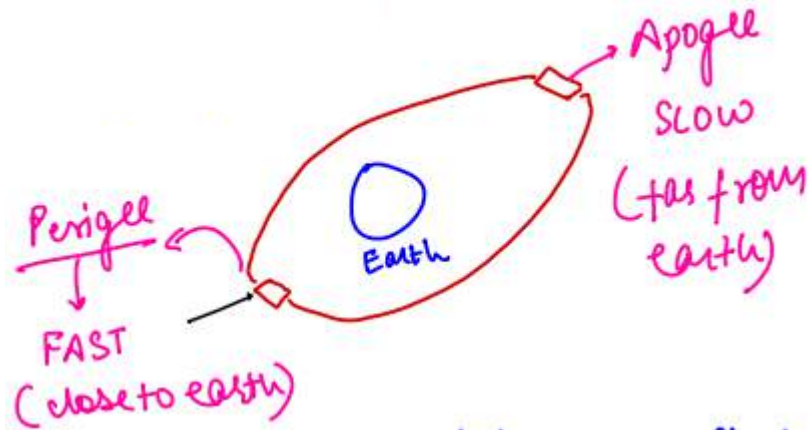
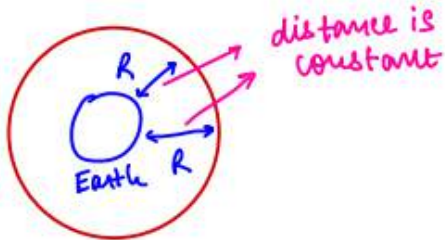
(a) Elliptical Orbit : orbit of planets

(b) Circular Orbit : eg: Geostationary

Classification of Orbits



Mv^2/R is constant



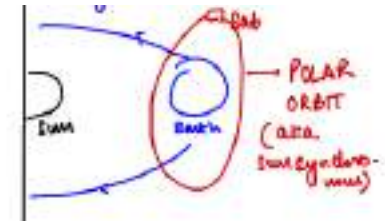


ELLIPTICAL	CIRCULAR
Distance variable speed variable velocity variable	Distance constant speed constant velocity variable (as direction is changing)

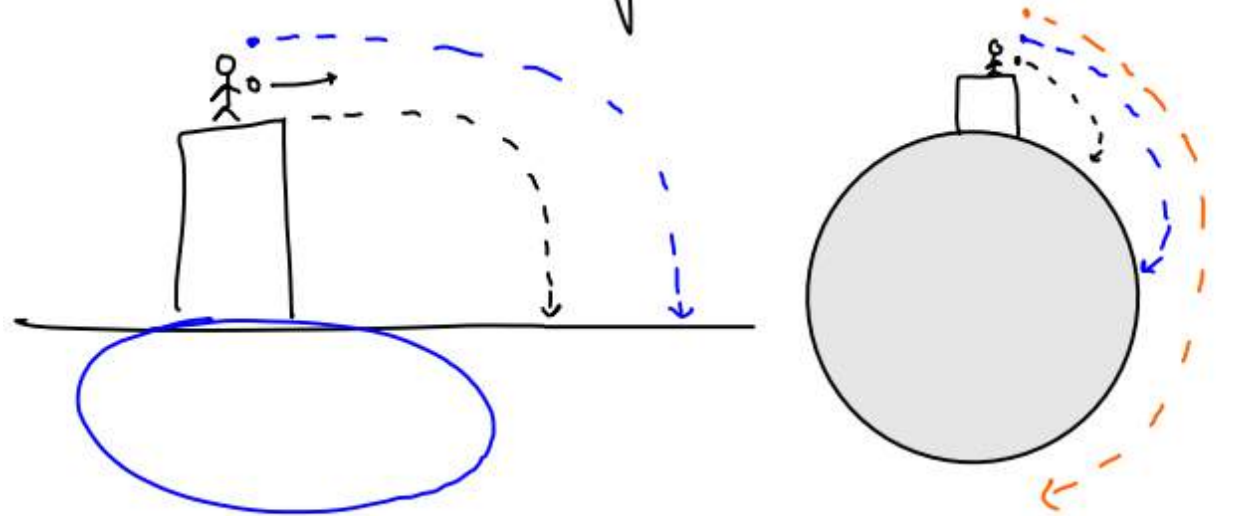
Classification : On Basis of Revolution



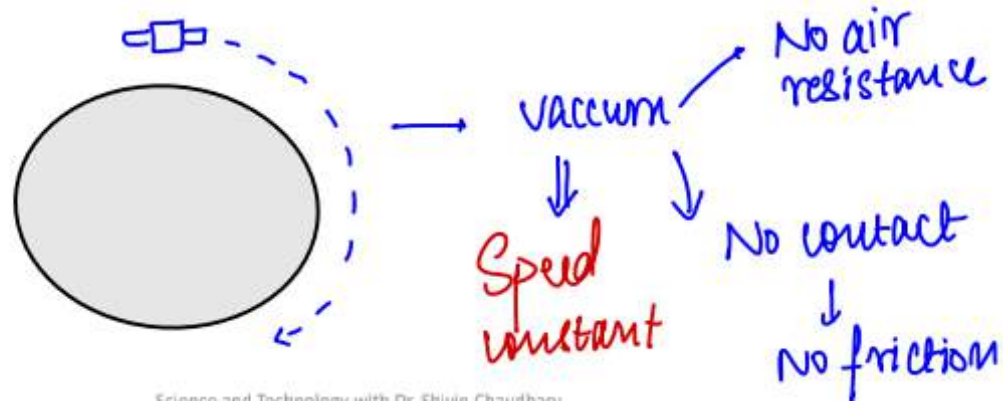
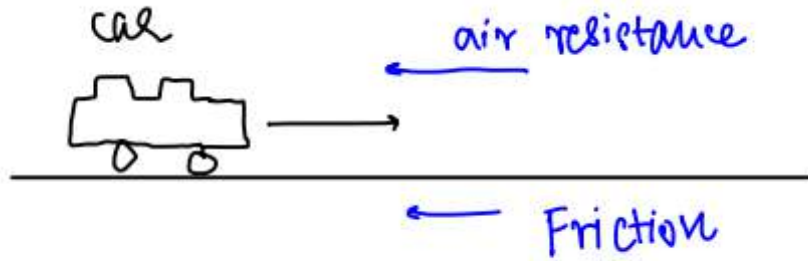
Geo Synchronous	Sun synchronous
(i) Satellite's revolutions is synchronised as per earth's Rotation (ii) Orbits in equatorial plane (iii) Application: (a) Communication (b) Navigation	(i) Satellites revolution is synchronised on basis of Earth's Revolution (ii) Orbits in Polar plane - Application (a) Earth Observation (b) Remote sensing



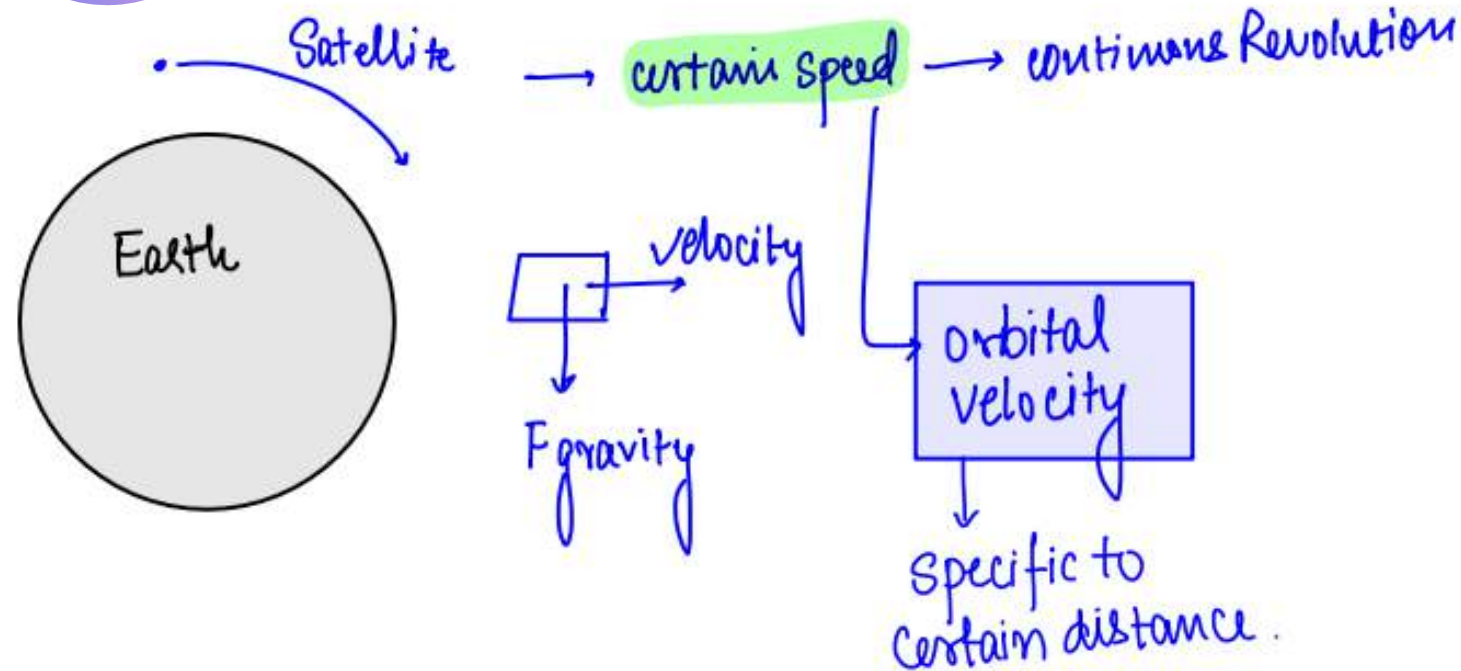
Why Satellites don't fall



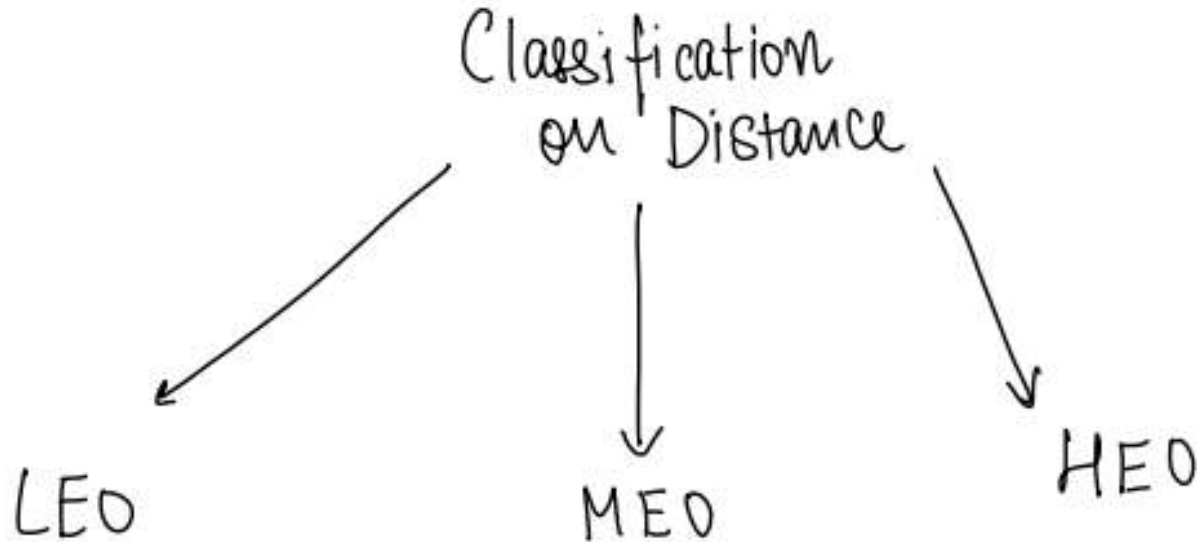
at certain speed → overcomes curvature



Science and Technology with Dr. Shivin Chaudhary



Orbital velocity is specific to a certain distance.



Low Earth Orbit



- O- 2000 km from earth's surface.
- Near earth orbit : 400-800 km
- Applications -
 - (i) Earth observation (Remote sensing)
 - (ii) Astronomical observation (Telescope) **why in LEO?**

Low Earth Orbit



- Astronomical observation : Easy servicing in case of defaults
- (ii) LEO allows to capture data such as radiation (cosmic rays, x rays, gamma rays)
 - Eg: Hubble, Kepler - LEO
 - Satellites : eg: Astrosat 650 Km (by India)

Hubble

- Launch in 1990
- Problem in Lens - NASA sent astronauts on space walk

Application Continued:



(iii) Space Station is conducted is LEO

Eg : International Space Station at - 384 km

by : NASA (USA) , Roskosman (Russia), ESA (Europe), JEXA (Japan), CSA (Canada)

- Tiangong - China (Independent)

Concerns : Weaponization, Spying, Control and Monopoly

Can you relate? Countries of ISS and global north



Applications continued : Space Stations



- (a) Microgravity experimentation (to test products at zero gravity)
 - Eg. Metallurgy, artificial photosynthesis
- b) Space observation
- c) Space Voyage - acts like terminal
- d) servicing workshop
- e) Space Tourism

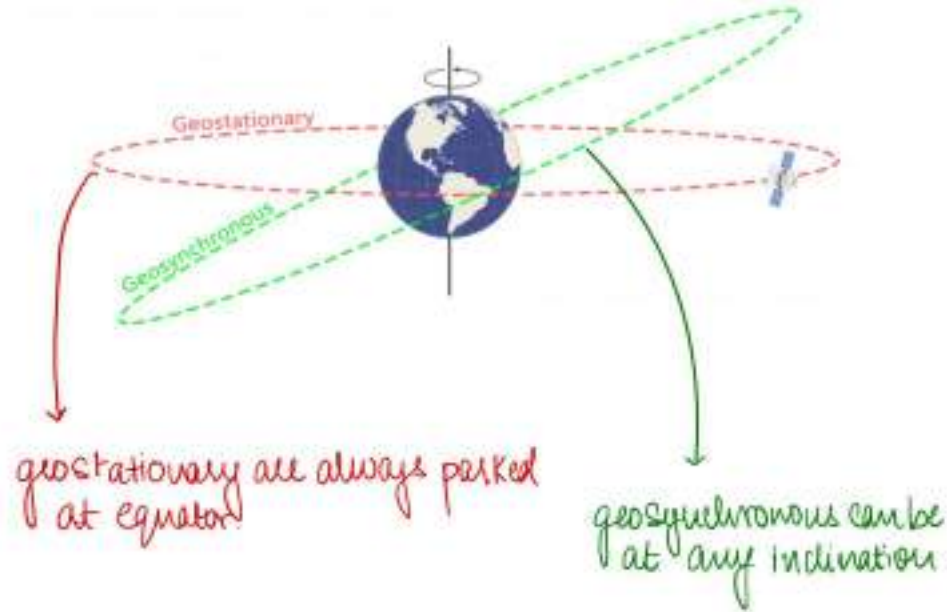
Medium Earth Orbit



- MEO : from 2000 km to 36000 km.
- Includes geostationary and geosynchronous orbits.

Geostationary v/s Geosynchronous

→ Both are at 35,786 km above surface



Centripetal force is provided by gravitational force

$$\frac{GM_e m_s}{(R + h)^2} = m_s (R + h) \omega^2$$

$$\frac{GM_e}{\omega^2} = (R + h)^3$$

$$\left(\frac{GM_e}{\omega^2} \right)^{\frac{1}{3}} = R + h$$

$$h = -R + \left(\frac{GM_e}{\omega^2} \right)^{\frac{1}{3}}$$

$$h = -6.4 \times 10^5 + \left(\frac{6.67 \times 10^{-11} \times 6 \times 10^{24}}{\frac{4\pi^2}{T^2}} \right)^{\frac{1}{3}}$$

$T = 24$ hours for geostationary satellite

$$= 24 \times 60 \times 60 \text{ second}$$

$$h = -6.4 \times 10^5 + 42.31 \times 10^6$$

$$= 35.911 \times 10^6 \text{ m}$$

$$= 36000 \text{ km}$$



Let's See A video and a demonstration



- Demonstration : Geostationary orbit and Geosynchronous orbit
- Video : Orbit from earth's perspective.

MEO : Navigation Systems



- GPS (global positioning system of USA)
- GLONASS (Global Navigation and satellite system of Russia)
- GALILEO (European union)
- BEIDOU -1 (China's regional navigation in central asia)
- BEIDOU - 2 (China's global navigation system)
- NAVIC - IRNSS (India's Regional positioning system)
- QZSS (Regional System of Japan)

India: NAVIC (Navigation Indian Constellation)



- 4 Satellites (Geosynchronous)
- 3 Satellites (Geostationary 36,000 km)
- Coverage: India + 1500 Km from Border.
- Resolution (10 meters).
- All satellites have been placed.

NAVIC (NAVIGATION WITH INDIAN CONSTELLATION)



- Centre is pushing smartphone makers to enable support for its NavIC navigation system in new devices from next year
- Independent stand-alone navigation satellite system Forthcoming change: To add the L1 Spectrum band into NavIC which is part of GPS and is the most used for civilian navigational use.

NAVIC (NAVIGATION WITH INDIAN CONSTELLATION)



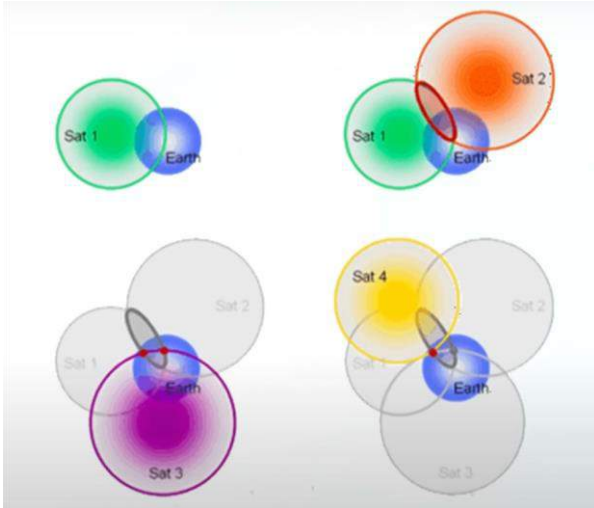
- Expansion will help to increase its use in civilian sector and ships, aircraft travelling far from the country's borders.
- Application: Public vehicle tracking in India, providing emergency warning alerts to fishermen, tracking, and providing information related to natural disasters.

GPS : NAVSTAR of USA

- Satellites are in semi synchronous orbit : half of geosynchronous.
- GPS has 24 satellites
- $R = 20,200$ km above surface
- It takes 12 hours to complete orbit

How does GPS work?

- Concept of Trilateralisation through signal testing and comparing time delays



Application of GPS systems

1. Location Services

Standard Positioning	Restricted Positioning
Civilians	Military (Use of NAVIK)
	In Kargil war, america denied GPS services to us, therefore a used of self reliance was felt.

Application of GPS systems

2. Timing Service

- Corroborates Local time.
- Use of Atomic Clocks

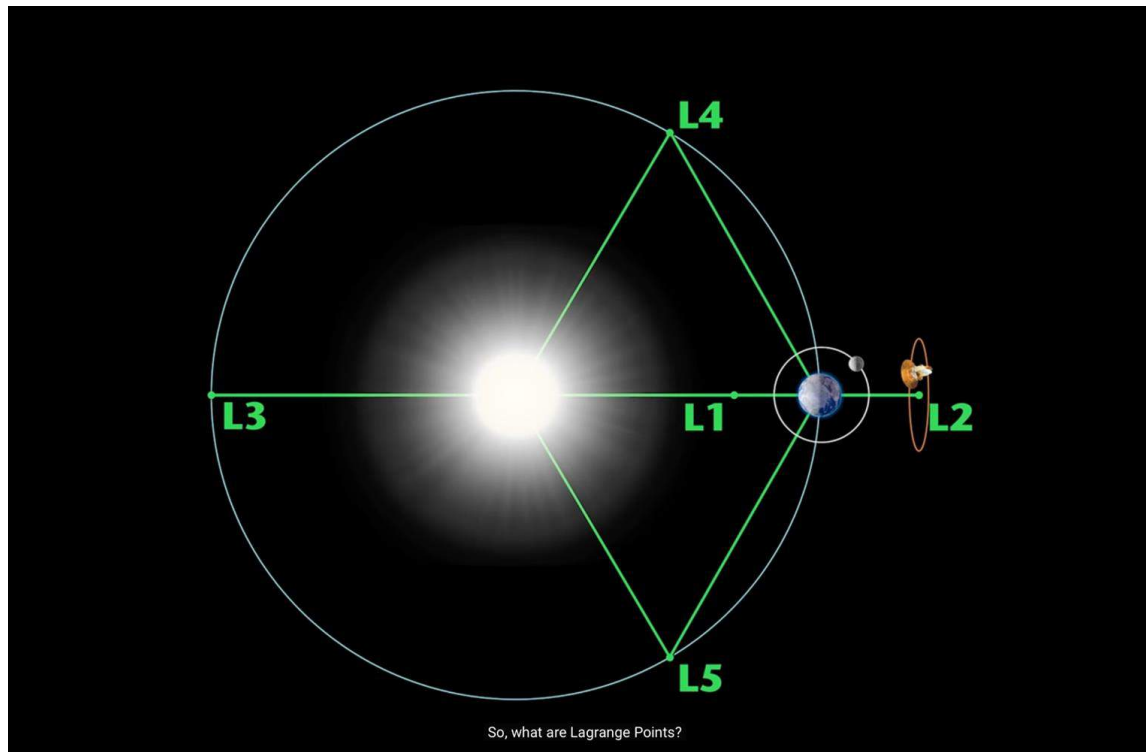
Application of MEO Continued:

Geo Stationary (Communication)

- (a) Circular orbit (35786)
- (b) 36000 km/22222 miles
- (c) 24 hours : time (Rotation and Revolution) : Speed of satellite matches that of earth.
- (d) Direction : West to East

High Earth Orbit

- Orbit more than 36,000 km.
- Applications : Lagrange points : lets see a video



High Earth Orbit

- L1 = 16 Lakh km from surface (Moon = 3.84 lakh km)
- L2 = Same distance otherside.
- Satellite at this point can constantly monitor the sun
- No impact of solar eclipse.

High Earth Orbit

- There are 5 Lagrange points in outer space. Here, The orbit, gravity of sun and the gravity of earth are in balance.
- If spacecraft is placed on such points, it will remain in a HALO orbit and Earth and the spacecraft together will go round the sun.
- L1, L2, and L3 points are in unstable equilibrium whereas L4 and L5 points are in stable equilibrium.
- L4 and L5 are extremely vulnerable to asteroids. Therefore even though they have stable equilibrium, L5 is preferred.

Equilibrium types



Stable



Unstable



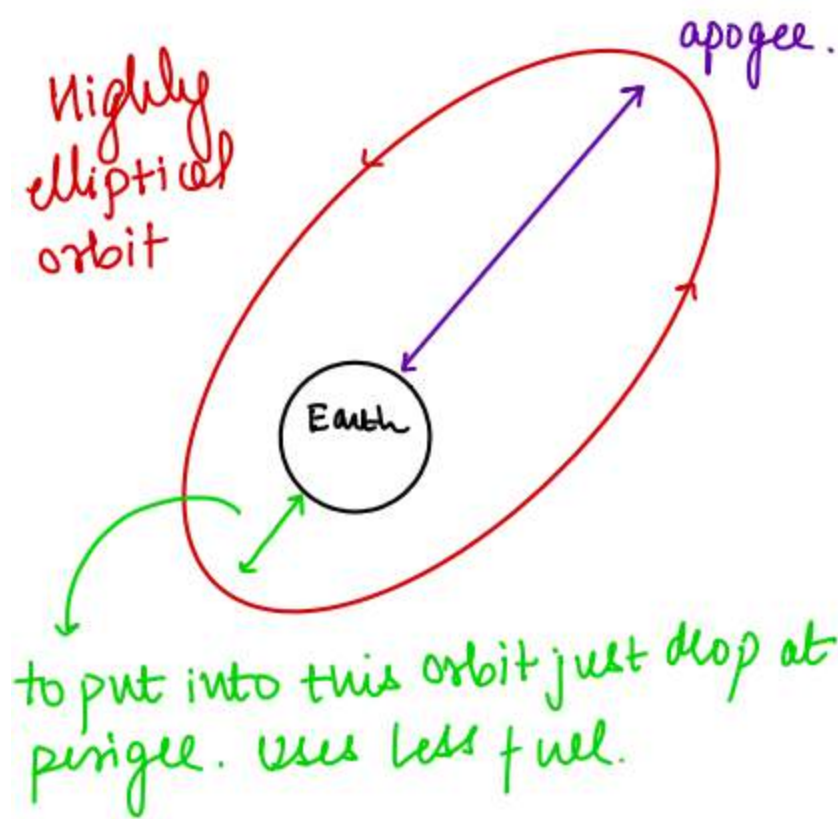
Neutral



Application of HEO: Continued.

Geosynchronous transfer orbit (GTO)

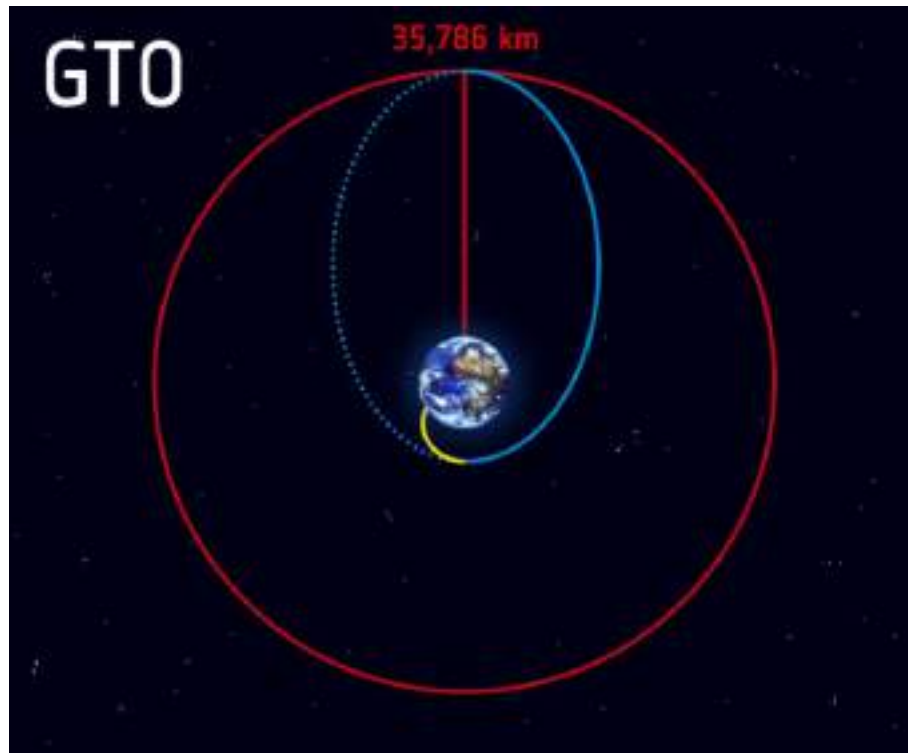
- A transfer orbit is HEO, geosynchronous with perigee = 300-400 km and apogee = 36000 km to 38000 km.
- This orbit is used for initial launch of a communication satellite or a spacecraft.
- This orbit is used by space agencies to manoeuvre a satellite for various applications.



Application of HEO: Continued.

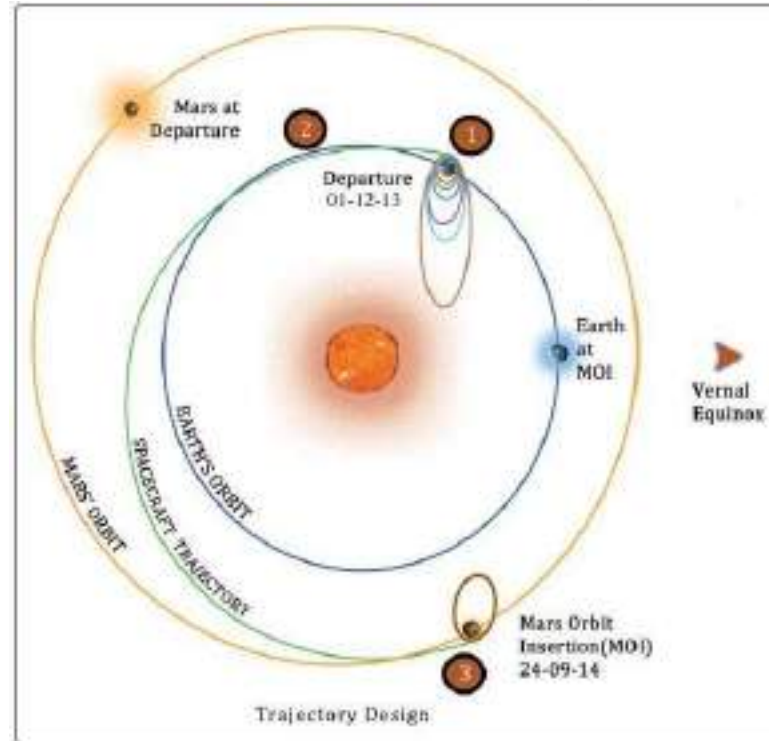
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- This orbit is used by space agencies to manoeuvre a satellite for various applications.



ISRO used geosynchronous transfer orbit in Mangalyaan Mission : Cost came to be close to ₹7 / Km only.

However : there was a risk : going from heliocentric to mars orbit was the most crucial step.



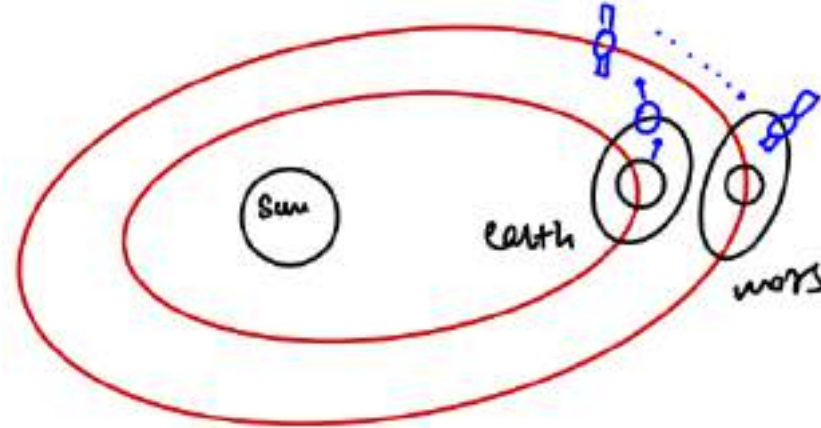
However, Certain risks are present

Geocentric - Heliocentric

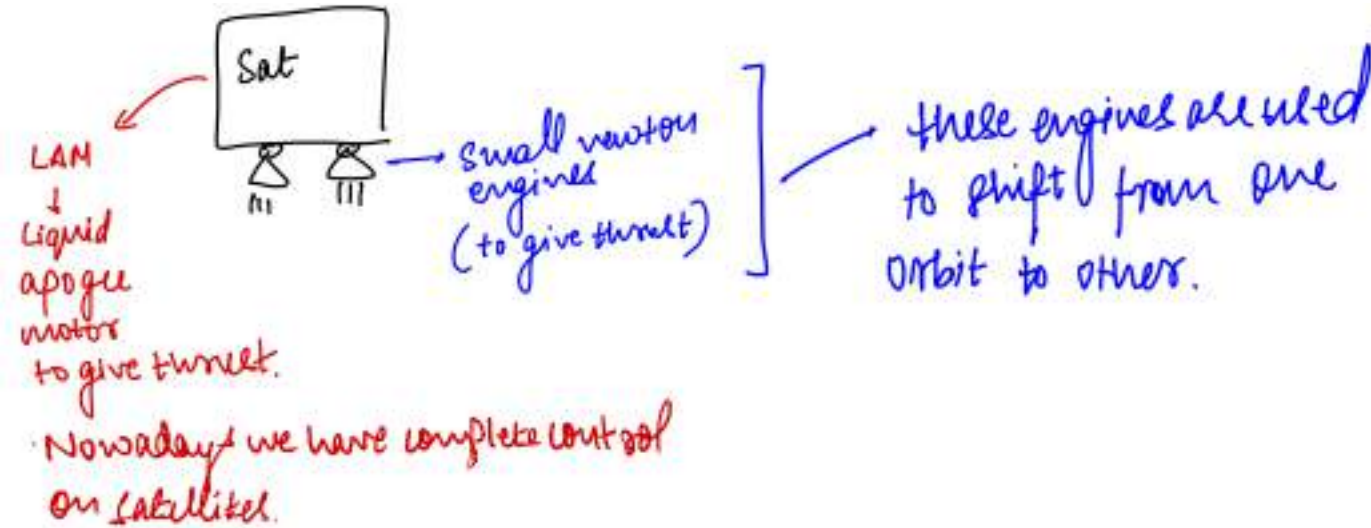
Issue - Shifting of orbits

NASA - Does not use GTO often

ISRO - Save cost



Shifting Orbit



LAUNCH VEHICLES

- Space vehicles to carry spacecraft or humans to outer space. (aka rocket)

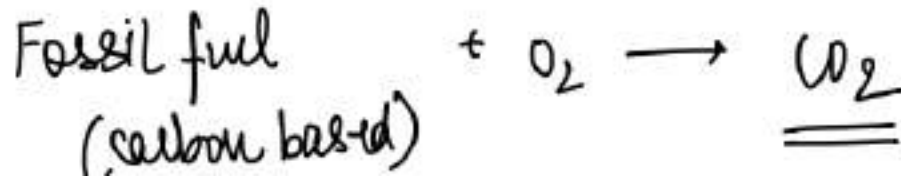
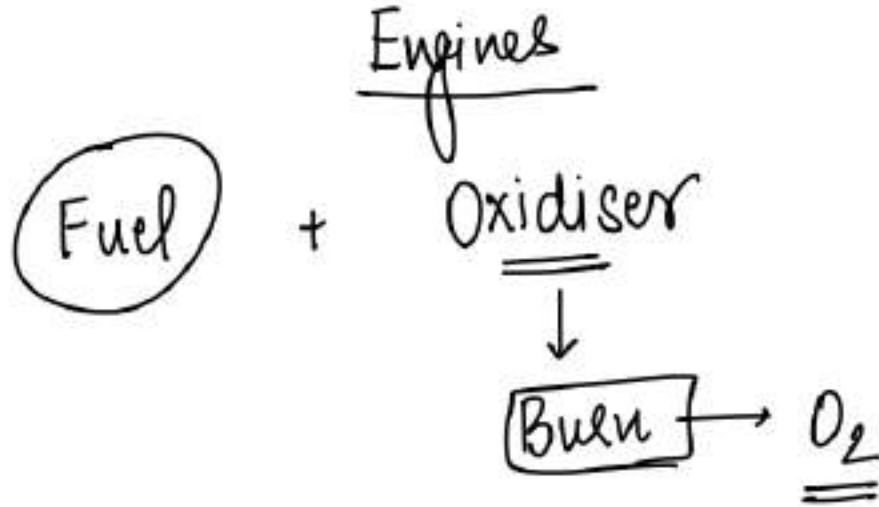
Types	
ELV (Expendable launch Vehicle)	Reusable launch vehicle
Used only once for a space mission Eg: PSLV GSLV	Can be used multiple times



ELV	RLV
• Space debris issues • Used for Unmanned missions • Indian only has ELV, working to develop RLV	• Returns back to earth • Can be used for both manned and unmanned mission • Advantage is that it is reusable and saves lot of money

Major Components of Launch Vehicles

- Engines
- Fuel
- Satellite / Payload
- Command and Control System



Components : Engines

Engines

Non air breathing propulsion system

Air breathing propulsion system (Ramjet/Scramjet)

Use of Oxidiser to burn fuel that is kept on the rocket

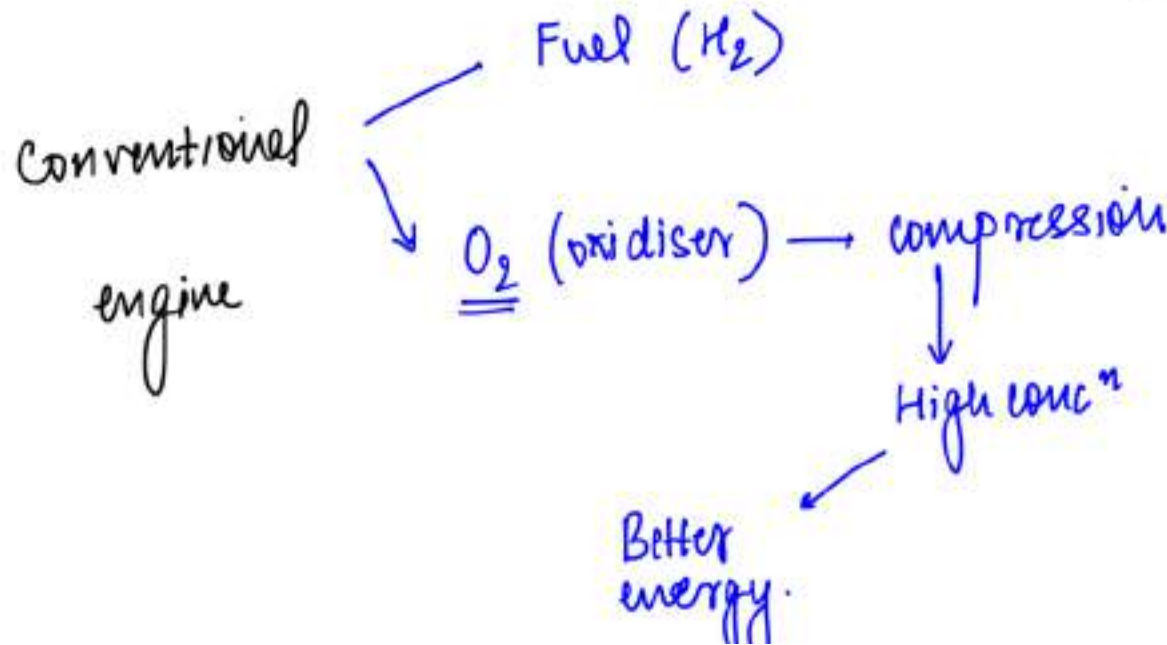
Air breathing : word breakdown

Oxidiser adds to mass of rocket

Oxygen from air to burn fuel

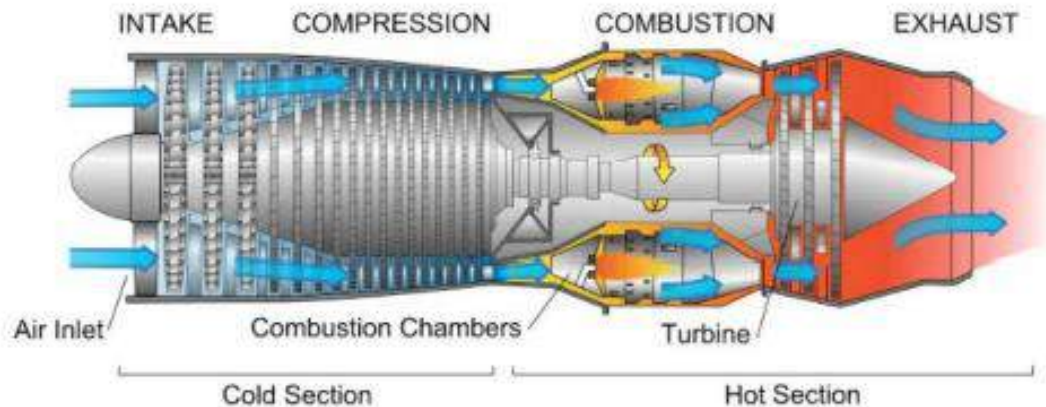
Heavier

Lighter



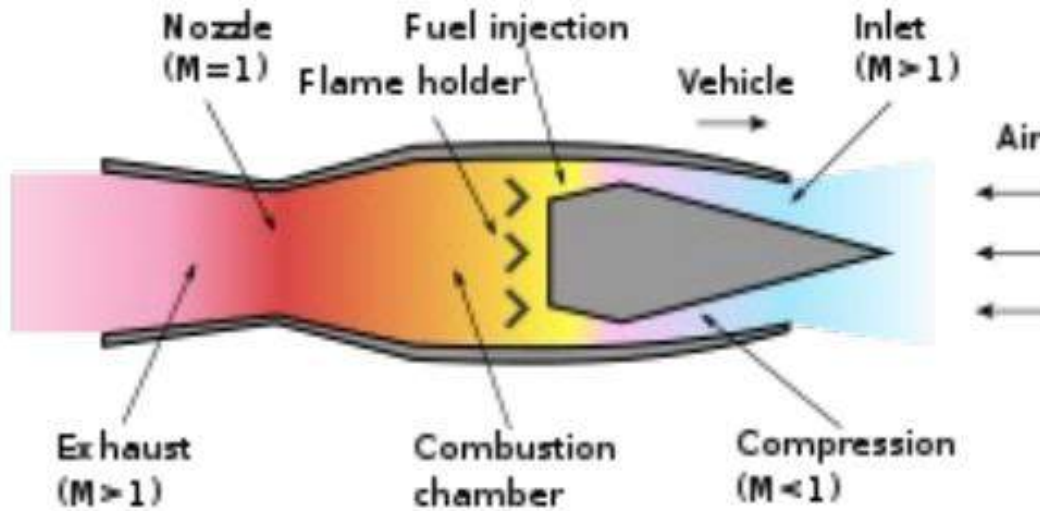
Jet Engines

- A jet engine is a type of reaction engine discharging a fast moving jet that generates thrust by jet propulsion.
- While this broad definition can include rocket, water jet, and hybrid propulsion, the term jet engine typically refers to an internal combustion airbreathing jet engine such as a turbojet, turbofan, ramjet, or pulse jet. In general, jet engines are internal combustion engines



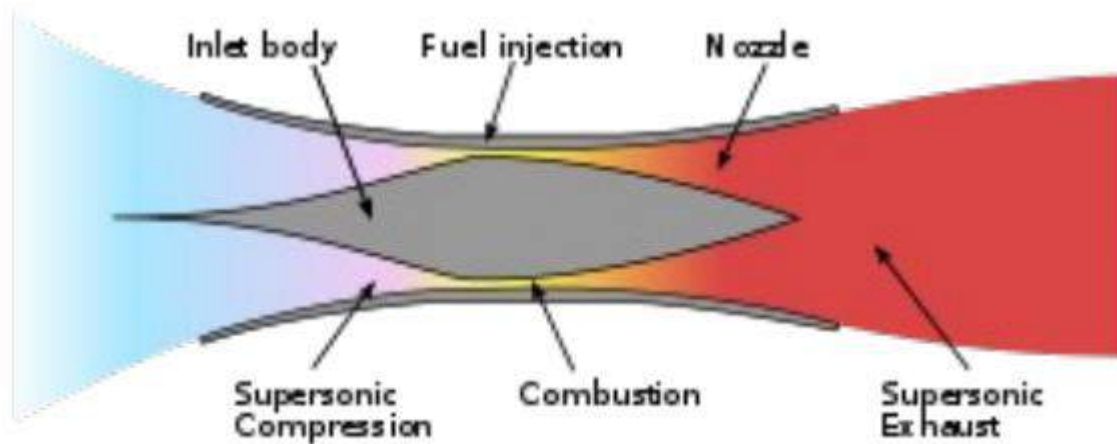
Ramjet Engine (word breakdown)

- A ramjet, or athodyd (aero thermodynamic duct), is a form of airbreathing jet engine that uses the forward motion of the engine to produce thrust. Since it produces no thrust when stationary (no ram air) ramjet-powered vehicles require an assisted take-off like a rocket assist to accelerate it to a speed where it begins to produce thrust. Ramjets work most efficiently at supersonic speeds around Mach 3 (2,300 mph; 3,700 km/h) and can operate up to speeds of Mach 6 (4,600 mph; 7,400 km/h).



Scramjet Engine

- A scramjet (supersonic combustion ramjet) is a variant of a ramjet airbreathing jet engine in which combustion takes place in supersonic airflow. As in ramjets, a scramjet relies on high vehicle speed to compress the incoming air forcefully before combustion (hence ramjet), but whereas a ramjet decelerates the air to subsonic velocities before combustion using shock cones, a scramjet has no shock cone and slows the airflow using shockwaves produced by its ignition source in place of a shock cone. This allows the scramjet to operate efficiently at extremely high speeds



SOLID FUEL DUCTED RAMJET (SFDR) TECHNOLOGY

- It is a missile propulsion system that includes a thrust modulated ducted rocket with a reduced smoke nozzle-less missile booster.
- First flight of SFDR, developed under a joint Indo-Russian R&D project, was tested in 2018. It had achieved the speed of Mach 3.
- Solid fuelled airbreathing Ramjet Engine
- Range : 70 - 340km

Dual Mode Ramjet (DMRJ)

- When a ramjet transforms into scramjet over mach 4-8 range.
- Operate both in subsonic and supersonic combustor modes.

2nd Component: Fuel (Propellant)

Solid Fuel

- Eg: HTPB : Hydroxy terminated poly- butadione (used in PSLV, GSLV)

Advantage

- Easy Storage and transfer
- All rockets use solid fuel in 1st stage as it gives more energy.

Disadvantage

- Heavy weight - Cost increase
- Non uniform beginning.

2nd Component: Fuel (Propellant)

Liquid Fuel

- UDMH (N2O4) (Unsymmetrical DiMethyl Hydrazine)
Used in PSLV and GSLV

Advantage

- Higher efficiency : Uniform Burning : **Demonstration**
- Less weight, So cheaper storage

Disadvantage

- Difficulty in storage (flammable/corrosive) and transport

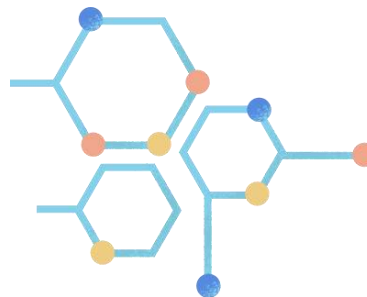
Cryogenic Engine

Indigenous developed by India!

- Stores fuel (Liquid hydrogen) and oxidiser (Liquid O₂) at very low temp.
- H₂ = 20K. (-253 degree C)
- O₂ = 90K (-183 degree C)
- Generates High energy thrust by burning Hydrogen
- Hydrogen is
 - Abundant
 - Environment Friendly (burns to produce H₂O)

Why Cryogenic Engine?

- Hydrogen and Oxygen are stored as Liquid. Why? If they were stored as gases, it would require a huge container and the chances of leakage would be very high.
- Why Oxygen and Hydrogen? : Highest efficiency
- Provides best thrust per kilogram of fuel combination.
- Challenge of American Obstructionism due to non membership of the MTCR.



UPSC Prelims

Science & Tech

Class No. 9

Space Technology

Indian Launch Vehicle

1. Historical

- SLV : Satellite launch vehicle
- ASLV : Augmented Satellite launch vehicle

2. Operational

- Sounding rockets : Every Wednesday tested in thumba (60s) :
Sounding rocket means Taking measurement
- PSLV (1993)
- GSLV (2001)
 - Mark - 3 developed recently

PSLV and GSLV

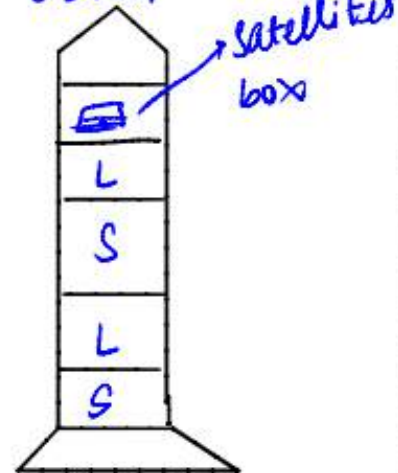
PSLV	GSLV
• Polar satellite launch vehicle • Can be used for lighter satellites only • Remote sensing satellites in polar sun synchronous orbit	• Geosynchronous satellite launch vehicle . • Can be used for heavier satellites also (2 ton-4 ton) • Heavier Communication Satellite in Geosynchronous transfer orbit

PSLV

4 Stage:

Solid → Liquid → S → L

HTPB — UDMH



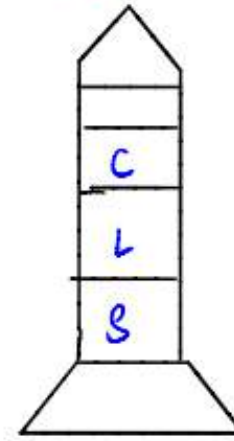
3 versions:

(1) core alone
(without strap on
motors)

GSLV

3 stage

Solid → Liquid →
cryogenic

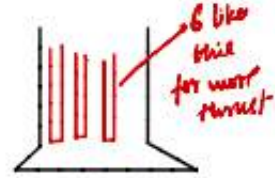


versions

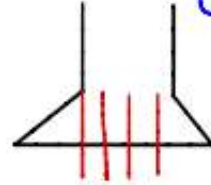
(1) Mk-I
(2 tons)



(i) standard config
↓
with strap on motors
(6)

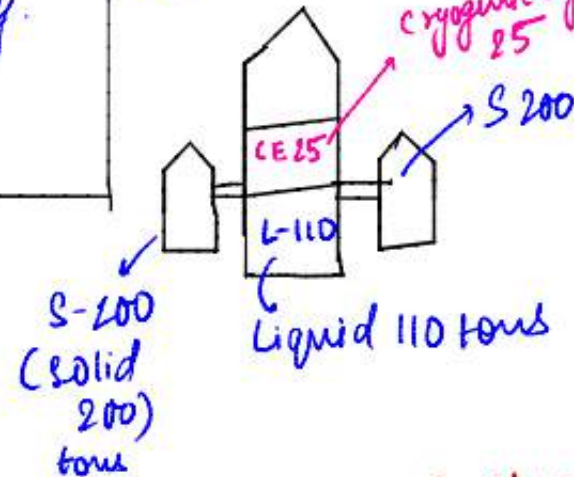


(ii) Mark - II
4 strap on motors
(3 ton)

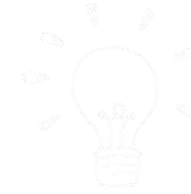


(iii) extra large
(6) strap on motors but
more fuel capacity.

(iii) Mark - III



these are heavy boosters.
using solid fuel.

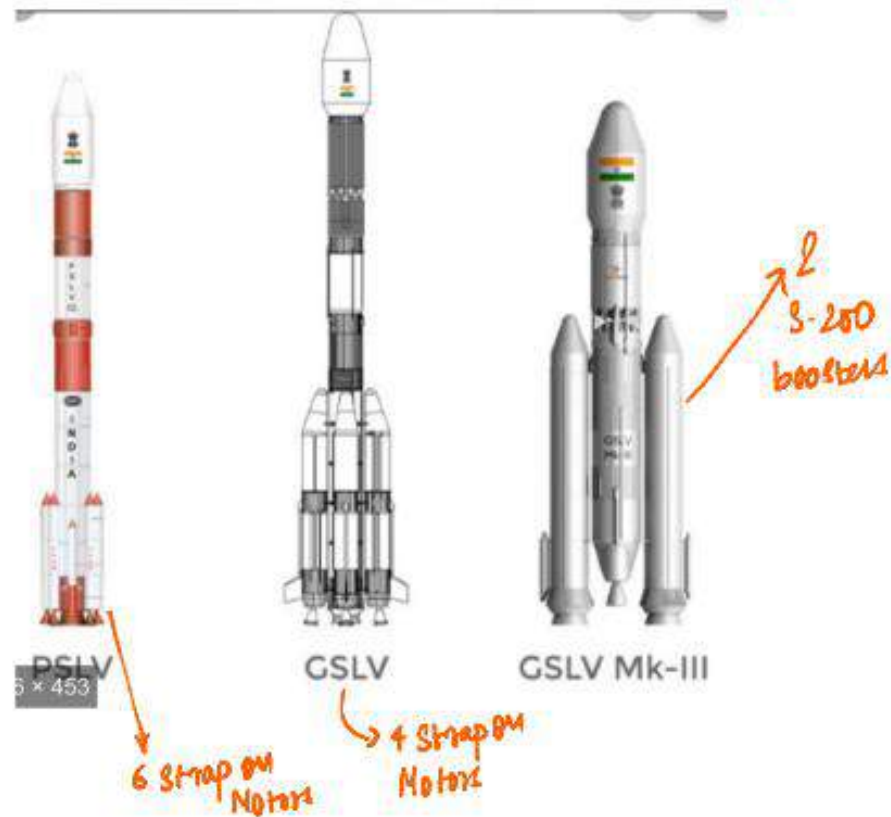
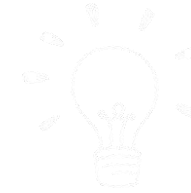


process of MK-3

S-200 → L-110 → CE-25

S-200

capacity of MK-III : 4 ton to 5 ton.



GSLV MK-III

- Heaviest launch vehicle of India (640 tones) (42 inter long)
- Can launch 4000-5000 kg class satellites
- 3 stages launch
 - S-200 - boosters (Solid)
 - L-110 - Liquid
 - CE - 25 Cryogenic

Importance

- Heavy communication Satellites (GSAT)
- Saves India's foreign exchange
 - Makes us self reliant
- Help each foreign exchange by launching foreign satellites
 - (Int space **sunset**) India **arriu** they each only 1/. Of world space revue .

PSLV in Current Affairs

- PSLV on its 55th mission successfully launched 3 Singaporean satellites in the second commercial mission of NSIL.
- Besides this, ISRO also achieved successful launch of PSLV orbital experimental module (POEM)

PSLV in Current Affairs

About POEM: A platform which allows in-orbit scientific experiments using the final (fourth) stage of PSLV.

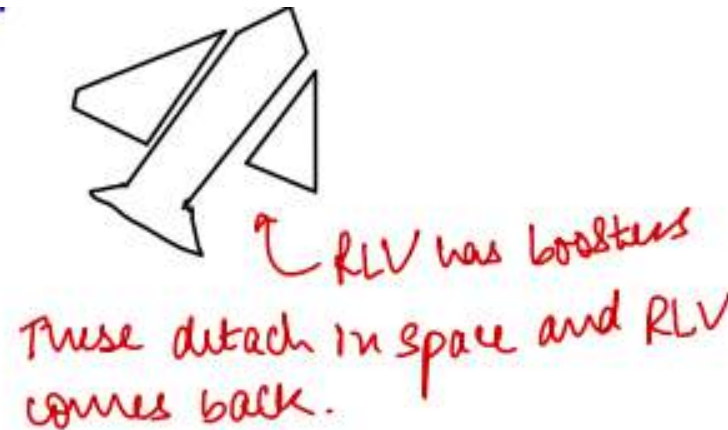
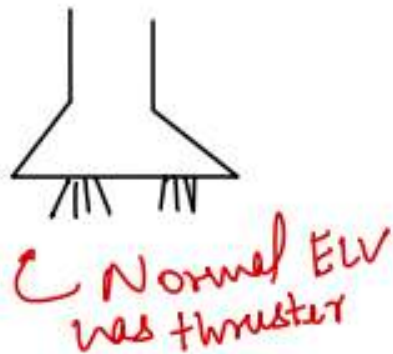
- Dedicated Navigation Guidance and Control system: Act as platform's brain to stabilize it with specified accuracy.
- Power source: Mounted solar panels and a Li-Ion battery.

ISRO's Launch Vehicles

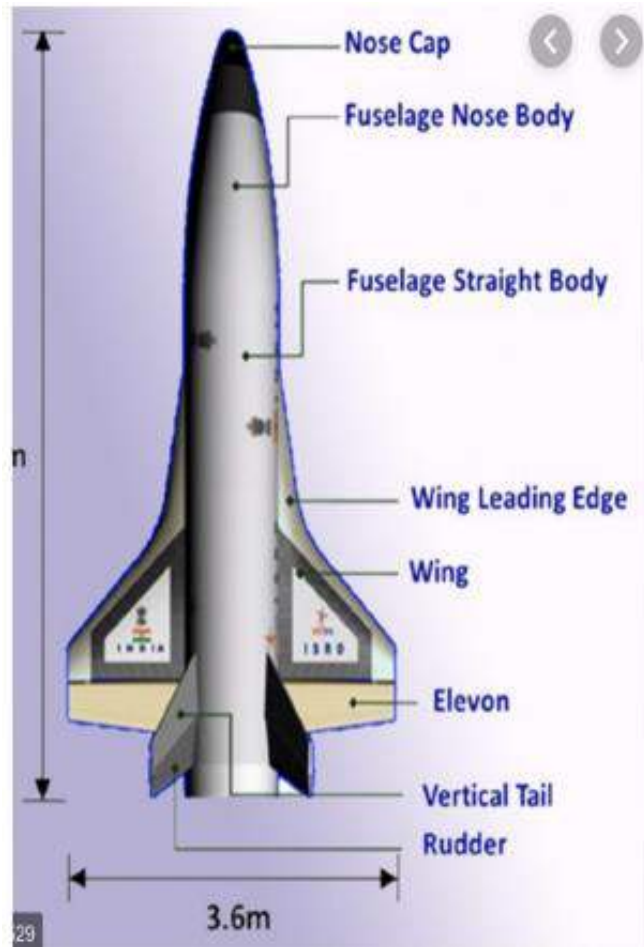
Launch Vehicle	Stages	Payload capacity and orbits
Small Satellite launch Vehicle (SSLV)	3 stages Launch Vehicle (Three Solid Propulsion stages and liquid propulsion based velocity module as a terminal stage)	Mini, Micro, or Nanosatellites (10 to 500 kg mass) to a 500 km orbit.
Polar Satellite Launch Vehicle (PSLV)	Four stage launch vehicle (1st & 3rd stage; Solid, 2nd & 4th stage; liquid)	Up to 1750 kg to Sun synchronous Polar orbits, 1425 into Geosynchronous and Geostationary orbits.
Geosynchronous Satellite Launch Vehicle (GSLV) Mark II	Three stage launch vehicle (1st solid, 2nd liquid, 3rd Cryogenic upper stage)	Up to 2250 into geosynchronous transfer orbits up to 6 tonne in low earth orbits
Geosynchronous Satellite Launch Vehicle MK-III (LVM3)	Three stage launch vehicle (1st liquid, 2nd solid, 3rd Cryogenic, Upper stage)	Up to 4 tonne into Geosynchronous Transfer orbits, 8 tonne into low earth orbits.

RLV : Reusable Launch Vehicle

- RLV uses a winged air-craft which re enters and returned back to the earth
- Use Boosters to go to outer space.
- 2 stage to orbit launch vehicle



ISRO Proposed RLV



Advantage of RLV for India:

It is like Nasa's space shuttle. NASA not using space shuttle and using private companies - Falcon - 9 by Space-X

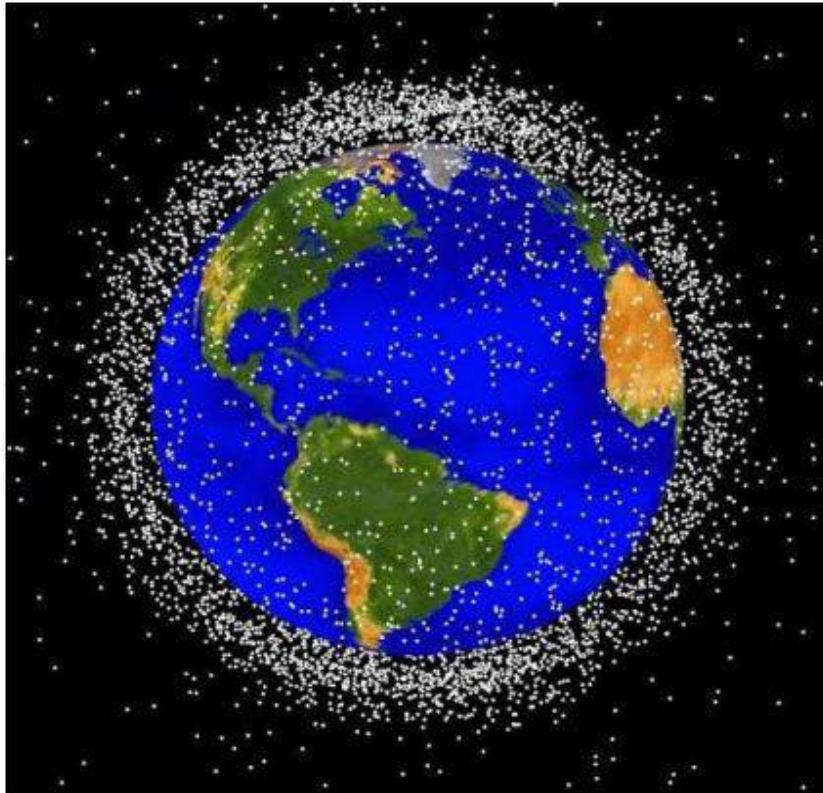
- Commercial launch of satellites will be low cost. (RLV is more cost effective wrt ELV)
- International space market - Capture foreign reserve.
 - Compete with space X
- Space tourism - India has cheaper technology than other nations

Challenges for RLV Development

For Re-Entry of RLV : 10km runway is needed.

- **Infrastructure** challenges: Long runway needed on coast to avoid inland disasters. India's undulating coastline will be an issue.
- **Cost benefit** of investing in RLV instead of Socio-Economic Programs will be questioned. As it is extremely expensive .
- **Logistic** Challenges : Higher Logistic costs
- **Training** Challenges : Training and maintenance cost.
- By the time India is able to develop its RLV, Space X might already capture the market. Competition from International space entities. (Falcon-9 Heavy)

Space Debris



Space Debris

- Why in News? : China rocket debris fell in Indian Ocean near Maldives.
- Parts of Long March 5B re-entered the atmosphere and landed in the Indian Ocean, with the bulk of its components destroyed upon re-entry into the Earth's atmosphere. Long March 5B was carrying components for the construction of China's space station, which will be completed by the end of next year.
- Space debris encompasses both natural (meteoroid) and artificial (man-made) particles. Meteoroids are in orbit about the sun, while most artificial debris is in orbit about the Earth which is commonly referred to as orbital debris.

Space Debris

- Much of the debris is in low Earth orbit (LEO), though some debris can be found in geostationary orbit.
- There are about 500,000 pieces of debris the size of a marble or larger orbiting the Earth, travelling at speeds up to 17,500 mph.
- International guidelines suggest removing space crafts from LEO within 25 years of the end of their mission. However, only 60 percent of missions follow the guidelines.

Tackling Space Debris

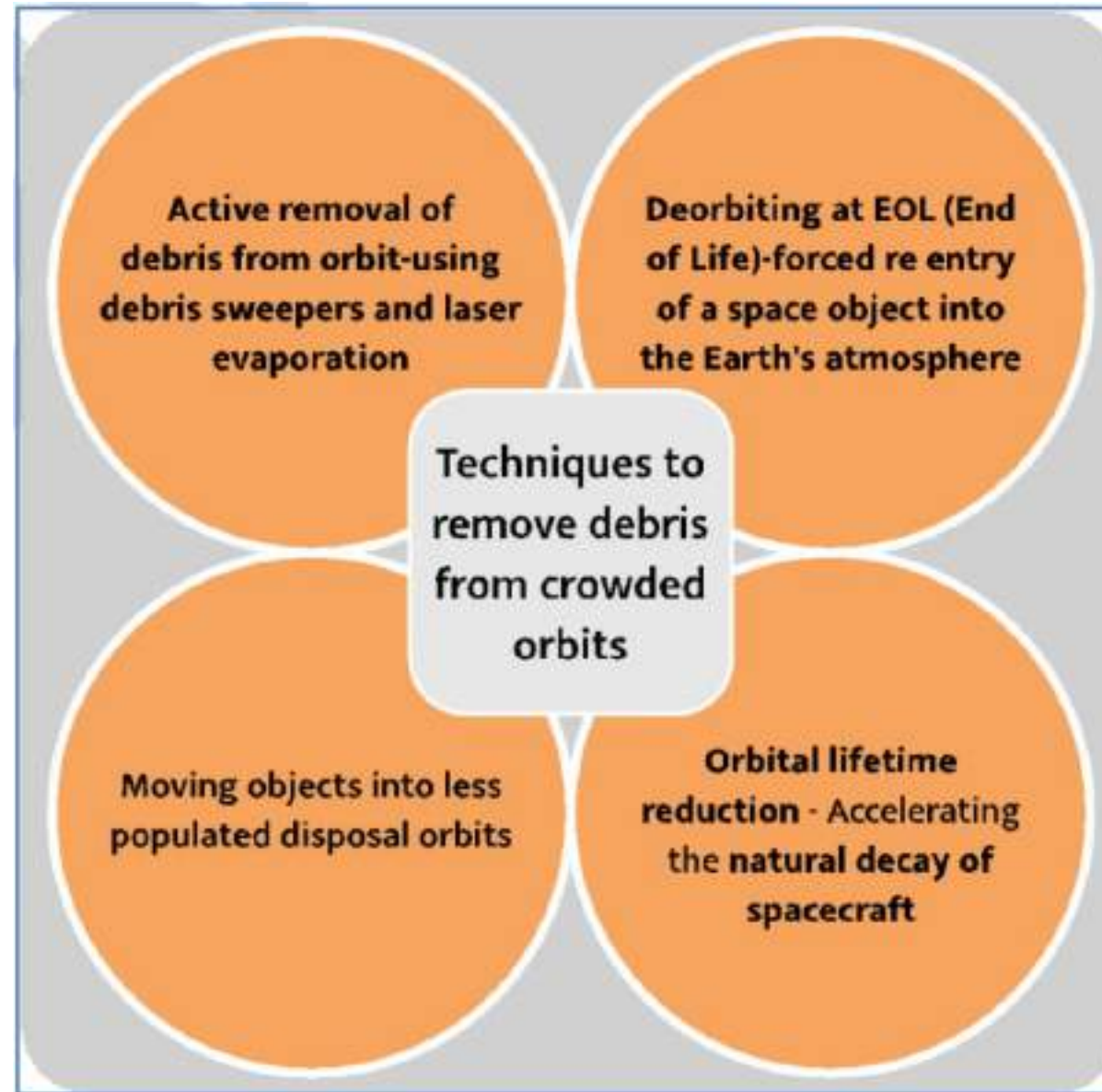
- Mitigating Damage: Space debris is tracked by a number of countries, including Germany, France, UK and USA.
- ISRO has come up with ‘Project NETRA’ - an early warning system in space to detect debris and other hazards to Indian satellites.
- Indraprastha Institute of Information Technology Delhi is working on a project titled “Orbit computation of Resident Space Objects for Space Situational Awareness” to predict collision from space debris.

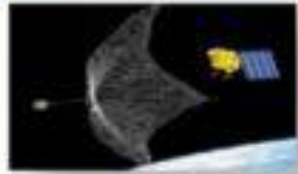
Tackling Space Debris

- Avoiding future debris: by adopting better designs of rockets and other objects. For example, making rockets reusable could vastly cut down waste. o UK's TechDemoSat-1 (TDS-1), was designed in such a way that once its mission is over, a system, would drag the satellite to re-enter the atmosphere and burn up.

Tackling Space Debris

- Removal of the debris:
- End-of-Life Services by Astroscale Demonstration (Elsa-D) was launched to locate and retrieve used satellites and other space junk.
- Remove Debris is an EU research project to give in orbit demonstrations of cost-effective technologies that can be used to observe, capture and dispose of space debris. It has performed key technology demonstrations including: Net Capture, Harpoon Capture,





Net Capture

It involves a net that will be deployed at the target CubeSat.



Harpoon Capture

Which will be launched at a target plate made of "representative satellite panel materials".



De-orbiting process

As it enters Earth's atmosphere, the spacecraft will burn up, leaving no debris behind.



Vision-Based Navigation

Using cameras and LiDAR, the platform will send data about the debris back to the ground for processing

Way forward for space junk

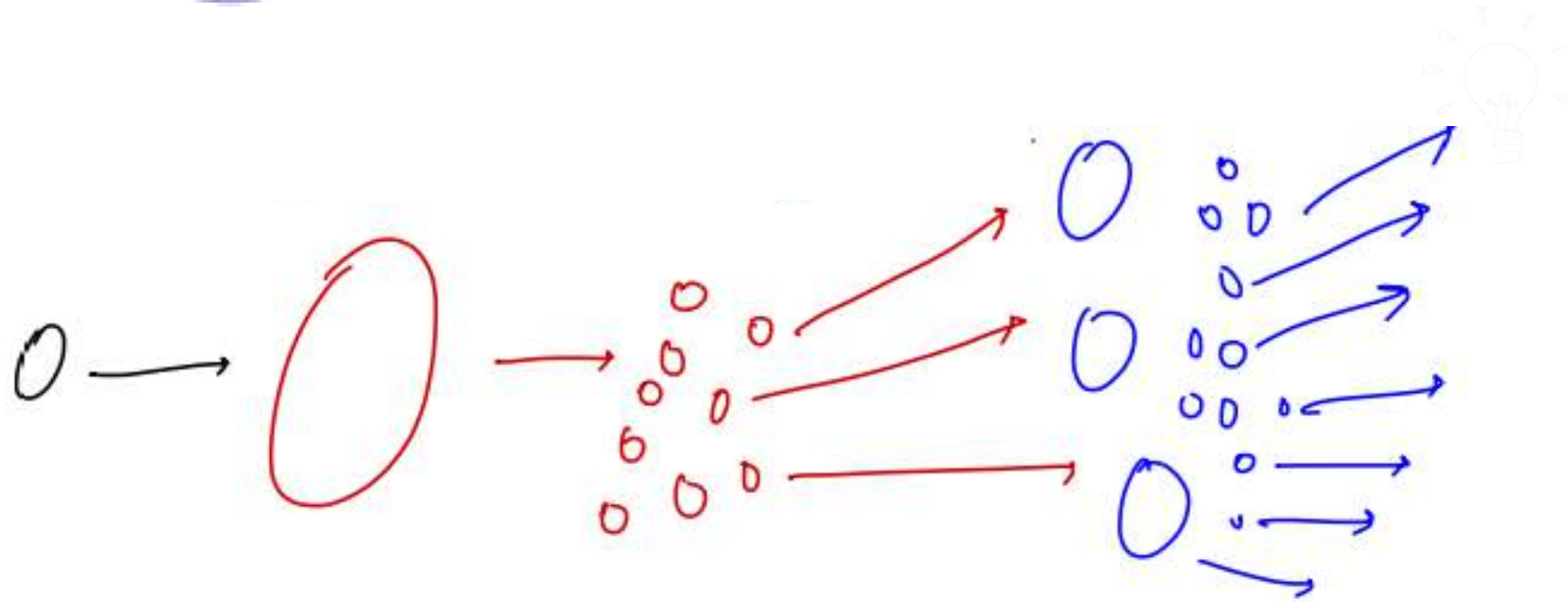
- 1. Kumotori Experiment (White Stork Jexa):** Use of tether to attach space junk magnetic tether) once tethered, it will be pulled down to atmosphere and burnt out by air friction. Japan tried it in 2017, but experiment FAILED.
- 2. Remove debris project :** Harpoon Satellite to catch debris (net capture - of space debris)
3. Providing high thrust and escape velocity to send to outer universe.
4. Production of Satellite that do not produce debris -
Ex. RLV even satellites need to be reusable.

Kessler Syndrome: Donald Kessler of NASA.

- Collision in space generates fragments that will collide again, leading to a cascade of collisions and proliferation of debris
- Outer space treaty by UN trying to clear space.

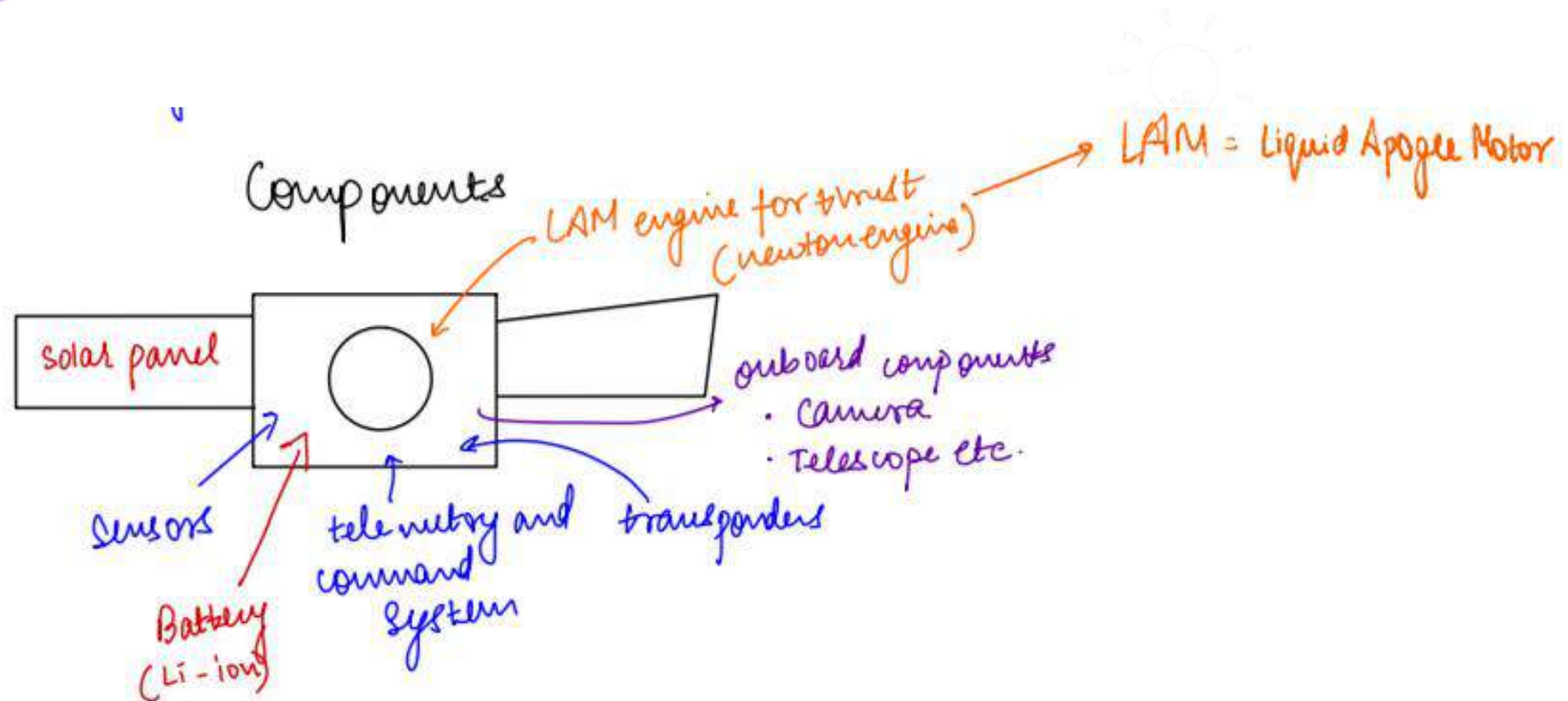
UNOSA : UN office on outer space activities was set up committee to investigate space debris.

- Space debris with large speed are the most dangerous as there is maximum chance of collisions.



SATELLITES

- Space crafts designed to carry payloads such as transponders , covers, instruments for various space applications.
- Different types of satellites for applications :
 - Communication
 - Meterology
 - Astronomy
 - Remote Sensing
 - Navigation
 - Defence and Reconnaissance
 - Anti Satellites



Transponders:

Transmitters and receiver.

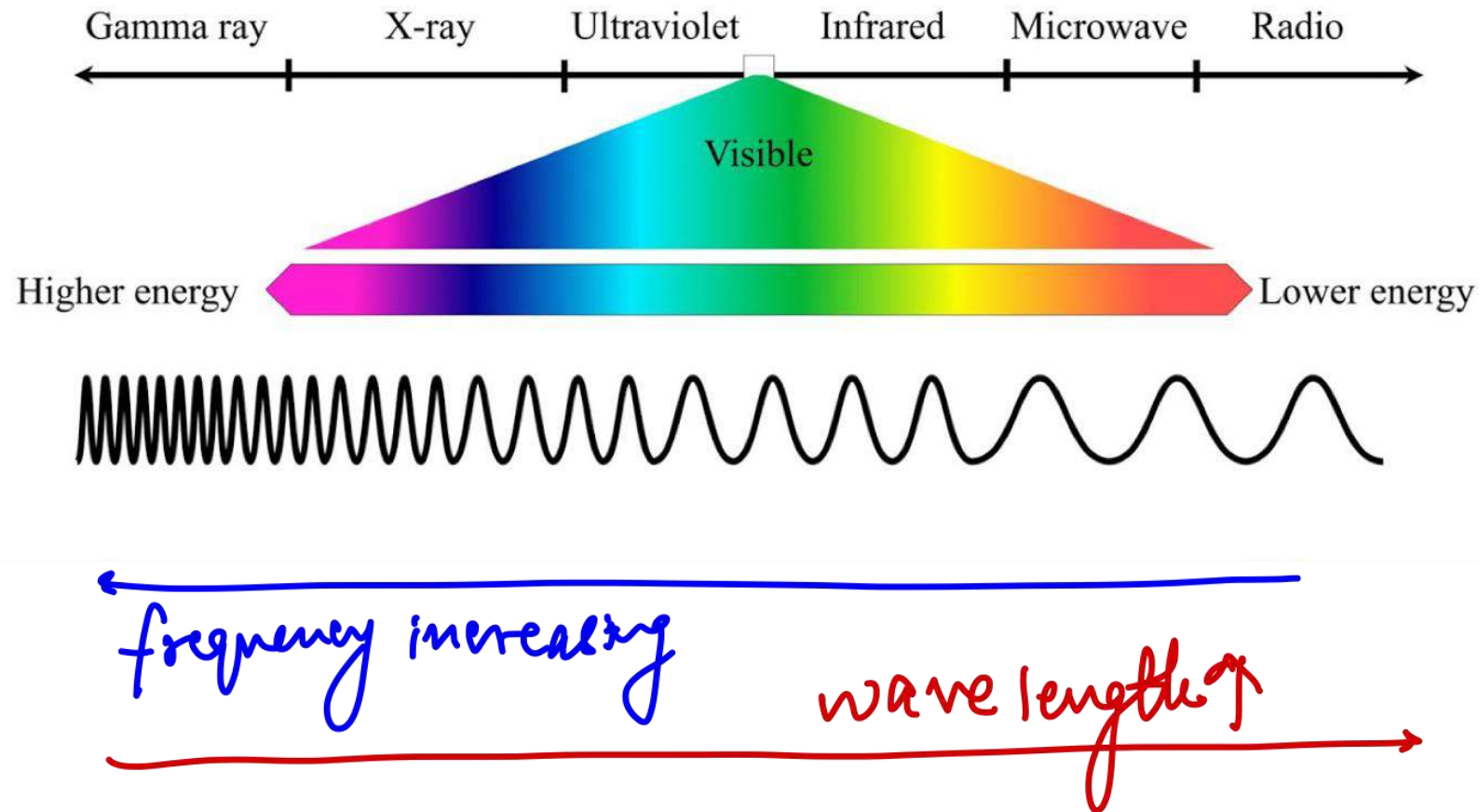
- a) Used for receiving and transmission on of signals.
- b) Many frequency bands : different forms of transmission due to different energy characteristics.
- c) Let's understand waves and the EM spectrum to understand this in detail.

Waves

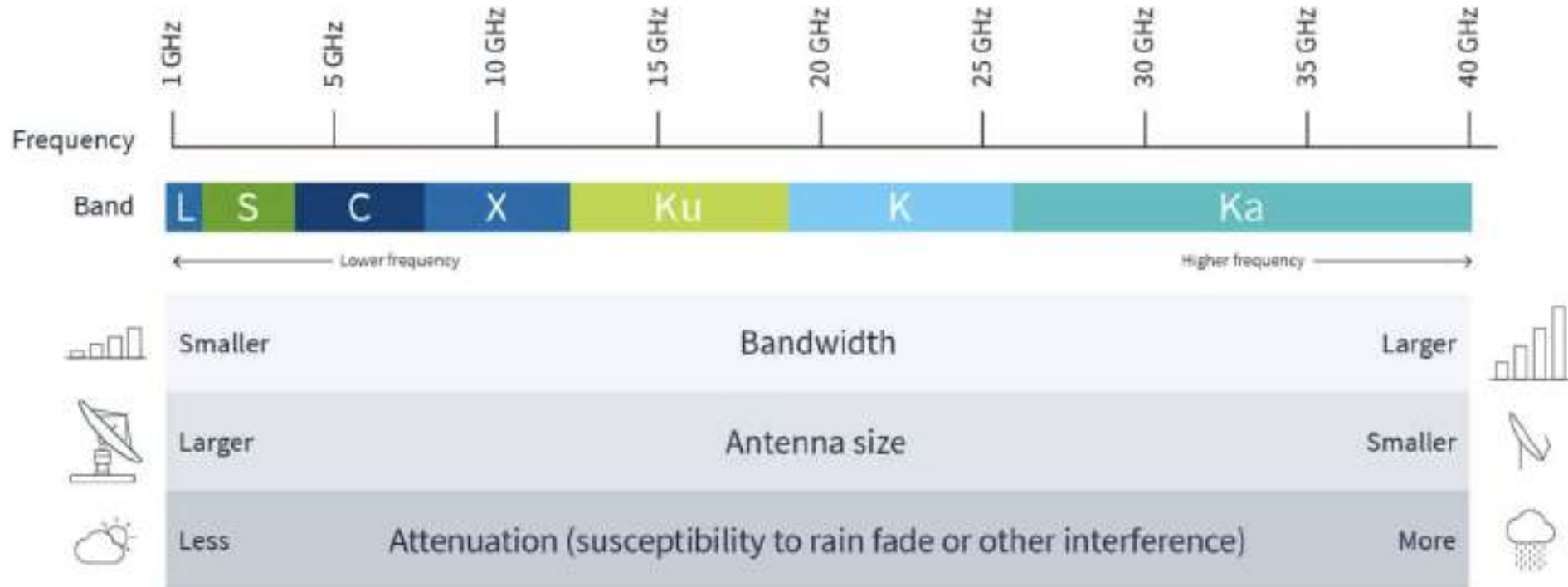
Some Basics :

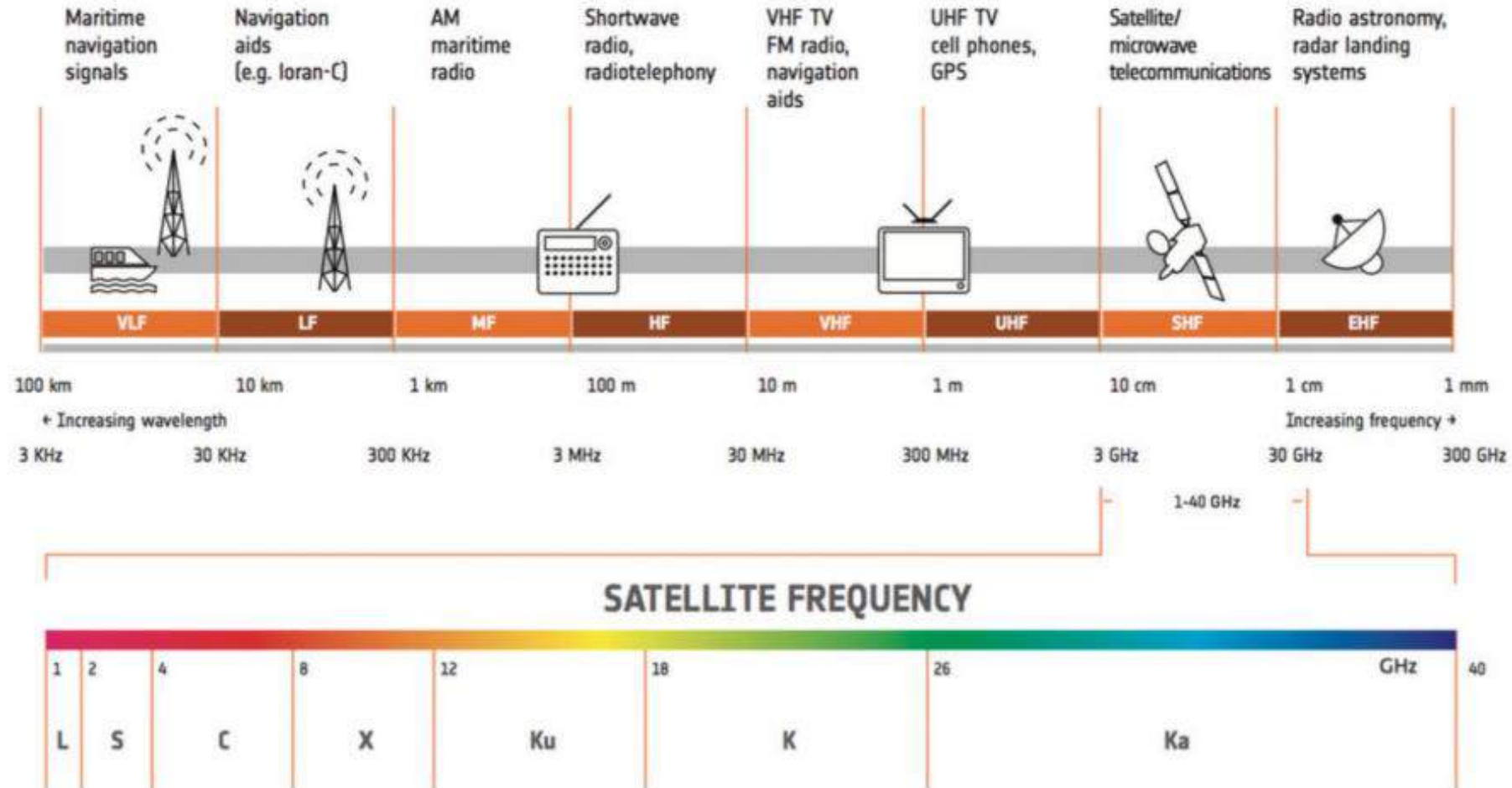
- Antenna Size is directly proportional to Wavelength.
- Frequency increase : more energy : more throughput (data transfer per unit time)
- Above 11Ghz : Wavelength are very small, Frequency is very high. Here there can be a phenomenon of RAIN FADE due to rain interfering with small wavelength particles.
- So we have to find the right balance.

Electromagnetic Spectrum



Satellite spectrum properties





Different Types of Bands

These are microwaves and radio waves

L-band (1–2 GHz)

Global Positioning System (GPS) carriers and also satellite mobile phones.

S-band (2–4 GHz)

Weather radar, surface ship radar, and some communications satellites, especially those of NASA for communication with ISS and Space Shuttle.

C-band (4–8 GHz)

Primarily used for satellite communications, for full-time satellite TV networks or raw satellite feeds. Commonly used in areas that are subject to tropical rainfall, since it is less susceptible to rain fade than Ku band

Different Types of Bands

X-band (8–12 GHz)

Primarily used by the military. X-band radar frequency sub-bands are used in civil, military and government institutions for weather monitoring, air traffic control, maritime vessel traffic control, defense tracking, and vehicle speed detection for law enforcement.

Ku-band (12–18 GHz)

Used for satellite communications

Ka-band (26–40 GHz)

Communications satellites for close up high resolutions applications, uplink in either the 27.5 GHz and 31 GHz bands, close-range targeting radars on military aircraft.

ISRO

Indian Remote Sensing Satellite (1988)

- (i) Survey and management of Natural resources. (Forest, agro, wetlands)
- (ii) Mineral prospecting : Locating minerals via Satellite (Resource SAT - 1, 2A, 2B)
- (iii) Cartography (Contour maps) : (Cartosat) (gis: geographical into system)

Bhuvan System : Collaboration between map my india and ISRO : Indigenous GeoSpatial Portal

IRS

(iv) Oceanography : mapping and study of geography.

Eg: OCEANSAT (4-5)

Scatsat-1 has replaced OCEANSAT 2-A

SARAL: Satellite for ARGUS and ALTIGA (ISRO) and CNES (france)

(v) Environment : Pollution management.

3 Countries have dedicate climate change satellites - US, China, Japan.

IRS

NISAR : observation of changes our earth (Land S band.)

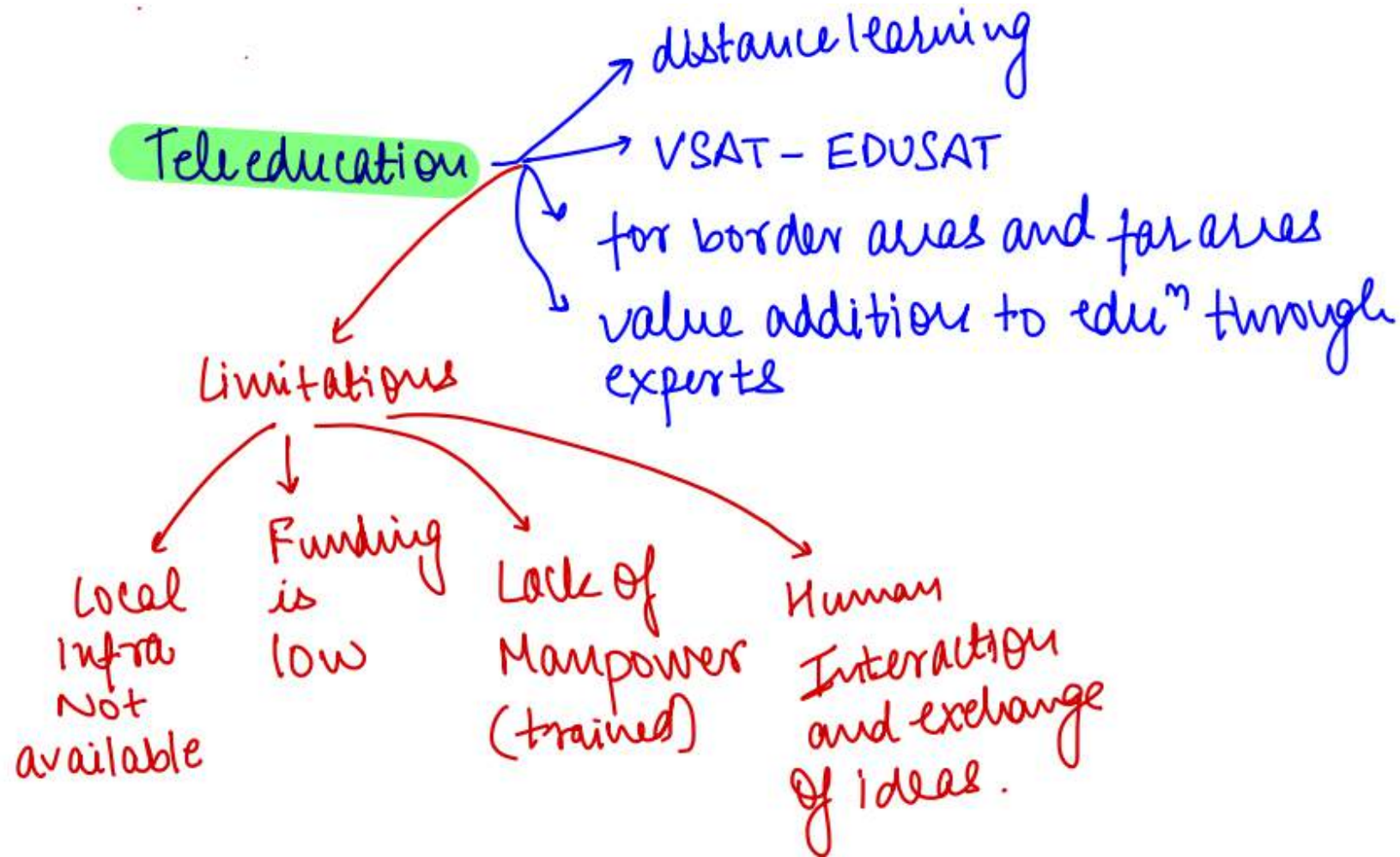
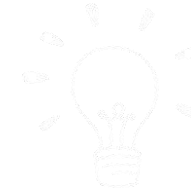
NISAR : Nasa ISRO Synthetic Aperture Radar : Collaboration between India and NASA for a dual frequency satellite.

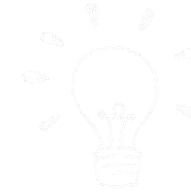
(vi) Disaster Management : Hazard Mapping of disaster prone areas

INSAT : GSAT (1983)

These satellites all in geostationary/geosynchronous orbit

- (i) Communication (all types)
- (ii) Weather forecasting (INSAT-3D) (INSAT-3DR)
- (iii) Tele education (Distance learning) (EDU-SAT)
- (iv) Telemedicine
- (v) Disaster Management





Telemedicine

: Healthcare services to rural via satellite communication



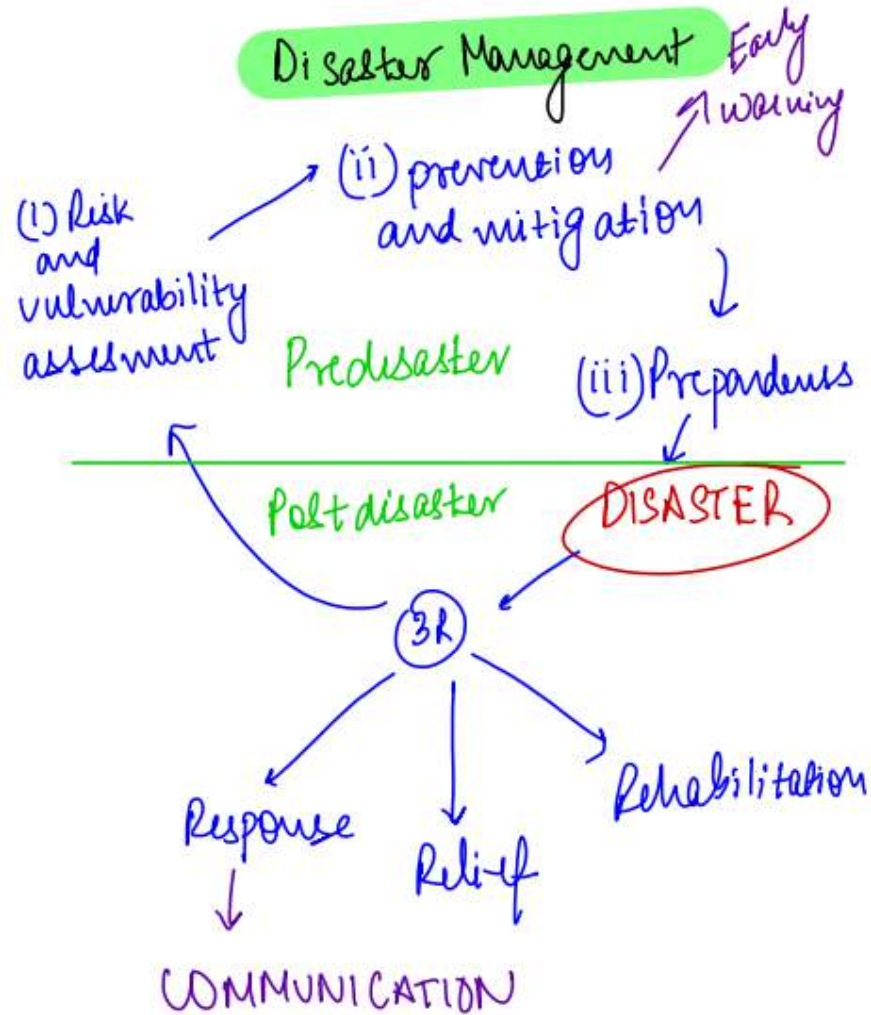
- 1st time done in Bangalore.
- already established system by ISRO.

Applications of Telemedicine

- Monitoring and treatment of conditions
- Continuing Medical education (CME).
- Emergency Situations: Trauma/RTA. Doctor can guide paramedics, (golden hour concept)
- Disaster Management.
- Aid to foreign countries - Outreach Programs
- **What is the Name of India's Flagship Telemedicine Program?**

Challenges

- No examination
- Psychology of Patients
- Digital Divide : Infrastructure issues
- Technical issues



- Mapping of hazard prone areas.
- Forecast and Warning
- Post Disaster Coordination
- DISANET : Network for Disaster management by Government of India.

GAGAN

- GPS aided geo augmented Navigation satellite system .
- AAI uses GAGAN for civil aviation and air traffic management safety management.

Use of 3 GSAT

Main Purpose is to prevent air line accidents and manage traffic.

Ex : MH 371 : Malaysian Airplane shot down.

Related News

- Airports Authority of India (AAI) successfully conducted flight trials using GAGAN based LPV approach procedure.
- India is the first country in the Asia Pacific Region to achieve such a landmark in field of Air Navigation Services (ANS).
- LPV (Localizer Performance with Vertical Guidance) permits aircraft guided approaches that are operationally nearly equivalent to Category 1- Instrument Landing System (Cat-1 ILS), without the need for ground-based navigational infrastructure.

About GAGAN (GPS Aided GEO Augmented Navigation)

- GAGAN is an Indian Satellite Based Augmentation System (SBAS) jointly developed by AAI and ISRO for India and neighbouring countries in the equatorial region.
- It is one among the only four Space-Based augmentation systems available in the world which also includes US(WAAS) Europe (EGNOS) and Japan (MSAS).
- Benefits of GAGAN: Air traffic control, manage road and railways transport, help farmers in crop spraying etc.

GPS aided geo augmented navigation

- ISRO+AAI
- Allows planes to land without ground markers :
 - Performance similar to cat-1 ILS
 - Instrument landing system
- Localizer performance with vertical guidance
- Only 4 countries : USA, Europe, Japan, India
- RUSSIA AND CHINA DONOT HAVE SUCH A SYSTEM

SPACE EXPLORATION

1. Orbit : Spacecraft orbits celestial body
2. Lander : Module used to land rover on a celestial body. (Soft landing) use of parachute.
3. Rover - Robotic device carries instruments as payload for scientific investigation of celestial bodies.

eg: Phoenix, Curiosity, Insight (for Mars)
by NASA

India - Chandrayan-2

- Lander (VIKRAM)
- Rover (PRAGYAN)

SPACE EXPLORATION

Fly-By Mission - Just pass by planet and collect data.

Eg: Voyager - I to jovian planets (1974). It was sent to study the outer edges of solar systems

Gravity Assist : Use of sun's gravity/other planets to propel space craft

ISRO Missions : Space Exploration

Earth's Moon : Chandrayaan 1 (2008), Chandrayaan 2 (2019) and Chandrayaan 3

Mars : Mangalyaan 1 and Mangalyaan 2

Astronomical Observation : AstroSat

Study of Sun : Aditya L1

Chandrayan - 1

- Orbiter only
- Had to be stopped due to battery issues
- Found loss of water are on surface of moon, confirmed by NASA
- Found landing site of apollo mission .
- Carried instruments from Multiple Countries.

Chandrayaan - 2 (3 parts)

- Orbiter
- Lander : Vikram
- Rover : Pragyan
- Assistance from Russia and France.
- Orbiter was successful, but lander and rover could not reach the surface of the moon.

Study of Moon

- Terrain and surface mapping
- Spectrometry of Soil of moon for Mineral Content
- Atmosphere of moon to check for any traces of gas.
- Presence of water on moon.

GSLV-Mk III-M1/ Chandrayaan-2 Mission

- India's Geosynchronous Satellite Launch Vehicle, GSLV Mk III-M1, successfully launched Chandrayaan-2 spacecraft on July 22, 2019 into its planned orbit with a perigee (nearest point to Earth) of 169.7 km and an apogee (farthest point to Earth) of 45,475 km. The launch took place from the Second Launch Pad at Satish Dhawan Space Centre SHAR Sriharikota.

GSLV-Mk III-M1/ Chandrayaan-2 Mission

- Chandrayaan-2 mission is a highly complex mission, which represents a significant technological leap compared to the previous missions of ISRO. It comprised an Orbiter, Lander and Rover to explore the unexplored South Pole of the Moon. The mission is designed to expand the lunar scientific knowledge through detailed study of topography, seismography, mineral identification and distribution, surface chemical composition, thermo-physical characteristics of top soil and composition of the tenuous lunar atmosphere, leading to a new understanding of the origin and evolution of the Moon.

GSLV-Mk III-M1/ Chandrayaan-2 Mission

- After the injection of Chandrayaan-2, a series of maneuvers were carried out to raise its orbit and on August 14, 2019, following Trans Lunar Insertion (TLI) maneuver, the spacecraft escaped from orbiting the earth and followed a path that took it to the vicinity of the Moon. On August 20, 2019, Chandrayaan-2 was successfully inserted into lunar orbit. While orbiting the moon in a 100 km hinar polar orbit, on September 02, 2019, Vikram Lander was separated from the Orbiter in preparation for landing. Subsequently, two de-orbit maneuvers were performed on Vikram Lander so as to change its orbit and begin circling the moon in a 100 km x 35 km orbit. Vikram Lander descent was as planned and normal performance was observed upto an altitude of 2.1 km. Subsequently communication from lander to the ground stations was lost.

GSLV-Mk III-M1/ Chandrayaan-2 Mission

- The Orbiter placed in its intended orbit around the Moon will enrich our understanding of the moon's evolution and mapping of the minerals and water molecules in Polar regions, using its eight state-of-the-art scientific instruments. The Orbiter camera is the highest resolution camera (0.3 m) in any lunar mission so far and will provide high resolution images which will be immensely useful to the global scientific community. The precise launch and mission management has ensured a long life of almost seven years instead of the planned one year.

Chandrayaan 3

- The space agency is aiming to achieve a soft landing on the south pole of the lunar surface, which is the least explored region of the Moon to date.
- Unlike its predecessor, Chandrayaan 3 will not carry an orbiter—but will include a lander and a rover to study the lunar surface

Countries with LUNAR missions

- (1) NASA= Apollo
- (2) Russia = LUNA Nussion
- (3) ESA
- (4) China : Change-E mission
- (5) Japan

Mangalyaan (MOM) - 1

- India's Mars Orbiter Mission (MOM) spacecraft has lost communication with the ground stations, bringing an end to its life after eight long years.

Mars Orbiter Mission

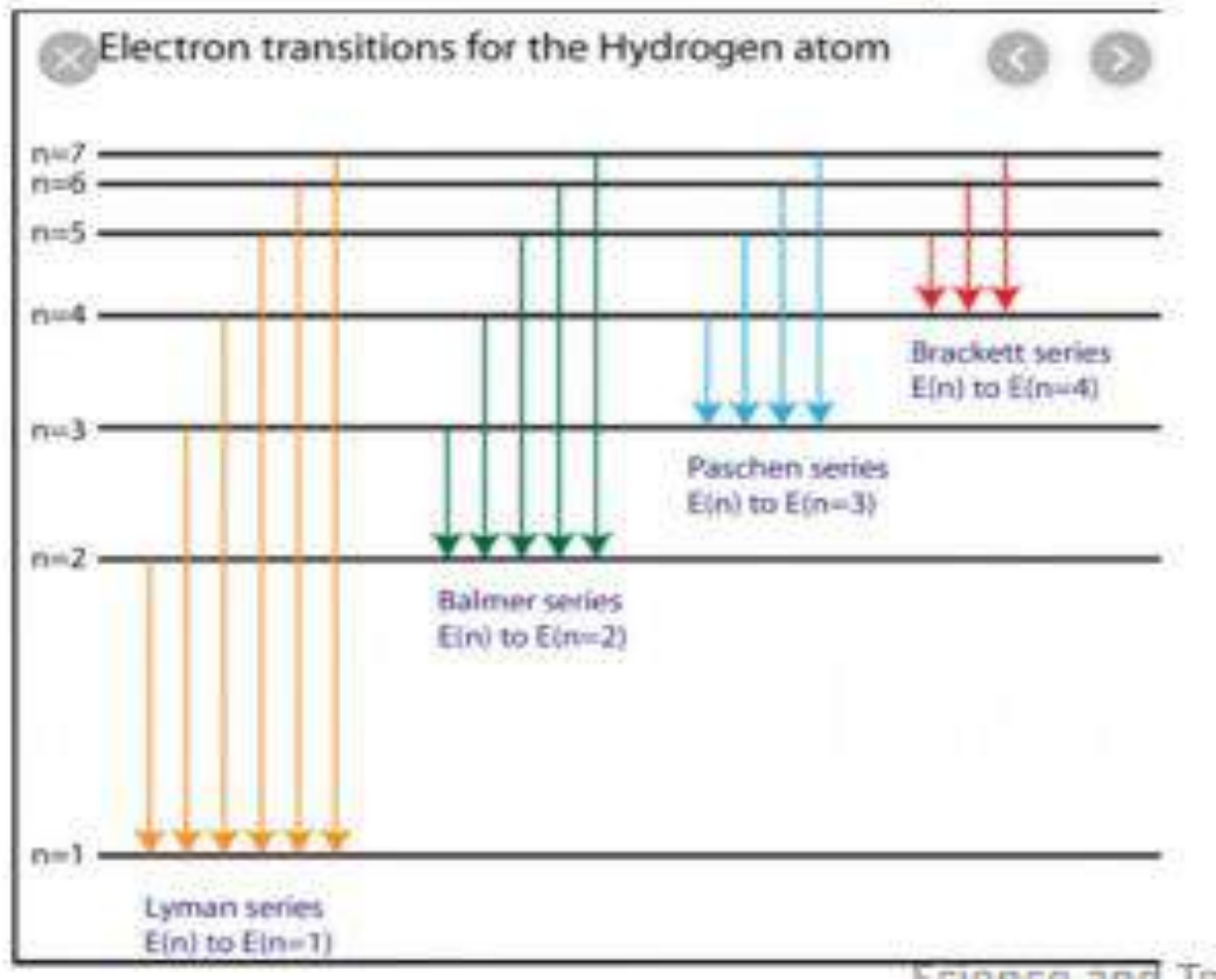
- (1) India's first interplanetary exploration mission/
- (2) Objectives
 - (a) Technology demonstration : interplanetary operation
 - (b) Scientific objective : data collection and exploration of martian surface features

Key fact about the mission

- Launched onboard PSLV-C25 (2013), It is India's first interplanetary mission
- Mission made India the first Asian country, and the fourth in the world (after Russia, US and EU) to get to the planet.
- Carried 850 kg of fuel and 5 science research payloads.
- Spacecraft is tracked by the Indian Deep Space Network (IDSN) located near Bengaluru

For scientific objective : 5 instruments

- (1) **Mars Color Camera** : Will take pictures of Mars' Surface and this will put other information into context.
- (2) **Lyman Alpha Photometer** : Ratio of Deuterium and Hydrogen. This will answer questions about water presence on Mars.
- (3) **Thermal Infrared imaging Spectrometer** : It will map the surface composition and Mineralogy of the planet by measuring thermal emission.
- (4) **MENCA** : mars exospheric neutral composition analyser. Seasonal Variations in the martian atmosphere.
- (5) **Methane Sensor** : Scan the entire martian disc in 6 minutes and measure very low levels of methane in parts per billion.

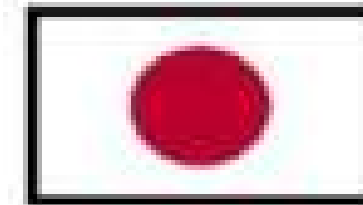
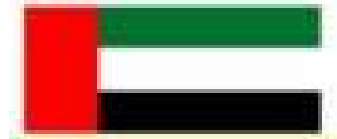


Technological Achievements of MOM

- Understanding of the composition of several gases in the martian exosphere including discovery of ‘suprathermal’ Argon-40 atoms.
- It Photographed, for the first time, the far side of Deimos, one of the natural satellites of Mars.
- Mission captured the time-variation of the Martian polar ice caps.

Other Mars Mission

- Perseverance (NASA, USA)
- Hope Mission (United Arab Emirates)
- Tianwen 1 (China)
- Mars Moons Exploration (MMX) Mission (Japan)

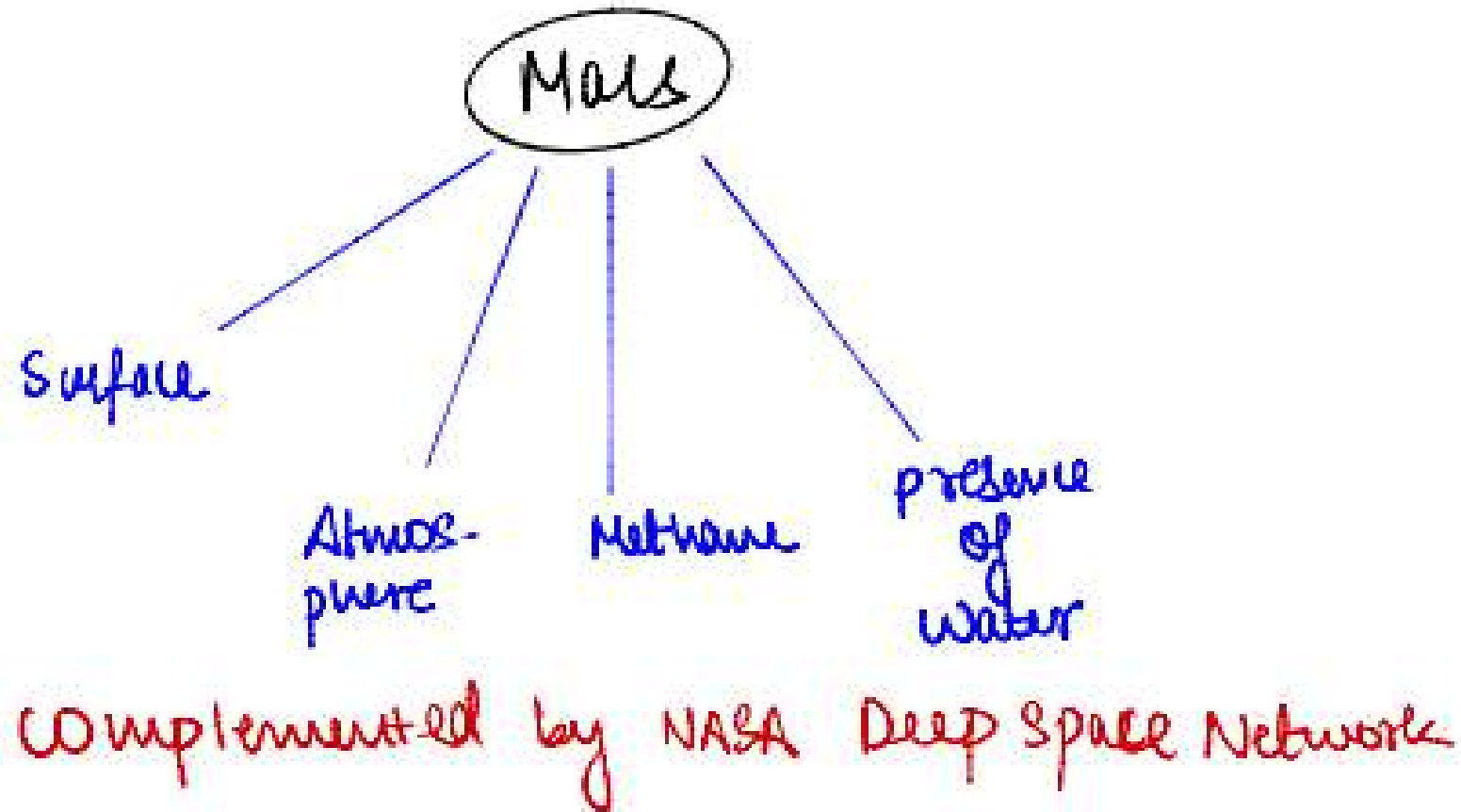


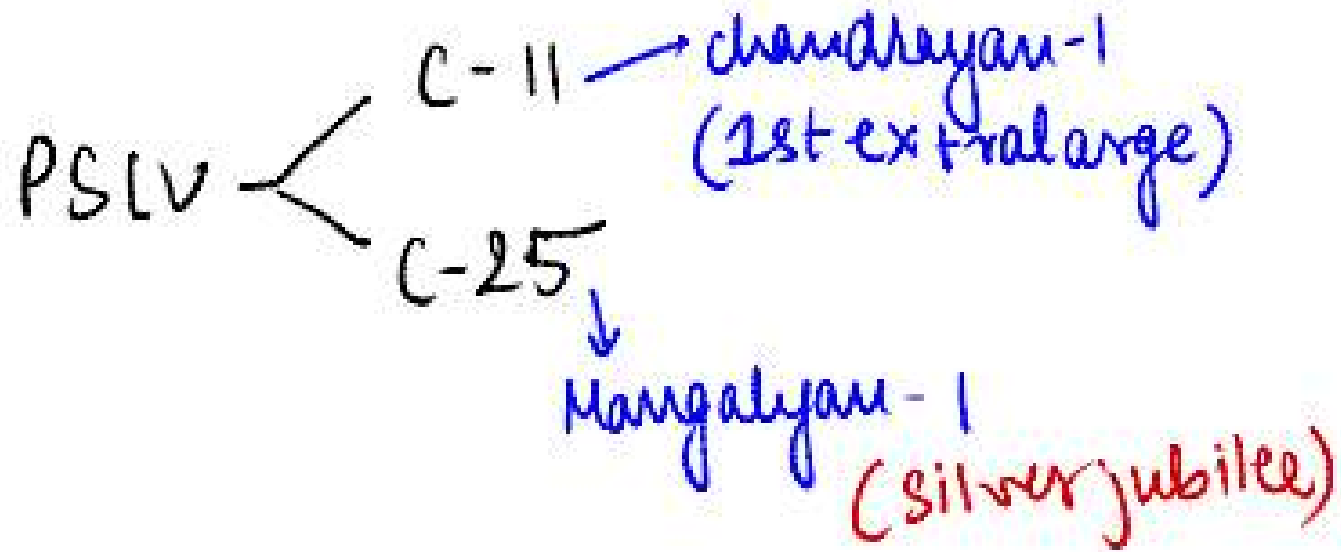
NASA's Perseverance Rover

- Perseverance rover mission is part of NASA's Mars Exploration Program, a long-term effort of robotic exploration of Mars.
- Recently, it placed a titanium tube containing a rock sample on the surface of Mars.
- Igneous rock sample was collected from Mars Jezero Crater called South Séítah.
- Samples are being placed at a location called "Three Forks," first such sample depot on another world.
- Depot will serve as a backup if Perseverance can't deliver its samples and subsequent NASA missions would

NASA's Perseverance Rover

- Recently, it also captured the solar eclipse on Mars featuring Phobos, one of Mars' two moons (other is Deimos).
- In a related news, Mars Oxygen In-Situ Resource Utilization Experiment (MOXIE) has produced oxygen at Mars with components from the planet's atmosphere.
- MOXIE was sent (by Massachusetts Institute of Technology) with NASA's Perseverance rover





Mangalyan-2 : GSLV

ASTROSAT 2015

India's astronomical Satellite

① Multiwavelength spacecraft.

- visible
- x-ray
- UV rays

Orbit = 650 Km (10 W earth Orbit)

Scientific Objectives:

1. High energy events of universe
 - Supernova
 - Collapsing black hole
2. Binary system study (2 star system) quasars and pulsars.
3. Mapping universe for black holes and study them
 - NASA has Chandra X ray observatory to map for black holes
Named after Indian scientist
 - Other countries with astronomical Satellite
 - Japan, ESA, Many More

Scientific Objectives:

- ASTROSAT is India's first dedicated multi wavelength space observatory. This scientific satellite mission endeavours for a more detailed understanding of our universe. One of the unique features of ASTROSAT mission is that enables the simultaneous multi-wavelength observations of various astronomical objects with a single satellite.

Scientific Objectives:

- ASTROSAT observes universe in the optical, Ultraviolet, low and high energy X-ray regions of the electromagnetic spectrum, whereas most other scientific satellites are capable of observing a narrow range of wavelength band. Multi-wavelength observations of ASTROSAT can be further extended with co-ordinated observations using other spacecraft and ground based observations. All major astronomy Institutions and some Universities in India are participating in these observations.

Scientific Objectives:

- ASTROSAT with a lift-off mass of about 1513 kg was launched into a 650 km orbit inclined at an angle of 6 deg to the equator by PSLV-C30. After injection into Orbit, the two solar panels of ASTROSAT were automatically deployed in quick succession. The spacecraft control centre at Mission Operations Complex (MOX) of ISRO Telemetry, Tracking and Command Network (ISTRAC) at Bangalore manages the satellite during its mission life.

Scientific Objectives:

- The science data gathered by five payloads of ASTROSAT are telemetered to the ground station at MCX. The data is then processed, archived and distributed by Indian Space Science Data Centre (SSDC) located at Byalalu, near Bangalore.

The scientific objectives of ASTROSAT mission are:

- To understand high energy processes in binary star systems containing neutron stars and black holes
- Estimate magnetic fields of neutron stars
- Study star birth regions and high energy processes in star systems lying beyond our galaxy
- Detect new briefly bright X-ray sources in the sky
- Perform a limited deep field survey of the Universe in the Ultraviolet region

Payloads of ASTROSAT:

- Five payloads of ASTROSAT are chosen to facilitate a deeper insight into the various astrophysical processes occurring in the various types of astronomical objects constituting our universe. These payloads rely on the visible, Ultraviolet and X-rays coming from distant celestial sources.

Payloads of ASTROSAT:

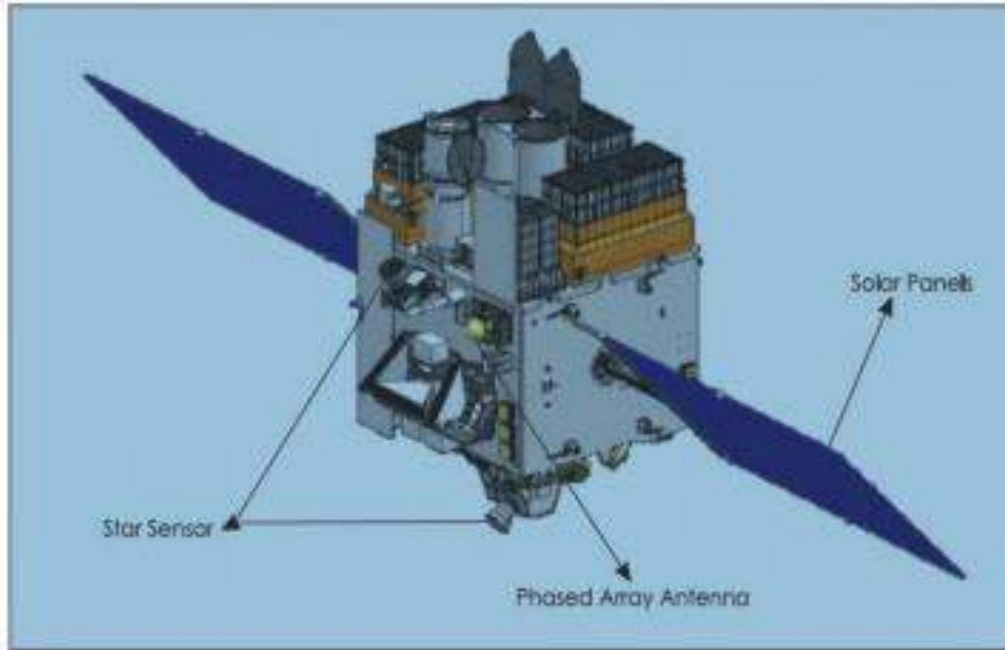
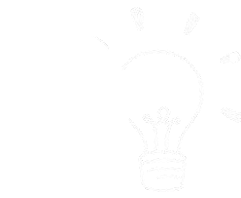
- 1. The Ultraviolet Imaging Telescope (UVM), capable of observing the sky in the Visible, Near Ultraviolet and Far Ultraviolet regions of the electromagnetic spectrum
- 2. Large Area X-ray Proportional Counter (LAXPC), is designed for study the variations in the emission of X-rays. from sources like X-ray binaries, Active Galactic Nuclei and other cosmic sources.

Payloads of ASTROSAT:

- 3. Soft X-ray Telescope (SXT) is designed for studying how the X-ray spectrum of 0.3-8 keV range coming from distant celestial bodies varies with time,
- 4. Cadmium Zinc Telluride Imager (CZTI), functioning in the X-ray region, extends the capability of the satellite to sense X-rays of high energy in 10-100 keV range.

Payloads of ASTROSAT:

- 5. Scanning Sky Monitor(SSM), is intended to scan the sky for long term monitoring of bright X-ray sources in binary stars, and for the detection and location of sources that become bright in X-rays for a short duration of time



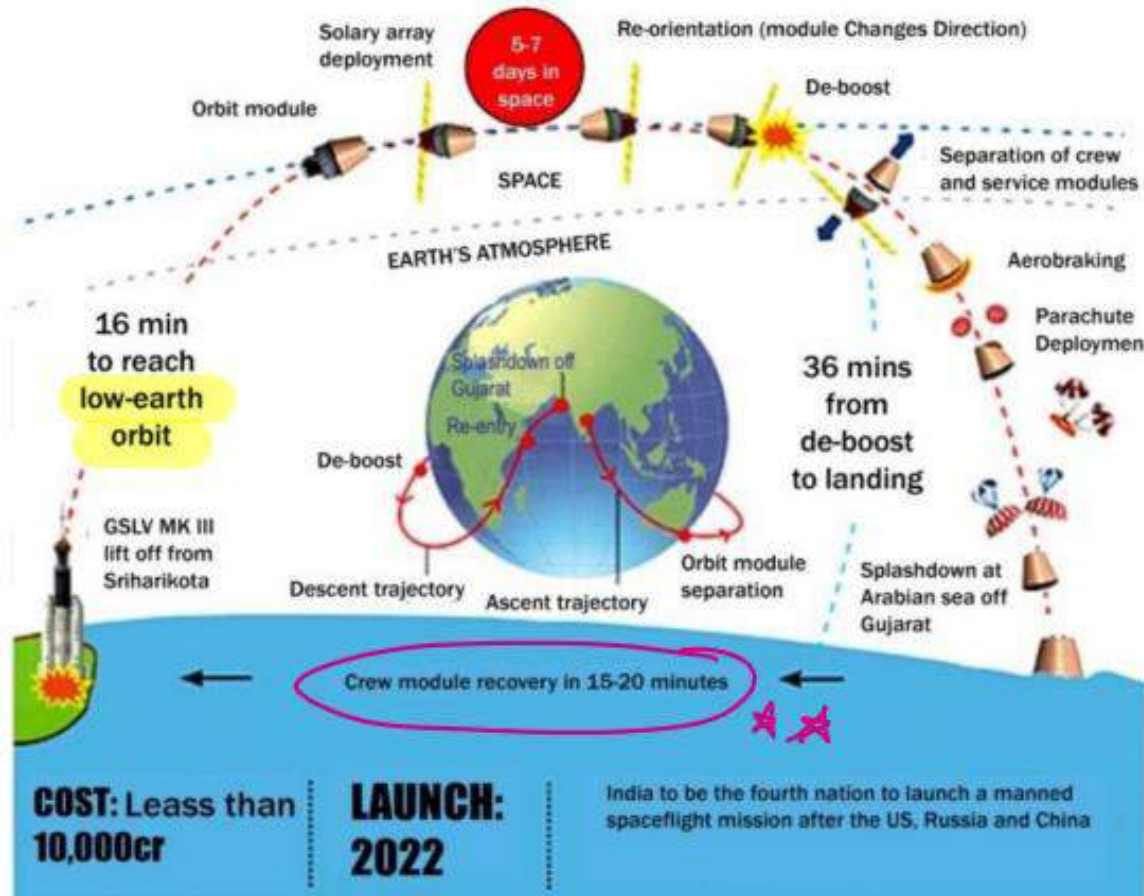
Gaganyaan

- The Gaganyaan orbital module has two parts the Crew Module (CM) and the Service Module (SM) and weighs about 8000 kg.
- The Crew Module (CM) is a double walled system and a habitat of astronauts. It has an ablative thermal protection system (TPS) to protect it during the intense aerodynamic heating during the flight.

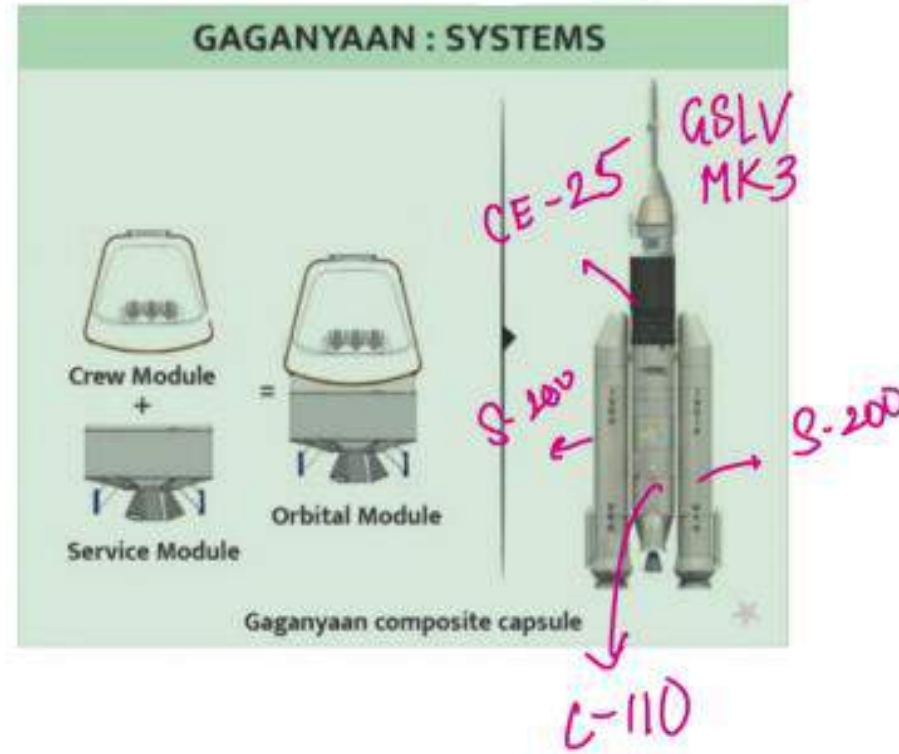
Gaganyaan

- The Orbital module (OM) will be launched by a human-rated GSLV MK-III vehicle and will orbit the Earth with a velocity of about 7,800 metre/second.
- Gaganyaan would be smaller in size than the current manned missions of Russian Soyuz, Chinese Shenzhou and NASA's planned Orion spacecraft.

MANNED MISSION



GAGANYAAN : SYSTEMS



SSLV

- Small Satellite Launch Vehicle (SSLV)
- It would help to meet the demand of the global launch services market for small satellites (nano, micro etc).
- New Space India Limited (NSIL), a PSU, will be the sole nodal agency responsible for providing end-to end SSLV Launch services.



Private Sector in Space

- Government has created Indian National Space Promotion and Authorization Centre (INSPACe) to boost private sector participation in entire range of space activities.
- About IN-SPACe It is the new entity of the Department of Space which will have its own chairperson and board. It will regulate and promote building of routine satellites, rockets and commercial launch services through Indian industry and startups. It will have its own directorates for technical, legal, safety and security, monitoring and activities promotion.

Private Sector in Space

- It will act as an interface between ISRO and private parties, and assess how best to utilise India's space resources and increase space-based activities.
- It will function autonomously and parallel to ISRO. ISRO will remain the basic body that decides what missions are to be undertaken but INSPACe will help fill the gaps

NSIL

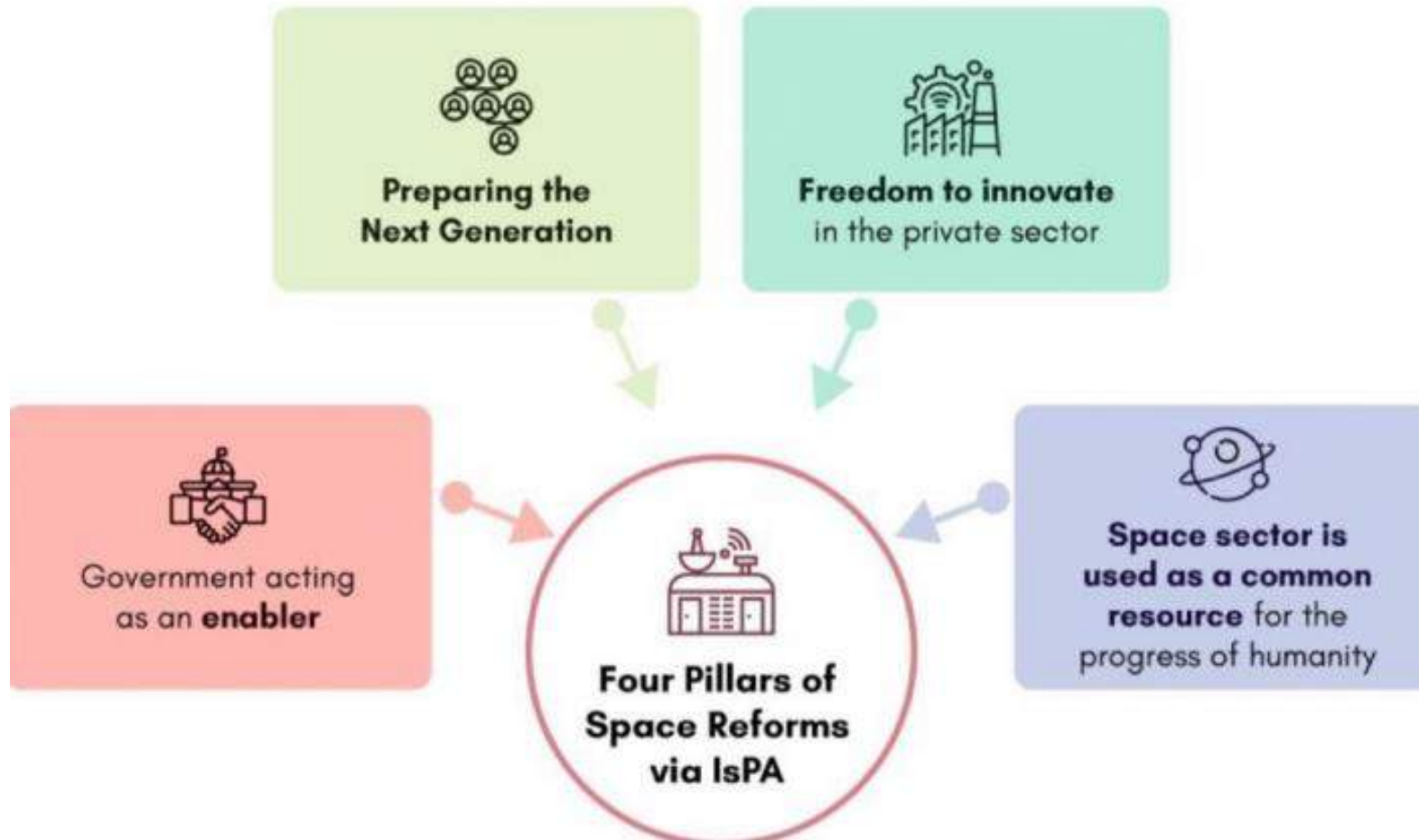
- About New Space India Limited
- It was founded in 2019, by Department of Space (DOS) to bridge gap between ISRO and private sector and facilitates transfer of ISRO technologies to industry.
- It is a wholly-owned Government of India company, under the administrative control of DOS.
- It will not replace ANTRIX, a private limited company owned by Government of India in 1992 as a Marketing arm of ISRO

Indian Space Association

- The industry association will act as an independent and a “single-window” agency for enabling the opening up of the space sector to start-ups and the private sector.
- ISpA will also work towards building global linkages for the Indian space industry to bring in critical technology and investments into the country to create more high-skill jobs
- The current global space economy is estimated at US\$ 360 billion, with India accounting for ~2% (US\$ 7 billion).

Indian Space Association

- In the global space market, rocket and satellite launch services, an area in which ISRO specialises, amount to only 5% share. Satellite-based services and ground-based systems account for the remaining 95%.
- Several private companies are developing plans to take paying customers to space on a regular basis leading to a rapid growth in space tourism industry.



Recent initiatives by Private Sector

- Hyderabad's Dhruva Space and Bengaluru's Digantara were authorized by IN-SPACe for launch of payloads onboard PSLV Orbital Experimental Module (POEM) of PSLV-C53.
- Mission Prarambh: Involved launch of Vikram-S (VKS), India's first privately built rocket by Hyderabadbased Skyroot Aerospace.
- VKS is solid propellant rocket.
- Payload Capacity: 290 kg- 560 kg support from ISRO and IN-SPACe

Recent initiatives by Private Sector

- India's first private launch pad unveiled at Sriharikota
- Agnikul and executed in support of ISRO and IN-SPACe.
- It is specifically built to support liquid-stage controlled launches.

Space Tourism

- Space tourism is space travel for recreational, leisure or business purposes. There are several different types of space tourism, including orbital, suborbital and lunar space tourism.
- To date, orbital space tourism has been performed only by the Russian Aviation and Space Agency- Roscosmos.
- Several private spaceflight companies are now working towards developing suborbital space tourism vehicles to take paying customers to space.

Space Tourism

- No international space law has defined space tourists. Existing space treaties such as Outer Space Treaty, Rescue Agreement etc. are only applicable to astronauts, envoys of mankind, or personnel of a spacecraft.
- To send a vehicle to space, it is compulsory for the national and international space law to authorize it according to Article VI of the Outer Space Treaty. However, there is lack of clarity on authorization of space travel with tourists aboard.

Space Travel

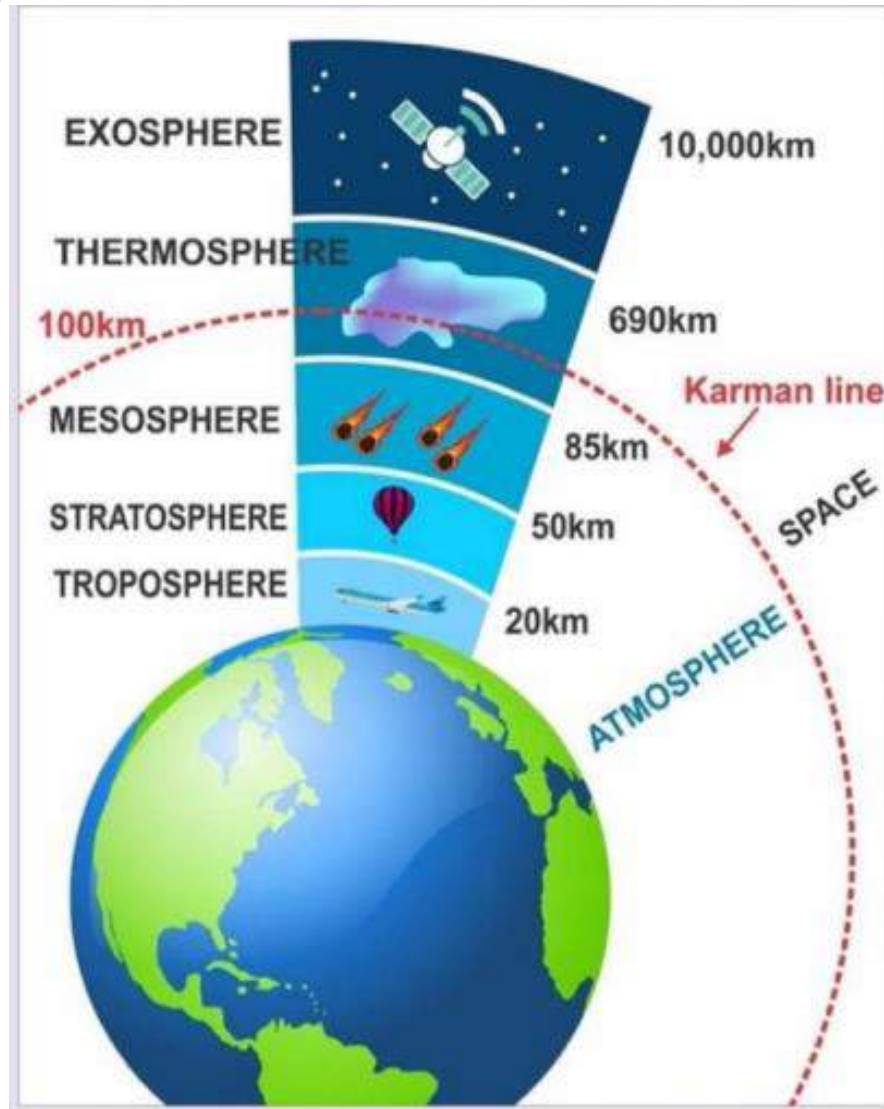
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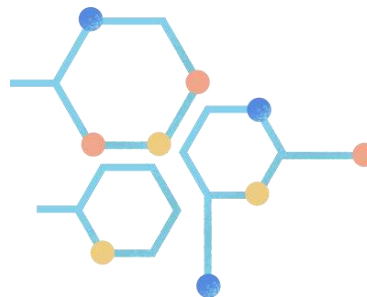
Space Travel

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- The main difference between orbital and suborbital flight is the speed at which a vehicle is traveling.
- An orbital spacecraft must achieve orbital velocity i.e. the speed that an object must maintain to remain in orbit around a planet. To orbit 125 miles (200 kilometers) above Earth for instance, a spacecraft must travel at a screaming 17,400 mph (28,000 km/h).

Space Travel

- Suborbital flight, in contrast, requires much lower speeds and doesn't have the power to achieve orbit. Instead, it will fly up to a certain height that depends on its speed, and then come back down once its engines are shut off.
- At the top of their flight arc in a suborbital flight, when the object is falling back toward Earth, passengers achieve a few minutes of weightlessness under zero gravity.





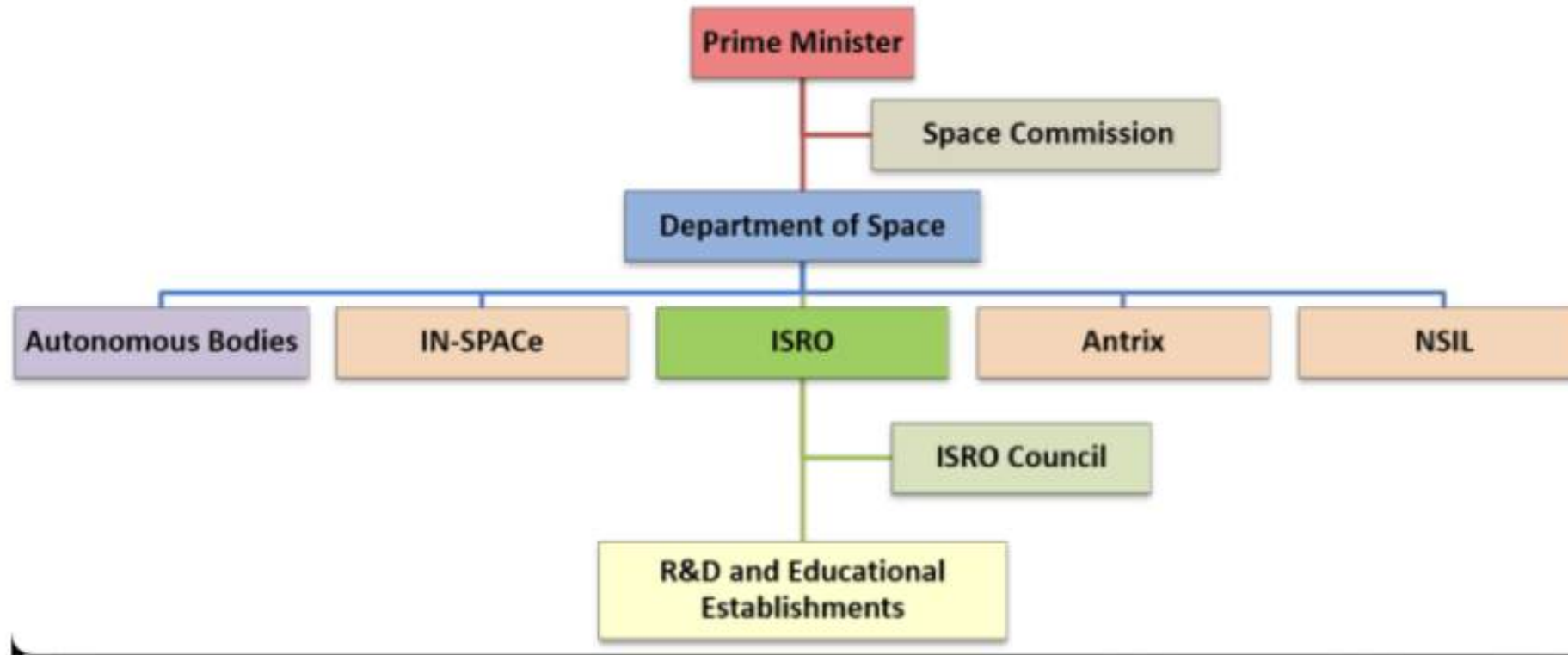
UPSC Prelims

Science & Tech

Class No. 10

Space Technology continued

Indian Space Architecture



ANTRIX

- Antrix—>Antrix Corporation Limited (ACL) is a PSU
- It is a wholly owned Government of India Company under the administrative control of the Department of Space.
- It was incorporated as a private limited company owned by Government of India in September 1992.
- It's mandate was to act as Marketing arm of ISRO for promotion and commercial exploitation of space products, technical consultancy services and transfer of technologies developed by ISRO.
- Another major objective is to facilitate development of space related industrial capabilities in India.
- As the commercial and marketing arm of ISRO, Antrix is engaged in providing Space products and services to international customers worldwide.

NSIL

- It was founded in 2019, by Department of Space (DOS) to bridge gap between ISRO and private sector and facilitates transfer of ISRO technologies to industry.
- It is a wholly-owned Government of India company, under the administrative control of DOS.
- It will not replace ANTRIX, a private limited company owned by Government of India in 1992 as a Marketing arm of ISRO

NSIL

- Central PSU under DoS
- Antrix will handle ISRO's commercial deals for satellites and launch vehicles with foreign customers.
- NSIL will deal with capacity building of local industry for space manufacturing.
- Vision : build launch vehicles, providing launch services, build satellites, providing space-based services, technology transfers,
- space-related products and services to global customers
- spur the growth of Indian Industry in undertaking technologically challenging space-related activities

NSIL

- NSIL's main purpose is to market the technologies developed by ISRO and bring it more clients that need space-based services.
- That role, incidentally, was already being performed by Antrix Corporation, another PSU working under the Department of Space, and which still exists.

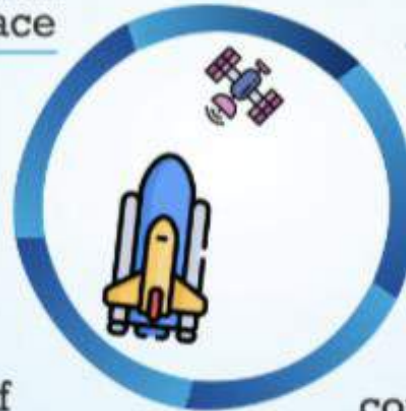


NEW SPACE INDIA LIMITED



NSIL - a Public Sector Enterprise - incorporated as new commercial arm of Department of Space

To harness India's space power commercially



To tap the benefits of R & D carried out by ISRO

To spearhead commercialization of space products including production of launch vehicles, transfer to technologies and marketing of space products

Private Sector in Space

- Government has created Indian National Space Promotion and Authorization Centre (INSPACe) to boost private sector participation in entire range of space activities.
- It is a single-window, independent, nodal agency which functions as an autonomous agency in the Department of Space (DOS).
- It is a single window agency for all space sector activities of private entities, IN-SPACe plays an important role in boosting the private space sector economy in India.



i.

Promote
and develop Indian space ecosystem accelerating space economy.

Enable
and nurture NGE's to accomplish their ventures in space sector.

Authorize
space operations and services in the country through well defined framework.

Supervise
the space activities of NGEs in the country



INDIAN NATIONAL SPACE PROMOTION AND AUTHORISATION CENTRE (IN-SPACE)



About: Autonomous nodal agency under Department of Space to promote, authorize and supervise space activities of Non-Governmental Private Entities in India.



Composition: A chairman, technical experts for space activities, safety experts, experts from Academia and Industries, members from Prime Minister's Office and Ministry of External Affairs.



Mandate:

- ⇒ Sharing of space infrastructure and premises under the control of ISRO.
- ⇒ Establishment of temporary facilities within premises under ISRO.
- ⇒ Establishment of new space infrastructure and facilities.



Other Key Information: Decision of IN-SPACE in context of this mandate, shall be final and binding on all stakeholders including ISRO.

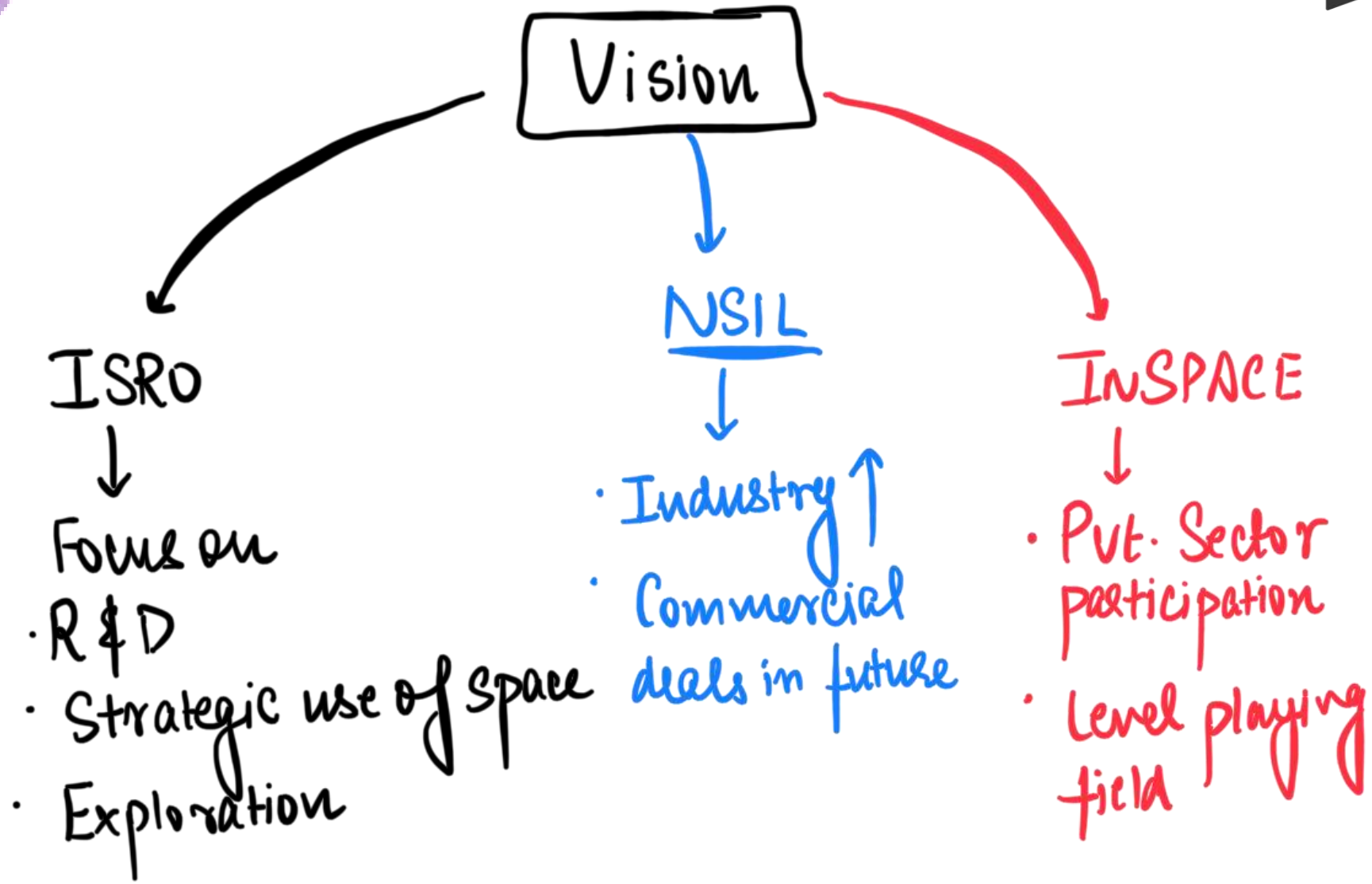


Private Sector in Space

- Interface—>agency acts as an interface between ISRO and Non-Governmental Entities (NGEs) and assesses how to utilize India's space resources better and increase space-based activities.
- Assess—>It also assesses the needs and demands of private players, including educational and research institutions, and explore ways to accommodate these requirements in consultation with ISRO.
- Accessibility—>Existing ISRO infrastructure, both ground- and space-based, scientific, and technical resources, and even data can be made accessible to interested parties to enable them to carry out their space-related activities through IN-SPACe.

Key Differences

- Antrix will handle ISRO's commercial deals for satellites and launch vehicles with foreign customers.
- NSIL will deal with capacity building of local industry for space manufacturing.
- The main objective of NSIL is to scale up industry participation in Indian space programmes in comparison to IN-SPACe which gives emphasis on the participation of the private sector.
- **However, NSIL is ALSO doing commercial deals.**

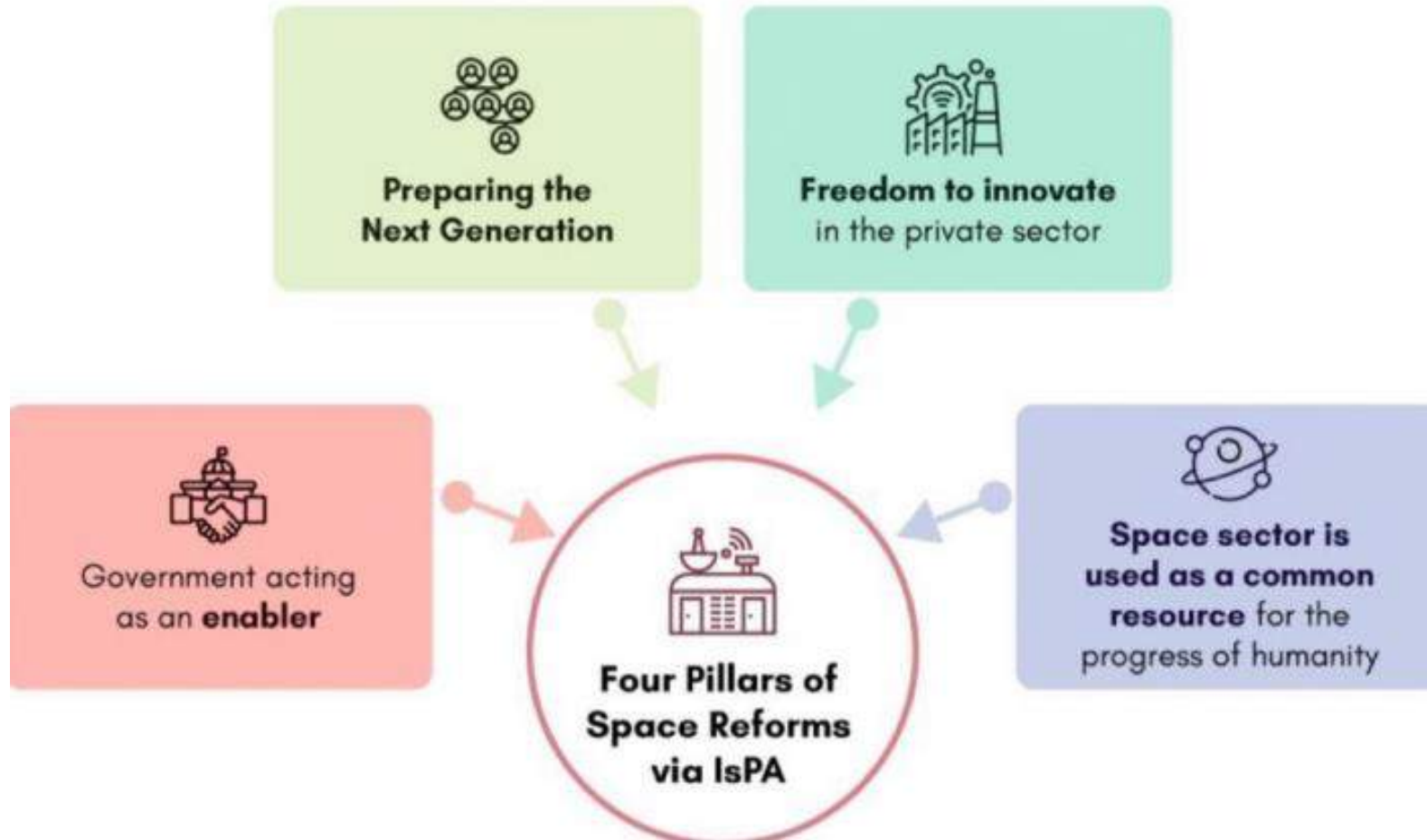


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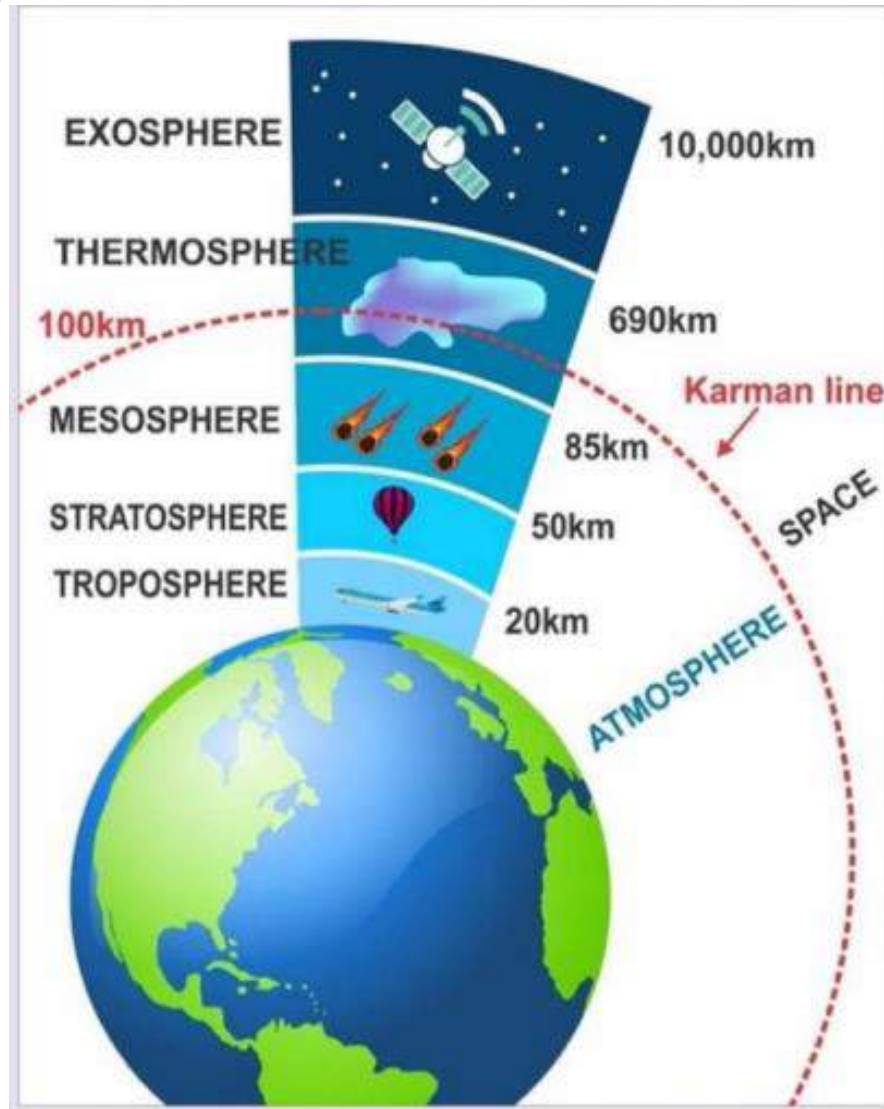
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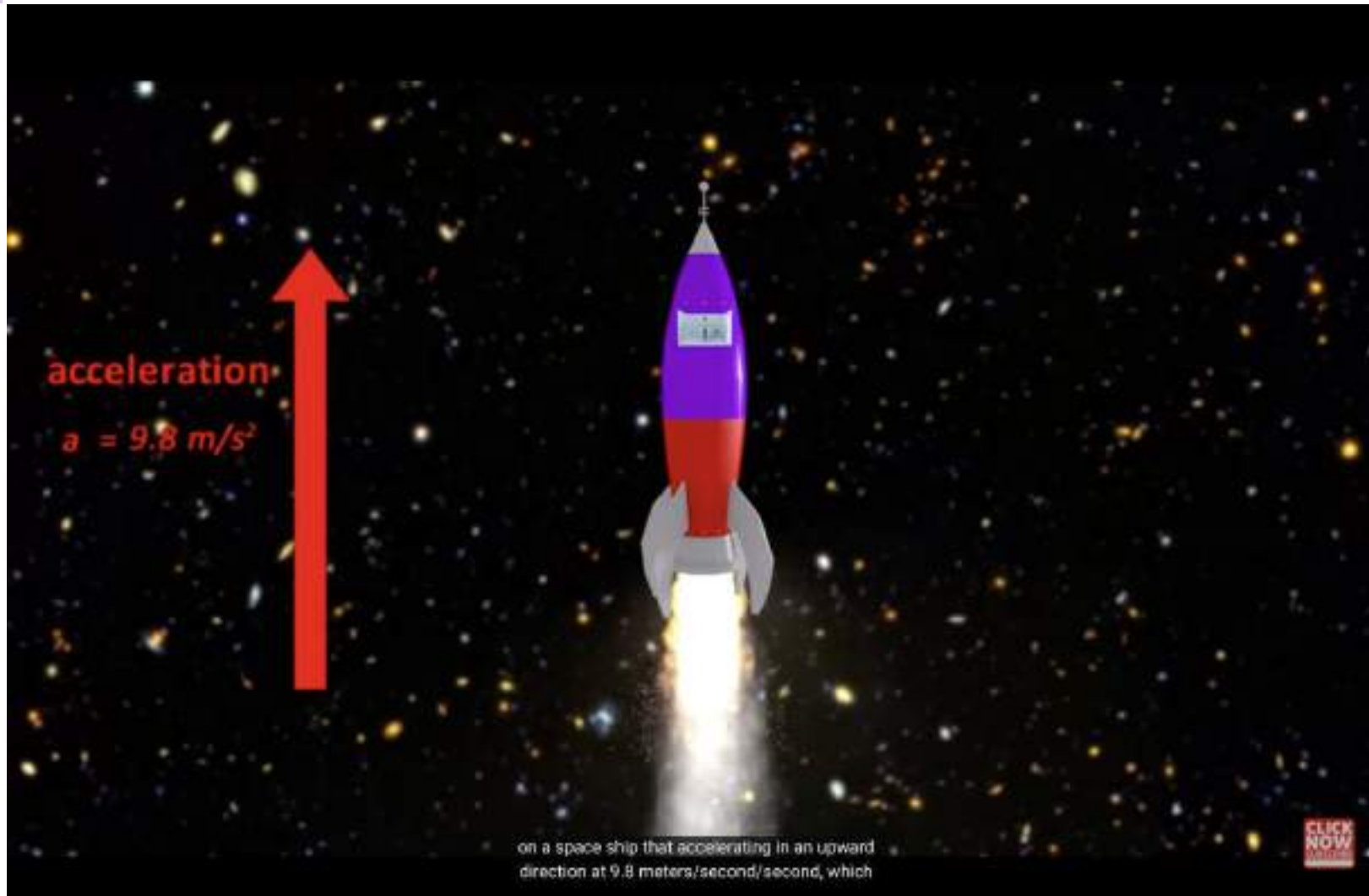
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- Inspiration 4 : first orbital trip where none of the people aboard is a professional astronaut. (by space X)



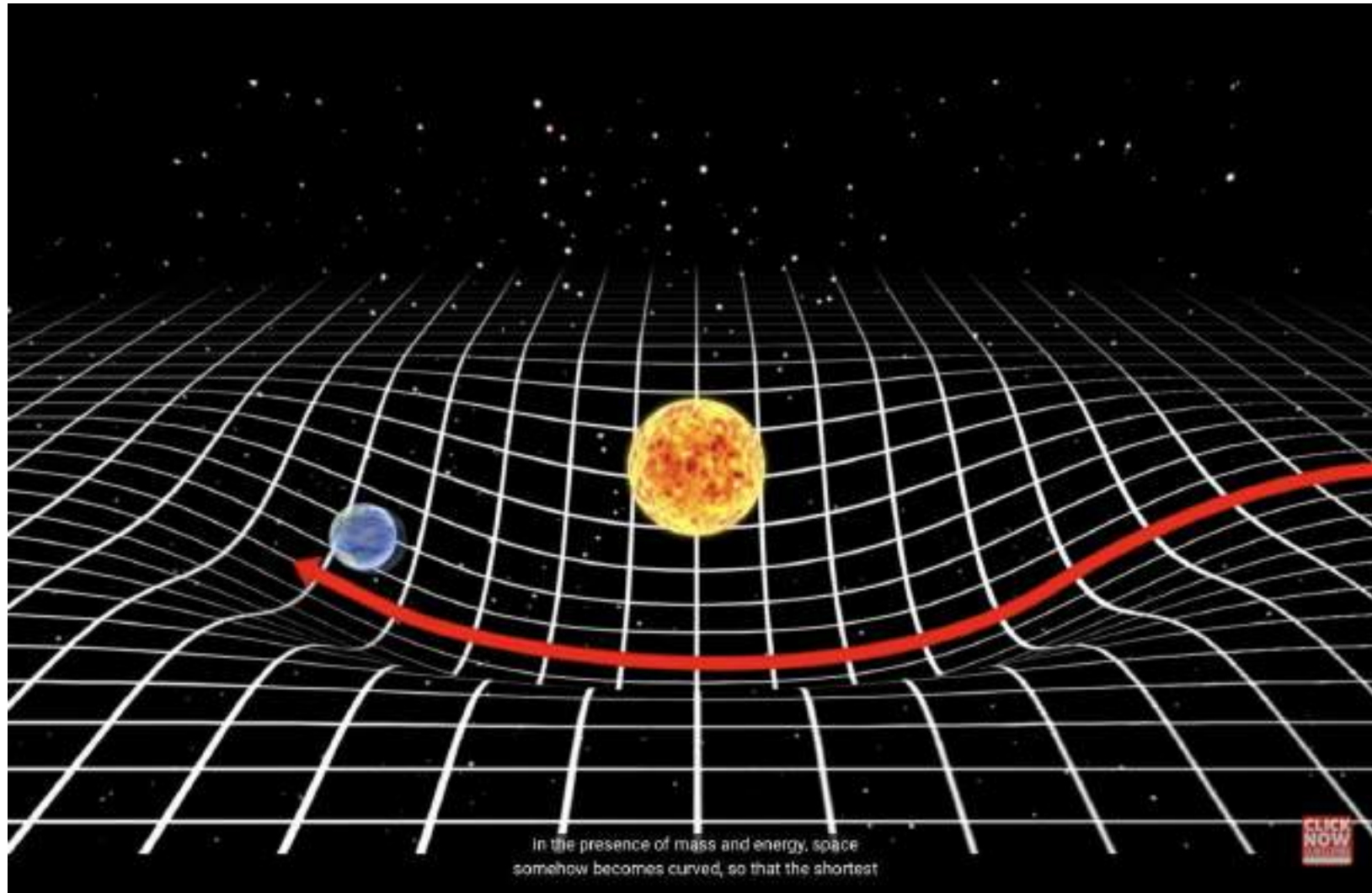
	 Suborbital tourism	 Orbital tourism
Altitude	About 100 kilometres	Over 400 kilometres
Duration in Space	Gives passengers a few minutes in space	Spend days or even more than a week in space.
Velocity	Sub-Orbital  • Requires much lower speeds (than orbital) and doesn't have the power to achieve orbit. • Instead, it will fly up to a certain height and then come back down once its engines are shut off.	Orbital  • Spacecraft must achieve orbital velocity i.e. the speed that an object must maintain to remain in orbit around a planet.

Black Hole

- Star Basics : Balance between radiation and gravity
- Radiation comes from nuclear fusion in the core
- Gravity we have already understood as a force : but now we will see it as bending of space time
- Let's understand these terms :







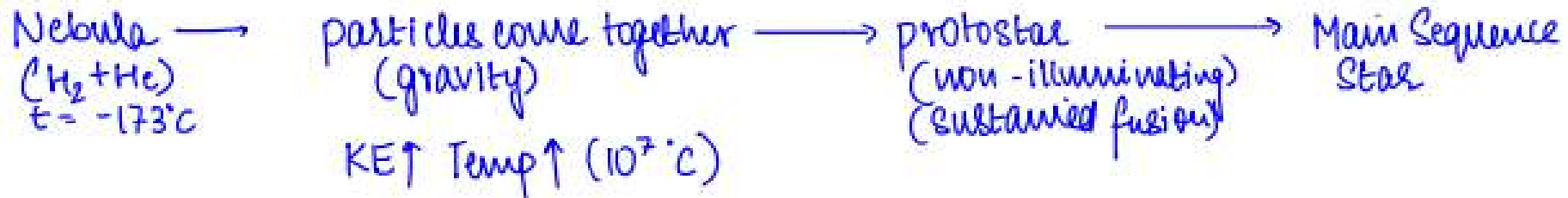
Let us see a video : how bending of space time explains gravitational motion

Links :

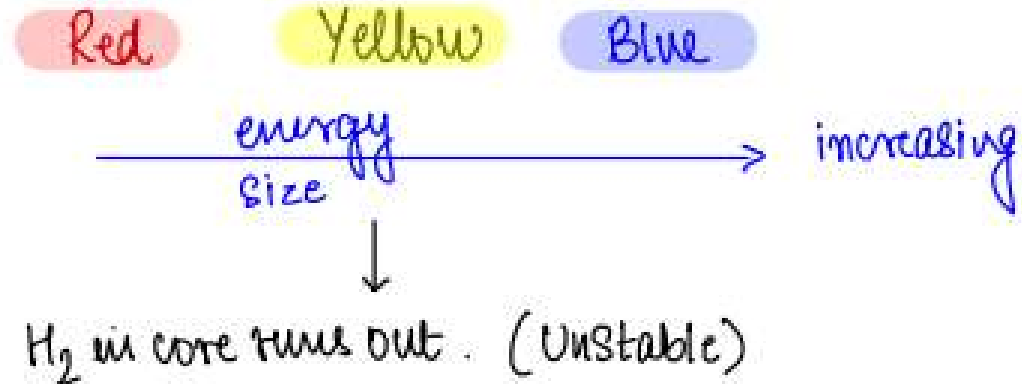
<https://www.youtube.com/watch?v=tzQC3uYL67U>

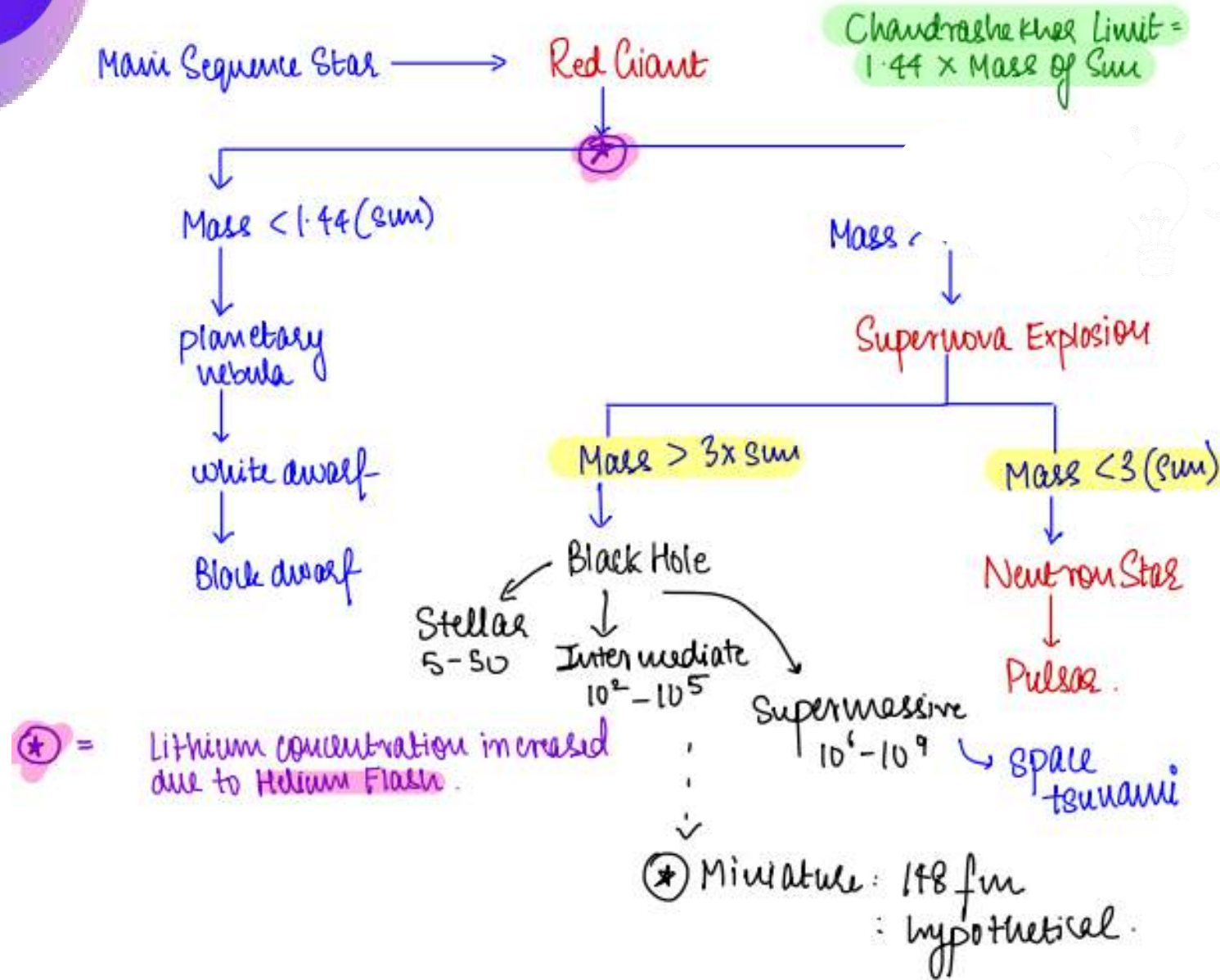
<https://www.youtube.com/watch?v=MTY1Kje0yLg>

What Happens to a Star?



Main Sequence Star: Core has fusion Rx^n





Black Holes

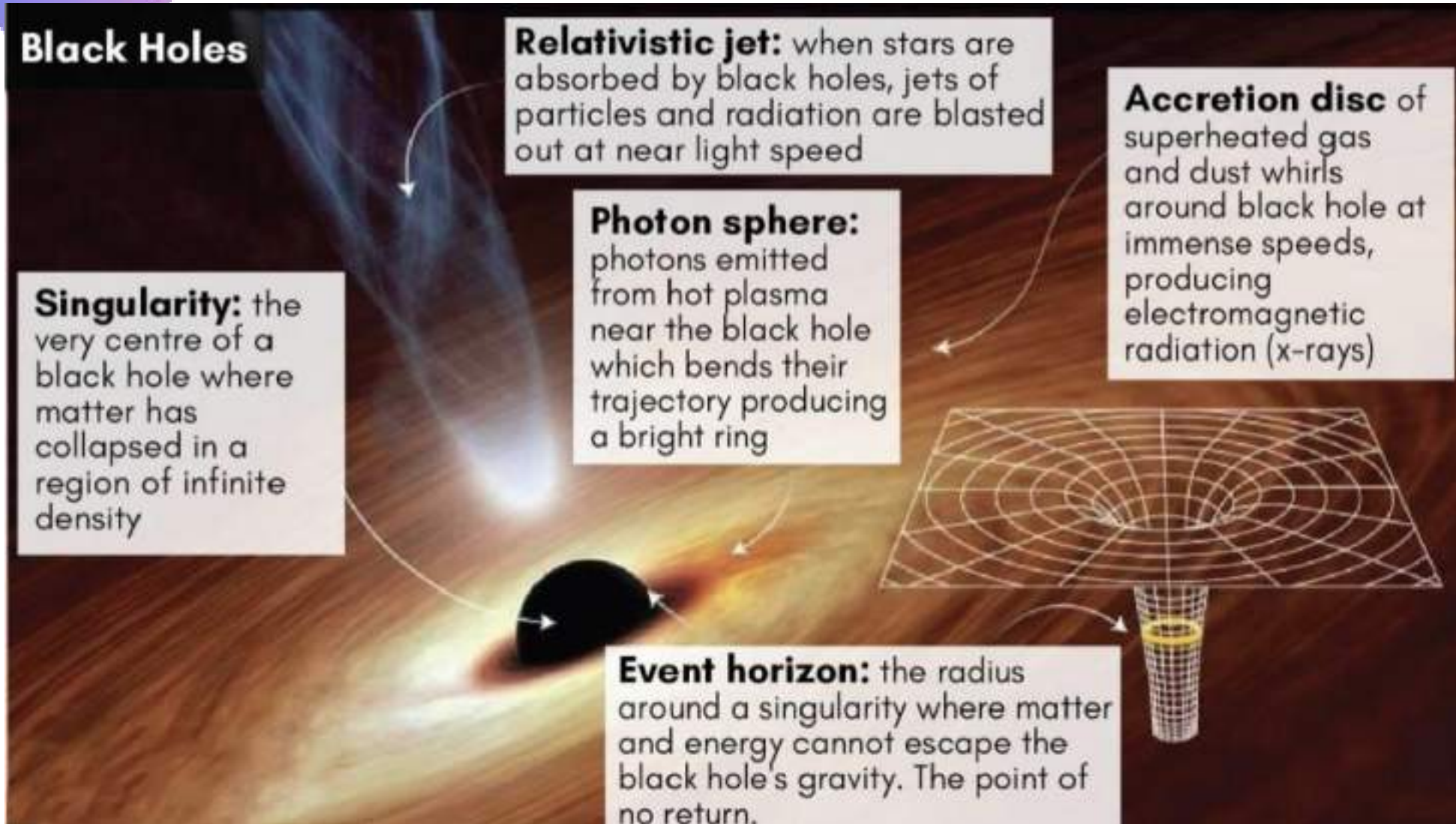
- Black hole is a place in space where gravity pulls so much that even light cannot get out. The gravity is so strong because matter has been squeezed into a tiny space.
- o This can happen when a big star is dying (our sun will never turn into a black hole as it is not big enough to make a black hole).
- o Because no light can get out, so they are invisible

Black Holes

- In the center of a black hole is a gravitational singularity, a one-dimensional point which contains a huge mass in an infinitely small space, where density and gravity become infinite and space-time curves infinitely, and the laws of physics as we know them cease to operate.

Black Holes

- Parts of Black hole :
- Accretion Disc
- Singularity
- Event Horizon
- Schwarzschild Radius : radius of event horizon
- ErgoSphere : Black hole is rotating, its mass causes the space time around black hole to rotate as well.



Categories of Black Hole



Stellar

Formed by the gravitational collapse of a star. Masses ranging from about 5 to several tens of solar masses.



Supermassive

The largest type of black hole, containing a mass of the order of hundreds of thousands, to billions of times, the mass of the sun.



Intermediate-mass

Mass in range $10^2 - 10^5$ solar masses; significantly more than stellar black holes but less than $10^5 - 10^9$ solar mass supermassive black holes.



Miniturn

Approximately 148 fm, Hypothetical tiny black holes, for which quantum mechanical efforts play an important role. The concept that black holes may exist that are smaller than stellar mass was introduced in 1971 by famous astrophysicist Stephen Hawking.

Black Holes

- Parts of Black hole :
- Accretion Disc
- Singularity
- Event Horizon
- Schwarzschild Radius : radius of event horizon
- ErgoSphere : Black hole is rotating, its mass causes the space time around black hole to rotate as well.

Taking Pictures of a black hole

- In 2019 scientists got the first optical image of a black hole through Event Horizon Telescope.
- It has captured the just outside region of a black hole, located 55 million light-years from Earth, at the centre of a galaxy named Messier 87. The image shows a photon (light quantum) can orbit the black hole without falling in. This is called the 'last photon ring'
- Sagittarius A* is the second black hole whose photographs have been captured by the Event Horizon Telescope project : at the center of milky way galaxy.

Taking Pictures of a black hole

- 2020 nobel in physics : Roger Penrose : black hole formation is based on general theory of relativity
- Reinhard Genzel and Andrea Ghez : discovery of supermassive compact object at the center of our galaxy : most likely a supermassive black hole : Sagittarius A*

Black holes and research

- Significance of black hole in space research
- Evolution of Galaxies
- Distribution of Dark Matter
- Star Formation
- Graviational Waves
- Testing general theory of relativity
- Light from behind the black hole : bending

Black holes and research

- Star formation: In particular, Supermassive black holes play an important role in star formation within galaxies.
- Gravitational waves: Scientists have detected that gravitational waves are generated when two black holes collide, and found that ringing pattern of the wave predicts the cosmic body's mass and spin.

Black holes and research

- General theory of relativity: The discovery of black hole has opened the door to several new lines of scientific investigation, enabling quantitative estimates of black hole related parameters. It has provided another laboratory to test the predictions of the Einstein's general theory of relativity.

Black holes and research

- Bending of light: The bending of light around a black hole is very significant because of its extremely large mass. Because of this bending, one can see light from material that is behind the black hole, which would not be seen if light were traveling in straight lines.

General theory of Relativity

- This theory was proposed by Albert Einstein in 1915.
- Essentially, it's a theory of gravity whose basic idea is that instead of being an invisible force that attracts objects to one another, gravity is a curving or warping of space. The more massive an object, the more it warps the space around it.
- The warping also affects measurement of time. Gravity can stretch space, it can also dilate time.

General theory of Relativity

- For example, the sun is massive enough to warp space across our solar system (a bit like the way a heavy ball resting on a rubber sheet warps the sheet). As a result, Earth and the other planets move in curved paths (orbits) around it.
- General theory of relativity explained the precession of Mercury

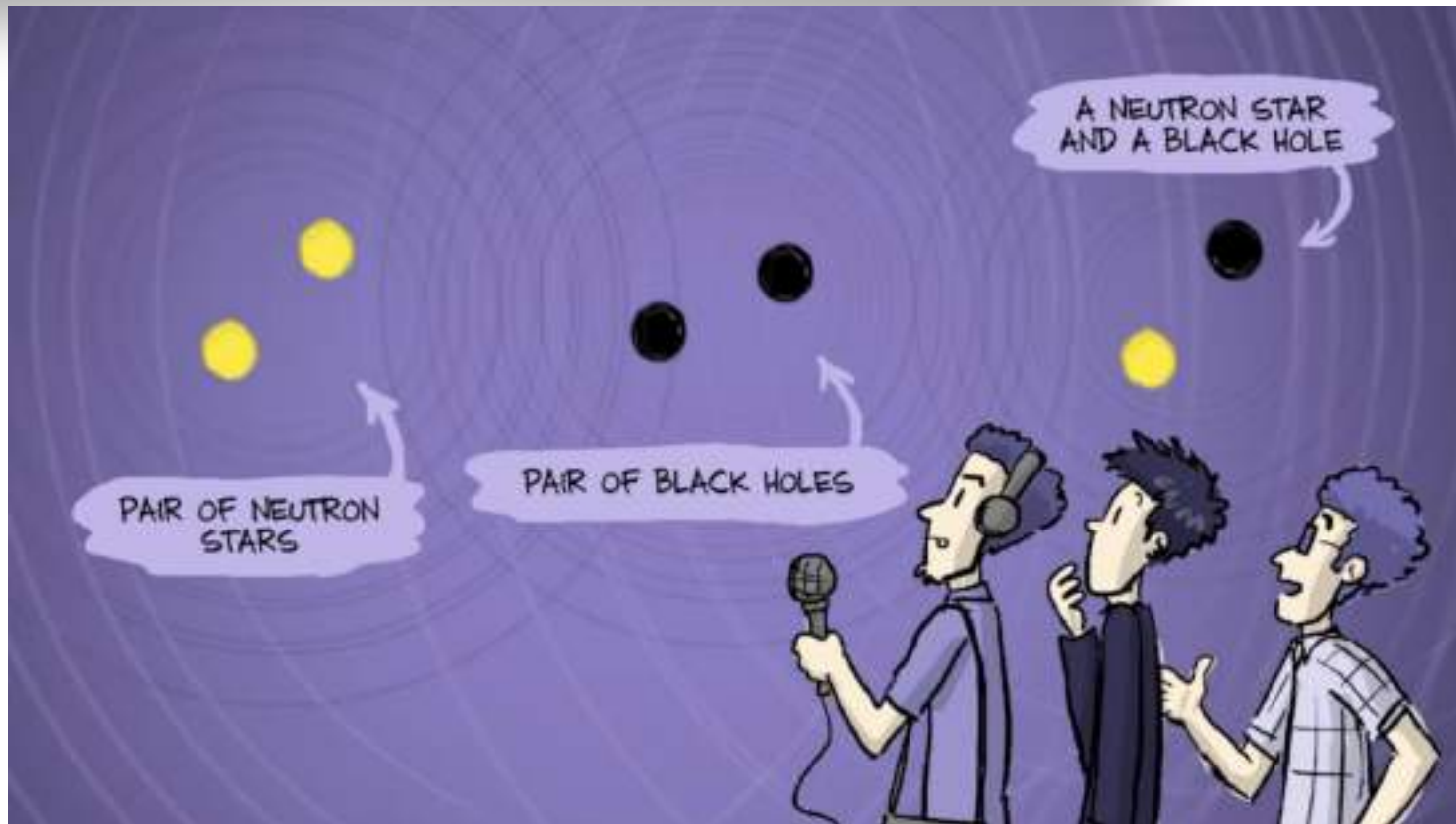
General theory of Relativity

- Confirmation : In the first major test of general relativity, astronomers in 1919 measured the deflection of light from distant stars as the starlight passed by our sun, proving that gravity does, in fact, distort or curve space.
- In 2016, the discovery of gravitational waves (subtle ripples in the fabric of spacetime) was another confirmation of general relativity.

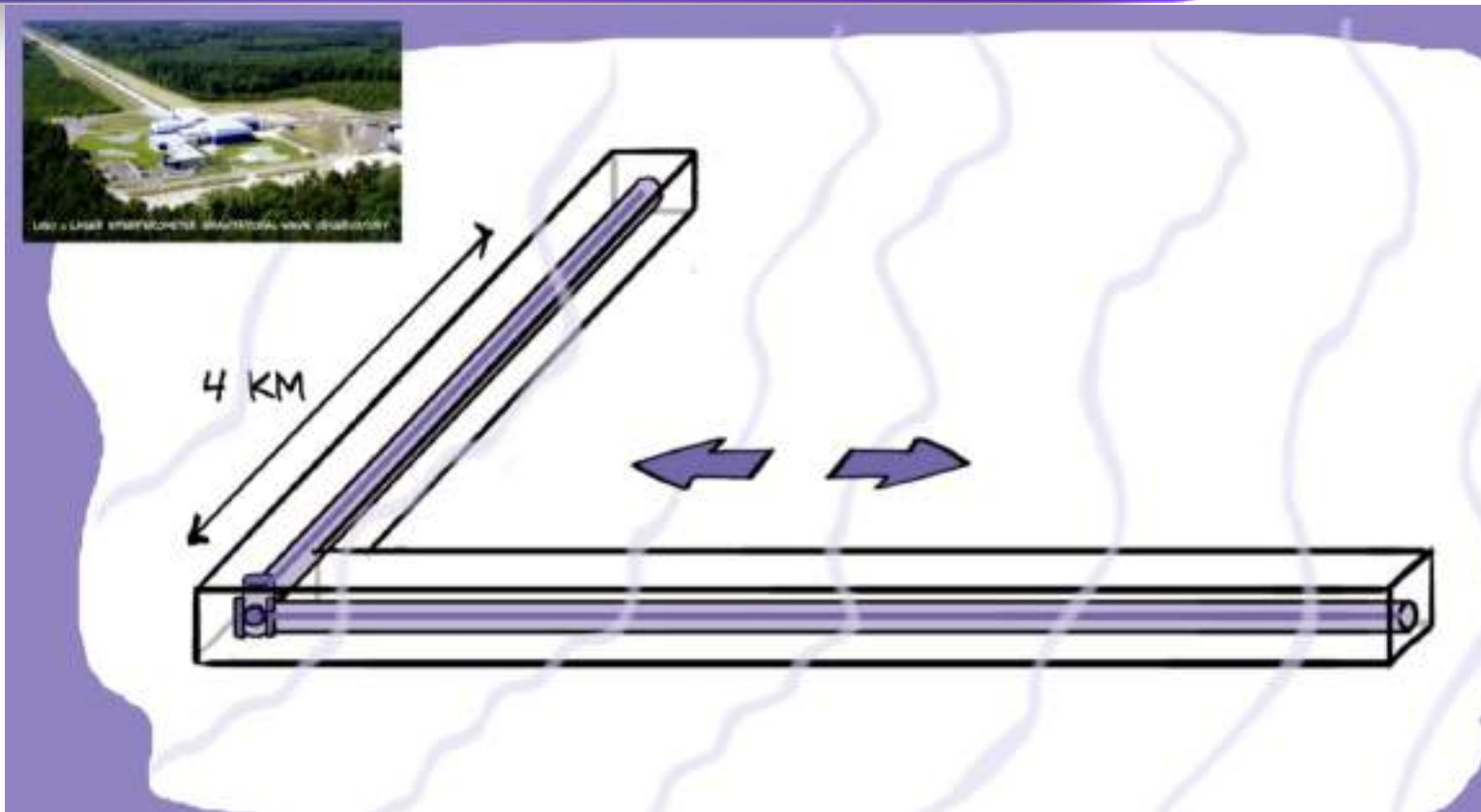
General theory of Relativity

- Gravitational waves are produced by cataclysmic events such as colliding black holes, supernovae (massive stars exploding at the end of their lifetimes), and colliding neutron stars. They travel at speed of light, squeezing and stretching anything in their path.
- Gravitational Waves were detected for the first time in the LIGO : Laser Interferometer Gravitational Wave Observatory

Gravitational Waves

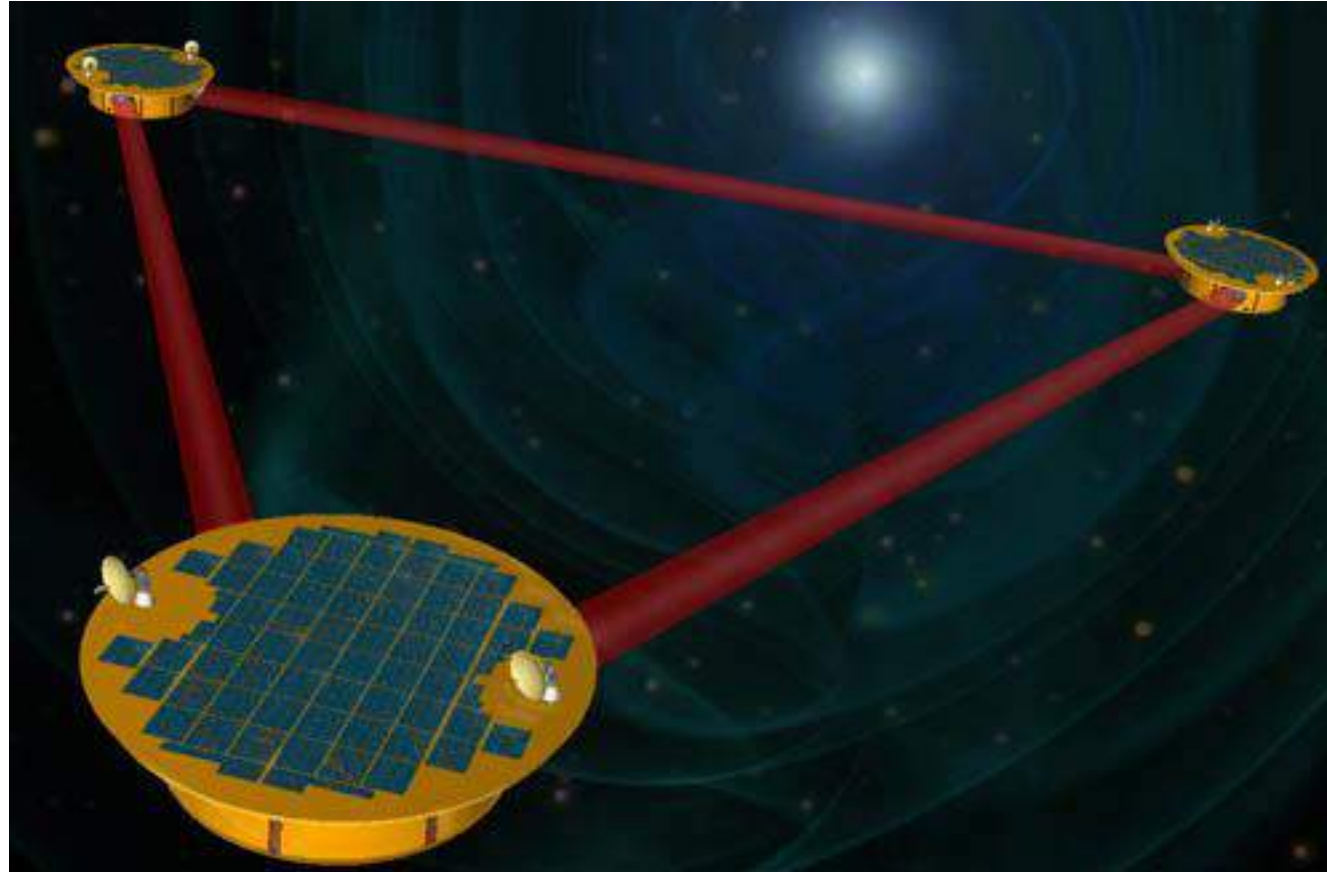


Gravitational Waves : LIGO



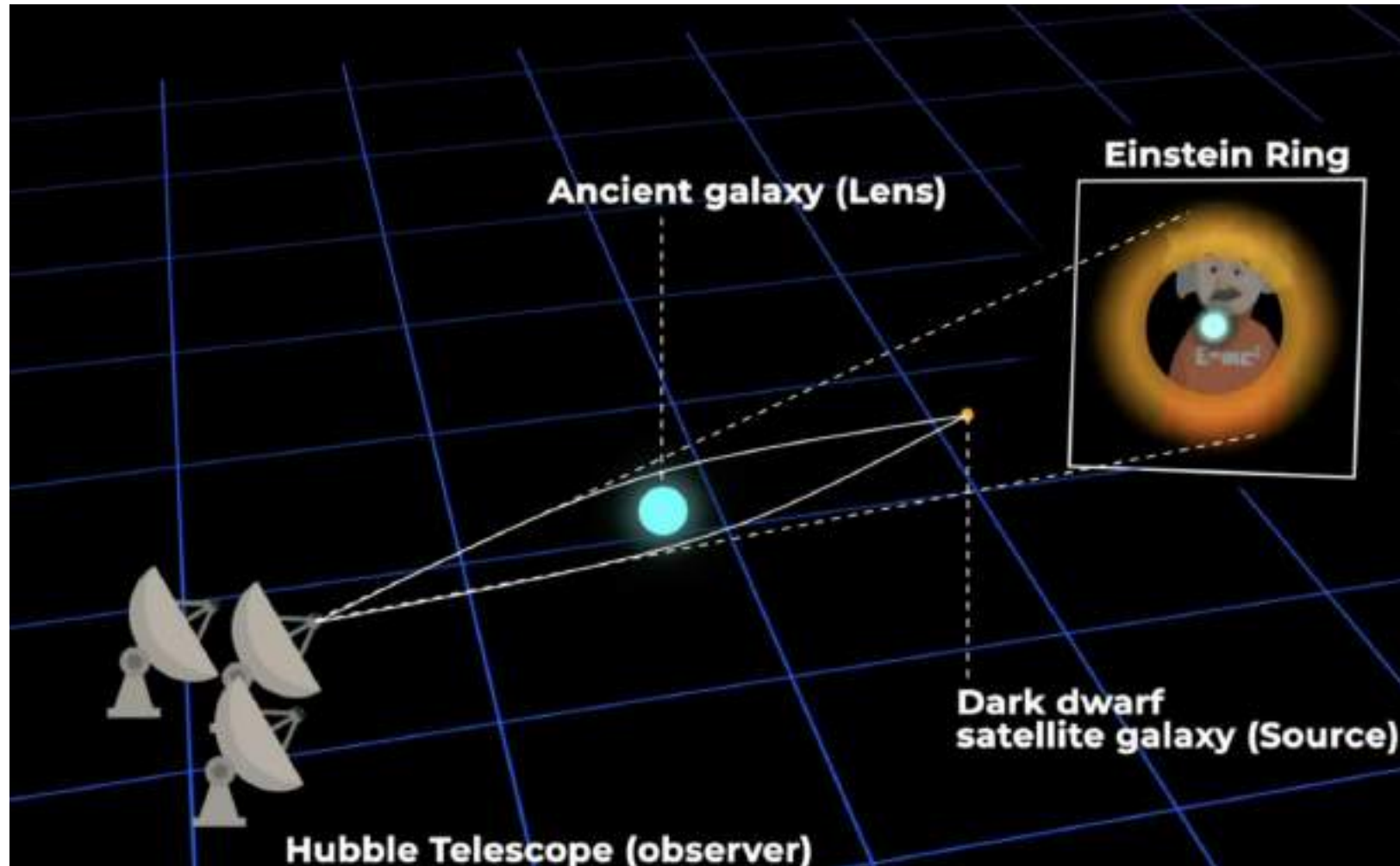
E-Lisa

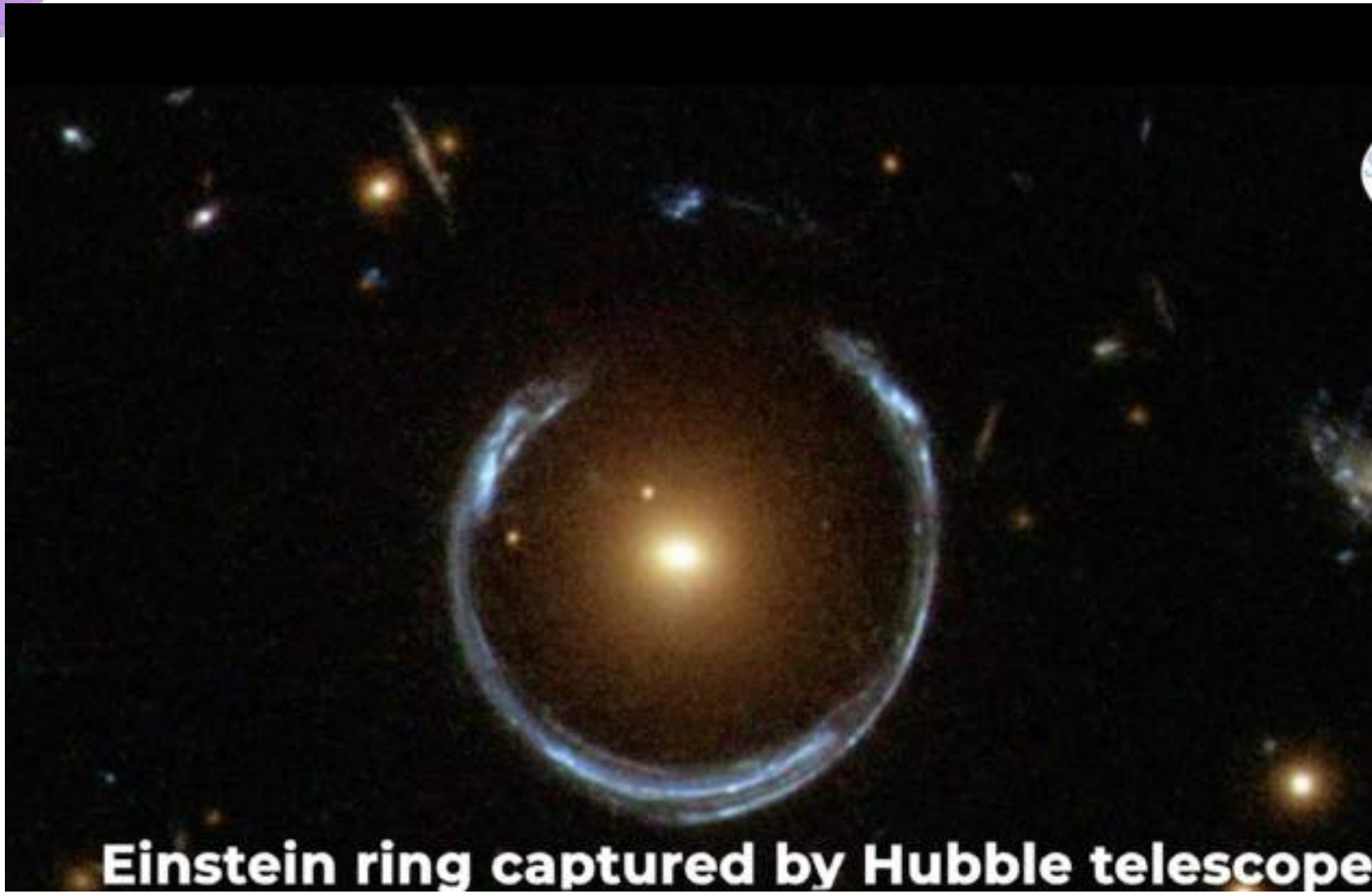
- eLISA's mission is to explore the Gravitational waves from space for the first time.
- The eLISA mission consists of a “Mother” and two “Daughter” spacecraft. These will orbit the Sun in a triangular configuration, as shown in the figure.
- The three satellites will form a precision interferometer, with the two Daughter spacecraft connected to the Mother one by 1 million km long laser beams.
- This interferometer will be capable of detecting gravitational waves at frequencies in the range of 0.1 mHz to 1 Hz.
- Such a frequency interval is not accessible on Earth due to arm length limitations and to noise caused by the terrestrial gravity gradient noise: in this sense, eLISA will complement the efforts of ground-based gravitational-wave detectors. Ex LIGO



Gravitational Lensing

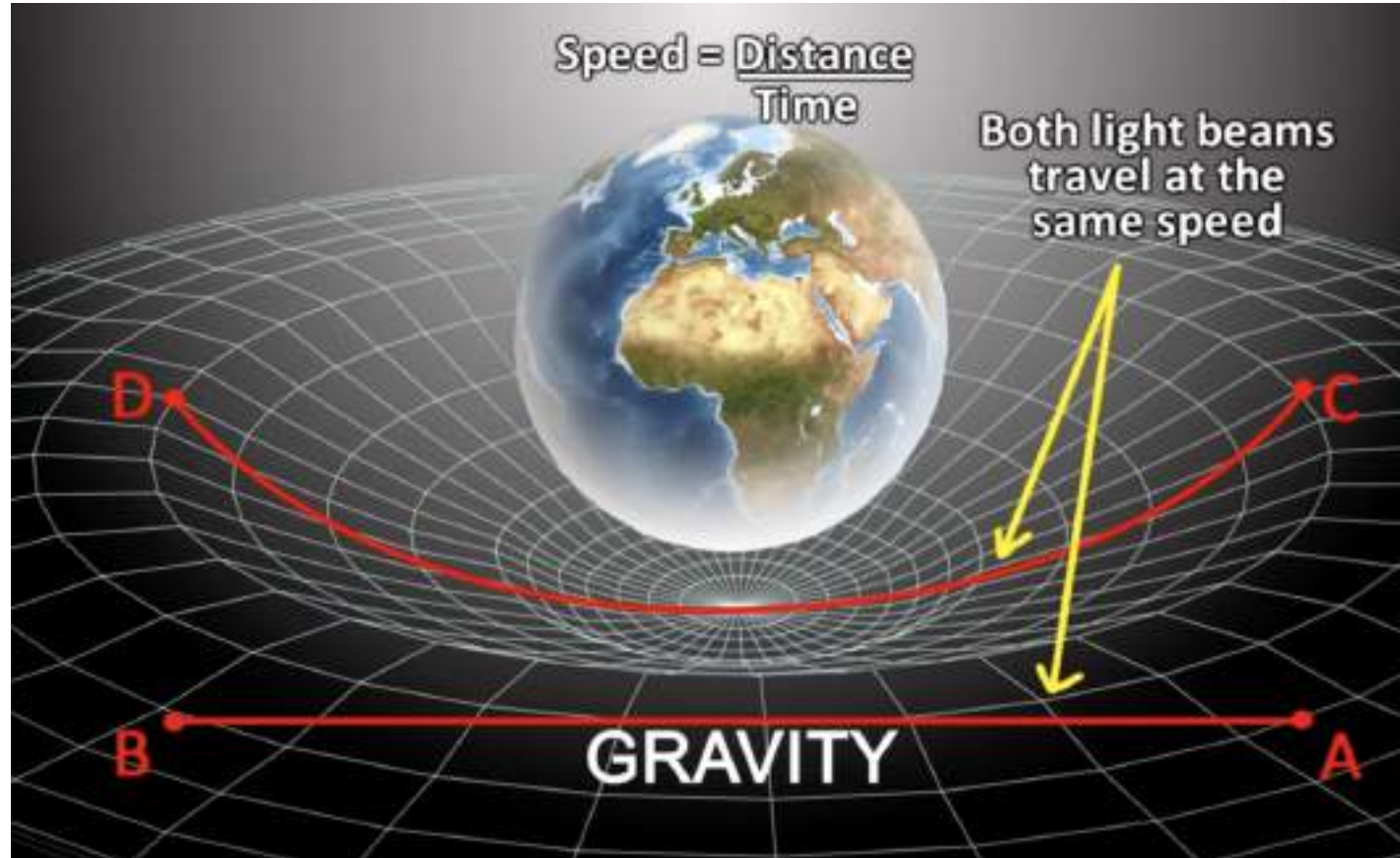
- Lens : anything that distorts light rays to make it easier for us to see.
- When gravity distorts light rays it is called gravitational lensing : gravity leads to curving of light allowing us to see things that we wouldn't see normally.





Special Theory of Relativity

- As per Einstein : the speed of light is always the same between two points.
- But if one place gravity curves / distorts objects and at the other place it doesn't, How will the speed remain same?
- This is because gravity can change the time also.



Since distance is longer here, time must also be longer to keep a constant speed



constant
Speed = $\frac{\text{Distance}}{\text{Time}}$ Goes Up



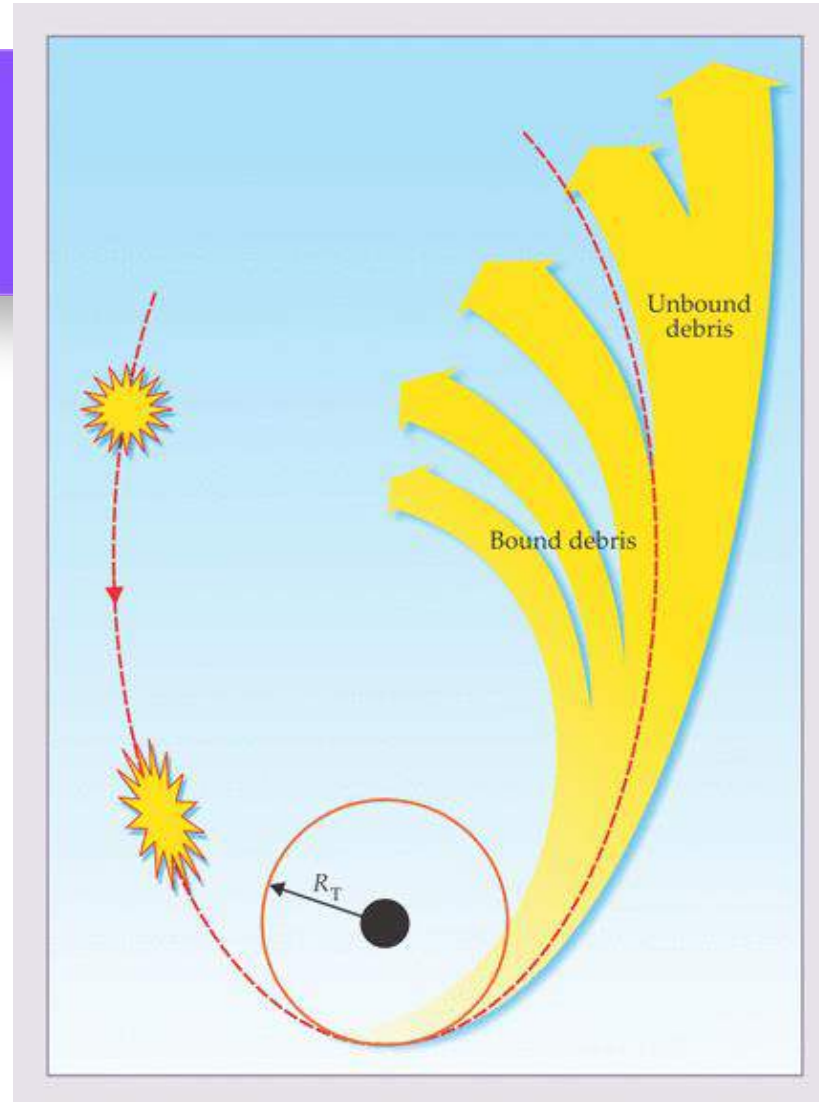
Conclusions

- Gravity is bending of space time : general theory of relativity
- Understanding black holes and how to photograph them
- Special theory of relativity : Space – time concept
- Gravitational Waves : in case of cataclysmic events in space

Tidal Disruption Events

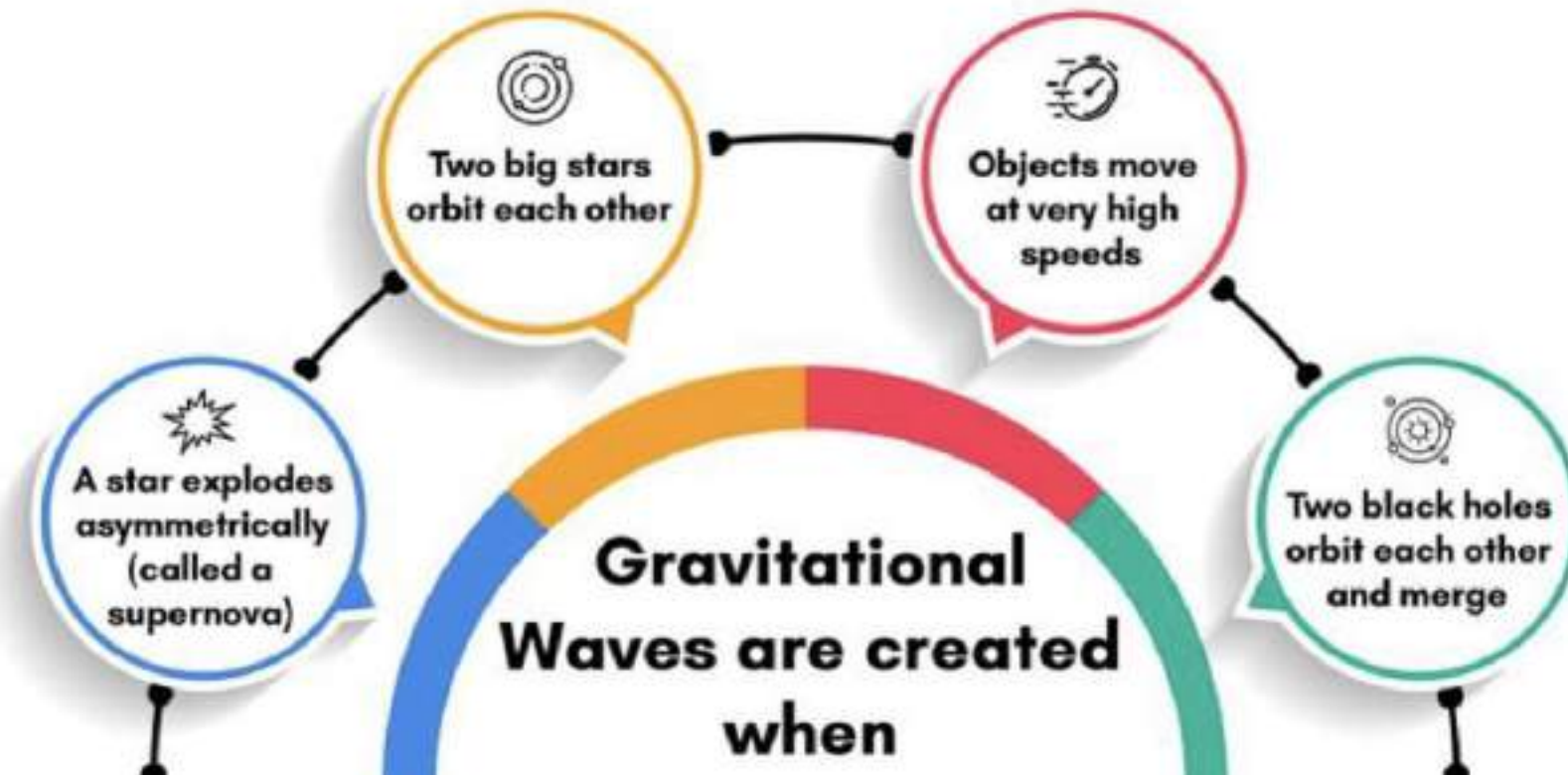
- 2018 : Scientists noticed that a black hole violently ripped apart a star when it got too close.
- Gravitational Forces stretch a star and then the star will be ripped apart.

Tidal Disruption Events



Binary Supermassive Black Hole

- Binary supermassive black hole system was discovered through gravitational lensing of blazars.
- Blazars : among the most luminous and energetic objects in the universe.
- Significance : will help in future detection of Graviational Waves



James Webb Space Telescope

- The James Webb Space Telescope (JWST) is NASA's infrared flagship observatory.
- It is an international collaboration between NASA, the European Space Agency (ESA), and the Canadian Space Agency (CSA).
- It was formerly called the Next Generation Space Telescope

James Webb Space Telescope

- Mission Goals are :
- Search for first galaxies or luminous objects formed after Big Bang.
- Big Bang is also called expanding universe hypothesis. It states that all of the current and past matter in the Universe came into existence at the same time, roughly 13.8 billion years ago.
- Determine how galaxies evolved.

James Webb Space Telescope

- Observe stars formation from the first stages to formation of planetary systems
- Measure physical and chemical properties of planetary systems, including our own Solar System, and investigate potential for life in those systems.

JSWT continued

- JWST is headed for Sun-Earth Lagrange Point 2 (L2).
- The JWST is being considered as the successor of the Hubble Space Telescope (HST).

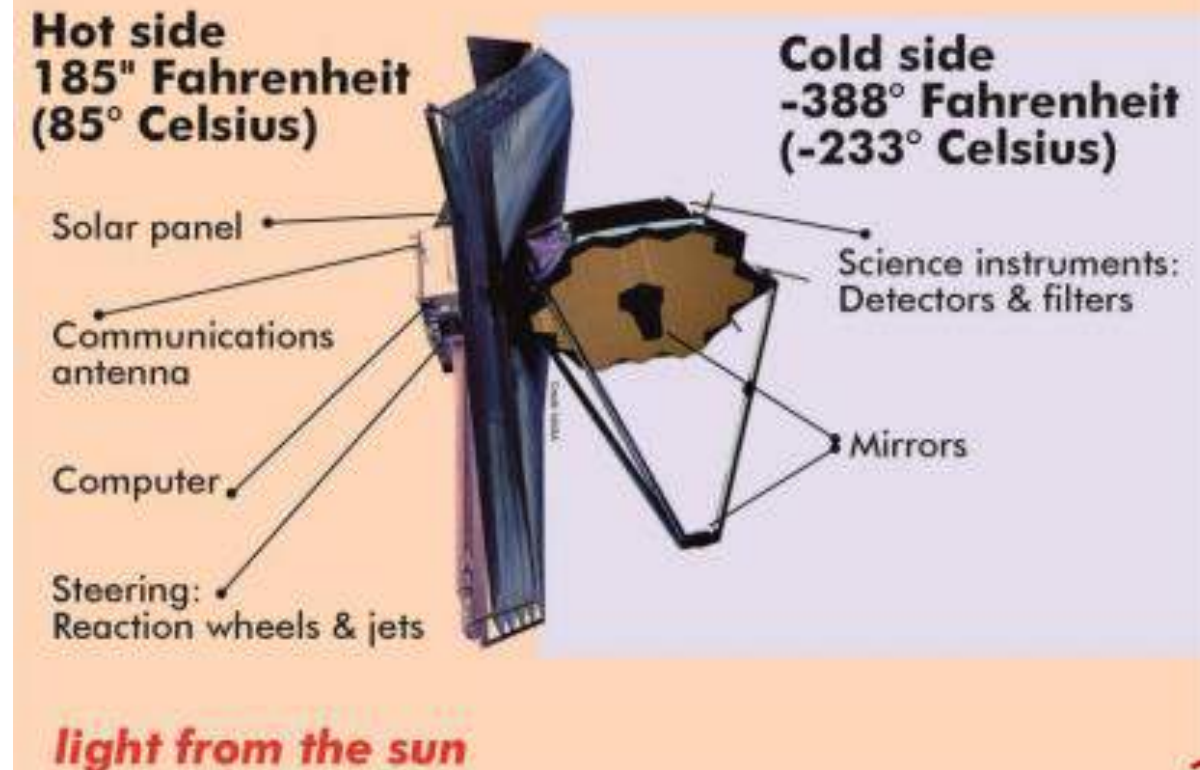
JSWT continued

- Advantages of point L2 :
- Spacecraft can communicate with earth
- Can observe the universe without chances of eclipse : clear view of deep space
- Solar power from the sun behind it

JAMES WEBB SPACE TELESCOPE

- James Webb Space Telescope has provided astronomers with a glimpse of the early universe in a new image
- Telescope captured an image of a galaxy cluster called MACS0647, as well as distant galaxy MACS0647-JD.
- Scientists believe that this might be an event of a galaxy merger in the early universe.
- The distant galaxy is visible because of gravitational lensing.

The Two Sides of the Webb Telescope



Key features of Webb

- Visibility Spectrum: Webb views the universe in infrared : **ONLY** specialized infrared telescope in space
- Mirrors: Its primary mirror is 6.5 meters in diameter
- A secondary mirror was also deployed
- L2 lets telescope stay in line with Earth as it moves around Sun.
- Allows satellite's large sunshield to protect telescope from light and heat of Sun and Earth

Major instruments:

- Contained within the Integrated Science Instrument Module (ISIM), which is one of the 3 major payloads.
- Optical Telescope Element (OTE) and Spacecraft Element are the other 2.
- ISIM is main payload. It houses four main instruments:
- Near-Infrared Camera
- Near-Infrared Spectrograph (NIRSpec),
- Mid-Infrared Instrument (MIRI) and
- Fine Guidance Sensor/ Near InfraRed Imager and Slit less Spectrograph (FGS/NIRISS)

How JWST will help in advancing understanding of Universe

Early Universe



JWST will allow us to see galaxies that formed just 100 million years after **Big Bang**

Exoplanets



JWST will scan atmospheres of **earth-like exoplanets**, searching for gases associated with life

Outer Planets



Will monitor Jupiter, Saturn, Uranus, and Neptune to see **how their atmospheres change** over time.

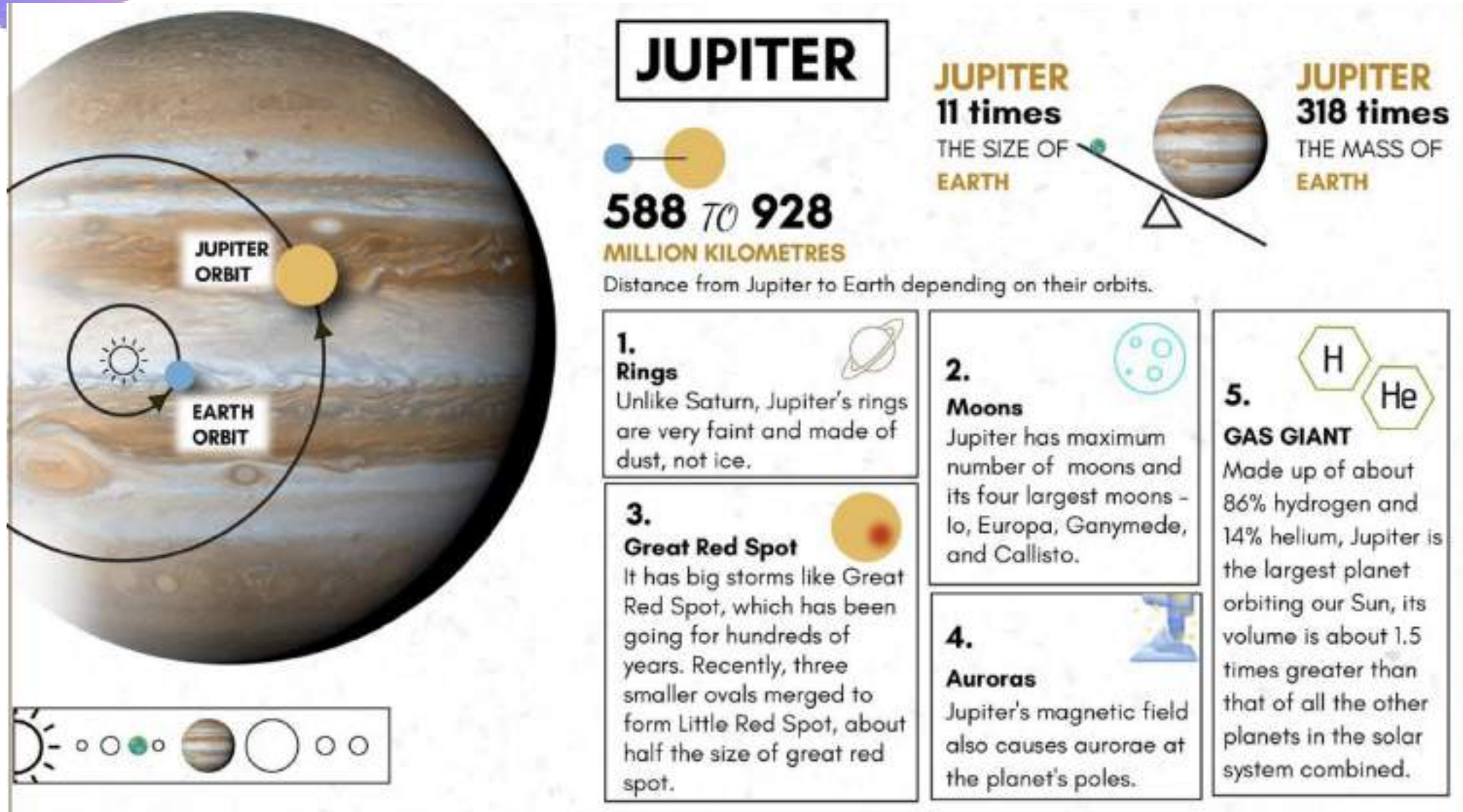
Small Worlds



From the **asteroid belt to Kuiper belt**, JWST will determine what makes up some of our solar system's smallest objects ★

OBSERVATIONS MADE BY JWST

- New images of Jupiter
- New images of Jupiter presented its massive storms, colourful auroras, faint rings and two small moons — Amalthea andAdrastea (Nearer to Jupiter).



Pillars of Creation (PoC) – Formation of new stars

- (JWST) has captured the iconic PoC where new stars are forming within dense clouds of gas and dust.
- The three-dimensional pillars are made up of cool interstellar gas and dust that appear – at times – semi-transparent in near-infrared light
- Located 6,500 light years from Earth, in the Eagle Nebula of our Milky Way galaxy
- It will help researchers revamp their models of star formation



Exo-Moons

- Scientists at Indian Institute of Astrophysics have developed a model to trace habitable exomoons with the help of the James Webb Space Telescope (JWST)
- They are natural satellites that revolve around exoplanets (planets orbiting stars other than the sun)
- So far, 5000 exoplanets have been discovered by using several telescopes
- However, the natural satellites or exo-moons around any of these planets still remain untraced.

Other observations made from images revealed by Webb

- SMACS 0723 (Called Webb's First Deep Field): It is a cluster teeming with thousands of galaxies
- WASP-96b ; planet outside our solar system reveals clear signature of water,
- Southern Ring Nebula: This planetary nebula is approximately 2,000 light-years away.

Other observations made from images revealed by Webb

- Stephan's Quintet: galaxies, located in constellation Pegasus.
- Carina Nebula : “Cosmic Cliffs” in the Carina Nebula unveils the earliest, rapid phases of star formation
- Einstein Ring: n light from a star or a galaxy passes another galaxy or a massive object on its way towards Earth.

Term

- Kulper Belt : A donut shaped region of icy bodies beyond orbit of Neptune.
- Asteroid belt : A region of space between the orbits of mars and jupiter.
- Exoplanet : Any planet beyond our solar system
- Nebula : It is a giant cloud of dust and gas in space. Some nebulae come from gas and dust thrown out by explosion of a dying star, such as a supernova. Other nebulae are regions where new start are beginning to form

	Hubble	Webb	Herschel Space Observatory
Distance Earth	507 Km	1.5 Million Km at Second Sun-Earth Lagrange point	Second Sun-Earth Lagrange Point
Primary Mirror Diameter	2.4 Meter	5.5 Meter	3.5 Meter
Searching for	Young Galaxies (12.5)Billion year ago)	New Born Galaxies (13.5 Billion years ago)	Most actively star -forming galaxies
Serviceable	Yes	No	No
Wavelengths	Visible, UV, Part of near-infrared	Near and mid infrared	Far-infrared and submillimeter

Parker Space Probe

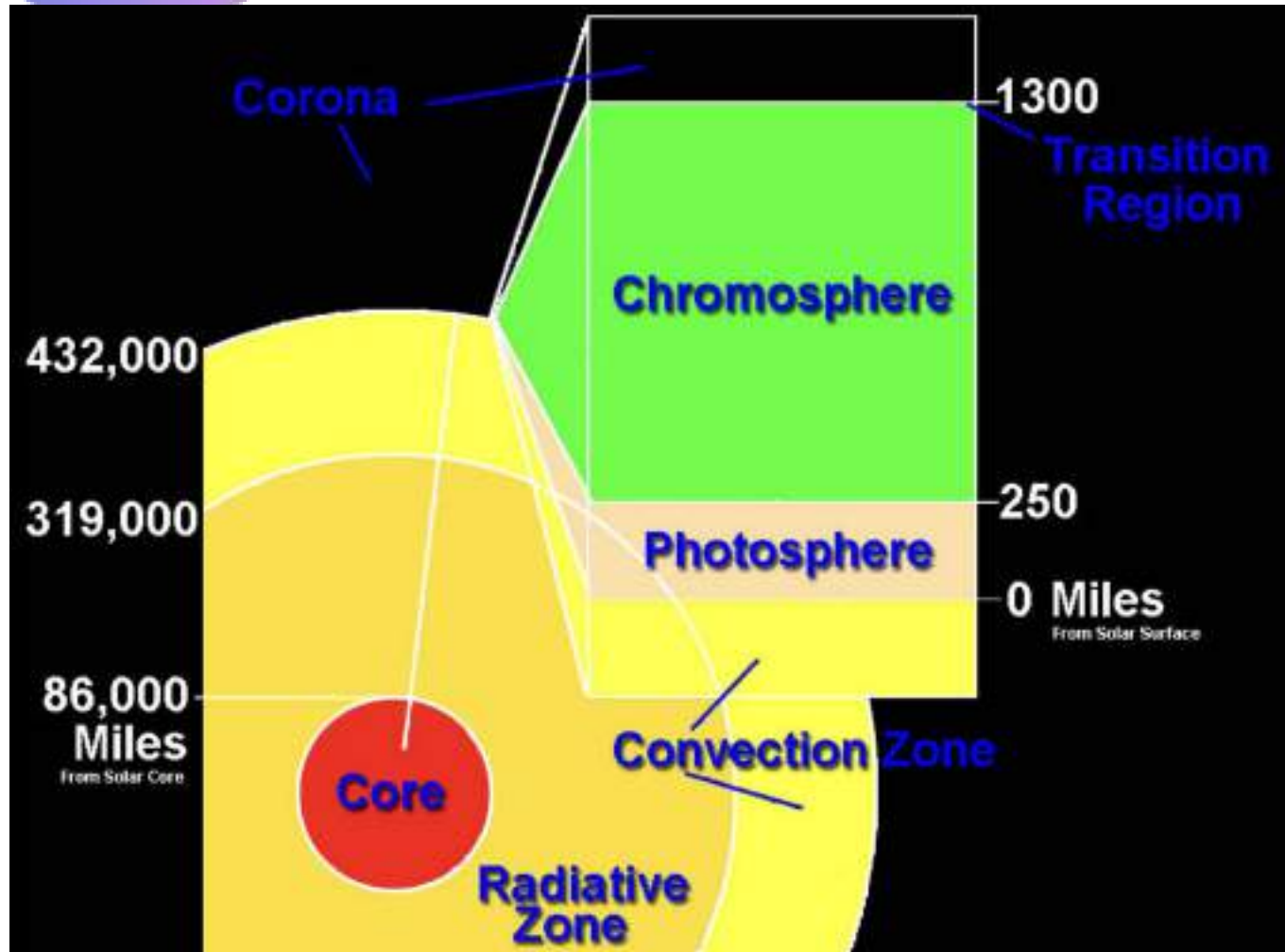
- NASA's Parker Solar Probe (PSP) has now flown through the Sun's upper atmosphere – the corona – and sampled particles and magnetic fields there.
- Launched in 2018, the PSP is part of NASA's Living with a Star program to explore aspects of the Sun-Earth system that directly affect life and society.

Parker Space Probe

- It is named after Eugene Newman Parker, a physicist who proposed concepts about how stars give off energy.
- It uses a carbon-composite shield, to withstand temperatures nearly 2,500 F (1,377 C), to protect itself from Sun's heat.

Parker Space Probe

- It became the first spacecraft in history to touch the Sun.
- It uses a series of gravitational encounters with the planet **Venus** to gradually lower the orbit of the spacecraft until it enters the outer atmosphere, or corona, of the Sun.



Mnemonic to remember layers of sun :

CoRaCoPh : Chrome and Corona

Relevance of PSP

- Deeper insights into sun's evolution.
- Predicting extreme weather events by helping in forecasting solar storms(or Geomagnetic storm).
- Solar storms are the biggest outbursts from the Sun can rattle Earth's magnetic field. In the process:
 - Communications may be disrupted,
 - Satellites can be knocked offline
 - Power grids will be vulnerable to electrical surges.

Solar Flares

- Solar flares are a sudden explosion of energy caused by reorganizing of magnetic field lines near sunspots.
- Solar magnetic cycle that works in deep interior of Sun creates regions that rise to surface and appear like dark spots. These are sunspots

Solar Flares

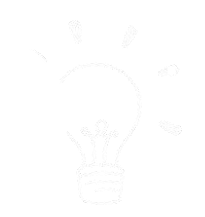
They appear dark because they are cooler than other parts of Sun's surface.

- In a solar flare, energy stored in sun's magnetic structures is converted into light and heat energy.
- This causes emission of high energy x-ray radiation and highly accelerated charged particles to leave the sun's surface.
- Sometimes solar flares also cause hot plasma to be ejected from Sun, causing a solar storm, and this is called Coronal Mass Ejection (CME).

Solar Flares

- Energy, radiation and high energy particles emitted by solar flares can trigger intense lights in the sky, called Auroras
- Auroras are caused by the interaction of energetic particles (electrons and protons) of the solar wind with atoms of the upper atmosphere occurring primarily in high latitudes of both hemispheres.
- Auroras in the Northern Hemisphere are called aurora borealis, aurora polaris, or northern lights, and in the Southern Hemisphere aurora australis, or southern lights.

Other Effects



Solar flare classification	Possible effects on Earth
B	none
C	Possible effects on space missions.
M	Blackout in radio transmissions and possible damages in astronauts outside spacecraft.
X	Damage to satellites, communication systems, power distribution stations and electronic equipment

Increasing Intensity



Solar Cycle

- Every 11 years, sun's magnetic field completely flips.
- This can be tracked using sunspots.
- Beginning of new solar cycle : Solar minimum.

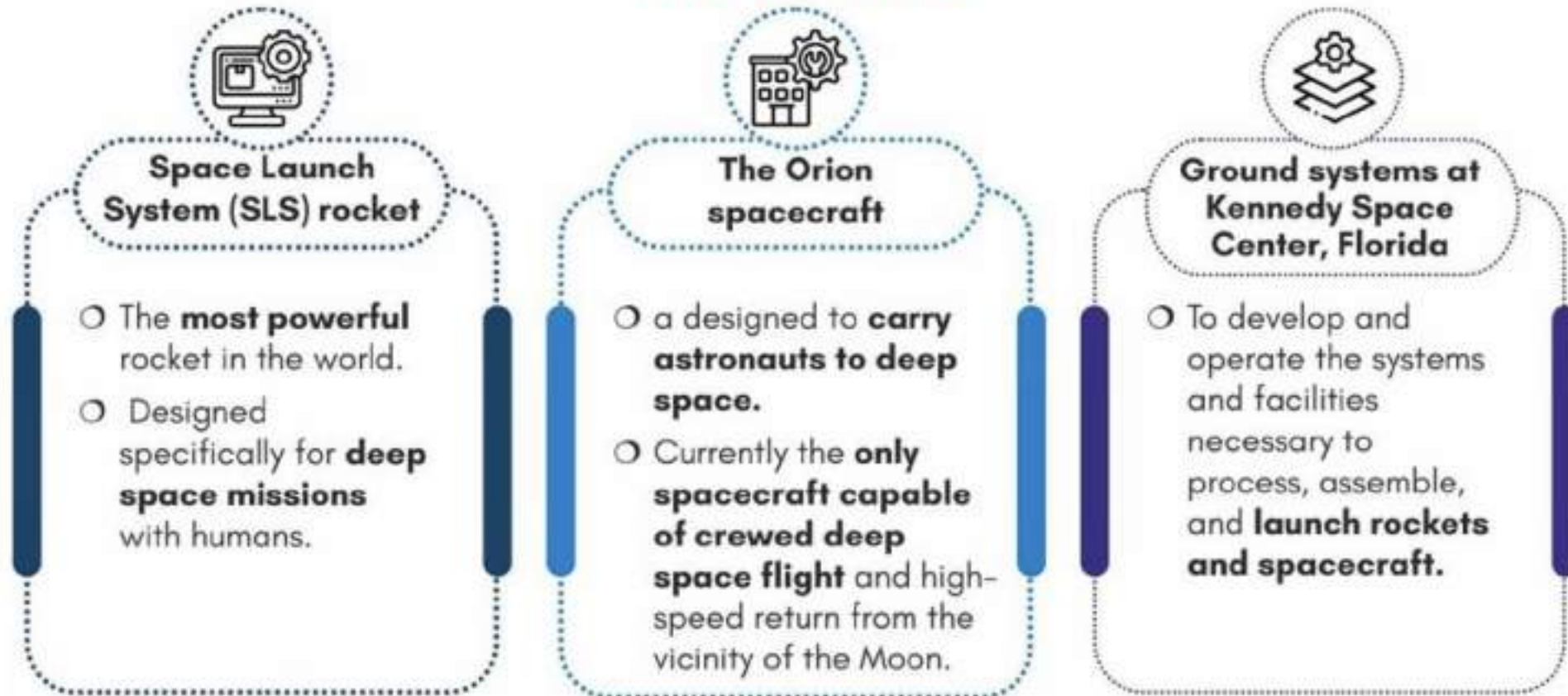
ARTEMIS I

- Recently, NASA's Artemis 1 mission successfully lifted off from the Kennedy Space Centre.
- Aim: To build a long term human presence at the Moon.

Mission specifications: An uncrewed flight test that will provide a foundation for human deep space exploration.

- With Artemis, NASA is planning to land the first woman on the Moon.
- First integrated test of NASA's deep space exploration systems,

Artemis 1



Artemis I Mission objectives

- Demonstrate Orion's heat shield can withstand the high speed only and high heat conditions when returning through Earth's atmosphere.
- Study the radiation environment of deep space
- Retrieving orion after splashdown
- Accomplish flight test objectives like certifying Orion's optical navigation system, deploying cube sats.
- Study the Moon to learn more about the origin and history Earth, the Moon, and our solar system.

USSR	• Luna 1, Luna 2, Luna 3. • Luna 2 was the first spacecraft to impact the Moon's surface.
USA	• Lunar Orbiter 1 • Apollo II: First humans to land on the moon. • Lunar Reconnaissance Orbiter (LRO)
Japan	• Hiten
China	• Change's 1
India	• Chandrayan - 1 • Chandrayaan-2

Retrograde orbit

- After getting closer to the moon, the Orion spacecraft used the gravitational kick it receives to enter a so-called "distant retrograde orbit."
- Retrograde means that it will orbit the moon in the opposite direction to the one in which the moon spins.
- Orion will stay in that orbit for some days. Then it will swing back down toward the moon for another kick to help power its journey back to Earth



WORD OF THE WEEK | PROGRADE vs RETROGRADE



The direction an object spins in relation to its orbit around the Sun. Prograde refers to an object that spins in the same direction as its orbit. Retrograde refers to an object that spins in the opposite direction of its orbit.

Earth has a prograde rotation and asteroid Bennu has a retrograde rotation, which means the two bodies rotate in opposite directions.



SPACE SUSTAINABILITY

- United Nations Office of Outer SPACE Affairs and government of AUSTRIA organized World Space Forum (WSF) 2022 under theme “Sustainability in Space for sustainability on Earth”.
- World Space Forum is a platform, to promote discussions
- Space sustainability
 - Ensuring that all humanity can continue to use outer space for peaceful purposes

Threats to space sustainability

- Orbital crowding and Space Debris:
- Militarisation and weaponisation of the space: Antisatellite (ASAT) weapons by countries like US, Russia, China, and India.
- Rendezvous and proximity operations (RPO): It involves altering the trajectory of one or more space objects to bring them into proximity to each other.
- Space weather can impact satellites

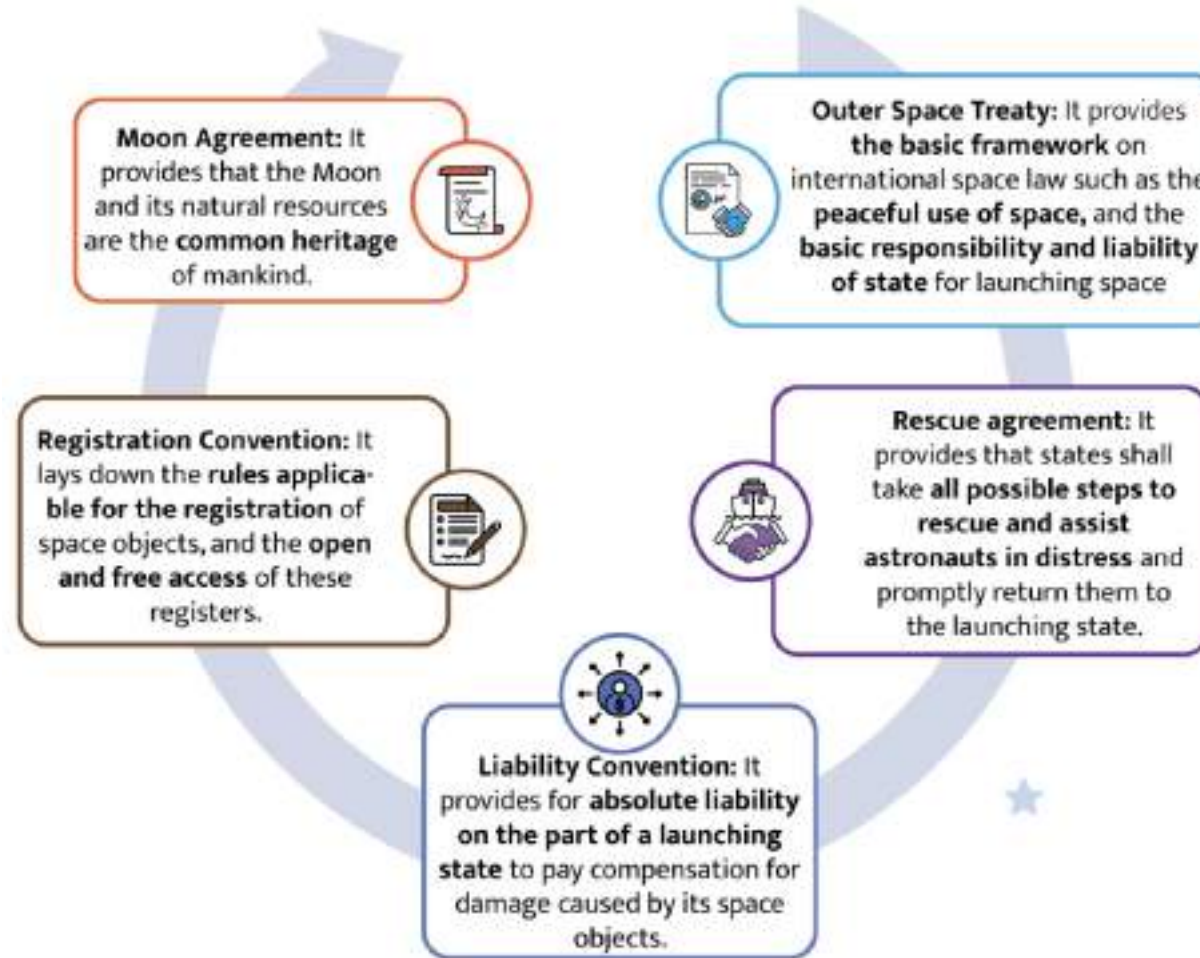
Initiatives taken towards space sustainability : Global

- U.N. Office for Outer Space Affairs (UNOOSA):
- Conference on Disarmament (CD): international forum with focus on prevention of an arms race in outer space (PAROS).
- Guidelines on space sustainability : by UNCOPOUS
- Space Sustainability Rating (SSR)
- ASAT test-ban resolution: A non-binding resolution (India abstained) passed by United Nations General Assembly (UNGA) calling for a ban on kinetic ASAT tests.

Indian Initiatives

- Space Situational Awareness : necessary mitigation measures to safeguard assets.
- It involves : SST : space surveillance and tracking
- Space Weather monitoring
- Near earth objects monitoring
- Project NETRA
- Digantra : space sector startup is going to set up India's first commercial SSA observatory.
- India USA bilateral SSA agreement.
- SPADEX : in orbit servicing : platform by ISRO

COPOUS oversees five international treaties underpinning space law:



Test of 13 Questions!

Cut off : 10/13



Q.1) PYQ (2013)

Q. The known forces of nature can be divided into four classes, viz, gravity, electromagnetism, weak nuclear force and strong nuclear force. With reference to them, which one of the following statements is not correct?

- a) Gravity is the strongest of the four
- b) Electromagnetism act only on particles with an electric charge
- c) Weak nuclear force causes radioactivity
- d) Strong nuclear force holds protons and neutrons inside the nuclear of an atom.

Q.2) PYQ (2016)

Q. With reference to 'Astrosat', the astronomical observatory launched by India, which of the following statements is/are correct?

1. Other than the USA and Russia, India is the only country to have launched a similar observatory into space.
2. Astrosat is a 2000 kg satellite placed in an orbit at 1650 km above the surface of the Earth.

Select the correct answer using the code given below

- A. 1 only

Q.3) PYQ (2016)

Q. Consider the following statements:

The Mangalyaan launched by ISRO

1. is also called the Mars Orbiter Mission
2. Made India the second country to have a spacecraft orbit Mars after the USA
3. Made India the only country to be successful in making its spacecraft orbits Mars in its very first attempt.

Which of the statements given above is/are correct?

Q.4) PYQ (2017)

Q. What is the purpose of ‘evolved Laser Interferometer Space Antenna (eLISA)’ project?

- a) To detect neutrinos
- b) To detect gravitational waves
- c) To detect the effectiveness of missile defence system
- d) To study the effect of solar flares on our communication systems

Q.5) PYQ (2017)

Q. The terms 'Event Horizon', 'Singularity', 'String Theory' and 'Standard Model' are sometimes seen in the news in the context of

- A. Observation and understanding of the Universe
- B. Study of the solar and the lunar eclipses
- c. Placing satellites in the orbit of the Earth
- D. Origin and evolution of living organisms on the Earth

Q.6) PYQ (2018)

Q. With reference to the Indian Regional Navigation Satellite System (IRNSS), consider the following statements :

1. IRNSS has three satellites in geostationary and four satellites in geosynchronous orbits.
2. IRNSS covers entire India and about 5500 sq. km beyond its borders.
3. India will have its own satellite navigation system with full global coverage by the middle of 2019.

Which of the statements given above is/are correct?

A. 1 only

Q.7) PYQ (2018)

Q. Consider the following phenomena :

1. Light is affected by gravity.
2. The Universe is constantly expanding.
3. Matter warps its surrounding space-time.

Which of the above is/are the prediction/predictions of Albert Einstein's General Theory of Relativity, often discussed in media?

- a) 1 and 2 only
- b) 3 only
- c) 1 and 2
- d) 1, 2 and 3

Q.8) PYQ (2018)

Q. In which of the following areas can GPS technology be used?

1. Mobile phone operations
2. Banking operations
3. Controlling the power grids

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only

Q.9) PYQ (2018)

Q. With reference to India's satellite launch vehicles, consider the following statements:

1. PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.
3. GSLV Mk III is a four-stage launch vehicle with the first and third stages using solid rocket motors, and the second and fourth stages using liquid rocket engines.

Which of the statements given above is/are correct?

- A. 1 only
- B. 2 and 3
- C. 1 and 2
- D. 3 only

Q.10) PYQ (2019)

Q. Recently, scientists observed the merger of giant ‘blackholes’ billions of light-years away from the Earth. What is the significance of this observation?

- a) ‘Higgs boson particles’ were detected.
- b) ‘Gravitational waves’ were detected.
- c) Possibility of intergalactic space travel through ‘wormhole’ was confirmed.
- d) It enabled the scientists to understand ‘singularity’.

Q.11) PYQ (2020)

Q. “The experiment will employ a trio of spacecraft flying in formation in the shape of an equilateral triangle that has sides one million kilometres long, with lasers shining between the craft”. The experiment in question refers to

- a) Voyager
- b) New Horizons
- c) LISA Pathfinder
- d) Evolved LISA

Q.12) PYQ (2021)

Q. Which one of the following is a reason why astronomical distances are measured in light-years?

- a) Distances among stellar bodies do not change.
- b) Gravity of stellar bodies does not change.
- c) Light always travels in a straight line.
- d) Speed of light is always the same.

Q.13) PYQ (2022)

Q. If a major solar storm (solar flare) reaches the Earth, which of the following are the possible effects on the Earth?

1. GPS and navigation systems could fail.
2. Tsunamis could occur at equatorial regions.
3. Power grids could be damaged.
4. Intense auroras could occur over much of the Earth.
5. Forest fires could take place over much of the planet.
6. Orbits of the satellites could be disturbed.
7. Shortwave radio communication of the aircraft flying over polar regions could be interrupted.

Select the correct answer using the code given below:

- | | |
|---------------------------|----------------------------|
| (a) 1, 2, 4 and 5 only | (b) 2, 3, 5, 6 and 7 only |
| (c) 1, 3, 4, 6 and 7 only | (d) 1, 2, 3, 4, 5, 6 and 7 |

Congratulations on giving the test!

Let's Discuss The Answers.



Q.1) PYQ (2013)

Q. The known forces of nature can be divided into four classes, viz, gravity, electromagnetism, weak nuclear force and strong nuclear force. With reference to them, which one of the following statements is not correct?

- a) Gravity is the strongest of the four**
- b) Electromagnetism act only on particles with an electric charge
- c) Weak nuclear force causes radioactivity
- d) Strong nuclear force holds protons and neutrons inside the nuclear of an atom.

Answer: A

- Statement 1 is incorrect since gravity is actually the weakest of the four fundamental forces for a given distance
- Electromagnetism acts only on charged particles, where the charge can be negative or positive
- The weak force is responsible for the existence and structure of atomic nuclei including radioactivity and nuclear fusion
- Strong nuclear force is responsible for holding protons and neutrons together inside the atomic nucleus and is 100 times stronger than electromagnetism

Q.2) PYQ (2016)

Q. With reference to 'Astrosat', the astronomical observatory launched by India, which of the following statements is/are correct?

1. Other than the USA and Russia, India is the only country to have launched a similar observatory into space.
2. Astrosat is a 2000 kg satellite placed in an orbit at 1650 km above the surface of the Earth.

Select the correct answer using the code given below

- A. 1 only

Answer: D

- ASTROSAT is **India's first dedicated multi-wavelength space observatory.**
- This scientific satellite mission endeavours for a more **detailed understanding of our universe.**
- **Before India, NASA, European Space Agency, JAXA, Roscosmos have launched similar observatory into the space.**
- **ASTROSAT with a lift-off mass of about 1513 kg was launched into a 650 km orbit inclined at an angle of 6 deg to the equator by PSLV-C30.**

Q.3) PYQ (2016)

Q. Consider the following statements:

The Mangalyaan launched by ISRO

1. is also called the Mars Orbiter Mission
2. Made India the second country to have a spacecraft orbit Mars after the USA
3. Made India the only country to be successful in making its spacecraft orbits Mars in its very first attempt.

Which of the statements given above is/are correct?

Answer: D

- The Mars Orbiter Mission (MOM), also called **Mangalyaan** is a **space probe orbiting Mars since 24 September 2014**.
- It was launched on 5 November 2013 **by the Indian Space Research Organisation (ISRO)**.
- It is **India's first interplanetary mission** and it made it the fourth space agency to reach Mars, after Roscosmos, NASA, and the European Space Agency.
- It is the **first Asian nation to reach Mars orbit**, and the first nation in the world to do so in its first attempt

Q.4) PYQ (2017)

Q. What is the purpose of 'evolved Laser Interferometer Space Antenna (eLISA)' project?

- a) To detect neutrinos
- b) To detect gravitational waves**
- c) To detect the effectiveness of missile defence system
- d) To study the effect of solar flares on our communication systems

Answer: B

- The Laser Interferometer Space Antenna (LISA) is a European Space Agency mission designed **to detect and accurately measure gravitational waves—tiny ripples in the fabric of space-time**—from astronomical sources. LISA would be the first dedicated space-based gravitational wave detector. It aims to measure gravitational waves directly by using laser interferometry.

Q.5) PYQ (2017)

Q. The terms 'Event Horizon', 'Singularity', 'String Theory' and 'Standard Model' are sometimes seen in the news in the context of

- A. Observation and understanding of the Universe**
- B. Study of the solar and the lunar eclipses
- c. Placing satellites in the orbit of the Earth
- D. Origin and evolution of living organisms on the Earth

Answer: A

- An **event horizon is most commonly associated with black holes.**
- Light emitted from inside the event horizon can never reach the outside observer.
- In physics, **string theory is a theoretical framework in which the point-like particles of particle physics** are replaced by one-dimensional objects called strings.
- It describes how these strings propagate through space and interact with each other

Q.6) PYQ (2018)

Q. With reference to the Indian Regional Navigation Satellite System (IRNSS), consider the following statements :

1. IRNSS has three satellites in geostationary and four satellites in geosynchronous orbits.
2. IRNSS covers entire India and about 5500 sq. km beyond its borders.
3. India will have its own satellite navigation system with full global coverage by the middle of 2019.

Which of the statements given above is/are correct?

A. 1 only

Answer: A

- The IRNSS is being developed parallel to the GAGAN (GPS Aided GEO Augmented Satellite Navigation) program, the ISRO SBAS (Satellite Based Augmentation System) version of an overlay system for GNSS signal corrections.
- The IRNSS system consists of a constellation of seven satellites and a supporting ground segment.
- Three of the satellites in the constellation will be placed in a geostationary orbit and the remaining four in a geosynchronous

Q.7) PYQ (2018)

Q. Consider the following phenomena :

1. Light is affected by gravity.
2. The Universe is constantly expanding.
3. Matter warps its surrounding space-time.

Which of the above is/are the prediction/predictions of Albert Einstein's General Theory of Relativity, often discussed in media?

- a) 1 and 2 only
- b) 3 only
- c) 1 and 2
- d) 1, 2 and 3**

Answer: D

- 2010: International Team of Astronomers confirmed Albert Einstein's theory of general relativity and that the expansion of the universe is accelerating after looking at data from the Hubble Space Telescope. There is an unknown source of energy in the universe which is causing the cosmic expansion to speed up.
- 2016: Japanese researchers' study finds that the expansion of the universe could be explained by a cosmological constant, as proposed by Einstein in his theory of general relativity.

Q.8) PYQ (2018)

Q. In which of the following areas can GPS technology be used?

1. Mobile phone operations
2. Banking operations
3. Controlling the power grids

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3**

Answer: D

- GPS is the Global Positioning System.
- It's a global navigation satellite system that uses at least 24 satellites, a receiver and algorithms to provide location, velocity and time synchronization for air, sea and land travel. GPS works at all times and in almost all weather conditions.
- Some examples of GPS applications include Agriculture, Astronomy, Automated vehicle, Cellular telephony, Clock synchronization, Disaster relief/emergency services, Radio occultation for weather and atmospheric science applications, Geofencing, GPS aircraft tracking, for mining, or for data mining, for navigation and surveying.

Q.9) PYQ (2018)

Q. With reference to India's satellite launch vehicles, consider the following statements:

1. PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.
3. GSLV Mk III is a four-stage launch vehicle with the first and third

Answer: A

- Polar Satellite Launch Vehicle (PSLV) is the third generation launch vehicle of India. It is the first Indian launch vehicle to be equipped with liquid stages.
- GSLV-Mk III is a three-stage vehicle with two solid motor strap-ons (S200), a liquid propellant core stage (L110) and a cryogenic stage (C25).
- Geostationary satellites orbit around the earth in 24 hours and since the earth rotates with the same period, the satellite would appear fixed from any point on earth.
- GSLV-Mk III is capable of launching 4 ton class of satellites to Geosynchronous Transfer orbit (GTO).

Q.10) PYQ (2019)

Q. Recently, scientists observed the merger of giant ‘blackholes’ billions of light-years away from the Earth. What is the significance of this observation?

a) ‘Higgs boson particles’ were detected.

b) ‘Gravitational waves’ were detected.

c) Possibility of intergalactic space travel through ‘wormhole’ was confirmed.

d) It enabled the scientists to understand ‘singularity’.

Answer: B

- It is possible for two black holes to collide.
- Once they come so close that they cannot escape each other's gravity, they will merge to become one bigger black hole.
- Such an event would be extremely violent.
- Even when simulating this event on powerful computers, we cannot fully understand it. However, we do know that a black hole merger would produce tremendous energy and send massive ripples through the space-time fabric of the Universe.
- These ripples are called gravitational waves.

Q.11) PYQ (2020)

Q. “The experiment will employ a trio of spacecraft flying in formation in the shape of an equilateral triangle that has sides one million kilometres long, with lasers shining between the craft”. The experiment in question refers to

- a) Voyager
- b) New Horizons
- c) LISA Pathfinder
- d) **Evolved LISA**

Answer: D

- Evolved Laser Interferometer Space Antenna (eLISA) is a spectacular plan of setting into space three spacecrafts, a mother and two daughter spacecrafts, which will fly in a triangular formation, trailing the earth in its orbit around the sun at a distance of over.
- 50 million km. Each arm of the imaginary triangle, from the mother to each daughter spacecraft, will measure about a million km.
- ELISA seeks to measure gravitational waves in the frequency range from 0.1 mHz to about 100 mHz. To achieve this, it is necessary for the interferometers to have an arm length of a million kilometres and that is impossible to achieve with an earth based setup.
- Therefore, option (d) is the correct answer.

Q.12) PYQ (2021)

Q. Which one of the following is a reason why astronomical distances are measured in light-years?

- a) Distances among stellar bodies do not change.
- b) Gravity of stellar bodies does not change.
- c) Light always travels in a straight line.
- d) **Speed of light is always the same.**

Answer: D

- A light-year is the distance a beam of light travels in a single Earth year, or 6 trillion miles (9.7 trillion kilometers).
- The reason for using light years to measure distance in space is because throughout the universe, light travels at exactly the same speed.
- Therefore, option (d) is the correct answer

Q.13) PYQ (2022)

Q. If a major solar storm (solar flare) reaches the Earth, which of the following are the possible effects on the Earth?

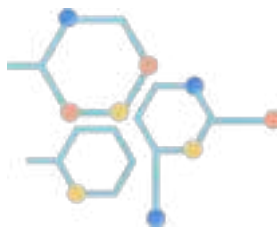
1. GPS and navigation systems could fail.
2. Tsunamis could occur at equatorial regions.
3. Power grids could be damaged.
4. Intense auroras could occur over much of the Earth.
5. Forest fires could take place over much of the planet.
6. Orbits of the satellites could be disturbed.
7. Shortwave radio communication of the aircraft flying over polar regions could be interrupted.

Select the correct answer using the code given below:

- (a) 1, 2, 4 and 5 only (b) 2, 3, 5, 6 and 7 only
(c) 1, 3, 4, 6 and 7 only (d) 1, 2, 3, 4, 5, 6 and 7

Answer: C

- Solar flares are large explosions from the surface of the sun that emit intense bursts of electromagnetic radiation.
- Solar flares occur when magnetic energy builds up in the solar atmosphere and is released suddenly. These outbursts are intrinsically linked to the solar cycle — an approximately 11-year cycle of solar activity driven by the sun's magnetic field.
- They can lead to failing of GPS and navigation systems and could damage power grids on earth. They could also result in intense auroras over much of the Earth. High intensity solar flares also could disturb space activities.



UPSC Prelims

Science & Tech

Class No. 11

Defence Technology



Goldilocks zone

The 'Goldilocks Zone,' or habitable zone, is the range of distance with the right temperatures for water to remain liquid.

Discoveries in the Goldilocks Zone, like Earth-size planet Kepler-186f, are what scientists hope will lead us to water—and one day life.

Ex : our earth is in the sun's goldilock's zone



Greased Lightning 10 : NASA

The GL-10 Greased Lightning is a hybrid diesel-electric tilt wing unmanned aircraft.

It is a battery-powered plane that has 10 engines and can take off like a helicopter and fly efficiently like an aircraft.

One other advantage to the GL-10 besides its versatile vertical take off and landing ability is its noise or lack of it.



Ice Cube Neutrino Observatory

The IceCube Neutrino Observatory is an array of 5,160 optical sensors buried in a cubic kilometer of ice beneath the South Pole.

- experiment detect light that is produced when neutrinos interact with a nucleus in the surrounding medium.
- neutrinos are weakly interacting, nearly massless fundamental particles might be produced (according to some theories) when particles of dark matter annihilate with each other. If these theories are true, neutrino detectors—like the IceCube Neutrino Observatory at the South Pole - can look for excess neutrinos coming from known dark matter hotspots.



Missile Tech

- A missile is a guided airborne ranged weapon capable of self-propelled flight usually by a jet engine or rocket motor.
- Missiles have five system components: targeting, guidance system, flight system, engine and warhead.
- A missile needs a flight system. Guided missile is a missile whose course may be altered during flight.
- Range: Land based missiles have longer range than air-based version. Submarine versions have least range.
- The Mysorean rockets were the first iron-cased rockets in world that were successfully deployed for military use. Hyder Ali and Tipu Sultan used the rockets effectively against the British East India Company during the 1780s and 1790s.

Missile Classification

Range

- a.) Short Range (≤ 500) : prithvi
- b.) Medium " (≤ 1500)
- c.) Intermediate (≤ 5000 km) : Agni
- d.) Long Range (> 5000 km) : ICBM. (Agni-V)

Missile Classification

Based on launch site
and target

- a) Surface to Surface Missile (prithvi)
- b) Surface to Air (Aakash) (Trishul) (Barak).
not used now
- c) Air to Air (Astra)
used by navy built in israel

Missile Classification

Speed

- a) Subsonic (< 1 Mach) < 332 m/s : Nirbhaya
- b) Supersonic (1-5 Mach) Brahmos
- c) Hypersonic (> 5 Mach) → proposed: Shaurya
→ (trying to upgrade Brahmos)



Missile Classification

Trajectory followed by missile :

- a) Ballistic
- b) Cruise



Ballistic v/s Cruise Missile

Ballistic : Predetermined Target

Most of the flight is unpowered and unguided after initial power by rocket.

Long Range : Outside atmosphere

ICBM : sub orbital trajectory

Tactical Ballistic Missile : Battlefield Range Ballistic Missile. Short range use <300 km

Cruise : Full flight is powered hence it is a self propelled guided missile

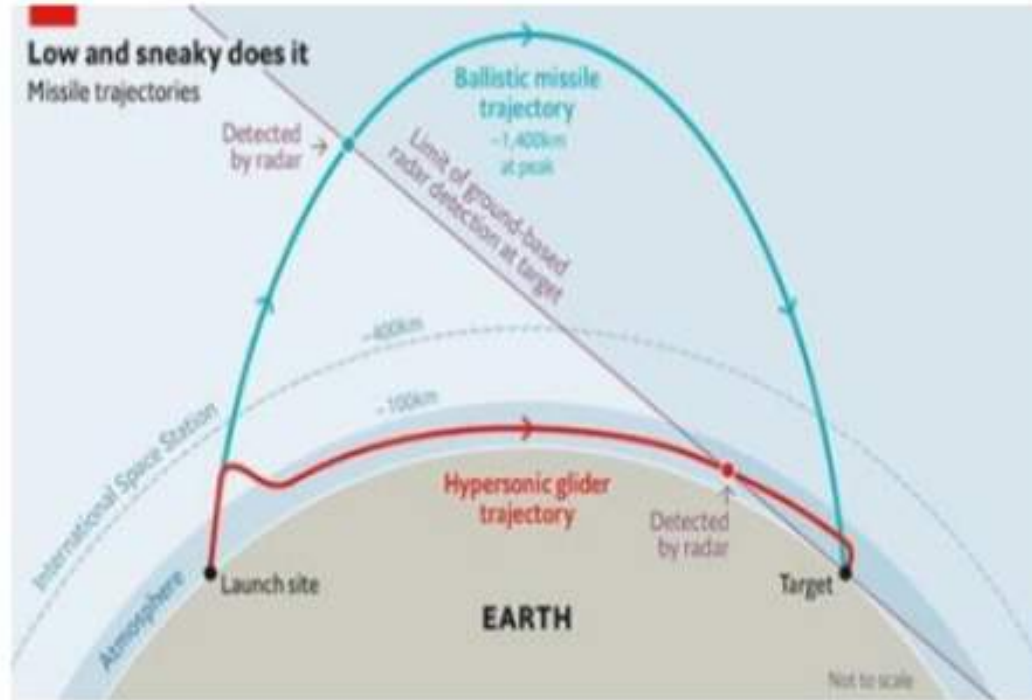
Within the earth's atmosphere : Jet Engine Technology

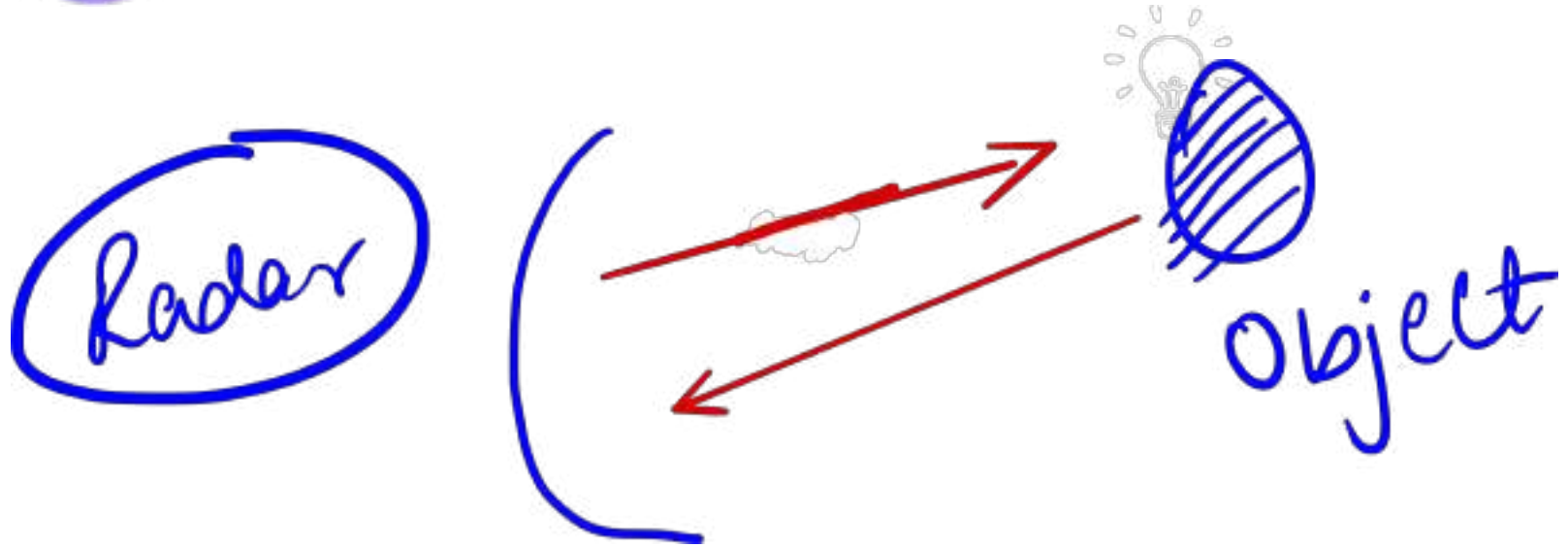
used against terrestrial targets. It flies the major portion of its flight path at approximately constant speed.

Examples of cruise missiles : Brahmos, Harpoon (USA), Exocet (France)



Ballistic	Cruise
Depends on Gravity to hit the target	Does not depend on gravity
2 phases : exo and endoatmospheric	Remains endoatmospheric
Advantages : Range is more, Saves Fuel	Advantages: Precise and High speed, Difficult to intercept by Radar.
Disadvantages : Can be intercepted by radar / Anti Ballistic Missile.	Disadvantages : Requires continuous guidance
Better for Fixed Targets but not for mobile ones	More fuel is needed
	Only short and medium Range





Stealth Technology

① designed to deflect radiation



② Surface coating (absorbs radiation)
→ use of material science



Classification : Launch Modes

- Surface to surface
- Surface-to-Air Missile
- Air-to-Air Missile
- Air-to-Surface Missile
- Sea-to-Sea Missile
- Torpedo
- Anti-Tank Missile



Ballistic Missiles

ICBM : intercontinental : $>5000\text{km}$

IRBM : intermediate range : $3000 - 5000\text{km}$

MRBM : Medium Range Ballistic Missile : $1000-3000\text{km}$

SRBM : Short Range Ballistic Missile : $<1000\text{km}$

Tactical : For use in the battlefield



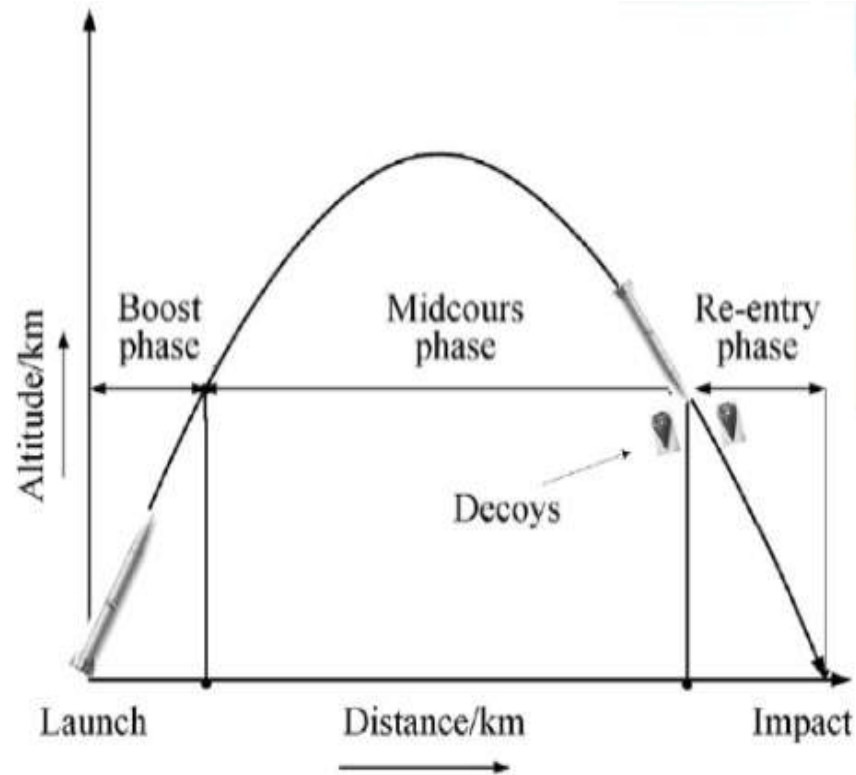
ICBM

- Inter continental Ballistic Missile
- It is a missile with a minimum range of 5,500 kilometres primarily designed for nuclear weapons delivery.
- India, Russia, the United States, China, the United Kingdom and France, Israel and North Korea are the only countries that have operational ICBMs.
- Modern designs support Multiple Independently Targetable Reentry Vehicles (MIRVs).
- A MIRV is an exo-atmospheric ballistic missile payload containing several warheads. Each one of them is capable of being aimed to hit a different target.



Test Question

- Which are the Non-NPT countries of the world?





Hypersonic Missiles

- DRDO has successfully tested a Hypersonic Technology Demonstrated Vehicle (HSTDV) in September 2020, and demonstrated its hypersonic air-breathing scramjet technology.
- The significance of developing the technology has come in the backdrop of development strides made by China, Russia and US in the domain.
- Hypersonic defines speed of about Mach 5 or at least at 1.6 km per second.
- Ballistic missiles, follow a fixed trajectory and travel outside the atmosphere to re-enter only near impact.
- On the contrary, hypersonic weapons travel within the atmosphere and can manoeuvre midway which combined with their high speeds makes their detection and interception extremely difficult.
- This means that radars and air defences cannot detect them till they are very close and little time to react.



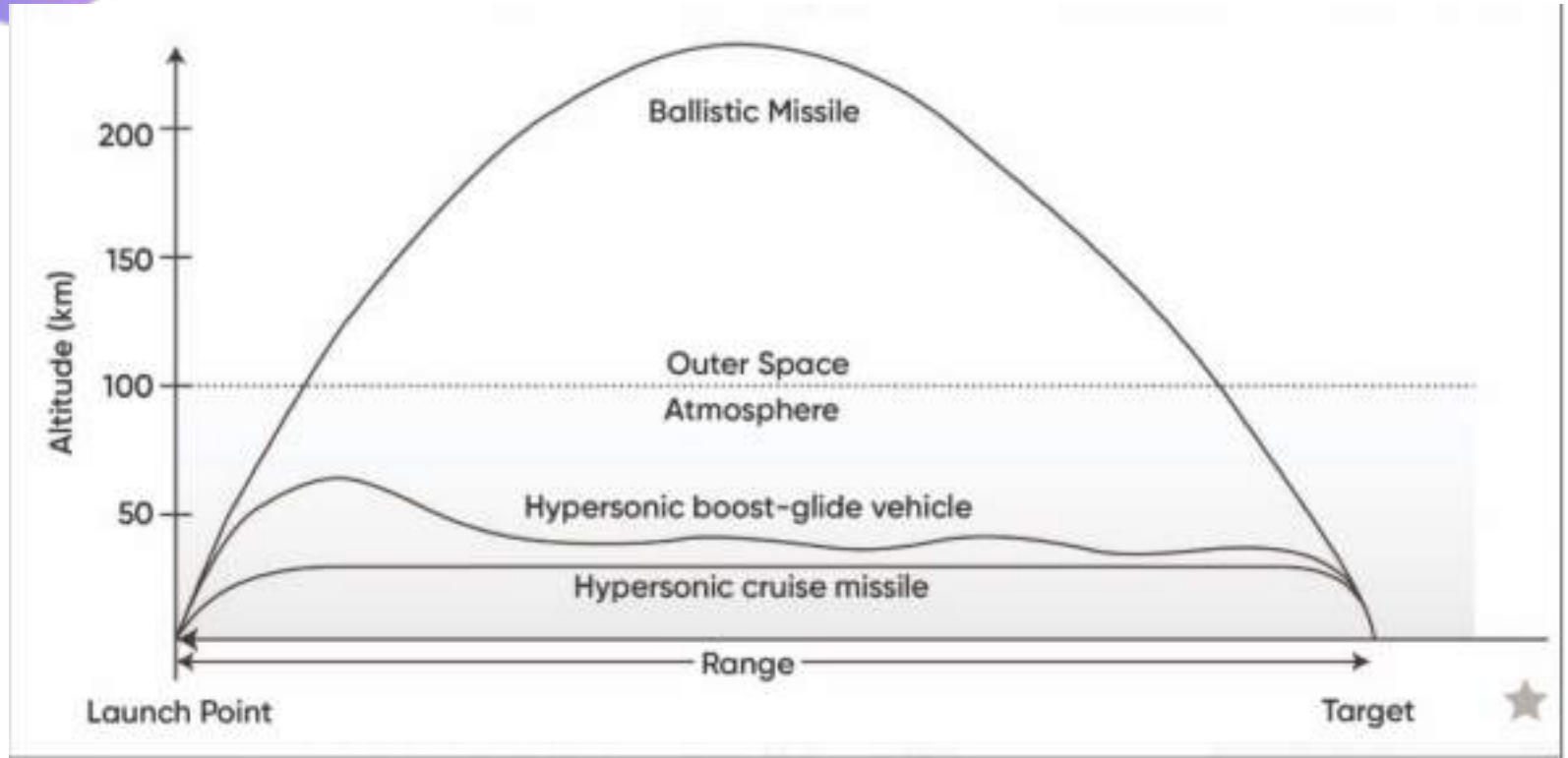
Hypersonic Missiles

2 Types of hypersonic platforms:

- o Hypersonic cruise missiles: Use rocket or jet propellant through their flight and are regarded as being faster versions of existing cruise missiles.

- o Hypersonic Glide Vehicle (HGV):

These missiles first go up into the atmosphere on a conventional rocket before being launched towards their target.



Hypersonic weapons

Considered the next generation of arms with conventional or nuclear warheads that are hard to detect and can travel more than five times the speed of sound

TRAJECTORY AND DETECTION

HYPERSONIC GLIDE VEHICLE

Speed: at least Mach 5

Launched from a rocket. Hard to detect, track and intercept

Hypersonic cruise missiles are powered by "scramjets", maneuvered at lower altitude toward target

Detected by radar

Limit of ground-based radar detection at target

CONVENTIONAL BALLISTIC MISSILE

Between Mach 1 and Mach 5

Easier to predict and intercept

Detected by radar

Atmosphere (100 km)

Launch

Target



Mach Number

Mach number : Speed relative to speed of sound.

Velocity Zone	Mach Number
Subsonic	Mach < 1
Trans-sonic	Mach ~ 1
Supersonic	Mach > 1.0
Hypersonic	Mach > 5.0



Hypersonic Weapons

- In August, China reportedly tested a nuclear-capable hypersonic glide vehicle that circled the globe before speeding towards its target.
- Chinese appear to have combined a Fractional Orbital Bombardment System, or FOBS, with a hypersonic weapon. FOBS sends missiles through a partial orbit around the earth to strike targets from an unexpected direction.
- According to the reports, weapon could, in theory, fly over the South Pole. That would pose a big challenge for the US military because its missile defence systems are focused on the northern polar route.
- This test was not in isolation. China is building up his strategic forces across the board, quadrupling the size of the arsenal and building new siloed intercontinental ballistic missiles (ICBMs), new mobile ICBMs, new nuclear bombers, new nuclear-armed submarines, new missile defences, and new hypersonic missiles.



Intercontinental Ballistic Missiles (ICBMs)	Hypersonic Cruise Missiles (HCMs)
• Traditional ballistic missiles and ICBMs arc along a predictable ballistic path, like a bullet, and lack the element of surprise. • Assessment by the Nuclear Threat Initiative premises that the President of the US has just 2-4 minutes to render a decision on intercepting incoming ballistic missiles fired by China/Russia and launch its own missiles against the latter.	• Hypersonic missile packs the manoeuvrability of much slower subsonic cruise missiles and the hypersonic speed greater than or equal to ICBMs, making it harder to track and defend against. • The detection-interception time in case of HCMs and HGVs would be far less. Besides, HGVs do not rise above 100-110 kms altitude and HCMs fly even lower, at about 20-30 kms altitude. ○ Their flatter trajectories vis-à-vis ICBMs, along with the Earth's curvature, further complicates the dynamics of detecting-intercepting.



Missile Defence System

Missile defense systems are intended to shield against incoming missiles, such as intercontinental ballistic missiles (ICBMs) or other ballistic missiles.

The United States, Russia, India, France, Israel, Italy, United Kingdom, China and Iran have developed missile defense systems.

Think : Easier to defend against which missile :

Ballistic or Cruise?

Supersonic or subsonic or Hypersonic?



Missile Defence System : Various Countries

France UK and Italy : PAAMS (principle anti air missile system). Also known as Sea Viper in the UK.

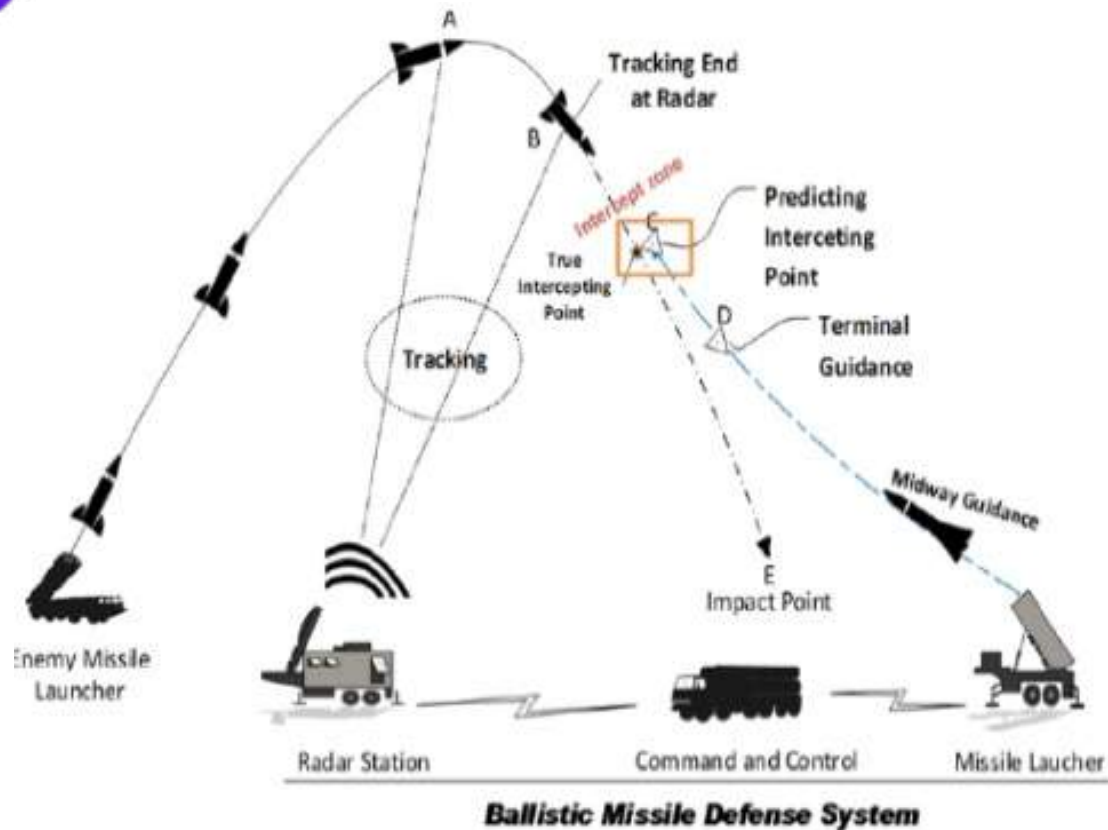
France and Italy also have Aster Missile System

Israel : Arrow missiles, Iron Dome, Iron Beam, Barak -8 Missiles

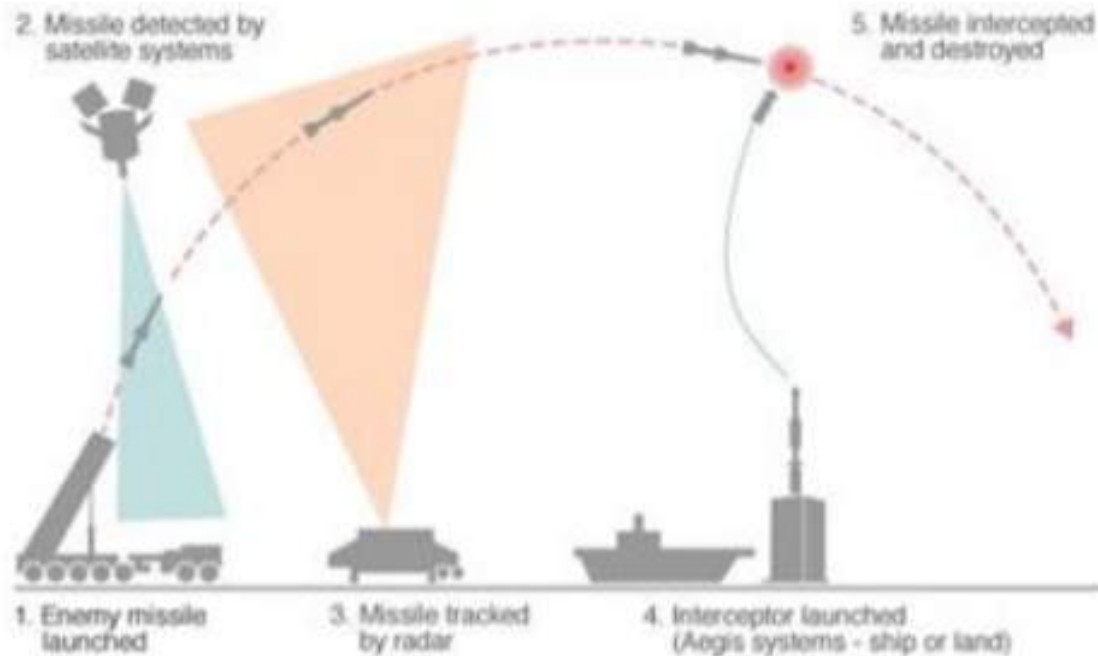
USA : Terminal High Altitude Aerial Defence (THAAD)

China : KT and HQ series of Missiles

Russia : A135, A35, S400, S300



How the missile defence system works



Source: Missile Defence Agency



India's Ballistic Missile Defence

It is an initiative to develop and deploy a multi-layered ballistic missile defense system to protect India from ballistic missile attacks.

Why? : Hostile Neighbors

It can intercept an incoming ballistic missile launched from 5,000 km.

It is a double-tiered system consisting of two interceptor missiles, namely PAD and AAD.

PAD is for high-altitude interception, and the Advanced Air Defence (AAD) Missile for lower altitude interception.

Pradhyuman : exoatmospheric

Ashwin : Endoatmospheric



Prithvi / Pradhyumna Air defence

The PAD is an anti-ballistic missile developed to intercept incoming ballistic missiles outside the atmosphere (exo- atmospheric, 80 km altitude).

It is based on the Prithvi missile, which is a tactical surface-to-surface short-range ballistic missile (SRBM)

PAD is a two-stage missile. The first stage is solid fuelled, while the second stage is liquid fuelled.

Guidance is provided by internal navigation system with active radar homing in the terminal phase.

PAD is fast enough to hit medium-range ballistic missiles and intermediate-range ballistic missiles.



Advanced Air Defence

AAD is an anti-ballistic missile designed to intercept incoming ballistic missiles in the endo-atmosphere at an altitude of 40 km.

AAD is a single stage, solid-fueled missile.

Guidance is similar to PAD

Difference: Indigenous RFID seeker



S 400

S 400 is Russia's 4th generation of long- range surface-to-air missile system.

The evolved version of the S-400 system is the S-500 Prometheus.

India has already procured a squadron of S- 300V systems which are in use as an anti- tactical ballistic missile screen’.

As per SIPRI : one of the most advanced air defense systems in the world

Stockholm International Peace Research Institute

capable of firing 3 types of missiles to create a layered defense.

anti-aircraft missile system : designed to destroy aircraft, cruise, and ballistic missiles. It can also be used against ground targets.

S 400

The S-400 is able to intercept cruise missiles at a range of about 40 km due to their low- altitude flight paths.

The anti-ballistic missile (ABM) capabilities of the S-400 system are near the maximum allowed under the Anti-Ballistic Missile Treaty.

Can engage all types of targets : UAV / Ballistic / Cruise

Range : 400km

Equipped with four different missiles, it can engage enemy aircraft, ballistic missiles, and Airborne Warning And Control System (AWACS) planes at 400km, 250km, medium-range 120km and short-range 40km.

It has the capability to engage 80 targets at one time with a response time of 9-10 seconds.



ABM treaty

It was an arms control treaty between the United States and the Soviet Union on the limitation of the anti-ballistic missile (ABM) systems used in defending areas against ballistic missile-delivered nuclear weapons. Under the terms of the treaty, each party was limited to two ABM complexes, each of which was to be limited to 100 anti-ballistic missiles. Signed in 1972

In 2002 : The US withdrew

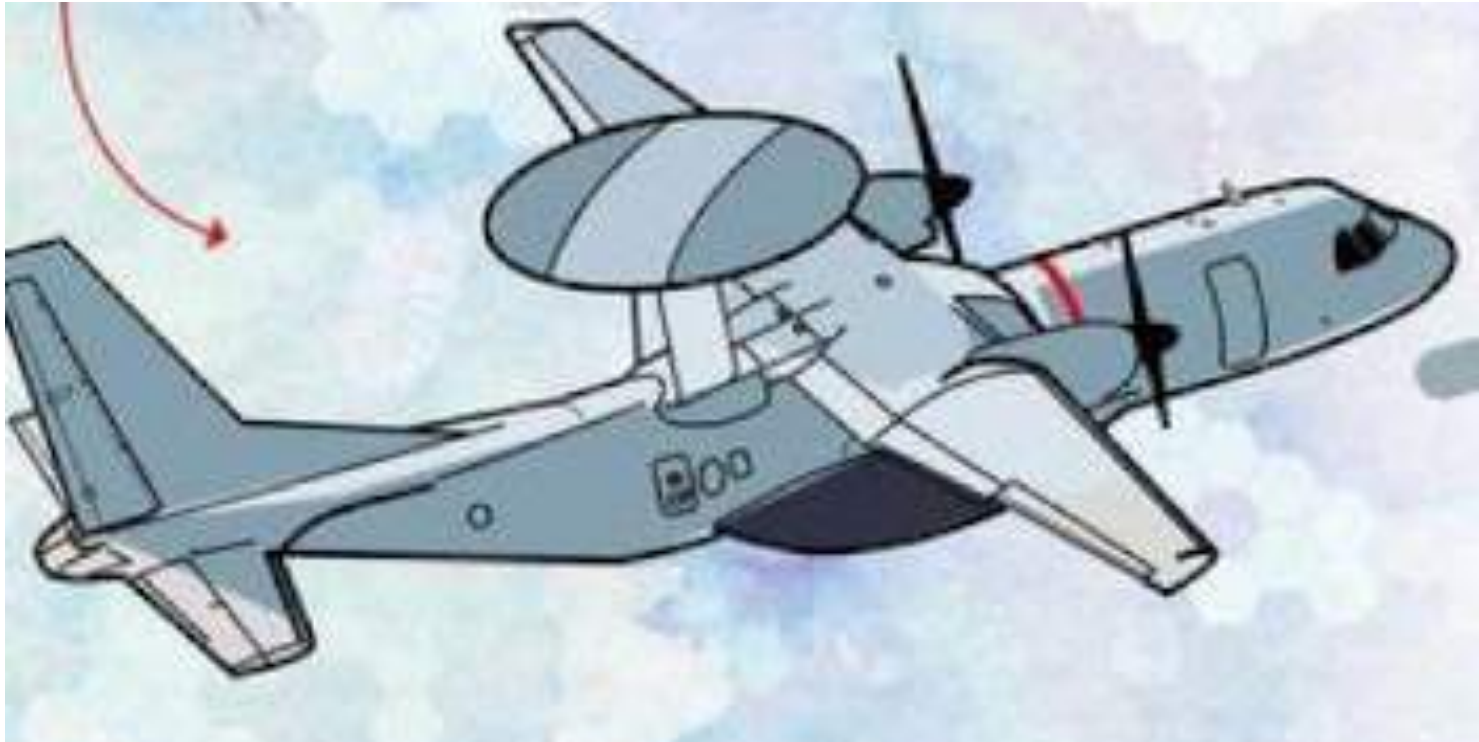


Cruise Missile Detection

Low trajectory : difficult to intercept via conventional methods

AAD : can intercept cruise missiles

Airborne early warning and control system :
AEW&CS



Defence Related Facts : Indian Missiles

Missile	Type and Details	From - to	Range (km)	Remarks
Prithvi I	SRBM	S2S	150	India's first indigenous missile, Nuclear capable, not in service
Prithvi II	SRBM	S2S	350	In service
Prithvi III	SRBM	S2S	600	In service
Dhanush	Naval variant of Prithvi II	Sea to Surface	350	In service
Agni I	MRBM, Single Stage	S2S	1250	In service
Agni II	IRBM, Two Stage	S2S	2000	In service
Agni III	IRBM, Two Stage	S2S	3000	In service
Agni IV	IRBM, Two Stage	S2S	4000	In service
Agni V	ICMB, 3 stage	S2S	5000	In service
Agni VI	ICBM, 4 stage	S2S	5500	In development
Agni P (prime)	maneuverable reentry vehicle, can be used as ASBM	S2S	Details pending	Tested in 2021
Trishul	Navy, short range	SAM	9km	Not in Service
Akash	Supersonic, Medium Range	SAM	30km	Recently, export authorised to friendly nations

Missile	Type and Details	From - to	Range (km)	Remarks
Astra	Active Radar Homing, beyond visual range	A2A	80	First A2A by DRDO, Mach 4.5 speed, Solid fuel, has radar transceiver
Shaurya	Canister based, Medium Range, Hypersonic, tactical	S2S	700-1900	In service, land based parallel of submarine launched k-15
Prahaar	Tactical Ballistic Missile, QR, Omnidirectional Warhead	S2S	150	In service, meant to replace Prithvi 1
K4	SLBM, intermediate range, in development	Underwater to Surface	3500	Named after Dr. Kalam, Nuclear triad and second strike capability
Sagarika (k-15)	SLBM	Under water to surface	700	SLBM version of Shaurya, to be integrated with INS arihant
Barak 1	Short range SAM, ship defense	Ship to air, surface	12km	Built with Israel, to be integrated with Shivalik Class frigates
Barak 8	SRSAM, ship defense	Ship to air, ship to surface	90km	Israel, to be integrated with Kolkata class destroyers.
Nirbhay	Stealth, subsonic, cruise, nuclear capable	Land, naval, air	1000km	In service, no MTCR issues



Brahmos

It is developed jointly by India and Russia. Named after Brahmaputra and Moskva rivers.

can be launched from Land, Air, Submarine and Ships.

is the world's fastest anti-ship cruise missile.

2 stage : first solid then liquid

As India became a member of MTCR in 2016, range was increased to 600km

Now the range of new land-attack version is extended to 400 km from 290 km. Speed is maintained at 2.8 Mach : supersonic



BRAHMOS

- BrahMos is a universal long range supersonic cruise missile system that can be launched from land, sea and air against surface and sea-based targets.
 - BrahMos is a collaboration between India (Defence Research and Development Organisation (DRDO) and Russia (NPO Mashinostroyeniya).
 - Name represents Brahmaputra and Moskva rivers



BRAHMOS

- A two-stage missile stage
- Fire and forget principle
- The range of the missile was originally capped at 290 km as per obligations of the Missile Technology
- Control Regime (MTCR)
- Following India's entry in MTCR in 2016, it was decided to extend the range to 450 km and to 600 km at a later stage.



Other similar missile systems

- Chinese HD-1 supersonic missile
- Tomahawk (used by U.S and its allies)
- Israel's Sea Breaker
- P-800 Oniks/Yakhont



Other Missiles

Python 5 : 5th Gen AAM, Israeli origin, integrate with Tejas, Beyond Visual Range, radar guided

Avangard : Russia deployed first hypersonic long range Nuclear-capable missiles, SCRAMJET engine



Agni Prime

DRDO successfully tested the new missile Agni Prime, advanced version of the ‘Agni-1’ missile.

Agni Prime’ is a short-range (surface to a surface) ballistic missile that will have a range of 1000 km to 1500 km and will have advanced features in agility and road mobility.

It can carry a payload of around 1,000 Kg or a nuclear warhead.

It is a two-stage canisterised solid propellant ballistic missile with dual redundant navigation and guidance system.



Agni Prime

o Since the missile is canisterised, it can be launched from rail and road and stored for a longer period and transported all across the country as per operational requirements.

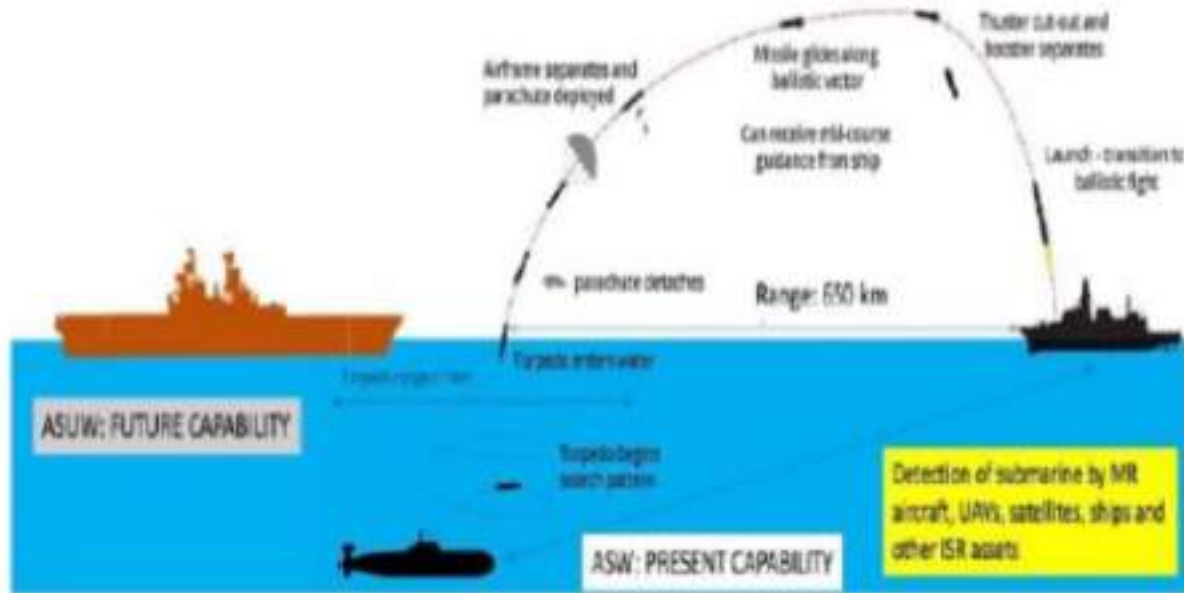
The double stage missile will be lighter and much sleeker than its predecessor 'Agni-1'.

Agni-1 is a short-range (surface to a surface) nuclear capable ballistic missile



Supersonic Missile Assisted Torpedo

- Recently, DRDO tested a long-range SMART from Wheeler Island in Odisha.
- SMART is a next-generation missile-based standoff torpedo delivery system.
- It has been designed to enhance anti-submarine warfare capability far beyond the conventional range of the torpedo.





Looking Ahead

India's progress towards developing hypersonic technology.

- **Brahmos II:** India has collaborated with Russia on the development of Brahmos II, a Mach 7 hypersonic cruise missile. It is scheduled to achieve initial operational capability between 2025 and 2028.
- **Mission Shakti:** To protect its space assets, India has already proved its capabilities through the test of indigenously developed anti-satellite (ASAT) missile technology.



Looking Ahead

- HSTDV: India is also developing an indigenous, dual capable hypersonic cruise missile as part of its Hypersonic (HSTDV) and successfully tested a mach 6 scramjet in June 2019 and September 2020.
- HWT test facility: An advanced Hypersonic Wind Tunnel (HWT) test facility of the DRDO was inaugurated last year in Hyderabad. It is a pressure vacuum-driven, enclosed jet facility that simulates Mach 5 to 12.



Missiles in News

- Agni 3 : IRBM
- S- 400
- Anti Tank Guided Missile : laser guided, placed on MBT arjun. Employs tandem high explosive anti tank (HEAT) warhead to counter explosive reactive armor, multi platform launch capable.
- Man Portable Anti Tank Guided Missile : indigenously developed, low weight, man portable launcher with fire and forget capability. Has miniaturized infrared seeker for on board control and guidance.
- Vertical Launch Short Range Surface to Air Missile : indigenous, ship borne, aerial threats at close range. Based on design of Astra missile (Beyond visual range air to air missile)
- QRSAM : Quick reaction SAM, indigenously designed, protective shield to moving armoured columns of the army from aerial attacks. Range : 25 – 30km



Missiles in News

- Very Short Range Air Defence System (VSHORADS) : Man portable air defence system
- HELINA : helicopter launched anti tank guided missile, Helicopter launched NAG, 3rd generation fire and forget missile mounted on ALH : All weather day and night capability
- Astra Mk-1 : BVR Air to Air missile
- Pralay Missile : Indigenously Developed Surface to Surface missile with range of 150-500km



Warships : classification

- Various types
- Destroyers
- Frigates
- Aircraft carrier
- Corvette



Warships : classification

- Destroyers: Combat operation equipped with missiles and torpedoes
- Big sized Ships : Kolkata class, Rajput , Delhi
- Frigates; Small size, stealth, high maneuverability



Warships : classification

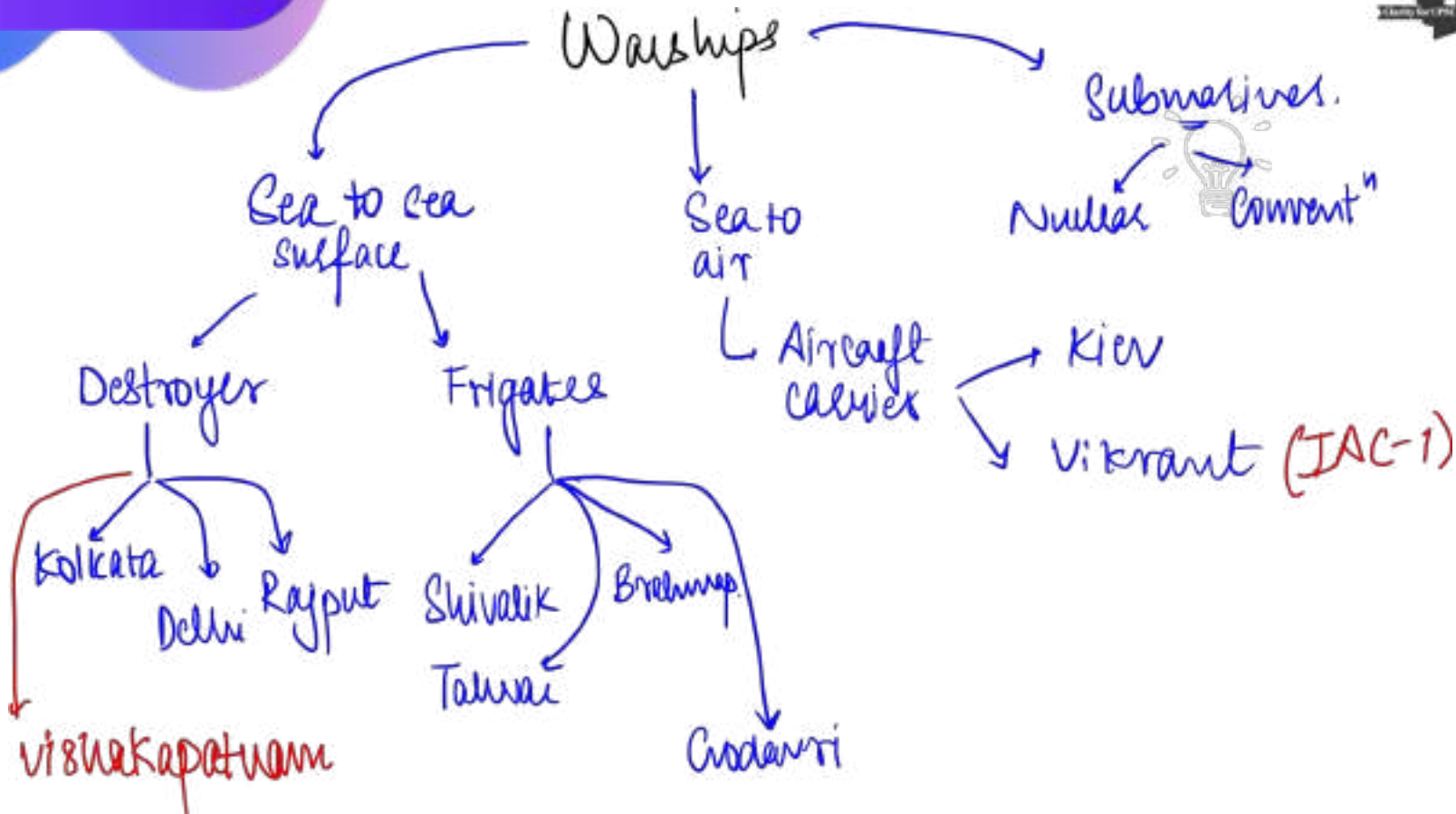
- Aircraft carriers
- Biggest warships
- Runway and takeoff platform
- carry fighters and helicopter



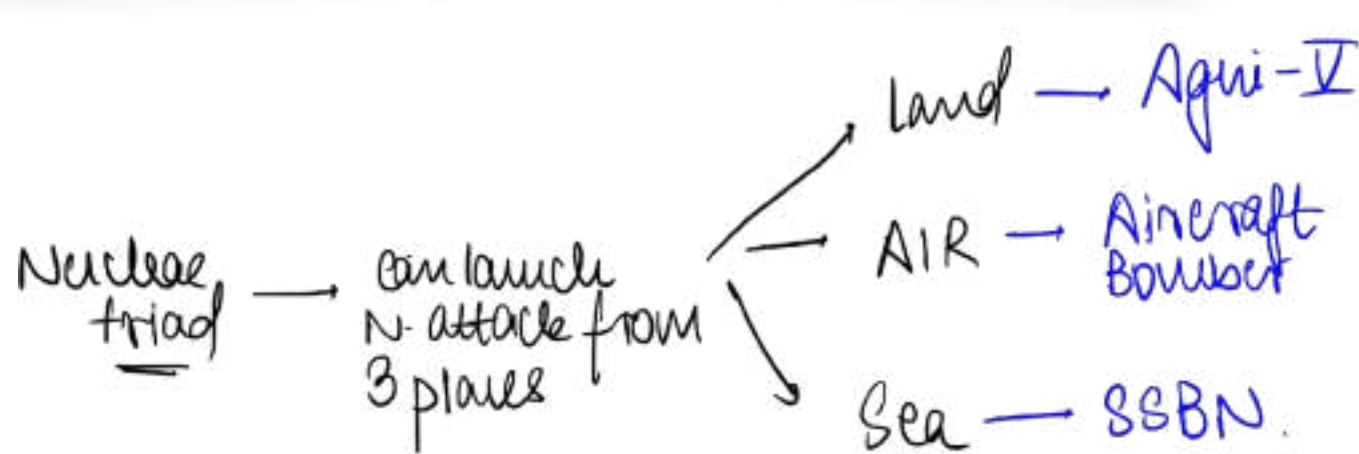
Warships : classification

Corvette

- Anti submarine warships (ASW) protect from submarine attack.



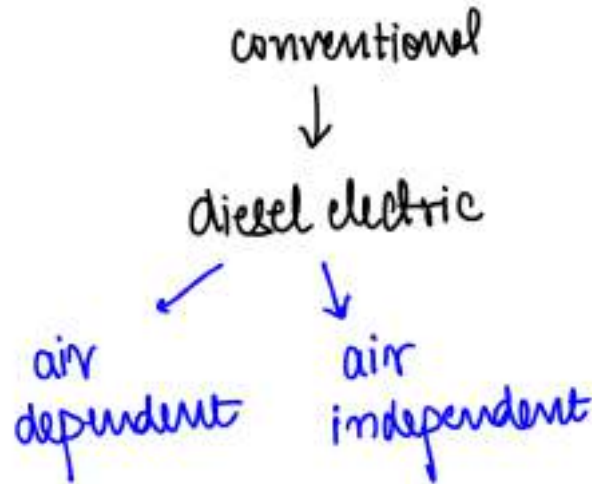
Understanding a Nuclear triad



- ① Second Strike Capability
- ② Mutually assured Destruction?

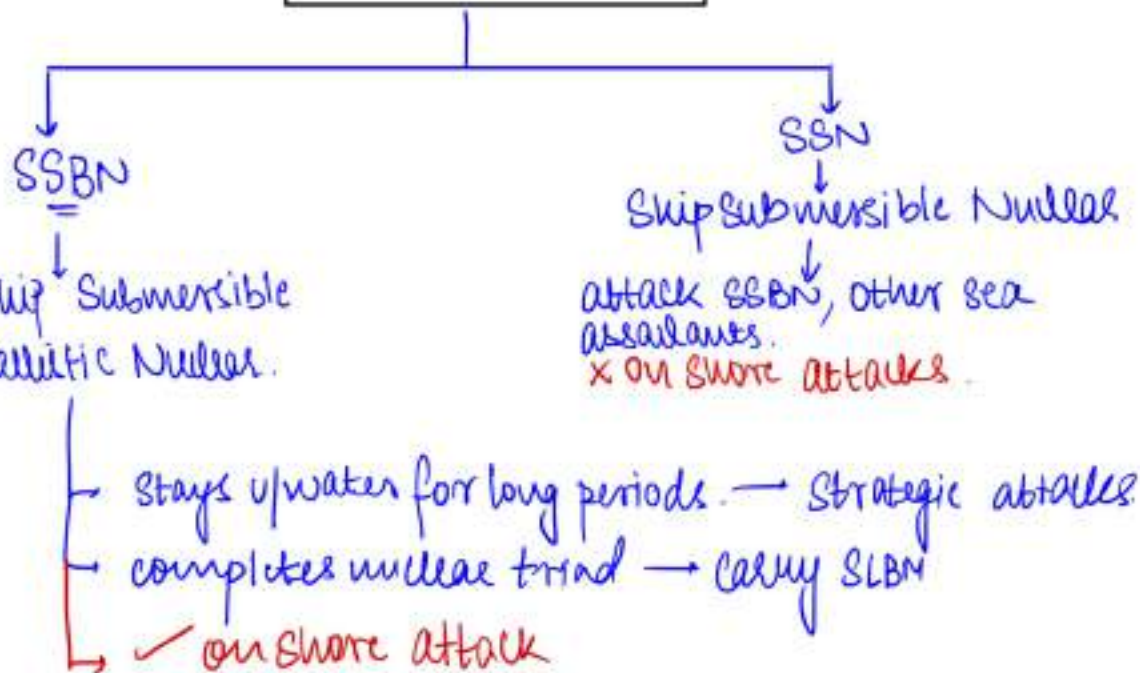
Submarines : types

- Conventional and Nuclear Submarine





Nuclear Submarine





Types : Based on capability

- An attack submarine is specifically designed for attacking and sinking other submarines, surface combatants and merchant vessels.
- They are also used to protect friendly surface combatants and missile submarines.
- Many modern attack submarines can launch cruise missiles (and dedicated anti-ship missiles) from their torpedo tubes, however main attack weapon remains the torpedo.



Ballistic Missile Subs

- A ballistic missile submarine is a submarine capable of deploying submarine-launched ballistic missiles (SLBMs) with nuclear warheads.
- The classification symbols for ballistic missile submarines are SSB and SSBN. The SS denotes submarine, the B denotes ballistic missile, and the N denotes that the submarine is nuclear powered.



Ballistic Missile Subs

- These submarines became a major weapon system in the Cold War because of their nuclear deterrence capability.
- They can fire missiles thousands of kilometers from their targets. Acoustic quieting makes them difficult to detect. Hence, they are a survivable deterrent in the event of a first strike and a key element of the mutual assured destruction policy of nuclear deterrence.



Ballistic Missile Subs

- The deployment of SSBNs is dominated by the United States and Russia. Smaller numbers are in service with France, the United Kingdom, China, and India.
- In 2009, India launched the first of its indigenously-built Arihant-class submarines.



Cruise missile submarine

- A cruise missile submarine is a submarine that carries and launches cruise missiles (SLCMs and anti-ship missiles) as its primary armament.
- The classification symbols for cruise missile submarines are SSG and SSGN. The SS denotes submarine, the G denotes guided missile, and the N denotes that the submarine is nuclear-powered.



Class of Submarine

- Vehicle class means the designation of motor vehicle types. It includes hatchback, sedans, sport utility vehicles, SUVs, trucks etc.
- A submarine class is a group of submarines of a similar design. Similarly, a ship class is a group of ships of a similar design.



Class of Submarine

- Submarines are built in classes, using a single design for a number of boats. Minor variations occur as improvements are incorporated into the design. However, in general, all boats of a class are noticeably similar.
- The 30 year's Indian Submarine Program (2000-2030) aims to build 24 submarines: 18 conventional and 6 nuclear powered (SSNs).



Indian Subs : Diesel electric submarines

- German subamrines : Shishumar class
- Russian Attack submarines : Sindhughosh class or kilo class
- Name of these submarines starts from Sindhu e.g. INS Sindhughosh (S55), INS Sindhudhvaj (S56), INS Sindhuraj (S57), INS Sindhuratna, (559), INS Sindhukesari (\$60), INS Sindhukirti (S61), INS Sindhuvijay (S62), INS Sindhurashtra (S65).

Indian Subs : Sindhu class





Indian Subs : Kalvari Class Scorpene

- Attack submarines, jointly developed by France and India.
- It has diesel propulsion and an additional air- independent propulsion (AIP) system
- Six submarines under this class: INS Kalvari, Khanderi, Karanj, Vela, Vagir, Vagsheer. INS Kalvari is also known as Tiger Shark (or S50).
- Developed under Project 75 or P 75 of the Indian Navy.
- Kalvari class of submarines design is a class of diesel-electric attack submarines.

Indian Subs : Scorpene and Kalvari





Nuclear Submarine

- It is a nuclear-powered ballistic missile submarine (SSBN). It can also launch missiles.
- INS Arihant (S2 or SSBN 80) is the lead ship of this class.
- INS Arighat (S3 or INS Aridhaman) is the second Arihant-class submarine.

Nuclear Submarine

INS ARIHANT



Akula Class

- Russian Powered Nuclear attack submarine : SSN
- INS chakra is an example





Vagir Submarine

- It is the fifth submarine (Kalvari, Khanderi, and Karanja, vela, vagsheer are other five) of the Project- 75 that was delivered to the Indian Navy recently.
- Project-75 includes construction of six submarines of Scorpene design.
- Scorpene submarines can undertake multifarious types of missions i.e Anti-Surface warfare, Anti-Submarine warfare, Intelligence gathering, Mine Laying, Area Surveillance etc.



Vagsheer Submarine

- Sixth and Last submarine of Project 75 has been launched.
- Diesel attack submarine.
- Status of P75 : kalvari, khanderi, karanj, vela have been commissioned.
- Vagir : Sea trials.



Project 75 I

- Project-75(I) envisages indigenous construction of six modern conventional Submarines with contemporary equipment, weapons & sensors including Fuel-Cell based AIP (Air Independent Propulsion Plant) etc.
- AIP technology allows conventional diesel-electric submarines to remain underwater for longer, enhancing its lethality.
- Other countries to have an AIP system include China, Germany, Sweden, France, Spain and Russia.
- P-75(I) submarine project is the first acquisition India has undertaken through its Strategic Partnership procurement model.



Project Sea Bird

- It involves creation of a naval base at Karwar (Karnataka) on the west coast of India.
- Upon completion, this will provide the Indian Navy with its largest naval base on the west coast and also the largest naval base east of The Suez Canal.



Types of Warships

- Aircraft Carrier :
- An aircraft carrier is a warship that serves as a dynamic seagoing airbase.
- It is equipped with a full-length flight deck.
- Typically, it is the capital ship of a naval fleet.



Types of Warships

- It allows a navy to project air power worldwide without depending on local air bases.
- By its diplomatic and tactical power, the aircraft carrier is the centerpiece of modern combat fleets.



Types of Ships

- Destroyer: It is a fast and maneuverable warship intended to defend larger vessels in a fleet.
- Frigates: it is a naval vessel intermediate between corvettes and destroyers.
- Corvette : Very small warship, fast and maneuverable
- Patrol vessels : Used to patrol coasts, example : ships of Indian Coast guard



Sea Denial and Sea Control

- Sea denial : word breakdown : no need of aircraft carrier
- Sea control : word breakdown : Needs aircraft carrier



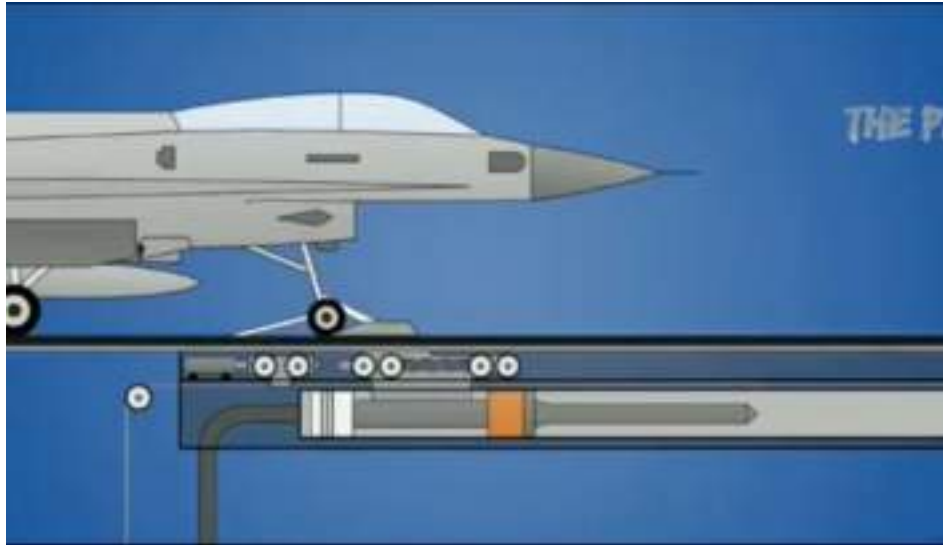
Carriers

- STOBAR system (Short Take-Off but Arrested Recovery)
- A system used for the launch and recovery of aircraft from the deck of an aircraft carrier.
- Combines elements of both: Short take-off and vertical landing (STOVL) and Catapult- assisted take-off but arrested recovery (CATOBAR).



Carriers

- EMALS system (Electromagnetic Aircraft Launch System)
- The system launches aircraft by a catapult employing a linear induction motor rather than the conventional steam piston.







Indian Carriers

- INS Vikrant : majestic Class, Decommissioned in 1997
- INS virat : centaur class, Decommissioned in 2016
- INS Vikramaditya : KIEV CLASS, active aircraft carrier in India.
- INS Vikrant (IAC -1)





INS Vikrant- First Indigenous Aircraft Carrier (IAC)

- INS Vikrant is commissioned in 2022.
- Before this, we had a solitary aircraft carrier : INS vikramaditya
- Vikrant, a STOBAR (short take-off but assisted recovery) aircraft carrier, has an indigenous content of 76%.
- Designed by the Indian Navy's Directorate of Naval Design, Vikrant has been built at the state-owned Cochin Shipyard Limited.



INS Vikrant- First Indigenous Aircraft Carrier (IAC)

- Apart from India, only US, UK, Russia, France and China have the capability to build aircraft carriers.
- It will operate MiG-29K fighter aircraft, Kamov-31 Air Early Warning Helicopters, the soon-to-be-inducted MH-60R multi-role helicopter, and the indigenously manufactured Advanced Light Helicopters.
- It will provide the ability to project air power over long distances, airborne anti- submarine warfare and airborne early warning etc.





INS vishal (IAC 2)

- EMALS CATOBAR system.
- Nuclear propulsion.
- It Will be India's 2nd Indigenous Aircraft Carrier (IAC-2) to be built in India after INS Vikrant (IAC-1).



Think : where will India deploy its
carriers?



Destroyers

- Rajput Class : oldest
- Guided Missile Destroyer
- INS Rajput, INS Rana, INS Ranveer, INS Ranvijay.

- Delhi Class : Project 15 : GMD
- Indigenous
- Delhi, Mysore, Mumbai





Destroyers

- Kolkata Class : project 15A
- Stealth GMD
- The class comprises three ships - INS Kolkata, INS Kochi and INS Chennai.
- Barak – 8 capable
- Vishakhapatnam class : Project 15B
- Stealth GMD
- Brahmos and MRSAM capable





Y 12704 (Visakhapatnam)

- It is the lead ship of the Project-15B state-of-the-art stealth guided missile destroyers.
- It is equipped with BrahMos supersonic cruise missiles and medium range Surface- to-Air missiles.
- Contract for four ships of Project 15B was signed in 2011.



Y 12704 (Visakhapatnam)

- Four ships are christened after major cities viz. Visakhapatnam, Mormugao, Imphal and Surat.
- India's indigenous Destroyer construction programme commenced in the late 1990s with the three Delhi class (P-15 class) warships and this was followed by three Kolkata class (P-15A) destroyers commissioned a decade later.



Project 15B

- Mormugao and Surat ship of project 15B have also been launched recently
- Built at Mazagaon Dock Shipbuilders Limited
- Stealth Guided missile destroyers



Frigates

- Godavri Class : Project 16
- Guided Missile frigates : Indigenous
- Godavri, Ganga , Gomti

- Shivalik Class : Project 17
- First Stealth warship in India
- Stealth GMF



Frigates

- Project 17A : Taragiri and Udaygiri ships of Project 17A have been launched. They are advanced version of P17 frigates (Shivalik Class)
- Designed by : Indian Navy's warship design bureau (indigenous)
- 75% orders from indigenous firms including MSMEs.



Frigates

- Talwar Class : Russian
 - Stealth Guided Missile Frigates
 - Talwar, trishul, Tabar, teg, tarkash, trikhand
-
- Brahmaputra class
 - Guided missile frigate
 - Brahmaputra, Beas, Godavri





Corvette

- Anti Submarine Warfare
- Indigenous
- INS Kamorta, INS Kadmatt, INS Kilatan, INS Kavaratti
- INS kilatan : NBC warfare, Carbon fibre composite material
- INS Kavaratti : indigenous, NBC capable



Patrol Vessels

- Car nicobar class patrol vessel : water jet propelled
- Started after 2008 Mumbai attacks
- INS tarmugli : highly maneuverable Water Jet Fast Attack Craft (WJFAC).



Patrol Vessels

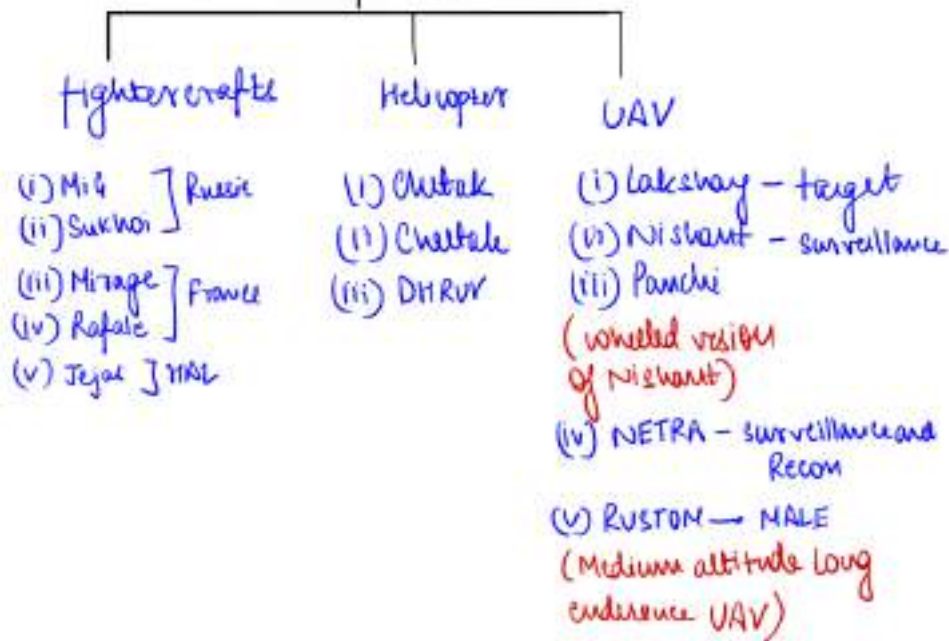
- Fast Patrol Vessel :
- Indian Coast guard, to control coasts
- Kankalata Barua, Priyadarshini , Annie Besant, Kamala Devi, Amrit Kaur

INS astradharini

- Torpedo launch and recovery
- Mine launch and recovery
- Indigenous



Air defense



Air Fleet

- Mig 21
- The Mikoyan-Gurevich MiG-21 is a Soviet designed supersonic jet fighter and interceptor.
- Interceptor: a fast aircraft for stopping or repelling hostile aircraft.



Air Fleet

- Rafale
- The Dassault Rafale is a French twin-engine, multirole, canard delta wing, fighter aircraft.
- Golden Arrows
- MMRCA : medium multi role combat aircraft



Rafale

Rafale advantages $\left\{ \begin{array}{l} \text{fast engine} \\ \text{air refuel} \\ \text{multimodal targeting} \end{array} \right.$

Air Fleet

- Sukhoi Su -30 Mki
- It is a twin-engine, multirole, all-weather, long-range fighter.
- It is developed by Russia's Sukhoi and built by India's Hindustan Aeronautics Limited



Air Fleet

- Mirage 2000
- Dassault (French)
- Balakot airstrike





Generations of fighter jets

- 1st generation : piston based propellers, no avionics, manually controlled guns.
- 2nd generation : afterburner, infrared homing missiles
- 3rd generation : multirole, BVR, turbofan engine
- 4th generation : fly by wire, compound delta wing, hypermaneuverability
- 4.5 generation : use of composite materials to reduce radar cross section, better radars
- 5th generation : C3 : command control and communication and digital computation, better battlefield awareness. Ex : F-35, Chengdu J-20, Su-57
- 6th generation : in development



Tejas

- Indigenously developed lightweight multi role, Single engine, tail-less tactical fighter aircraft with compound delta wing. **4.5 generation.**

Special features.

- Compound delta platform
- Static stability
- Composite structure
- Fly-By-Wire flight control system



Meaning of these terms

- Fly by wire : Fly-by-wire (FBW) is a system that replaces the conventional manual flight controls of an aircraft with an electronic interface. The movements of flight controls are converted to electronic signals transmitted by wires, and flight control computers determine how to move the actuators at each control surface to provide the ordered response.



Meaning of these terms

- **Afterburner** : An afterburner (or reheat in British English) is an additional combustion component used on some jet engines, mostly those on military supersonic aircraft. Its purpose is to increase thrust, usually for supersonic flight, takeoff, and combat. The afterburning process injects additional fuel into a combustor in the jet pipe behind (i.e., "after") the turbine, "reheating" the exhaust gas. Afterburning significantly increases thrust as an alternative to using a bigger engine with its attendant weight penalty, but at the cost of increased fuel consumption (decreased fuel efficiency) which limits its use to short periods



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Relaxed Static Stability

relaxed stability refers to low or negative stability as a characteristic of aircraft.

An aircraft with negative stability will have a tendency to change its pitch and bank angles spontaneously.



Tejas

- Glass cockpit

It can carry A2A missile, A2G missiles, anti - ship missiles, laser guided bombs, drop tanks, Night targeting pod, electronic warfare system etc.

- Speed : 1.8 mach
- Kaveri Engine : GE 404 turbofan with afterburner available



Tejas MK2

Cabinet Committee on security has cleared tejas mk 2 which is a more capable version of indigenous tejas LCA.

4.5 generation aircraft with 70% indigenization.

GE 414 engine : longer range and greater weapon capacity than Tejas Mk-1



Other Aircraft

- Ilyushin IL-78 : Mid air refuel tanker
- Heron UAV : MALE Medium Altitude Long Endurance
- Chinook Helicopter : American made, received by India : used for HADR
- Apache Helicopter : Multi Role Heavy attack helicopter : Flying tank.





Aircraft in News

- Dornier 228 : The 19 seater HAL - DO - 228 aircraft is a highly versatile multi-purpose light transport aircraft. It has been developed specifically to meet the manifold requirements of utility and commuter transport, third level services and air-taxi operations, coast guard duties and maritime surveillance. It has been delivered to alliance air for civil operations in north east. Part of UDAN.
- LCH - Prachand : Indigenously developed and adopted by the Indian airforce. India has become the 7th country to make attack helicopters. Range : 550km, 6.5 km service ceiling (maximum altitude of flight). Only combat helicopter in the world that can take off and land at an altitude of 5000 meters.



Aircraft in News

- Indigenous stealth drone aka : Unmanned combat airborne vehicle has taken its maiden flight. UCAV is the precursor to the Ghatak armed stealth drone programmed called AURA : Autonomous Unmanned Research Aircraft.
- Chinook : US army has grounded chinook after finding that helicopter can be at risk of engine fires. India inducted chinook in 2019. Used for HADR.



Aircraft in News

- Kamikaze drones : suicide drones packed with explosives.
Also called switchblades. Army is inducting in eastern Ladakh.
They were used in Russia Ukraine war extensively.
- Hermes Starliner unmanned system : Israel has allowed
drones in civilian airspace (first country to do so)



Various types of detectors

(i) Hull Mounted sonar (HMSA)

- 3rd generation, indigenous, ship-borne, well mounted Sonar system designed and developed by DRDO.
- Active-cum passive integrated system, can be installed on various naval platforms. Active : sound sent by SONAR, Passive : Present signals are intercepted.
 - Frigates
 - Destroyers
 - ASW corvettes.



USHUS

Sonar for submarines with target classification facilities. It can be interfaced with onboard weapon systems.

Range > 30km



Various types of detectors

Dunking Sonar

- Advanced sonar used to survey under sea targets with the help of helicopter.
- The chopper is used for dunking sonar underwater, Range ~ 1 km



Various types of detectors

Battlefield Surveillance Radar

- Man portable battery operated surveillance radar. Simple, easy to use and can be installed anywhere in the battlefield.
- Audio and visual aids help identify and classify target.



Various types of detectors

Rohini

- Indigenous, stand alone, medium Range, 3D surveillance radar for air force. It can scan the airspace and use multi-beams technology to identify targets accurately.
- Range = 200 km.



Various types of detectors

Bharani

- Low level, Lightweight, L-Band, 2D radar used in mountainous terrain against hostile arial target (UAV, chopper, Low and medium altitude aircraft)



Various types of detectors

Revathi

- 1st Indigenous Radar and also 1st in Navy to use digital Infrared frequency tech.
- Radar can auto trace upto 150 targets with Range = 200 km.



Various types of detectors

Ashlesha Radar : ground based, S band, 3D, low level leight weight radar with range of 60km.

Swathi Radar : Weapon locating radar, automatically locates hostile artillery, mortars and rocket launchers by using signal processing technique.

CHIEF OF DEFENCE STAFF (CDS)

The emblem of the Chief of Defence Staff, featuring a golden anchor and a sword.

Chief of Defence Staff (CDS)

About: Approved in 2019, is the head of Army, Navy and Air Force and is a four-star military officer.

Eligibility: Any serving or retired Lieutenant General, Air Marshal and Vice Admiral under age of 62 years.

Proposed by: Kargil Review Committee (2000), Naresh Chandra Committee on defence and security (2011) and Shekatkar Committee (2016).

Tenure: Central government may extend tenure to maximum of 65 years.

Roles and powers of CDS

- Heads Department of Military Affairs within Ministry of Defence and acts as its secretary.
- Permanent chairman of existing Chiefs of Staff Committee.
- Member of Defence Acquisition Council and Defence Planning Committee.
- Principal military advisor to Defence Minister and military advisor to Nuclear Command Authority.



GSAT 7 SERIES SATELLITES

- GSAT 7 series satellites are advanced satellites developed by ISRO to meet the communication needs of the defence services.
 - GSAT 7 satellite : used by the Indian Navy
 - GSAT 7A : boosting the connectivity between the ground radar stations, airbases and the airborne early warning and control aircraft (AEW&C) of the IAF (Indian Air Force).
- GSAT 7B : needs of the Army.



Operation 'Skylight'

- It involved the army testing the readiness of its satellite based systems.
- Unlike the Indian Air Force and the Navy, the army currently does not have a dedicated satellite.
 - Currently, the Army is using 30% of the communication capabilities of the GSAT 7A.
 - Defence Acquisition Council has cleared an army proposal for a GSAT-7B satellite.



Other news

- Dirty bomb : non highly refined nuclear material like uranium obtained from hospitals, nuclear power stations or research labs. Cheaper, quicker to make.
- Sweden has announced plans to manufacture Carl Gustaf M4 in India. It is a man portable recoil less rifle with ability to fire variety of ammunition.
- Pinaka Multi Barrel Rocket Launcher : Armenia has placed first ever export order of India's MBRL, indigenous rocket system developed with DRDO. It is a mobile rocket launch system with range 40km (mk 1) to 75km (Pinaka ER : extended range)



Other news

- Submersible platform for acoustic characterization and evaluation : testing and evaluation facility for SONAR systems by DRDO.
- Abhyas : High speed expendable aerial target developed by DRDO.
- High Energy Laser Systems : SFO technologies is partnering with DRDO to develop HELS. They emit thermal energy to directly render a target ineffective without using a conventional munition. It is a type of Directed Energy weapon. India has launched various projects for HELS : Durga, KALI, Aditya.



Q.1 PYQ (2014)

Q. Which reference to Agni-IV Missile, which of the following statements is/are correct?

1. It is a surface-to-surface missile.
2. It is fuelled by liquid propellant only.
3. It can deliver one-tonne nuclear warheads about 7500 km away.

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3



Q.2 PYQ (2016)

Q. Which one of the following is the best description of 'INS Astradharini', that was in the news recently?

- a) Amphibious warfare ship
- b) Nuclear-powered submarine
- c) Torpedo launch and recovery vessel
- d) Nuclear-powered aircraft carrier



Q.3 PYQ (2022)

Q. Orbital Bombardment System” often talked about in media?

- (a) A hypersonic missile is launched into space to counter the asteroid approaching the Earth and explode it in space.
- (b) A spacecraft lands on another planet after making several orbital motions.
- (c) A missile is put into a stable orbit around the Earth and deorbits over a target on the Earth.
- (d) A spacecraft moves along a comet with the same surface. speed and places a probe on its



Q4) PYQ (2018)

Q) What is "Terminal High Altitude Area Defense (THAAD)", sometimes seen in the news?

- a) An Israeli radar system
- b) India's indigenous anti-missile programme
- c) An American anti-missile system
- d) A defence collaboration between Japan and South Korea



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c) 1 and 3 only

d) 1, 2 and 3



Answer: A

- Agni-IV is the fourth in the Agni series of missiles which were earlier known as Agni II prime.
- The missile is light-weight and has two stages of solid propulsion and a payload with re-entry heat shield.
- The strategic **surface-to-surface missile** was flight tested from launch complex-4 of the Integrated Test Range (ITR) at Dr Abdul Kalam Island, earlier known as Wheeler Island,
- The indigenously developed sophisticated **Agni-IV having 4,000-km strike** range is a two-stage missile



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- d) Nuclear-powered aircraft carrier



Answer: C

- INS Astradharini is an indigenously designed and **built torpedo launch and recovery vessel built by Shoft Shipyard for the Indian Navy.**
- She was commissioned to Navy Service on 6 October 2015 at the Naval Base in Visakhapatnam.
- After commissioning, the ship entered the Eastern Naval Command.



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Answer: C

- In 1960s, the Fractional Orbital Bombardment System (FOBS) was developed by the Soviet Union, an orbital bombardment system that could carry a nuclear weapon into orbit at a lower trajectory than a traditional fixed trajectory ballistic missile.
- The idea behind FOBS is, a warhead is put into a stable orbit and it deorbits over the target.
- If the target and the launch position is lined up and the warhead keeps going round, it will complete a circle.
- The real point is that missile can be flown directly to the target, or it is fired in the opposite direction the other way around the planet and still arrive at the target from the opposite direction.



Q4) PYQ (2018)

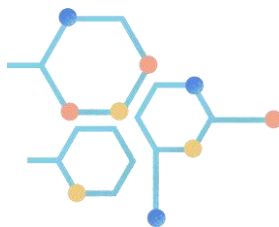
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Answer: C

- Terminal High Altitude Area Defense (THAAD), formerly Theater High Altitude Area Defense is an American anti-ballistic missile defence system designed to shoot down short, medium, and intermediate-range ballistic missiles.
- It is used by USA



UPSC Prelims

Science & Tech

Class No. 13

Quantum Computing

Quantum Computing

- Quantum effects : we read in nanotech terms
- Quantum scale : scale at atomic level
- What is the difference?
- At the quantum level, laws of classical physics and mechanics do not apply.
- Quantum computing is the area of study focused on developing computer technology based on the principles of quantum theory, which explains the nature and behaviour of energy and matter on the quantum (atomic and subatomic) level.

Classical Computing

- classical computers operate using binary physical state, meaning its operations are based on one of two positions (1 or 0).
- 1 : electricity is being passed through the wire
- 0: no electricity is being passed through the wire
- Classical computers process information in a binary format, called bits, which can represent either a 0 or 1.

Classical Computing

- Quantum computers, in contrast, use logical units called quantum bits, or qubits for short, that can be put into a quantum state where they can simultaneously represent both 0 and 1 and their correlations.

2 key principles of Quantum Computing

- Superposition
- Entanglement
- Let's understand these 2 in detail

Superposition : Schrodinger's cat

- Is the cat dead or alive?



Superposition : Schrodinger's cat

- The cat is both dead and alive at the same time
- Only when you open the box will you know if it is dead or alive
- This thought experiment is used to understand the quantum state of electrons.
- Superposition means that each quantum bit (basic unit of information in a quantum computer) can represent both a 1 and a 0 at the same time.

2nd Property : Entanglement

- In quantum entanglement, subatomic particles become “entangled” (linked) in such a way that any change in one disturbs the other even if both are at opposite ends of the universe.

Let me take you to a mcdonald's drive through

- You order a mc chicken and a mc veggie
- However, they give you both the burgers in an exact same box.
- Now, when you open the first box, and you see a mc chicken, you know the other box has a mc veggie.
- The state of both these burgers is entangled : change in one means the change in other.

Entanglement in Quantum Computing

- Here we are talking about electrons at subatomic level
- Electrons have a spin : either positive or negative
- They aren't actually spinning : this is to understand their orientation
- Superposition of spin : think?
- Imagine a coin being flipped on a table : demonstration
- When there is interaction with the outside world : there is collapse of the wave function and electron shows its spin

Entanglement

- 2 electrons close to each other : they will get opposite spin
- Their spin states will become entangled / linked
- Let's watch a video and understand

Entanglement leads to Superlinkages

- Despite a huge / astronomical distance, disruption in one electron will lead to a change in the other electron
- This has huge applications in Communication and Cryptography
- In quantum entanglement, subatomic particles become “entangled” (linked) in such a way that any change in one disturbs the other even if both are at opposite ends of the universe

Quantum Computing

- Recap :
- 2 principles : Superposition and Entanglement
- Much Faster than Other Computers
- Quantum principles will be used for engineering solutions to extremely complex problems in communications, sensing, chemistry, cryptography, imaging and mechanics.

Basics

- Qubits: basic unit of quantum computing; can contain both values “1” and “0” simultaneously.
- Superposition: ability to be in multiple states simultaneously → provides exponential speed.
- Entanglement: ability of two members of a pair (Qubits) to exist in a single quantum state.

Basics

- Quantum supremacy: state when a quantum device can solve a problem that no classical computer can solve in any feasible amount of time (irrespective of the usefulness of the problem).

Advantages of Quantum Computing

- **Speed**: perform quick and easy calculations that are time consuming on conventional computers.
- **Storage**: drastically bring down the space needed to store data.
- **Simulation**: doing data simulation involving multiple parameters like weather forecasting, chemical simulation.
- **Privacy**: impossible to break the security of quantum entanglement-based cryptography

Applications

- **Secure Communication**: quantum cryptography → significant to satellites, military and cyber security; promises unimaginably fast computing and safe, unhackable satellite communication to its users. E.g. China demonstrated secure quantum communication between terrestrial stations and satellites; E.g. March 2021 ISRO demonstrated free space Quantum communication.

Applications

- **Research**: solving some of the fundamental questions in physics related to gravity, black hole etc; could give a big boost to the Genome India project. E.g. CERN (God particle)
- **Agriculture**: precision agriculture – predicting rainfall, type of soil, crop and fertilizers; make precise long-term large-scale production plans and helps to increase yield and efficiency; detecting weed through an invasive weed optimization algorithm.

Applications

- **Pharmaceutical**: Enhanced data processing abilities - develop vaccines faster, predict epidemics, simulate drug/vaccine responses/side effects, genome analysis and gene-editing; could reduce the time frame of the discovery of new molecules; E.g. Covid 19 variants

Applications Continued

- **Augmenting Industrial revolution 4.0**: Quantum computing is an integral part of Industrial revolution 4.0.; help leveraging other IR 4.0 technologies like the Internet-of-Things, machine learning, robotics, and AI.
- **Artificial intelligence and big data**: could empower machine learning by enabling AI programs to search through the gigantic datasets, consumer behavior, financial markets, etc.

Applications Continued

- **Weather: Forecasting and Climate Change**: Enhancing weather system modeling allowing scientists to predict the changing weather patterns with excellent accuracy; meteorologists can generate and analyze more detailed climate models - will provide greater insight into climate change and ways to mitigate it.
- **Disaster Management**: collection of data regarding climate change can be streamlined; overlaying this weather modelling data; Tsunamis, drought, earthquakes and floods may become more predictable

Quantum Supremacy

- It's the point at which a quantum computer can complete a mathematical calculation that is beyond the reach of even the most powerful supercomputer.
- Recently, Sycamore (Google's quantum computer) took 200 seconds to perform a calculation that the world's fastest supercomputer, Summit, would have taken 10,000 years to do.

Challenges

- **Quantum physics** – programmers will also need to know and learn particle physics;
- **Quantum states** – Quantum decoherence : the Qubits lose the quantum state easily
- **Designing quantum algorithm** – Need specialized algorithm, totally different from conventional algorithm.
- **Cyber Security**: QC can decrypt all the data on cyber space
- **Limited working**: The tasks like word processing and email are not suitable for quantum computers.

Quantum Key Distribution

- In traditional cryptography, security is based on that an adversary is unable to solve a certain mathematical problem.
- In QKD, security is achieved through laws of quantum physics.
- 2 most important laws are: Superposition and Entanglement.

Quantum Key Distribution

- News Context: A satellite-based communication between 2 ground stations was activated by entangled-based quantum key distribution (OKD).
- This was achieved by Micius (Quantum Experiments at Space Scale), World's 1st quantum-enabled satellite.
- Micius was launched by China in 2016

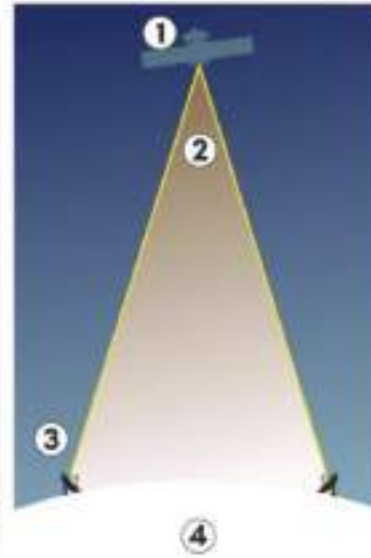
Quantum Key Distribution

- News Context: Bengaluru based startup Qnu labs recently innovated advanced secured communication through QKD.
- Qnu labs under aegis of Innovation for Defence Excellence (iDEX)
- iDEX is the operational framework of Defence Innovation Organization, a special purpose vehicle under Ministry of Defence.

How does Quantum Key Distribution works?

Quantum key distribution allows user to agree on a way of transmitting their data without the worry that someone is listening in

1. Sender instructs satellite to generate 2 entangled photons of particular quantum state
2. Photons are beamed to both ground stations
3. Sender and receiver compare the quantum states of the photons to check if they have been intercepted. If not they use the photons to create a code to encrypt the data
4. Encrypted data can then be sent securely via conventional means



Quantum Key Distribution

- Two main categories of QKD are:
- Prepare-and-measure protocols : focus on measuring unknown quantum states. This type of protocol can be used to detect eavesdropping, as well as how much data was potentially intercepted.
- Entanglement-based protocols focus on quantum states in which two objects are linked together, forming a combined quantum state. In this method, if an eavesdropper accesses a previously trusted node and changes something, the other involved parties will know.

Quantum Key Distribution

- Quantum cryptography is considered future proof : it is believed that no future changes in technology can crack quantum cryptography.

QKD Challenges

- Transmission Loss
- Lower Communication rates
- Infrastructure shortfalls

Quantum Computing and India

- National Mission on Quantum Technologies and Applications (NM-QTA): Budget 2020 allocated Rs 8000 Crore to the mission for a period of five years.
- Quantum Information and Computing (QuIC) lab at the Raman Research Institute, Bangalore.
- Quantum-Enabled Science & Technology (QuEST) a research program by the Department of Science & Technology.

Quantum Computing and India

- Quantum Frontier mission: It is an initiative of the Prime Minister's Science, Technology, and Innovation Advisory Council (PM-STIAC) which aims to initiate work in the understanding and control of quantum mechanical systems.
- DRDO Young Scientist Laboratory for Quantum Technologies (DYSL-QT), Mumbai.

Supercomputers

- Supercomputers are large systems that are specifically designed to solve complex scientific and industrial challenges.
- Unlike traditional computers, supercomputers use more than one central processing unit (CPU).
- Supercomputing is measured in floating-point operations per second (FLOPS).
- Petaflops are a measure of a computer's processing speed equal to a thousand trillion flops.

Supercomputers

- Floating-point is a method of encoding real numbers within the limits of finite precision available on computers.
- Supercomputers can be one million times more processing power than the fastest laptop
- They work on parallel computing or parallelism where many processors work together.
- World Fastest Supercomputer is Fugaku of Japan with speed of 415 Peta-Flops

Supercomputers

- It is a common benchmark measurement for rating speed of microprocessors.
- For comparison, a desktop computer has
- performance in the range of hundreds of giga FLOPS to tens of tera FLOPS.
- 1 Mega FLOPS: Equal to 1 million FLOPS.
- 1 Giga FLOPS: Equal to 1 billion FLOPS.
- 1 Tera FLOPS: Equal to 1 trillion FLOPS.
- 1 Peta FLOPS: It can be measured as 1.000 teraflops.

National Supercomputing Mission

- NSM, launched in 2015, is being steered jointly by Ministry of Electronics & Information Technology and Department of Science and Technology and implemented by C-DAC and Indian Institute of Science, Bangalore.
- Mission envisages empowering our national academic and R&D institutions spread over the country by installing a vast supercomputing grid comprising of more than 70 high-performance computing (HPC) facilities.

National Supercomputing Mission

- Till now C-DAC has deployed 11 such facilities under NSM Phase-1 and Phase-2 with a CCP of more than 20 Petaflops. Eg: PARAM Shivay [IIT (BHU)], PARAM Shakti (IIT Kharagpur), PARAM Brahma (IISER, Pune) etc.
- These supercomputers will also be networked on the National Supercomputing Grid over National Knowledge Network (NKN).
- 4 major pillars of NSM are Infrastructure, Applications, R&D, Human resource development.

National Supercomputing Mission

- NSM is being used for Genomics and Drug Discovery, Urban Modelling, Flood Early Warning and Prediction System for River Basins, seismic Imaging to aid Oil and Gas Exploration etc.
- Recently : PARAM Ganga

India's Supercomputers

- Pratyush : Located in Indian Institute of Meteorology.
- Mihir: Located in National Centre for Medium Range Weather Forecasting.
- PARAM Siddhi : It will be India ' s largest High-Performance Computing and Artificial Intelligence (HPC-AI) supercomputer. The supercomputer will have speed o f 210 AI Petaflops.
- Param Shakti and Param Brahma: Installed at IIT-Kharagpur and IISER, Pune.

Blockchain Technology and Cryptocurrency

- Blockchain : a chain of blocks that have certain information
- Decentralised Ledger.

Cryptocurrency and Blockchain

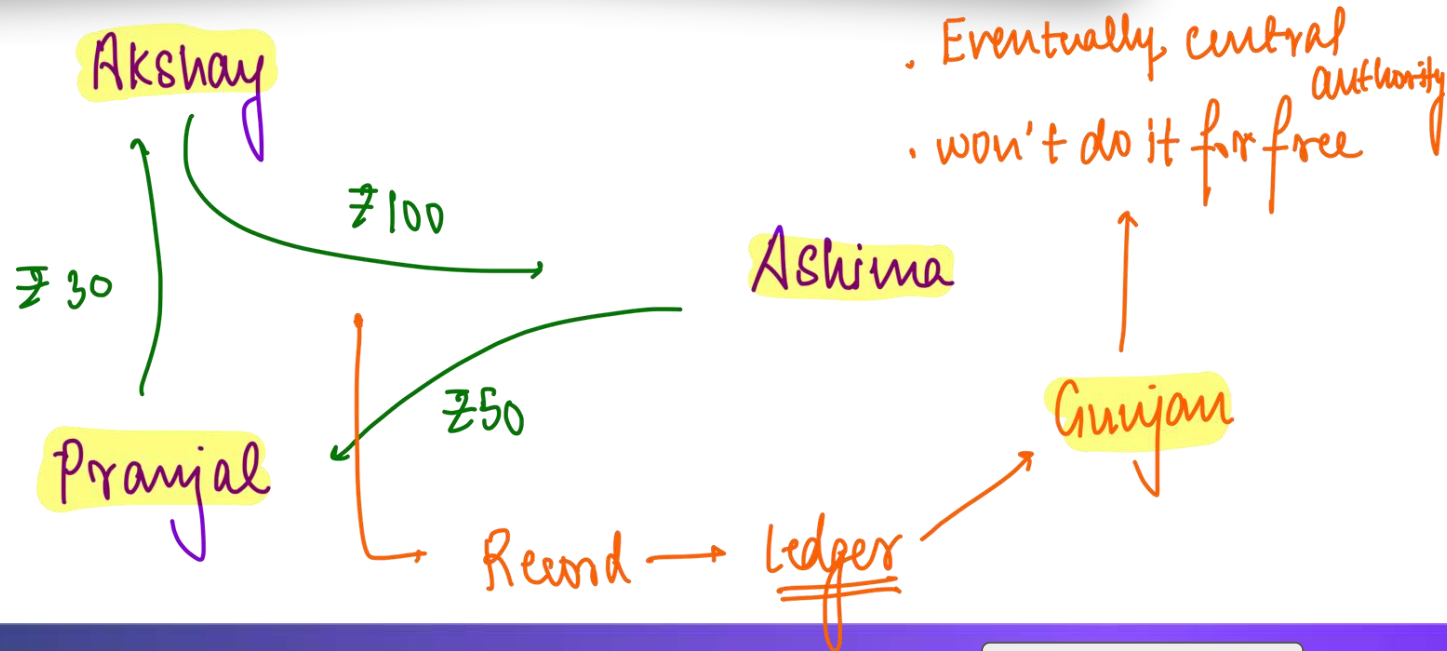
- Think : why can money not be digitized ?

Cryptocurrency and Blockchain

- The internet is centralised
 - Eg: Google.
 - Trust deficit
- Internet has abundance : no scarcity
- Hacking might be possible.
- These 3 problems are why money cannot be digitized

Sol = Blockchain

Normal Transaction



How to Decentralise?

- We will maintain the ledger ourselves.
- How?

How to Decentralise?

- Ashima pays Akshay Rs. 100
- Akshay pays Pranjal Rs. 30
- Pranjal Pays Ashima Rs. 10

At the end of the month you get together and tally.

How to Decentralise?

- However , if you start with a certain amount
- You won't need to tally till you don't overspend.

How to Decentralise?

- Akshay has Rs. 100
 - Ashima has Rs. 100
 - Pranjal has Rs. 100
- 
- Ashima pays akshay Rs. 100
 - Akshay pays pranjal Rs. 30
 - Pranjal pays Ashima Rs. 10
- 
- To insure this you need record of all transactions
 - Ashima pays Pranjal Rs. 50
- 

How to Decentralise?

- Each set of transaction - Recorded on a block.
 - You need record of previous transactions to validate future ones.



Chain of blocks

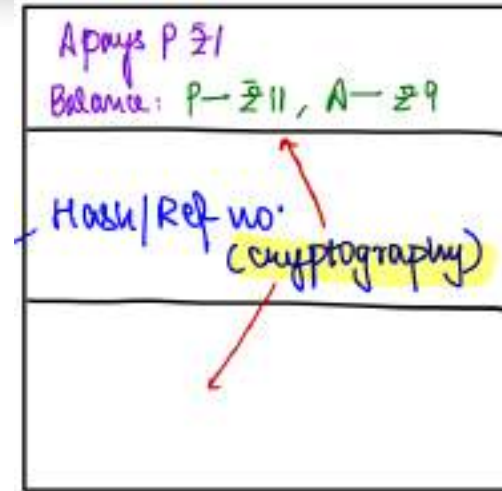
How to Decentralise?

Akshay (A) has Rs. 10

Pranjal (P) has Rs. 10

This is the fingerprint of that block

Change data in any compartment the hash changes.



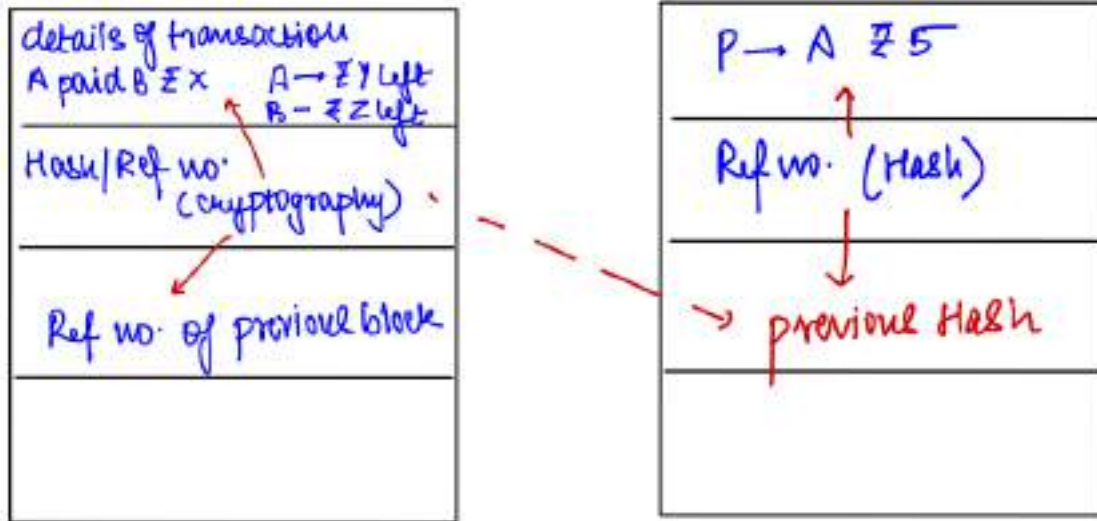
Cryptography : one way function

A pays P 21
 Balance: P → 211, A → 29
 → 147 132 989 7425
 ↖
 — X —

New Transaction :

- Your friends will check the previous ledger
- Validate
- Add to the chain

How to link the new block to the previous one ?



Block - Chain

- It exists on your computer. Not on the servers of a centralized tech giant like google. : Decentralised
- What is a block chain? : Record of transactions : Ledger
- Therefore, Block chain is a decentralized Ledger system.

Block

- Hash (#)/ref no. (unique fingerprint);
 - Generated on basis of 1 and 3 compartment
- New transaction
 - If any illegitimate content : Ref # will change completely
 - Validated - added to chain of blocks.
- Block chain
 - Not centralised
 - Everyone has copy, and any one can validate (What if you change a block) ?

A



Block #6



Block #7



Block #8



Block #9

P



Block #6



Block #7



Block #8



Block #9

AB



Block #6



Block #7



Block #8



Block #9

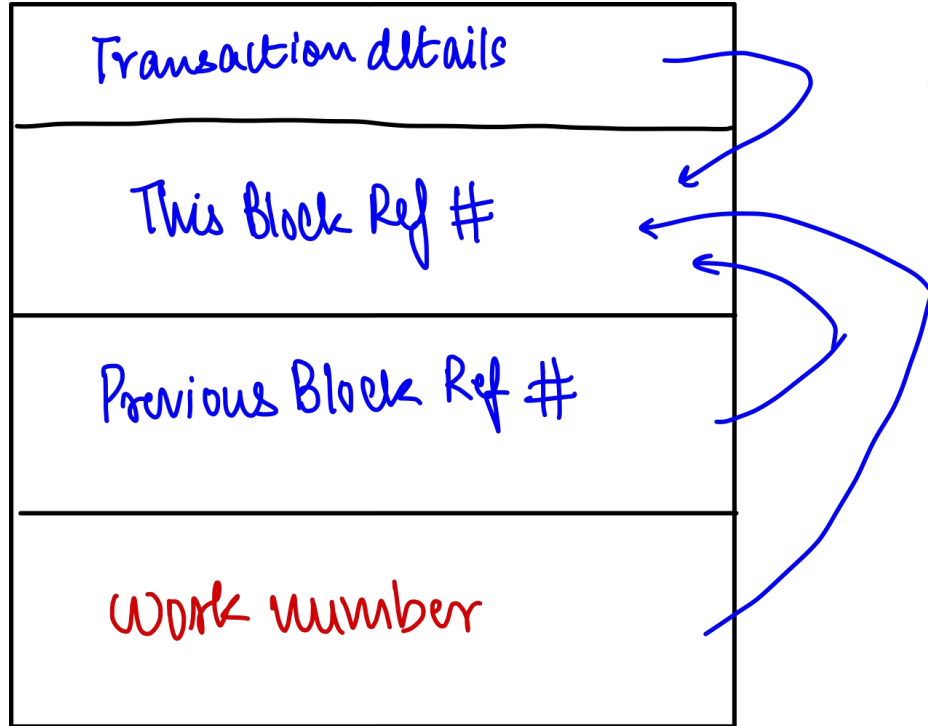
- Decentralisation
 - Change Resistant
 - Hack Proof

Blockchain - Decentralised

- Since all are linked you can't randomly change any record.
 - Change resistant Hack proof.
- Can be validated/invalidated by anyone
 - Digital scarcity + Decentralization
- Missing = Accountability

What if 10000 people blockchains simultaneously

- You will end up with 10000 different chains
- Work needs to be done to change any block chain.
- Changing anything becomes tough



What if 10000 people blockchains simultaneously

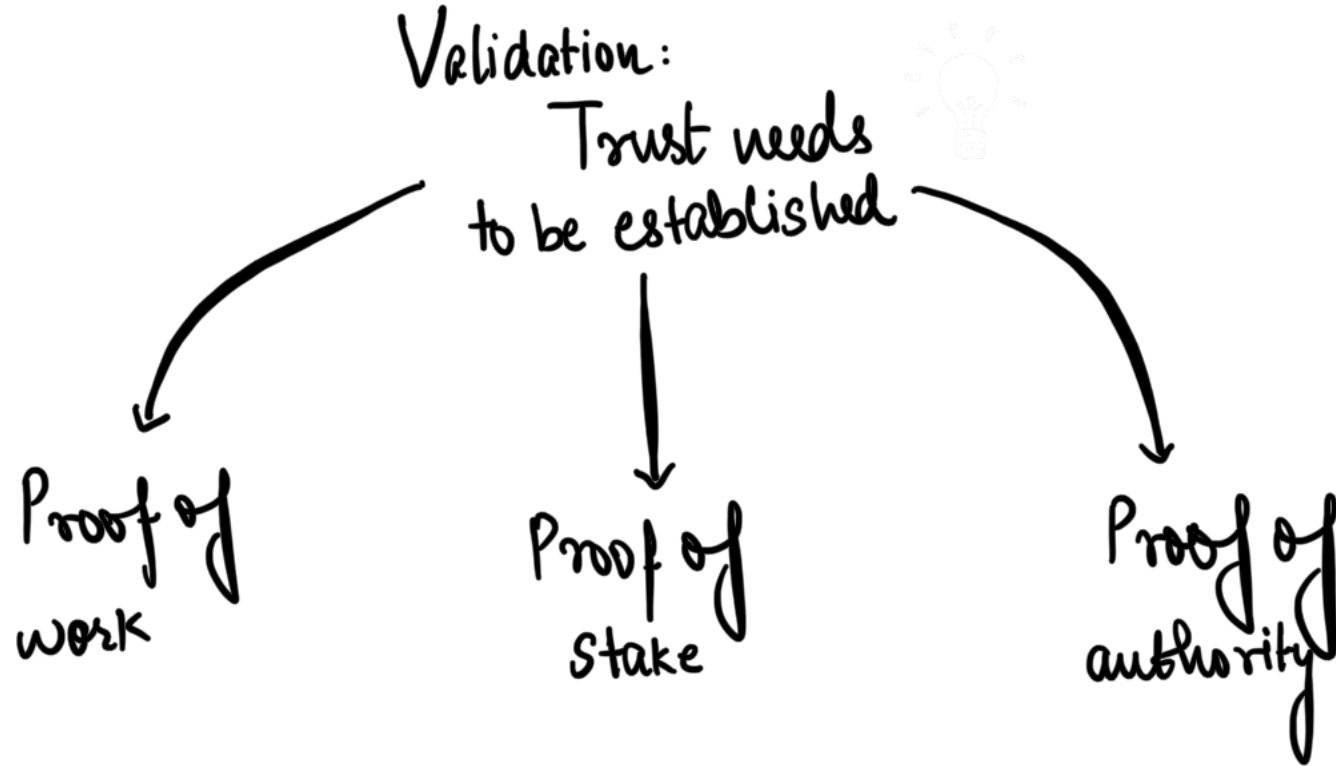
- Lets say rule that all Ref # must start with 0000
- To get 0000 - manually center no. from 1 to 1 million, at some point you'll get 0000
- Lot of computational power required Because work number of all block used to change.
- If before this someone sees that ref# of last block is off : it has no zeroes in the beginning, there would be a problem.

What if 10000 people blockchains simultaneously

- Moreover, new blocks are constantly being added.
Therefore it becomes very difficult to change a block chain.
- The longest chain would obviously be more secure.

What if 10000 people blockchains simultaneously

- New transactions need to be validated. Why would someone validate transactions / remain a good actor?
- Validation here is calculation of work number for a particular block. This is also known as mining.
- Reasons for remaining a good actor :
 1. You need a lot of computational power for validation.
 2. Every time you validate new currency : you will get tokens. These tokens have value only till the time your chain has validity.



Proof of work : Bitcoin

- Proof of work means calculation of work number for validation of new transactions. It is a highly complex problem requiring lots of computational power and electricity. You will get certain tokens for demonstrating proof of work.
- Problem : every validation is a competition between miners, therefore lots of computational power gets wasted. This leads to environmental problems / water wastage in cooling systems.

Proof of work : Bitcoin

- 51 % attack : if one actor has more computational power than all the others in the system, (rest have 49%) then this actor can choose to include or exclude any transaction it wants.
- This person can solve the computational puzzle faster than all others and this person will constantly have the longest chain.

Proof of stake : ethereum

- Every person who wishes to validate a new transaction needs to put certain tokens they have on stake. Usually there is no competition as the block creator is the one who has maximum tokens.
- Since almost always the same person gets to validate, there is no reward for making a block, so the block creator earns only a transaction fee.
- PoS replaces miners with validators and rewards only the top stakeholders, resulting in less energy consumption.

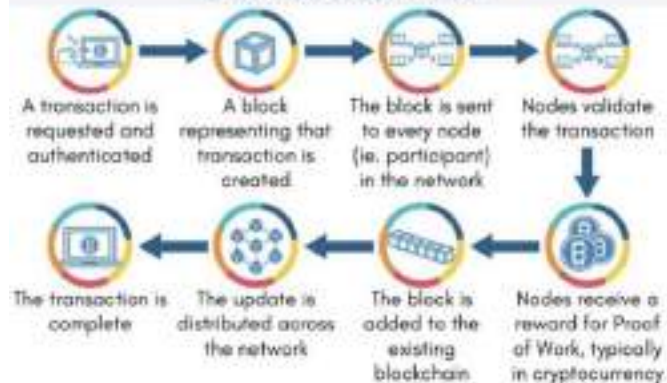
Proof of authority

- Consensus protocol is based on value of identity. There are limited number of moderators and it is close to a centralized system.
- Reputation based system.
- No mining / validation.

Recent Development

- Ethereum switched to merge software mechanism that uses proof of stake (PoS) mechanism. This will help slash energy requirements.
- Ethereum is a decentralized blockchain platform used to build decentralized Apps and Smart Contracts as well as a cryptocurrency.
- From proof of work to proof of stake consensus mechanism.

HOW DOES A TRANSACTION GET INTO THE BLOCKCHAIN?



	Proof-of-Work (PoW)	Proof-of-Stake (PoS)
About	Mechanism bitcoin uses to regulate creation of blocks through the process of mining.	An alternative consensus mechanism that delegates control of the network to the owners of a given token.
 Working Mechanism	PROOF OF WORK  To add each block to the chain, miners must compete to solve a difficult puzzle using their computers processing power.  In order to add a malicious block, you'd have to have a computer more powerful than 51% of the network.  The first miner to solve the puzzle is given a reward for their work.	PROOF OF STAKE  There is no competition as the block creator is chosen by an algorithm based on the user's stake.  In order to add a malicious block, you'd have to own 51% of all the cryptocurrency on the network.  There is no reward for making a block, so the block creator takes a transaction fee.
 Energy Consumption	PoW consumes more energy since it allows all miners on a network to try and validate a transaction.	PoS replaces miners with validators and rewards only the top stakeholders, resulting in less energy consumption.

Blockchain

- Blockchain is a distributed or decentralised ledger technology which was first introduced in the design and development of cryptocurrency, Bitcoin in 2009.
- Blockchain technology is an amalgamation of various technologies such as distributed systems, cryptography, etc
- Blockchain is a series of blocks, where each block contains details of transactions executed over the network, hash(address) of the previous block, timestamp etc.

Blockchain

- Data and transactions stored in blocks are secured against tampering using cryptographic hash algorithms and are validated and verified through consensus (consensus protocols) across nodes of the Blockchain network.
- Blockchain platforms are being developed to offer Blockchain based digital transaction platforms. Popular blockchain platforms include Hyperledger, Cosmos, Polkadot, Redbelly, Ethereum etc.

BlockChain : National Strategy

- National Blockchain Framework : strategy for the next 5 years
- **Need:** Reports predicts that by 2030, Blockchain would be used as a foundational technology for 30% of the global customer base.
- **Plan:** MeitY has initiated a project on design and development of a **NBF** for creation of a shared Blockchain infrastructure and offering Blockchain as-a-Service (BaaS). Initially, NBF would be used for e-governance domain transitioning to incorporate various use cases over time



Objectives and Vision of National Strategy on Blockchain

To create trusted digital platforms through shared Blockchain infrastructure

Promoting research and development, capacity, skill and innovation, technology and application development

Making India a global leader in Blockchain Technology

Facilitating state of the art, transparent, secure and trusted digital service delivery to citizens and businesses

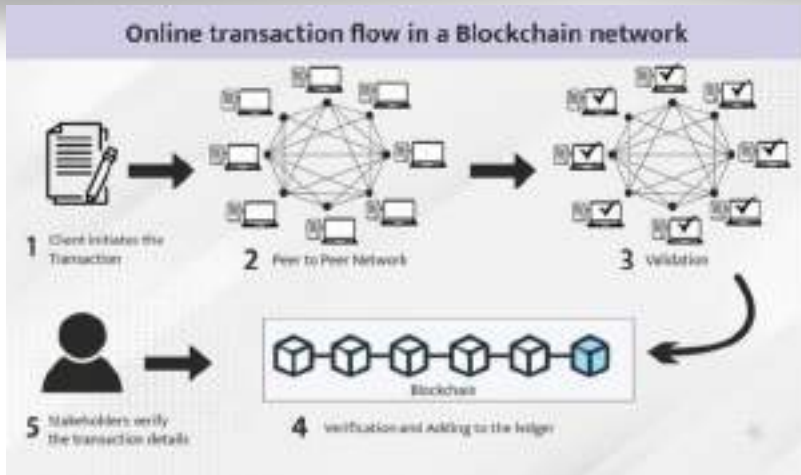
Blockchain National Strategy

- Components :
- Geographically distributed nodes
- R and D
- Indigenous blockchain platform
- Awareness creation
- Integration with online services such as digi locker
- Awareness Creation

Blockchain : Related news

- Presidio Principles : Foundational Values for a decentralized future : released by world economic forum
- 4 pillars : transparency and accessibility, privacy and security, accountability and governance, and Agency and interoperability
- First country to give legal tender status to bitcoin : El Salvador

Blockchain : Potential Applications



Potential Blockchain Applications

- | | |
|---|---|
| <ul style="list-style-type: none">• Transfer of Land Records (Property Record Management);• Digital Certificates Management (Education, Death, Birth, agreements, etc);• Pharmaceutical supply chain;• e-Notary Service (Blockchain enabled e-Sign Solution);• Farm Insurance;• Identity management;• Power distribution; | <ul style="list-style-type: none">• Duty payments;• Agriculture and other supply chains;• eVoting;• Electronic Health Record Management;• Digital Evidence Management System;• Public Service Delivery;• IoT Device Management and Security;• Vehicle lifecycle management;• Chit fund operations administration;• Microfinance for Self-Help Groups (SHG) |
|---|---|

Blockchain : Potential Applications

- Currency : crypto
- Banking and Finance : Transaction ledgers
- Inventory : Supply Chain Management
- Health : patient records : digital health mission
- Property and ownership records
- Smart contracts : update regularly
- IPR and asset protection
- Good governance : India Chain
- Corruption : audits

Blockchain : Cryptocurrency

- Cryptocurrency → digital currency secured by cryptography → medium of exchange where the ownership data is stored in a computerized database; Based on → Blockchain technology
- Cryptography- The science of protecting information by transforming it into a secure format through encryption.
- Decentralized; Security; Anonymity; Speed; Ease of transfer; Cross border trade;
- Has certain advantages and disadvantages

Blockchain : Cryptocurrency

- **Wallet**- the account addresses of the user which is accessible only by a private and a public key; the private key is unique and known only to the owner of the wallet.
- Instantaneous transaction → help save time of both the parties; everything is being done on the internet.
- International monetary market : work in countries with unstable economy; poor institutional strength;

Blockchain : Cryptocurrency

- Disadvantages
- Vulnerability: It's hidden nature of transaction → illegal activities → terror-funding, money laundering and tax-evasion.
- The RBI & Ministry of Finance issued statements on cryptocurrencies ;
- Frauds and scams – Squid cryptocurrency
- Payments in one go: Payments cannot be reversed back; no rollback and no central legal authority;

Blockchain : Cryptocurrency

- Acceptance: Not legal in many countries → holds different values at different places; almost 90% of the currencies are scams.
- Risky investment: Not issued by the government agency of most countries → makes it less reliable;
- Reports of hacking of several exchanges; Highly volatile;
- Price: depends on demand and supply; high volatile; cannot be relied as medium of exchange of value

Blockchain : Cryptocurrency

- Disadvantages in obtaining cryptocurrency
- 3 ways to obtain

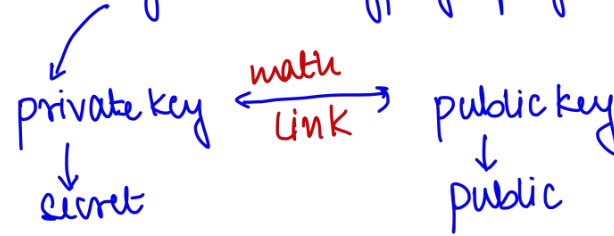
1. Mining : use powerful computers, issues : graphics card demand and generation of e waste

2. Selling goods to owner of CC : public and private key, untraceable, money laundering and terror finance, government will be deprived of taxes; **Sovereign function of money creation**

Blockchain : Cryptocurrency

1. Exchanging legal tender : no protection under FEMA act, tax haven issues, illegal trade and terror finance
 - Which organization helps curb terror finance?

Bitcoin → asymmetric cryptography



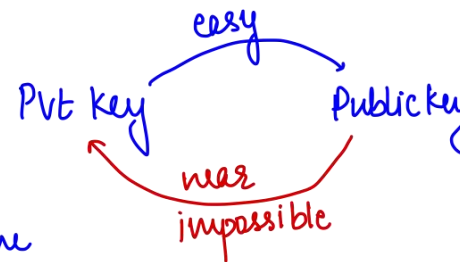
- any change to ledger — pvt key used →

verified by
public key

creates a
digital
signature

One way function

- Mathematical principle
- to derive pvt key from Public key, astronomical time needed.



Shor's Quantum Algorithm

- Break the one way function.
- $\therefore$ Large enough Quantum Computer $\rightarrow$ digital signature can be falsified.

Bitcoin

- decentralised generation of pvt and public key
 - for transaction
 - $\rightarrow$ use pvt key $\rightarrow$ validate transactⁿ
 - $\rightarrow$ spend your BTC.
- Quantum computer $\rightarrow$ public key

A $\longrightarrow$ B

transaction reveals
public key

↓
till it is not mined

↓
vulnerable to Quantum
attack.



Mining → competition b/w
validators.
Quantum co. takes
over.



during this time, your a/c
is vulnerable

current
mining time
||
10 min

Quantum
Shor's algo
predicted
time
↓
30 min - 8h.

Web 3.0

- Web 1.0, also called the Static Web, which enabled easy access to information. However, the information was largely disorganised and difficult to navigate.
- The Social Web, or Web 2.0, made the internet a lot more interactive thanks to advancements in web technologies like Javascript, HTML5, CSS3, etc., which enabled startups to build interactive web platforms such as YouTube, Facebook, Wikipedia and many more.

Web 3.0

- Facebook.com : different for all of us
- Brainstorm : Web2.0 problems?
- Web 3.0 : third stage in internet evolution

Web 3.0

- Web 2.0 : you are the product
- Web 3.0 : you are the owner
- new iteration of the World Wide Web which would incorporate concepts such as decentralization, distributed ledger such as block chain technologies, and token-based economics.
- How?

THREE STAGES OF INTERNET CONSUMPTION



	 Web1	 Web2	 Web3
Time period*	1990-2005	2005-till date	2021
Where data is stored	Server's file system	On-premise/Cloud	Blockchain, distributed across multiple networks
Examples	Static web pages	User generated content like Social media, and web applications like e-commerce etc...	NFTs, cryptocurrency transaction
Who owns data	Companies running the webpages	Companies that host application, cloud service providers	No one owns the data
Transacting	No transaction possible	Payment gateways for currency transactions	Transaction happens using crypto tokens

Web 3.0 explained

- Its like a big torrent network
- Data is stored in blockchain over many computers around the world
- Token based economics : Example : odyssey website : alternative to YouTube : creators earn library tokens for making others watch
- Post on many computers around the world : cannot be taken down or censored.

Web 3.0 explained

- Who runs it : DAO for every company
- Decentralised autonomous organization : no CEO, no one controlling authority
- Most tokens : decision on rules etc
- Anonymized identity

Web 3.0 advantages

- Decentralization: core tenet of Web 3.0; could be stored in multiple locations simultaneously; break down massive databases currently held by internet giants like Meta and Google;
- Democratize net: Greater control to user; data security;
- Trustless and permissionless: run on decentralized peer-to-peer networks – Trustless (i.e., the network will allow participants to interact directly without going through a trusted intermediary); permissionless (meaning that anyone can participate without authorization from a governing body);

Web 3.0 advantages

- dApp- Decentralized applications (dApps): applications built on blockchain; use smart contracts to enable service delivery in a programmatic approach;
- DeFi - Decentralized finance (DeFi): use blockchain for enabling financial services
- DAOs - providing structure and governance in a decentralized approach.

Web 3.0 advantages

- Artificial intelligence (AI) and machine learning: Computers will be able to understand information similarly to humans; will use machine learning, - data and algorithms to imitate how humans learn;
- Others: NFT, Crypto currency;

Web 3.0 challenges

- Consensus approach: slow development, decisions;
- Regulation
- Data privacy
- Malafide use
- Not decentralised: controlled by capitalists, big firms;
- Business models: averse to decentralization; stakeholders, assets drive economy;
- Finances: acceptance and illegality of crypto e.g. torrent

Non Fungible Tokens

- Digital identifier, similar to a certificate of ownership that represents a digital or physical asset; financial security consisting of digital data stored in a blockchain, a form of distributed ledger;
- NFT market grew exponentially from 2020–2021: the trading of NFTs in 2021 increased to more than \$17 billion, up by 21,000% over 2020's total of \$82 million; Work entitled Merge by artist Pak was one the most expensive NFT, with an auction price of US\$91.8 million.
- What is fungibility?

Fungibility

- Fungibility means subdivision into units as well as mutual substitution
- Example : Rs. 2000 note.
- Subdivided : yes, 500 x 4 notes
- Mutual substitution : yes, both have the same value
- Other cases :
- 1 kg gold bar
- 100 gm diamond
- NFT : not possible

NFT

- NFT is a Digital file such as photo file(JPEG),Animated image(GIF),music file (MP3) etc. Stored using blockchain Technology.
- They cannot be sub divided and/or their individual sub-units can not be exchanged with one another. Because their values are different based on buyer's preference. So NFTs are non- fungible.
- How does it work?

NFT : working explained (in class)

NFT

- Advantages :
- Multimedia creators : trading of digital assets
- Easier to verify opwnership records
- Paper less administrations : digital torage of tickets / records
- Challenges :
- mass hysteria of investment bubble might collapse leading to bankruptcy of gullible citizens
- Money laundering and terror finance : same as bitcoin
- Electricity consumption and environmental challenges

Internet Of Things

- It is a seamless connected network of embedded objects/ devices, with identifiers, in which Machine to Machine (M2M) communication without any human intervention is possible using standard and interoperable communication protocols.
- In general terms, it includes any object or thing that can be connected to an Internet network, from factory equipment and cars to mobile devices and smart watches.
- Phones, Tablets and PCs are not included as part of IoT.

Internet Of Things

- It specifically comes to mean **connected things that are equipped with sensors, software, and other technologies** that allow them to transmit and receive data.
- It is widely being used to create **smart infrastructure in various verticals such as Power, Automotive, Safety and Surveillance, Remote Health Management, Agriculture, Smart Homes and Smart Cities** etc, using connected devices.

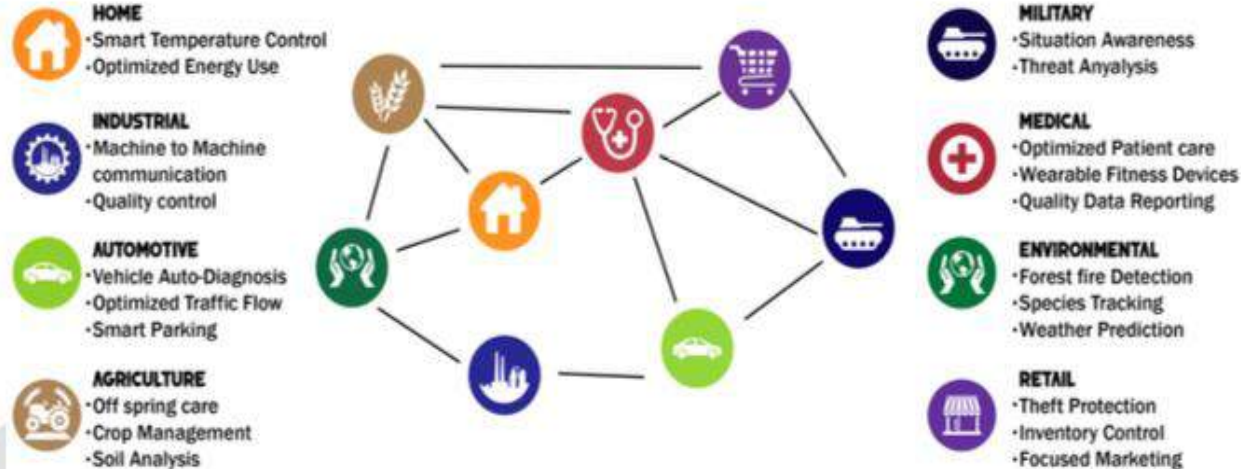
How does it work?

- An IoT ecosystem consists of web-enabled smart devices use embedded systems, such as processors, sensors and communication hardware, to collect, send and act on data they acquire from their environments.
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How does it work?

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Internet of Things Uses



Significance and challenges

- IoT encourages the communication between devices, also known as M2M communication.
- M2M communication helps to maintain transparency in the processes. It leads to uniformity in the tasks and maintain the quality of service.
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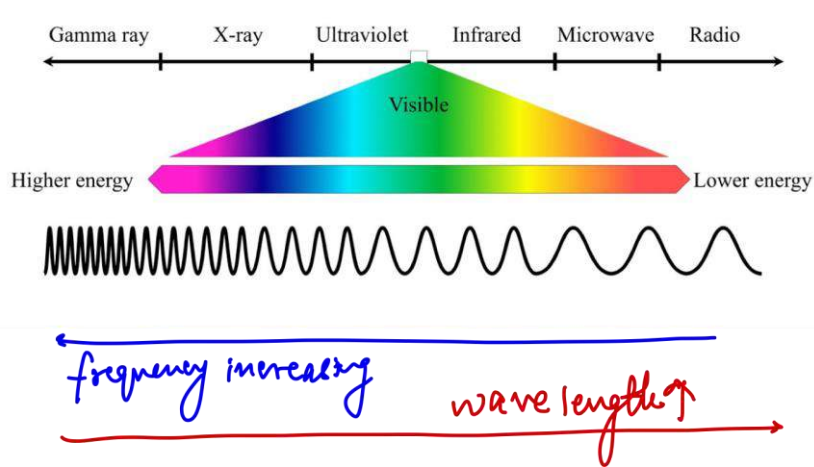
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- Further challenges : internet consumption and bandwidth issues
- Solutions : Lifi , 5g

Electromagnetic Spectrum Explained

Electromagnetic Spectrum

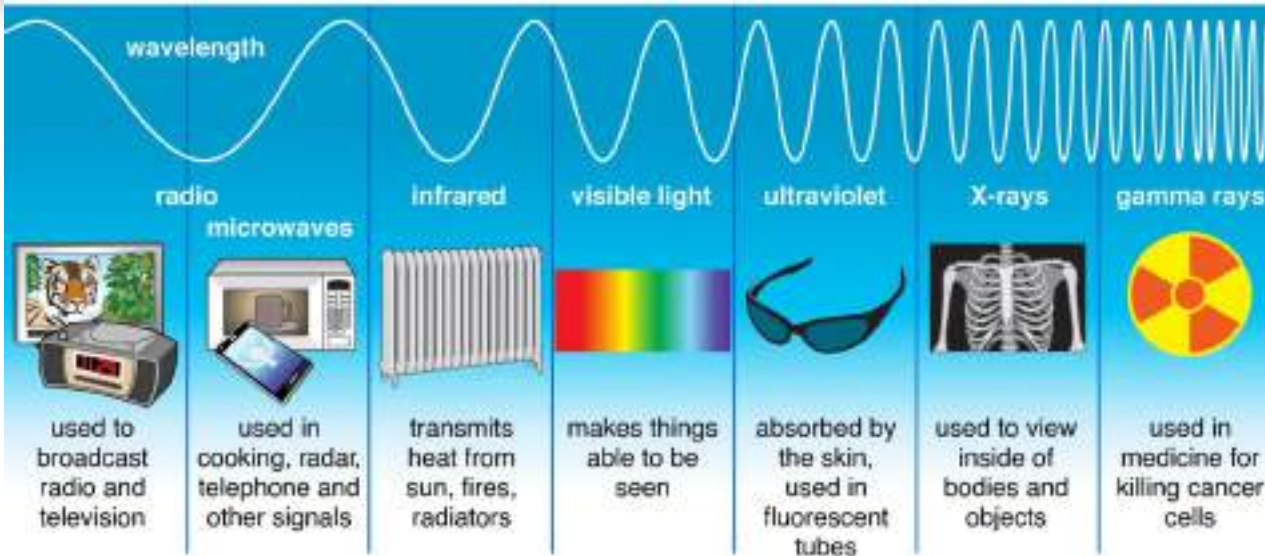
Full 'spectrum' of radiation emitted from a star.



$$E = h \frac{c}{\lambda} = h\nu$$

Electromagnetic Spectrum Explained

Types of Electromagnetic Radiation



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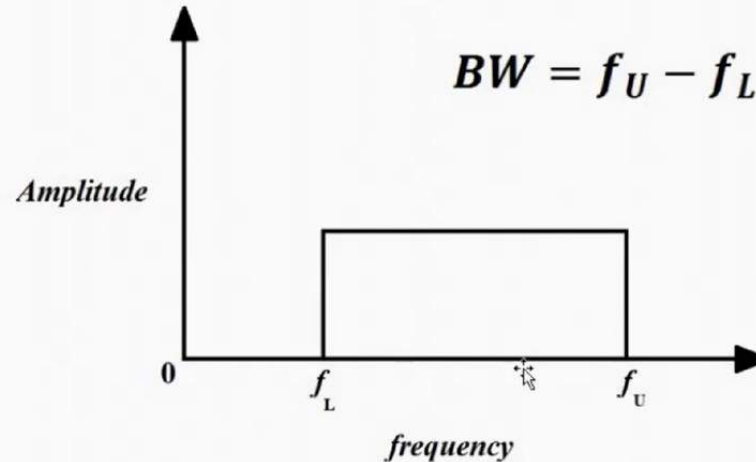
Electromagnetic Spectrum Explained

EM Wave	Frequency Range
Extremely high frequencies (EHF)	30 – 300 GHz
Super high frequencies (SHF)	3 – 30 GHz
Ultra high frequencies (UHF)	300 MHz – 3 GHz
Very high frequencies (VHF)	30 – 300 MHz
High frequencies (HF)	3 – 30 MHz
Medium frequencies (MF)	300 KHz – 3 MHz
Low Frequencies (LF)	30 – 300 KHz
Very low frequencies (VLF)	3 – 30 KHz
Voice frequencies (VF)	300 – 3000 Hz
Extremely low frequencies (ELF)	30 – 300 Hz

Electromagnetic Spectrum Explained

What is Bandwidth ?

- *Bandwidth* is the *portion of the electromagnetic spectrum occupied by a signal.*
- It is the *frequency range over which signal is transmitted.*



LiFi technology

- LiFi is a wireless optical networking technology, which uses light emitting diodes (LEDs) to transmit data.
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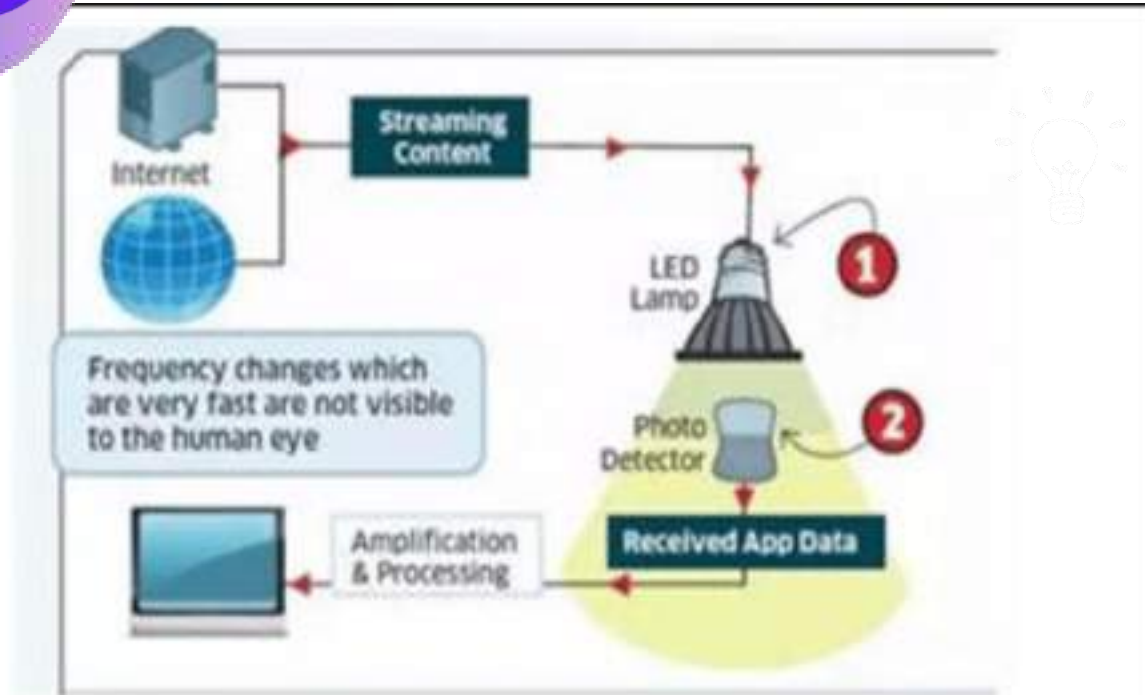
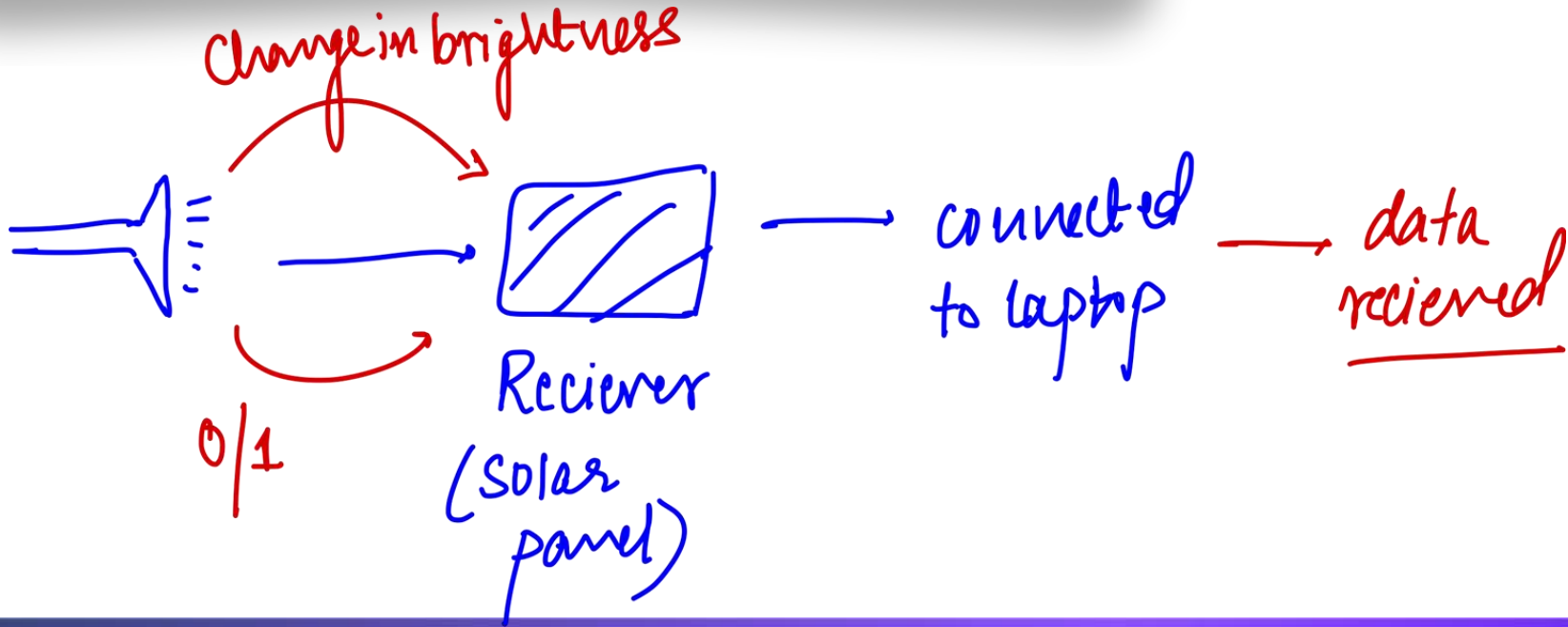


Figure 2: Procedure of Li-fi Technology

LIFI explained



LIFI explained : visible light

Metaverse

- Metaverse projects refer to blockchain-based games and applications that are set in virtual worlds.
- Such games follow a play-to-earn model, where gamers are rewarded for being invested in the game.
- Metaverse can be understood as a combination of multiple elements of technology, including virtual reality, augmented reality and video where users "live" within a digitally enhanced surrounding.

Metaverse

- **These technologies** are currently being developed by Apple, Google, Amazon, and Microsoft in addition to Facebook.
- Metaverse uses the following technologies : (**extended realities**)
- Virtual Reality
- Augmented Reality
- Hologram
- Avatar
- Let's see a video on how it works

Metaverse : Technologies

- **Virtual Reality (VR):** VR can be understood as the use of computer modelling and simulation that enables a person to interact with an artificial three-dimensional (3-D) visual or other sensory environment. E.g., games like World of Warcraft. **VR closes the world**, and transposes an individual, providing complete immersion experience.

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- **Avatar:** An avatar in the metaverse is a representation of an individual in the virtual world, this digital avatar enables the person to function like an actual human being in a digitally created world.
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Metaverse : Technologies : Images



5G technology

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- **5G operates at higher frequencies** to offer a new kind of network that is **designed to connect virtually everyone and everything together** including machines, objects, and devices.

5G technology

- It will also have an **enhanced throughput to handle more simultaneous connections** at a time than current-generation networks.
- 5G **mainly works in 3 bands**, namely low, mid and high frequency spectrum.

3G vs 4G vs 5G vs 6G					
		3G	4G	5G	6G
	Deployment	2004-06	2006-10	2020	2028-2030
	Bandwidth	2 mbps	200 mbps	>1 gbps	1 tbps
	Latency	100-150 millisecond	20-50 millisecond	<10 millisecond	<1 microsecond
	Average Speed	144 kbps	25 mbps	200-400 mbps	About 50 times faster than 5G

5G technology

Spectrum	Uses/Advantage	Limitations
Low Band	• Great promise in terms of coverage and speed of internet and data exchange, the maximum speed is limited to 100 Mbps (Megabits per second) • For commercial cellphone users who may not have specific demands for every high speed internet.	• May not be optimal for specialised needs of the industry.
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5G technology : challenges in India

Challenges to 5G evolution in India



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(inadequate optical fiber infrastructure and backhaul, interrupted power supply ,etc)



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Telco issues

(Low Average revenue per user, Lack of skillset etc.)



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(Network coverage issue, handset availability, data privacy etc)



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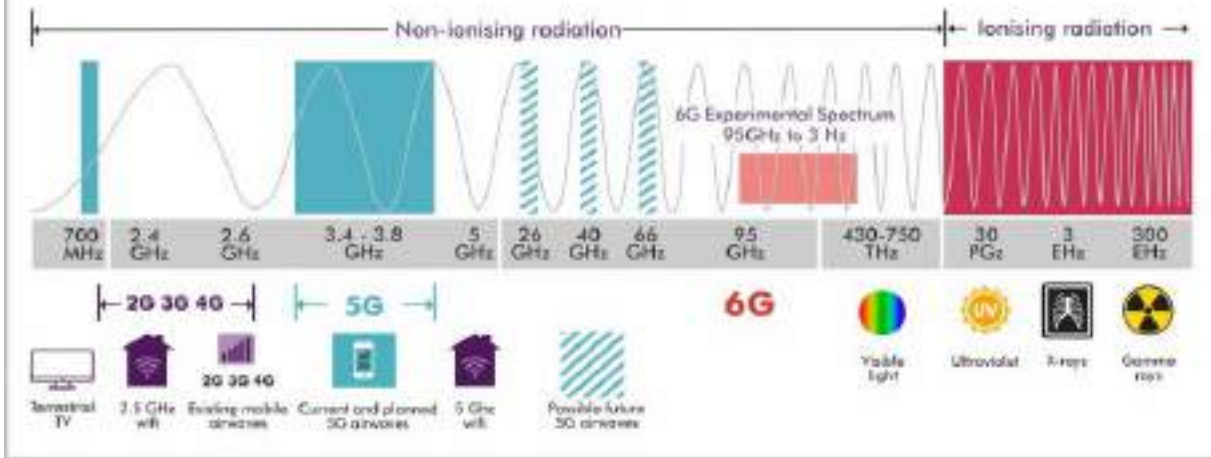


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5G and EM spectrum

Electromagnetic Spectrum and 6G Spectrum



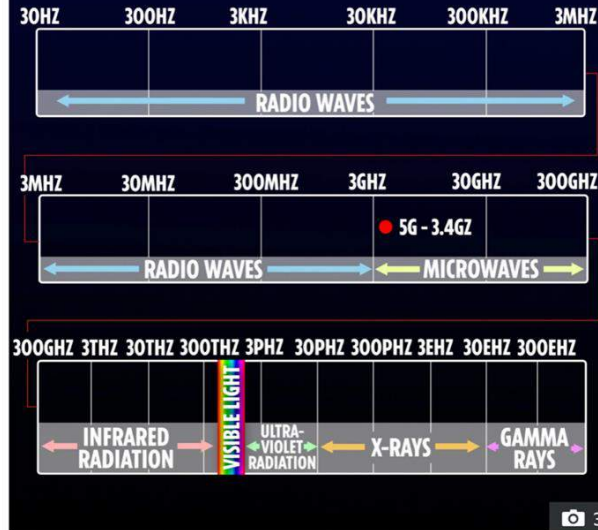
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ELECTROMAGNETIC SPECTRUM



5G is a very low-frequency form of radiation – far below visible light and infrared

Long term Evolution (LTE) and Voice over LTE (VoLTE)

- VoLTE is a **technology update to the LTE protocol** used by mobile phone networks.
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- Such as slow internet speeds and poor voice clarity.
- LTE is commonly marketed as 4G LTE.
- **VoLTE allows voice calls to be ‘packaged’ and carried through LTE networks.** This would mean 4G data accessibility even during calls.

Dark Net

- Imagine you get an email containing your bank statement :
- It is on the internet
- It is not accessible to me.
- Also known as Dark Web, it is that part of Internet which cannot be accessed through traditional search engines like Google nor is it accessible by normal browsers like Chrome or Safari.

Dark Net

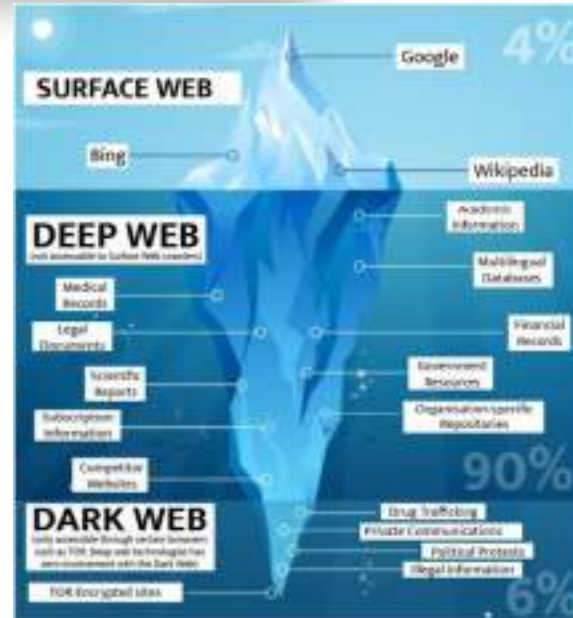
- It generally uses non-standard communication protocols which make it inaccessible to internet service providers (ISPs) or government authorities.
- Content on Dark Net is encrypted and requires specific browser such as TOR (The Onion Router) browser to access those pages.

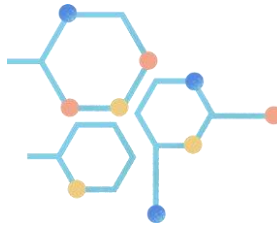
Dark Net

- Dark Net itself is only a part of Deep Web that is a broader concept, which includes sites that are protected by passwords.
- Part of internet that is readily available to general public and searchable on standard search engines is called as Surface Web.
- It is used by journalists and citizens working in oppressive regimes, researchers and students to do research on sensitive topics, law enforcement agencies etc.

Dark Net

- Concerns over its use: Anonymity, Haven for illicit activity, Privacy and ethical concerns, drug dealing, communication by terrorists etc.





UPSC Prelims

Science & Tech

Class No. 14

Emerging Technology

Non Fungible Tokens

- Digital identifier, similar to a certificate of ownership that represents a digital or physical asset; financial security consisting of digital data stored in a blockchain, a form of distributed ledger;
- NFT market grew exponentially from 2020–2021: the trading of NFTs in 2021 increased to more than \$17 billion, up by 21,000% over 2020's total of \$82 million; Work entitled Merge by artist Pak was one the most expensive NFT, with an auction price of US\$91.8 million.
- What is fungibility?

Fungibility

- Fungibility means subdivision into units as well as mutual substitution
- Example : Rs. 2000 note.
- Subdivided : yes, 500 x 4 notes
- Mutual substitution : yes, both have the same value
- Other cases :
- 1 kg gold bar
- 100 gm diamond
- NFT : not possible

NFT

- NFT is a Digital file such as photo file(JPEG),Animated image(GIF),music file (MP3) etc. Stored using blockchain Technology.
- They cannot be sub divided and/or their individual sub-units can not be exchanged with one another. Because their values are different based on buyer's preference. So NFTs are non- fungible.
- How does it work?

NFT

- Advantages :
- Multimedia creators : trading of digital assets
- Easier to verify opwnership records
- Paper less administrations : digital torage of tickets / records
- Challenges :
- mass hysteria of investment bubble might collapse leading to bankruptcy of gullible citizens
- Money laundering and terror finance : same as bitcoin
- Electricity consumption and environmental challenges

Internet Of Things

- It is a **seamless connected network of embedded objects/ devices, with identifiers, in which Machine to Machine (M2M) communication without any human intervention is possible** using standard and interoperable communication protocols.
- In general terms, it includes **any object or thing that can be connected to an Internet network**, from factory equipment and cars to mobile devices and smart watches.
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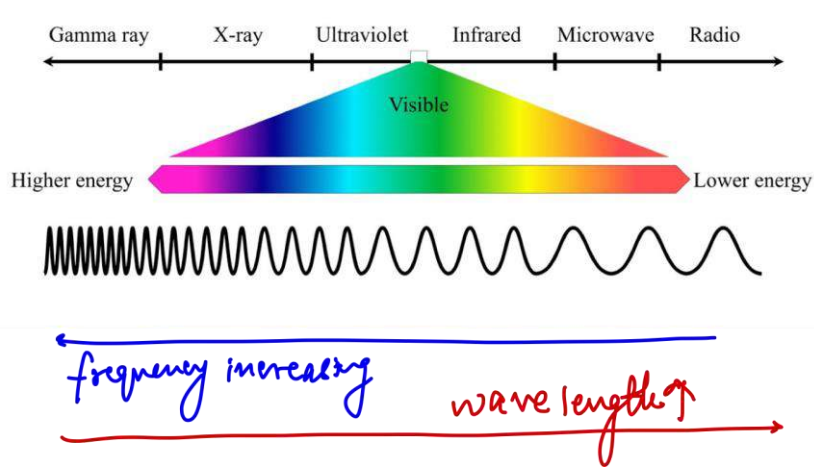
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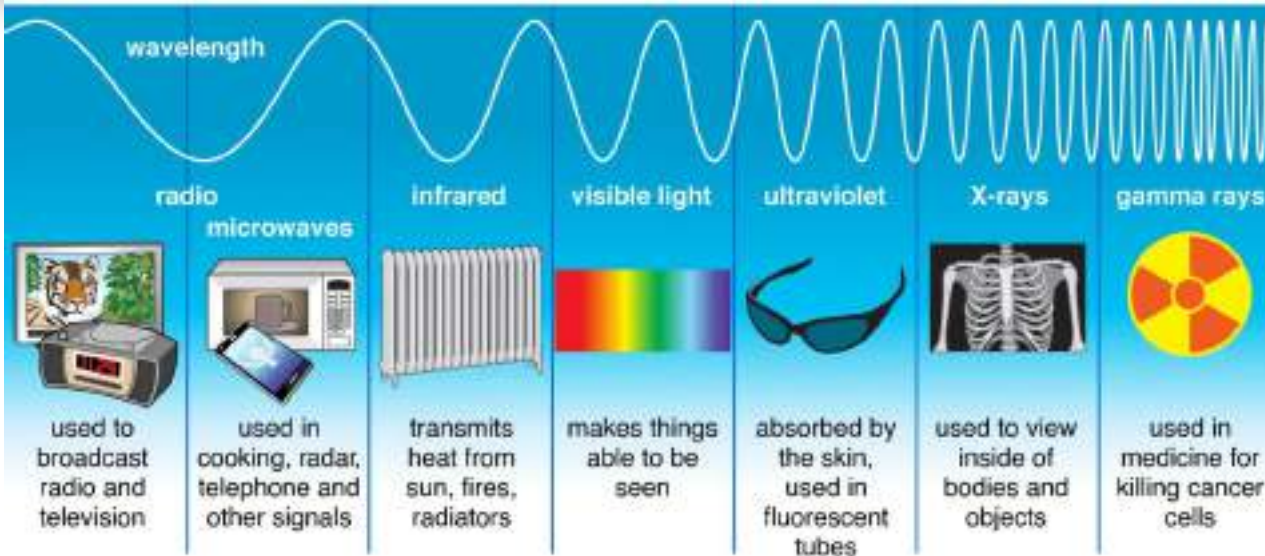
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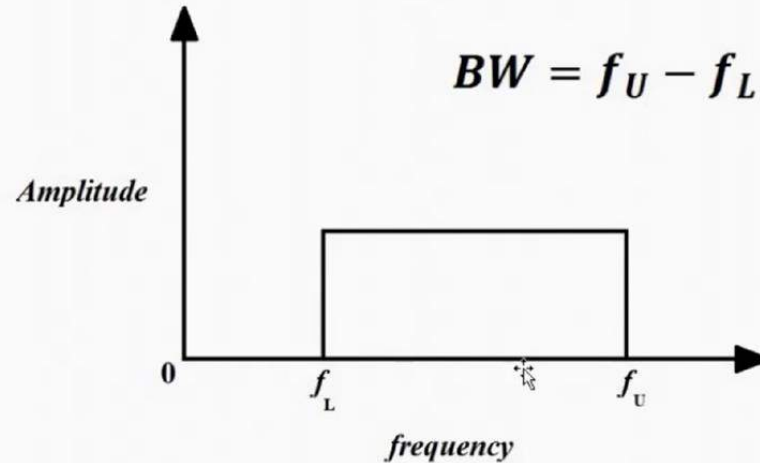
Electromagnetic Spectrum Explained

<u>Region</u>	<u>Frequency</u>	<u>Wavelength</u>
Radio	$< 3 \times 10^9 \text{ Hz}$	$> 10 \text{ cm}$
Microwave	$3 \times 10^9 - 3 \times 10^{11} \text{ Hz}$	$10 - 0.1 \text{ cm}$
Infrared	$3 \times 10^{11} - 4 \times 10^{14} \text{ Hz}$	$1000 - 0.7 \mu\text{m}$
Visible	$4 \times 10^{14} - 7.5 \times 10^{14} \text{ Hz}$	$700 - 400 \text{ nm}$
Ultraviolet	$7.5 \times 10^{14} - 3 \times 10^{16} \text{ Hz}$	$400 - 10 \text{ nm}$
X-ray	$3 \times 10^{16} - 3 \times 10^{19} \text{ Hz}$	$10 - 0.01 \text{ nm}$
γ -ray	$> 3 \times 10^{19} \text{ Hz}$	$< 0.01 \text{ nm}$

Electromagnetic Spectrum Explained

What is Bandwidth ?

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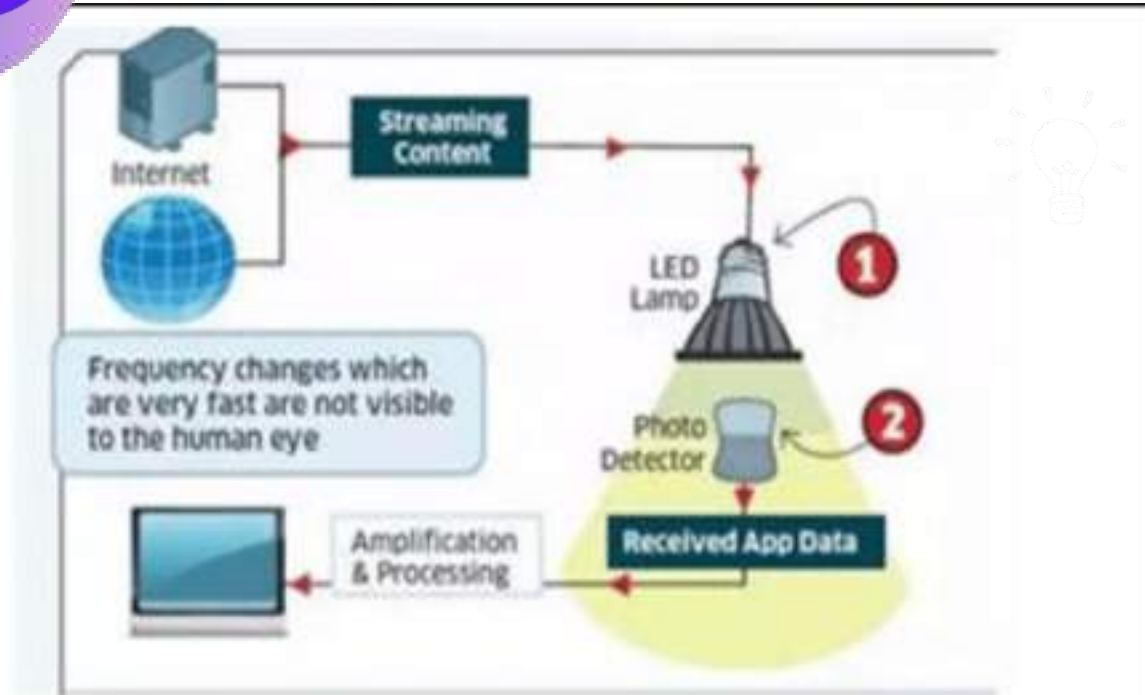
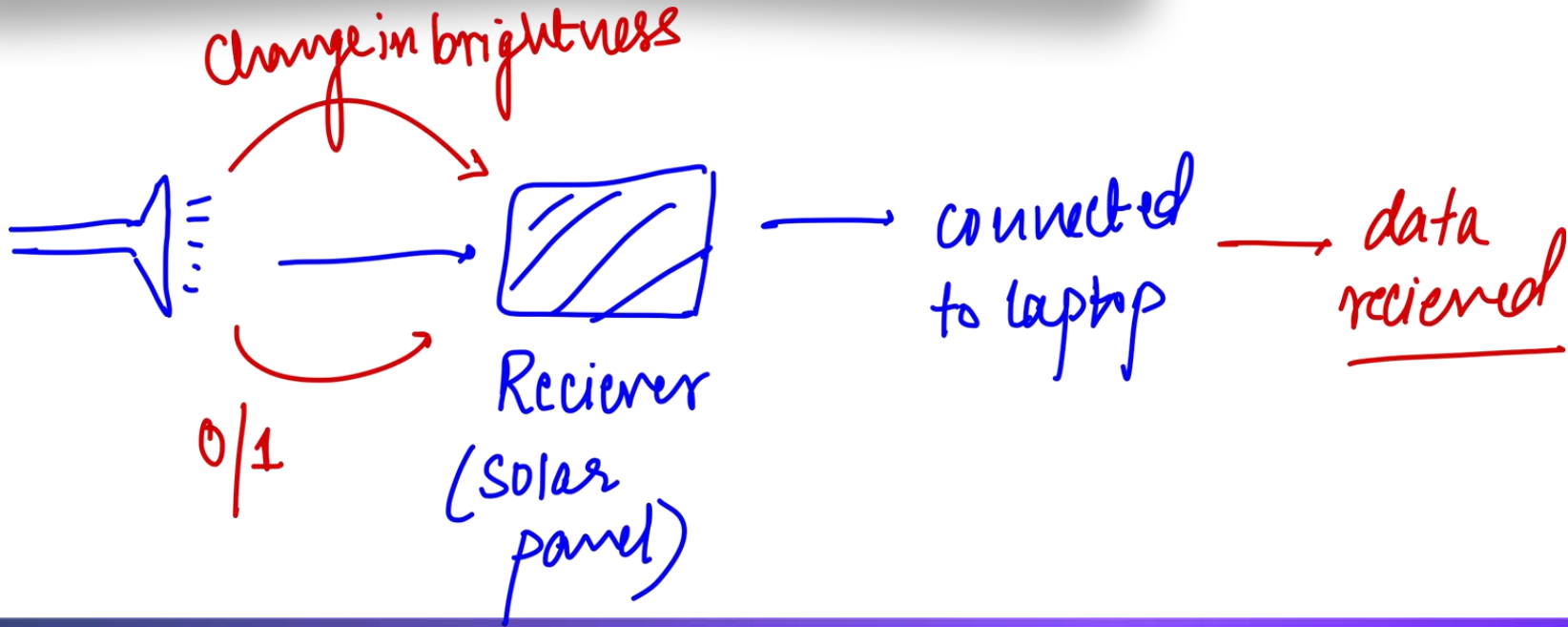


Figure 2: Procedure of Li-fi Technology

LIFI explained



LIFI technology

- Not a strictly line of sight technology : data rate is not dependent on line of sight but on signal quality of device.
- **Wifi : uses radio waves, Lifi : uses light.**

LIFI technology

Visible Light Communication (VLC) technology



VLC uses **electromagnetic spectrum wavelengths 375 to 780 nm.**



It is possible to **achieve illumination and data transfer simultaneously** by means of LEDs that is the prominent lighting equipment lately.



VLC **can transmit large amounts of data** faster than Bluetooth.



VLC has no **electromagnetic interference.**



Feature	LiFi	WiFi
Operation	LiFi transmits data using light with the help of LED bulbs.	WiFi transmits data using radio waves with the help of WiFi router.
Interference	Do not have any interference issues similar to radio frequency waves.	Will have interference issues from nearby access points (routers).
Privacy	In LiFi, light does not pass through the walls and hence will provide a much secure data transfer	In WiFi, Radio frequencies signal pass through the walls and hence there is a need to employ techniques to achieve secure data transfer.
Data transfer speed	About 1 Gbps	About 150Mbps.
Frequency of operation	10 thousand times frequency spectrum of the radio	2.4GHz, 4.9GHz and 5GHz
Coverage distance	About 10 meters	About 32 meters (vary based on transmit power and antenna type)
Data density	work with high dense environment	work in less dense environment due to interference related issues

5G technology

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- **5G operates at higher frequencies** to offer a new kind of network that is **designed to connect virtually everyone and everything together** including machines, objects, and devices.

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3G vs 4G vs 5G vs 6G

	Deployment	Bandwidth	Latency	Speed Average
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5G	2020	>1 gbps	<10 millisecond	200-400 mbps
6G	2028-2030	1 tbps	<1 microsecond	About 50 times faster than 5G

5G is the Fifth generation of cellular networks.

5G Spectrum Types

5G Spectrum	Features
Low-band	• Lower than 1 GHz on the spectrum chart. • Wider coverage but limited improvement in speed and latency. • For thinly populated areas and provides indoor coverage in built-up areas.
Mid-band	• 1 GHz - 6 GHz range. • Ideal for 5G because it can carry plenty of data while also traveling significant distances.
High-band	• 24 GHz band and higher. • It delivers super-fast speeds over short distances.

5G technology

Spectrum	Uses/Advantage	Limitations
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5G technology

Remember : What is the problem of having a high frequency communication band?

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(inadequate optical fiber infrastructure and backhaul, interrupted power supply, etc)



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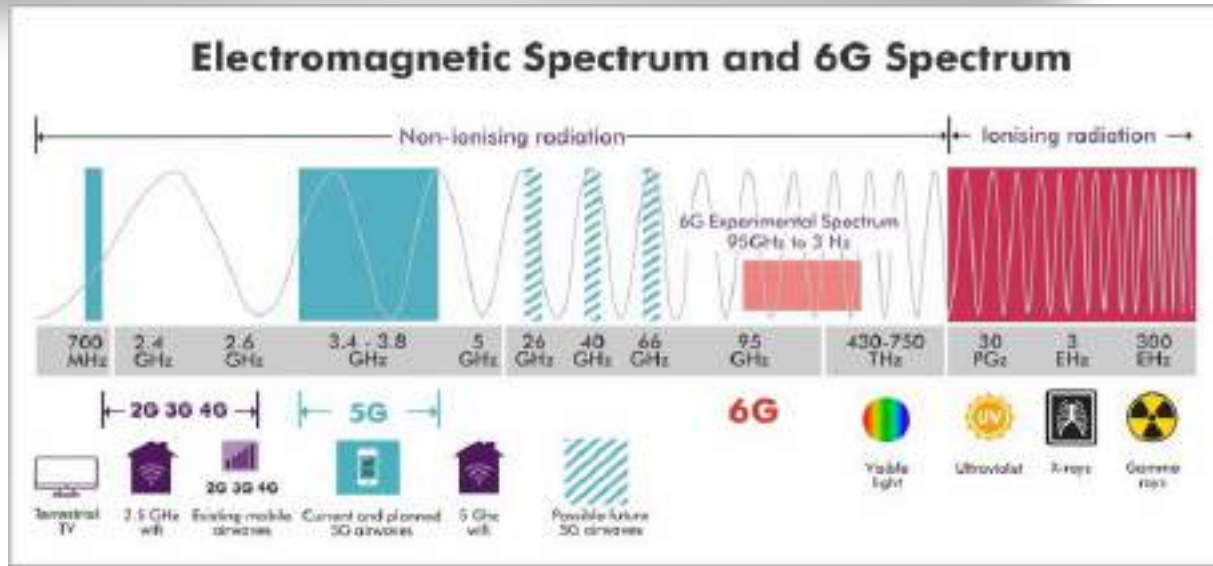
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5G and EM spectrum



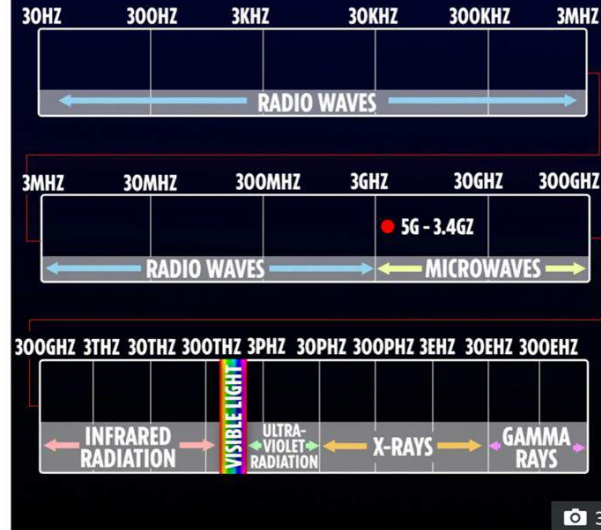
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ELECTROMAGNETIC SPECTRUM



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5G in news

- Hon'ble PM has inaugurated country's first 5G testbed.
- Multi- Institute collaborative project lead by IIT madras.
- 5G has an amalgamation of various technologies :
- Massive Multi User MIMO (multiple input multiple output) enabled network.
- Small Cell Stations : connect base stations and users.
- Mobile Edge computing.
- Beamforcing

5G : related terms

- Network Slicing : multiple networks to be created on top of a common physical infrastructure. Each 'slice' can be allocated based on the specific needs of the application, use case or customer.
- Fiberisation : Connecting radio towers with each other via optical fiber cables. Connects core of network to edge facilitating large amounts of data used in 5G services.
- 5G airwave interference : 5G uses the same spectrum as aircraft radio altimeters leading to interference.

5G : spectrum auction

- Auction of 5G spectrum was held recently worth ₹1.5L crore.
- It relates to the radio frequencies allocated to the mobile industry and other sectors for communication over the airwaves.
- Spectrum is a sovereign asset.
- Use of mix of low band, mid band and high band spectrum to deliver tailored 5G experience.

5G : open radio access network

- Centre for development of telematics along with 2 private networks has signed agreement for developing 5G open RAN.
- RAN : part of network that connects main network infrastructure to end users.
- Connects individual devices to other parts of a network through radio connections. It acts as a bridge to access all key applications on the web.
- Current RAN is integrated platform of both hardware and software, therefore it is difficult to mix vendors.
- Open RAN will allow vendors to mix and match components from different vendors, leading to a multi supplier RAN solution.

Long term Evolution (LTE) and Voice over LTE (VoLTE)

- VoLTE is a **technology update to the LTE protocol** used by mobile phone networks.
- **Under LTE**, the infrastructure of telecom players **only allows transmission of data while voice calls are routed to their older 2G or 3G networks.**
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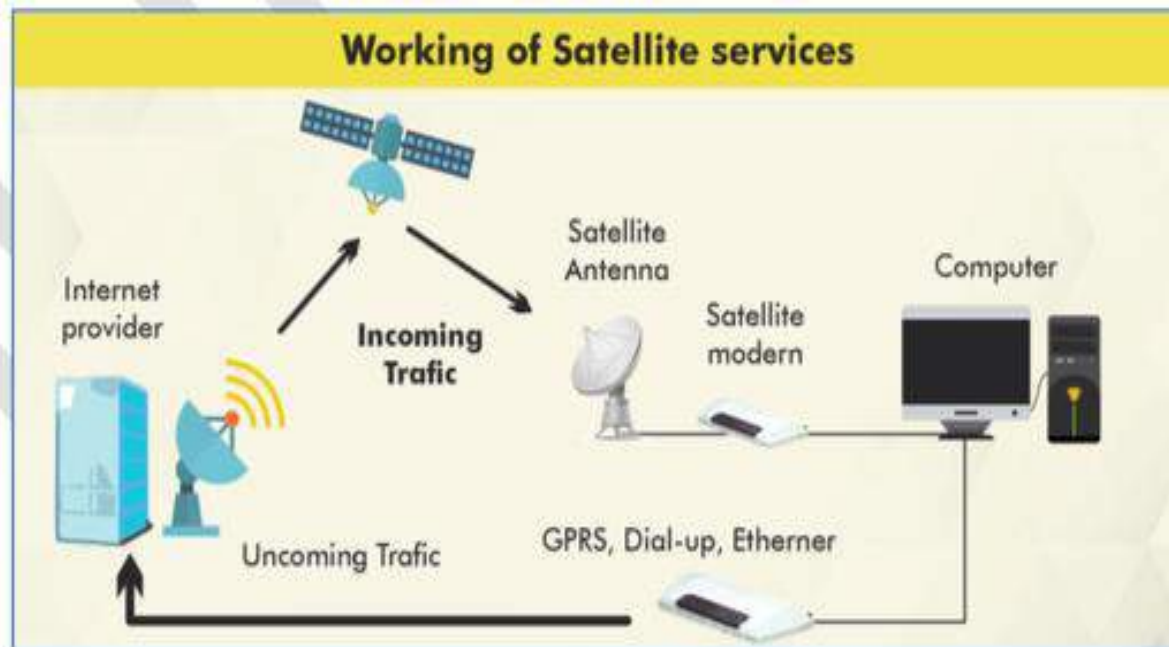


Satellite based Internet Connectivity

- Satellite Internet Services is a **wireless internet connection** which uses satellites in space to get an internet signal from the Internet Service Provider (ISP) to users.

Satellite based Internet Connectivity

- It operates using two way satellites like **VSAT (Very Small Aperture Terminal)** or telecommunication satellites to liaison between Internet Service Providers (ISPs) and Users with satellite dishes.
- **Direct to Satellite Model:** allows devices to directly communicate with the satellite without the need of any intermediate ground gateway.
- 2 types : satellite in LEO or satellite in GEO



Satellite based Internet Connectivity

- **Advantages of Space-based internet systems using satellites in LEO compared to GEO:** low Latency in signal transmission.
- **Advantages of Space-based internet systems using satellites in GEO compared to LEO:** Serviceable area on Earth by each satellite is more, easier to establish a transmission link with a satellite.

Models for Satellite based connectivity

- **Hybrid : LPWAN + Satellite** : each sensor first connects with a sink node, (LPWAN) then to Satellite.
- **Direct to Satellite** : Devices to directly communicate with satellite without need of ground gateway.



Satellite-based connectivity is possible using **Geo Stationary, Medium and Low Earth orbit Satellites**



It is **different from satellite television** as Internet signals go **both ways** with a much **greater amount of bandwidth** to move all that data



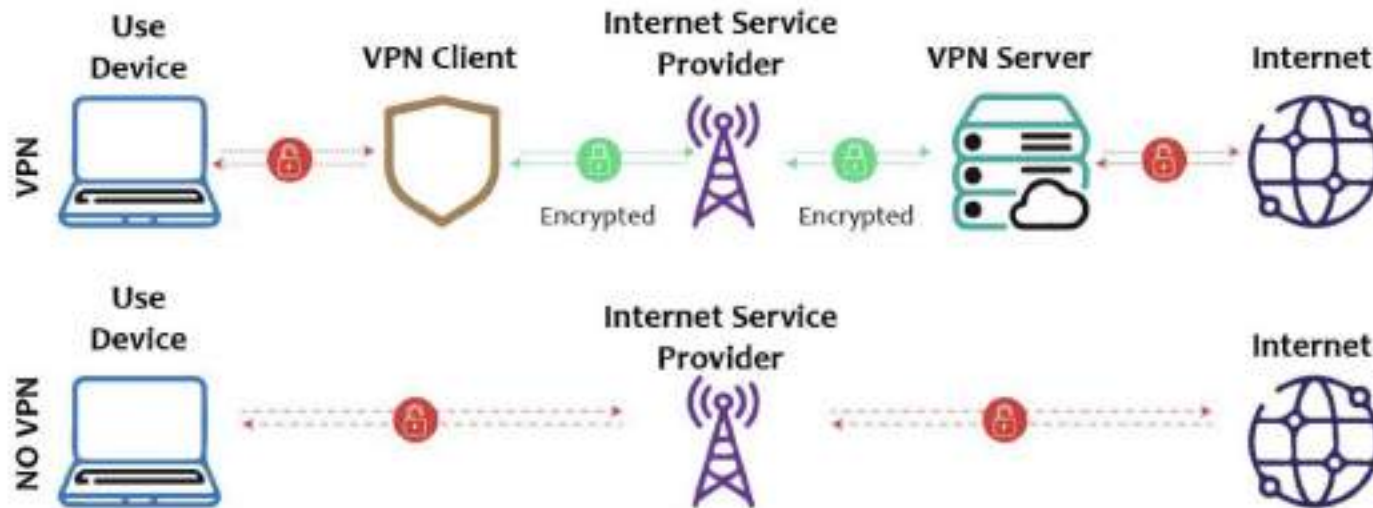
Broadband signal is sent and received to a satellite in space rather than using an underground copper/fibre network

Satellite based Internet Connectivity

Benefits of Satellite internet Services	Challenges in the Satellite internet Services
• Cheap and easy internet option for rural area, hilly terrain, and islands for last mile connectivity. • Faster than DSL connection with continuously improving technology with present speeds reaching around 100 mbps. • Can play a significant role in emergency or disaster recovery due to limited ground infrastructure.	• Low speed and high latency - In comparison to fiber optic cables, • High cost as compared to wireless cellular network and others. • Limitations of Bandwidth with high impact of weather aberrations • Potential to damage spacecrafts/satellites or interrupt of higher satellites frequency because of its large satellite network need, • Potential of increase space junk. • A VPN gives online privacy anonymity by creating a private network from a public internet connection

Virtual Private Networks

- A VPN is a computer network across a public internet that provides user access to their organization's network while maintaining the security of information provided
- It gives online privacy and anonymity by creating a private network from a public internet connection.



Types of VPN



Secure Sockets Layer (SSL) VPN

It enables individual users to **access an organization's network** and client-server applications **without the need for specialized software**



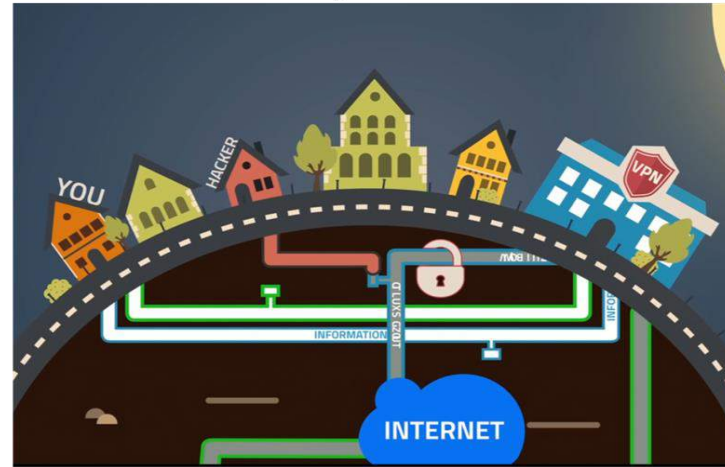
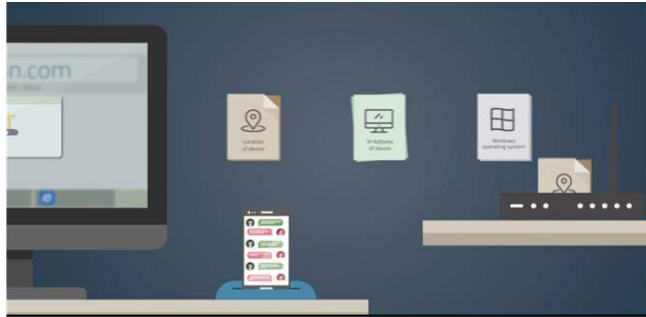
Site-to-site VPN

A **private network designed to hide private intranets** and allow users of these secure networks to access each other's resources.



Client-to-Server VPN

User is not connected to the **internet via his own ISP** (internet service provider), but establishing a **direct connection through VPN provider**.



Dark Net

- Imagine you get an email containing your bank statement :
- It is on the internet
- It is not accessible to me.
- This is the deep web, it is that part of Internet which cannot be accessed through traditional search engines like Google nor is it accessible by normal browsers like Chrome or Safari.
- Dark Net is a part of deep web

Dark Net

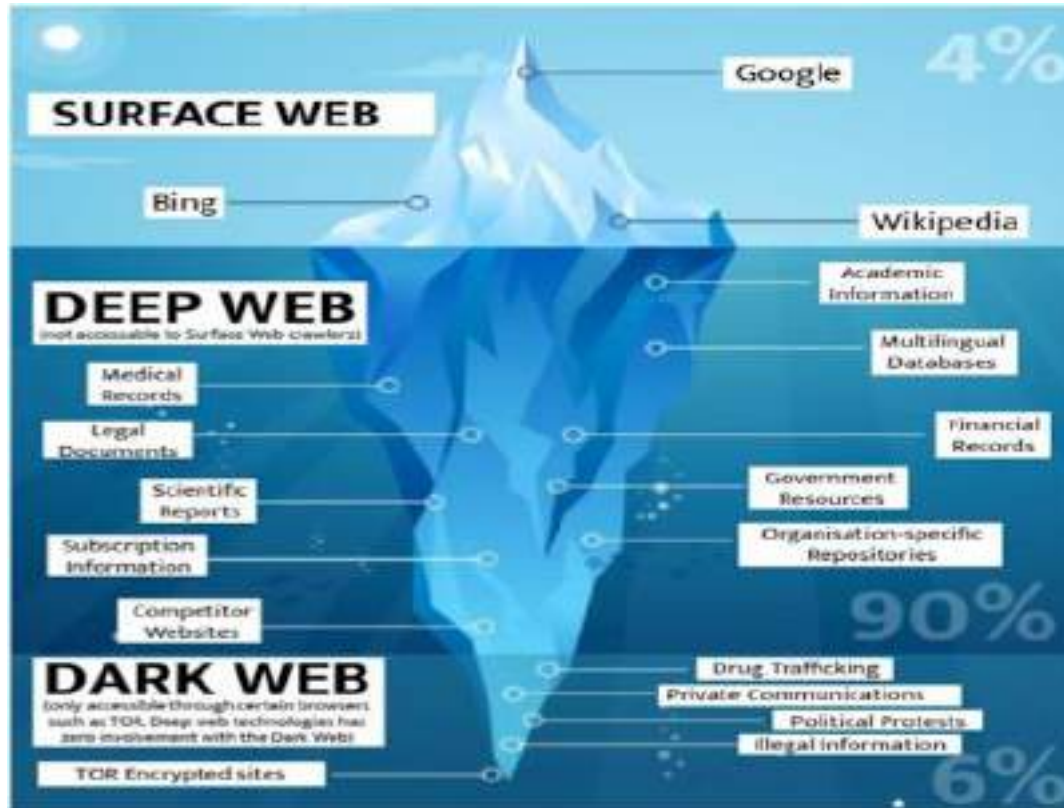
- It generally uses non-standard communication protocols which make it inaccessible to internet service providers (ISPs) or government authorities.
- Content on Dark Net is encrypted and requires specific browser such as TOR (The Onion Router) browser to access those pages.

Dark Net

- Dark Net itself is only a part of Deep Web that is a broader concept, which includes sites that are protected by passwords.
- Part of internet that is readily available to general public and searchable on standard search engines is called as Surface Web.

Dark Net

- Concerns over its use: Anonymity, Haven for illicit activity, Privacy and ethical concerns, drug dealing, communication by terrorists etc.

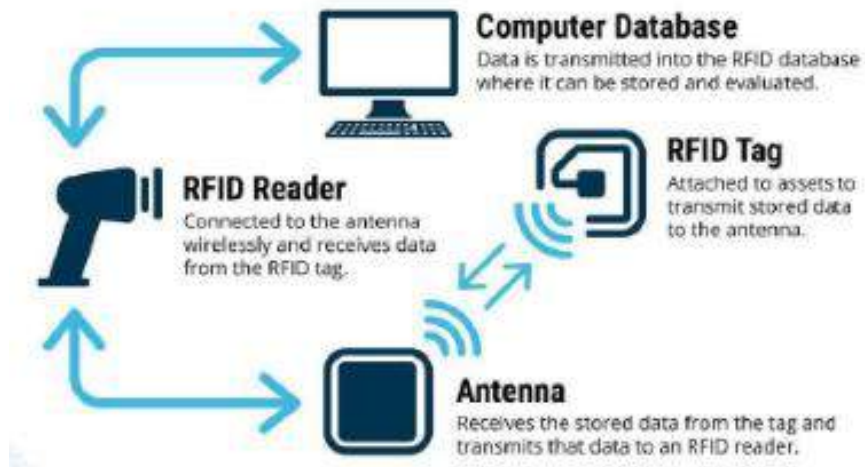


RFID

- It is an electronic tag that can exchange data with an RFID reader through radio waves.
- In this, an antenna broadcasts energy to tag, which in turn returns modulated energy back in form of back-scatter.
- It is faster connection, and operational range is about 100m.
- RFID tag typically sends out information hence only one-way information.

RFID

Basic RFID System

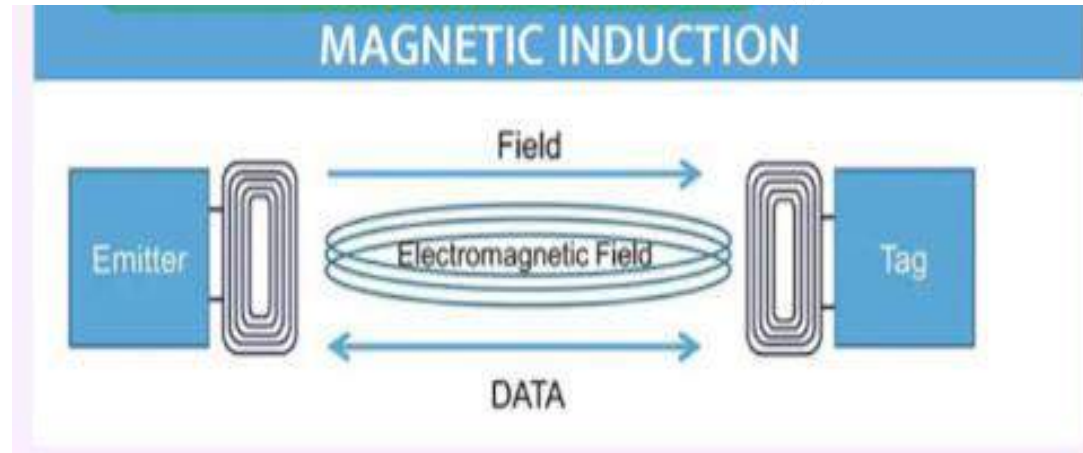


Near Field Communication

- NFC is the technology that allows compatible devices to establish radio communication with each other by touching them together or bringing them into a distance maximum of 10 cm.
- NFC works on Radio Frequency Identification (RFID) technology and Magnetic induction principle
- However, RFID and NFC also have many differences

Near Field Communication

- When electrons flow through a conductor, they create a magnetic field. And when magnetic fields change, they can cause electrons to flow through a conductor. This relationship -- known as inductive coupling.
- In NFC also the reader or the mobile acts as the active device whose antenna or the coil releases a sinusoidal carrier wave and creates a magnetic field. Then the passive tags present in that field gets energy from the reader and change the properties of the receiving frequency.



Basis	RFID	NFC
Communication	 Unidirectional—typically an RFID tag sending data to an RFID reader	 Both unidirectional and bidirectional. This allows NFC to be used for more complex interactions such as peer-to-peer (P2P) data sharing as well as card emulation.
Frequency range	RFID FREQUENCY RANGES: Low Frequency (LF): 125-134 kHz  (LF) Range: Up to 10 m High Frequency (HF): 13.56 MHz NFC  (HF) Range: Up to 30 m Ultra High Frequency (UHF): 856 MHz TO 950 MHz  (UHF) Range: Up to 100 m	
Application	 Asset tracking  Race timing  Attendee tracking  Inventory management	 Contactless payment  Information/data sharing  e-store check-ins  wearables like smartwatch  Post-sale product experiences (such as scanning QR codes on posters for additional product information)

	Radiofrequency Identification (RFID) 	Quick Response (QR) code 	Near Field Communication Technology (NFC) 
About	A wireless system comprised of two components: tags and readers.	A two-dimensional version of a barcode that can be read easily by a digital device	A short-range wireless connectivity technology that allows NFC-enabled devices to communicate with each other and transfer information quickly and easily with a single touch
Working Range	Read distance depends upon the frequency (Ranging from 300 gigahertz to as low as 9 kilohertz) used	Ideal scanning size to distance ratio is 1:10.	NFC-enabled devices must be either physically touching or within a few centimeters from each other for data transfer.
Other Key Information	Do not require a direct line of sight to be read. - Data stored in an RFID tag can be updated in real-time.	- Require a line of sight to scan. No real-time tracking.	Transfer information quickly and easily with a single touch.

Other Similar Technology

- Infrared : short connection and transmission of information through infrared light.
- Needs line of sight between 2 devices.
- Ex : Remote control devices.

Edge Computing

- It enables data to be analysed, processed, transferred at edge of a network.
- It means, data is analysed locally, closer to where it is stored, in real-time without latency.
- Basic difference between edge computing and cloud computing lies in where data processing takes place

Edge Computing

- Cloud Computing: It means storing and accessing data and programs over Internet instead of your computer's hard drive.
- Existing Internet of Things (IoT) systems perform all their computations in cloud using data centres.
- Edge Computing: It manages massive amounts of data generated by IoT devices by storing and processing data locally.

Edge Computing

Benefits :

- High speed
- Reduced latency
- Better reliability and security : difficult for any single disruption to take down a network
- Very important in upcoming internet of things technology

4th IR

- 1st IR : ?
- 2nd IR ?
- 3rd IR ?
- 4th IR :
- Building on the 3rd Industrial Revolution
- Includes cyber-physical systems, the Internet of things, big data analytics, cloud computing, cognitive computing, artificial intelligence, 3-D printing, and autonomous vehicles etc.
- Impact : all of the above

Big Data

- Big data is a term applied to datasets whose size or type is beyond the ability of traditional relational databases to capture, manage and process the data with low latency.
- Big data analytics is the use of advanced analytic techniques against very large, diverse data sets that include structured, semi-structured and unstructured data, from different sources, and in different sizes from terabytes to zettabytes.



Big Data : Advantages

- Improves the decision making of businesses
- Protects company and client information
- Effective marketing and targeted advertising
- Cost and time reduction through streamlining operations
- Better Products designing by getting info and analysis
- Applications in various sectors :
 - Banking : NPA prevention and trend analysis
 - Education : identify at risk students etc.
 - Government : traffic congestion and utilities management
 - Health care : insights to improve patient care
 - Manufacturing : boost quality output while minimizing waste processes
 - Agriculture : Optimizing crop efficiency

Big Data

- Challenges :
- Human resource : lack of data scientists
- Privacy Issues : reveal sensitive and personal information of users
- Security issues and outsourcing of data analytics
- Lack of availability of quality data
- Ethics and issues of informed consent of users

Additive Manufacturing (3d printing)

- Three-dimensional printing (3D printing), also known as Additive manufacturing (AM) is a process of joining materials to make objects from 3D model data, usually layer upon layer.
- Different from conventional manufacturing
- It is a process in which a three-dimensional object is built from a computer-aided design (CAD) model, usually by successively adding materials in a layer-by-layer fashion.
- The addition of material can happen in multiple ways, namely power deposition, resin curing, filament fusing.

Additive Manufacturing (3d printing)

- Thermoplastics, Metals, Ceramics as well as biomaterials can be used for additive manufacturing.
- Traditional vs. AM: Unlike AM, traditional manufacturing methods, by contrast, are subtractive in nature.
- Subtractive manufacturing involves removing parts of a block of material in order to create the desired shape. Cutting wood into useful shapes, for instance, is a very simple example of a subtractive process.

3D printing

Basic Additive Manufacturing Process

CAD Model.....3D Object



3D Card Model



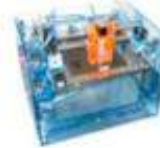
.STL File



Slicing Software



Layer Silces & Tool Path



AM Process



3D Object

Benefits

- Low cost
- Less time
- Efficiency
- Customization
- Increased Productivity
- Reduced Wastage

Issues

- Size is limited
- Limited raw materials
- Employment destruction
- Copyright infringement concerns
- Dangerous items can be produced

Artificial Intelligence

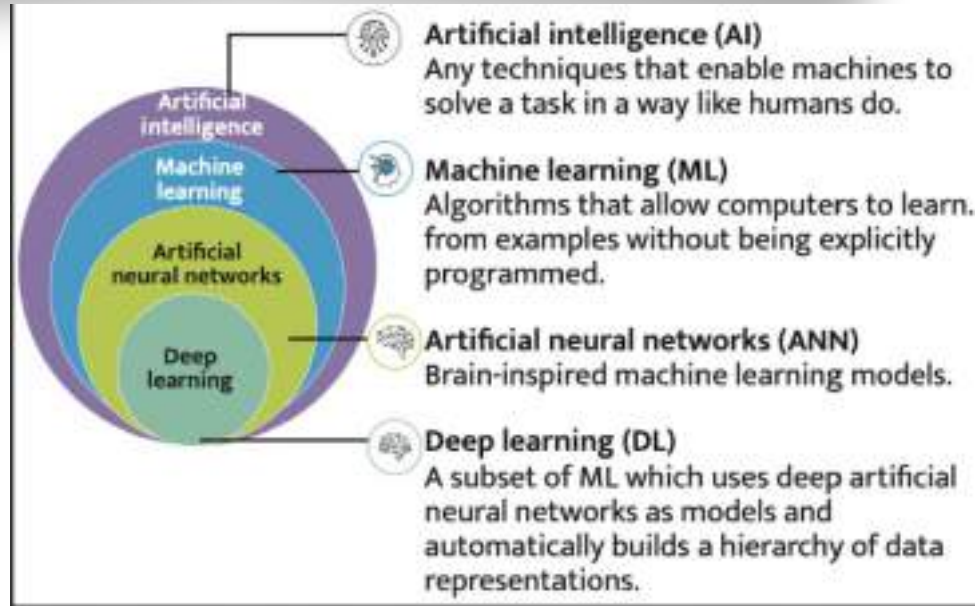
Simulation of Human Intelligence processes by machines

- Cognitive skills
 - Learning
 - Reasoning
 - Self correction

Artificial Intelligence

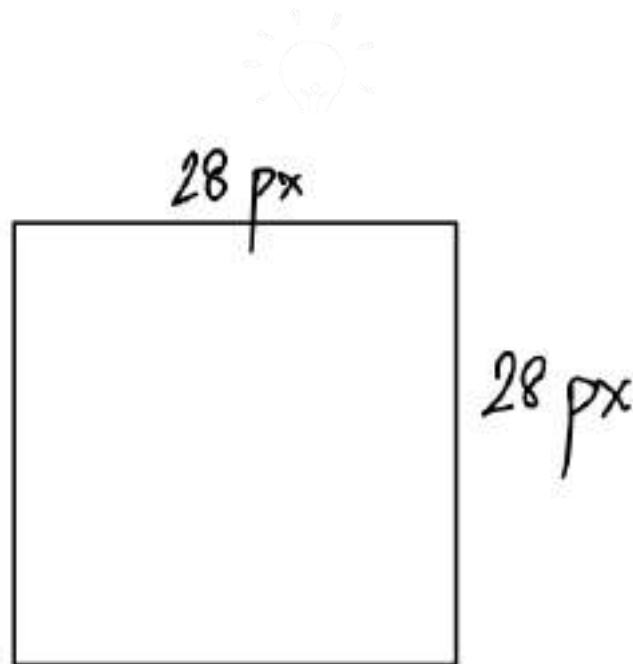
- **Learning** : Acquire data of create rules : Algorithms
- **Reasonings** : Choosing the right algorithm
- **Self Correction**: Five time algorithms to ensure most accurate result

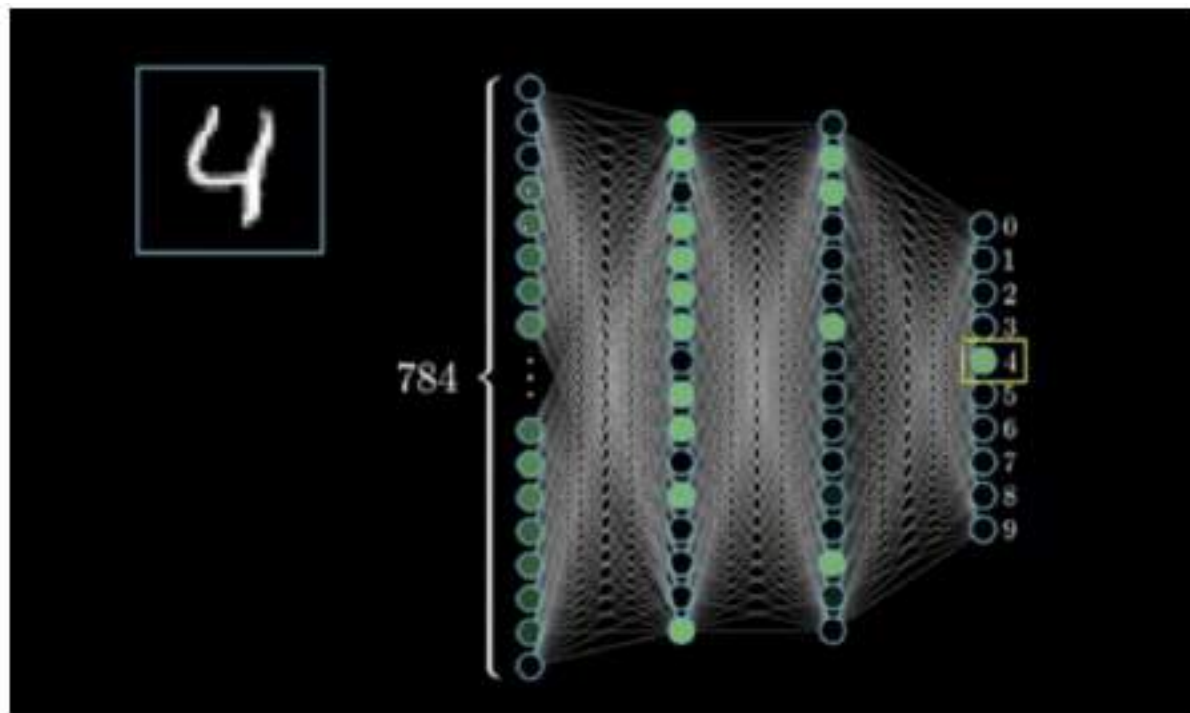
Artificial Intelligence

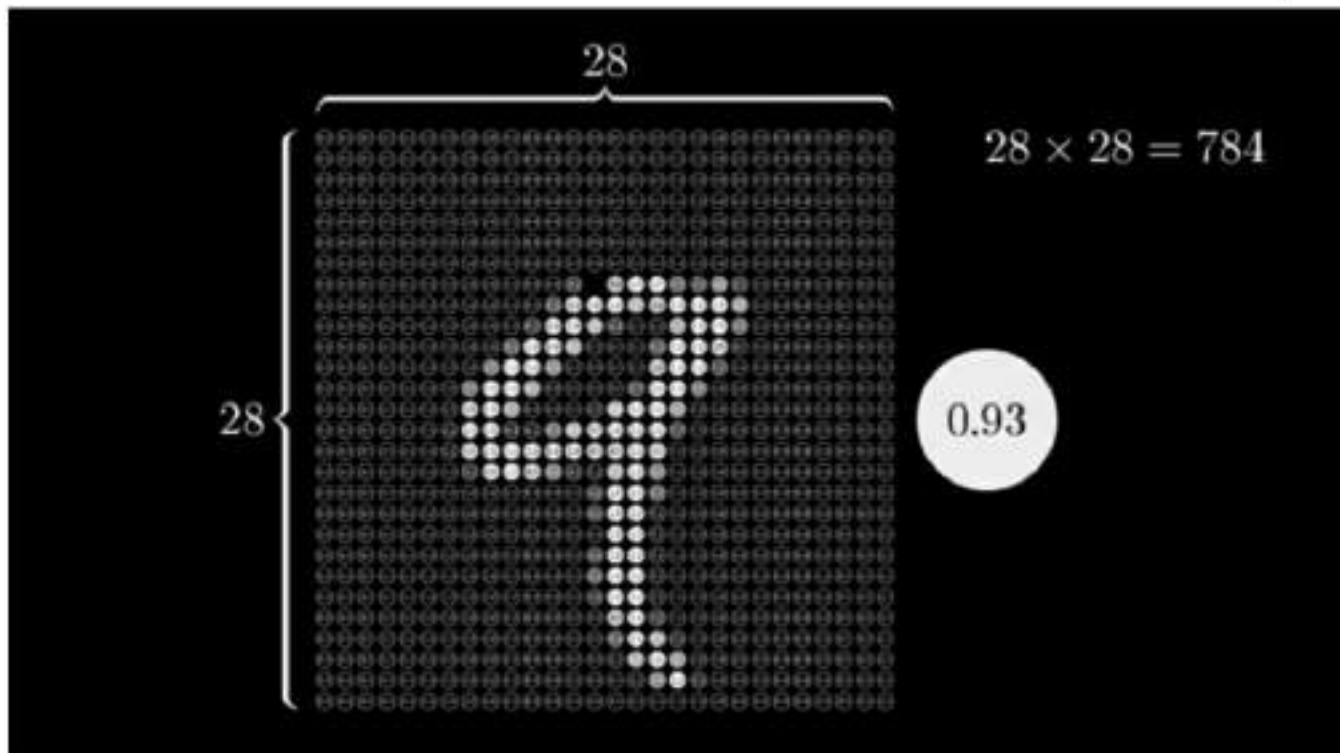


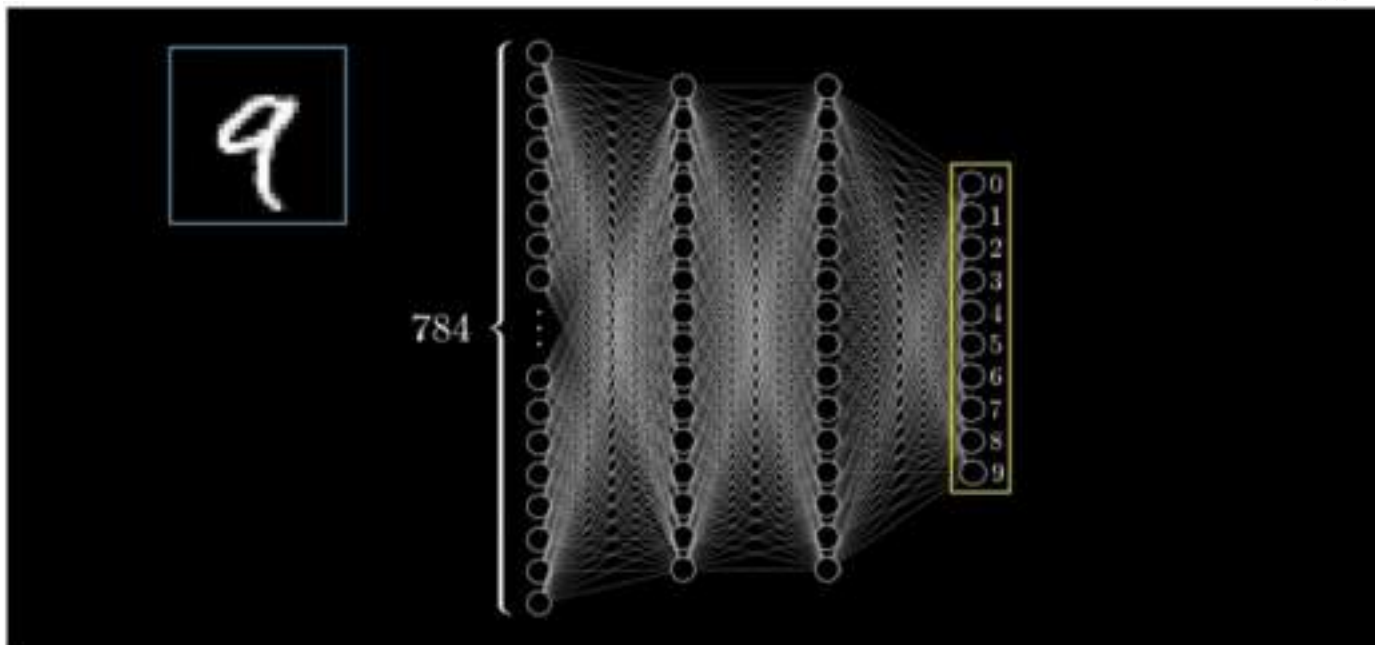
Artificial Intelligence

- **Understanding Intelligence**
 - **Neural Network**
 - **Inspired by the brains.**
 - **Computer learns the way we do.**



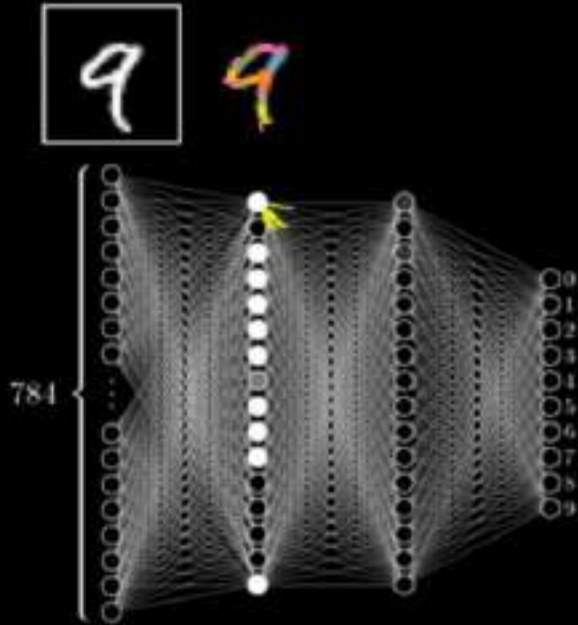


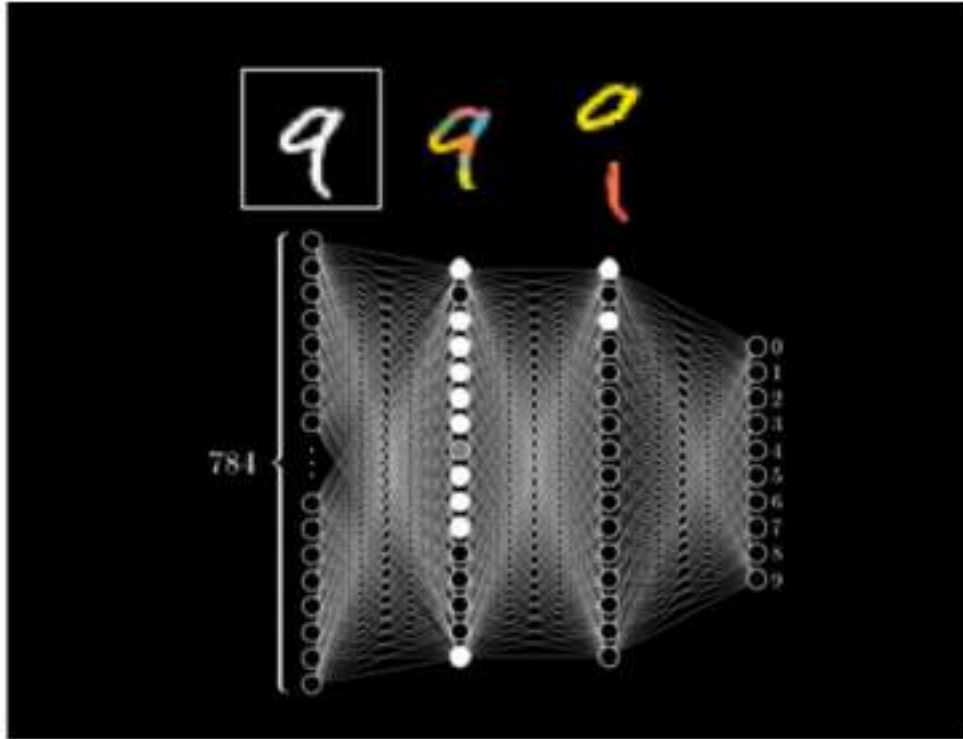


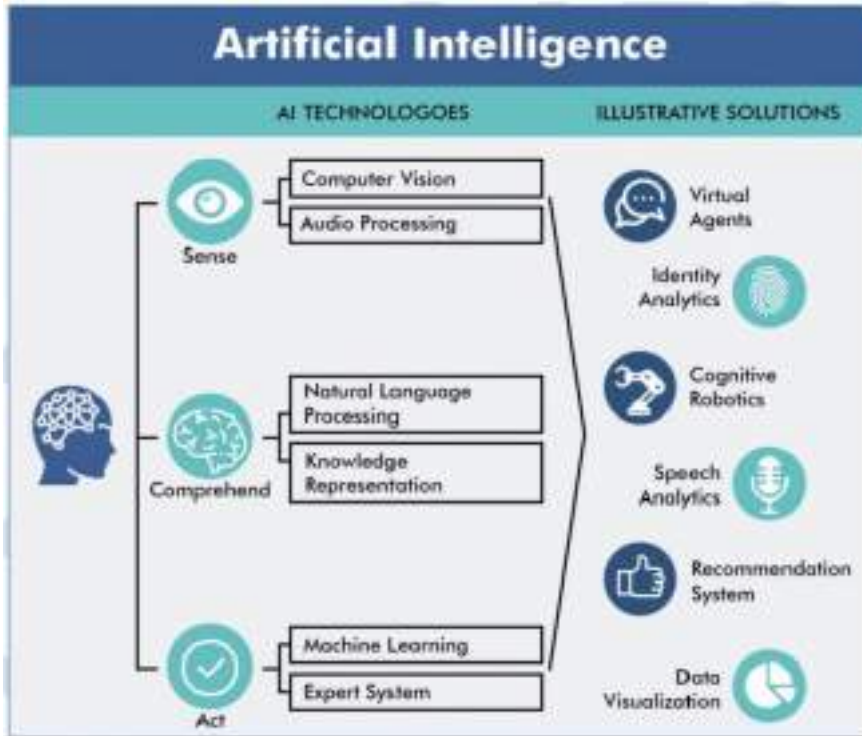


$9 = 0 + 1$
 $8 = 0 + 0$
 $4 = 1 + 1 + -$









Generative AI



- Type of artificial intelligence that focuses on creating new and original content that is indistinguishable from human-created content
- Uses algorithms and machine learning techniques to generate everything from images and music to text and video
- Trained on large datasets of existing content, such as images, text, music and videos

Type of Generative AI frameworks



- Generative Adversarial Networks (GANs): Uses two neural networks generator and a discriminator. The generator produces new data based on a given set of data, while the discriminator evaluates the data, determining if it is real or generated.
- VAEs: Uses an encoder-decoder architecture. The encoder encodes input data into a latent space, and the decoder decodes it back into the original data.
- Transformer Based Models: Read words and sentences in books, or articles. They learn from things they read and use that knowledge to create new things, like stories or poems.

Transformer Based Models (TBM)



- Uses a deep neural network.
- Trained on large dataset, allowing it to generate human like text or videos
- The transformer model consists of an encoder, which reads the input sequence of words and produces a representation of it, and a decoder, which produces the output sequence of words.
- Used for a wide range of tasks, question answering, text summarization, writing poems and stories.
- Programs based on TBM: OpenAI ChatGPT, Google BARD AI, Dall-E
- **Uses natural language processing (NLP) to understand and generate text data**

Transformer Based Models (TBM)



- Many words are synonyms of each other.
- **To understand words that are similar, to remove words that don't carry a substantive meaning and to simplify sentences this model uses NLP.**
- **So NLP helps the model be trained better.**

Natural Language Processing (NLP)



- Used to analyze and understand natural language text data
- Used by OpenAI ChatGPT, Google Bard AI to understand the data
- Steps involved:
 - Segmentation/ Tokenization: Breaking up a sentence or text into individual units, such as words, numbers. Each unit is referred to as a token.

0 1 2 3 4 5 6
I am learning about ChatGPT right now

Natural Language Processing (NLP)



- Removing Stop words: Removing words that don't contribute to the meaning of the sentence
This way it understands that 2 sentences have the same meaning for better pattern analysis.

0	1	2	3	4	5	6
i	am	learning	about	chatgpt	right	now

Natural Language Processing (NLP) - Cont'd



- Stemming: Removing the suffix or prefix of a word to obtain its root form.

0	1	2	3	4
i	am	learn	ing	chatgpt now

Natural Language Processing (NLP) - Cont'd



- Lemmatization: Normalizing a word to its base form, such as mapping a verb to its infinitive form.

0	1	2	3	4
i	be	learn	chatgpt	now

Natural Language Processing (NLP) - Cont'd



- Part of Speech Tagging: Assigning a part of speech to each token, such as noun, verb, adverb, etc.

0	1	2	3	4
i	be	learn	chatgpt	now
<i>pronoun</i>	<i>auxiliary verb</i>	<i>verb</i>	<i>noun</i>	<i>adverb</i>

Natural Language Processing (NLP) - Cont'd



- Named Entity Recognition: Identifying and labeling entities, such as people, organizations, locations, dates, etc.

Ousted **WeWork** founder **Adam Neumann** lists his **Manhattan** penthouse for **\$37.5 million**

[organization] [person] [location] [monetary value]

CHAT GPT

- **Chat Generative Pre-trained Transformer** is an artificial intelligence chatbot developed by OpenAI and launched in November 2022.
- OpenAI has created ChatGPT, a start-up focused on artificial intelligence and its potential use cases.
- According to OpenAI's description, ChatGPT can answer “follow-up questions”, and can also “admit its mistakes, challenge incorrect premises, and reject inappropriate requests.”

CHAT GPT

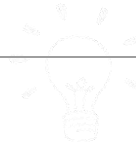
- It is based on the company's GPT 3.5 series of **Language Learning Models (LLM)**.
- GPT stands for **Generative Pre-trained Transformer 3** and this is a kind of *computer language model* that relies on **deep learning techniques** to produce human-like text based on inputs.
- The model is trained to predict what will come next, and that's why one can technically have a 'conversation' with ChatGPT.

CHAT GPT

- chatbot was also trained using “**Reinforcement Learning from Human Feedback (RLHF)**.”
- Interestingly, ChatGPT has been trained to decline ‘inappropriate’ requests, presumably those which are ‘illegal’ in nature. However, it should be noted that ChatGPT has limitations, as it may generate incorrect information, and create “biased content.” More importantly, the chatbot’s knowledge of the world and events after 2021 is limited.
- Other products of Open AI
 - **Dall-E** — the AI-based text-to-image generator

CHAT GPT

- Google had recently showcased how its **LaMDA chatbot** is being used to help with fiction writing
 - LaMDA is a machine learning language model created by Google as a chatbot that is supposed to mimic humans in conversations
 - LaMBDA is built on neural network architecture



Human-AI Collaboration

Supportive Mode	Reimagining the Business	Ai Replaces the Humans
AI performs alongside humans by facilitating human judgments by providing resources such as predictive outputs.	AI performance activities that go beyond the cognitive abilities of humans. For e.g., large-scale genome study in bioinformatics.	Useful in potentially harmful situations for humans such as environments and rapid system response in nuclear reactors.

Benefits and applications of AI



Healthcare

- Address issues particularly in rural areas that suffer from poor connectivity and limited supply of healthcare professional .
- This can be achieved through AI driven diagnostics, early identification of potential pandemics, and imaging diagnostics.

Education and skilling

- Solve problem of quality and access issues observed in the Indian education sector.
- Augmenting and enhancing the learning experience through personalised learning, automating and expediting administrative tasks.

Benefits and applications of AI



Agriculture

- Driving a food revolution to meet the increased demand for food by improvement in crop yield.
- Address challenges in irrigation, misuse of pesticides and fertilisers through real time advisory.

Manufacturing

- Manufacturing industry is expected to be one of the biggest beneficiaries of AI based solutions in field of engineering, supply chain management, production and quality assurance etc.

Benefits and applications of AI



Energy

- AI can enable storage of energy through intelligent grids enable by smart meters, and also improve the reliability and affordability of photovoltaic energy.
- AI may also be deployed for predictive maintenance of grid infrastructure.

Financial Services

- AI can ensure early detection of financial risk and systemic failures, and automation of reduce malicious intent in financial system, such as market manipulation, anomalous trading etc.

Benefits and applications of AI



Defence and Security

- AI can be used for intelligence gathering, cyber defence, risk terrain analysis, detecting anomalous behaviour in human etc.

Law Enforcement

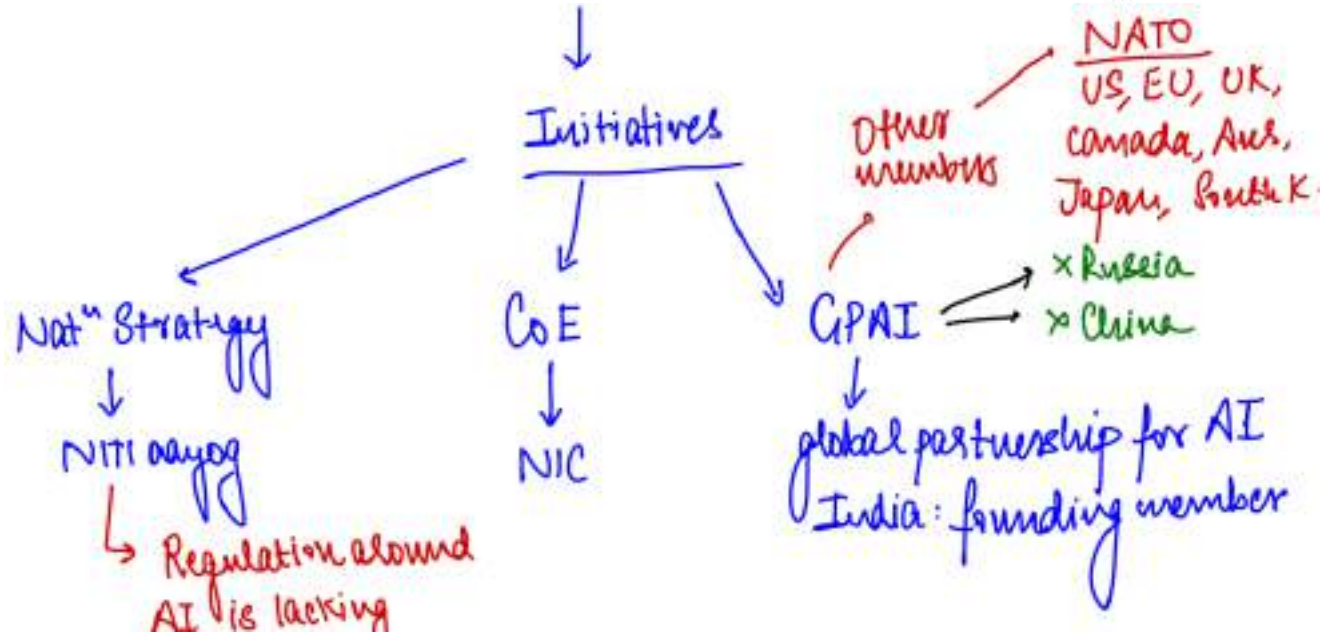
- AI technologies can be used by law enforcement which include facial recognition, speech recognition, predictive analytics etc.

Disaster Management

- AI has potential in providing remedial measures and control in aftermath of man-made recognition can be used in infrastructure damage assessments.

Issue	
General Context	● Indian Context ● Awareness regulation ● Funding ● HR ● Infra.

Indian Scenario



Introducing Bard, an experimental conversational AI service powered by LaMDA

You can use Bard to —

Plan a friend's baby shower

Compare two Oscar nominated movies

Get lunch ideas based on what's in your fridge

What's on your mind?

It may give inaccurate or inappropriate information. Your feedback matters. It and more helpful over time.

E-SIM

- an eSIM is an embedded SIM
 - It has same hardware of a regular SIM card chip
 - It is a permanently embedded part of the motherboard of a device.
- e-SIM acting as a unique identifier for telecom operators to reach the exact smartphone when they make a call or send a text.
- eSIMs were first established a decade ago in 2012.

E-SIM

Benefits

- users switch operators without having to replace any physical SIM cards.
- **Convenience:** The ability to store multiple SIM profiles in your eSIM also means you can switch between profiles easily, without the need of activating a SIM repeatedly or physically switching cards repeatedly.

E-SIM

Benefits

- **Security:** In case of theft of phone, SIM card can not be misused as it is embedded part of smart phone.
- **One less opening on your phone:** Having an eSIM also means there is one less opening on the frame of your phone, in theory, which should reduce the likelihood of elements like dust and water entering the phone from yet another slot.

E-SIM

Disadvantages

- **Emergencies:** If your phone stops working, runs out of battery or simply, falls and gets a cracked screen, your communication is brought to a complete standstill with eSIMs.
- **Unusable in countries with no eSIM support:** You cannot use an eSIM phone in a country where the telecom operators simply don't support the technology yet.
- **Support only available in premium phones:** In India, eSIM support is currently available on more expensive devices.

D2M Technology

- Direct To Mobile Tech—>broadcast video and other forms of multimedia content directly to mobile phones
- It does not require an active internet connection

D2M Technology

Concept

- D2M is based on the convergence of broadband and broadcast, using which mobile phones can receive terrestrial digital TV.
- It would be similar to how people listen to FM radio on their phones, where a receiver within the phone can tap into radio frequencies.
- Using D2M, multimedia content can also be beamed to phones directly.

D2M Technology

Benefits

- It can be used to directly broadcast content related to citizen-centric information
- It can be used to counter fake news, issue emergency alerts and offer assistance in disaster management
- It can be used to broadcast live news, sports etc. on mobile phones.

D2M Technology

Impact

- Consumers
 - Can access multimedia content from Video on Demand (VoD) or Over The Top (OTT) content platforms without having to exhaust their mobile data at a nominal rate.
 - The technology will also allow people from rural areas, with limited or no internet access, to watch video content.

D2M Technology

Businesses

- It can enable telecom service providers to offload video traffic from their mobile network onto the broadcast network, thus helping them to **decongest valuable mobile spectrum**.
- This will also **improve usage of mobile spectrum** and free up bandwidth which will help reduce call drops, increase data speeds etc.

D2M Technology

Government Measures

- The Department of Telecommunications (DoT) has set up a committee to study the feasibility of a spectrum band for offering broadcast services directly to users' smartphones

D2M Technology

Challenges

- bringing key stakeholders like mobile operators onboard will be the “biggest challenge” in launching D2M technology on a wide scale.
- mass roll out of the technology will entail changes in infrastructure and some regulatory changes.

CrysXPP

- Researchers from **IIT Kharagpur** in collaboration with the **Indo-Korea Science and Technology Center (IKST)** have developed a method called **CrysXPP**.
- It is a **machine learning (ML) system** that enables rapid prediction of various **material properties** with high precision.

CrysXPP

- ML is a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed to do so
- Machine learning algorithms use historical data as input to predict new output values
- Applications of ML: Recommendation engines, fraud detection, spam filtering, malware threat detection, business process automation (BPA) and Predictive maintenance.

CrysXPP

Applications

- **CrysXPP made large-scale testing easier**, because until now, it has been difficult to test crystalline materials on a large scale.
- It also helps determine the electronic, **magnetic and elastic properties of crystals**, which are often time- consuming, expensive and computation-intensive.

Web 5.0

- Former Twitter CEO recently announced his vision for a new decentralized web platform that is being called Web 5.0.
- Objective—> “ownership of data and identity to individuals”.
- It is a **Decentralized Web Platform** that enables developers to leverage Decentralized Identifiers, Verifiable Credentials, and Decentralized Web Nodes to write Decentralized Web Apps.
- Web 5.0 is Web 2.0 plus Web 3.0 that will allow users to own their identity on the Internet and control their data.
- It aimed at building an extra decentralized web that puts one in control of your data and identity.

Web 5.0

Web 1.0 : Read only web

Web 2.0 : Social Web

Web 3.0 : Semantic Web

Web 5.0

Web 4.0 (Mobile Web)

- It is already a work in progress, with no clear understanding of what it would entail.
- It is often referred to as the **symbiotic web**.
 - It will make contact between humans and computers in symbiosis.

Geospatial Data

- **The Ministry of Science and Technology** has notified a citizen-centric **National Geospatial Policy (NGP) 2022**.
- **Geospatial Data Promotion and Development Committee (GDPDC):** At national level, an apex body for formulating and implementing appropriate guidelines, strategies, and programs.
 - **GDPDC would replace the National** Spatial Data Committee (NSDC) and Geospatial Data Promotion and Development Committee.

Geospatial Data

Geospatial Data Infrastructure

- GDPDC will **develop 14 global geospatial data themes** recognized by the United Nations Statistics Division.
- **National Geospatial Data Registry (NGDR):** Accessible by all stakeholders.
- **Unified Geospatial Interface (UGI):** To provide consumer-oriented products and solutions using Geospatial data.
- **Survey of India (SoI):** Will help in developing NGDR and UGI in collaboration with Bhask`atics (BISAG-N)-under MeitY, private sector etc.

Geospatial Data

- **Mapping infrastructure:** Policy shall replace the National Map Policy, 2005.
- **Geospatial Knowledge Infrastructure (GKI):** Will be enabled by the integration of Geospatial data with Fourth Industrial Revolution technologies and the growing digital infrastructure.
- **Geospatial Education and Skill Development:** National Institute for Geo-informatics Science and Technology (NIGST), Indian Institute of Remote Sensing (IIRS), and/or any suitable institute(s), public or private, will be developed into Centre(s) of Excellence.
- **Geospatial Industrial Development Board (GIDB):** Advisory body under GDPDC.

National Data Governance Policy

Aim—>

- To **enhance access, quality, and use of data**
- It will **provide datasets access to Indian researchers and startups.**
- **NDGFP does not apply to the private players**, but they can voluntarily contribute their datasets.
- No provision for monetization of such data.

National Data Governance Policy

Components

- **Indian Datasets Platform:** Consisting of anonymized non-personal datasets from Government entities collected from Indian citizens or those in India
- **Indian Data Management Office (IDMO)**
 - It will be set up under Digital India Corporation (DIC), under MeitY
- **Data Management Units (DMUs):** In every Ministry/Department to work closely with IDMO for ensuring implementation of the Policy

National Data Governance Policy

Objectives of the NDGFP are:

- To accelerate Digital Governance.
- To have standardized data management and security standards across whole of Government;
- To accelerate creation of common standard based public digital platforms while ensuring privacy, safety, and trust; (d) To have standard APIs and other tech standards for Whole of Government Data management and access.

National Data Governance Policy

Objectives of the NDGFP are:

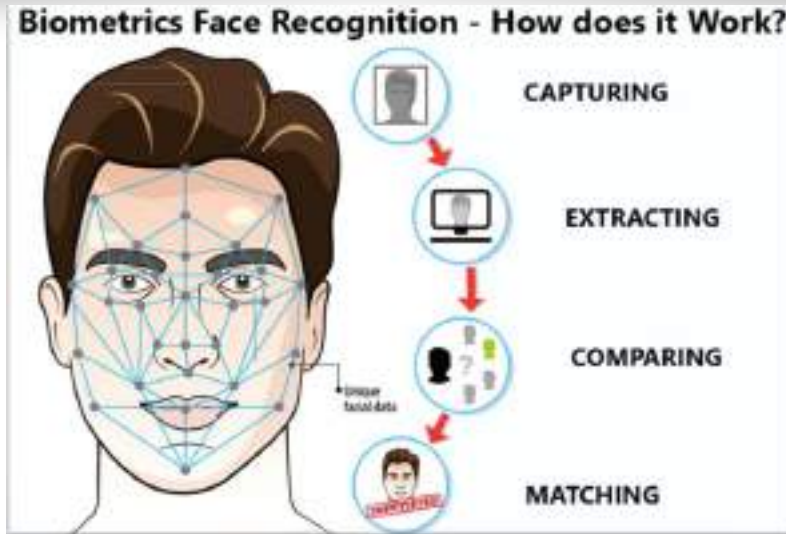
- To promote transparency, accountability, and ownership in Non-personal data and Datasets access. For purposes of safety and trust, any non-personal data sharing by any entity can be only via platforms designated and authorised by IDMO. (1) To build a platform that will allow Dataset requests to be received and processed.

National Data Governance Policy

Objectives of the NDGFP are:

- To build Digital Government goals and capacity, knowledge and competency in Government departments and entities.
- To set quality standards and promote expansion of India Datasets program and overall non-personal Datasets Ecosystem.
- To ensure greater citizen awareness, participation, and engagement.

Facial Recognition System



Facial Recognition System

- FRS is a way of identifying or confirming an individual's identity using their face.
- It can be used to identify people in photos, videos, or in real-time.
- Computer algorithms map unique **facial-landmarks** such as shape of cheekbones, contours of lips etc. and convert these into a numerical code— termed a **faceprint**.

Facial Recognition System

Applications

- FRS is part of the **Digi Yatra initiative** to promote seamless and hassle-free experience at airports and simultaneously improving the security.
- Government is looking to make ticket booking, airport entry and boarding pass security check-in digital.

DIFFERENT TYPES OF BIOMETRICS:



Typing Style



Navigation Style

Behavioral

Biometric Identifiers:



Interaction Style



Physical Style



Signature



Face Recognition

Physical

Biometric Identifiers:



Eye Scanners



Fingerprints



Voice Recognition



DNA

Chemical & Vein

Biometric Identifiers:



Vein Recognition

India Stack

- India stack is a collection of open APIs (**Application Programming Interface**) and digital public goods
- IndiaStack includes **APIs of Aadhaar, Unified Payment Interface (UPI), Co-Win, DigiLocker, Aarogya Setu, eSanjeevani, UMANG, DIKSHA**, etc
- Its objective is to unlock the economic opportunities of **identity, data, and payments** at population scale.

Remaining Terms

- **Digital Signature**
- **Software as a service**

Test of 20 Questions!

Cut off : 15/20



Q.1 (2016)

Q. With reference to 'LiFi', recently in the news, which of the following statements is/are correct?

1. It uses light as the medium for high-speed data transmission.
2. It is a wireless technology and is many times faster than 'WiFi'.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.2 (2016)

Q. 'Project Loon', sometimes seen in the news, is related to

- a) Waste management technology
- b) Wireless communication technology
- c) Solar power production technology
- d) Water conservation technology

Q.3 (2018)

Q. Consider the following pairs :

Terms sometimes seen in news

Context /Topic

1. Belle II experiment -

Artificial Intelligence

2. Blockchain technology -

Digital/Cryptocurrency

3. CRISPR — Cas9 -

Particle Physics

Which of the pairs given above is/are correctly matched ?

a) 1 and 3 only

b) 2 only

c) 2 and 3 only

d) 1, 2 and 3

Q.4 (2018)

Q. When the alarm of your smartphone rings in the morning, you wake up and tap it to stop the alarm which causes your geyser to be switched on automatically. The smart mirror in your bathroom shows the day's weather and also indicates the level of water in your overhead tank. After you take some groceries from your refrigerator for making breakfast, it recognises the shortage of stock in it and places an order for the supply of fresh grocery items. When you step out of your house and lock the door, all lights, fans, geysers and AC machines get switched off automatically. On your way to the office, your car warns you about traffic congestion ahead and suggests an alternative route, and if you are late for a meeting, it sends a- message to your office accordingly.

In the context of emerging communication technologies, which one of the following term" best applies to the above scenario?

- a) Border Gateway Protocol
- c) Internet Protocol

- b) Internet of Things
- d) Virtual Private Network

Q.5 (2018)

Q. "3D printing" has applications in which of the following?

1. Preparation of confectionery items
2. Manufacture of bionic ears
3. Automotive industry
4. Reconstructive surgeries
5. Data processing technologies

Select the correct answer using the code given below :

- | | |
|--------------------|---------------------|
| a) 1, 3 and 4 only | b) 2, 3 and 5 only |
| c) 1 and 4 only | d) 1, 2, 3, 4 and 5 |

Q.6 (2019)

Q. With reference to communication technologies, what is/are the difference/differences between LTE (Long-Term Evolution) and VoLTE (Voice over Long-Term Evolution)?

1. LTE is commonly marketed as 3G and VoLTE is commonly marketed as advanced 3G.
2. LTE is data-only technology and VoLTE is voice-only technology.

Select the correct answer using the code given below.

- | | |
|-----------------|--------------------|
| a) 1 only | b) 2 only |
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Q.7 (2019)

Q. In the context of digital technologies for entertainment, consider the following statements:

1. In Augmented Reality (AR), a simulated environment is created and the physical world is completely shut out.
2. In Virtual Reality (VR), images generated from a computer are projected into real-life objects or surroundings.
3. AR allows individuals to be present in the world and improves the experience using the camera of smart-phone or PC.
4. VR closes the world, and transposes an individual, providing complete immersion experience.

Which of the statements given above is/are correct?

- | | |
|-----------------|------------|
| a) 1 and 2 only | b) 3 and 4 |
| c) 1, 2 and 3 | d) 4 only |

Q.8 (2019)

Q. Consider the following statements: A digital signature is

1. an electronic record that identifies the certifying authority issuing it
2. used to serve as a proof of identity of an individual to access information or server on Internet
3. an electronic method of signing an electronic document and ensuring that the original content is unchanged

Which of the statements given above is/are correct?

- | | |
|-----------|-----------------|
| a) 1 only | b) 2 and 3 only |
| c) 3 only | d) 1, 2 and 3 |

Q.9 (2019)

Q. In the context of wearable technology, which of the following tasks is/are accomplished by wearable devices?

1. Location identification of a person
2. Sleep monitoring of a person
3. Assisting the hearing impaired person

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Q.10 (2020)

Q. Consider the following activities:

1. Spraying pesticides on a crop field
 2. Inspecting the craters of active volcanoes
 3. Collecting breath samples from spouting whales for DNA analysis
- At the present level of technology, which of the above activities can be successfully carried out by using drones?
- a) 1 and 2 only
 - b) 2 and 3 only
 - c) 1 and 3 only
 - d) 1,2 and 3

Q.11 (2020)

Q. In India, the term “Public Key Infrastructure” is used in the context of

- a) Digital security infrastructure
- b) Food security infrastructure
- c) Health care and education infrastructure
- d) Telecommunication and transportation infrastructure

Q.12 (2020)

Q. With the present state of development, Artificial Intelligence can effectively do which of the following?

1. Bring down electricity consumption in industrial units.
2. Create meaningful short stories and songs.
3. Disease diagnosis.
4. Text-to-Speech Conversion.
5. Wireless transmission of electrical energy.

Select the correct answer using the code given below:

- | | |
|-----------------------|---------------------|
| a) 1, 2, 3 and 5 only | b) 1, 3 and 4 only |
| c) 2, 4 and 5 only | d) 1, 2, 3, 4 and 5 |

Q.13 (2020)

Q. With reference to Visible Light Communication (VLC) technology, which of the following statements are correct?

1. VLC uses electromagnetic spectrum wavelengths 375 to 780 nm.
2. VLC is known as long-range optical wireless communication.
3. VLC can transmit large amounts of data faster than Bluetooth
4. VLC has no electromagnetic interference.

Select the correct answer using the code given below:

- a) 1, 2 and 3 only
- b) 1, 2 and 4 only
- c) 1, 3 and 4 only
- d) 2, 3 and 4 onl

Q.14 (2020)

Q. With reference to “Blockchain Technology”, consider the following statements:

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. The structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that depend on basic features of blockchain can be developed without anybody’s permission.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 only
- d) 1 and 3 only

Q.15 (2022)

Q. With reference to Web 3.0, consider the following statements:

1. Web 3.0 technology enables people to control their own data.
2. In Web 3.0 world, there can be blockchain-based social networks.
3. Web 3.0 is operated by users collectively rather than a corporation

Which of the following given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.16 (2022)

Q. With reference to “Software as a Service”, Consider the following statements.

1. SaaS buyers can customise the user interface and can change data fields.
2. SaaS users can access their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.17 (2022)

Q. Which one of the following is the context in which the term “qubit” is mentioned?

- (a) Cloud Services
- (b) Quantum Computing
- (c) Visible Light Communication Technologies
- (d) Wireless Communication Technologies

Q.18 (2022)

Q. Consider the following communication technologies:

1. Closed-circuit Television
2. Radio Frequency Identification
3. Wireless Local Area Network

Which of the above are considered of Short-Range devices/technologies?

- (a) 1 and 2 only.
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.19 (2014)

Q. In addition to fingerprint scanning, which of the following can be used in the biometric identification of a person?

1. Iris scanning
2. Retinal scanning
3. Voice recognition

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.20 (2015)

Q. With reference to 'Near Field Communication (NFC) Technology', which of the following statements is/are correct?

1. It is a contactless communication technology that uses electromagnetic radio fields
2. NFC is designed for use by devices which can be at a distance of even a metre from each other
3. NFC can use encryption when sending sensitive information.

Select the correct answer using the code given below.

- | | |
|-----------------|--------------|
| a) 1 and 2 only | b) 3 only |
| c) 1 and 3 only | d) 1,2 and 3 |

Congratulations on giving the test!

Let's Discuss The Answers.



Q.1 (2016)

Q. With reference to 'LiFi', recently in the news, which of the following statements is/are correct?

1. It uses light as the medium for high-speed data transmission.
2. It is a wireless technology and is many times faster than 'WiFi'.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2**
- d) Neither 1 nor 2

Answer: C

- LiFi is a wireless optical networking technology that uses light-emitting diodes (LEDs) for data transmission.
- LiFi is designed to use LED light bulbs similar to those currently in use in many energy-conscious homes and offices. However, LiFi bulbs are outfitted with a chip that modulates the light imperceptibly for optical data transmission.
- LiFi data is transmitted by the LED bulbs and received by photoreceptors.
- In terms of its end use the technology is similar to Wi-Fi.
- The key technical difference is that Wi-Fi uses radio frequency to transmit data.
- Using light to transmit data allows Li-Fi to offer several advantages like working across higher bandwidth, working in areas susceptible to electromagnetic interference (e.g. aircra

Q.2 (2016)

Q. 'Project Loon', sometimes seen in the news, is related to

- a) Waste management technology
- b) Wireless communication technology**
- c) Solar power production technology
- d) Water conservation technology

Answer: B

- Loon LLC is an Alphabet Inc.(Parent company of Google) subsidiary working on providing Internet access to rural and remote areas.
- The company uses high-altitude balloons placed in the stratosphere at an altitude of about 18 km (11 mi) to create an aerial wireless network with up to 4G-LTE speeds.
- It was named Project Loon since even Google itself found the idea of providing Internet access to the remaining 5 billion population unprecedented and "loony".
- It may also be a reference to the balloons used.

Q.3 (2018)

Q. Consider the following pairs :

Terms sometimes seen in news

Context /Topic

1. Belle II experiment -

Artificial Intelligence

2. Blockchain technology -

Digital/Cryptocurrency

3. CRISPR — Cas9 -

Particle Physics

Which of the pairs given above is/are correctly matched ?

a) 1 and 3 only

b) 2 only

c) 2 and 3 only

d) 1, 2 and 3

Answer: B

- The blockchain is an undeniably ingenious invention which created the backbone of a new type of internet originally devised for the digital currency, Bitcoin.
- Genome editing (also called gene editing) is a group of technologies that give scientists the ability to change an organism's DNA. These technologies allow genetic material to be added, removed, or altered at particular locations in the genome.
- Several approaches to genome editing have been developed.
- A recent one is known as CRISPR-Cas9, which is short for clustered regularly interspaced short palindromic repeats and CRISPR-associated protein 9.
- The CRISPR-Cas9 system has generated a lot of excitement in the scientific community because it is faster, cheaper, more accurate, and more efficient than other existing genome editing methods.

Q.4 (2018)

Q. When the alarm of your smartphone rings in the morning, you wake up and tap it to stop the alarm which causes your geyser to be switched on automatically. The smart mirror in your bathroom shows the day's weather and also indicates the level of water in your overhead tank. After you take some groceries from your refrigerator for making breakfast, it recognises the shortage of stock in it and places an order for the supply of fresh grocery items. When you step out of your house and lock the door, all lights, fans, geysers and AC machines get switched off automatically. On your way to the office, your car warns you about traffic congestion ahead and suggests an alternative route, and if you are late for a meeting, it sends a- message to your office accordingly.

In the context of emerging communication technologies, which one of the following term" best applies to the above scenario?

- a) Border Gateway Protocol
- c) Internet Protocol

b) Internet of Things

- d) Virtual Private Network

Answer: B

- Internet of things is a seamless connected network system of embedded objects/ devices, with identifiers, in which communication without any human intervention is possible using standard and interoperable communication protocols. With this technology, each and every device that we use in daily life (including refrigerators, cars and even toothbrush) can be made intelligent and smart by using device-specific operating systems and connectivity to the Internet

Q.5 (2018)

Q. "3D printing" has applications in which of the following?

1. Preparation of confectionery items
2. Manufacture of bionic ears
3. Automotive industry
4. Reconstructive surgeries
5. Data processing technologies

Select the correct answer using the code given below :

a) 1, 3 and 4 only

b) 2, 3 and 5 only

c) 1 and 4 only

d) 1, 2, 3, 4 and 5

Answer: D

- There are endless possibilities for the utilization of 3d Printing. We can do many things with 3d printing machines. All the above things can be done with the help of 3D Printing

Q.6 (2019)

Q. With reference to communication technologies, what is/are the difference/differences between LTE (Long-Term Evolution) and VoLTE (Voice over Long-Term Evolution)?

1. LTE is commonly marketed as 3G and VoLTE is commonly marketed as advanced 3G.
2. LTE is data-only technology and VoLTE is voice-only technology.

Select the correct answer using the code given below.

- a) 1 only
b) 2 only
c) Both 1 and 2
d) Neither 1 nor 2

Answer: D

LTE and VoLTE

- VoLTE is an HD voice calling Service over 4G LTE rather than 2G/3G network present we are using.
- VoLTE is a technology wherein you can simultaneously send voice and data over the network without diminishing quality of voice. In the case of LTE, if you make a voice call and you also keep your data connection on, the quality of voice will reduce.
- So, in order to make a good quality voice call, you will have to switch off data. You may have noticed that in case of 2G and 3G, while you are making a phone call, some phones will automatically stop data transfer so as to preserve the voice quality.
- But in case of VoLTE, the voice quality will not reduce even if your data connection is on. With VoLTE it is very easy to transmit telephone conversation over the data network.

Q.7 (2019)

Q. In the context of digital technologies for entertainment, consider the following statements:

1. In Augmented Reality (AR), a simulated environment is created and the physical world is completely shut out.
2. In Virtual Reality (VR), images generated from a computer are projected into real-life objects or surroundings.
3. AR allows individuals to be present in the world and improves the experience using the camera of smart-phone or PC.
4. VR closes the world, and transposes an individual, providing complete immersion experience.

Which of the statements given above is/are correct?

- a) 1 and 2 only **b) 3 and 4**
c) 1, 2 and 3 d) 4 only

Answer: B

- Augmented reality (AR) is an interactive experience of a real-world environment where the objects that reside in the real-world are enhanced by computer-generated perceptual information, sometimes across multiple sensory modalities, including visual, auditory, haptic, somatosensory and olfactory
- The origin of the word augmented is augment, which means to add or enhance something. In the case of Augmented Reality (also called AR), graphics, sounds, and touch feedback are added into our natural world to create an enhanced user experience.

- Cell phones apps and business applications by companies using augmented reality are a few of the many applications driving augmented reality application development.
- The key point is that the information provided is highly topical and relevant to what you want you are doing.
- Virtual reality (VR) is an experience taking place within simulated and immersive environments that can be similar to or completely different from the real world.
- Applications of virtual reality can include entertainment (i.e. gaming) and educational purposes (i.e. medical or military training). Other, distinct types of VR style technology include augmented reality and mixed reality.
- 1st and 2nd are wrong statement while, 3rd and 4th are the correct.

Q.8 (2019)

Q. Consider the following statements: A digital signature is

1. an electronic record that identifies the certifying authority issuing it
2. used to serve as a proof of identity of an individual to access information or server on Internet
3. an electronic method of signing an electronic document and ensuring that the original content is unchanged

Which of the statements given above is/are correct?

- | | |
|------------------|-----------------|
| a) 1 only | b) 2 and 3 only |
| c) 3 only | d) 1, 2 and 3 |

Answer: C

- A digital signature is basically a way to ensure that an electronic document (e-mail, spreadsheet, text file, etc.) is authentic. Authentic means that you know who created the document and you know that it has not been altered in any way since that person created it.
- Digital signatures rely on certain types of encryption to ensure authentication.
- Encryption is the process of taking all the data that one computer is sending to another and encoding it into a form that only the other computer will be able to decode.
- Authentication is the process of verifying that information is coming from a trusted source.
- These two processes work hand in hand for digital signatures.

Q.9 (2019)

Q. In the context of wearable technology, which of the following tasks is/are accomplished by wearable devices?

1. Location identification of a person
2. Sleep monitoring of a person
3. Assisting the hearing impaired person

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 3 only

d) 1, 2 and 3

Answer: D

- The terms “wearable technology”, “wearable devices”, and “wearables” all refer to electronic technologies or computers that are incorporated into items of clothing and accessories which can comfortably be worn on the body
- Examples of wearable devices include watches, glasses, contact lenses, e-textiles and smart fabrics, headbands, beanies and caps, jewelry such as rings, bracelets, and hearing aid-like devices that are designed to look like earrings.
- The implications and uses of wearable technology are far reaching and can influence the fields of health and medicine, fitness, aging, disabilities, education, transportation, enterprise, finance, gaming and music.

Q.10 (2020)

Q. Consider the following activities:

1. Spraying pesticides on a crop field
2. Inspecting the craters of active volcanoes
3. Collecting breath samples from spouting whales for DNA analysis

At the present level of technology, which of the above activities can be successfully carried out by using drones?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3**

Answer: D

- Without crop protection, including pesticides, more than half of the world's crops would be lost to insects, diseases and weeds. Pesticides are important.
- A crater is a bowl-shaped depression, or hollowed-out area, produced by the impact of a meteorite, volcanic activity, or an explosion. Craters produced by the collision of a meteorite with the Earth (or another planet or moon) are called impact craters.

Q.11 (2020)

Q. In India, the term “Public Key Infrastructure” is used in the context of

- a) Digital security infrastructure**
- b) Food security infrastructure
- c) Health care and education infrastructure
- d) Telecommunication and transportation infrastructure

Answer: A

- Public Key Infrastructure (PKI) is a technology for authenticating users and devices in the digital world. Under this system, one or more trusted parties digitally sign documents certifying that a particular cryptographic key belongs to a particular user or device. The key can then be used as an identity for the user in digital networks.
- Therefore, option (a) is the correct answer

Q.12 (2020)

Q. With the present state of development, Artificial Intelligence can effectively do which of the following?

1. Bring down electricity consumption in industrial units.
2. Create meaningful short stories and songs.
3. Disease diagnosis.
4. Text-to-Speech Conversion.
5. Wireless transmission of electrical energy.

Select the correct answer using the code given below:

a) 1, 2, 3 and 5 only

b) 1, 3 and 4 only

c) 2, 4 and 5 only

d) 1, 2, 3, 4 and 5

Answer: B

- Statement 1 is correct→Bring down electricity consumption in industrial units by using efficient usage of electricity supply by use of data analytics
- Statement 2 is incorrect→Creating meaningful short stories and songs is not possible by AI because of 'Meaningful' keyword in stories and songs can not be made by AI
- Statement 3 is correct→Disease diagnosis by Machine Learning models
- Statement 4 is correct→Text-to-Speech Conversion is possible by help of Natural Language Processing in AI
- Statement 4 is incorrect→Wireless transmission of electrical energy is unrelated to AI

Q.13 (2020)

Q. With reference to Visible Light Communication (VLC) technology, which of the following statements are correct?

1. VLC uses electromagnetic spectrum wavelengths 375 to 780 nm.
2. VLC is known as long-range optical wireless communication.
3. VLC can transmit large amounts of data faster than Bluetooth
4. VLC has no electromagnetic interference.

Select the correct answer using the code given below:

- a) 1, 2 and 3 only
- b) 1, 2 and 4 only
- c) 1, 3 and 4 only**
- d) 2, 3 and 4 onl

Answer: C

- Visible light communication (VLC) is a data communications variant which uses visible light between 400 and 800 THz (780–375 nm). VLC is a subset of optical wireless communications technologies. 1 is correct.
- The technology uses fluorescent lamps (ordinary lamps, not special communications devices) to transmit signals at 10 kbit/s, or LEDs for up to 500 Mbit/s over short distances.
- Systems such as RONJA can transmit at full Ethernet speed (10 Mbit/s) over distances of 1–2 kilometres (0.6–1.2 mi). It is not long range. 2 is incorrect.
- VLC has the characteristic, to resist electromagnetic interference. 4 is correct.
- It carries more data than WiFi and bluetooth. 3 is correct.

Q.14 (2020)

Q. With reference to “Blockchain Technology”, consider the following statements:

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. The structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that depend on basic features of blockchain can be developed without anybody’s permission.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 only

d) 1 and 3 only

Answer: D

- Blockchain technology is a structure that stores transactional records, also known as the block, of the public in several databases, known as the “chain,” in a network connected through peer-to-peer nodes. Typically, this storage is referred to as a ‘digital ledger.’
- In simpler words, the digital ledger is like a Google spreadsheet shared among numerous computers in a network, in which, the transactional records are stored based on actual purchases.
- The fascinating angle is that anybody can see the data, but they can’t corrupt it. 1 is correct.
- Blockchain can be used for many purposes like Secure sharing of medical data, Music royalties tracking. 2 is incorrect. It is an extreme statement also.
- Applications that depend on basic features of the blockchain can thus be developed without asking anybody for permission or paying anyone. 3 is correct.

Q.15 (2022)

Q. With reference to Web 3.0, consider the following statements:

1. Web 3.0 technology enables people to control their own data.
2. In Web 3.0 world, there can be blockchain-based social networks.
3. Web 3.0 is operated by users collectively rather than a corporation

Which of the following given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Web 3.0 is in an internet where users own their own data, not corporations
- Global digital transactions are secure, online exchanges of information and value are decentralized.
- Instead of being owned by centralized institutions, the community will own the network.
- This means the governance, rule sets, and general operation of the platform will move back to the users, as will the larger share of the potential profits.
- Blockchain is imperative to this, working as a digital ledger.
- Users will have their profile on their own Web 3.0 wallet which can be utilized across any decentralized platform.

Q.16 (2022)

Q. With reference to “Software as a Service”, Consider the following statements.

1. SaaS buyers can customise the user interface and can change data fields.
2. SaaS users can access their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Software as a service (SaaS) is a software distribution model in which a cloud provider hosts applications and makes them available to end users over the internet.
- The public cloud provider manages all the hardware and traditional software, including middleware, application software, and security.
- SaaS makes it easy to "mobilise" the workforce because users can access SaaS apps and data from any Internet-connected computer or mobile device.
- Common examples are email, calendaring and office tools (such as Microsoft Office 365).
- Buyers can customize the user interface (UI) to change the look and feel of the program, as well as modify specific areas, such as data fields, to alter what data

Q.17 (2022)

Q. Which one of the following is the context in which the term “qubit” is mentioned?

(a) Cloud Services

(b) Quantum Computing

(c) Visible Light Communication Technologies

(d) Wireless Communication Technologies

Answer: B

- A qubit (or quantum bit) is the quantum mechanical analogue of a classical bit.
- In classical computing the information is encoded in bits, where each bit can have the value zero or one.
- In quantum computing the information is encoded in qubits.
- Qubits can be in a state of 0 or 1 or (unlike a classical bit) in a linear combination of both states.

Q.18 (2022)

Q. Consider the following communication technologies:

1. Closed-circuit Television
2. Radio Frequency Identification
3. Wireless Local Area Network

Which of the above are considered of Short-Range devices/technologies?

- (a) 1 and 2 only.
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Closed Circuit Television or CCTV is a short range device.
- Radio-frequency identification (RFID) that is being used for Toll collection is a short range device.
- A Wireless Local Area Network (WLAN) is a wireless computer network that links two or more devices using wireless communication within a limited area such as a home, school, computer laboratory, campus, or office building.

Q.19 (2014)

Q. In addition to fingerprint scanning, which of the following can be used in the biometric identification of a person?

1. Iris scanning
2. Retinal scanning
3. Voice recognition

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only

d) 1, 2 and 3

Answer: D

- Biometric verification is any means by which a person can be uniquely identified by evaluating one or more distinguishing biological traits.
- Unique identifiers include fingerprints, hand geometry, earlobe geometry, retina and iris patterns, voice waves, DNA, and signatures.

Q.20 (2015)

Q. With reference to 'Near Field Communication (NFC) Technology', which of the following statements is/are correct?

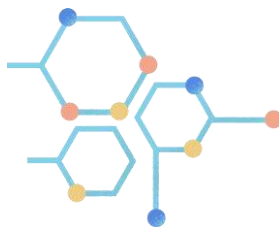
1. It is a contactless communication technology that uses electromagnetic radio fields
2. NFC is designed for use by devices which can be at a distance of even a metre from each other
3. NFC can use encryption when sending sensitive information.

Select the correct answer using the code given below.

- | | |
|------------------------|--------------|
| a) 1 and 2 only | b) 3 only |
| c) 1 and 3 only | d) 1,2 and 3 |

Answer: C

- Near-field communication is a wireless technology that allows users to make payments by placing a compatible device like a smartphone or payment card within a few centimetres of another compatible device like a terminal, tablet or another smartphone.
- Near-field communication transmits data through electromagnetic radio fields and is the technology behind payment services like Google Wallet and Apple Pay.
- Both devices must contain NFC chips for an NFC transaction to take place.



UPSC Prelims

Science & Tech

Class No. 15

IPR and Role of Scientists

CHAT GPT

- **Chat Generative Pre-trained Transformer** is an artificial intelligence chatbot developed by OpenAI and launched in November 2022.
- Open AI has created Chat GPT, a start-up focused on artificial intelligence and its potential use cases.
- According to OpenAI's description, ChatGPT can answer “follow-up questions”, and can also “admit its mistakes, challenge incorrect premises, and reject inappropriate requests.”

CHAT GPT

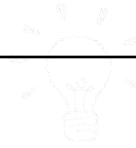
- It is based on the company's GPT 3.5 series of Language Learning Models (LLM).
- GPT stands for Generative Pre-trained Transformer 3 and this is a kind of computer language model that relies on deep learning techniques to produce human-like text based on inputs.
- The model is trained to predict what will come next, and that's why one can technically have a 'conversation' with ChatGPT.

CHAT GPT

- chatbot was also trained using “**Reinforcement Learning from Human Feedback (RLHF)**.”
- Interestingly, ChatGPT has been trained to decline ‘inappropriate’ requests, presumably those which are ‘illegal’ in nature. However, it should be noted that ChatGPT has limitations, as it may generate incorrect information, and create “biased content.” More importantly, the chatbot’s knowledge of the world and events after 2021 is limited.
- Other products of Open AI
 - **Dall-E** — the AI-based text-to-image generator

CHAT GPT

- Google had recently showcased how its **LaMDA chatbot** is being used to help with fiction writing
 - LaMDA is a machine learning language model created by Google as a chatbot that is supposed to mimic humans in conversations
 - LaMBDA is built on neural network architecture



Human-AI Collaboration		
Supportive Mode	Reimagining the Business	Ai Replaces the Humans
AI performs alongside humans by facilitating human judgments by providing resources such as predictive outputs.	AI performance activities that go beyond the cognitive abilities of humans. For e.g., large-scale genome study in bioinformatics.	Useful in potentially harmful situations for humans such as environments and rapid system response in nuclear reactors.

Benefits and applications of AI



Healthcare

- Address issues particularly in rural areas that suffer from poor connectivity and limited supply of healthcare professional .
- This can be achieved through AI driven diagnostics, early identification of potential pandemics, and imaging diagnostics.

Education and skilling

- Solve problem of quality and access issues observed in the Indian education sector.
- Augmenting and enhancing the learning experience through personalised learning, automating and expediting administrative tasks.

Benefits and applications of AI



Agriculture

- Driving a food revolution to meet the increased demand for food by improvement in crop yield.
- Address challenges in irrigation, misuse of pesticides and fertilisers through real time advisory.

Manufacturing

- Manufacturing industry is expected to be one of the biggest beneficiaries of AI based solutions in field of engineering, supply chain management, production and quality assurance etc.

Benefits and applications of AI



Energy

- AI can enable storage of energy through intelligent grids enable by smart meters, and also improve the reliability and affordability of photovoltaic energy.
- AI may also be deployed for predictive maintenance of grid infrastructure.

Financial Services

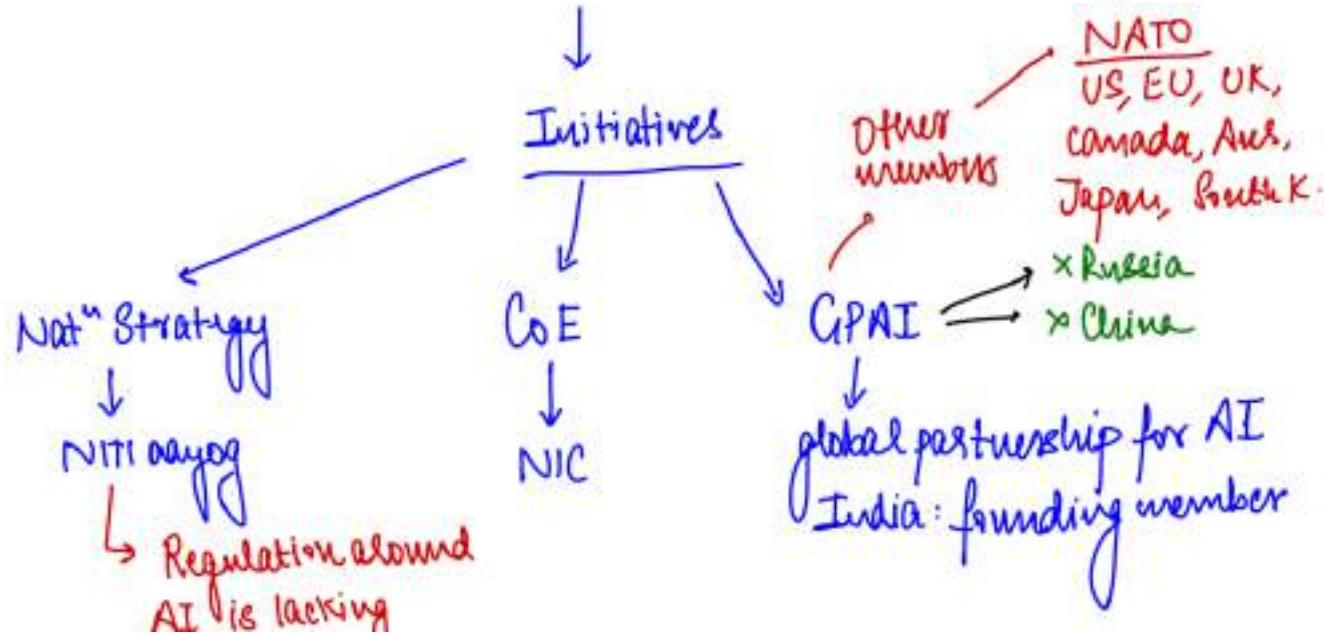
- AI can ensure early detection of financial risk and systemic failures, and automation of reduce malicious intent in financial system, such as market manipulation, anomalous trading etc.

Benefits and applications of AI



Defence and Security	AI can be used for intelligence gathering, cyber defence, risk terrain analysis, detecting anomalous behaviour in human etc.
Law Enforcement	AI technologies can be used by law enforcement which include facial recognition, speech recognition, predictive analytics etc.
Disaster Management	AI has potential in providing remedial measures and control in aftermath of man-made recognition can be used in infrastructure damage assessments.

Indian Scenario



Introducing Bard, an experimental conversational AI service powered by LaMDA

You can use Bard to —

Plan a friend's baby shower

Compare two Oscar nominated movies

Get lunch ideas based on what's in your fridge

What's on your mind?

It may give inaccurate or inappropriate information. Your feedback matters. It and more helpful and safe.

Metaverse

- Metaverse projects refer to blockchain-based games and applications that are set in virtual worlds.
- Such games follow a play-to-earn model, where gamers are rewarded for being invested in the game.
- Metaverse can be understood as a combination of multiple elements of technology, including virtual reality, augmented reality and video where users "live" within a digitally enhanced surrounding.

Metaverse

- **These technologies** are currently being developed by Apple, Google, Amazon, and Microsoft in addition to Facebook.
- Metaverse uses the following technologies : (**extended realities**)
- Virtual Reality
- Augmented Reality
- Hologram
- Avatar
- Let's see a video on how it works : <https://www.youtube.com/watch?v=SAL2JZxpoGY>

Metaverse : Technologies

- **Virtual Reality (VR):** VR can be understood as the use of computer modelling and simulation that enables a person to interact with an artificial three-dimensional (3-D) visual or other sensory environment. E.g., games like World of Warcraft. **VR closes the world**, and transposes an individual, providing complete immersion experience.

Metaverse : Technologies

- **Augmented Reality (AR):** It is an enhanced version of the real physical world that is achieved through the use of digital visual elements, sound, or other sensory stimuli delivered via technology. E.g., games like Pokémon Go.
- **Hologram:** Holograms are virtual three-dimensional images created by the interference of light beams that reflect real physical objects.

Metaverse : Technologies

- **Avatar:** An avatar in the metaverse is a representation of an individual in the virtual world, this digital avatar enables the person to function like an actual human being in a digitally created world.
- **Platform/Developer/User distinction:** In metaverse, digital engagement will get very personal and tailormade for the user. In this context, it is important that role played by the platform, the developers and the users is clearly understood.

Metaverse : Technologies : Images

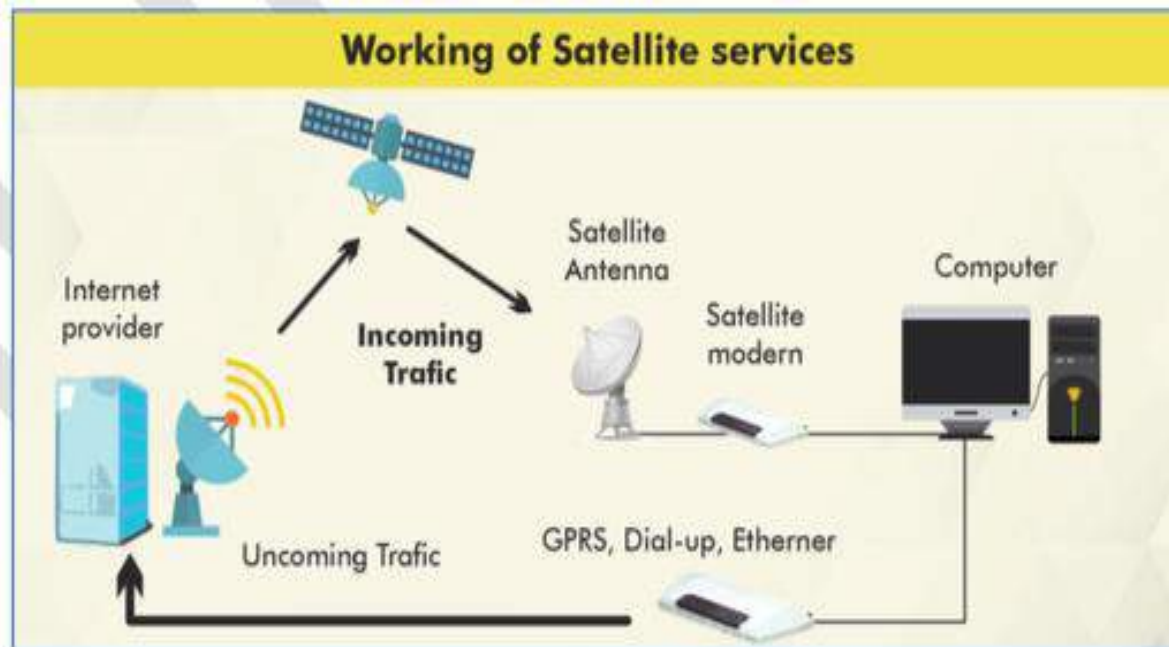


Satellite based Internet Connectivity

- Satellite Internet Services is a **wireless internet connection** which uses satellites in space to get an internet signal from the Internet Service Provider (ISP) to users.

Satellite based Internet Connectivity

- It operates using two way satellites like **VSAT (Very Small Aperture Terminal)** or telecommunication satellites to liaison between Internet Service Providers (ISPs) and Users with satellite dishes.
- **Direct to Satellite Model:** allows devices to directly communicate with the satellite without the need of any intermediate ground gateway.
- 2 types : satellite in LEO or satellite in GEO



Satellite based Internet Connectivity

- **Advantages of Space-based internet systems using satellites in LEO compared to GEO:** low Latency in signal transmission.
- **Advantages of Space-based internet systems using satellites in GEO compared to LEO:** Serviceable area on Earth by each satellite is more, easier to establish a transmission link with a satellite.

Models for Satellite based connectivity

- **Hybrid : LPWAN + Satellite** : each sensor first connects with a sink node, (LPWAN) then to Satellite.
- **Direct to Satellite** : Devices to directly communicate with satellite without need of ground gateway.



Satellite-based connectivity is possible using **Geo Stationary, Medium and Low Earth orbit Satellites**



It is **different from satellite television** as Internet signals go **both ways** with a much **greater amount of bandwidth** to move all that data



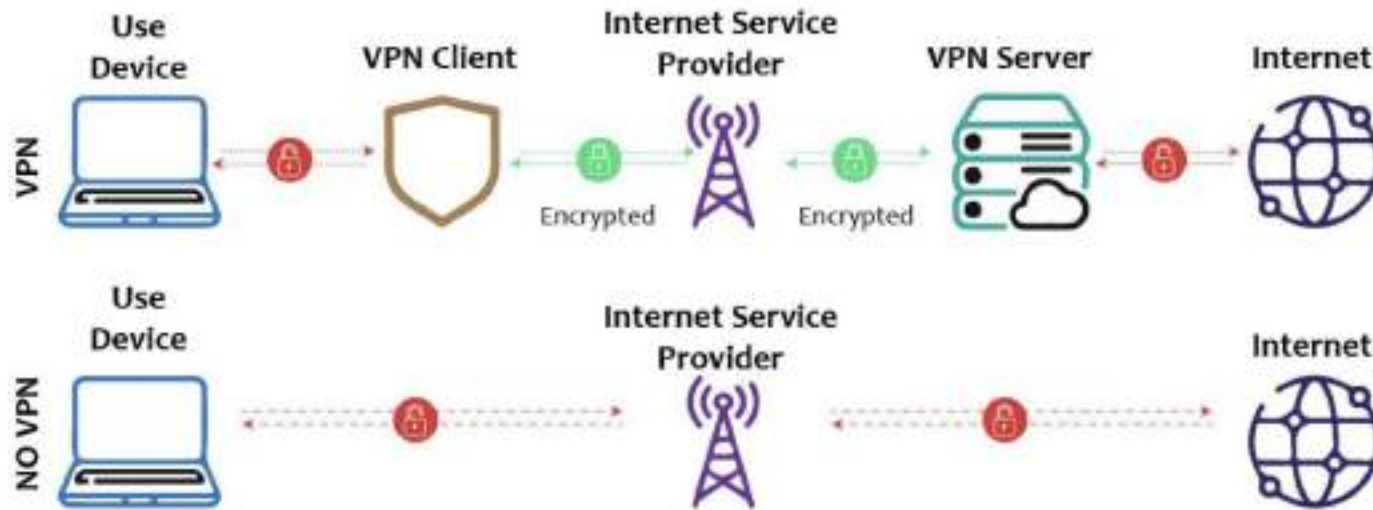
Broadband **signal is sent and received to a satellite in space** rather than using an **underground copper/fibre network**

Satellite based Internet Connectivity

Benefits of Satellite internet Services	Challenges in the Satellite internet Services
• Cheap and easy internet option for rural area, hilly terrain, and islands for last mile connectivity. • Faster than DSL connection with continuously improving technology with present speeds reaching around 100 mbps. • Can play a significant role in emergency or disaster recovery due to limited ground infrastructure.	• Low speed and high latency - In comparison to fiber optic cables, • High cost as compared to wireless cellular network and others. • Limitations of Bandwidth with high impact of weather aberrations • Potential to damage spacecrafts/satellites or interrupt of higher satellites frequency because of its large satellite network need, • Potential of increase space junk. • A VPN gives online privacy anonymity by creating a private network from a public internet connection

Virtual Private Networks

- A VPN is a computer network across a public internet that provides user access to their organization's network while maintaining the security of information provided
- It gives online privacy and anonymity by creating a private network from a public internet connection.



Types of VPN



Secure Sockets Layer (SSL) VPN

It enables individual users to **access an organization's network** and client-server applications **without the need for specialized software**



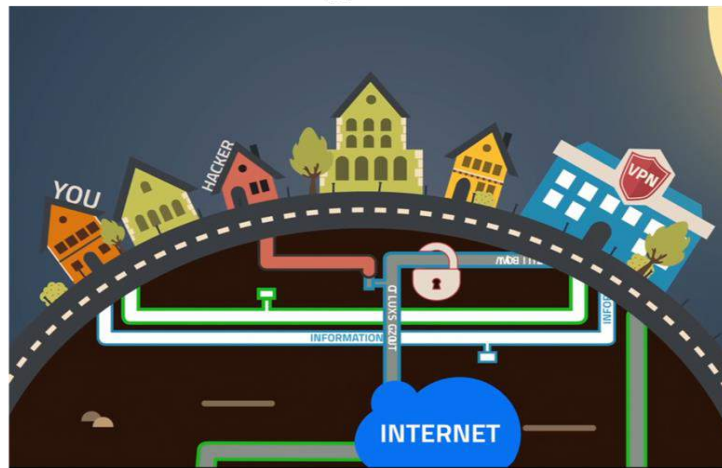
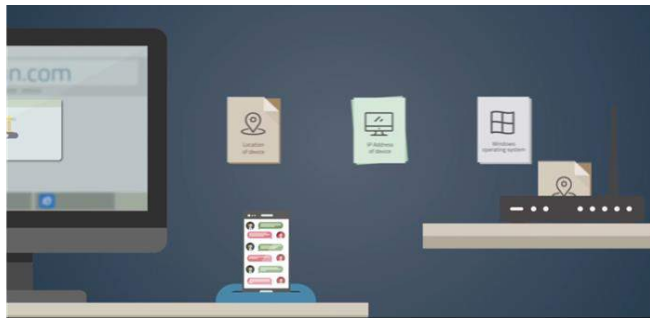
Site-to-site VPN

A **private network designed to hide private intranets** and allow users of these secure networks to access each other's resources.



Client-to-Server VPN

User is not connected to the **internet via his own ISP** (internet service provider), but establishing a **direct connection through VPN provider**.



Long term Evolution (LTE) and Voice over LTE (VoLTE)

- VoLTE is a **technology update to the LTE protocol** used by mobile phone networks.
- **Under LTE**, the infrastructure of telecom players **only allows transmission of data while voice calls are routed to their older 2G or 3G networks.**
- This is why, under LTE, you cannot access your 4G data services while on a call. This leads to problems

Long term Evolution (LTE) and Voice over LTE (VoLTE)

- Such as slow internet speeds and poor voice clarity.
- LTE is commonly marketed as 4G LTE.
- **VoLTE allows voice calls to be ‘packaged’ and carried through LTE networks.** This would mean 4G data accessibility even during calls.

Dark Net

- Imagine you get an email containing your bank statement :
- It is on the internet
- It is not accessible to me.
- Also known as Dark Web, it is that part of Internet which cannot be accessed through traditional search engines like Google nor is it accessible by normal browsers like Chrome or Safari.

Dark Net

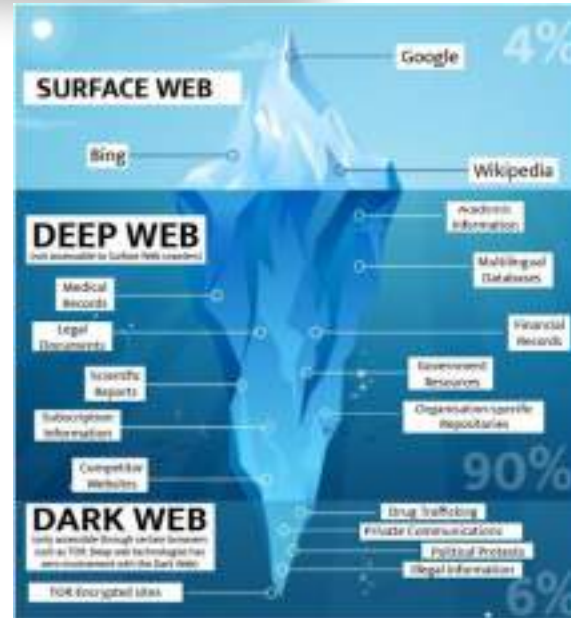
- It generally uses non-standard communication protocols which make it inaccessible to internet service providers (ISPs) or government authorities.
- Content on Dark Net is encrypted and requires specific browser such as TOR (The Onion Router) browser to access those pages.

Dark Net

- Dark Net itself is only a part of Deep Web that is a broader concept, which includes sites that are protected by passwords.
- Part of internet that is readily available to general public and searchable on standard search engines is called as Surface Web.
- It is used by journalists and citizens working in oppressive regimes, researchers and students to do research on sensitive topics, law enforcement agencies etc.

Dark Net

- Concerns over its use: Anonymity, Haven for illicit activity, Privacy and ethical concerns, drug dealing, communication by terrorists etc.

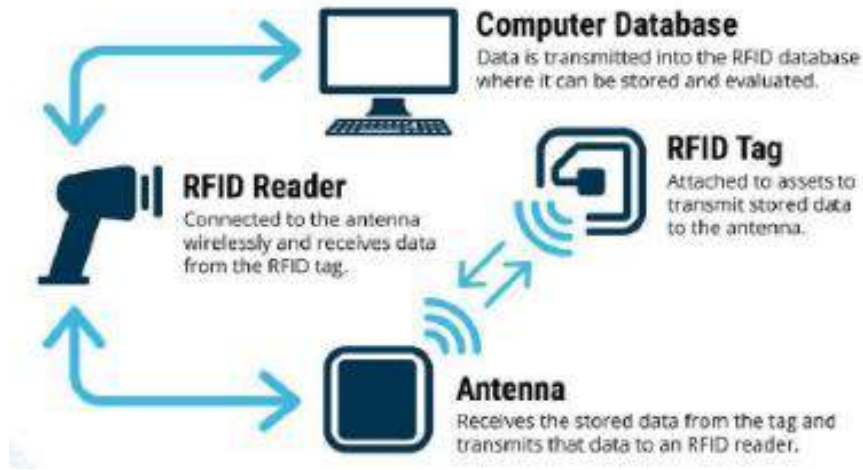


RFID

- It is an electronic tag that can exchange data with an RFID reader through radio waves.
- In this, an antenna broadcasts energy to tag, which in turn returns modulated energy back in form of back-scatter.
- It is faster connection, and operational range is about 100m.
- RFID tag typically sends out information hence only one-way information.

RFID

Basic RFID System

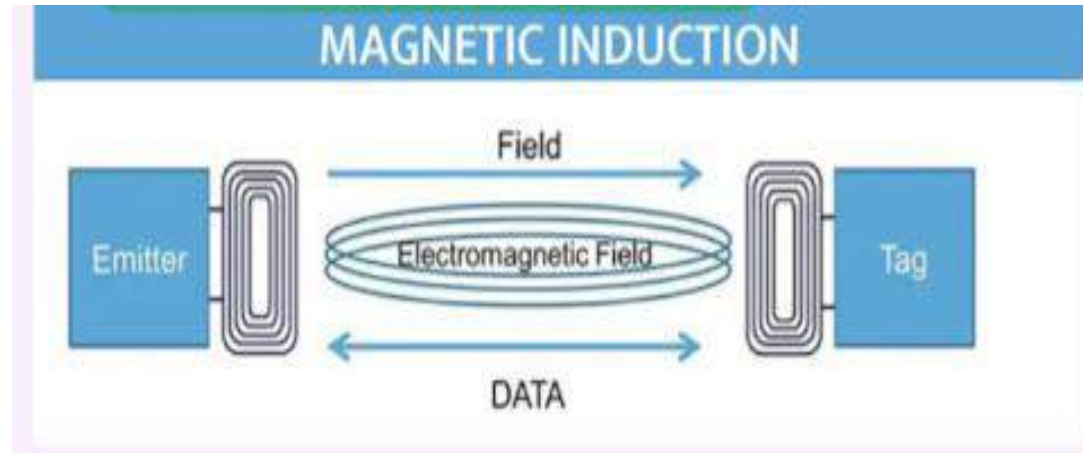


Near Field Communication

- NFC is the technology that allows compatible devices to establish radio communication with each other by touching them together or bringing them into a distance maximum of 10 cm.
- NFC works on Radio Frequency Identification (RFID) technology and Magnetic induction principle
- However, RFID and NFC also have many differences

Near Field Communication

- When electrons flow through a conductor, they create a magnetic field. And when magnetic fields change, they can cause electrons to flow through a conductor. This relationship -- known as inductive coupling.
- In NFC also the reader or the mobile acts as the active device whose antenna or the coil releases a sinusoidal carrier wave and creates a magnetic field. Then the passive tags present in that field gets energy from the reader and change the properties of the receiving frequency.



Basis	RFID	NFC
Communication	 Unidirectional—typically an RFID tag sending data to an RFID reader	 Both unidirectional and bidirectional. This allows NFC to be used for more complex interactions such as peer-to-peer (P2P) data sharing as well as card emulation.
Frequency range	RFID FREQUENCY RANGES: Low Frequency (LF): 125-134 kHz  (LF) Range: Up to 50 m High Frequency (HF): 13.56 MHz NFC  (HF) Range: Up to 30 m Ultra High Frequency (UHF): 856 MHz TO 950 MHz  (UHF) Range: Up to 300 m	
Application	 Asset tracking  Race timing  Attendee tracking  Inventory management	 Contactless payment  Information/data sharing  e-store check-ins  Wearables like smartwatch  Post-sale product experiences (such as scanning QR codes on posters for additional product information)

	Radiofrequency Identification (RFID) 	Quick Response (QR) code 	Near Field Communication Technology (NFC) 
About	A wireless system comprised of two components: tags and readers.	A two-dimensional version of a barcode that can be read easily by a digital device	A short-range wireless connectivity technology that allows NFC-enabled devices to communicate with each other and transfer information quickly and easily with a single touch
Working Range	Read distance depends upon the frequency (Ranging from 300 gigahertz to as low as 9 kilohertz) used	Ideal scanning size to distance ratio is 1:10.	NFC-enabled devices must be either physically touching or within a few centimeters from each other for data transfer.
Other Key Information	Do not require a direct line of sight to be read. - Data stored in an RFID tag can be updated in real-time.	- Require a line of sight to scan. No real-time tracking.	Transfer information quickly and easily with a single touch.

Other Similar Technology

- Infrared : short connection and transmission of information through infrared light.
- Needs line of sight between 2 devices.
- Ex : Remote control devices.

Edge Computing

- It enables data to be analysed, processed, transferred at edge of a network.
- It means, data is analysed locally, closer to where it is stored, in real-time without latency.
- Basic difference between edge computing and cloud computing lies in where data processing takes place

Edge Computing

- Cloud Computing: It means storing and accessing data and programs over Internet instead of your computer's hard drive.
- Existing Internet of Things (IoT) systems perform all their computations in cloud using data centres.
- Edge Computing: It manages massive amounts of data generated by IoT devices by storing and processing data locally.

Edge Computing

Benefits :

- High speed
- Reduced latency
- Better reliability and security : difficult for any single disruption to take down a network
- Very important in upcoming internet of things technology

4th IR

- 1st IR : ?
- 2nd IR ?
- 3rd IR ?
- 4th IR :
- Building on the 3rd Industrial Revolution
- Includes cyber-physical systems, the Internet of things, big data analytics, cloud computing, cognitive computing, artificial intelligence, 3-D printing, and autonomous vehicles etc.
- Impact : all of the above

Big Data

- Big data is a term applied to datasets whose size or type is beyond the ability of traditional relational databases to capture, manage and process the data with low latency.
- Big data analytics is the use of advanced analytic techniques against very large, diverse data sets that include structured, semi-structured and unstructured data, from different sources, and in different sizes from terabytes to zettabytes.



Big Data : Advantages

- Improves the decision making of businesses
- Protects company and client information
- Effective marketing and targeted advertising
- Cost and time reduction through streamlining operations
- Better Products designing by getting info and analysis
- Applications in various sectors :
 - Banking : NPA prevention and trend analysis
 - Education : identify at risk students etc.
 - Government : traffic congestion and utilities management
 - Health care : insights to improve patient care
 - Manufacturing : boost quality output while minimizing waste processes
 - Agriculture : Optimizing crop efficiency

Big Data

- Challenges :
- Human resource : lack of data scientists
- Privacy Issues : reveal sensitive and personal information of users
- Security issues and outsourcing of data analytics
- Lack of availability of quality data
- Ethics and issues of informed consent of users

Additive Manufacturing (3d printing)

- Three-dimensional printing (3D printing), also known as Additive manufacturing (AM) is a process of joining materials to make objects from 3D model data, usually layer upon layer.
- Different from conventional manufacturing
- It is a process in which a three-dimensional object is built from a computer-aided design (CAD) model, usually by successively adding materials in a layer-by-layer fashion.
- The addition of material can happen in multiple ways, namely power deposition, resin curing, filament fusing.

Additive Manufacturing (3d printing)

- Thermoplastics, Metals, Ceramics as well as biomaterials can be used for additive manufacturing.
- Traditional vs. AM: Unlike AM, traditional manufacturing methods, by contrast, are subtractive in nature.
- Subtractive manufacturing involves removing parts of a block of material in order to create the desired shape. Cutting wood into useful shapes, for instance, is a very simple example of a subtractive process.

3D printing

Basic Additive Manufacturing Process

CAD Model.....3D Object



3D Card Model



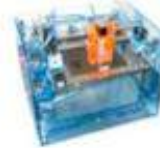
.STL File



Slicing Software



Layer Silces & Tool Path



AM Process



3D Object

Benefits

- Low cost
- Less time
- Efficiency
- Customization
- Increased Productivity
- Reduced Wastage

Issues

- Size is limited
- Limited raw materials
- Employment destruction
- Copyright infringement concerns
- Dangerous items can be produced

Software as a service

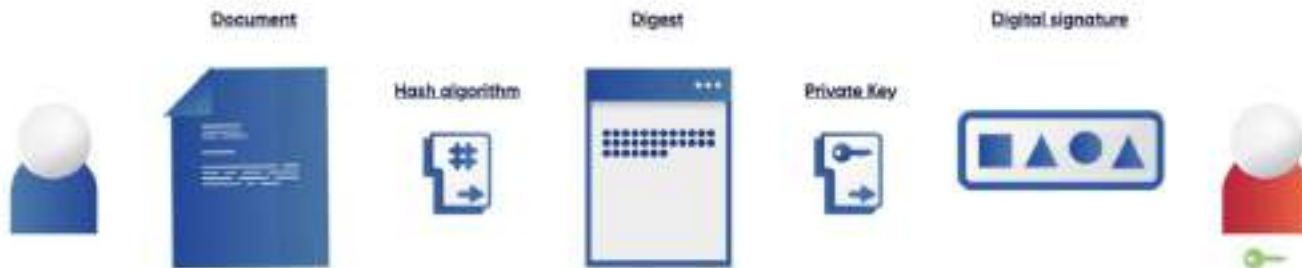
- On demand access to application software usually requiring a licensing or subscription by the user.
- For example : when you use google docs to edit a document, you do not need to worry about installation / configuration of a software.
- User is charged per usage and can eliminate the need for huge investment upfront for a new or evolving organization.

Software as a service

- Examples : Microsoft office 365, Drop Box, Outlook, Gmail.
- If you want, you can customize the user interface and labels.

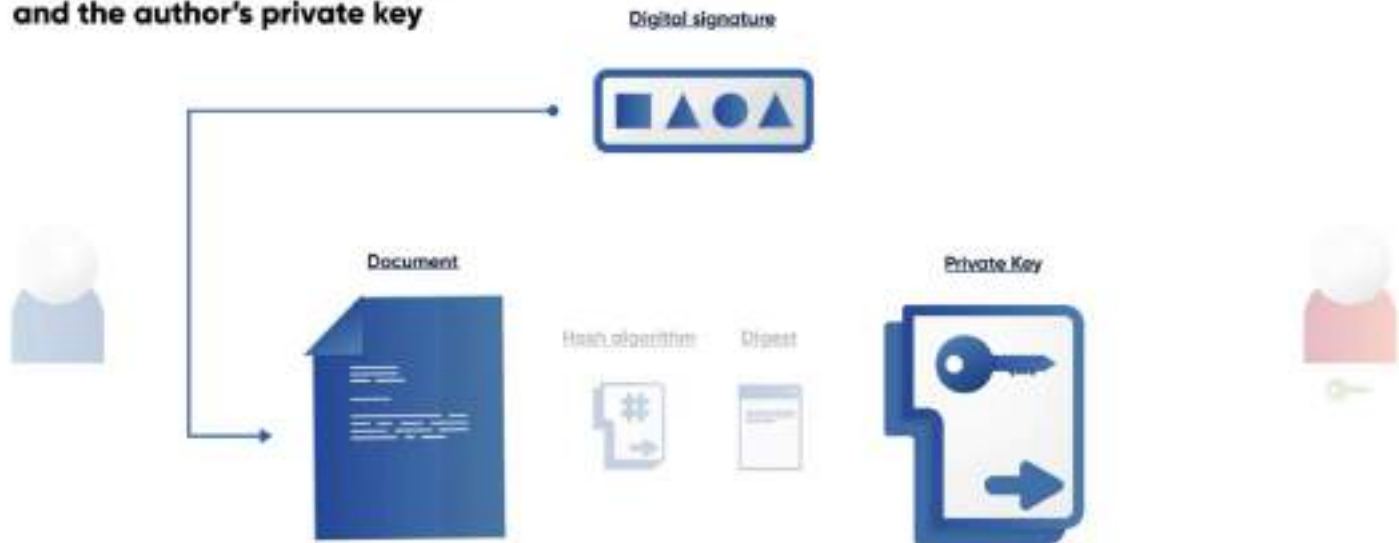
Digital Signature

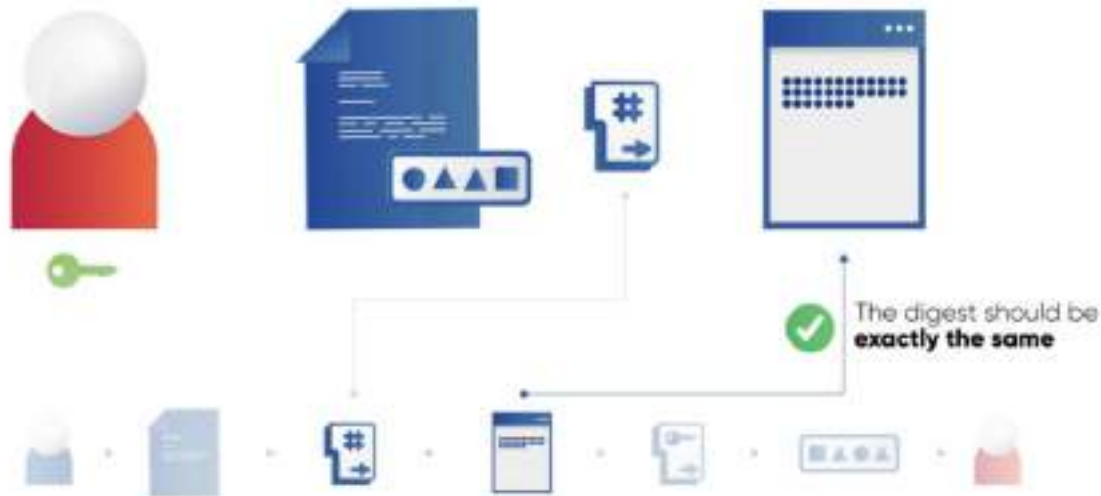
- It is a way to ensure integrity of data recorded onto a blockchain. Akin to a physical signature on a paper, digital signature is for digital documents. It binds a person / entity to a digital document.



A digital signature is

**a combination of the document
and the author's private key**





Digital Certificate

- Verification by public key and hash algorithm : ensures that document has not been altered.
- But you have no idea about the true identity of the sender as it depends on the public key which is publically available.
- Solution : digital certificate : electronic credentials issued by a trusted third party. It not only verifies the identity of the owner but also verifies that owner owns public key.

Digital Certificate

- Digital certificates are electronic credentials issued by a trusted third party. It not only verifies the identity of the owner, but also verifies that owner owns the public key.
- It contains the certificate owner's name, owner's public key, certificate issuer's name and certificate issuer's digital signature.
- Trusted 3rd party : issues digital certificates.
- You trust my passport issuer : therefore you trust my passport to tell my identity.

E-SIM

- an eSIM is an embedded SIM
 - It has same hardware of a regular SIM card chip
 - It is a **permanently embedded part** of the motherboard of a device.
- e-SIM acting as a unique identifier for telecom operators to reach the exact smartphone when they make a call or send a text.
- eSIMs were first established a decade ago in 2012.

E-SIM

Benefits

- users switch operators without having to replace any physical SIM cards.
- **Convenience:** The ability to store multiple SIM profiles in your eSIM also means you can switch between profiles easily, without the need of activating a SIM repeatedly or physically switching cards repeatedly.

E-SIM

Benefits

- **Security:** In case of theft of phone, SIM card can not be misused as it is embedded part of smart phone.
- **One less opening on your phone:** Having an eSIM also means there is one less opening on the frame of your phone, in theory, which should reduce the likelihood of elements like dust and water entering the phone from yet another slot.

E-SIM

Disadvantages

- **Emergencies:** If your phone stops working, runs out of battery or simply, falls and gets a cracked screen, your communication is brought to a complete standstill with eSIMs.
- **Unusable in countries with no eSIM support:** You cannot use an eSIM phone in a country where the telecom operators simply don't support the technology yet.
- **Support only available in premium phones:** In India, eSIM support is currently available on more expensive devices.

D2M Technology

- Direct To Mobile Tech—>broadcast video and other forms of multimedia content directly to mobile phones
- It does not require an active internet connection

D2M Technology

Concept

- D2M is based on the convergence of broadband and broadcast, using which mobile phones can receive terrestrial digital TV.
- It would be similar to how people listen to FM radio on their phones, where a receiver within the phone can tap into radio frequencies.
- Using D2M, multimedia content can also be beamed to phones directly.

D2M Technology

Benefits

- It can be used to directly broadcast content related to citizen-centric information
- It can be used to counter fake news, issue emergency alerts and offer assistance in disaster management
- It can be used to broadcast live news, sports etc. on mobile phones.

D2M Technology

Impact

- Consumers
 - Can access multimedia content from Video on Demand (VoD) or Over The Top (OTT) content platforms without having to exhaust their mobile data at a nominal rate.
 - The technology will also allow people from rural areas, with limited or no internet access, to watch video content.

D2M Technology

Businesses

- It can enable telecom service providers to offload video traffic from their mobile network onto the broadcast network, thus helping them to **decongest valuable mobile spectrum**.
- This will also **improve usage of mobile spectrum** and free up bandwidth which will help reduce call drops, increase data speeds etc.

D2M Technology

Government Measures

- The Department of Telecommunications (DoT) has set up a committee to study the feasibility of a spectrum band for offering broadcast services directly to users' smartphones

D2M Technology

Challenges

- bringing key stakeholders like mobile operators onboard will be the “biggest challenge” in launching D2M technology on a wide scale.
- mass roll out of the technology will entail changes in infrastructure and some regulatory changes.

CrysXPP

- Researchers from **IIT Kharagpur** in collaboration with the **Indo-Korea Science and Technology Center (IKST)** have developed a method called **CrysXPP**.
- It is a **machine learning (ML) system** that enables rapid prediction of various **material properties** with high precision.

CrysXPP

- ML is a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed to do so
- Machine learning algorithms use historical data as input to predict new output values
- Applications of ML: Recommendation engines, fraud detection, spam filtering, malware threat detection, business process automation (BPA) and Predictive maintenance.

CrysXPP

Applications

- **CrysXPP made large-scale testing easier**, because until now, it has been difficult to test crystalline materials on a large scale.
- It also helps determine the electronic, **magnetic and elastic properties of crystals**, which are often time- consuming, expensive and computation-intensive.

Web 5.0

- Former Twitter CEO recently announced his vision for a new decentralized web platform that is being called Web 5.0.
- Objective—> “ownership of data and identity to individuals”.
- It is a **Decentralized Web Platform** that enables developers to leverage Decentralized Identifiers, Verifiable Credentials, and Decentralized Web Nodes to write Decentralized Web Apps.
- Web 5.0 is Web 2.0 plus Web 3.0 that will allow users to own their identity on the Internet and control their data.
- It aimed at building an extra decentralized web that puts one in control of your data and identity.

Web 5.0

Web 1.0 : Read only web

Web 2.0 : Social Web

Web 3.0 : Semantic Web

Web 5.0

Web 4.0 (Mobile Web)

- It is already a work in progress, with no clear understanding of what it would entail.
- It is often referred to as the **symbiotic web**.
 - It will make contact between humans and computers in symbiosis.

Geospatial Data

- **The Ministry of Science and Technology** has notified a citizen-centric **National Geospatial Policy (NGP) 2022**.
- **Geospatial Data Promotion and Development Committee (GDPDC)**: At national level, an apex body for formulating and implementing appropriate guidelines, strategies, and programs.
 - **GDPDC would replace the National** Spatial Data Committee (NSDC) and Geospatial Data Promotion and Development Committee.

Geospatial Data

Geospatial Data Infrastructure

- GDPDC will **develop 14 global geospatial data themes** recognized by the United Nations Statistics Division.
- **National Geospatial Data Registry (NGDR):** Accessible by all stakeholders.
- **Unified Geospatial Interface (UGI):** To provide consumer-oriented products and solutions using Geospatial data.
- **Survey of India (SoI):** Will help in developing NGDR and UGI in collaboration with Bhask`atics (BISAG-N)-under MeitY, private sector etc.

Geospatial Data

- **Mapping infrastructure:** Policy shall replace the National Map Policy, 2005.
- **Geospatial Knowledge Infrastructure (GKI):** Will be enabled by the integration of Geospatial data with Fourth Industrial Revolution technologies and the growing digital infrastructure.
- **Geospatial Education and Skill Development:** National Institute for Geo-informatics Science and Technology (NIGST), Indian Institute of Remote Sensing (IIRS), and/or any suitable institute(s), public or private, will be developed into Centre(s) of Excellence.
- **Geospatial Industrial Development Board (GIDB):** Advisory body under GDPDC.

National Data Governance Policy

Aim—>

- To **enhance access, quality, and use of data**
- It will **provide datasets access to Indian researchers and startups.**
- **NDGFP does not apply to the private players**, but they can voluntarily contribute their datasets.
- No provision for monetization of such data.

National Data Governance Policy

Components

- **Indian Datasets Platform:** Consisting of anonymized non-personal datasets from Government entities collected from Indian citizens or those in India
- **Indian Data Management Office (IDMO)**
 - It will be set up under Digital India Corporation (DIC), under MeitY
- **Data Management Units (DMUs):** In every Ministry/Department to work closely with IDMO for ensuring implementation of the Policy

National Data Governance Policy

Objectives of the NDGFP are:

- To accelerate Digital Governance.
- To have standardized data management and security standards across whole of Government;
- To accelerate creation of common standard based public digital platforms while ensuring privacy, safety, and trust; (d) To have standard APIs and other tech standards for Whole of Government Data management and access.

National Data Governance Policy

Objectives of the NDGFP are:

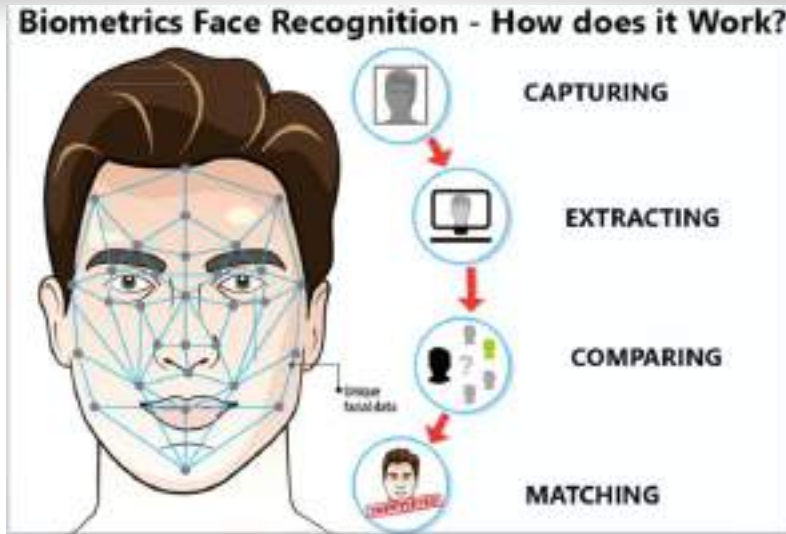
- To promote transparency, accountability, and ownership in Non-personal data and Datasets access. For purposes of safety and trust, any non-personal data sharing by any entity can be only via platforms designated and authorised by IDMO. (1) To build a platform that will allow Dataset requests to be received and processed.

National Data Governance Policy

Objectives of the NDGFP are:

- To build Digital Government goals and capacity, knowledge and competency in Government departments and entities.
- To set quality standards and promote expansion of India Datasets program and overall non-personal Datasets Ecosystem.
- To ensure greater citizen awareness, participation, and engagement.

Facial Recognition System



Facial Recognition System

- FRS is a way of identifying or confirming an individual's identity using their face.
- It can be used to identify people in photos, videos, or in real-time.
- Computer algorithms map unique **facial-landmarks** such as shape of cheekbones, contours of lips etc. and convert these into a numerical code— termed a **faceprint**.

Facial Recognition System

Applications

- FRS is part of the **Digi Yatra initiative** to promote seamless and hassle-free experience at airports and simultaneously improving the security.
- Government is looking to make ticket booking, airport entry and boarding pass security check-in digital.

DIFFERENT TYPES OF BIOMETRICS:



Typing Style



Navigation Style

Behavioral

Biometric Identifiers:



Interaction Style



Physical Style



Signature



Face Recognition

Physical

Biometric Identifiers:



Eye Scanners



Fingerprints



Voice Recognition



DNA

Chemical & Vein

Biometric Identifiers:



Vein Recognition

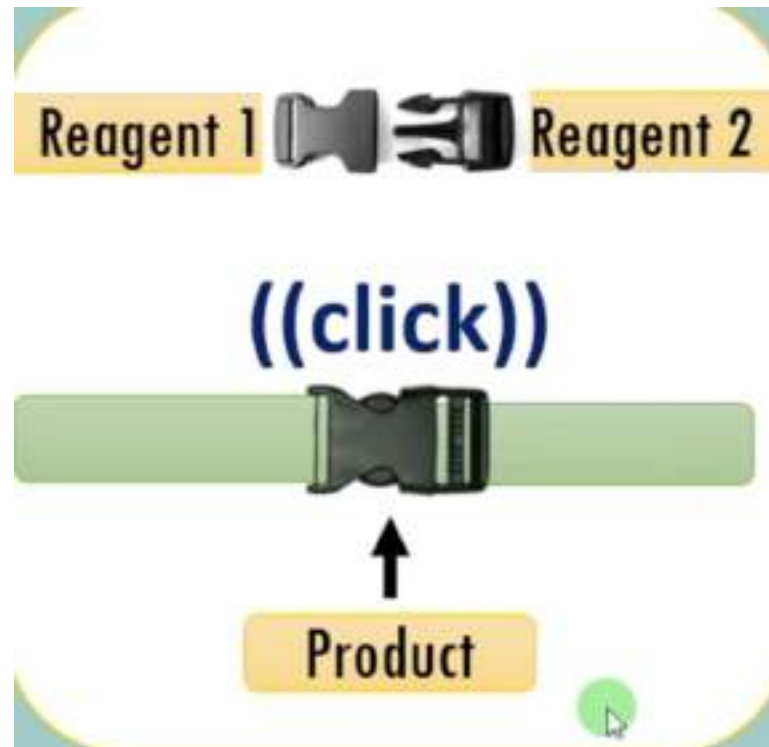
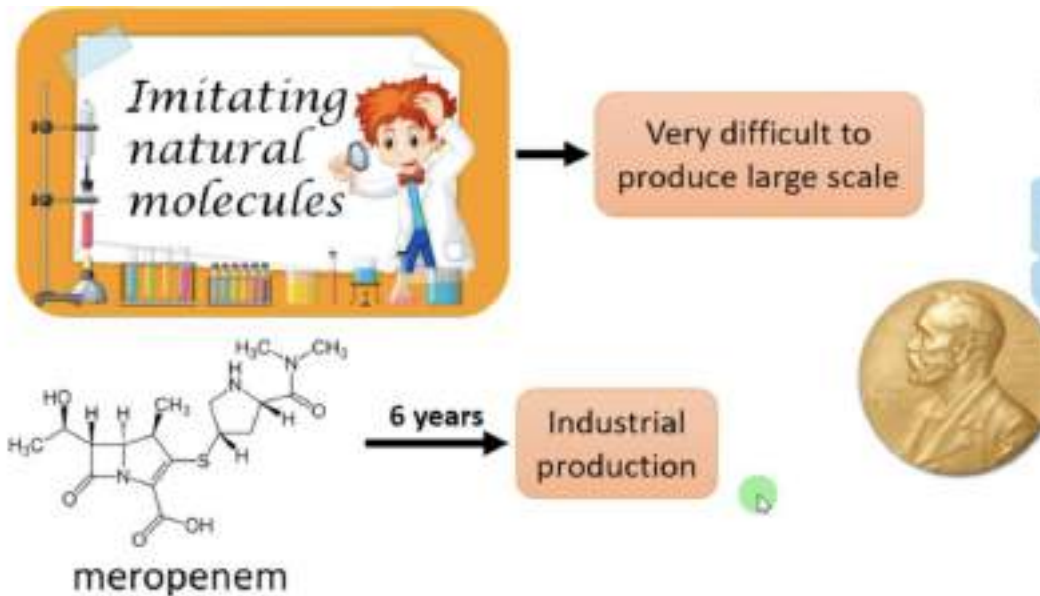
India Stack

- India stack is a collection of open APIs (**Application Programming Interface**) and digital public goods
- IndiaStack includes **APIs of Aadhaar, Unified Payment Interface (UPI), Co-Win, DigiLocker, Aarogya Setu, eSanjeevani, UMANG, DIKSHA**, etc
- Its objective is to unlock the economic opportunities of **identity, data, and payments** at population scale.

Role of Scientists

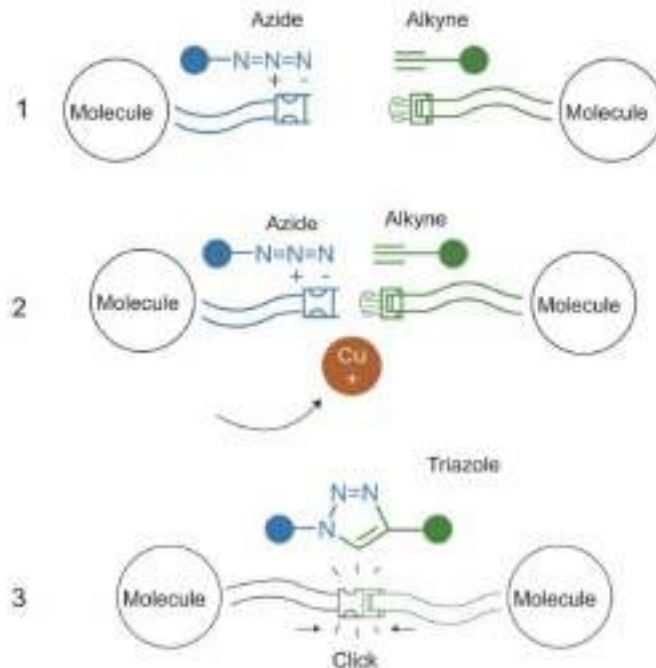
Nobel Prize in Chemistry 2022

- Click Chemistry and biorthogonal reactions
- Earlier, Biomolecules were being made from artificial copying of natural processes.
- This was leading to unwanted intermediate products and the process was expensive.
- This year the award is given to **Carolyn Bertozzi, Morten Meldal and K Barry Sharpless** for Click Chemistry.



The Click Reaction that changed chemistry

Azides and alkynes react very efficiently when copper ions are added. This reaction is now used globally to link molecules together in a simple manner.



Click Chemistry

- Rather than trying to make carbon atoms react with each other, click chemistry focuses on smaller molecules that already have a complete carbon frame. They are added together using click chemistry.
- One such reaction is copper catalyzed azide – alkyne cycloaddition.

Bioorthogonal Chemistry

- Carolyn Bertozzi developed click chemistry that could be used inside living organisms.
- Copper ions are cyto-toxic, so a different process was needed to be developed.
- These reactions help us tag 'glycans' which are present on the cell membrane.



Carolyn Bertozzi

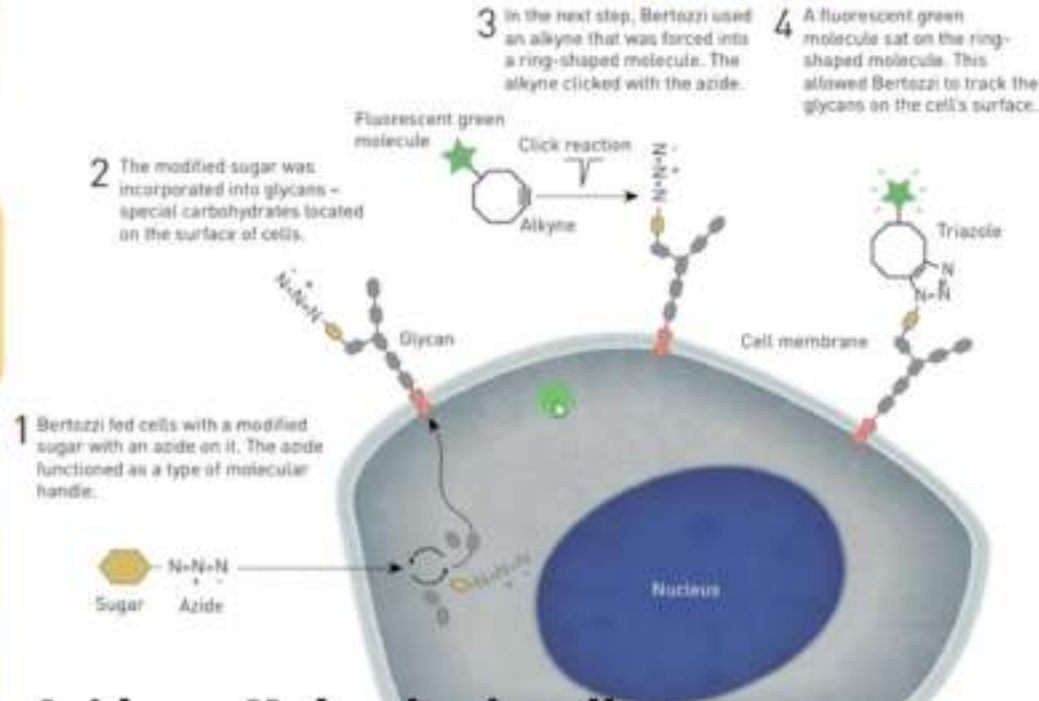
Mapping of
genes and
proteins to
understand
how cells work

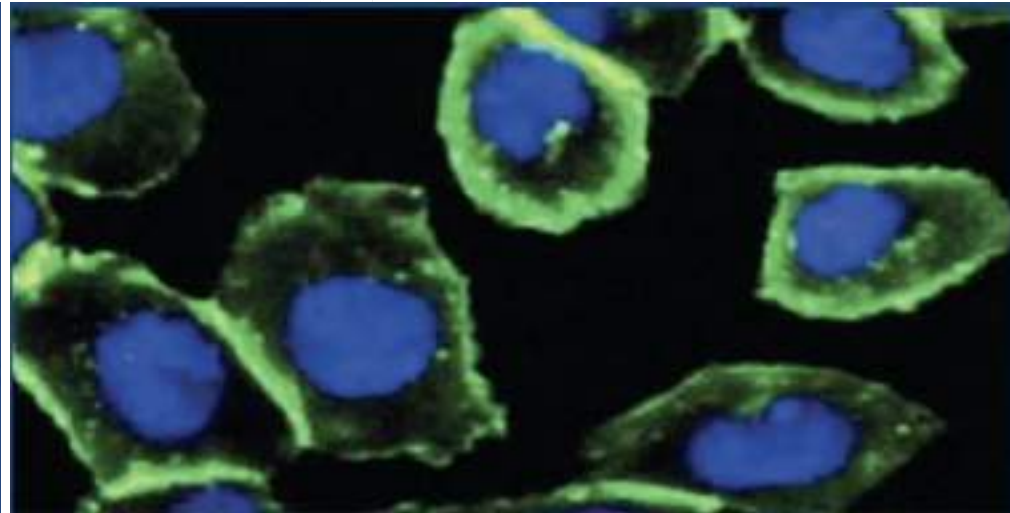
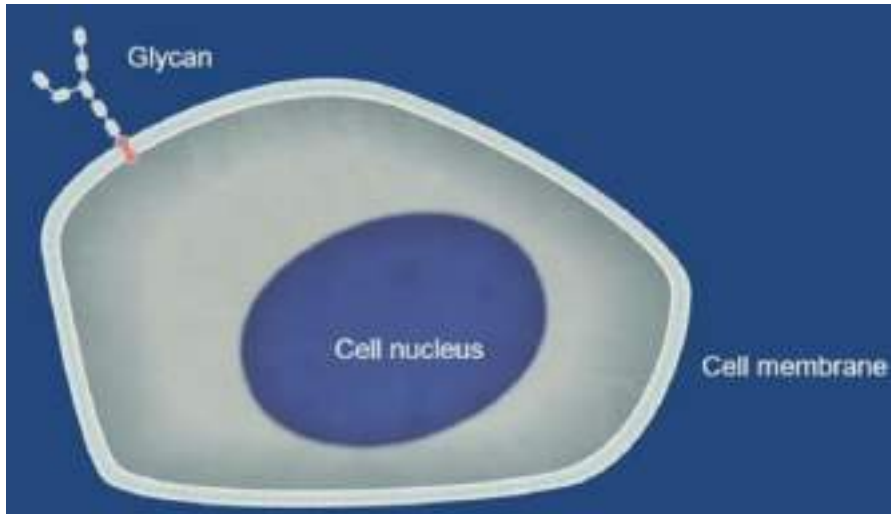
Glycans are
complex
carbohydrates
often sit on
the surface of
proteins and
cells.

Bioorthogonal chemistry illuminates the cell

strain-promoted alkyne-azide cycloaddition

The reaction between the
handle and the fluorescent
molecule had to be
bioorthogonal





Applications

- Development of pharmaceuticals, herbicides, photostabilisers etc.
- Mapping of complex biological processes like DNA, exploring how cells function.
- Targeted drug delivery for cancer cells.

Nobel Prize in Physics 2022

- Given to Alain Aspect (France), John F. Clauser (USA) and Anton Zeilinger (Austria)
- Potential to investigate and control particles in entangled states.
- Remember : what is entanglement?
- Change / measure the quantum state of one particle and it will automatically lead to a measurement of the other particle regardless of the distance between them.

Nobel Prize in Physics 2022

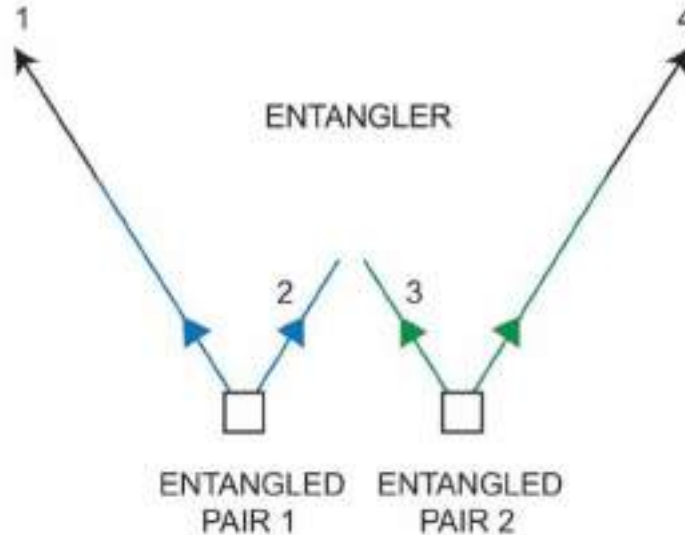
- They also demonstrated 3 other things :
- Quantum Teleportation
- Quantum Entanglement Swapping
- Theoretical work on bell inequalities : A set of Maths principles made by scientist JS Bell to mathematically prove entanglement and its existence.
- Measurement of 2 quantum entangled photons can be called bell measurement.

Nobel Prize in Physics 2022

- Quantum Teleportation : Transferring the quantum state of 1 photon to another through the help of quantum entanglement.
- Related words : No cloning, No deleting theorem.
- Entanglement Swapping

Entangled particles that never met

- Two pairs of entangled particles 1 & 2, 3 & 4 are emitted from different sources.
- One particle from each pair (2 and 3) is brought together in a special way that entangles.
- The two other particles (1 and 4 in the diagram) are then also entangled.
- In this way, two particles that have never been in contact can become entangled.



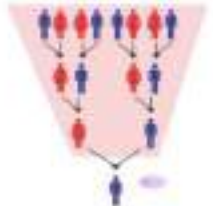
Applications

- Quantum Computing, Quantum Networks, secure quantum cryptography.
- Research in Quantum information science.

Nobel Prize in Physiology / Medicine 2022

- Awarded to Swedish geneticist Svante Paabo
- He sequenced the genome of a neanderthal. He used mitochondrial dna for his analysis as there are less changes in mitochondrial dna from one generation to the next.
- He also discovered a previously unknown hominin called denisova, discovered in 2008 from southern part of Siberia. (denisova cave)

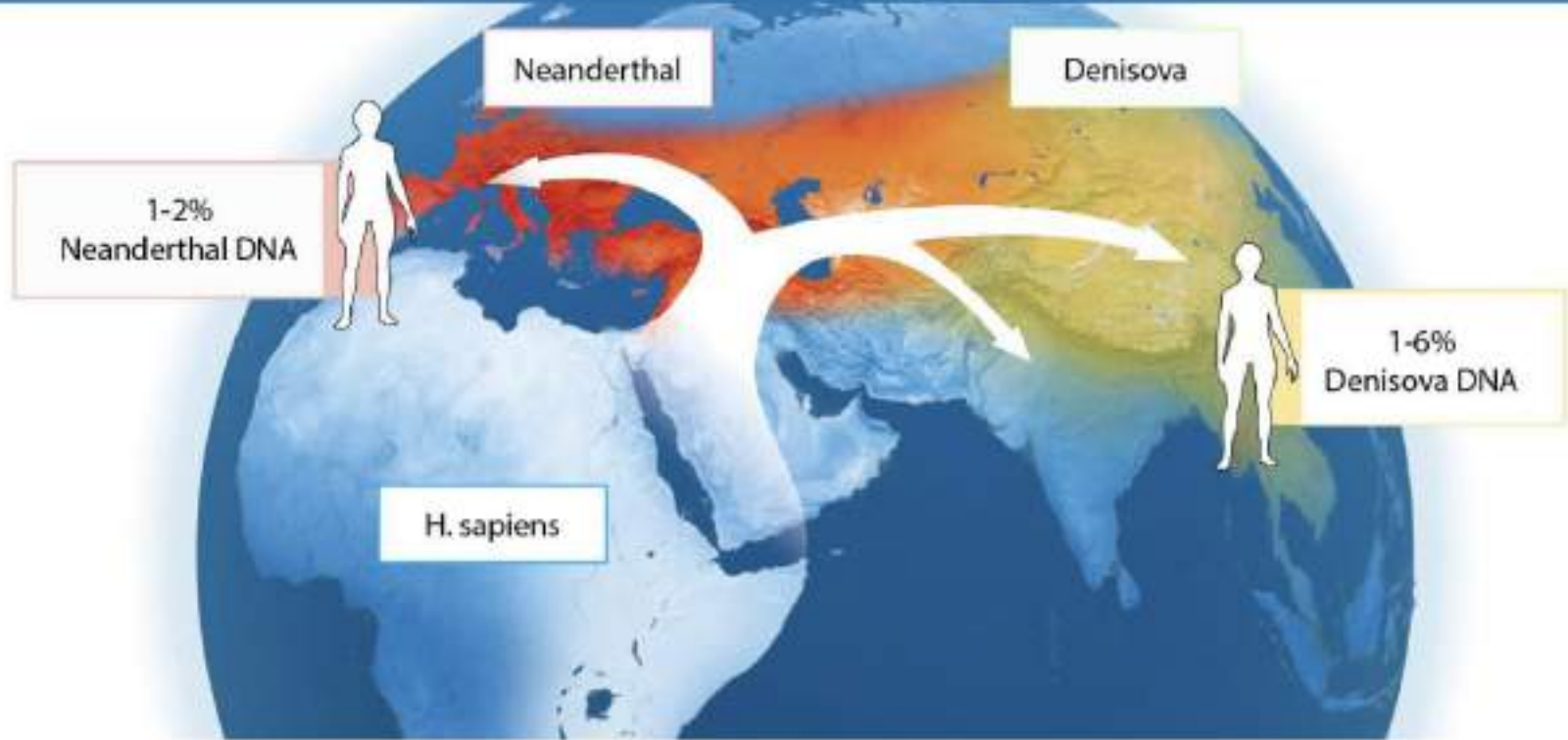
nDNA VS. mtDNA

	Inheritance	Shape	Consists of	Share in genetic makeup	Replication
Nuclear DNA (nDNA) 	Nuclear DNA inherited from all ancestors 	Linear 	Cell's genome (including mtDNA) 	99.75% (~16,500 base pairs) 	Replicates only during the cell creation 
Mitochondrial DNA (mtDNA) 	Nuclear DNA inherited from all ancestors 	Mitochondrial DNA 	Mitochondrial genome 	0.25% (~3.3 billion base pairs) 	Replicates independently 

Svante Paabo

- He found how gene transfer has occurred from extinct hominins to Homo Sapiens following the migration out of Africa.
- Related terms : Out of Africa theory, Multiregional Hypothesis.
- This study is a new field called paleogenomics : analysis of genes of ancient or extinct organisms.

SPREADING OF DNA



Neanderthals and AMHS

- Neanderthals developed outside Africa in Europe and west asia, lived from 400,000ya to 30,000 ya.
- Anatomically modern Homo Sapiens first appeared in Africa 300,000 years ago. 70,000ya they migrated outside Africa and had genetic exchanges with neanderthals and denisovans, getting some of their genes into our genome.

Neanderthals and AMHS

- This study helps us better understand human evolution and human diversity
- European populations : closer to neanderthals : different immune responses.
- Denisovans : EPAS-1 gene : survival at high altitude : Tibetans.

Neanderthals and AMHS

- Related terms
- Neanderthal types: Classical, La chapelle aux saints.
- Homo Sapien types : Cro magnon, Grimaldi, Chancaledé.

Facts related to Nobel Prizes

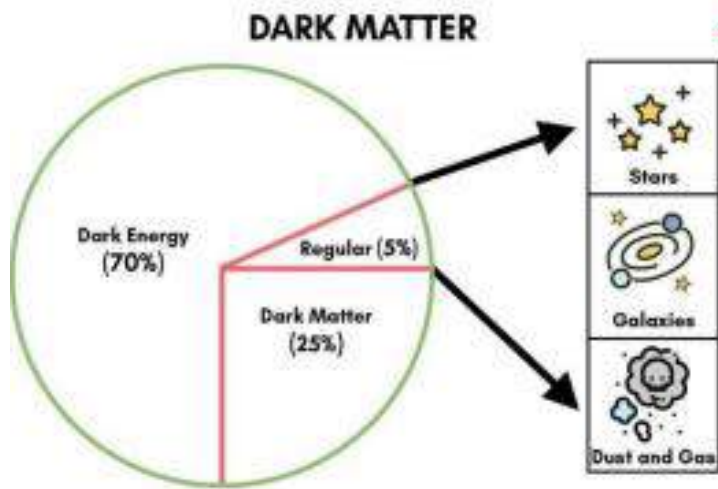
- 5 people have won the prize twice : Madam Curie, Linus Pauling, John Bardeen, Fredrick Sanger, K Barry Sharpless (recent).
- International Committee of the red cross : won it thrice.
- UNHCR : won twice.
- Cannot be awarded posthumously.

DARK MATTER

- A test run of LZ detector in U.S. has shown it to be the most sensitive dark matter detector yet created
- It is designed (underground) to capture dark matter in the form of weakly interacting massive particles (WIMPs)
- Centre of LZ is one of the purest places on Earth (free of radiation and dust).
- Collaboration of scientists/institutions from U.S., U.K., Portugal, and Korea.

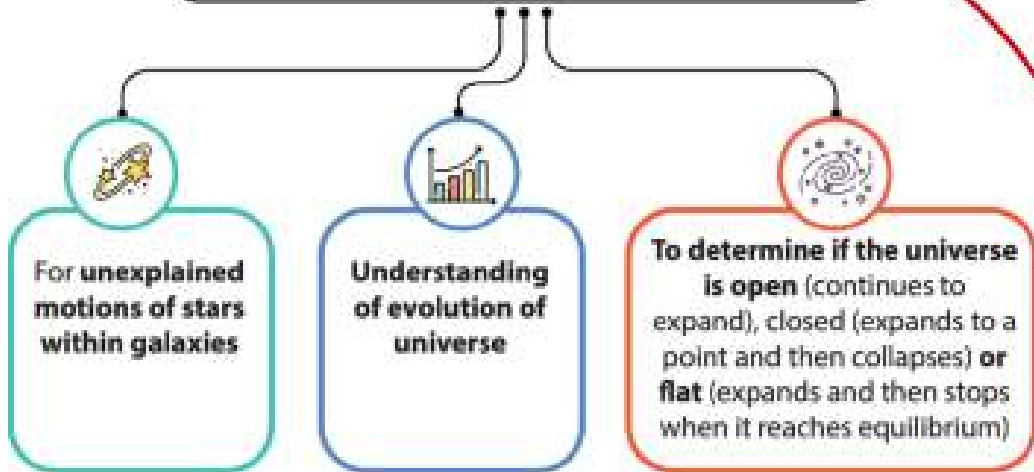
DARK MATTER

- dark matter does not interact with electromagnetic force. This means it does not absorb, reflect, or emit light, making it extremely hard to spot.
- Scientists study dark matter by looking at the effects it has on visible objects.
- It is believed that dark matter is what gives galaxies extra mass, generating extra gravity they need to stay intact.





Significance of Dark Matter Study



About Dark Energy

- Universe has been known to be expanding as a consequence of the Big Bang.
- First hypothesized by Albert Einstein
- Discovered that this expansion is accelerating (Nobel Prize in Physics 2011)
- Dark energy is the name given to the mysterious force that's causing the rate of expansion of universe to accelerate over time.

Standard Model of Particle Physics

- We want to find the fundamental particle of the universe.
- Fundamental : cannot be divided further.
- Earlier : we thought this is an atom
- Then we discovered that Atoms can be divided into protons, neutrons and electrons.
- Further division of neutrons and protons are also possible.

Standard Model of Particle Physics

- Further division : Quarks, Leptons.
- 6 types of Quarks : up, down, charm, strange, top, bottom.
- 3 types of charged Leptons : electron, muon, tau ;
- 3 types of uncharged leptons : muon neutrino, tau neutrino, electron neutrino

Standard Model of Particle Physics

- Forces : there are 4 fundamental forces.
- Gravity : weakest, not part of standard model.
- Other 3 are part of standard model.
- Electromagnetism : light etc.
- Weak nuclear force : responsible for radioactivity
- Strong Nuclear Force : holds quarks together.

Forces in standard model

- Imagine throwing things off a boat : it will generate a force.
- There are some small particles that are exchanged to carry out the fundamental forces.
- Gluon : strong Nuclear Force.
- Photon : Electromagnetic Force
- W/ Z boson : weak nuclear force.
- Gravity : we don't know yet.

Standard Model

- How did these particles get mass?
- How was the universe formed?
- Higgs boson particle : also known as god particle : explains how these fundamental particles got mass. It was first demonstrated in the large hadron collider at cern.

CERN



European Organization for Nuclear Research (CERN)



Geneva, Switzerland

About: Founded in 1954, researchers at CERN are probing fundamental structure of the universe and study basic constituents of matter – the fundamental particle.

Members: 23 Members. India is an Associate Member.



Mission: To perform research in fundamental physics.

- To provide a unique range of particle accelerator facilities.
- To use its flagship accelerator, Large Hadron Collider (LHC), to its maximum potential.

Key Achievements: Higgs Boson, W Boson, Z Boson, LHC, World Wide Web etc.





Large Hadron Collider



It is the world's largest and most powerful particle accelerator.



LHC is pursuing the elusive dark matter



Started in 2008 and remains the latest addition to CERN's accelerator complex.



How it works?



Acceleration

It consists of a 27-kilometre ring of superconducting magnets with a number of accelerating structures to boost the energy of the particles along the way



Collision

Inside the accelerator, two high-energy particle beams travel at close to the speed of light before they are made to collide.



Detection

Four particle detectors record the tracks left by debris from the collisions.



Four particle detectors

**ATLAS (A Toroidal
LHC ApparatuS)**

**CMS (the
Compact Muon
Solenoid)**

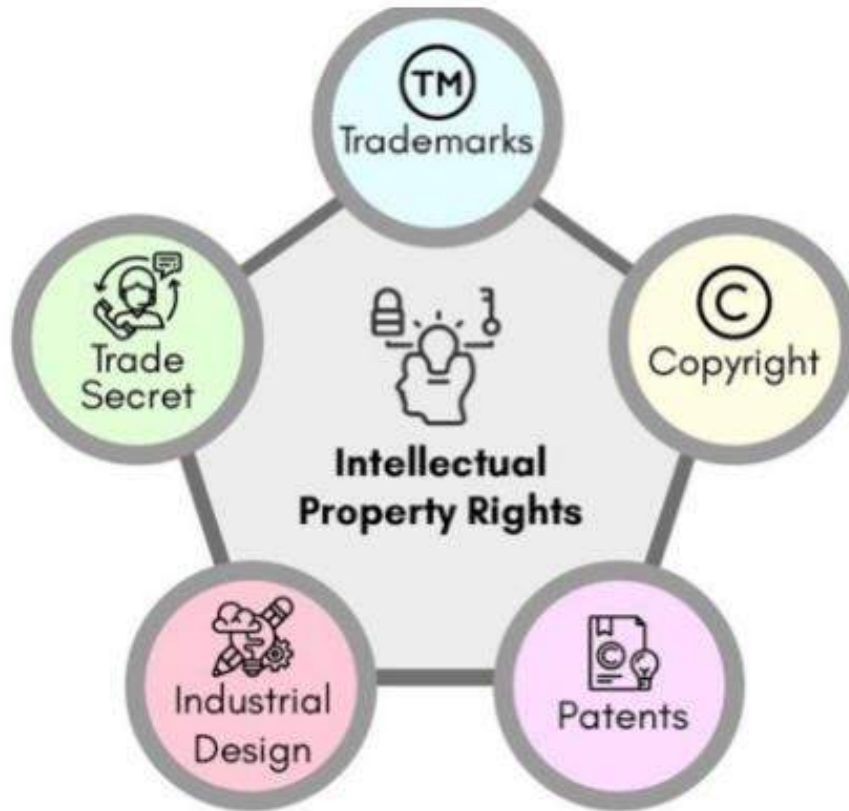
**ALICE (A Large
Ion Collider
Experiment)**

**LHCb (study
of physics in
B-meson
decays at LHC)**



IPR

- IPRs are the rights given to persons over the creations of their minds. They usually give the creator an exclusive right over the use of his/her creation for a certain period of time.
- It can include creations such as a new drug composition, business module, product, software and so on.



Copyright

- Copyright is a legal term used to describe the rights that creators have over their literary and artistic works.
- Range from books, music, paintings, films, programs, advertisements, maps and technical drawings.
- The general rule is that copyright lasts for 60 years. In the case of original literary, dramatic, musical and artistic works the 60- year period is counted from the year following the death of the author.

Patents

- A patent is an exclusive right granted for an invention. It provides the patent owner with the right to decide how - or whether - the invention can be used by others.
- Patents should be obtained in each country where the applicant requires protection of his invention.

Patents

- The product or process should provide in general, a new technical solution to a problem of any field. An invention relating either to a product or process that is new, involving an inventive step and capable of industrial application can be patented.
- The term of every patent in India is twenty years from the date of filing the patent application.

What cannot be patented ?

- Frivolous inventions. E.g. schoolbag, laptop bag
- Anything seriously harmful to public order, morality, humans, animals, plants or environment.
- Atomic energy related inventions.
- Mere discovery of a scientific principle or the formulation of an abstract theory or discovery of a living thing
- Plants-seeds varieties, biological process to create them

What cannot be patented ?

- Animals breeding; Medical treatment of humans and animals
- Mere discovery of a new form of a known substance which does not result in the increased efficiency.
- Mere re-arrangement or duplication of known devices.
- Mere aggregation of properties of the components e.g. lemon + sugar + water = juice

What cannot be patented ?

- Traditional knowledge. E.g. Ayurvedic information.
- Mathematical formula, business method, method of playing game for ex :helicopter shot.
- computer program, presentation of information, integrated circuit
- literary, dramatic, musical or artistic work; cinema-TV shows

Compulsory Licensing

- It enables a competent government authority to license the use of a patented invention to a third party or government agency without the consent of the patentholder.
- Section 92 of the Patent Act, 1970, provides for issuing CL by the central government (permitted under the WTO's TRIPS (IPR) Agreement) in circumstances of “national emergency or in
- circumstances of extreme urgency or in case of public non-commercial use”.

Parallel Import

- It is importation without the consent of the patentholder of a patented product marketed in another country either by the patent holder or with the patentholder's consent.
- For example : USA based company Pfizer sells a drug Lipitor in congo for \$1 / pill.
- Same drug is needed by India but India gets the drug at \$5 / pill
- India can use parallel import to get the drug.

Evergreening of patent

- Ever-greening of patent: Section 3(d) Indian Patent Act doesn't allow the renewal of patent over a product by introducing minor changes to it. It has been an issue of concern for pharmaceutical companies (Ex: Novartis issue on the issue of patent of cancer drug Glivec).
- Done by pharma companies to extend patent term beyond 20 years

TKDL

- Traditional Knowledge Digital Library (TKDL) is a pioneering initiative of India to prevent misappropriation of the country's traditional medicinal knowledge at International Patent Offices on which healthcare needs of more than 70% population and livelihood of millions of people in India is dependent. Its genesis dates back to the Indian effort on revocation of the patent on wound healing properties of turmeric at the USPTO.

TKDL

- Besides, in 2005, the TKDL expert group estimated that about 2000 wrong patents concerning Indian systems of medicine were being granted every year at international level, mainly due to the fact that India's traditional medicinal knowledge which exists in local languages such as Sanskrit, Hindi, Arabic, Urdu, Tamil etc. is neither accessible nor comprehensible for patent examiners at the international patent offices. So, it can be said TKDL is a vital tool to fight against biopiracy and erroneous filing of patents.

Advantages

- Informed decisions
- Available to every citizen
- Economic Benefits
- Cheap Medicines
- Acceptance
- Indigenous Research

Disadvantages

- Prone to data theft
- IPR issues
- Discourage investment
- Indigenous communities deprived of their rights

Utility Model

- A Utility Model just like a Patent also protects inventions/innovations but for a shorter period.
- The main difference between a Patent and Utility Model is that the requirements for granting a Utility Model are less stringent than for Patents.
- A product or process which is Novel and Industrial applicable passes for Utility Model, inventive step is not a requirement.
- The term of protection for utility models is shorter than for patents,(usually between 6 and 15 years).

Trademarks

- A trademark is a sign capable of distinguishing the goods or services of one enterprise from those of other enterprises.
- A trademark can be any distinctive word, symbol, slogan, logo, brand label, name, signature, letter, numeral or any combination of them.
- The trademark is initially registered for a period of 10 years, which is calculated from the date of filing of the application.

Industrial Design

- An industrial design constitutes the ornamental or aesthetic aspect of an article.
- A design may consist of three-dimensional features, such as the shape or surface of an article, or of two- dimensional features, such as patterns, lines or color.
- In India the maximum validity of a registration under the (Indian) Designs Act, 2000 can be 15 years

Geographical Indications

- Signs used on goods that have a specific geographical origin and possess qualities, a reputation or characteristics that are essentially attributable to that place of origin.
- Most commonly, a geographical indication includes the name of the place of origin of the goods.
- It is registered for a period of 10 years and the registration may be renewed from time to time for a period of 10 years at a time.

Trade Secrets

- Trade secrets are IP rights on confidential information which may be sold or licensed.
- Examples include formulae, recipes, pattern, technique, compilation, method, program, process, device or product mechanism.
- Trade secret remains valid as long as one does not discover it independently.

EARTH RECORDS SHORTEST DAY

- 29 July 2022 has set the record for the shortest day.
- Possible reason for increased speed: Changes in the processes in the inner or outer layers of the core, oceans, tides, or even changes in climate To combat the long-term slowdown in the Earth's rotation which is caused by the constant melting and refreezing of ice caps, leap second to our calendars were added occasionally.
- Leap second was first introduced in 1972 by the International Earth Rotation and Reference Systems Service.

Negative leap seconds:

- If Earth continues to rotate at an increasing rate, negative leap seconds would need to be introduced
- Atomic clocks are designed to measure the precise length of a second,
- Issues with introduction of Negative leap seconds: Can confuse smartphones, computers, and communications systems.

About Chandler Wobble

- Change in the spin of Earth as it rotates on its axis, much as a toy top wobble as it spins.
- Time Period ; 433 days Wobble amounts to about 20 feet at the North Pole.
- Possible Cause: Fluctuating pressure on the bottom of the ocean, caused by temperature and salinity changes and wind-driven changes in circulation of the oceans

Oral Rehydration Solution



- Dr Dilip Mahalanabis who pioneered the proper practical, emergency use of oral rehydration solution, for diarrhoeal diseases passed away recently.
- Dr Dilip Mahalanabis was working in refugee camps during the 1971 Bangladesh Liberation war
- ORS: mixture of electrolytes (salts) and carbohydrates (in the form of sugar) dissolved in water. Used to replace salts and water that the body loses during dehydration caused by gastroenteritis, diarrhoea or vomiting. Contains potassium and sodium.
- These components maximize fluid absorption in the gastrointestinal tract.
- Gastrointestinal tract relies on sodium-glucose cotransporters (SGLTs)
- Cotransporters help move substances across membranes.

Indian Biological Data Center



- Indian Biological Data Center (IBDC) was inaugurated at Faridabad, Haryana
- First national repository for life science data
- Storage: Four-petabyte supercomputer '**Brahm**'
- Supported by: Department of Biotechnology (DBT).
- **Data sharing:** FAIR (Findable, Accessible, Interoperable and Reusable)
- **FAIR Data Principles:** support the reusability of digital assets.

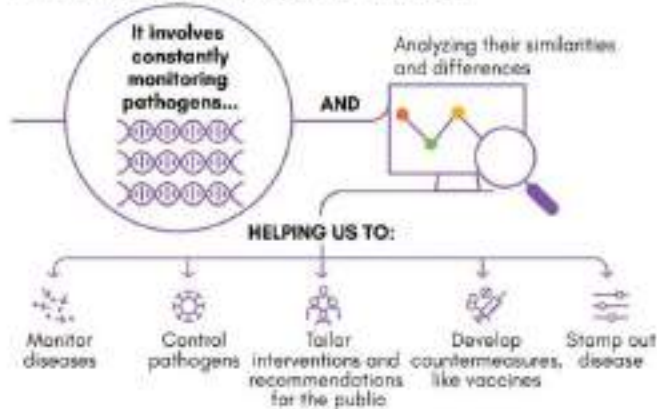
Indian Biological Data Center - Cont'd



- Also contains genomes sequenced by the Indian Sars-CoV-2 Genomic Consortium (INSACOG)
- Two sections have been developed:
 - Indian Nucleotide Data Archive (INDA) shall archive data generated from publicly funded research projects in India and provide internationally accepted data accession numbers
 - Indian Nucleotide Data Archive – Controlled Access (INDA-CA)

GENOMIC SURVEILLANCE

WHAT IS GENOMIC SURVEILLANCE?



APPLICATIONS OF GENOMICS:

- Human genomics for medical purposes** (Helping to treat genetic disorders)
- Microbial genomics for medical purposes** (Identification of infectious agents, vaccine design etc.)
- Agriculture and aquaculture genomics** (Identification of new traits and disease susceptibility etc.)
- Biological and medical research** (drug development, discovery of genes influencing disease etc.)
- Others** (forensic science, assessment of ancestry etc.)

GENOMIC SURVEILLANCE IN INDIA

Regional Genome Sequencing Laboratories (ROSL) serve as the regional hub for genome sequencing.

Indian SARS-CoV-2 Genomics Consortium (INSACOG) was established to expand Whole Genome Sequencing of SARS-CoV-2 across the nation*.

Indian Tuberculosis Genomic Surveillance Consortium (InTGS) is proposed on the lines of INSACOG.

*INSACOG, jointly initiated by Ministry of Health & Family Welfare, and Department of Biotechnology (DBT) with Council for Scientific & Industrial Research (CSIR) and Indian Council of Medical Research (ICMR), is a consortium of over 50 laboratories.

Dengue

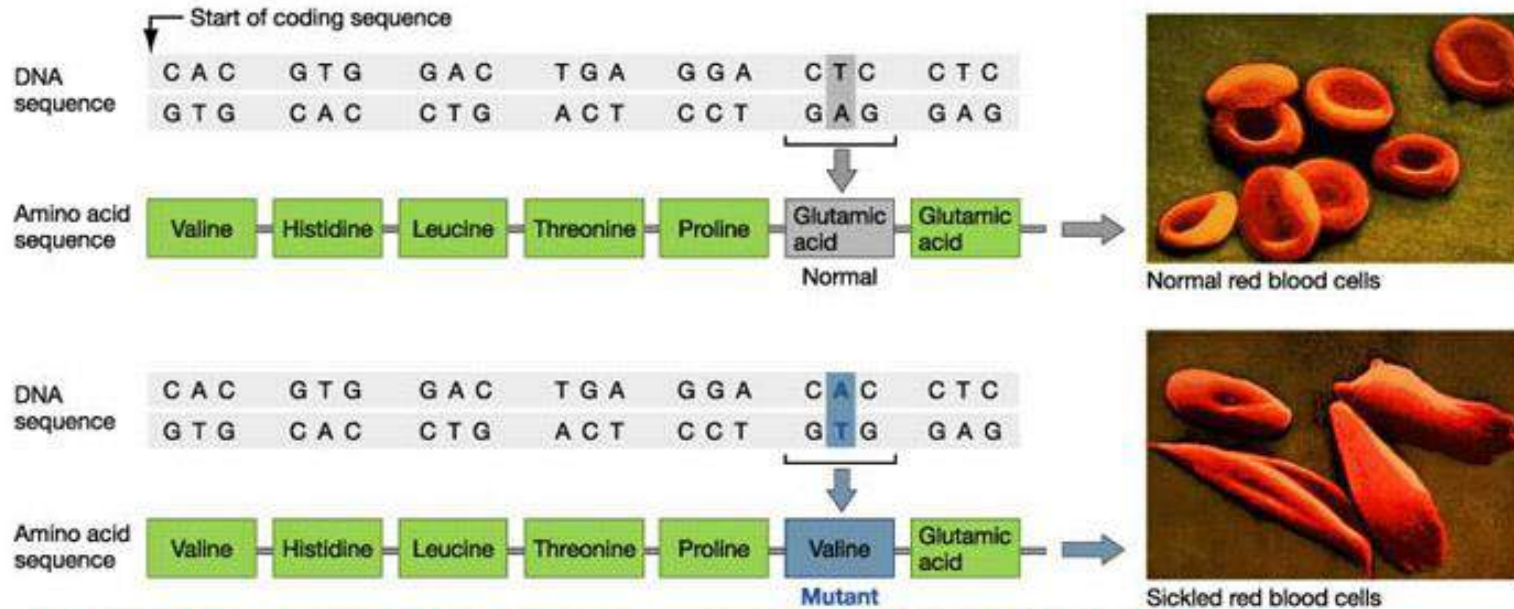


- Mosquito borne tropical disease caused by dengue virus (flavivirus)
- Spread through bite of infected aedes mosquito.
- First dengue vaccine : dengvaxia : approved by USFDA in 2019

Sickle Cell Anaemia



- Important this year : National Mission to Eliminate sickle cell anaemia launched in budget.
- Occurs due to genetic mutation that causes hemoglobin in RBCs to clump together.
- RBC shape changes to Sickle shape : flexibility lost.



The change in amino acid sequence causes hemoglobin molecules to crystallize when oxygen levels in the blood are low. As a result, red blood cells sickle and get stuck in small blood vessels.

Other Important News



- Trachoma : first country in southern Africa to eliminate trachoma is Malawi. India eliminated trachoma in 2017. Trachoma is an infection of the eye caused by Chlamydia Trachomatis. Can cause blindness.
- Diabetes : recent case of infection from diabetes medication. Diabetes is of 2 types : type 1 and type 2.
- Type 1 : insulin dependent, juvenile or childhood onset.
- Type 2 : Non insulin dependent : adult onset. (most common)
- Shigella : bacteria causing shigellosis. Leads to diarrhoea in children. No vaccine available.



Some Final Words on Science and Technology

Environment Starts Tomorrow



- Format will be the same : slides and one pagers.
- Class timings : 4pm – 6:30 pm
- Access will be granted by tonight to everyone.
- Should Take around 13 classes, will be easily done by 20th April (this was the original commitment)
- Will share the class plan tomorrow.

Test of 20 Questions!

Cut off : 15/20



Q.1 (2016)

Q. With reference to 'LiFi', recently in the news, which of the following statements is/are correct?

1. It uses light as the medium for high-speed data transmission.
2. It is a wireless technology and is many times faster than 'WiFi'.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2
- d) Neither 1 nor 2

Q.2 (2016)

Q. 'Project Loon', sometimes seen in the news, is related to

- a) Waste management technology
- b) Wireless communication technology
- c) Solar power production technology
- d) Water conservation technology

Q.3 (2018)

Q. Consider the following pairs :

Terms sometimes seen in news

Context /Topic

1. Belle II experiment -

Artificial Intelligence

2. Blockchain technology -

Digital/Cryptocurrency

3. CRISPR — Cas9 -

Particle Physics

Which of the pairs given above is/are correctly matched ?

a) 1 and 3 only

b) 2 only

c) 2 and 3 only

d) 1, 2 and 3

Q.4 (2018)

Q. When the alarm of your smartphone rings in the morning, you wake up and tap it to stop the alarm which causes your geyser to be switched on automatically. The smart mirror in your bathroom shows the day's weather and also indicates the level of water in your overhead tank. After you take some groceries from your refrigerator for making breakfast, it recognises the shortage of stock in it and places an order for the supply of fresh grocery items. When you step out of your house and lock the door, all lights, fans, geysers and AC machines get switched off automatically. On your way to the office, your car warns you about traffic congestion ahead and suggests an alternative route, and if you are late for a meeting, it sends a- message to your office accordingly.

In the context of emerging communication technologies, which one of the following term" best applies to the above scenario?

- a) Border Gateway Protocol
- b) Internet of Things
- c) Internet Protocol
- d) Virtual Private Network

Q.5 (2018)

Q. "3D printing" has applications in which of the following?

1. Preparation of confectionery items
2. Manufacture of bionic ears
3. Automotive industry
4. Reconstructive surgeries
5. Data processing technologies

Select the correct answer using the code given below :

- | | |
|--------------------|---------------------|
| a) 1, 3 and 4 only | b) 2, 3 and 5 only |
| c) 1 and 4 only | d) 1, 2, 3, 4 and 5 |

Q.6 (2019)

Q. With reference to communication technologies, what is/are the difference/differences between LTE (Long-Term Evolution) and VoLTE (Voice over Long-Term Evolution)?

1. LTE is commonly marketed as 3G and VoLTE is commonly marketed as advanced 3G.
2. LTE is data-only technology and VoLTE is voice-only technology.

Select the correct answer using the code given below.

- | | |
|-----------------|--------------------|
| a) 1 only | b) 2 only |
| c) Both 1 and 2 | d) Neither 1 nor 2 |

Q.7 (2019)

Q. In the context of digital technologies for entertainment, consider the following statements:

1. In Augmented Reality (AR), a simulated environment is created and the physical world is completely shut out.
2. In Virtual Reality (VR), images generated from a computer are projected into real-life objects or surroundings.
3. AR allows individuals to be present in the world and improves the experience using the camera of smart-phone or PC.
4. VR closes the world, and transposes an individual, providing complete immersion experience.

Which of the statements given above is/are correct?

- | | |
|-----------------|------------|
| a) 1 and 2 only | b) 3 and 4 |
| c) 1, 2 and 3 | d) 4 only |

Q.8 (2019)

Q. Consider the following statements: A digital signature is

1. an electronic record that identifies the certifying authority issuing it
2. used to serve as a proof of identity of an individual to access information or server on Internet
3. an electronic method of signing an electronic document and ensuring that the original content is unchanged

Which of the statements given above is/are correct?

- | | |
|-----------|-----------------|
| a) 1 only | b) 2 and 3 only |
| c) 3 only | d) 1, 2 and 3 |

Q.9 (2019)

Q. In the context of wearable technology, which of the following tasks is/are accomplished by wearable devices?

1. Location identification of a person
2. Sleep monitoring of a person
3. Assisting the hearing impaired person

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 3 only
- d) 1, 2 and 3

Q.10 (2020)

Q. Consider the following activities:

1. Spraying pesticides on a crop field
 2. Inspecting the craters of active volcanoes
 3. Collecting breath samples from spouting whales for DNA analysis
- At the present level of technology, which of the above activities can be successfully carried out by using drones?
- a) 1 and 2 only
 - b) 2 and 3 only
 - c) 1 and 3 only
 - d) 1,2 and 3

Q.11 (2020)

Q. In India, the term “Public Key Infrastructure” is used in the context of

- a) Digital security infrastructure
- b) Food security infrastructure
- c) Health care and education infrastructure
- d) Telecommunication and transportation infrastructure

Q.12 (2020)

Q. With the present state of development, Artificial Intelligence can effectively do which of the following?

1. Bring down electricity consumption in industrial units.
2. Create meaningful short stories and songs.
3. Disease diagnosis.
4. Text-to-Speech Conversion.
5. Wireless transmission of electrical energy.

Select the correct answer using the code given below:

- | | |
|-----------------------|---------------------|
| a) 1, 2, 3 and 5 only | b) 1, 3 and 4 only |
| c) 2, 4 and 5 only | d) 1, 2, 3, 4 and 5 |

Q.13 (2020)

Q. With reference to Visible Light Communication (VLC) technology, which of the following statements are correct?

1. VLC uses electromagnetic spectrum wavelengths 375 to 780 nm.
2. VLC is known as long-range optical wireless communication.
3. VLC can transmit large amounts of data faster than Bluetooth
4. VLC has no electromagnetic interference.

Select the correct answer using the code given below:

- a) 1, 2 and 3 only
- b) 1, 2 and 4 only
- c) 1, 3 and 4 only
- d) 2, 3 and 4 onl

Q.14 (2020)

Q. With reference to “Blockchain Technology”, consider the following statements:

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. The structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that depend on basic features of blockchain can be developed without anybody’s permission.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 only
- d) 1 and 3 only

Q.15 (2022)

Q. With reference to Web 3.0, consider the following statements:

1. Web 3.0 technology enables people to control their own data.
2. In Web 3.0 world, there can be blockchain-based social networks.
3. Web 3.0 is operated by users collectively rather than a corporation

Which of the following given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.16 (2022)

Q. With reference to “Software as a Service”, Consider the following statements.

1. SaaS buyers can customise the user interface and can change data fields.
2. SaaS users can access their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.17 (2022)

Q. Which one of the following is the context in which the term “qubit” is mentioned?

- (a) Cloud Services
- (b) Quantum Computing
- (c) Visible Light Communication Technologies
- (d) Wireless Communication Technologies

Q.18 (2022)

Q. Consider the following communication technologies:

1. Closed-circuit Television
2. Radio Frequency Identification
3. Wireless Local Area Network

Which of the above are considered of Short-Range devices/technologies?

- (a) 1 and 2 only.
- (b) 2 and 3 only
- (c) 1 and 3 only
- (d) 1, 2 and 3

Q.19 (2014)

Q. In addition to fingerprint scanning, which of the following can be used in the biometric identification of a person?

1. Iris scanning
2. Retinal scanning
3. Voice recognition

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1, 2 and 3

Q.20 (2015)

Q. With reference to 'Near Field Communication (NFC) Technology', which of the following statements is/are correct?

1. It is a contactless communication technology that uses electromagnetic radio fields
2. NFC is designed for use by devices which can be at a distance of even a metre from each other
3. NFC can use encryption when sending sensitive information.

Select the correct answer using the code given below.

- | | |
|-----------------|--------------|
| a) 1 and 2 only | b) 3 only |
| c) 1 and 3 only | d) 1,2 and 3 |

Congratulations on giving the test!

Let's Discuss The Answers.



Q.1 (2016)

Q. With reference to 'LiFi', recently in the news, which of the following statements is/are correct?

1. It uses light as the medium for high-speed data transmission.
2. It is a wireless technology and is many times faster than 'WiFi'.

Select the correct answer using the code given below:

- a) 1 only
- b) 2 only
- c) Both 1 and 2**
- d) Neither 1 nor 2

Answer: C

- LiFi is a wireless optical networking technology that uses light-emitting diodes (LEDs) for data transmission.
- LiFi is designed to use LED light bulbs similar to those currently in use in many energy-conscious homes and offices. However, LiFi bulbs are outfitted with a chip that modulates the light imperceptibly for optical data transmission.
- LiFi data is transmitted by the LED bulbs and received by photoreceptors.
- In terms of its end use the technology is similar to Wi-Fi.
- The key technical difference is that Wi-Fi uses radio frequency to transmit data.
- Using light to transmit data allows Li-Fi to offer several advantages like working across higher bandwidth, working in areas susceptible to electromagnetic interference (e.g. aircra

Q.2 (2016)

Q. 'Project Loon', sometimes seen in the news, is related to

- a) Waste management technology
- b) Wireless communication technology**
- c) Solar power production technology
- d) Water conservation technology

Answer: B

- Loon LLC is an Alphabet Inc.(Parent company of Google) subsidiary working on providing Internet access to rural and remote areas.
- The company uses high-altitude balloons placed in the stratosphere at an altitude of about 18 km (11 mi) to create an aerial wireless network with up to 4G-LTE speeds.
- It was named Project Loon since even Google itself found the idea of providing Internet access to the remaining 5 billion population unprecedented and "loony".
- It may also be a reference to the balloons used.

Q.3 (2018)

Q. Consider the following pairs :

Terms sometimes seen in news

Context /Topic

1. Belle II experiment -

Artificial Intelligence

2. Blockchain technology -

Digital/Cryptocurrency

3. CRISPR — Cas9 -

Particle Physics

Which of the pairs given above is/are correctly matched ?

a) 1 and 3 only

b) 2 only

c) 2 and 3 only

d) 1, 2 and 3

Answer: B

- The blockchain is an undeniably ingenious invention which created the backbone of a new type of internet originally devised for the digital currency, Bitcoin.
- Genome editing (also called gene editing) is a group of technologies that give scientists the ability to change an organism's DNA. These technologies allow genetic material to be added, removed, or altered at particular locations in the genome.
- Several approaches to genome editing have been developed.
- A recent one is known as CRISPR-Cas9, which is short for clustered regularly interspaced short palindromic repeats and CRISPR-associated protein 9.
- The CRISPR-Cas9 system has generated a lot of excitement in the scientific community because it is faster, cheaper, more accurate, and more efficient than other existing genome editing methods.

Q.4 (2018)

Q. When the alarm of your smartphone rings in the morning, you wake up and tap it to stop the alarm which causes your geyser to be switched on automatically. The smart mirror in your bathroom shows the day's weather and also indicates the level of water in your overhead tank. After you take some groceries from your refrigerator for making breakfast, it recognises the shortage of stock in it and places an order for the supply of fresh grocery items. When you step out of your house and lock the door, all lights, fans, geysers and AC machines get switched off automatically. On your way to the office, your car warns you about traffic congestion ahead and suggests an alternative route, and if you are late for a meeting, it sends a- message to your office accordingly.

In the context of emerging communication technologies, which one of the following term" best applies to the above scenario?

- a) Border Gateway Protocol
- c) Internet Protocol

b) Internet of Things

- d) Virtual Private Network

Answer: B

- Internet of things is a seamless connected network system of embedded objects/ devices, with identifiers, in which communication without any human intervention is possible using standard and interoperable communication protocols. With this technology, each and every device that we use in daily life (including refrigerators, cars and even toothbrush) can be made intelligent and smart by using device-specific operating systems and connectivity to the Internet

Q.5 (2018)

Q. "3D printing" has applications in which of the following?

1. Preparation of confectionery items
2. Manufacture of bionic ears
3. Automotive industry
4. Reconstructive surgeries
5. Data processing technologies

Select the correct answer using the code given below :

- a) 1, 3 and 4 only b) 2, 3 and 5 only
c) 1 and 4 only **d) 1, 2, 3, 4 and 5**

Answer: D

- There are endless possibilities for the utilization of 3d Printing. We can do many things with 3d printing machines. All the above things can be done with the help of 3D Printing

Q.6 (2019)

Q. With reference to communication technologies, what is/are the difference/differences between LTE (Long-Term Evolution) and VoLTE (Voice over Long-Term Evolution)?

1. LTE is commonly marketed as 3G and VoLTE is commonly marketed as advanced 3G.
2. LTE is data-only technology and VoLTE is voice-only technology.

Select the correct answer using the code given below.

- | | |
|-----------------|---------------------------|
| a) 1 only | b) 2 only |
| c) Both 1 and 2 | d) Neither 1 nor 2 |

Answer: D

LTE and VoLTE

- VoLTE is an HD voice calling Service over 4G LTE rather than 2G/3G network present we are using.
- VoLTE is a technology wherein you can simultaneously send voice and data over the network without diminishing quality of voice. In the case of LTE, if you make a voice call and you also keep your data connection on, the quality of voice will reduce.
- So, in order to make a good quality voice call, you will have to switch off data. You may have noticed that in case of 2G and 3G, while you are making a phone call, some phones will automatically stop data transfer so as to preserve the voice quality.
- But in case of VoLTE, the voice quality will not reduce even if your data connection is on. With VoLTE it is very easy to transmit telephone conversation over the data network.

Q.7 (2019)

Q. In the context of digital technologies for entertainment, consider the following statements:

1. In Augmented Reality (AR), a simulated environment is created and the physical world is completely shut out.
2. In Virtual Reality (VR), images generated from a computer are projected into real-life objects or surroundings.
3. AR allows individuals to be present in the world and improves the experience using the camera of smart-phone or PC.
4. VR closes the world, and transposes an individual, providing complete immersion experience.

Which of the statements given above is/are correct?

- a) 1 and 2 only **b) 3 and 4**
c) 1, 2 and 3 d) 4 only

Answer: B

- Augmented reality (AR) is an interactive experience of a real-world environment where the objects that reside in the real-world are enhanced by computer-generated perceptual information, sometimes across multiple sensory modalities, including visual, auditory, haptic, somatosensory and olfactory
- The origin of the word augmented is augment, which means to add or enhance something. In the case of Augmented Reality (also called AR), graphics, sounds, and touch feedback are added into our natural world to create an enhanced user experience.

- Cell phones apps and business applications by companies using augmented reality are a few of the many applications driving augmented reality application development.
- The key point is that the information provided is highly topical and relevant to what you want you are doing.
- Virtual reality (VR) is an experience taking place within simulated and immersive environments that can be similar to or completely different from the real world.
- Applications of virtual reality can include entertainment (i.e. gaming) and educational purposes (i.e. medical or military training). Other, distinct types of VR style technology include augmented reality and mixed reality.
- 1st and 2nd are wrong statement while, 3rd and 4th are the correct.

Q.8 (2019)

Q. Consider the following statements: A digital signature is

1. an electronic record that identifies the certifying authority issuing it
2. used to serve as a proof of identity of an individual to access information or server on Internet
3. an electronic method of signing an electronic document and ensuring that the original content is unchanged

Which of the statements given above is/are correct?

- | | |
|------------------|-----------------|
| a) 1 only | b) 2 and 3 only |
| c) 3 only | d) 1, 2 and 3 |

Answer: C

- A digital signature is basically a way to ensure that an electronic document (e-mail, spreadsheet, text file, etc.) is authentic. Authentic means that you know who created the document and you know that it has not been altered in any way since that person created it.
- Digital signatures rely on certain types of encryption to ensure authentication.
- Encryption is the process of taking all the data that one computer is sending to another and encoding it into a form that only the other computer will be able to decode.
- Authentication is the process of verifying that information is coming from a trusted source.
- These two processes work hand in hand for digital signatures.

Q.9 (2019)

Q. In the context of wearable technology, which of the following tasks is/are accomplished by wearable devices?

1. Location identification of a person
2. Sleep monitoring of a person
3. Assisting the hearing impaired person

Select the correct answer using the code given below:

- a) 1 only
- b) 2 and 3 only
- c) 3 only

d) 1, 2 and 3

Answer: D

- The terms “wearable technology”, “wearable devices”, and “wearables” all refer to electronic technologies or computers that are incorporated into items of clothing and accessories which can comfortably be worn on the body
- Examples of wearable devices include watches, glasses, contact lenses, e-textiles and smart fabrics, headbands, beanies and caps, jewelry such as rings, bracelets, and hearing aid-like devices that are designed to look like earrings.
- The implications and uses of wearable technology are far reaching and can influence the fields of health and medicine, fitness, aging, disabilities, education, transportation, enterprise, finance, gaming and music.

Q.10 (2020)

Q. Consider the following activities:

1. Spraying pesticides on a crop field
2. Inspecting the craters of active volcanoes
3. Collecting breath samples from spouting whales for DNA analysis

At the present level of technology, which of the above activities can be successfully carried out by using drones?

- a) 1 and 2 only
- b) 2 and 3 only
- c) 1 and 3 only
- d) 1,2 and 3**

Answer: D

- Without crop protection, including pesticides, more than half of the world's crops would be lost to insects, diseases and weeds. Pesticides are important.
- A crater is a bowl-shaped depression, or hollowed-out area, produced by the impact of a meteorite, volcanic activity, or an explosion. Craters produced by the collision of a meteorite with the Earth (or another planet or moon) are called impact craters.

Q.11 (2020)

Q. In India, the term “Public Key Infrastructure” is used in the context of

- a) Digital security infrastructure**
- b) Food security infrastructure
- c) Health care and education infrastructure
- d) Telecommunication and transportation infrastructure

Answer: A

- Public Key Infrastructure (PKI) is a technology for authenticating users and devices in the digital world. Under this system, one or more trusted parties digitally sign documents certifying that a particular cryptographic key belongs to a particular user or device. The key can then be used as an identity for the user in digital networks.
- Therefore, option (a) is the correct answer

Q.12 (2020)

Q. With the present state of development, Artificial Intelligence can effectively do which of the following?

1. Bring down electricity consumption in industrial units.
2. Create meaningful short stories and songs.
3. Disease diagnosis.
4. Text-to-Speech Conversion.
5. Wireless transmission of electrical energy.

Select the correct answer using the code given below:

- | | |
|-----------------------|---------------------------|
| a) 1, 2, 3 and 5 only | b) 1, 3 and 4 only |
| c) 2, 4 and 5 only | d) 1, 2, 3, 4 and 5 |

Answer: B

- Statement 1 is correct→Bring down electricity consumption in industrial units by using efficient usage of electricity supply by use of data analytics
- Statement 2 is incorrect→Creating meaningful short stories and songs is not possible by AI because of 'Meaningful' keyword in stories and songs can not be made by AI
- Statement 3 is correct→Disease diagnosis by Machine Learning models
- Statement 4 is correct→Text-to-Speech Conversion is possible by help of Natural Language Processing in AI
- Statement 4 is incorrect→Wireless transmission of electrical energy is unrelated to AI

Q.13 (2020)

Q. With reference to Visible Light Communication (VLC) technology, which of the following statements are correct?

1. VLC uses electromagnetic spectrum wavelengths 375 to 780 nm.
2. VLC is known as long-range optical wireless communication.
3. VLC can transmit large amounts of data faster than Bluetooth
4. VLC has no electromagnetic interference.

Select the correct answer using the code given below:

- a) 1, 2 and 3 only
- b) 1, 2 and 4 only
- c) 1, 3 and 4 only**
- d) 2, 3 and 4 onl

Answer: C

- Visible light communication (VLC) is a data communications variant which uses visible light between 400 and 800 THz (780–375 nm). VLC is a subset of optical wireless communications technologies. 1 is correct.
- The technology uses fluorescent lamps (ordinary lamps, not special communications devices) to transmit signals at 10 kbit/s, or LEDs for up to 500 Mbit/s over short distances.
- Systems such as RONJA can transmit at full Ethernet speed (10 Mbit/s) over distances of 1–2 kilometres (0.6–1.2 mi). It is not long range. 2 is incorrect.
- VLC has the characteristic, to resist electromagnetic interference. 4 is correct.
- It carries more data than WiFi and bluetooth. 3 is correct.

Q.14 (2020)

Q. With reference to “Blockchain Technology”, consider the following statements:

1. It is a public ledger that everyone can inspect, but which no single user controls.
2. The structure and design of blockchain is such that all the data in it are about cryptocurrency only.
3. Applications that depend on basic features of blockchain can be developed without anybody’s permission.

Which of the statements given above is/are correct?

- a) 1 only
- b) 1 and 2 only
- c) 2 only
- d) 1 and 3 only**

Answer: D

- Blockchain technology is a structure that stores transactional records, also known as the block, of the public in several databases, known as the “chain,” in a network connected through peer-to-peer nodes. Typically, this storage is referred to as a ‘digital ledger.’
- In simpler words, the digital ledger is like a Google spreadsheet shared among numerous computers in a network, in which, the transactional records are stored based on actual purchases.
- The fascinating angle is that anybody can see the data, but they can’t corrupt it. 1 is correct.
- Blockchain can be used for many purposes like Secure sharing of medical data, Music royalties tracking. 2 is incorrect. It is an extreme statement also.
- Applications that depend on basic features of the blockchain can thus be developed without asking anybody for permission or paying anyone. 3 is correct.

Q.15 (2022)

Q. With reference to Web 3.0, consider the following statements:

1. Web 3.0 technology enables people to control their own data.
2. In Web 3.0 world, there can be blockchain-based social networks.
3. Web 3.0 is operated by users collectively rather than a corporation

Which of the following given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Web 3.0 is in an internet where users own their own data, not corporations
- Global digital transactions are secure, online exchanges of information and value are decentralized.
- Instead of being owned by centralized institutions, the community will own the network.
- This means the governance, rule sets, and general operation of the platform will move back to the users, as will the larger share of the potential profits.
- Blockchain is imperative to this, working as a digital ledger.
- Users will have their profile on their own Web 3.0 wallet which can be utilized across any decentralized platform.

Q.16 (2022)

Q. With reference to “Software as a Service”, Consider the following statements.

1. SaaS buyers can customise the user interface and can change data fields.
2. SaaS users can access their data through their mobile devices.
3. Outlook, Hotmail and Yahoo! Mail are forms of SaaS.

Which of the statements given above are correct?

- (a) 1 and 2 only
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Software as a service (SaaS) is a software distribution model in which a cloud provider hosts applications and makes them available to end users over the internet.
- The public cloud provider manages all the hardware and traditional software, including middleware, application software, and security.
- SaaS makes it easy to "mobilise" the workforce because users can access SaaS apps and data from any Internet-connected computer or mobile device.
- Common examples are email, calendaring and office tools (such as Microsoft Office 365).
- Buyers can customize the user interface (UI) to change the look and feel of the program, as well as modify specific areas, such as data fields, to alter what data

Q.17 (2022)

Q. Which one of the following is the context in which the term “qubit” is mentioned?

(a) Cloud Services

(b) Quantum Computing

(c) Visible Light Communication Technologies

(d) Wireless Communication Technologies

Answer: B

- A qubit (or quantum bit) is the quantum mechanical analogue of a classical bit.
- In classical computing the information is encoded in bits, where each bit can have the value zero or one.
- In quantum computing the information is encoded in qubits.
- Qubits can be in a state of 0 or 1 or (unlike a classical bit) in a linear combination of both states.

Q.18 (2022)

Q. Consider the following communication technologies:

1. Closed-circuit Television
2. Radio Frequency Identification
3. Wireless Local Area Network

Which of the above are considered of Short-Range devices/technologies?

- (a) 1 and 2 only.
- (b) 2 and 3 only
- (c) 1 and 3 only

(d) 1, 2 and 3

Answer: D

- Closed Circuit Television or CCTV is a short range device.
- Radio-frequency identification (RFID) that is being used for Toll collection is a short range device.
- A Wireless Local Area Network (WLAN) is a wireless computer network that links two or more devices using wireless communication within a limited area such as a home, school, computer laboratory, campus, or office building.

Q.19 (2014)

Q. In addition to fingerprint scanning, which of the following can be used in the biometric identification of a person?

1. Iris scanning
2. Retinal scanning
3. Voice recognition

Select the correct answer using the code given below.

- a) 1 only
- b) 2 and 3 only
- c) 1 and 3 only

d) 1, 2 and 3

Answer: D

- Biometric verification is any means by which a person can be uniquely identified by evaluating one or more distinguishing biological traits.
- Unique identifiers include fingerprints, hand geometry, earlobe geometry, retina and iris patterns, voice waves, DNA, and signatures.

Q.20 (2015)

Q. With reference to 'Near Field Communication (NFC) Technology', which of the following statements is/are correct?

1. It is a contactless communication technology that uses electromagnetic radio fields
2. NFC is designed for use by devices which can be at a distance of even a metre from each other
3. NFC can use encryption when sending sensitive information.

Select the correct answer using the code given below.

- | | |
|------------------------|--------------|
| a) 1 and 2 only | b) 3 only |
| c) 1 and 3 only | d) 1,2 and 3 |

Answer: C

- Near-field communication is a wireless technology that allows users to make payments by placing a compatible device like a smartphone or payment card within a few centimetres of another compatible device like a terminal, tablet or another smartphone.
- Near-field communication transmits data through electromagnetic radio fields and is the technology behind payment services like Google Wallet and Apple Pay.
- Both devices must contain NFC chips for an NFC transaction to take place.